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JSSG-2007A
29 January 2004

SUPERSEDING
JSSG-2007
30 October 1998

DEPARTMENT OF DEFENSE
JOINT SERVICE SPECIFICATION GUIDE

ENGINES, AIRCRAFT, TURBINE

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JSSG RELEASE NOTE

This specification guide supports acquisition reform initiatives, and is predicated on a performance-based business environment approach to product development. As such it is intended to be used in the preparation of performance specifications. It is one in a set of specification guides. This document will continue to be improved as the development program is accomplished.

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JOINT SERVICE SPECIFICATION GUIDE

ENGINES, AIRCRAFT, TURBINE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope.

This Aircraft Turbine Engines Joint Service Specification Guide (JSSG) and its handbook appendices provide rationale, guidance, background, lessons learned, and instructions necessary to tailor sections 3 and 4 of JSSG-2007A for a specific application. These applications include fixed and rotary wing, wide body, manned, and unmanned air vehicles.

1.2 Purpose.

This specification is intended for Government and Industry program teams as guidance in developing program-unique specifications.

1.3 Use.

This document consists of three parts. Part 1 is the JSSG that is a template for developing the program-unique performance specification. As a generic document, it contains requirement statements for the full range of aviation sector applications. It must be tailored to delete nonapplicable requirements to form the program-unique specification. In addition, where blanks exist, these blanks must be filled in for the program-unique specification to form a complete and consistent set of requirements to meet program objectives. Part 2 is the appendix A handbook which provides the rationale, guidance, and lessons learned relative to each statement in Part 1. The section 4, verification requirements, must be tailored to reflect an understanding of: (1) the design solution; (2) the identified program milestones; (3) the associated level of maturity which is expected to be achieved at those milestones; and (4) the specific approach to be used in the design and verification of the required products and processes. It must be recognized that the rationale, guidance, and lessons learned are not only generic in nature, but also document what has been successful in past programs and practices. This must not be interpreted to limit new practices, processes, methodologies, or tools. Part 3 is the appendix B handbook that provides rationale, guidance, and lessons learned to establish an engine model specification for the production phase of the engine program.

1.3.1 Program-unique applications.

Different program applications require different levels of requirements. Manned systems will often include additional requirements having to do with aircrew safety and survivability where an unmanned system would not. Rotary wing systems have unique components and subsystems not found on fixed wing applications. Wide body systems (cargo, tanker, transport) usually have more benign missions than fighters. Through careful tailoring of section 3 and 4 requirements and table XXXI, JSSG-2007A can be used to develop a comprehensive, performance-based engine specification for any air system application.

DoD is placing increased emphasis on the development of unmanned air vehicle (UAV) weapon systems. The requirements contained in JSSG-2007A can be tailored for UAV programs using table XXXI (i.e., high-value UCAVs and UAVs such as Global Hawk). AFGS-87271 may be used for additional tailoring guidance for UAV systems, (expendable missile/target engines). AFGS-87271 is a controlled document (Distribution D) with distribution limited to Department of Defense and DoD Contractors only.

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1.3.1.1 Commercial Off The Shelf (COTS) applications.

The use of Commercial Off The Shelf (COTS) systems may rely heavily on alternate verification and certification approaches such as those used by the FAA or Joint Aviation Authority (JAA). Every effort should be made to ensure all qualified missions of COTS applications meet military applications. Differences may exist between commercial and military missions and usage. These differences may force re-verifications of some JSSG-2007A requirements reflected in table XXXI. Differences may force the user to use information available other that what was used by the civilian airworthiness authority to meet military requirements. Fleet usage information in addition to the qualification information should be utilized as an aide to evaluate the differences for applicable requirements in table XXXI with moderate/high risk.

1.4 Performance Based Product Definition Guide (PBPDG).

For additional guidance on data and deliverable characteristics, the specification developer should consult the Performance Based Product Definition Guide (PBPDG). The PBPDG is a product of the Joint Aeronautical Commanders Group (JACG). A copy may be requested from the JACG.

1.5 Interoperability and Joint Technology Architecture.

Joint Technology Architecture (JTA) is to improve and facilitate the ability of systems to support joint and combined operations. The air vehicle JTA will be used to achieve interoperability for new and upgraded systems (and the interfaces to such systems). Where the JTA mandates a standard, this standard should be considered for incorporation in this document. Additional information on JTA may be found online at http://www-JTA.ITSI.DISA.mil.

2. APPLICABLE DOCUMENTS.

2.1 General.

The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks.

The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMERCIAL ITEM DESCRIPTIONS

A-A-52557	Fuel Oil, Diesel; for Posts, Camps and Stations

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-5624	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-PRF-7808	Lubricating Oil, Aircraft Turbine Engine, Synthetic Base
MIL-PRF-16884	Fuel, Naval Distillate

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MIL-PRF-23699	Lubricating Oil, Aircraft Turbine Engine, Synthetic Base, NATO Code Number O-156
MIL-PRF-25017	Inhibitor, Corrosion/Lubricity Improver, Fuel Soluble
MIL-DTL-83133	Turbine Fuels, Aviation, Kerosene Types, NATO F-34(JP-8), NATO F-35, and JP-8 + 100
MIL-DTL-85470	Inhibitor, Icing, Fuel System, High Flash, NATO Code Number S-1745

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-810	Environmental Engineering Considerations and Laboratory Tests

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-310	Global Climatic Data for Developing Military Products
MIL-HDBK-1783	Engine Structural Integrity Program (ENSIP)

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or www.dodssp.daps.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia PA 19111-5094; phone (215) 697-2667.)

2.2.2 Other Government documents, drawings, and publications.

The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

UNITED STATES ARMY

FM 3-5	NBC Decontamination

(Application for copies should be addressed to USA AMCOM, Engineering Directorate, AMSAM-AR-E, Redstone Arsenal AL 35898.)

DEPARTMENT OF DEFENSE

DODD 5000.1	The Defense Acquisition System
DODI 5000.2	Operation of the Defense Acquisition System

(These documents are available online at http://www.dtic.mil/whs/directives/. Application for paper copies should be addressed to DAPS, ATTN: DODSSP, Building 4D, 700 Robbins Avenue, Philadelphia PA 19111-5094.)

2.3 Non-Government publications.

The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D1655	Aviation Turbine Fuels
ASTM D4814	Automotive Spark-Ignition Engine Fuel
ASTM D5001	Standard Test Method for Measurement of Lubricity of Aviation Turbine Fuels by the Ball-on-Cylinder Lubricity Evaluator (BOCLE)
ASTM D6615	Jet B Wide-Cut Aviation Turbine Fuel

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(Application for copies should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken PA 19428-2959; www.astm.org.)

SAE INTERNATIONAL

SAE AS 681	Gas Turbine Engine Steady-State and Transient Performance Presentation for Digital Computer Programs
SAE AS 755	Aircraft Propulsion System Performance Station Designation and Nomenclature

(Application for copies should be addressed to SAE International, 400 Commonwealth Drive, Warrendale PA 15096-0001; www.sae.org.)

MISCELLANEOUS

Defence Standard 91-86	Turbine Fuel, Aviation Kerosine Type: High Flash Type, Containing Fuel System Icing Inhibitor; NATO Code: F-44; Joint Service Designation: AVCAT/FSII
Defence Standard 91-87	Turbine Fuel, Aviation Kerosine Type, Containing Fuel System Icing Inhibitor; NATO Code: F-34; Joint Service Designation: AVTUR/FSII
Defence Standard 91-88	Turbine Fuel, Aviation: Wide Cut Type, Containing Fuel System Icing Inhibitor; NATO Code: F-40; Joint Service Designation: AVTAG/FSII
Defence Standard 91-91	Turbine Fuel, Aviation Kerosine Type, Jet A-1; NATO Code: F-35; Joint Service Designation: AVTUR

(Ministry of Defence standards are available online at www.dstan.mod.uk.)

GOST 10227	Russian State Standard for TS-1, Kerosene-Type Fuel

(Application for copies should be addressed to National Technical Information Service (NTIS), 5285 Port Royal Rd, Springfield VA 22161; www.ntis.gov.)

2.4 Order of precedence.

In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

2.5 Streamlining.

This specification has been streamlined. Those documents listed in Appendix A which are required for acquisition have the same status as those referenced directly in section 2 (first tier documents). All other documents referenced through tiering may be used for guidance and information.

3. REQUIREMENTS.

The paragraphs of this specification and all references herein comprise the complete set of engine requirements for the gas turbine engine(s).

3.1 Design and construction.

An open system approach (see 6.3) shall be followed for the engine design where feasible.

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3.1.1 Item and interface definition.

3.1.1.1 Interface and installation.

Engine and air vehicle interfaces shall be as shown in ____. The input and output from the air vehicle to engine shall be defined in ____. The control system interface shall be compatible with and shall receive and process air vehicle thrust or power or torque requests via input signals for the control system. Control system interpretation of power lever angle signals shall be defined in _____.

3.1.1.2 Installation hardware.

The engine shall be suitable for installation in any air vehicle configuration for multi-engine aircraft with different engine locations. Parts requiring reconfiguration shall be minimized and controls and externals shall not have to be relocated for different engine installations on multi-engine aircraft. Parts that require reconfiguration are ___.

3.1.1.3 Interface loads.

The maximum allowable loads for all parts that interface with the air vehicle shall be as shown in ____.

3.1.1.4 Mounts.

3.1.1.4.1 Main mounts.

The engine mounts shall retain the engine, including retained fluids and externals, at all flight, takeoff and landing, and ground conditions and shall withstand elastic limit loads of __(a)__ without permanent deformation and ultimate tensile strength loads of __(b)__ without complete fracture. The main mounts shall meet the requirements of 3.4.1.7. The locations of all engine mounts shall be as specified in __(c)__.

3.1.1.4.2 Ground handling mounts.

The ground handling mounts shall support the engine, including all engine mounted equipment and externals, components, and operating fluids, under the following maximum inertia load conditions, without deformation to the mounts or damage to the engine: _(a)_ acting in combination at the engine center of gravity.

The locations and descriptions for the individual ground handling mounts shall be as specified in _(b)_. The arrangement shall be compatible with _(c)_.

3.1.1.4.3 Engine stiffness.

For turboshaft and turboprop engines, the estimated stiffness of the engine in resisting loads and moments applied at the outboard end of the output shaft, relative to the engine mounting points, shall be specified in ___. The first “free-free” lateral and vertical engine bending modes shall be specified therein.

3.1.1.5 Control lever and signals.

3.1.1.5.1 Power command lever and signal.

The power command lever positions and signals shall be specified in the _____. Engine power response shall be continuous and proportionally constant in relation to engine power lever input.

For multi-power section applications a signal shall be provided to indicate the power section on-off status.

3.1.1.5.2 Load demand command and signal.

If required for turboshaft engines, a means shall be provided at the control interface to adjust for rapid power absorber load changes. The control system interface shall provide for compatibility between the air vehicle load demand output signal and the control system. Engine power response shall be continuous and proportionally constant in relation to engine power lever input.

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3.1.1.5.3 Output speed lever and signal.

If required for turboprop or turboshaft engines, a separate shaft speed lever and signal, if required, shall be provided to adjust and control the output shaft speed. The control system interface shall provide for compatibility between the air vehicle output speed lever output signal and the control system. No abrupt change in output shaft speed shall occur with linear shaft speed demand input.

3.1.1.5.4 Fuel shutoff lever and signal.

The fuel control shall process air vehicle signals as required in the _____ for fuel shutoff.

3.1.1.5.5 Lever torque.

For systems with mechanical interfaces, the torque required to operate the engine thrust or power level or any additional levers necessary to control the engine through their range of travel shall be no greater than _____ throughout all flight loads and conditions. The lever(s) shall not move with the engine operating unless external torque is applied. Movement of the lever(s) throughout the operating range shall be free of abrupt changes in actuating torque.

3.1.1.6 Subsystem interfaces.

The engine shall receive and transmit signals as required by the air vehicle and ground support systems. The signals, signal characteristics and physical interface shall be shown herein. Engine subsystems that interface with the air vehicle and ground support system are:

_____ Subsystem (Contractor to specify subsystem):

Input	Parameter	Function

Output	Parameter	Function

Digital message format (data) and timing requirements (transmission rate) communications between the air vehicle and engine control and monitoring computers shall be specified herein.

3.1.1.7 Bleed air interface.

The engine shall provide customer bleed air extraction for aircraft use. The capabilities and maximum flow of each bleed port, including the acceleration or start bleed air ports, shall be as shown in ___(a)___. Bleed duct external surface temperatures shall meet the requirements of 3.1.8.2. The bleed port internal pickup points shall be located at positions which have low susceptibility to FOD, and ingestion of sand, dust, ice, moisture, and any other foreign materials contained in the air. Provisions shall be made to prevent high-pressure bleed air from entering lower pressure bleed ports. A failure in the air vehicle bleed system shall result in __(b)____ engine operation. If compressor venting is necessary, the temperature, pressure and airflow inflow and outflow conditions for which venting provisions must be made shall be specified in ___(c)____. The ducting attachment detail shall be shown on the engine interface and installation drawings.

3.1.1.7.1 Customer bleed air contamination.

Engine generated substances contained in customer bleed air shall be below the threshold limit of __.

3.1.1.8 Drains and fluid collection devices.

Fluid drains, drain characteristics, and collection requirements shall be shown in ___. There shall be no leakage beyond ___ cc/hr of fluids from the engine except at the drains provided for this purpose. The

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flow rate into all drains and the maximum amount of combustible fluid drainage following shutdown after normal operation shall be as shown in: ___. Drains shall function statically and throughout the operating envelope. Drains for different fluid types shall not be interconnected.

3.1.1.9 Power absorber interface.

For turboshaft and turboprop engines, the allowable range of characteristics of the shaft power absorber at the power absorber to engine interface shall be as shown in: __(a)___. No resonant frequency shall be transmitted to or from the power absorber through the engine.

The output drive shaft shall be capable of continuously absorbing thrust in either direction of an amount not less than __(b)__ percent of the circumferential force acting on the output drive shaft spline at the steady-state torque limit as specified in the engine specification.

The engine shall provide a signal for operation of a torque indicator throughout the operating envelope of the engine. The signal shall be continuous as a function of torque. The specified accuracy shall be maintained without recalibration after the torque sensor is disassembled and reassembled. The accuracy of the torque sensor, from __(c)__ percent rated delivered shaft speed to __(d)__ percent rated delivered shaft speed, shall be:

a. Within plus or minus __(e)__ percent of the torque at maximum continuous rated power from zero delivered shaft torque to the torque at maximum continuous rated power.

b. Within plus or minus __(f)__ percent of the torque being measured from the torque at maximum continuous rated power to the transient torque limit.

3.1.1.10 Power takeoff (PTO).

When PTO pads and drives are provided for driving aircraft accessories, the ratings, clearance envelopes, dimensions and other pertinent information shall be as shown in: ____. The design and configuration of the PTO pads and drives shall be consistent with the requirements of 3.7.16.

3.1.1.11 Starting drivetrain.

The engine starting torque acceptance capability shall be specified in _____, shall be sufficient to prevent inadvertent starter drivetrain failure with impact loading, and shall meet the requirements of 3.1.1.11.1 and 3.2.2.3.1. Failure of the air vehicle engine starting drivetrain shall not result in loss of the remaining gearbox functions. The starter drive pad characteristics shall be included in table I. The direction of rotation when facing the starting pad on the engine shall be specified herein. No resonant frequency shall be transmitted to or from the starting drivetrain through the engine interface.

3.1.1.11.1 Starting torque and speed.

The required starting torques and drive speeds shall be specified in _____ and shall include engine drag and engine gearbox drag. Figure 2 shall show the effects, singly and in combination, of ambient temperatures, of altitudes from sea level to the maximum of 3.2.2.3.3, and for air starts with starter assist, using the worst-case fuels of the specification. Figure 2 shall be presented for each of the following:

a. No customer bleed air extraction, customer power extraction as specified in the model specification.

b. Maximum allowable customer bleed air extraction, customer power extraction as specified in the model specification.

3.1.1.12 Thrust reverser interface.

The thrust reverser shall deploy and stow after receiving the appropriate command signals. Positive mechanical locking shall prevent reverser deployment in the event of system failure. Failure of the locking system shall not cause reverser deployment. No resonant frequency shall be transmitted to or from the thrust reverser interface.

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3.1.1.13 Exhaust system interface.

Maximum allowable static (1g) shear, axial and moment loads in all primary axes (x, y, z, as applicable) for any interface connections between the engine and parts of the exhaust system not supplied with the engine shall be specified in ___. The maximum allowable loads at any exhaust interface shall also be specified for the maximum allowable maneuver loads as defined in 3.4.1.2.1. No hot exhaust gas leakage is allowed at the interface. No detrimental resonant frequency shall be transmitted to or from the exhaust system through the interface.

3.1.1.14 Compatibility - engine and air vehicle.

The interface of the engine with the air vehicle shall allow the engine to perform as required in the specification.

3.1.1.15 Control and external components list.

The control, external components, and component functional subsystems of the engine, which require qualification testing separate from the engine, shall be listed, see 4.11.2.

3.1.2 Physical characteristics.

3.1.2.1 Dry weight of complete engine.

The weight(s) for the engine shall not exceed:

CONFIGURATION	WEIGHT KG (LBS)	CG
complete engine, dry		
engine exhaust system (if a separate system), dry		
residual fluids (draining attitude ___° from horizontal)		
operating fluids		
items part of the engine parts list not mounted on the engine (identify and list each item)		

3.1.2.2 Mass moment of inertia of complete engine.

The maximum effective mass moment of inertia kg-m^2 (lbm - ft^2) of the complete operational engine about three mutually perpendicular axes with the origin at the center of gravity, shall be specified in _____. The maximum effective mass moment of inertia kg-m^2 (lbm - ft^2) about the resultant rotational axis of each engine rotor, together with the direction of rotation when viewed looking forward from the exhaust exit, shall also be specified in _____.

3.1.3 Materials, processes, and parts.

The materials, coatings, and processes used in the design and manufacture of the complete engine shall be chosen, applied and utilized such that the engine shall meet the requirements of this specification.

A.3.1.3.1 Hazardous materials.

The use of any banned material, as listed in the specification, shall not be permitted in the construction, operation, maintenance, storage, transportation or disposal of engine delivered hardware.

The use of any highly-restricted materials, as listed in the specification, shall be subject to procuring activity approval prior to use in the construction, operation, maintenance, storage, transportation or disposal of engine delivered hardware.

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3.1.4 Fasteners.

The loss of a fastener shall not affect safety of flight or ability to control the engine.

3.1.5 Nameplate and product marking.

Equipment, assemblies, modules, and parts shall be marked legibly and such that component removal is not necessary to read the data plate. The type and length of product identification characters on the data plate shall be compatible with the Using Service's maintenance tracking system. The color red shall not be used for marking unless adjacent non-red color marking is easily readable in red light conditions.

3.1.5.1 Engine data plate marking.

The engine data plate shall include: manufacturer's identification, engine serial number, purchase order or contract number and engine model designation.

3.1.5.2 Critical parts/critical safety items identification and tracking.

The engine life critical parts/critical safety items and subassemblies shall be identified by serial numbers marked legibly on the part. The parts shall have multiple, non-wearing reference surfaces and have a space designated for marking the number of cycles and time accumulated between each overhaul period.

Engine critical parts shall be identified and managed through a Critical Item Management Plan.

3.1.6 Transportability.

The engine and individual components shall be suitable for transportation as follows:

Features shall be provided to permit installation on and use of appropriate static and mobile ground equipment.

3.1.7 Interchangeability.

All parts and modules having the same manufacturer's part number shall be functionally and dimensionally interchangeable. Matched parts or selective fits shall be permitted upon Using Service approval. Complete engines shall be interchangeable in multi-engine aircraft (i.e., right side to left side).

3.1.8 Safety.

3.1.8.1 Flammable fluid systems.

All exterior lines, fittings and components which contain fuel shall be fire resistant (for at least five-minutes at a minimum flame temperature of 1093°C (2000°F)). The lubrication oil system exterior lines, fittings and components and hydraulic fluid system exterior lines, fittings and components shall be fireproof (for at least 15-minutes at a minimum flame temperature of 1093°C (2000°F)). During exposure to the above conditions, the lines and components shall be conveying fluids under the worst possible operational combination of fluid parameters (i.e.; flow rate, pressure, and temperature) encountered throughout the complete environmental conditions and operating envelope of the engine.

3.1.8.2 Fire prevention.

Where engine surface area temperatures are sufficient to ignite flammable fluid that could impinge on the engine, these areas shall be specifically identified in _____. If a fire shield is required, it shall be identified in _____.

3.1.8.3 Explosion-proof.

All electrical components (except igniter plug electrodes and devices intended to ignite fuel air mixtures) shall not ignite any explosive mixture surrounding the equipment.

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3.1.8.4 Combustible fluid drains.

Provisions shall be made to prevent entrapment of fuel after each false start and for preventing excess combustible fluids from entering the combustion areas after shutdown with the engine in a range of positions from _(a)_ degrees nose up to _(a)_ degrees nose down. For turboshaft engines, no pilot or maintenance action shall be required to clear the engine of fuel after each false start or following engine shutdown.

For engine applications in V/STOL, STOVL aircraft requiring engine attitude changes, the drain limits of 3.1.1.8 shall not be exceeded for combustible fluids after engine shutdown with the engine in a range of positions from __(b)___ nose up to __(b)___ nose down. Provisions shall also be made for clearing all vent areas and other pockets or compartments where combustible fluids may collect during or subsequent to operation of the engine. The engine shall be designed so that combustible fluids cannot enter the combustion area when the engine is in, or turned to, a vertical position for maintenance.

The fluid drain system shall be configured to collect and route fuel and oil as specified in the __(c)____.

3.1.8.5 Air and gas leakage.

There shall be no air and gas leakage of sufficient temperature and concentrated impingement to present a safety hazard or affect installation requirements or adversely affect aircraft nacelle equipment.

3.1.8.6 Ground safety.

Warning notices shall be provided, where applicable, for high voltage ignition systems and other high voltage electrical sources, radioactive devices, explosive devices, and high-powered optical sources.

3.1.8.7 Jet wake.

Jet wake temperature and velocity profiles at sea level static standard day conditions when operating at idle, intermediate, and maximum shall be as shown on figure 4. Areas hazardous to personnel and equipment shall be appropriately defined and marked on figure 4.

3.1.9 Design control.

3.1.9.1 Parts list.

The parts list for the engine including software version that successfully completes the designated engine milestone verifications shall constitute the approved parts list. Changes to the approved parts list shall require approval by the Using Service prior to parts incorporation. Prior to the initiation of IFR, FFR, ISR, and OCR, the contractor shall submit a proposed parts list, which shall include hardware and software configuration. Hardware configuration shall be identified by part number and associated drawings. Software configuration shall be identified by a specific source code listing and associated logic diagrams. The parts list for the engine configuration that successfully completes the IFR, FFR, ISR, and OCR verifications shall be the approved parts list for the respective engine model. The parts list shall be updated, as required, by changes incorporated in the parts list approved by the Using Service.

3.1.9.2 Assembly of components and parts.

Equipment, parts, and components that are not structurally or functionally interchangeable with other parts on the same engine shall not be physically interchangeable unless approved by the Using Service. Parts and components shall be designed such that it is impossible to install them incorrectly (e.g.; backwards, upside down, reversed in an assembly, or installed in the wrong location in an assembly.) Connections located in close proximity to each other shall be made physically non-interchangeable.

3.1.9.3 Changes in vendors or fabrication process.

Changes in any vendor, fabrication process, or fabrication source for any component or part shall be subject to all requirements specified herein.

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3.1.9.4 Standardization.

Standardization principles, standard parts, materials, processes, tools, subsystems, and components shall be used to the maximum extent possible without compromise in design, performance, operability, or economic life of the engine. All parts, materials, and processes, whether or not identified as a Government, industry, or contractor standard shall be qualified for the intended use as a part of the qualification specified herein. Items already in the Government inventory shall be used to the maximum extent possible where suitable for the intended purpose. Variation in similar components or parts shall be held to the absolute minimum. Proprietary designs shall be kept to a minimum. Under conditions wherein economics of production conflict with standardization objectives, the Using Service shall be requested to select the component desired for use. Military and industry standards and parts developed for engines in air vehicles shall be used unless they are unsuitable for the intended purpose. Open Systems (see 3.1 and 6.3 for definition) shall be used to the maximum extent possible.

3.2 Performance and operability.

3.2.1 Performance characteristics.

3.2.1.1 Steady-state performance.

The minimum engine steady-state performance requirements shall be defined by the steady-state performance computer program (a) dated (a), within the range of effects specified in 3.2.1.3. The specification computer program shall calculate minimum performance for both new engines and engines deteriorated for the time period specified in 3.2.1.4. Cycle calculations for the minimum engine computer program shall be based on an average engine with thrust or power and fuel flow related output adjusted to minimum engine performance as follows: (b). The relationship between minimum and average performance shall be specified throughout the operating envelope of 3.2.2.1. Tables IIa, IIb, IIc, and table III shall provide reference performance points from the performance computer program for new and deteriorated engines (c). In the event of conflict between the tables and the performance computer program, (d) shall take precedence.

For turboprop and turboshaft engines, the first stage turbine rotor inlet temperature for each rating established at (e) conditions shall be constant, for all atmospheric and Mach number conditions except when limited by fuel flow, compressor aerodynamics, or torque limits.

3.2.1.2 Transient performance.

The status or average engine transient performance requirements shall be modeled by the transient performance computer program (a) dated (a), per the conditions of 3.2.1.3. Engine thrust or power transient times from the transient computer program shall be specified in 3.2.2.7, as required therein.

3.2.1.3 Performance computer program.

The specification performance computer program shall be prepared per the practices of SAE AS 681. In addition, the program shall be capable of representing the effects of the following parameters on the engine cycle performance, singly and in any combination:

a. Fuel lower heating value at constant pressure of (a) BTU/lbm, and otherwise conforming to the fuel specified in 3.7.3.1.

b. Fuel specific gravity of (b).

c. Standard and non-standard atmospheric conditions of (c).

d. Inlet pressure recovery of (d).

e. Exhaust pipe and jet nozzle, fan exit duct and fan nozzle, and engine front and rear frame features specified in (e).

f. Customer bleed air extraction (f).

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g. Customer power extraction (g).

h. Using the engine control system specified in (h).

i. Humidity (i).

j. For turboshaft/turboprop engines, the shaft torque absorber characteristics specified in 3.1.1.9 (j).

k. Stall margin and operating line consistent with requirements when matched with vehicle flight exhaust and distortion requirements as specified in (k).

l. Inlet distortion of (l)

3.2.1.4 Performance retention.

Unless otherwise specified, during and at the completion of one cold parts life, the engine thrust or power shall not fall more than (a) below the minimum engine thrust or power requirements specified in 3.2.1.1 and 3.2.1.2. at (b) temperature. During and at the completion of the specified hot section life, the engine thrust or power shall be no less than (c) percent of the actual thrust or power at the beginning of the hot section life at (b) temperature. The specific fuel consumption (SFC) shall be no more than (d) percent of the actual SFC at the same thrust or power at the beginning of the hot section life at (b) temperature.

3.2.2 Operating characteristics and limits.

All engine steady-state and transient operating limits (maximum, minimum) shall be specified in table IV and figure 6. The specified limits shall be predicated on the most critical tolerances of the engine.

3.2.2.1 Operating envelope.

The engine shall meet all the requirements of the specification throughout the complete operating envelope without exceeding any limits. The engine operating limits shall be specified for the environment of figure 5, per the format shown on figures 6 and 7 and in tables IIa, IIb, IIc, and III. If applicable, the thrust augmentation-operating envelope shall be included on figures 6 and 7.

3.2.2.2 Operating attitude and conditions.

The engine operating attitude limits shall be shown on figure 8. The engine shall meet the requirements of the specification when operating in the unshaded area of figure 8, and operate continuously at least (a) seconds in the shaded areas of figure 8. Operation in the shaded area shall not degrade engine performance or cause any damage. The engine shall start, stop, and be stowed in any of the attitudes shown in the unshaded area of figure 8. Engine stowing capability outside of the unshaded area of figure 8 shall be specified herein. The engine shall function satisfactorily for at least (b) seconds in negative g and for at least (c) seconds in zero g conditions. The engine shall meet all requirements of this specification with unlimited occurrences of negative and zero g flight defined above.

3.2.2.3 Starting.

The engine shall satisfactorily make ground and air starts, and shall restart under the conditions, limitations, and procedures, specified in this section. The engine shall start with inlet air at any temperature between and including the extreme hot and cold atmospheres of figure 5, with fuel inlet temperatures as specified in 3.7.3.2.2. Starts shall be made with the primary fuel(s) of 3.7.3.1.1 and oil(s) of 3.7.8.1, unless specified otherwise. A start shall be considered satisfactory when the engine rotor is accelerated from either rest or wind milling speed to idle, when the minimum assist torque is provided by a starter or ram air, as specified in 3.1.1.11, and the engine stays within the engine starting and operating limits of figure 6. Starting time shall be measured from the initiation of the starting sequence to the attainment of a stabilized uncorrected engine idle rotor speed.

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3.2.2.3.1 Ground starts.

The engine shall start at the specified ambient temperatures and altitudes, in times equal to or less than those shown on figure 9a. The engine shall be capable of _(a)_ consecutive start cycles, with a maximum time interval of _(b)_ seconds between the completion of one cycle and the initiation of the next cycle. The wind condition envelopes for ground starting shall be shown on figure 9b.

Special starting requirements apply as follows:

a. Hot temperature soak start. The engine shall start, after a _(c)_ hour hot soak at the temperatures specified on figure 5, using _(d)_ fuel, and with inlet air, fuel, and oil at the maximum hot ambient temperatures shown on figure 5. Following shutdown, the engine shall start, after a resoak period of _(e)_ minutes at the temperatures specified on figure 5, using _(d)_ fuel, and with inlet air, fuel, and oil at the maximum hot ambient temperature shown on figure 5.

b. Cold temperature soak start. The engine shall start, using _(f) fuel_ and _(g)_ oil, after a cold soak for a period of _(h)_ hours with inlet air, fuel, and oil at the minimum cold ambient temperature shown on figure 5.

3.2.2.3.2 Air starts.

Windmilling, spooldown, and starter assisted airstart envelopes and maximum start times, with customer extractions shall be defined in _(a)._ Air starts within figure 10 shall be considered acceptable when the engine achieves idle speed within the air start times specified and no starting or operating limits are exceeded. The engine shall be capable of consecutive start cycles, with a maximum time interval of _(b)_ seconds between the completion of one cycle and the initiation of the next cycle. Starting time shall be measured from the initiation of the starting sequence to the attainment of _(c)_ percent of stabilized uncorrected engine idle rotor speed. During air starts the engine rotor load shall be at least _(d)_ percent of the total maximum accessory drive load specified for the drives shown in table I.

For turboprop engines that are shut down in flight on patrol aircraft, the engine shall start at the post loiter altitude start envelope of figure 7 after being inoperative for at least _(e)_ hours.

3.2.2.3.3 Starting limits.

The starting altitude for ground starts shall be up to _(a)_ km, (_(a)_ ft). The maximum turbine temperature for ground and air starts shall be specified _(b)._ Differences in the altitude starting limits for “hot” and “cold” engines shall be specified _(b)_. Limits with maximum customer bleed air and maximum customer power extraction shall be specified _(b)_. For turboprop applications, the post loiter altitude start limit shall be specified _(b)_.

3.2.2.3.4 Starting procedure.

The starting procedure shall be simple and shall not require critical timing. The engine shall be capable of starting with the thrust or power demand at idle or above and, after initiation of the starting sequence, being accelerated immediately to any power request. This shall be accomplished within specified engine starting limits and without stall.

3.2.2.3.5 Automatic relight.

The engine shall incorporate an automatic rapid relight system that shall immediately reestablish engine operation (without any pilot action) after a main combustor flameout or augmentor blowout has occurred. The automatic relight system shall detect any flameout that may have occurred and initiate a sequence for automatic recovery anywhere in the automatic relight operating envelope shown on figure 7.

3.2.2.4 Stopping.

Upon receiving a signal from the air vehicle for stopping, the engine shall terminate fuel flow in any operating condition.

Stopping of the engine from any power setting (including augmentor) or at any rate shall not:

a. result in immediate or subsequent exceedance of any engine limits

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b. adversely impact engine durability, structural integrity, or operational capability

c. cause unacceptable rotor bowing or rotor seizure, and

d. delay a satisfactory start or augmentor light.

The engine shall not experience any post-shutdown fires. No damage to the engine shall result from shutting off the fuel supply by the foregoing means or from shutting off the fuel supply to the engine inlet connection during any engine operating condition.

Provisions for stopping the engine by means other than a completely mechanical system shall be subject to approval of the Using Service.

3.2.2.5 Lower power conditions.

3.2.2.5.1 Idle thrust or power - (ground or flight idle).

When the engine incorporates provisions for an additional idle power setting (e.g., ground idle), the performance shall be included in table II, and the means of accomplishment shall be specified herein or in the ICD.

3.2.2.5.2 No-load condition.

For turboprop and turboshaft engines, the no-load condition shall be an engine operating condition of zero delivered output shaft torque while maintaining constant steady-state governed output shaft speed with a specified power absorber and operating at altitudes up to _____ km (____ ft). Engine gas generator speed at no-load shall be specified in the performance computer program of 3.2.1.3.

The engine(s) and torque absorber system shall be capable of transitioning between load and no-load conditions without unacceptable mismatches in rotor speeds or torques defined in 3.2.2.7, and without exceeding gas generator operating limits.

3.2.2.6 Stability.

Under steady-state operating conditions, engine thrust or power fluctuations shall not exceed (a) percent of the intermediate or maximum continuous between idle and intermediate or maximum continuous conditions, or (b) percent of the thrust or power available at the power level position and operating condition, whichever is less. During operation above intermediate up to maximum augmentor, fluctuations shall not exceed (c) percent of the thrust or power available at that condition. During steady-state operating conditions, the period between major discrete frequency fluctuations, as defined below, (see 3.7.11 Guidance Background) shall not be less than (d) seconds. Main combustor and augmentor blowout margins shall be sufficient to protect against blowouts or flameouts during rapid throttle movements under worst-case conditions, and engine control and main burner and augmentor stackups. Bleed air extraction shall not require thrust or power demand modulation to maintain engine stability and limits. The maximum allowable variation shall never be less than (e) thrust or power.

3.2.2.7 Transients.

The engine shall satisfactorily perform any thrust or power transient throughout the operating envelope while meeting the requirements of the specification. Thrust or power requests in any sequence and at any rate for both primary and backup control modes shall not result in exceeding any engine operating limit (including overspeed and overtemperature), result in unstable operation of the engine, or cause any mechanical damage.

For engines operating in backup control modes, thrust or power transient times shall be within (a) of the primary control mode.

For engines operating in primary control mode, thrust or power transient times in all backup control modes shall be within (a) of the primary control mode. Thrust or power transient times for installed engine conditions, including the effects of customer bleed air, customer power extraction, anti-icing bleed, inlet distortion, and the exhaust nozzle, singly or in any combination, for non-standard days throughout

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the complete environmental conditions and operating envelope, shall not exceed __(b)__ percent of the standard day thrust or power transient times.

For thrust or power lever movements of _(c)_ degrees/second, the transient times shall be the slowest thrust or power transient times for standard day conditions with no customer power or bleed air extractions and no engine anti-icing bleed air operation but with all other engine system bleed air requirements (e.g.; acceleration bleed air, cooling bleed air).

The times required for the engine to accomplish _(d)_ % of the thrust or power change, and the total time required to accomplish the full transient and reach transient stability shall be specified below.

For a turboshaft engine in a helicopter application, the times required for the engine to accomplish _(d)_ % of the power change and the total time required to accomplish the full transient and reach transient stability when the power absorber load is applied or removed per figure 11 shall be specified below.

Continuous sinusoidal throttle transients with peak-to-peak thrust or power change demands of up to _(e)_ % with a frequency up to _(f)_ cycle per second shall not result in a divergent sinusoidal thrust or power response. Thrust or power response phase lag shall be no more than _(g)_ degrees and thrust or power peaks shall be at least _(h)_ % of demand. Symmetrical throttle demands shall not result in divergent symmetrical throttle responses.

3.2.2.7.1 Overshoot and undershoot.

The engine thrust or fan, core, rotor speed overshoot following any acceleration demand shall not exceed _(a)_ percent of the final stabilized steady-state thrust or speed value. The engine thrust or fan, core, rotor speed undershoot following any deceleration demand shall not exceed _(b)_ percent of the final stabilized steady-state thrust or speed value.

For turboshaft engines, during the transient to the stabilized value, output shaft speed and torque shall not exceed the limits of table IV.

3.2.2.7.2 Droop.

During any acceleration, no thrust or power droop below _(a)_ percent of the demanded thrust or power value shall occur between the attainment of _(b)_ percent thrust or power change and achievement of transient stability as defined by 3.2.2.7. In addition, there shall be no more than a _(c)_ percent thrust or power droop from the point of transient stability up to attainment of steady-state stability defined in 3.2.2.6.

For turboshaft engines in helicopter applications, the output shaft speed shall not deviate more than the following for the table XLVII power transients at the specified ambient conditions. This shall be verified through computer modeling using either a full, linearized rotor model or a lumped inertia model (attached to the power turbine) derived from the full, linearized rotor model. The actual value for 100% aircraft torque shall be determined using 100% engine output shaft speed and the value of the power required for each engine as specified by the Using Service through coordination with the air vehicle manufacturer.

Table XLVII points:

(d11) The value shall not exceed __(d)__.

(d12) The value shall not exceed __(e)__.

(d13) The value shall not exceed __(f)__.

(d14) The value shall not exceed __(g)__.

(d15) The value shall not exceed __(h)__.

(d16) The value shall not exceed __(i)__.

(d18) The value shall not exceed __(i)__.

(d21) The value shall not exceed __(g)__.

(d22) The value shall not exceed __(h)__.

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(d23) The value shall not exceed __(h)__.

(d24) The value shall not exceed __(i)__.

(d25) The value shall not exceed __(i)__.

(d26) The value shall not exceed __(i)__.

(d28) The value shall not exceed __(i)__.

(d31) The value shall not exceed __(j)__.

(d32) The value shall not exceed __(k)__.

(d33) The value shall not exceed __(l)__.

(d34) The value shall not exceed __(m)__.

(d35) The value shall not exceed __(n)__.

(d36) The value shall not exceed __(n)__.

(d38) The value shall not exceed __(n)__.

(d41) The value shall not exceed __(l)__.

(d42) The value shall not exceed __(m)__.

(d43) The value shall not exceed __(n)__.

(d44) The value shall not exceed __(n)__.

(d45) The value shall not exceed __(n)__.

(d46) The value shall not exceed __(n)__.

(d47) The value shall not exceed __(n)__.

3.2.2.8 Windmilling.

The engine shall be capable of continuous windmilling for ___ hours, without damage to the engine, and without affecting air restart and operational capability.

The following information shall be specified (for an uninstalled engine):

a. The limits of windmilling operation, including curves showing windmilling rotor speed and windmilling drag vs. flight Mach number.

b. The oil consumption rate during windmilling operation.

c. In the event of lubrication failure, the time duration and limits of windmilling operation after depletion of oil supply. This is not required for turboshaft applications.

d. The power extraction available during windmilling operation as shown on figure 12a. This is not required for turboshaft applications.

e. The customer bleed air extraction available during windmilling operation as shown on figure 12b. This is not required for turboshaft applications.

3.2.2.9 Reverse thrust.

For turbofan and turbojet engines, the operating envelope, limitations, and description for thrust reverser operation shall be specified herein. The percentage of total engine thrust used for reverse thrust shall be ___.

3.2.2.10 Steady-state and transient airflow.

The engine maximum allowable steady-state and transient airflow limits shall be shown in table IV. The transient airflow limits specified shall include augmentor ignition and termination, as applicable, as well as

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thrust or power demand increase and decrease. The steady-state airflow of any engine shall not vary by more than (a) percent of the steady-state intermediate airflow computed by the steady-state computer program of 3.2.1.1 and shown in tables IIa, IIb, IIc, and III. The transient airflow of any engine shall not vary by more than (b) percent of the transient airflow computed by the transient computer program of 3.2.1.2.

Transient airflow shall not deviate from the maximum steady-state airflow by more than (c) percent and shall be controlled to assure compatible engine inlet operation. The engine steady-state and transient airflow limits shall include engine deterioration, control tolerances (including temperature lags and transients), engine-to-engine variations and shall apply to the nonstandard atmospheres of this specification as shown on figure 5.

The effects of customer air bleed and power extraction shall be identified.

3.2.2.11 Inlet airflow distortion.

The engine shall not surge, stall, flameout, or incur any damage with the steady-state or time variant inlet distortion (pressure, temperature, or any combination of both) shown on figure 13. The aerodynamic interface plane shall be defined as (a). Engine stability and performance assessments shall use the methodology and inlet distortion descriptors defined in (b) for total pressure distortion. Temperature distortion methodology and the methodology for combination of both pressure and temperature distortion shall be specified.

3.2.2.11.1 Pressure and temperature rate of change.

The engine shall withstand the following rate-of-change of pressure and temperature, separately or in combination, without surge, stall, flameout, or damage:

Parameters	Rate-of-change	Duration
Engine inlet pressure	(a) kPa/sec (psi/sec)	(b) sec
Engine inlet temperature	(c) °C/sec (°F/sec)	(d) sec

3.2.2.11.2. Engine recoverability.

The engine shall recover from any stalls encountered during steady-state operation within, or any transient exceedence of, the operating envelope of 3.2.2.2, and/or with transient exceedence of inlet pressure and temperature distortion limits as defined in 3.2.2.11 and 3.2.2.11.1 without pilot action. After the destabilizing influence(s) have been removed, the engine shall meet the requirements of this specification, and shall automatically recover to a safe, steady-state operating condition in a maximum of ___ seconds longer than the sum of the transient times and start times as defined in 3.2.2.7 and 3.2.2.3.

3.2.2.12 Gas path and measurement plane temperature limits.

The first stage high or low pressure turbine rotor inlet gas temperature limit(s) shall be _____.

The maximum allowable steady-state gas temperature shall be shown in table IV. The location of the measurement plane (including station identification number) shall be shown in table II. No engine shall operate above the maximum allowable measuring plane turbine temperature. When a direct measuring metal temperature sensing system is provided at the measurement plane, the maximum allowable steady-state metal temperature(s) shall be shown in table IV. The maximum allowable transient measured temperature during starting and acceleration shall also be specified herein or shown in table IV.

3.2.2.13 Surface temperature and heat rejection.

The maximum operating surface temperatures and heat rejection rates of the engine shall be shown on figure 14a. The engine surface emissivity shall be presented as shown on figure 14b. Accessory pad

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loading, compressor bleed air conditions, oil system cooling requirements, and air and gas leakage from engine case flanges and split lines shall be used in establishing engine heat rejection rates. The conditions surrounding the engine, the engine power condition, and the oil system temperatures for which the surface temperatures are applicable shall be shown. For components and externals on the surface of the engine, the specified component and externals surface temperature and heat generation curves may differ from the engine temperature as shown on figure 14a.

3.2.2.13.1 Controls and external components limiting temperatures.

Engine components mounted on the engine shall not exceed their allowable temperatures when surrounded by still air under the following conditions:

A tabulation of the maximum ambient air temperature surrounding each component, the limiting surface temperatures for all components, the time period(s) applicable, and the limiting cooling fuel inlet temperature for fuel cooled components, for the most severe engine condition(s), shall be specified in table V. The specific points of measurement, where the ambient air temperature and surface temperature with respect to the three coordinate axes of the component are maximum, shall be specified. The engine components requiring special cooling and ventilation shall be listed in table V. Length of operation of these components without special cooling shall be defined in table V.

Off-engine components shall not exceed their allowable temperatures when surrounded by still air at _____ degrees with no special cooling such as forced ventilation or refrigeration. The specific points of measurement, where the ambient air temperature and surface temperature with respect to the three coordinate axes of the component are maximum, shall be specified.

3.3 Environmental conditions.

3.3.1 Atmospheric conditions.

3.3.1.1 Humidity.

The engine shall meet all requirements of this specification when subjected to _____% humidity at _____ °C (°F).

3.3.1.2 Fungus.

The engine shall meet all specification requirements when subjected to fungus.

3.3.1.3 Corrosive atmosphere.

The engine shall meet all requirements of this specification during and after exposure to the corrosive atmosphere of table VI for (a) corrosion cycles. The engine shall not deteriorate more than (b) percent in thrust or power, or gain (c) percent in specific fuel consumption. Engine parts shall show no corrosion which impairs function, component integrity, or prescribed maintenance. Corrosion protected parts shall show no corrosion effects after cleaning and stripping of corrosion protection coatings.

3.3.1.4 Icing conditions.

The engine shall meet the requirements of this specification while operating in the meteorological conditions shown on figures 15a and 15b and table VII, with not more than (a) percent total loss in thrust or power and (b) percent total increase in specific fuel consumption at all operating conditions. After removal of the icing conditions, the engine shall perform the same as prior to encountering the icing conditions. No damage to the engine shall be allowed as a result of operating in the icing environment.

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3.3.2 Ingestion capability (hazard resistance).

3.3.2.1 Bird ingestion.

The engine shall continue to operate and perform during and after the ingestion of birds as specified in table VIII.

3.3.2.2 Foreign object damage (FOD).

The engine shall have foreign object damage requirements specified per _____.

3.3.2.3 Ice ingestion.

The engine shall operate and perform per table IX, during and after ingestion of hailstones and sheet ice at the take-off, cruise, and descent aircraft speeds. The engine shall not be damaged beyond field repair capability after ingesting the hailstones and ice.

3.3.2.4 Sand and dust ingestion.

The engine shall meet all requirements of the specification during and after the sand and dust ingestion event specified herein. The engine shall ingest air containing sand and dust particles in a concentration of (a) mg sand/m^3. The engine shall ingest the specified coarse and fine contaminant distribution for (b) and (c) hours, respectively. The engine shall operate at intermediate thrust for TJ/TFs or maximum continuous power for TP/TSs with the specified concentration of sand and dust particles, with no greater than (d) percent loss in thrust or power, and (e) percent gain in specific fuel consumption (SFC). Helicopter engines shall ingest the 0-80 micron (0 - 3.15 x 10^{-3} in.) fine sand and dust of table XXXV in a concentration of 53 mg/m^3 (3.3 x 10^{-6} lb/ft^3) air for 54 hours and 0-1000-micron coarse sand and dust in table XXXV for 50 hours and inspection shall reveal no impending failure.

3.3.2.5 Atmospheric liquid water ingestion.

The engine shall start and operate satisfactorily with (a) percent of the total airflow weight in the form of water (liquid and vapor), with (b) percent of the liquid water entering the inlet through a segment equivalent to (c) of the inlet area.

3.3.2.6 Armament gas ingestion.

The engine shall meet all the requirements of the specification during and after Armament Gas Ingestion (AGI), except as follows.

If any performance loss occurs during AGI, the engine shall recover within (a) seconds and without pilot intervention, to the thrust or power available immediately prior to the AGI event. In addition, when an AGI signal is present, engine thrust or power shall never fall below (b) percent of the thrust or power available immediately prior to the AGI.

3.3.2.7 Steam ingestion.

The engine shall meet all specification requirements without stall, surge, flameout, mechanical/structural damage, or permanent performance loss, when the total inlet mass flow is at least (a) steam, and has a minimum temperature ramp (increase) of (b) °C/sec ((b) °F/sec) with a (c) °C ((c) °F) temperature rise in a region less than (d) percent of the engine inlet flow area. After a steam ingestion event, the engine shall meet all requirements of the specification.

3.3.3 Electromagnetic environmental effects (E^3).

The engine shall meet the electromagnetic environmental effects requirements specified in this section.

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3.3.3.1 Electromagnetic interference (EMI).

All engine electrical and electronic equipment and subsystems shall meet their performance requirements when subjected to the susceptibility requirements of (a) requirement (b) and shall comply with the emission requirements of (a) requirement (c) . These requirements shall be met for engine operation in all environmental conditions within the operating envelope for all control system operating modes.

3.3.3.2 Intrasystem electromagnetic compatibility (EMC).

The engine, installed in the air vehicle, shall meet the applicable requirements of: _____.

3.3.3.3 Intersystem electromagnetic compatibility (EMC).

The engine as installed in the air vehicle shall meet the requirements of the specification when exposed to the following external electromagnetic environment:

3.3.3.4 Lightning.

The engine shall meet the applicable lightning requirements of: _____.

3.4 Integrity.

3.4.1 Structural integrity.

3.4.1.1 Design service life.

The engine shall have a design service life specified per _____.

3.4.1.2 Design usage.

The engine structure shall be capable of withstanding the design usage specified per ____.

3.4.1.2.1 Externally applied forces.

The engine shall function satisfactorily under externally applied forces specified per _____.

3.4.1.3 Material characterization.

Material properties shall be based per_____:

3.4.1.4 Parts classification.

All engine parts shall be classified as fracture critical, durability critical, and durability non critical. Fracture critical parts/critical safety items shall be further classified as safety and mission critical per _____.

3.4.1.5 Durability.

The engine shall be durable for the design service life when subjected to the usage of 3.4.1.2 and the requirements specified per _____.

3.4.1.5.1 High cycle fatigue (HCF) life.

Engine parts shall have the high cycle fatigue life specified per _____.

3.4.1.5.2 Low cycle fatigue (LCF) life.

All engine parts shall have their minimum LCF lives specified per _____.

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3.4.1.5.3 Creep.

All engine parts shall not creep to the extent that the engine does not meet the requirements specified per _____.

3.4.1.6 Strength.

The engine shall have the strength requirements specified per _____.

3.4.1.6.1 Factors of safety.

Factors of safety shall be applied to design usage induced loads to establish limit and ultimate conditions specified per _____.

3.4.1.6.2 Blade and disk deflection.

The engine shall have blade and disk deflection requirements specified per _____.

3.4.1.6.3 Containment.

The engine shall have containment requirements specified per _____.

3.4.1.6.4 Blade out.

The engine shall have requirements for blade out specified per _____.

3.4.1.6.5 Overspeed and overtemperature.

The engine shall have overspeed and overtemperature requirements specified per _____.

3.4.1.6.6 Disk burst speed.

The minimum loaded disk burst speed of the complete disk assembly shall have the requirements specified per _____.

3.4.1.6.7 Output shaft torque limits.

For turboprop and turboshaft engines the maximum allowable steady-state delivered shaft torque (mechanical) limit shall have the requirements specified per _____.

3.4.1.6.8 Output shaft speed limits.

For turboprop and turboshaft engines the maximum allowable steady-state delivered shaft speed (mechanical) limit shall have the requirements specified per _____.

3.4.1.6.9 Bearing load.

The engine bearings shall react the loads applied during engine operation and meet the design service life of 3.4.1.1. The engine bearings shall react the loads applied during flight when the engine is not operating (i.e., shutdown and windmilling). The engine bearings shall have any applicable load requirements specified per _____.

3.4.1.6.10 Gyroscopic moments.

The engine shall have gyroscopic moment requirements specified per _____.

3.4.1.7 Damage tolerance.

Fracture and safety and mission critical engine parts shall have damage tolerance requirements specified per _____.

3.4.1.7.1 Residual strength.

All damage tolerant designed parts and components shall have residual strength requirements specified per _____.

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3.4.1.7.2 Initial production and in-service flaw size.

All fracture critical parts/critical safety items shall have initial flaw sizes specified per _____.

3.4.1.7.3 Flaw growth and inspection intervals.

The initial flaw size as specified in 3.4.1.7.2 shall not grow to critical size and cause failure of the part due to the application of the required residual strength load within the inspection intervals as specified per __.

3.4.1.8 Vibration and dynamic response.

The engine shall have vibration and dynamic response requirements specified per _____.

3.4.1.8.1 Vibration limits.

The engine shall have vibration limits specified per _____.

3.4.1.8.2 Critical speeds.

The engine shall be free of resonance conditions at all shaft rotational speeds that would cause the engine not to meet all the requirements of the specification. Critical speeds existing below the engine operating range shall be at least _(a)_ percent below idle. Critical speeds existing above the maximum operating speed shall be at least _(b)_ percent above the maximum allowable transient shaft rotational speed. The natural frequencies of the mounting system with the engine installed shall be at least _(c)_ percent below idle shaft rotational speed(s) in all damaging modes of vibration which can be excited by the rotor imbalances.

3.4.1.9 External surface foreign object damage.

The engine external surface, protruding from the air vehicle, shall meet all requirements of this specification, without repair, after FOD that produces damage equivalent to a stress concentration factor (Kf) of _____.

3.4.2 Mechanical equipment and subsystems integrity.

The engine's mechanical equipment and subsystems shall meet the integrity requirements specified per _____.

3.4.3 Avionics and electronic integrity.

The engine shall meet the electronic system integrity requirements specified per _____.

3.4.4 Software integrity.

The software integrity shall be obtained through the use of ___.

3.5 Reliability and maintainability (R&M).

3.5.1 Reliability.

3.5.1.1 Reliability quantitative requirements.

The engine design reliability shall be capable of meeting engine or air system availability goals for the mission and mission mix profile of table XI. The engine reliability shall be as defined in table XIII.

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3.5.2 Maintainability.

3.5.2.1 Maintainability quantitative requirements.

The engine design maintainability shall be capable of meeting engine or air system availability goals for the mission and mission mix profile of table XI. Maintainability requirements shall be specified in tables XIV and XV.

The diagnostic capabilities necessary to meet engine maintainability requirements shall be specified herein.

3.5.2.1.1 Excluded maintenance functions.

All excluded maintenance and repair functions shall be listed in table XVI.

3.5.2.2 Maintainability qualitative requirements.

The engine shall meet the qualitative maintainability requirements designated in table XVII.

3.5.2.3 Maintenance.

3.5.2.3.1 Modules.

The structural components that define each engine module, and the length of time to remove and replace each module, shall be specified in table XVIII. All modules shall be capable of removal and replacement at the ____ maintenance level.

3.5.2.3.2 Maintenance, inspection, and repair cycle.

The maintenance, inspection, and repair periods for the total engine and each module shall be specified in tables XIXa-c, together with the estimated time in man-hours required to perform these functions.

3.5.2.3.2.1 Maintenance inspection techniques.

Engine design shall permit maximum use of non-destructive inspection techniques and multi-purpose test and inspection equipment. Inspection provisions, including access envelopes, shall be shown on the engine configuration and envelope figure.

3.5.2.3.3 Tools.

Required tooling shall be held to a minimum. The engine shall be capable of being maintained with standard hand tools, as identified in table XX. Where provisions for standard hand tools are not feasible, the design should provide, wherever possible and cost effective, for special tools and test equipment that are available and in use on other in-service engines, as identified in table XX.

3.5.2.4 Battle damage repair.

The engine design shall support the battle damage repair requirements of the air vehicle. Battle damage repairs shall not take longer than _(a)_ hours and shall be of sufficient strength to sustain at least _(b)_ flight hours of operation.

3.5.3 Human performance and human engineering.

The engine shall be designed to be maintained by at least the central ninety percent (90%) of the maintainer population (5th percentile female stature through 95th percentile male stature) wearing a full complement of personal protective equipment, under all environmental conditions (including day and night), defined in the specification and all operational scenarios at operating bases and deployed locations.

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3.6 Combat survivability.

The engine shall apply a combat survivable design concept that includes state of the art survivability technology (vulnerability and susceptibility) which insures survival in the intended threat environment.

3.6.1 Susceptibility.

3.6.1.1 Noise.

The near and far field noise levels for an uninstalled engine, vectored or non-vectored, operating at sea level static shall not exceed the levels of figures 17a, 17b, and 17c.

3.6.1.2 IR signature.

The engine shall not exceed the IR signature levels contained in table XXI while meeting the requirements specified herein at any point during the engine life.

3.6.1.3 Smoke.

The engine shall not emit smoke trails or smoke puffs defined by ___, during any transient or steady-state operation when using any primary or alternate fuel.

3.6.1.4 Gaseous emissions.

The engine shall not produce byproduct exhaust emissions greater than the levels specified on figures 18a and 18b when using any primary or alternate fuel.

3.6.1.5 Fuel streaming and vapor puffing.

The engine shall not emit visible, unburned fuel (fuel streams or vapor puffs) as defined by ___, at any power setting at or above flight idle or during any transient or steady-state operation when using any primary or alternate fuel.

3.6.1.6 Water vapor contrails.

The engine shall not emit water vapor contrails during any transient or steady-state operation when using any primary or alternate fuel.

3.6.1.7 Radar cross-section.

The engine shall not exceed the Radar Cross Section (RCS) signature requirements of tables XXII and XXIII. This requirement is not applicable to turboprop engines with no externally visible rotating parts.

3.6.1.8 Radar absorbent materials and coatings.

If Radar Absorbent Materials (RAM) or radar absorbent coatings are used, the ingestion requirements of 3.3.1.3 and 3.3.2.4 combined shall not deteriorate the RCS signature level by more than __ dB.

3.6.2 Vulnerability.

3.6.2.1 Ballistic weapons.

The engine, including externally mounted components, shall tolerate ballistic weapon threats as follows:

3.6.2.1.1 Static structure.

No case, frame or gas pressure loaded non-rotating component, except airfoils, shall collapse or rupture when subject to its maximum operating pressure and a round hole of ____ diameter placed anywhere on the structure.

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3.6.2.2 Nuclear weapons effects.

The engine shall have the following operational capabilities and balanced hardness, in a nuclear environment that includes blast and gust, dust, thermal, transient radiation effects, and electromagnetic pulse:

3.6.2.3 Chemical and biological agent effects.

The engine shall survive contamination from chemical and biological agents, and decontamination without losing mission capability as follows: (a) . The engine shall be capable of being decontaminated as defined in FM 3-5, NBC decontamination, to reduce the hazard to a level where personnel operating under Mission Orientated Protective Posture (MOPP) level 0 suffer no physical impairment or injury within (b) of completing the procedures and decontamination agents defined by (c) . Engine materials shall be compatible with the decontamination agents. The engine shall be capable of being operated and maintained by personnel in nuclear, biological and chemical (NBC) MOPP-4 protective ensembles.

3.6.2.4 Fuel ingestion effects.

3.6.2.4.1 Fuel ingestion - steady-flow.

The engine shall be capable of withstanding the effects of steady flow fuel ingestion via the inlet duct with the fuel and plume characteristics (a) and initial flow rates as follows. The engine shall not suffer more than (b) % thrust loss for ingestant flow rates up to (c) kg/hr (PPH) for 30 seconds, and shall be capable of operating for 30 minutes at the residual thrust level, at (d) flight condition. The swirl displacement angles that would be experienced by entrained fuel in core and bypass flow paths shall be described by the contractor for intermediate and flight idle throttle settings, and the windmill state, at (d) flight condition.

3.6.2.4.1.1 Fuel ingestion - cooling air contamination.

Contamination of engine internal cooling air from fuel ingestion shall not degrade engine life nor engine parts life below that specified in 3.4.1.1.

3.6.2.4.2 Fuel ingestion - transient (“quick dump”).

The engine shall withstand ingestion of transient fuel with no more than (a) % permanent thrust loss under operating conditions (b) . The fuel shall be injected from the duct wall into the center of the inlet duct at a flow rate sufficient to carburet to twice the stoichiometric ratio the complete inlet air column for a length corresponding to the amount of air contained between engine face and exhaust nozzle throat. Engine instability and total loss of thrust is permitted during the event, but normal engine function shall be reestablished within (c) seconds of the ingestion event with no pilot input. Normal throttle agility may be degraded, but the thrust must be capable of modulation down to flight idle with gentle throttle input. The engine shall withstand ingestion of transient “quick dump” fuel that may occur during aerial refueling operations.

3.7 Subsystems.

3.7.1 Anti-icing and de-icing system.

Within ____ seconds following termination of the anti-icing or de-icing system, engine performance shall return to the level prior to system actuation.

The anti-icing or de-icing system shall allow (no/no detrimental) accumulation of ice on any engine part while operating under the icing condition specified in 3.3.1.4. Operation of the anti-icing or de-icing system throughout the operating envelope shall not damage the engine. The performance and operability effects of the anti-icing or de-icing system operation in a non-icing environment shall be specified. The anti-icing or de-icing system shall provide a signal for indicating that the anti-icing or de-icing system is operating. The fail-safe mode for the ice protection system shall allow safe return to base in icing conditions and non-icing conditions without safety of flight damage to the engine. If an inlet particle

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separator (IPS) is used, ice formation on the IPS shall not be permitted. Limited ice accumulation shall be permitted in the IPS scavenge system if it does not affect IPS operation.

3.7.2 Control system.

The control system shall be fault tolerant and automatically control the engine to meet all specification requirements.

Regions of control limiting functions, as a function of altitude, mach number, thrust and power setting for standard, hot, and cold day atmospheres of figure 5, shall be shown in the format of figure 19 in _____. The limiting values used to establish the various regions shall be specified in _____.

For turboshaft applications, the engine control system shall provide for load sharing and torque matching for multiple turboshaft engines.

3.7.2.1 Control system performance.

The control system fault accommodation shall detect failures that shall result in fail-operational, fail-degraded, and fail-safe engine operation, during and after control system malfunction(s) or failure(s). The engine control system shall provide fully modulated thrust or power response, and prevent exceedance of aero thermodynamic and mechanical limits when subjected to unrestricted thrust or power demand inputs. Thrust or power signal demand versus engine thrust or power output shall be defined in (a) and free of abrupt changes between ground idle and intermediate thrust or maximum power. A step change in thrust at augmentor light off should not exceed (b). The relationship between control demand and power output shall be shown on figure 20 and in the ICD if available. For thrust reversers, the relationships between reverse thrust and thrust demand input signals shall be defined in (a) and free of abrupt changes.

For turboshaft and turboprop engines, the system shall allow the selection of any output shaft speed between (c) and (c) percent, inclusive, at all powers equal to and greater than no load. For turboshaft engines, the system shall have the capability to automatically match the output shaft torque of both engines of a twin engine installation within (d) percent of the torque available from a single engine at maximum continuous power.

For US Army turboshaft multi-engine applications; engine power loss shall be sensed and power automatically increased on the other engine(s) to provide power required up to contingency power (one engine inoperative) capability. The control system shall automatically enable an emergency power capability from a contingency power condition when the engine is operating at contingency $T_{4.5}$ and gas generator speed and rotor droop is detected.

3.7.2.1.1 Backup control.

If required by the Using Service, a backup control in conjunction with a redundant control system shall be required for single engine aircraft unless sufficient statistical evidence on the proposed system shows acceptable levels of risk and the backup control shall be considered part of the control system. During transfer from the primary mode to the backup mode and during operation, the backup control shall provide, as a minimum, a safe return to base. The capabilities and limitations of the engine when operating with the backup control shall be specified. Transfer from the primary system to the backup control and back to the primary system shall be (a) at any power setting without engine instability or change to the control system input. Transfer from primary control to backup control shall be within (b) seconds after system malfunction or failure. Thrust or power demand input signal relationships shall be similar to those of the primary control system, but cut-off, idle and intermediate thrust or power signal positions shall be the same as for the primary mode. The backup control shall prevent the engine from exceeding any limits.

3.7.2.2 Control system adjustments.

No adjustment of the engine control system shall be permitted except as follows:

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3.7.2.3 Overspeed protection system.

The control system shall detect all main engine rotor overspeed conditions and take corrective action to prevent overspeed limits from being exceeded.

For turboshaft and turboprop applications, the measurement of the power rotor system speeds shall be sensed directly with no intermediate mechanical devices such as gears, flexible shafts, or clutches and the rotor speeds at which the overspeed protection occurs shall be specified.

3.7.3 Fuel system.

3.7.3.1 Fuels.

The engine shall meet all applicable requirements when supplied with the fuels specified per _____ .

3.7.3.1.1 Primary fuel.

The primary fuels shall be as specified in table XXIV. The engine shall meet the requirements of the specification when using any of the fuels conforming to and having any of the variations in characteristics permitted by the specifications listed in table XXIV. The engine shall meet all requirements of the specification after transition between any primary fuels with no engine damage. The altitude relight envelop for each of the three major fuel types (High Flash Kerosene, Kerosene, and Wide-Cut) shall be identified.

3.7.3.1.2 Alternate fuel.

The engine shall be capable of starting, and operating satisfactorily when using the alternate fuels. The operating limits, thrust outputs, and thrust transients specified herein shall not be adversely affected when using alternate fuels. There shall be no adverse effect on aircraft mission(s). Those external adjustments permitted in 3.7.2.2 shall be allowed in order to meet these requirements. Alternate fuels shall be the commercial jet fuels specified in table XXIV. Any limitations on the engine when using alternate fuels shall be specified herein.

3.7.3.1.3 Restricted fuel.

The restricted fuel(s) the engine shall be capable of using shall be specified in table XXIV. Any limitations (fuel temperature limits, altitude, etc.) on the engine when using restricted fuels shall be specified herein.

3.7.3.1.4 Emergency fuel.

The engine shall function satisfactorily for a time period of at least six hours from sea level to 35,000-ft altitude, at least throughout a range from idle to 90 percent of maximum thrust and at no greater than 120 percent of the specification rated or estimated specific fuel consumption when using automotive and/or marine diesel fuels (A-A-52557, MIL-PRF-16884, and ASTM D4814). Those external adjustments permitted in 3.7.2.2 shall be allowed in order to meet this requirement. If applicable, operating limitations, special inspection or maintenance actions required as a result of using this fuel shall be specified in the model specification. If applicable, emergency fuel shall be specified herein. If applicable, emergency fuel shall be specified herein.

3.7.3.2 Fuel system performance.

3.7.3.2.1 Fuel contamination.

The engine shall meet specification requirements when supplied with fuel contaminated in an amount up to the extent specified in table XXVa, and when supplied with _(a)_ liters (_(a)_ gallons) of fuel, contaminated per table XXVb, with the filter in full bypass. Engines for land based rotorcraft and engines with unassisted (suction) fuel systems shall meet specification requirements with fuel contaminated in amounts specified in table XXVc, with fuel filter not in bypass mode.

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3.7.3.2.2 Fuel system performance with external assistance.

The fuel system shall supply the required amount of fuel at the required pressure and temperatures for operation, including starting and augmentation (if applicable), with the following conditions at the fuel inlet connection:

Normal Operation:

a. Fuel temperature: From a minimum of __(a)__ °C, (__(a)__ °F) when using worst case allowable viscosity fuels of 3.7.3.1.1, to a maximum of no less than __(b)__ °C, (__(b)__ °F).

b. Fuel pressure: From a minimum of __(c)__ kPa, (__(c)__ psi) above the true vapor pressure of the fuel to a maximum of __(d)__ kPa, (__(d)__ psig) with a vapor/liquid volume ratio of __(e)__ as defined at worst case operating conditions.

c. Fuel pressure surges: The fuel system shall accept air vehicle induced inlet pressure surges up to __(f)__ kPa, (__(f)__ psig) occurring at a maximum rate of change of up to __(g)__ kPa, (__(g)__ psi) per second. Engine induced pressure surges at the fuel inlet flange shall not exceed __(h)__ kPa, (__(h)__ psi) above the steady-state fuel inlet pressure.

3.7.3.2.3 Fuel system performance with no external assistance.

The fuel system shall supply the required amount of fuel at the required pressure and temperatures for operation, including starting and augmentation (if applicable), with the following conditions at the fuel inlet connection:

a. Fuel Temperature: From a minimum of __(a)__ °C (__(a)__ °F) when using worst case primary fuel or __(b)__ centistoke, whichever is worse, to a maximum of no less than __(c)__ °C (__(c)__ °F).

b. Fuel Pressures: From a minimum of __(d)__ above the true vapor pressure of the fuel to a maximum of __(e)__.

c. Vapor/liquid volume ratio from zero to __(f)__.

d. For turboshaft engines, vapor shall be introduced continuously as vapor only (bubbles) equivalent to 300 mm (11.8 in.) of fuel line of the same area as the engine fuel inlet followed by liquid fuel only equivalent to 300 mm (11.8 in.) of fuel line of the same area as the engine fuel inlet.

e. If required, quantity of air to be ingested __(g)__.

3.7.3.2.4 Fuel system performance under conditions of excessive fuel vapor.

When the vapor to liquid ratio at the fuel inlet is greater than _____, it shall not be necessary for the fuel system to supply fuel for engine operation. However, after the conditions of 3.7.3.2.2 or 3.7.3.2.3 have been reestablished, it shall be possible to accomplish normal starts and operate satisfactorily thereafter.

3.7.3.2.5 Fuel pump priming.

The fuel pump shall be self priming and be capable of a dry lift of at least __(a)__ meter, (__(a)__ feet) at a minimum absolute pump inlet pressure of __(b)__ kPa, (__(b)__ psia), using the fuels of 3.7.3.1 with the RVP not to exceed __(c)__ kPa, (__(c)__ psi), and a fuel temperature not to exceed __(d)__ °C (__(d)__ °F).

3.7.3.2.6 Fuel lubricity.

The engine shall operate satisfactorily when using fuel with a lubricity value equivalent to a 0.75 - 0.90-mm (0.030 - 0.035 inch) Wear Scar Diameter (WSD) as defined by ASTM D5001.

3.7.3.2.7 Fuel system performance with water saturated fuel.

The engine shall meet the requirements of this specification when supplied with fuel saturated with water up to ____ at worst case fuel pressure and temperature conditions.

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3.7.3.2.8 Fuel filter.

Fuel filtration on the engine shall be provided, as needed, for the engine to meet the requirements of the specification. The fuel filter shall consist of (a) filter(s), and shall be made of a material that does not cause fuel contamination due to media migration. The filtration ratings of the filter elements, contaminated per table XXVa, shall be per table XXVI.

The filter assembly shall incorporate a pressure relief bypass and shall prevent the discharge of filter contaminant through the bypass. The filter assembly shall have (b) indications of both impending and actual filter bypass. The indication shall remain activated until the filter is removed for inspection. The filter shall be removable without spills or draining of the fuel lines. Fuel pressure surges during operation within the limits specified in 3.7.3.2.2 shall not cause false activation of the impending or actual bypass indicators.

3.7.3.2.9 Fuel flow limit.

The maximum and minimum fuel flow shall be specified in _____.

3.7.4 Electrical system.

3.7.4.1 Electrical power.

After reaching compressor speeds specified herein**,** engine electrical power shall be uninterruptible and supplied by the (engine or air vehicle) as specified herein. Engine electrical shall have (b) indications equipment shall be capable of accepting externally supplied air vehicle electrical power as a (primary or alternate) power source as specified herein.

Transfer between primary and alternate power sources shall not interrupt engine operation.

3.7.4.1.1 Generator.

If an engine generator is required it shall provide power to satisfy the full electrical load of the engine. The generator performance shall not degrade during and after continuous operation at a speed equivalent to _____ percent of the maximum allowable steady-state engine speed under full electrical load. The generator performance shall not degrade following the overspeed of 3.4.1.6.5, with no electrical load. The generator housing shall completely contain all damage if a mechanical failure should occur when operating at or below maximum transient rotor speed.

3.7.4.2 Alternate electrical power.

The engine shall automatically accept alternate power when the source is either air vehicle generated power or external (ground) power or a combination thereof as defined in (a).

Engine performance and operability shall not be degraded with electrical power transients of: (b).

3.7.4.3 Electrical connectors and cables.

Electrical connectors shall be specified herein. Engine interface connectors shall be specified in _____. At the sea level cold atmospheric temperature of figure 5, it shall be possible to flex electrical cable and conductors during routine maintenance without damage to these items and to connect or disconnect electrical connectors using normal maintenance procedures.

3.7.4.4 Electrical bonding.

The engine shall have a clearly defined electrical bonding interface with the air vehicle which shall be identified in the ____. All electrical and electronic components on the engine shall have a clearly defined electrical bonding interface to the engine structure that shall be identified in the design documentation including drawings.

3.7.4.5 Ground isolation.

The engine to aircraft interface shall not use aircraft structure for power or signal return.

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3.7.4.6 Potting compounds.

The use of potting compounds shall not be permitted except where specifically approved by the Using Service. When approved, the potting compounds and their application(s) shall be described.

3.7.5 Ignition system(s).

The engine ignition system(s) shall ignite the combustor and if applicable the augmentor. An external means shall be provided to deactivate the engine ignition system(s), and it shall be described in _____.

The engine ignition system(s) shall ignite the combustor(s) under carbon and water fouling conditions.

a. Main ignition system. The main engine ignition system shall be: _____.

Each separate igniter and circuit shall release sufficient energy for all ground and air starting requirements.

b. Augmentor ignition system. If applicable, the augmentor ignition system shall be: _____.

3.7.5.1 Ignition system fouling.

3.7.5.1.1 Carbon fouling.

The spark igniters of the ignition system shall provide sparking performance with spark gaps completely covered or bridged with a material simulating carbon fouling. With the minimum power input of section 3.7.5, and the carbon fouling conditions, the sparking rate shall not be less than the minimum design value.

3.7.5.1.2 Water fouling.

The spark igniters of the ignition system shall provide sparking performance when thoroughly drenched with water or frosted over. With the minimum power input of section 3.7.5, and the water fouling conditions, the sparking rate shall not be less than the minimum design value.

3.7.6 Engine health monitoring system (EHMS).

The EHMS as defined in the _____ shall provide, monitor, and coordinate the recorded appropriate parameters for maintenance, performance trending, parameter limit exceedance, and life usage. The malfunction of any EHMS hardware or software shall not affect engine performance or operability.

3.7.6.1 EHMS fault detection and isolation.

The EHMS fault detection and isolation capability shall be described herein. The EHMS functions required to achieve this level of detection and isolation faults shall be described herein. The onboard engine diagnostics shall provide fault detection and isolation, to the faulty component level, to meet the reliability and maintainability requirements as specified in section 3.5. Onboard engine diagnostics system false alarms shall be considered as failures and shall be used in the reliability and maintainability calculations per section 3.5. The mission critical equipment and functions of the onboard diagnostic system, which monitor mission and safety critical parameters, shall be defined herein. The onboard engine diagnostics system shall not cause a failure of any other mission or safety critical system.

3.7.6.2 Onboard engine diagnostic functions.

The EHMS shall acquire the following data ___.

3.7.6.2.1 Built-in-test and inspectability.

Built-in-Test (BIT) shall be implemented to accomplish failure detection and failure isolation to the WRA and LRU without the assistance of support equipment at the organizational level. The testability of the engine system electronics shall be per _____. If required by the air vehicle, failure status to determine mission capability shall be provided to the air vehicle. A nonvolatile record of BIT failure data

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shall be retained in memory. The BIT information shall include provisions to allow both Read Only Memory (ROM) and Random Access Memory (RAM) contents of the subject computer to be inspected (monitored).

3.7.7 Optical systems.

Switching from the engine to the air vehicle or external (ground) electrical power system shall not affect the operation of the optical system. Optical sensors (e.g.; augmentor flame detectors, optical pyrometers) shall maintain accuracy and calibration, within established limits, during the life of the engine module in which it is installed.

3.7.7.1 Fiber optic cables and connectors.

Fiber optic cables and connectors shall be specified herein. At the sea level cold temperature of figure 5, it shall be possible to flex fiber optic cables and connect or disconnect optical connectors using normal maintenance procedures, without damage to the fiber cables or connectors.

It shall be possible to remove contaminants from the ends of the transmissive elements without causing damage to them.

3.7.8 Lubrication system.

3.7.8.1 Lubrication oil.

The engine shall meet the requirements of the specification when using the following lubricating oil(s):

3.7.8.1.1 Oil pressure and temperature limits.

The maximum and minimum operating oil pressure limits and maximum transient and maximum allowable steady-state oil temperature limits shall be listed in table IV. The maximum and minimum oil pressures during starting and initial operation at the sea level cold atmospheric temperature of figure 5 or (a) centistoke oil viscosity, whichever is worse, shall be listed in table IV.

Oil system temperatures and pressures following in-flight shutdown and subsequent in-flight air re-starts shall not be detrimental to the lubrication system components nor the system components requiring oil.

3.7.8.1.2 Oil consumption limits.

The maximum oil depletion rate shall be specified in table IV.

3.7.8.1.3 Oil flow interruption or depletion.

The engine shall meet the requirements of the specification during and after operation at (a) thrust or power for a period of at least (b) seconds with no oil (only air) supplied to the oil pump inlet.

Engines in fixed wing air vehicle applications shall be capable of operating without oil for (c) minutes at (d) thrust or power without rotor seizure.

For rotary wing air vehicle applications, the engine shall be capable of operation at (e) power for a period of at least (f) minutes with loss of all oil supplied to the engine bearing sumps followed by (g) minutes of operation at (h) power with lubrication restored.

3.7.8.2 Lubrication system components and features.

3.7.8.2.1 Oil reservoir.

Unvented oil reservoirs shall meet the factor of safety pressure requirements of 3.4.1.6.1. If the lubrication and hydraulic systems share a common reservoir, the lubrication system shall be protected from total depletion in the event of a hydraulic failure.

The capacity of the oil reservoir shall be as shown in table XXVII.

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3.7.8.2.1.1 Oil reservoir external features.

The oil reservoir shall contain features to determine the oil level in the reservoir for the attitudes of 3.2.2.2. An oil quantity signal (a) be provided to the air vehicle. The reservoir shall have a (b) system(s) and shall have features for overfill protection. Oil servicing shall be possible with the engine in a range of positions from 15 degrees nose-up to 20 degrees nose-down.

3.7.8.2.2 Oil drains.

Drain ports shall be provided at appropriate low points in the oil system for draining the oil and taking oil samples with the engine in a range of positions from (a) degrees nose-up to (b) degrees nose-down.

3.7.8.2.3 Oil filters.

Oil filter type(s), micron size, capacity, and filtration ratio rating shall be specified herein. Primary filter assemblies shall incorporate a pressure relief bypass and (a) provide direct and/or remote indication of filter capacity used and/or remaining. The bypass indicators shall incorporate provisions to preclude activation during low temperature starting. Bypass shall remain indicated until reset. The oil filter element shall have a life of at least (b) hours prior to impending bypass indication and at least (c) hours between impending bypass and bypass.

3.7.8.2.4 Oil debris monitor(s)

Oil debris monitor(s) shall be installed in locations where wear particles in the oil would most likely be detected and shall detect oil debris as specified herein.

3.7.8.3 Breather mist.

Lubrication system breather exhaust shall not inhibit ground handling or maintenance personnel from performing tasks around and underneath the installed engine during ground operations at approved ground maintenance power settings. If oil mist is emitted, it shall not exceed the limits specified in the ICD and the quantity shall not exceed the current OSHA health and safety Threshold Limit Values (TLV).

The oil mist exhaust ports shall preclude foreign material (rain, particulates, debris matter, etc) from intruding back into the system during both operation and non-operation.

3.7.9 Hydraulic system.

The hydraulic system fluid(s) shall be (a). The hydraulic system shall withstand a proof pressure of (b) times the maximum working pressure. Hydraulic systems that are integral to the fuel system shall meet the requirements of 3.7.3.

3.7.9.1 Hydraulic ground test provisions.

Ground test provisions shall be provided if required by the Using Service. Fluid leakage shall not occur during ground test.

3.7.9.2 Hydraulic system air removal.

Entrapped air shall not cause sustained loss of system pressure or degradation of system operational performance. There shall be provisions for bleeding air from the system at critical points for maintenance purpose.

3.7.9.3 Hydraulic fluid filters.

Filter(s), type(s), location(s), and filtration rating(s) in microns (nominal and absolute particle sizes) shall be specified herein. Each filter housing shall incorporate an automatic shutoff device to prevent fluid drainage during filter element removal. Each filter element shall be accessible, easily removed and provide (visual/engine monitoring system) indication of both impending and actual bypass. The bypass indicators shall incorporate provisions to preclude activation during low temperature operation. Whenever

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activation of the indicator is a result of the exceedance of the differential pressure limit, the indicator shall remain in the fault mode until removal, cleaning, or replacement of the filter element is accomplished.

The filter shall have a life of at least (a) hours to impending bypass indication and at least (b) hours between impending bypass and bypass.

3.7.10 Exhaust nozzle system.

Any exhaust nozzle system failure shall not prevent safe return.

Failure of any single part of the actuating mechanism on a controllable nozzle shall not cause asymmetric thrust relative to the nozzle. The operational mode, and limits resulting from any single part failure in the actuating mechanism of a variable nozzle, shall be specified herein. Provisions shall be made to actuate the nozzle for ground maintenance checkouts, rigging, and adjustment without the engine running. Life of the engine exhaust system, special components, features, materials, and applied coatings shall be specified.

3.7.10.1 Exhaust nozzle external asymmetrical air pressure loads.

The engine shall meet the requirements of the specification when subject to the asymmetrical and all other air pressure loads resulting from the external flow field pressure distribution on the nozzle and exhaust systems. The limits shall be specified: _____.

3.7.10.2 Vectoring nozzle.

The engine thrust vectoring system shall revert to the fail-safe mode after receiving a fail-safe or standby signal. Nozzle feedback signals, if required, shall be provided to the air vehicle and shall include: _____.

3.7.10.2.1 Vectoring nozzle angle and rate.

The vectoring nozzle shall have a maximum slew rate of at least (a) degree/sec at (b) flight condition. The nozzle shall provide a maximum vectored deflection angle of (c) degrees from the symmetric axial thrust position with an accuracy of (d) degrees over the entire deflection range. The nozzle shall vector without interruption, in a constant, steady movement.

3.7.10.2.2 Vectoring nozzle failure accommodation.

Failure accommodation for the nozzle system shall be described. Any system failure shall not preclude the nozzle from reverting to a fail-safe position.

3.7.11 Augmentation system.

The engine shall provide thrust or power augmentor operation, as required, including initiation and termination, throughout the augmentor operating envelope. The amount of augmentation shall be controllable, and transition from nonaugmented to augmented operation shall be free of large abrupt thrust or power changes. The augmentor shall not induce an unacceptable loss of engine operability, either in operation or during initiation or cancellation over the required envelope.

3.7.11.1 Water injection system fluid.

For engines that use a water injection system, the engine shall meet all requirements specified herein with the water solution consisting of distilled or demineralized water and contaminated up the following levels:

Impurities	Quality
Total solids (max) PPM	35
Dissolved solids (max) PPM	25
pH Factor	6.5 - 7.5

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3.7.12 Wash system.

An integral wash system shall be provided with the engine if required by the Using Service. The wash system shall not allow standing water to accumulate in the engine. Wash system interface connections shall be compatible with existing Using Service support equipment and be described in the _____.

3.7.13 Brake system.

For turboprop engines, if required by the air vehicle, when the air vehicle is parked and the engine is shut off, the brake shall prevent (a) when exposed to winds up to (b) from any direction with the lower half of the propeller area blanked off. If required for air vehicle mission performance during flight, the brake shall be capable of being shut off during flight and shall prevent rotation at (c). For free turbine engines, the brake shall be capable of preventing rotation of the power turbine at the highest ground idle engine torque, if required by the Using Service.

3.7.14 Negative torque limiter.

For turboshaft or turboprop engines, following an engine failure or malfunction, a means to automatically prevent excessive negative torque shall be provided. The propulsion system shall return to normal operation when the malfunction is corrected and positive torque is requested.

3.7.14.1 Multiple power section engines.

For multiple power section turboshaft engines, a means shall be provided to automatically prevent an inoperative power section from absorbing power from the remaining operating power section(s) and the shaft power absorber at any flight condition or power setting. Inoperative power sections shall not result in the operating power section exceeding design or operating limits. An override mechanism shall be provided. A signal shall be provided to identify the section that has been separated. Separation and reconnection of a power section shall not cause any adverse affects on the performance and durability of the engine(s) and power train, and shall not affect safety of flight.

Multiple power section engines shall provide for load sharing and torque matching required by the engine control system in 3.7.2.

When re-starting an inoperative power section with the other power section(s) running, the subsequent reconnection shall be automatic.

3.7.15 Power absorber (engine supplied).

The engine supplied power absorber, which is used to propel the air vehicle as installed in the aircraft, shall meet the requirements of this specification. The power absorber shall include any remote engine supplied mechanism that absorbs power from the engine cycle.

3.7.15.1 Propeller feathering and reverse pitch.

Engines using propellers shall not exceed any operating limit when the propeller is operated in the feather or reverse pitch mode. If fuel shutoff is required, it shall be accomplished automatically by the control system. If engine and gearbox lubrication oil shutoff is required, it shall not occur until rotation has ceased.

3.7.16 Gearbox.

The gearbox shall provide mechanical power to components required for engine and aircraft operation and, if applicable, as a means to transmit torque to the engine for starting. The mechanical power characteristics at each pad shall be specified in table I.

The gearbox shall be capable of simultaneous operation of all the drives when each drive is subjected to (a) times the maximum permissible torque or power rating specified for the individual drive.

All gears shall be free from damaging resonance at all speeds up to (b) percent of maximum allowable steady-state speed.

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3.8 Software resources.

3.8.1 Software performance and design.

Software shall perform all computations, data processing, and interfacing functions necessary for the engine to meet the performance and operability requirements of the specification. All programming language shall be per _____.

3.8.1.1 Computer reprogramming.

The engine system shall have the computer capability for making software changes without requiring removal of memory hardware. Safety design provisions shall be provided to prevent unauthorized reprogramming. Each software version shall be identified by a unique software version number, which shall be verified during the loading process.

3.8.2 Spare resources.

Prior to the completion of the _(a)_ engine milestone, the electronic fuel control and engine monitoring system shall utilize no greater than _(b)_ and _(c)_ percent, respectively, of the available throughput during the worst case (most heavily loaded) execution of the operational software for each minor frame. The memory reserve shall be at least _(d)_ percent for control systems, and at least _(e)_ percent for engine monitoring systems.

3.9 Production facilities, capabilities, and processes.

The engine design shall be producible using the contractor's manufacturing system. The manufacturing system shall have the facilities, capabilities, processes, and process controls to provide products of consistent quality that meet performance requirements. Production processes shall have the stability, capability, and controls to maintain product characteristics within design tolerances and to reduce the variability of key product characteristics.

3.10 Production cost requirement (PCR).

The production cost shall not exceed that specified in the table:

Subsystem or Equipment	Cost to be Controlled	Dollar Amount	Type of Dollars	Volume	Rate	Elements

3.11 Engine system and controls and externals verification.

For the purposes of qualifying the engine system and Controls and Externals (C&E) during System Design and Development (Engineering and Manufacturing Development), the engine system and controls and externals shall be verified per 4.11.

If testing is used as the verification method, then the verification methods described in 4.11 shall be used.

3.12 Engine qualification.

For the purposes of qualifying the engine during System Design and Development (formerly Engineering and Manufacturing Development), Production and Deployment and Operations and Support (formerly Production, Fielding/Deployment and Operational Support), the engine shall be verified per 4.12.

Note: 3.12 shall not contain requirements. 4.12 shall contain verification requirements for engine tests that have not been specified in other sections of this guide.

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4. VERIFICATION.

The verifications established in Section 4 for the requirements specified in Section 3 are intended to result in a progressive in-process verification of design maturity that will be consistent with key milestones of the Government Acquisition Process relating to System Development and Demonstration (Engineering and Manufacturing Development) program schedule. The Verification Matrices (tables XXXI a and b) provide a cross-reference between the requirements and the associated method and timing of the verification. Table XXXIa may be used on programs employing a Spiral Development approach. Table XXXIb may be used for programs not using a Spiral Development approach but adhering to 5000 Series Acquisition System requirements.

All verification activities are required to provide adequate evidence, as determined by the Using Service, that the requirement has been verified. Detected failures or deficiencies during testing constitute non-compliance with the verification paragraph and corrective action and reverification shall be performed to the satisfaction of the Using Service. Verification procedures and success criteria not specified in this section, shall be established by the Contractor and be subject to review and approval or disapproval by the Using Service, prior to the initiation of qualification milestones.

4.1 Design and construction.

The acceptability of the open design shall be verified by completion of the qualification requirements.

4.1.1 Item and interface definition.

4.1.1.1 Interface and installation.

The requirements of 3.1.1.1 shall be verified by demonstration and inspection.

4.1.1.2 Installation hardware.

The requirements of 3.1.1.2 shall be verified by:

4.1.1.3 Interface loads.

The requirements of 3.1.1.3 shall be verified by:

4.1.1.4 Mounts.

4.1.1.4.1 Main mounts.

The requirements of 3.1.1.4.1 shall be verified by:

4.1.1.4.2 Ground handling mounts.

The requirements of 3.1.1.4.2 shall be verified by:

4.1.1.4.3 Engine stiffness.

The requirements of 3.1.1.4.3 shall be verified by:

4.1.1.5 Control lever and signals.

4.1.1.5.1 Power command lever and signal.

The requirements of 3.1.1.5.1 shall be verified by:

4.1.1.5.2 Load demand command and signal.

The requirements of 3.1.1.5.2 shall be verified by:

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4.1.1.5.3 Output speed lever and signal.

The requirements of 3.1.1.5.3 shall be verified by:

4.1.1.5.4 Fuel shutoff lever and signal.

The requirements of 3.1.1.5.4 shall be verified by:

4.1.1.5.5 Lever torque.

The requirements of 3.1.1.5.5 shall be verified by:

4.1.1.6 Subsystem interfaces.

The requirements of 3.1.1.6 shall be verified by:

4.1.1.7 Bleed air interface.

The requirements of 3.1.1.7 shall be verified by:

4.1.1.7.1 Customer bleed air contamination.

The requirement of 3.1.1.7.1 shall be verified by:

4.1.1.8 Drains and fluid collection devices.

The requirements of 3.1.1.8 shall be verified by:

4.1.1.9 Power absorber interface.

The requirements of 3.1.1.9 shall be verified by _____.

The maximum allowable misalignment of the power absorber shall be tested.

4.1.1.10 Power takeoff (PTO).

The requirements of 3.1.1.10 shall be verified by:

4.1.1.11 Starting drivetrain.

The requirements of 3.1.1.11 shall be verified by:

4.1.1.11.1 Starting torque and speed.

The requirements of 3.1.1.11.1 shall be verified by:

4.1.1.12 Thrust reverser interface.

The requirements of 3.1.1.12 shall be verified by:

4.1.1.13 Exhaust system interface.

The requirements of 3.1.1.13 shall be verified by:

4.1.1.14 Compatibility - engine and air vehicle.

The requirement of 3.1.1.14 shall be verified by:

4.1.1.15 Control and external components list.

The requirement of 3.1.1.15 shall be verified by:

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4.1.2 Physical characteristics.

4.1.2.1 Dry weight of complete engine.

The requirement of 3.1.2.1 shall be verified by test.

4.1.2.2 Mass moment of inertia of complete engine.

The requirement of 3.1.2.2 shall be verified by:

4.1.3 Materials, processes, and parts.

The requirements of 3.1.3 shall be verified by:

4.1.4 Fasteners.

The requirements of 3.1.4 shall be verified by:

4.1.5 Nameplate and product marking.

The requirement of 3.1.5 shall be evaluated by inspection.

4.1.5.1 Engine data plate marking.

The requirement of 3.1.5.1 shall be verified by inspection.

4.1.5.2 Critical parts/critical safety items identification and tracking.

The requirement of 3.1.5.2 shall be verified by inspection.

4.1.6 Transportability.

The requirements of 3.1.6 shall be verified by:

4.1.7 Interchangeability.

The requirements of 3.1.7 shall be verified by:

4.1.8 Safety.

4.1.8.1 Flammable fluid systems.

The requirements of 3.1.8.1 shall be verified as follows:

4.1.8.2 Fire prevention.

The requirements of 3.1.8.2 shall be verified by:

4.1.8.3 Explosion-proof.

The requirements of 3.1.8.3 shall be verified by:

4.1.8.4 Combustible fluid drains.

The requirements of 3.1.8.4 shall be verified by:

4.1.8.5 Air and gas leakage.

The requirements of 3.1.8.5 shall be verified by:

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4.1.8.6 Ground safety.

The requirements of 3.1.8.6 shall be verified by analysis and inspection.

4.1.8.7 Jet wake.

The requirements of 3.1.8.7 shall be verified by:

4.1.9 Design control.

4.1.9.1 Parts list.

The requirements of 3.1.9.1 shall be verified by inspection.

4.1.9.2 Assembly of components and parts.

The requirement of 3.1.9.2 shall be verified by:

4.1.9.3 Changes in vendors or fabrication process.

The requirements of 3.1.9.3 shall be verified by:

4.1.9.4 Standardization.

The requirements of 3.1.9.4 shall be verified by:

4.2 Performance and operability.

4.2.1 Performance characteristics.

4.2.1.1 Steady-state performance.

The requirements of 3.2.1.1 shall be verified by:

4.2.1.2 Transient performance.

The requirements of 3.2.1.2 shall be verified by:

4.2.1.3 Performance computer program.

The requirements of 3.2.1.3 shall be verified by:

4.2.1.4 Performance retention.

The requirements of 3.2.1.4 shall be verified by:

4.2.2 Operating characteristics and limits.

The requirements of 3.2.2 shall be verified by:

4.2.2.1 Operating envelope.

The requirements of 3.2.2.1 shall be verified by:

4.2.2.2 Operating attitude and conditions.

The requirements of 3.2.2.2 shall be verified by:

4.2.2.3 Starting.

The requirements of 3.2.2.3 shall be verified by:

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4.2.2.3.1 Ground starts.

The requirements of 3.2.2.3.1 shall be verified by:

4.2.2.3.2 Air starts.

The requirements of 3.2.2.3.2 shall be verified by:

4.2.2.3.3 Starting limits.

The requirements of 3.2.2.3.3 shall be verified by:

4.2.2.3.4 Starting procedure.

The requirements of 3.2.2.3.4 shall be verified by:

4.2.2.3.5 Automatic relight.

The requirements of 3.2.2.3.5 shall be verified by:

4.2.2.4 Stopping.

The requirements of 3.2.2.4 shall be verified by:

4.2.2.5 Low power conditions.

4.2.2.5.1 Idle thrust or power - (ground or flight idle).

The requirements of 3.2.2.5.1 shall be verified by:

4.2.2.5.2 No-load condition.

The requirements of 3.2.2.5.2 shall be verified by:

4.2.2.6 Stability.

The requirements of 3.2.2.6 shall be verified by:

4.2.2.7 Transients.

The requirement of 3.2.2.7 shall be verified by:

4.2.2.7.1 Overshoot and undershoot.

The requirement of 3.2.2.7.1 shall be verified by:

4.2.2.7.2 Droop.

The requirements of 3.2.2.7.2 shall be verified by:

4.2.2.8 Windmilling.

The requirements of 3.2.2.8 shall be verified by:

4.2.2.9 Reverse thrust.

The requirements of 3.2.2.9 shall be verified by:

4.2.2.10 Steady-state and transient airflow.

The requirements of 3.2.2.10 shall be verified by:

4.2.2.11 Inlet airflow distortion.

The requirements of 3.2.2.11 shall be verified by:

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4.2.2.11.1 Pressure and temperature rate of change.

The requirement of 3.2.2.11.1 shall be verified by:

4.2.2.12 Gas-path and measurement plane temperature limits.

The requirements of 3.2.2.12 shall be verified by:

4.2.2.13 Surface temperature and heat rejection.

The requirements of 3.2.2.13 shall be verified by:

4.2.2.13.1 Controls and external components limiting temperatures.

The requirement of 3.2.2.13.1 shall be verified by:

4.3 Environmental conditions.

4.3.1 Atmospheric conditions.

4.3.1.1 Humidity.

The requirements of 3.3.1.1 shall be verified by:

4.3.1.2 Fungus.

The requirements of 3.3.1.2 shall be verified by:

4.3.1.3 Corrosive atmosphere.

The requirements of 3.3.1.3 shall be verified by:

4.3.1.4 Icing conditions.

The requirements of 3.3.1.4 shall be verified by:

4.3.2 Ingestion capability (hazard resistance).

4.3.2.1 Bird ingestion.

The requirements of 3.3.2.1 shall be verified by:

4.3.2.2 Foreign object damage (FOD).

The requirements of 3.3.2.2 shall be verified by:

4.3.2.3 Ice ingestion.

The requirements of 3.3.2.3 shall be verified by:

4.3.2.4 Sand and dust ingestion.

The requirements of 3.3.2.4 shall be verified by:

4.3.2.5 Atmospheric liquid water ingestion.

The requirements of 3.3.2.5 shall be verified by:

4.3.2.6 Armament gas ingestion (AGI).

The requirements of 3.3.2.6 shall be verified by:

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4.3.2.7 Steam ingestion.

The requirements of 3.3.2.7 shall be verified by:

4.3.3 Electromagnetic environmental effects (E^3).

N/A.

4.3.3.1 Electromagnetic interference (EMI).

The requirements of 3.3.3.1 shall be verified by:

4.3.3.2 Intrasystem electromagnetic compatibility (EMC).

The requirement of 3.3.3.2 shall be verified by analysis and test on the engine as installed on the aircraft.

4.3.3.3 Intersystem electromagnetic compatibility (EMC).

The requirement of 3.3.3.3 shall be verified by analysis or test.

4.3.3.4 Lightning.

The requirement of 3.3.3.4 shall be verified by:

4.4 Integrity.

4.4.1 Structural integrity.

4.4.1.1 Design service life.

The requirements of 3.4.1.1 shall be verified by:

4.4.1.2 Design usage.

The requirements of 3.4.1.2 shall be verified by:

4.4.1.2.1 Externally applied forces.

The requirement of 3.4.1.2.1 shall be verified by:

4.4.1.3 Material characterization.

The requirement of 3.4.1.3 shall be verified by:

4.4.1.4 Parts classification.

The requirement of 3.4.1.4 shall be verified by:

4.4.1.5 Durability.

The requirements of 3.4.1.5 shall be verified by:

4.4.1.5.1 High cycle fatigue (HCF) life.

The requirements of 3.4.1.5.1 shall be verified by:

4.4.1.5.2 Low cycle fatigue (LCF) life.

The requirement of 3.4.1.5.2 shall be verified by:

4.4.1.5.3 Creep.

The requirements of 3.4.1.5.3 shall be verified by:

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4.4.1.6 Strength.

The requirements of 3.4.1.6 shall be verified by:

4.4.1.6.1 Factors of safety.

The requirements of 3.4.1.6.1 shall be verified by:

4.4.1.6.2 Blade and disk deflection.

The requirements of 3.4.1.6.2 shall be verified by:

4.4.1.6.3 Containment.

The requirements of 3.4.1.6.3 shall be verified by:

4.4.1.6.4 Blade out.

The requirements of 3.4.1.6.4 shall be verified by:

4.4.1.6.5 Overspeed and overtemperature.

The requirements of 3.4.1.6.5 shall be verified by:

4.4.1.6.6 Disk burst speed.

The requirements of 3.4.1.6.6 shall be verified by:

4.4.1.6.7 Output shaft torque limits.

The requirements of 3.4.1.6.7 shall be verified by:

4.4.1.6.8 Output shaft speed limits.

The requirements of 3.4.1.6.8 shall be verified by:

4.4.1.6.9 Bearing load.

The requirement of 3.4.1.6.9 shall be verified by:

4.4.1.6.10 Gyroscopic moments.

The requirements of 3.4.1.6.10 shall be verified by:

4.4.1.7 Damage tolerance.

The requirements of 3.4.1.7 shall be verified by:

4.4.1.7.1 Residual strength.

The requirements of 3.4.1.7.1 shall be verified by:

4.4.1.7.2 Initial production and in-service flaw size.

The requirements of 3.4.1.7.2 shall be verified by:

4.4.1.7.3 Flaw growth and inspection intervals.

The requirements of 3.4.1.7.3 shall be verified by:

4.4.1.8 Vibration and dynamic response.

The requirements of 3.4.1.8 shall be verified by:

4.4.1.8.1 Vibration limits.

The requirements of 3.4.1.8.1 shall be verified by:

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4.4.1.8.2 Critical speeds.

The requirements of 3.4.1.8.2 shall be verified by:

4.4.1.9 External surface foreign object damage.

The requirement of 3.4.1.9 shall be verified by:

4.4.2 Mechanical equipment and subsystems integrity.

The requirements of 3.4.2 shall be verified by:

4.4.3 Avionics and electronic integrity.

The requirements of 3.4.3 shall be verified by:

4.4.4 Software integrity.

The requirements of 3.4.4 shall be verified by:

4.5 Reliability and maintainability (R&M).

4.5.1 Reliability.

4.5.1.1 Reliability quantitative requirements.

The requirements of 3.5.1.1 and table XIII shall be verified by:

4.5.2 Maintainability.

4.5.2.1 Maintainability quantitative requirements.

The requirements of 3.5.2.1 shall be verified by:

4.5.2.1.1 Excluded maintenance functions.

The requirements of 3.5.2.1.1 shall be verified by:

4.5.2.2 Maintainability qualitative requirements.

The requirements of 3.5.2.2 shall be verified by:

4.5.2.3 Maintenance.

4.5.2.3.1 Modules.

The requirements of 3.5.2.3.1 shall be verified by:

4.5.2.3.2 Maintenance, inspection, and repair cycle.

The requirements of 3.5.2.3.2 shall be verified by:

4.5.2.3.2.1 Maintenance inspection techniques.

The requirements of 3.5.2.3.2.1 should be verified by:

4.5.2.3.3 Tools.

The requirements of 3.5.2.3.3 should be verified by:

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4.5.2.4 Battle damage repair.

The requirements of 3.5.2.4 should be verified by:

4.5.3 Human performance and human engineering.

The requirements of 3.5.3 shall be verified by:

4.6 Combat survivability.

The engine combat survivability design features that include state of the art survivability technology (vulnerability and susceptibility) to insure survival in the intended threat environment shall be demonstrated.

4.6.1 Susceptibility.

4.6.1.1 Noise.

The requirements of 3.6.1.1 shall be verified by:

4.6.1.2 IR signature.

The requirements of 3.6.1.2 shall be verified by:

4.6.1.3 Smoke.

The requirements of 3.6.1.3 shall be verified by:

4.6.1.4 Gaseous emissions.

The requirements of 3.6.1.4 shall be verified by:

4.6.1.5 Fuel streaming and vapor puffing.

The requirements of 3.6.1.5 shall be verified by:

4.6.1.6 Water vapor contrails.

The requirements of 3.6.1.6 shall be verified by:

4.6.1.7 Radar cross section.

The requirements of 3.6.1.7 shall be verified by:

4.6.1.8 Radar absorbent materials and coatings.

The requirements of 3.6.1.8 shall be verified by:

4.6.2 Vulnerability.

4.6.2.1 Ballistic weapons.

The requirements of 3.6.2.1 shall be verified by:

4.6.2.1.1 Static structure.

The requirements of 3.6.2.1.1 shall be verified by:

4.6.2.2 Nuclear weapons effects.

The requirement of 3.6.2.2 shall be verified by:

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4.6.2.3 Chemical and biological agent effects.

The requirements of 3.6.2.3 shall be verified by:

4.6.2.4 Fuel ingestion effects.

4.6.2.4.1 Fuel ingestion - steady-flow.

The requirements of 3.6.2.4.1 shall be verified by:

4.6.2.4.1.1 Fuel ingestion - cooling air contamination.

The requirements of 3.6.2.4.1.1 shall be verified by:

4.6.2.4.2 Fuel ingestion - transient (“quick dump”).

The requirement of 3.6.2.4.2 shall be verified by:

4.7 Subsystems.

4.7.1 Anti-icing and de-icing system.

The requirement of 3.7.1 shall be verified by:

4.7.2 Control system.

The requirements of 3.7.2 shall be verified by:

4.7.2.1 Control system performance.

The requirements of 3.7.2.1 shall be verified by:

4.7.2.1.1 Backup control.

The requirements of 3.7.2.1.1 shall be verified by:

4.7.2.2 Control system adjustments.

The requirements of 3.7.2.2 shall be verified by:

4.7.2.3 Overspeed protection system.

The requirements of 3.7.2.3 shall be verified by:

4.7.3 Fuel system.

4.7.3.1 Fuels.

The requirements of 3.7.3.1 shall be verified by:

4.7.3.1.1 Primary fuel.

Unless noted otherwise within the specification, engines shall be verified using the worst-case fuel(s) below:

4.7.3.1.2 Alternate fuel.

The requirements of 3.7.3.1.2 shall be verified by:

4.7.3.1.3 Restricted fuel.

The requirements of 3.7.3.1.3 shall be verified by:

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4.7.3.1.4 Emergency fuel.

The requirements of 3.7.3.1.4 shall be verified by:

4.7.3.2 Fuel system performance.

4.7.3.2.1 Fuel contamination.

The requirements of 3.7.3.2.1 shall be verified by:

4.7.3.2.2 Fuel system performance with external assistance.

The requirements of 3.7.3.2.2 shall be verified by:

4.7.3.2.3 Fuel system performance with no external assistance.

The requirements of 3.7.3.2.3 shall be verified by:

4.7.3.2.4 Fuel system performance under conditions of excessive fuel vapor.

The requirements of 3.7.3.2.4 shall be verified by:

4.7.3.2.5 Fuel pump priming.

The requirements of 3.7.3.2.5 shall be verified by:

4.7.3.2.6 Fuel lubricity.

The requirement of 3.7.3.2.6 shall be verified by:

4.7.3.2.7 Fuel system performance with water saturated fuel.

The requirements of 3.7.3.2.7 shall be verified by:

4.7.3.2.8 Fuel filter.

The requirements of 3.7.3.2.8 shall be verified by:

4.7.3.2.9 Fuel flow limit.

The requirements of 3.7.3.2.9 shall be verified by:

4.7.4 Electrical system.

4.7.4.1 Electrical power.

The requirements of 3.7.4.1 shall be verified by:

4.7.4.1.1 Generator.

The requirements of 3.7.4.1.1 shall be verified by:

4.7.4.2 Alternate electrical power.

The requirements of 3.7.4.2 shall be verified by:

4.7.4.3 Electrical connectors and cables.

The requirements of 3.7.4.3 shall be verified by:

4.7.4.4 Electrical bonding.

The requirements of 3.7.4.4 shall be verified by:

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4.7.4.5 Ground isolation.

The requirements of 3.7.4.5 shall be verified by:

4.7.4.6 Potting compounds.

The requirements of 3.7.4.6 shall be verified by:

4.7.5 Ignition system(s).

The requirement of 3.7.5 shall be verified by:

4.7.5.1 Ignition system fouling.

4.7.5.1.1 Carbon fouling.

The requirement of 3.7.5.1.1 shall be verified by:

4.7.5.1.2 Water fouling.

The requirement of 3.7.5.1.2 shall be verified by:

4.7.6 Engine health monitoring system (EHMS).

The requirement of 3.7.6 shall be verified by:

4.7.6.1 EHMS fault detection and isolation.

The requirement of 3.7.6.1 shall be verified by:

4.7.6.2 Onboard engine diagnostic functions.

The requirement of 3.7.6.2 shall be verified by:

4.7.6.2.1 Built-in-test and inspectability.

The requirements of 3.7.6.2.1 shall be verified by: (a) . The equipment fault detection and isolation capabilities shall be verified per: (b) .

4.7.7 Optical systems.

The requirements of 3.7.7 shall be verified by:

4.7.7.1 Fiber optic cables and connectors.

The requirements of 3.7.7.1 shall be verified by:

4.7.8 Lubrication system.

4.7.8.1 Lubrication oil.

The requirement of 3.7.8.1 shall be verified by:

4.7.8.1.1 Oil pressure and temperature limits.

The requirement of 3.7.8.1.1 shall be verified by:

4.7.8.1.2 Oil consumption limits.

The requirement of 3.7.8.1.2 shall be verified by:

4.7.8.1.3 Oil flow interruption or depletion.

The requirement of 3.7.8.1.3 shall be verified by:

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4.7.8.2 Lubrication system components and features.

4.7.8.2.1 Oil reservoir.

The requirements of 3.7.8.2.1 shall be verified by:

4.7.8.2.1.1 Oil reservoir external features.

The requirements of 3.7.8.2.1.1 shall be verified by:

4.7.8.2.2 Oil drains.

The requirements of 3.7.8.2.2 shall be verified by:

4.7.8.2.3 Oil filters.

The requirements of 3.7.8.2.3 shall be verified by:

4.7.8.2.4 Oil debris monitor.

The requirements of 3.7.8.2.4 shall be verified by:

4.7.8.3 Breather mist.

The requirements of 3.7.8.3 shall be verified by:

4.7.9 Hydraulic system.

The requirements of 3.7.9 shall be verified by:

4.7.9.1 Hydraulic ground test provisions.

The requirements of 3.7.9.1 shall be verified by:

4.7.9.2 Hydraulic system air removal.

The requirements of 3.7.9.2 shall be verified by:

4.7.9.3 Hydraulic fluid filters.

The requirements of 3.7.9.3 shall be verified by:

4.7.10 Exhaust nozzle system.

The requirements of 3.7.10 shall be verified by:

Actuator scraper and seal design shall be verified for effectiveness and wear from engine generated debris in a heated environment.

4.7.10.1 Exhaust nozzle external asymmetrical air pressure loads.

The requirements of 3.7.10.1 shall be verified for each engine and air vehicle combination by:

4.7.10.2 Vectoring nozzle.

The requirements of 3.7.10.2 shall be verified by:

4.7.10.2.1 Vectoring nozzle angle and rate.

The requirements of 3.7.10.2.1 shall be verified by:

4.7.10.2.2 Vectoring nozzle failure accommodation.

The requirement of 3.7.10.2.2 shall be verified by:

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4.7.11 Augmentation system.

The requirement of 3.7.11 shall be verified by:

4.7.11.1 Water injection system fluid.

The requirements of 3.7.11.1 shall be verified by:

4.7.12 Wash system.

The requirement of 3.7.12 shall be verified by:

4.7.13 Brake system.

The requirement of 3.7.13 shall be verified by:

4.7.14 Negative torque limiter.

The requirement of 3.7.14 shall be verified by:

4.7.14.1 Multiple power section engines.

The requirement of 3.7.14.1 shall be verified by:

4.7.15 Power absorber (engine supplied).

The requirement of 3.7.15 shall be verified by:

4.7.15.1 Propeller feathering and reverse pitch.

The requirement of 3.7.15.1 shall be verified by:

4.7.16 Gearbox.

The requirement of 3.7.16 shall be verified by:

4.8 Software resources.

4.8.1 Software performance and design.

The requirements of 3.8.1 shall be verified by: _____. The functional performance of the complete computer system shall be demonstrated after integration into the computer system hardware during all engine testing.

4.8.1.1 Computer reprogramming.

The requirements of 3.8.1.1 shall be verified by:

4.8.2 Spare resources.

The requirements of 3.8.2 shall be verified by:

4.9 Production facilities, capabilities, and processes.

The requirements of 3.9 shall be verified by:

a. The following Incremental Verifications shall be accomplished prior to the Preliminary Design Review (PDR):

b. The following Incremental Verifications shall be accomplished prior to Critical Design Review (CDR):

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c. The following Incremental Verifications shall be accomplished prior to Initial Service Release (ISR):

d. The following Incremental Verifications shall be accomplished prior to Operational Capability Release:

4.10 Production cost requirement (PCR).

This requirement shall be verified by analysis using a Government agreed to production cost estimating method. The production cost estimating method consists of the procedures, assumptions, tools, database(s), and model(s) agreed to by the government and contractor and is hereafter referred to as the "Production Cost Model" (PCM).

a. The following Incremental Verifications shall be accomplished prior to the Preliminary Design Review:

b. The following Incremental Verifications shall be accomplished prior to Critical Design Review:

c. The following Incremental Verifications shall be accomplished prior to Initial Service Release (ISR):

4.11 Engine system and controls and externals verification.

The engine system and controls and externals (C&E) that require a test for verification shall be verified by the test procedures per the following paragraphs.

4.11.1 Engine systems tests.

4.11.1.1 Endurance tests.

The endurance tests in the subparagraphs below shall be required for the class of engines specified herein.

4.11.1.1.1 Accelerated mission test (AMT).

For turbofan and turbojet engines the following tests shall apply:

For IFR:

An engine conforming to the IFR configuration shall be subjected to an AMT of at least (a) hours or equivalent cycles. The test shall consist of the engine accelerated mission duty cycles defined on figure 21 and in table XL and table XXVIII.

A minimum of (b) starts shall be made on the engine. There shall be (c) false starts and (d) restarts. All test starts shall be performed with a starter as specified in 4.12.3.3.4. At least (e) starts shall be regulated to provide shutdown periods between starts of (f) minutes increasing at (g) minute intervals. Each of the regulated shutdown periods shall be preceded by immediate engine shutdown, without being held at idle, after operation for a duration of not less than 30 seconds at intermediate power at the discretion of the Using Service. The remaining starts shall be conducted per the temperature stabilization procedure specified below, but otherwise the shutdown period need not be controlled. The test shall be preceded and followed by the stair-step (HCF) test schedule as specified in 4.4.1.5.1.

For FFR:

An engine conforming to the FFR configuration shall be subjected to an AMT of at least (a) hours or equivalent cycles. The test shall consist of the accelerated mission duty cycles defined on figure 21 and in table XL and table XXVIII. A minimum of (b) starts shall be made on the engine. There shall be (c) false starts and (d) restarts. All test starts shall be performed with a starter as specified in 4.12.3.3.4. At least (e) starts shall be regulated to provide shutdown periods between starts of (f) minutes increasing at (g) minute intervals. Each of the regulated shutdown periods shall be preceded by immediate engine shutdown, without being held at idle, after engine operation for a duration of not less than 30 seconds at

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intermediate power at the discretion of the Using Service. The remaining starts shall be conducted per the temperature stabilization procedure specified below, but otherwise the shutdown period need not be controlled.

The test shall be preceded and followed by the stair-step (HCF) test schedule as specified in 4.4.1.5.1.

For ISR:

An engine conforming to the ISR configuration shall be subjected to an AMT of at least (a) hours or equivalent cycles. The test shall consist of the accelerated mission duty cycles defined on figure 21 and in table XL and table XXVIII. A minimum of (b) starts shall be made on the engine. There shall be (c) false starts and (d) restarts. All test starts shall be performed with a starter as specified in 4.12.3.3.4. At least (e) starts shall be regulated to provide shutdown periods between starts of (f) minutes increasing at (g) minute intervals. Each of the regulated shutdown periods shall be preceded by immediate engine shutdown, without being held at idle, after engine operation for a duration of not less than 30 seconds at intermediate power at the discretion of the Using Service. The remaining starts shall be conducted per the temperature stabilization procedure specified below, but otherwise the shutdown period need not be controlled.

The test shall be preceded and followed by the stair-step (HCF) test schedule as specified in 4.4.1.5.1.

For OCR:

An engine conforming to the OCR configuration, but containing the same cold parts used in the ISR test, shall be subjected to an AMT of at least (a) hours or equivalent cycles. The test shall consist of the accelerated mission duty cycles defined on figure 21 and in table XL and table XXVIII. A minimum of (b) starts shall be made on the engine. There shall be (c) false starts and (d) restarts. All test starts shall be performed with a starter as specified in 4.12.3.3.4. At least (e) starts shall be regulated to provide shutdown periods between starts of (f) minutes increasing at (g) minute intervals. Each of the regulated shutdown periods shall be preceded by immediate engine shutdown, without being held at idle, after engine operation for a duration of not less than 30 seconds at intermediate power at the discretion of the Using Service. The remaining starts shall be conducted per the temperature stabilization procedure specified below, but otherwise the shutdown period need not be controlled.

The test shall be preceded and followed by the stair-step (HCF) test schedule as specified in 4.4.1.5.1.

The test operation for all engine milestones shall be conducted as follows:

A fixed level of engine inlet air temperature, fuel inlet temperature, and pressure shall be set for each mission cycle as defined by the Strength and Life Analysis Report and defined on figure 21 and in table XL and table XXVIII.

Air passing over the engine shall correspond to the engine inlet air temperature for that mission condition, within facility constraints.

Upon completion of each mission cycle, the engine shall be shutdown prior to the start of a subsequent cycle. During the shutdown period the engine shall be motored or wind-milled until the (h) measured temperature stabilizes below (i) °C ((i) °F) when the engine is not rotating.

Condition monitoring and diagnostic system installed engine hardware and software shall be in the production configuration. The condition monitoring and diagnostic system shall be operated in a manner analogous to its proposed service operation throughout the test and the output data shall be correlated to observations made during the test.

The customer power takeoff drives and customer bleed air extraction level shall be representative of the maximum required at that condition.

The engine anti-icing or de-icing system shall be actuated at periods during the test to simulate mission operation.

The exhaust duct, inlet duct, and bleed air connections shall be loaded to the maximum allowable loads specified in 3.1.1.3 and in the pretest data of 4.12.4.1.1 respectively, at the discretion of the Using Service.

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Fuel inlet temperature shall be maintained at the maximum fuel inlet temperature specified in 3.7.3.2.2 and 3.7.3.2.3 for at least _(j)_ percent of the duration of the mission oriented schedule.

The thrust or power demand shall be advanced or retarded in 0.50 seconds or less unless otherwise specified. The time for changing thrust shall be charged to the duration of the lower thrust setting.

The oil pressure, if adjustable, shall be adjusted at the beginning of the test to the minimum steady-state value specified in 3.7.8.1.1. No further adjustments shall be permitted during the test except when authorized by the Using Service.

The total quantity of fluid leakage from the engine drains shall be recorded and shall not exceed values specified in 3.1.1.8.

Special engine control system features (e.g.; overspeed, overtemperature) shall be cycled through their functions periodically as specified in the pre-test data of 4.12.4.1.1.

At the completion of the test, the engine oil drain provisions specified in 3.7.8.2.2 shall be demonstrated.

If the engine provides special features (i.e.; IR suppressors, fuel heaters, etc.), these items shall be activated during the test to simulate mission operation.

This test is to be run using the distortion pattern(s) of 3.2.2.11.

The test exhaust system (if different from vehicle flight exhaust) shall produce the same aerodynamic and thermodynamic cycle match as the flight exhaust of the primary using application.

4.11.1.1.2a Missionized durability test.

The following tests apply for turbofan, turbojet, turboprop, and turboshaft engines:

For IFR:

An engine conforming to the IFR configuration shall be tested to the missionized cycles of figure 21 and table XL, for _(a)_ hours or cycles. The test shall be conducted so that the _(b)_ turbine temperature shall be equal to _(c)_ °C (_(c)_ °F) at thrust or power settings of intermediate power and above.

For turboprop and turboshaft engines: not applicable

For FFR:

An engine conforming to the FFR configuration shall be tested to the missionized cycles of figure 21 and table XL, for _(d)_ hours or cycles. The test shall be conducted so that the _(e)_ turbine temperature shall be equal to _(f)_ °C (_(f)_ °F) at thrust or power settings of intermediate power and above.

For turboprop and turboshaft engines: not applicable

For ISR:

An engine conforming to the ISR configuration shall be tested to the missionized cycles of figure 21 and table XL, for _(g)_ hours or cycles. The test shall be conducted so that the _(h)_ turbine temperature shall be equal to _(i)_ °C (_(i)_ °F) at thrust or power settings of intermediate power and above.

For turboprop and turboshaft engines: not applicable

For OCR:

An engine conforming to the OCR configuration shall be tested to the missionized cycles of figure 21, table XL, and table XXIX for _(j)_ hours or cycles. The test shall be conducted so that the _(k)_ turbine temperature shall be equal to _(l)_ °C (_(l)_ °F) at thrust or power settings of intermediate power and above.

For turboprop and turboshaft engine Life Verification Test. Two engines each conforming to the OCR configuration shall be tested to the missionized cycles of figure 21 and table XL for _(m)_ hours. The test shall be conducted so that the _(k)_ turbine temperature shall be equal to _(l)_ °C (_(l)_ °F) at maximum continuous and above.

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4.11.1.1.2b Cyclic durability test.

The following tests shall apply for turboshaft and turboprop engines:

For IFR:

An engine conforming to the IFR configuration shall be tested for (a) hours of the durability cycles of table XXIX. The engine shall be tested such that the (b) turbine temperature shall be equal to the maximum operating or control limiting temperature for each power setting.

The durability test shall be performed on one engine using the fuels and oils of 4.12.3.4.2.1 and 4.12.3.4.2.2 respectively, and the shaft torque absorber characteristics of 3.7.16.

For FFR:

An engine conforming to the FFR configuration shall be tested for (c) hours of the durability cycles of table XXIX. The engine shall be tested such that the (d) turbine temperature shall be equal to the maximum operating or control limiting temperature for each power setting.

The durability test shall be performed on one engine using the fuels and oils of 4.12.3.4.2.1 and 4.12.3.4.2.2, respectively, and the shaft torque absorber characteristics of 3.7.16.

For ISR:

An engine conforming to the ISR configuration shall be tested for (e) hours of the durability cycles of table XXIX. The engine shall be tested such that the (f) turbine temperature shall be equal to the maximum operating or control limiting temperature for each power setting.

The durability test shall be performed on two engines, using the fuels and oils of 4.12.3.4.2.1 and 4.12.3.4.2.2, respectively, and the shaft torque absorber characteristics representative of different vehicle applications, as applicable.

For OCR:

An engine conforming to the OCR configuration shall be tested for (g) hours of the durability cycles of table XXIX. The engine shall be tested such that the (h) turbine temperature shall be equal to the maximum operating or control limiting temperature for each power setting.

The durability test shall be performed on one engine using the fuels and oils of 4.12.3.4.2.1 and 4.12.3.4.2.2 respectively, and the shaft torque absorber characteristics representative of 3.7.16.

4.11.1.2 Altitude test.

The performance requirements of 3.2 shall be verified by test. An engine conforming to the same parts list and configuration as the endurance test engine of 4.11.1.1, per 4.12.3.2, shall be subjected to altitude tests at several selected conditions within the operating limits envelope specified in figure 6 (P_{T1} and T_{T1}). Pre-test data shall be per 4.12.4.1.1. The test points shall include the effects of power extraction, bleed air extraction, inlet recovery, inlet distortion, anti-icing or de-icing, and windmilling on engine performance and operability. No adjustments shall be made to the engine without approval of the Using Service. The altitude tests shall be accomplished using the fuels and oils of 3.7.3.1.1 and 3.7.8.1, respectively. Fuel temperature shall be varied over a range sufficient to encompass all anticipated engines operating environments. If the engine has a continuous ignition system, it shall be in operation at all times after the start sequence has been completed. Data to be taken and recorded during the test shall be as specified in 4.12.4. Overall true root mean square (RMS) velocity measurements and acceleration spectrograms shall be obtained for each transducer mounted on the engine case and engine gearbox case at the engine speeds and thrusts or power settings selected for the test. The points selected shall include at least the altitude representative points and the point in the complete environmental conditions and operating envelope where the highest engine vibration levels are generated. Critical components of the engine shall be identified on each spectrogram.

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The following altitude test procedures apply to each of the following engine qualification milestones.

For IFR:

a. Altitude Performance Points. The requirements of 3.2.1.1 and 3.2.1.2 shall be verified at no fewer than (a) flight altitude condition performance points selected by the Using Service. An adequate number of engine thrust or power settings as determined by the Using Service shall be selected for each specified altitude test condition to establish operating and performance characteristics at each flight altitude condition tested. Operation at each test point shall be of sufficient duration to stabilize the engine.

b. Transient Operation and Functional Test. The requirements of 3.2.2.6 and 3.2.2.7 shall be verified at no fewer than (b) flight altitude conditions selected by the Using Service. Effects of bursts, chops, and bodies throughout the thrust or power range shall be demonstrated. The time elapsed versus engine speeds, measured pressures and temperatures, fuel flows, thrust or power and facility operating conditions shall be obtained for stability verification over the entire thrust or power range. Transients shall be repeated at selected flight conditions to explore control system lags, repeatability, and performance droop. Augmentor ignition limits of 3.2.2.1 shall be tested.

c. Inlet distortion. The requirements of 3.2.2.11 shall be verified by engine operation with simulated inlet airflow distortion at selected operating conditions chosen by the Using Service. Engine stability shall be verified and stall margin remaining defined by engine test at a minimum of (c) operating conditions and (d) distortion patterns selected by the Using Service that combine worst case engine destabilizing influences. Effects of bursts, chops, and bodies throughout the thrust or power range shall be demonstrated.

d. Starts and Restarts. Sufficient engine start and restart testing shall be accomplished to verify the starting requirements of 3.2.2.3 and 3.2.2.3.2. If the automatic relight feature of 3.2.2.3.5 is provided, it shall be verified. The automatic relight feature may be disabled to allow verification of restart requirements with the approval of the Using Service. Each spooldown airstart test condition shall be accomplished at no fewer than (e) core turnaround speeds as defined by the Using Service. A success rate of (f) shall be used to establish successful air starting and restarting at any given condition.

e. Windmilling Test. The requirements of 3.2.2.8 shall be verified at (g) altitude and Mach number condition(s) as selected by the Using Service.

f. Stopping. The requirements of 3.2.2.4 shall be verified each time the engine is stopped and at (h) flight test condition(s) selected by the Using Service.

For FFR (If there is no configuration change from IFR, this test may be waived by the Using Service.)

a. Altitude Performance Points. The requirements of 3.2.1.1 and 3.2.1.2 shall be verified at no less than (aa) flight altitude condition performance points selected by the Using Service. An adequate number of engine thrust or power settings as determined by the Using Service shall be selected for each specified altitude test condition to establish operating and performance characteristics at each flight altitude condition tested. Operation at each test point shall be of sufficient duration to stabilize the engine.

b. Transient Operation and Functional Test. The requirements of 3.2.2.6 and 3.2.2.7 shall be verified at no fewer than (bb) flight altitude conditions selected by the Using Service. Effects of bursts, chops, and bodies throughout the thrust or power range shall be demonstrated. The time elapsed versus engine speeds, measured pressures and temperatures, fuel flows, thrust or power, and facility operating conditions shall be obtained for stability verification over the entire thrust or power range. Transients shall be repeated at selected flight conditions to explore control system lags, repeatability, and performance droop. Augmentor ignition limits of 3.2.2.1 shall be tested.

c. Inlet distortion. The requirements of 3.2.2.11 shall be verified by engine operation with simulated inlet airflow distortion at selected operating conditions chosen by the Using Service. Engine stability shall be verified and stall margin remaining defined by engine test at a minimum of (cc) operating conditions and (dd) distortion patterns selected by the Using Service that combine worst case

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engine destabilizing influences. Effects of bursts, chops, and bodies throughout the thrust or power range shall be demonstrated.

d. Starts and Restarts. Sufficient engine start and restart testing shall be accomplished to verify the starting requirements of 3.2.2.3 and 3.2.2.3.2. If the automatic relight feature of 3.2.2.3.5 is provided, it shall be verified. The automatic relight feature may be disabled to allow verification of restart requirements with the approval of the Using Service. Each spooldown airstart test condition shall be accomplished at no fewer than (ee) core turnaround speeds as defined by the Using Service. A success rate of (ff) shall be used to establish successful air starting and restarting at any given condition.

e. Windmilling Test. The requirements of 3.2.2.8 shall be verified at (gg) altitude and Mach condition(s) as selected by the Using Service.

f. Stopping. The requirements of 3.2.2.4 shall be verified each time the engine is stopped and at (hh) flight test condition(s) selected by the Using Service.

For ISR (If there is no configuration change from FFR, this test may be waived by the Using Service)

a. Altitude Performance Points. The requirements of 3.2.1.1 and 3.2.1.2 shall be verified at no less than (aaa) flight altitude condition performance points selected by the Using Service. An adequate number of engine thrust or power settings as determined by the Using Service shall be selected for each specified altitude test condition to establish operating and performance characteristics at each flight altitude condition tested. Operation at each test point shall be of sufficient duration to stabilize the engine.

b. Transient Operation and Functional Test. The requirements of 3.2.2.6 and 3.2.2.7 shall be verified at no fewer than (bbb) flight altitude conditions selected by the Using Service. Effects of bursts, chops, and bodies throughout the thrust or power range shall be demonstrated. The time elapsed versus engine speeds, measured pressures and temperatures, fuel flows, thrust or power, and facility operating conditions shall be obtained for stability verification over the entire thrust or power range. Transients shall be repeated at selected flight conditions to explore control system lags, repeatability, and performance droop. Augmentor ignition limits of 3.2.2.1 shall be tested.

c. Inlet distortion. The requirements of 3.2.2.11 shall be verified by engine operation with simulated inlet airflow distortion at selected operating conditions chosen by the Using Service. Engine stability shall be verified and stall margin remaining defined by engine test at a minimum of (ccc) operating conditions and (ddd) distortion patterns selected by the Using Service that combine worst case engine destabilizing influences. Effects of bursts, chops, and bodies throughout the thrust or power range shall be demonstrated.

d. Starts and Restarts. Sufficient engine start and restart testing shall be accomplished to verify the starting requirements of 3.2.2.3 and 3.2.2.3.2. If the automatic relight feature of 3.2.2.3.5 is provided, it shall be verified. The automatic relight feature may be disabled to allow verification of restart requirements with the approval of the Using Service. Each spooldown airstart test condition shall be accomplished at no fewer than (eee) core turnaround speeds as defined by the Using Service. A success rate of (fff) shall be used to establish successful air starting and restarting at any given condition.

e. Windmilling Test. The requirements of 3.2.2.8 shall be verified at (ggg) altitude and Mach condition(s) as selected by the Using Service.

f. Stopping. The requirements of 3.2.2.4 shall be verified each time the engine is stopped and at (hhh) flight test condition(s) selected by the Using Service.

For OCR (If there is no configuration change from ISR, this test may be waived by the Using Service)

a. Altitude Performance Points. The requirements of 3.2.1.1 and 3.2.1.2 shall be verified at no fewer than (aaaa) flight altitude condition performance points selected by the Using Service. An adequate number of engine thrust or power settings as determined by the Using Service shall be selected for each specified altitude test condition to establish operating and performance

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characteristics at each flight altitude condition tested. Operation at each test point shall be of sufficient duration to stabilize the engine.

b. Transient Operation and Functional Test. The requirements of 3.2.2.6 and 3.2.2.7 shall be verified at no fewer than (bbbb) flight altitude conditions selected by the Using Service. Effects of bursts, chops, and bodies throughout the thrust or power range shall be demonstrated. The time elapsed versus engine speeds, measured pressures and temperatures, fuel flows, thrust or power, and facility operating conditions shall be obtained for stability verification over the entire thrust or power range. Transients shall be repeated at selected flight conditions to explore control system lags, repeatability, and performance droop. Augmentor ignition limits of 3.2.2.1 shall be tested.

c. Inlet distortion. The requirements of 3.2.2.11 shall be verified by engine operation with simulated inlet airflow distortion at selected operating conditions chosen by the Using Service. Engine stability shall be verified and stall margin remaining defined by engine test at a minimum of (cccc) operating conditions and (dddd) distortion patterns selected by the Using Service that combine worst case engine destabilizing influences. Effects of bursts, chops, and bodies throughout the thrust or power range shall be demonstrated.

d. Starts and Restarts. Sufficient engine start and restart testing shall be accomplished to verify the starting requirements of 3.2.2.3 and 3.2.2.3.2. If the automatic relight feature of 3.2.2.3.5 is provided, it shall be verified. The automatic relight feature may be disabled to allow verification of restart requirements with the approval of the Using Service. Each spooldown airstart test condition shall be accomplished at no fewer than (eeee) core turnaround speeds as defined by the Using Service. A success rate of (ffff) shall be used to establish successful air starting and restarting at any given condition.

e. Windmilling Test. The requirements of 3.2.2.8 shall be verified at (gggg) altitude and Mach condition(s) as selected by the Using Service.

f. Stopping. The requirements of 3.2.2.4 shall be verified each time the engine is stopped and at (hhhh) flight test condition(s) selected by the Using Service.

4.11.1.3 Engine systems tests success criteria.

The test shall be satisfactorily completed when the following conditions are met:

a. The engine shall meet the pass or fail criteria of the test plan.

b. The engine shall meet the inspection criteria of 4.12.3.5.4.

c. The engine and engine components meet the service limits of 4.12.3.5.3.

d. There are no part failures or any parts failing integrity requirements of 3.4.

e. All failures and problems encountered during the test and any additional testing that may be required has been analyzed and resolved to the satisfaction of the Using Service (i.e., penalty runs).

f. The test engine and all initial (start of test) components are operated satisfactorily to the end of the test and an analysis utilizing engine calibration results establishes that the performance characteristics of 3.2 and the starting requirements of 3.2.2.3 have been met.

g. Inspection of engine parts and engine components throughout the test, including the final inspection, shows that the parts life requirements of the milestone have been met.

h. Test results indicate compliance with the reliability and maintainability requirements specified in 3.5.

i. The engine weight meets the requirements of 3.1.2.1.

j. The engine has not exceeded any limits of table IV.

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4.11.2 Controls and externals component tests.

4.11.2.1 Controls and externals component operational tests.

Controls and external component specification compliance shall be verified by analysis, test, and inspection. Table XXX specifies required component tests.

4.11.2.1.1 Controls and externals component calibration.

If C&E testing is performed, these components shall be calibrated and the calibration shall be recorded.

4.11.2.1.2 Controls and externals component test procedures.

If component testing is required, then prior to the start of any testing, all components shall be cleaned of any corrosion preventive compounds used for storage protection to simulate the condition of the part as installed on the engine. Test assemblies or components shall be subjected to operating loads simulating those encountered on the engine. Instrumentation shall be provided to indicate the performance of each component and to indicate that the functional relationships of components are maintained as required by the applicable test schedule. Functional checks shall be performed at the end of each test or group of tests and at other times, at the option of the Contractor, to indicate that no calibrated component has changed its calibration beyond allowable service limits and that the function of uncalibrated components is unimpaired. All components shall be tested with the primary operating fluids of 3.7.3.1.1, 3.7.8.1, and 3.7.9. All items driven by the engine gearbox shall be operated at maximum allowable axial and angular misalignment conditions at the drive pad. Component subassemblies or components shall be the correct part number intended for use during the appropriate milestone.

4.11.2.1.3 Controls and externals component missionized tests.

If C&E component missionized tests are required, then all engine components, shall be subjected to a simulated mission operational test (or cycles) intended to produce the same equivalent component damage as would be achieved by real time tests based on the air system mission and mission mix of table XI. The simulated mission operational test cycle shall be defined by the contractor. The test shall include maximum and minimum fluid temperatures expected during engine operation and shall include maximum hot and cold soak temperatures of figure 5.

Components utilizing bleed air or requiring pneumatic input signals shall be subjected to air at pressure and temperature values corresponding to those occurring throughout the range of engine operation. _____ test cycles, interspersed throughout the test, shall include air contaminated as follows:

a. The amount of engine lubricating oil shall be consistent with what has leaked into the flow path in previous engine testing, but not less than 3 parts per million engine lubricating oil by mass.

b. A salt concentration of 200-ppb (0.2-ppm) salt in air injected into the engine using a 4-percent salt-water solution per table VI.

c. Humidity levels consistent with 3.3.1.1

d. Sand and dust composed of quartz and other material per table XXXV with a particle size range of 0-80 micron (0.00315-inches).

4.11.2.1.4 Controls and externals component recalibration, disassembly, and inspection.

Upon completion of the preceding tests, component calibrations of 4.11.2.1.1 shall be repeated. Components not subjected to calibration shall be operated under normal operating conditions to demonstrate satisfactory functioning and compatibility with other system components. During recalibration, the same fluids, inputs, and operating cycles shall be used as in the calibration. Each recalibration shall be recorded. All components shall then be completely disassembled and inspected for indications of failure or excessive wear.

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4.11.2.1.5 Controls and externals component environmental test.

All engine components or sub-components shall be subjected to environmental component testing per table XXX and as noted herein. These tests may be conducted on test assemblies or individual components, either of which may be new or not previously subjected to any testing. When a single hermetically sealed component is used in a series of the tests below, the component need not be disassembled for inspection until the last test of such series has been completed. In addition, hermetically sealed components need not be subjected to the explosion proof, sand and dust, and fungus tests. Prior to disassembly, a test to determine hermetic seal integrity shall be performed. Failure of the hermetic seals during any test shall disqualify that component. The components shall be inspected for defects or damage that may have been incurred during any of the tests performed.

4.11.2.1.5.1 Controls and externals component sand and dust.

Components shall be subjected to a sand and dust test per MIL-STD-810, Method 510. Components need not be operated during the test; however, at the conclusion of the test, and prior to any cleaning of the components, four power supply transients, covering the transient range, shall be applied to the component. Criteria for passing the test shall be as defined in MIL-STD-810, Method 510.

4.11.2.1.5.2 Controls and externals component vibration.

All components shall be verified to be operational when exposed to vibrations as when the components are installed and in an operational environment. During the verification the component shall be subjected to its maximum limiting component temperature specified in table V.

4.11.2.1.6 Controls and externals component test success criteria.

The component tests shall be considered to be satisfactorily completed when, in the judgment of the Using Service:

a. During the tests, component performance and function were within established limits.

b. During the tests, there was no fluid leakage from any component other than that of a nature and rate specified in the specification.

c. During the tests, there was no binding or hesitation of any component.

d. Recalibrations indicate that no component has changed its calibration beyond allowable service limits.

e. The component teardown inspection shows no indication of failed, excessively worn, and distorted parts. Measurements shall be taken and compared with the contractor's drawing dimensions and tolerances or with similar measurements made prior to the test.

4.12 Engine qualification.

The engine shall be verified during each engine milestone per this section, and table XXXI. Each verification in each milestone must pass its respective criteria and gain approval of the Using Service prior to proceeding to the next milestone. All verifications shall have quantified success criteria and these criteria shall be approved by the Using Service prior to initiation of each milestone.

4.12.1 Qualification milestones.

The acceptability of the engine design for each milestone shall be predicated on satisfactory completion of the verification requirements of section 4 per a or b and the approval of the Using Service.

4.12.2 Verification responsibility.

The contractor is responsible for the performance of all verification methods specified herein.

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4.12.3 Manner of testing.

Unless otherwise noted, the following testing methods, conditions, and procedures shall be required for full engine system verification and qualification per table XXXI a or b.

The test procedures shall be in compliance with table XXXII.

4.12.3.1 Test surveillance.

Each test and demonstration herein shall be subject to witnessing by authorized Using Service representatives. At convenient times prior to the tests and during teardown inspections, the engine and components shall be examined to determine if they conform to all requirements of the contract and specifications under which they were built. At no time shall any part of the engine or component be disassembled, adjusted, cleaned, replaced, or removed without prior approval of the Using Service representative.

4.12.3.2 Test article configuration.

Test article configuration and the differences from the milestone parts list of 3.1.9.1 shall be identified and justified in the pretest data of 4.12.4.1.1 and shall be included in the detailed test procedure. The mixing of parts of segmented stator assembly or different designed blades in a disc is not allowed without Using Service approval. All parts shall be considered as having zero time at the start of a test, as appropriate, unless otherwise specified by the Using Service. While all parts do not have to be new, any part that fails the success criteria during the test shall be cause for rejection of that part or test article as determined by the Using Service.

4.12.3.2.1 Test article.

The particular article intended for a specific milestone test or demonstration shall be officially designated by the contractor prior to the start of the test.

4.12.3.2.2 Installation interfaces.

Inlet, exhaust, and air bleed connections shall be as defined in 3.1.1.1. The engine gearbox shall duplicate the overhung moments of table I.

4.12.3.3 Test equipment.

4.12.3.3.1 Automatic recording equipment.

Automatic steady-state and transient recording equipment shall be used to record data during the execution of those parts of the tests required to meet the objectives of each verification. Automatic continuous recording equipment shall be used for tests requiring the verification of test article variables versus time.

4.12.3.3.2 Vibration measuring equipment and response characteristics.

The vibration measurement and analysis equipment shall operate over a frequency band of at least _(a)_ to _(a)_ and produce acceleration spectrograms with a confidence level of _(b)_.

4.12.3.3.3 Test stand dynamic characteristics.

The test stand shall not induce damaging or detrimental resonances into the engine at any test or operating condition.

4.12.3.3.4 Starter torque characteristic.

Starting shall be performed with a starter that has torque characteristics within _____ percent of the minimum required torque shown on figure 2.

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4.12.3.4 Test conditions.

4.12.3.4.1 Oil servicing and consumption.

The oil system shall be serviced at the start of each test or demonstration. All additions of oil added after initial servicing shall be recorded. Oil shall be drained only when authorized by the Using Service. The use of external oil filters shall not be permitted. Oil consumption shall be measured and the average consumption rate determined.

4.12.3.4.2 Fluid properties for test.

4.12.3.4.2.1 Fuel properties.

Each test and demonstration shall be conducted using a primary or alternate fuels specified in 3.7.3.1.1 and 3.7.3.1.2, unless otherwise specified. If more than one primary fuel is specified, the fuel used shall be the fuel, which makes the test the most difficult as determined by the Using Service. As directed by the Using Service and consistent with the model specification, testing will be repeated with other fuels. Fuel samples shall be taken at the start and completion of Verification and Qualification tests, and other tests as applicable. The fuel samples shall be analyzed for physical and chemical properties to determine conformance with applicable specifications. The results of these analyses shall be included as part of the applicable reports.

4.12.3.4.2.2 Oil properties.

Each test and demonstration shall be conducted using the oils specified in 3.7.8.1. If more than one oil is specified, the oil used will be the oil that makes the test most difficult as determined by the Using Service. Oil samples shall be taken at the start and completion of Verification and Qualification tests and other tests as applicable. The oil samples shall be analyzed for physical and chemical properties to determine conformance with applicable specifications. The results of these analyses shall be included as part of the applicable reports.

4.12.3.4.2.3 Hydraulic fluid properties.

Each test and demonstration shall be conducted using the hydraulic fluids specified in 3.7.9. Hydraulic fluid samples shall be taken at the start and completion of Verification and Qualification tests and other tests as applicable. The fluid samples shall be analyzed for physical and chemical properties to determine conformance with applicable specifications. The results of these analyses shall be included as part of applicable [test] reports.

4.12.3.5 Test calibrations and procedures.

4.12.3.5.1 Calibration.

4.12.3.5.1.1 Control and engine monitoring system calibration.

Prior to the initiation of the engine calibration all components necessary for proper control and engine monitoring systems operation and scheduling shall be checked on the bench and on the engine to establish proper functioning and calibration over the applicable ranges of engine operating conditions.

4.12.3.5.1.2 Prior to the initiation of the engine calibration all components necessary for proper control and engine monitoring systems operation and scheduling shall be checked on the bench and on the engine to establish proper functioning and calibration over the applicable ranges of engine operating conditions.

Prior to the initiation of the engine calibration all components necessary for proper control and engine monitoring systems operation and scheduling shall be checked on the bench and on the engine to establish proper functioning and calibration over the applicable ranges of engine operating conditions.

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4.12.3.5.1.2 Engine calibration.

The engine calibration shall be accomplished to establish the performance characteristics of the complete engine. Prior to the beginning of the calibration the engine may be cleaned using the wash procedure specified in 3.7.12 and engine controls may be adjusted but shall not be readjusted throughout the calibration. Calibrations shall be made initially with no customer power extraction and no bleed air extraction other than that required for continuous engine operation. Data required in 4.12.4.1.3 and 4.12.4.1.4 shall be recorded. During calibration, conformance with the leakage and shutdown drainage requirement of 3.1.1.8 shall be demonstrated. The engine shall be calibrated using fuel per 3.7.3.1.1 and oil per 3.7.8.1. The following shall be obtained:

a. Data to establish compliance with 3.2.1 and sea level performance of _(a)_. The use of outside ambient temperature (OAT), to calibrate the engine in lieu of _(a)_ conditions is acceptable with Using Service approval. _(b)_ power settings shall be included in the performance calibration. For turboprop and turboshaft engines, sufficient data to establish the accuracy of the torque sensor signals shall be recorded.

b. Data to establish compliance with 3.2.2.6, 3.2.2.7 and 3.2.2.3 requirements at sea level static conditions. All transient performance data shall be used to update the transient status performance computer program of 3.2.1.2 and 3.2.1.3.

c. Repeat items “a” and “b” conditions with maximum permissible bleed air extraction for that condition. The maximum permissible bleed air extraction includes customer and anti-icing bleed airflow, if applicable.

d. Repeat item “c” with customer or accessory power extraction as specified by _(a)_, except for starting.

e. Data required characterizing engine performance as a function of time as defined by the Using Service.

f. Data showing adequate distorted stall or surge margin remaining with _(c)_ distortion pattern (screen) selected by the Using Service. In addition, data establishing the distorted stall or surge margin shall be determined at _(d)_ engine thrust setting selected by the Using Service.

Additionally, for turboprop engines, the following shall be obtained with the engine fitted with the power-absorbing device specified in the specification:

g. Data to establish compliance with 3.7.15.1 and load and no load operation.

4.12.3.5.1.2.1 Customer bleed air analysis calibration.

Prior to and during the _____ test, a customer bleed air analysis shall be performed. The customer bleed air shall be sampled from each bleed air outlet during an intermediate or maximum continuous power run. A sample of air entering the engine inlet shall be taken at the same time the bleed air samples are obtained. The samples shall be properly identified and processed through laboratory analysis to determine whether the contaminant levels are within the limits specified in 3.1.1.7.1.

4.12.3.5.2 Inspection, maintenance and calibration.

The following inspections, maintenance, and calibrations shall be performed:

a. Using established procedures, inspections shall be performed at intervals equivalent to the inspection periods specified in 3.4.1.7.3. Inspections and partial disassembles at other times during the test are permitted when warranted by performance deterioration and after approval from the Using Service is obtained.

b. Prior to and following each inspection interval, an abbreviated engine calibration may be performed. If performance deterioration exceeds 3.2.1.4, the engine may be cleaned. If cleaning is necessary, the washing system and procedure specified in 3.7.12 shall be used prior to initiation of the next test segment.

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c. Maintenance shall be performed on a basis equivalent to the maintenance procedures specified in 3.5.2.3. Parts replaced during the test, except for consumables defined by maintenance instructions, shall not be permitted without approval of the Using Service. A failure analysis shall be performed for any failures resulting in the replacement or repair of non-consumable parts. Non-consumable parts replaced during the test shall require retest equivalent to the severity and test hours of the full engine test.

4.12.3.5.3 Recalibration.

After completion of the tests specified in 4.11.1, a recalibration per the requirements of 4.12.3.5.1.2 shall be made on the engine. The recalibration may be preceded by a run during which the cleaning procedure of 3.7.12 may be used. The fuel and oil used shall be the same as those used during the initial calibration.

After completion of the engine recalibration, all components necessary for proper control and engine monitoring systems operation and scheduling shall be rechecked to establish proper functioning and calibration per 4.12.3.5.1.1.

The performance of these components shall meet their design service limits.

4.12.3.5.4 Engine disassembly and inspection.

For tests requiring engine disassembly the Using Service shall be notified of the inspection commencement date prior to each inspection. Prior to cleaning, the engine parts shall be given a “dirty” inspection for abnormal conditions. The dirty inspection shall be completed before any parts are cleaned. Parts shall then be cleaned and a “clean” inspection shall be performed. Part measurements shall be taken, as necessary, to determine excessive wear and distortion. Those measurements shall be compared with the manufacturer’s drawing dimensions and tolerances and with similar measurements made prior to the test. Non-Destructive Inspection (NDI) techniques shall be per 3.4.1.7.2. During the clean inspection a visual examination and condition assessment shall be conducted. The contractor shall maintain all data and results of non-destructive inspection tests. All NDI inspection data and findings shall be made available for Using Service review during the inspections. All data of 4.12.4 shall be made available for Using Service review upon request.

As a result of the inspection, parts shall be categorized as follows:

Cat. A. No repair required.

Cat. B. Repair required before further use.

Cat. C. Outside repair limits.

4.12.4 Data and reporting.

Data and reporting for tests conducted in this specification shall be per table XXXIII and as required below.

4.12.4.1 Types of data.

4.12.4.1.1 Pretest data.

Pretest data shall be made available to the Using Service for approval prior to the initiation of each test or demonstration. The data shall include:

a. Detailed test and maintenance plans. The detailed test and maintenance plans to be used by test and maintenance personnel during the conduct of the test. Test procedures shall be per the test plan.

b. Service and wear limits. Service limits to be used by maintenance personnel in the field and repair facilities.

c. The parts list and configuration identification of the engine, component or assembly and justification for any differences per 4.12.3.2.

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4.12.4.1.2 Preliminary data.

Photographs and other pertinent data shall be obtained and recorded at the time the test article is prepared for the test. For engines, dry mass (weight), as specified in 3.1.2.1 and center of gravity as specified in 3.1.2.2 shall be provided by the contractor.

4.12.4.1.3 Data - steady-state.

During operation at each specified steady-state condition and after performance stabilization, as a minimum the data specified in table XXXIV shall be recorded.

4.12.4.1.4 Data - transient.

As a minimum, the data specified in table XXXIV shall be recorded.

4.12.4.1.5 Starting data.

During each start, to the extent defined in the individual test verification requirement derived from table XXXIV, data shall be recorded to verify the requirements of 3.1.1.11.1, 3.2.2.3.3, and 3.2.2.3, as a minimum.

4.12.4.1.6 Accreditable test time.

Total test time for a period less than fifteen minutes shall not be credited to the official run time, except when shorter periods are a test requirement.

4.12.4.1.7 Miscellaneous data and notes.

The date, test title, test article designation, and serial number shall be recorded on each page of the log sheet. Notes shall be added to the log sheets to record all incidents and irregular functioning of the test article and equipment and the corrective measures taken. All test article software changes shall be recorded on the log sheets.

4.12.4.2 Accuracy of data.

For all test article calibrations and tests or demonstrations, the reported data shall have steady-state accuracy within the tolerances specified herein. The accuracy of transient data and the corresponding instrument calibration methods shall be described in the test plans. No credit or debit shall be permitted for measurement uncertainty to establish compliance with the requirements of this specification. The instrument calibration methods shall be subject to Using Service approval prior to the initiation of any testing. The “as tested” accuracies as defined by the test facility shall be included in the test report.

4.12.4.2.1 Measurement uncertainty analysis.

A detailed measurement uncertainty analysis addressing all elemental random and systematic errors shall be performed per:

As a minimum, the analysis shall cover the three categories of errors: calibration errors, data acquisition errors, and data reduction errors. Analysis will be performed for the following measured parameters: airflow, fuel flow, thrust, torque, temperature, pressure, rotor speeds (i.e., Ng & Np), SFC, output power, engine weight, and vibration.

4.12.4.3 Atmospheric data.

4.12.4.3.1 Barometer reading and relative humidity.

The barometric reading shall be corrected for temperature and shall be recorded at intervals of _(a)_ hours. Relative humidity data readings shall be recorded at intervals of _(b)_ if required for performance calculations. The relative humidity shall be measured in the vicinity of the test article.

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5. PACKAGING.

5.1 Packaging.

For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES.

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use.

Engines covered by this specification are intended for military-unique air vehicle propulsion. Fighter and attack missions, combat environment, carrier suitability, rotary wing, transports, UAVs, etc., are a few examples of the different design requirements for military engines.

6.2 Acquisition requirements.

Acquisition documents should specify the following:

a. Title, number, and date of the specification.

b. Preparation of the engine for storage and shipment: the level of preservation, the type of shipping container, and delivery of engine historical records and a packing list for each engine which will include all components and tools which are not installed but are shipped with the engine will be provided per contract requirements. Packaging guidance may be found in MIL-STD-2073.

Acquisition documents will be tailored based on the information in the JSSG-2007A tailoring handbook (Appendix A) and peculiar program requirements, which may include spiral development.

6.3 Definitions.

Definitions for terms used in this specification follow.

a. General definitions:

Acceptance Limit. Criterion of acceptance at bench test after calibration of any new or overhauled component. This criterion should apply before shipment from the component manufacturer or overhaul facility.

Accessories. Items, which are engine-mounted equipment not furnished by the engine contractor or air vehicle mounted equipment, which are required for aircraft operation or as auxiliaries for engine operation.

Alternate Fuel. An alternate fuel is one on which the aircraft can be flown without operational restrictions but which can have long term durability or maintainability impact if used for continuous operation (multiple flights). Alternate fuels are used only on an occasional or intermittent basis.

Analysis. The verification process of studying or reviewing the nature and relationship of the component, environment, strength or any other aspect of the design.

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Article. An individual item or particular unit which may be a component (e.g., fuel pump) or a system when considered as a whole (e.g., engine).

Case. A case is any static member of the engine structure, which carries a load or acts as a pressure vessel.

Cold Parts. Those parts not listed as hot parts.

Component. An integral element. Synonymous with part.

Component Test. A test of engine subsystem components. This includes components such as fuel or lubrication pumps as well as cases, gearboxes, disks, and blades.

Continuous Ignition. A low energy level, low spark rate (relative to the system used to ignite the combustor during an engine start) ignition system that is constantly fired regardless of operating condition or power setting. No pilot action is needed to activate the system.

Control Limiting Temperature. The maximum value built into the engine control to limit the operating value at a commanded power setting and flight condition.

Controls and Externals. All engine parts and components which are engine manufacturer provided that are required for engine operation. The controls and externals may be off engine mounted parts or components. Components contained within the subsystems listed below would be considered controls and externals:

> Lubrication, Hydraulic, Fuel, Electrical (including cables), Ignition, Tubing and Brackets, Actuation, Drive, Gearbox, Bleed, Sensing (Speed(s), Temperature(s), Torque, etc.,), Monitoring, and Optical.

Critical Design Review (CDR). The transition point between detailed design and fabrication of a configuration item or aggregate of configuration items. The primary focus is on the completed detailed design documentation and draft production specifications. The detail design on the hardware drawings is reviewed before the contractor manufactures actual test items. See DODI 5000.2

Critical Size. The crack size at which unstable growth occurs.

Damage Tolerance. The ability of the engine to resist failure due to the presence of flaws, cracks or other damage for a specified period of unrepaired usage. See MIL-HDBK-1783.

Damage Tolerant. Resistant to failure due to the presence of flaws, cracks or other damage for a specified period of unrepaired usage. See MIL-HDBK-1783.

Demonstration. The verification procedure where the functional capabilities of a component, system, or procedure are measured and reviewed.

Design Service Life. The life defined by the Using Service (used for design purposes) that the engine is expected to attain during operational service. See MIL-HDBK-1783.

Design Usage. The engine usage conditions (used for design purposes) specified in 3.4.3. See MIL-HDBK-1783.

Development Test. A non-qualification test used to show risk reduction, proof of concept, or maturity of design. Rig tests and bench tests are examples of development tests.

Diagram. A diagram is a sketch or outline giving only interface or mounting information for a part or part of an assembly without all the details of the item shown (i.e.; diagrams show bolt hole patterns, flanges, electrical plug sockets, etc.). A diagram shows the relationship between the parts of a whole assembly. It does not show many details.

Documentation. Any media which provides a record of the design or design process that can be reviewed.

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Drawing. A drawing is a document that delineates a part or assembly showing dimensions, clearances, and details of a part or assembly. It can show every detail needed to manufacture the part, assembly, or engine. Materials of composition may be shown on the drawing as a related document.

Durable. Resistant to cracking, corrosion, deterioration, thermal degradation, delamination, wear, etc., for a specified period of time. (See MIL-HDBK-1783 definitions.)

Durability. The ability of the engine to resist cracking, corrosion, deterioration, thermal degradation, delamination, wear, etc., for a specified period of time. (See MIL-HDBK-1783.)

Durability Critical. Failure results in a significant economic impact to the system but will not necessarily impair flight safety or mission capability. (See MIL-HDBK-1783.)

Duty Cycle. A composite derived from the mission profiles and mission mix.

Dynamic Response. Response due to forced functions. Forcing functions include: throttle transients, surge, stall, flutter, resonance, etc.

Emergency Fuel. An emergency fuel is one which significantly limits the overhaul life of the engine and imposes operational restrictions on the aircraft as well.

Engine. The complete propulsion system (including all parts, components, and expendables) delivered by the engine contractor including the exhaust nozzle, propeller(s), transmissions and rotor systems, etc..

Engine Component. Items of equipment, furnished as part of and qualified with the engine, whose size, conformation, and dynamic and static characteristics are essential to attain the engine performance specified in the engine specification. Fuel pumps, engine controls, jet nozzle and actuators, anti-icing valves, and the temperature sensing system or devices are included in this category. Components may require separate qualification, calibration, and adjustment. Some engine components (e.g., exhaust nozzles) may be delivered by the airframer.

Engine Part. A piece, or two or more pieces joined together, that are not normally subject to disassembly without destruction of the designed use.

Engine Stiffness. The total deflection of the engine at the outboard end of the output shaft or propeller relative to air vehicle mounting points.

Engine System Level Ground Test. A complete engine test in either a sea level open air cell or a test cell that can be conditioned to simulate altitude conditions.

False Start. A starting sequence without benefit of light-off which immediately after the permissible engine draining time is followed by a successful start.

Flaw Growth Interval. The predicted typical propagation life required to grow an assumed initial flaw to its critical size. (See MIL-HDBK-1783.)

Flight Hours. (See A.3.4.1.1 Guidance in Appendix A.)

Flight Idle. The minimum engine rotational speed necessary during flight to maintain engine acceleration time requirements (go-around at low altitude), minimum combustor pressure limits (combustor blow-out limits or minimum ECS bleed), or minimum inlet airflow requirements (inlet buzz avoidance at supersonic speeds).

Flight Test. The engine is installed in an aircraft which is then tested in flight to acquire verification data not obtainable through engine system level ground test.

Foreign Object Damage. Foreign object damage (FOD) is caused by non-engine parts striking the engine. (See domestic object damage (DOD).)

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Fracture Critical. Failure will result in probable loss of the aircraft or degradation in mission capability. There are two categories under fracture critical: safety critical and mission critical. (See MIL-HDBK-1783.)

Full Flight Release (FFR). This milestone establishes the acceptability of the engine to power the aircraft throughout its full envelope. The engine is not required to meet full verification requirements for durability and reliability and is not required to be the final production configuration. The engine should complete a limited durability demonstration as defined in 4.11.1.1.1. (See DODI 5000.2.)

Gas Temperature. The actual steady-state temperature of the gas averaged over the gas path at the engine station designated.

Gas Temperature Margin. Gas temperature margin is the difference between the maximum allowable steady-state temperature limit specified in table IV and the highest first stage turbine rotor inlet temperature (T4.1) which can occur within the engine operating envelope at the highest engine rating.

Government Furnished Equipment. Equipment provided by the government.

Ground Idle. The engine rotational speed necessary to limit the maximum allowable thrust or power for safe ground handling of the aircraft, to provide maximum customer horsepower extraction, or to maintain minimum accessory generator speed.

Hazardous Materials. Hazardous materials are substances that may pose an environmental or personnel health hazard. (See 3.1.3.1.)

Hermetically Sealed. Any component designated as hermetically sealed will be constructed such that final seal closure will be via fusion bond (i.e.; weld, glass to metal bond, etc). Soft solder bonds are not considered to be a fusion in nature. The final component assembly must be capable of seal verification by helium leak detection or equivalent methods.

Idle. The minimum engine rotational speed necessary for stable low thrust or power operation. The aircraft in which the engine is installed may require idle thrust or power to be greater than that required by the engine.

Impending Failure. The condition of a part where loss of performance or function would take place if the part was continued to be used. This could occur within the part life required by the specification.

Initial Service Release (ISR). This milestone establishes the acceptability of the engine for low rate production release. (See DODI 5000.2).

Inspection. The physical examination of an article, item, drawing, or part.

Intermediate Thrust (Augmented engines). Intermediate is defined as the maximum thrust without afterburning. The engine should be capable of continuously operating at this setting.

Intermediate Thrust (Non-augmented engines). Intermediate is the maximum thrust and may have an incremental duration time limit, but this limit should be at least 30 minutes.

Limit Load. The maximum load expected to be encountered when operated for the design service life at design usage conditions. The factor of safety associated with this load is defined as the limit load factor.

Line Replaceable Unit. See Weapons Replaceable Assembly.

Loaded Disk. The condition where all blades are attached to the disk.

Maximum Allowable Speed. For any given engine operating condition within the operating envelope, the maximum allowable rotor speed for the low pressure and high pressure rotor systems, as specified for either steady-state or transient operation, is the highest mechanical rotor speed at which any production engine is allowed to operate. Exceedance of this speed is not permitted during service operation of the engine. The maximum allowable rotor speed specified should also take into account speed measurement

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inaccuracies, control system tolerances, engine deterioration, customer bleed air and power extraction, and all other factors that can influence the rotor speed level achieved.

Maximum Allowable Temperature. The maximum allowable gas temperature, as specified for either steady-state or transient condition, is the limit beyond which operation of the engine is not allowed.

Minimum Engine. A performance level having the lowest acceptable thrust or power output over a time period and environment specified by the Using Service without exceeding the highest SFC and considering all effects due to control system variations, engine-to-engine build variations, deterioration and flight condition, without exceedance of engine pressure, temperature, torque, and speed operating limits.

Mission. A mission is the period beginning with the start of engine prior to flight and ending with the engine shutdown at the completion of the flight. Specific missions include: familiarization, air combat maneuvers, navigation, and air-to-ground weapons.

Mission Critical. Failure will generate a significant operational impact by degrading mission completion capabilities and may create an indirect safety impact on the weapon system or result in less than Level II handling qualities. (See MIL-HDBK-1783).

Mission Mix. The frequency each mission type is flown in a finite period of time (e.g., 5% for familiarization and 25% for air combat maneuvers).

Mission Profile. Mission profile is a representation of a specific mission in terms of flight conditions and usage parameters.

Monitoring Systems. Monitoring systems are diagnostic and prognostic systems that monitor the health of the engine and its components.

Open System. A system that implements sufficient open specifications (for interfaces, services, and supporting formats to enable properly engineered components to be used across a wide range of systems with minimal changes) to interoperate with other components on local and remote systems, and to interact with users in a style that facilitates portability.

Open System Approach. A physical and logical organization of system functions, structures, and operations based on open system interface standards to meet requirements. Usually applies to computer resources or electronic components.

Operational Capability Release (OCR). The OCR Milestone establishes the production engine configuration for unlimited production release. (See DODI 5000.2)

Parts List. See 3.1.9.1.

Preliminary Design Review (PDR). This review represents the approval to begin detailed design. The primary focus is on the adequacy of top-level design documentation for hardware and software configuration items (CI). The PDR is a check to verify that the allocated baseline requirements for the engine, including interfaces, have been addressed and that they can most likely be met or exceeded by the contractor's proposed functional design.

Primary Fuels. Primary fuels are the fuels on which the engine is designed to operate continuously without restrictions.

Recovery time. The period it takes the engine to return to stable operation after an ingestion event.

Residual Strength. The load carrying capability of a part at any time during the design service life. Residual strength accounts for the presence of damage and the growth of damage as a function of design usage and must maintain limit load capability. (See MIL-HDBK-1783.)

Restart. A start within a maximum of 14 minutes time from shutdown.

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Restricted Fuel. A restricted fuel is one which imposes operational restrictions such as altitude, climb rate, fuel temperature limits, etc., on the aircraft when used.

Safety Critical. Failure will result in the probable loss of usable engine power or aircraft or hazard to personnel due to direct part failure or by causing other progressive part failures. (See MIL-HDBK-1783.)

Satisfactorily. The words ‘‘satisfactorily” or ‘‘satisfactory” as used in this specification in conjunction with words or terms relating to operation or performance of the engine described in this model specification mean: Under the condition specified, throughout the entire operating envelope, the engine operating characteristics and performance are not affected, and the operating and physical limits shown in the specification are not exceeded and no permanent deformation or other damage to the engine occurs.

Satisfactory. The words ‘‘satisfactorily” or ‘‘satisfactory,” as used in this specification in conjunction with words or terms relating to operation or performance of the engine described in the model specification, mean: Under the condition specified, throughout the complete environmental conditions and operating envelope, the engine operating characteristics and performance are not affected, and the operating and physical limits shown in the specification are not exceeded and no permanent deformation or other damage to the engine occurs.

Smoke Puff. A smoke puff is visually detectable unburned fuel droplets that are produced when a combustor or augmentor is initiated (ignition) or terminated (shutdown).

Spiral Development. In this process, a desired capability is identified, but the end-state requirements are not known at program initiation. Those requirements are refined through demonstration and risk management; there is continuous user feedback; and each increment provides the user the best possible capability. The requirements for future increments depend on feedback from users and technology maturation.

Stall. Stall is an engine compression system flow instability which does not result in the loss of engine control. A condition in the engine compression system where one or more blades or stages is operating with separated flow.

Steady-state Stability. Steady-state stability is the condition at a constant power setting in which thrust fluctuations do not exceed a stated margin.

Strength. The ability of the engine or engine component to withstand limit loading conditions while maintaining operational capability. (See MIL-HDBK-1783.)

Surge. Surge is the loss of engine control resulting from an engine compression system flow instability.

Surge Margin. Surge margin is the calculated and/or demonstrated tolerance of the engine to adverse operating conditions while maintaining the required steady-state and transient performance capability.

Test. A test is an empirical effort performed to show that contractual requirements have been met. Documented procedures, instrumentation, and known environmental conditions are normally applicable or noncompliance is determined by observation, where practical, and evaluation of collected data. Most ground and flight empirical efforts associated with this procurement and acquisition qualify as tests.

Throughput. The time required for a computer circuit to process the input signal or to accomplish a required calculation.

Transient. Non-steady-state operation including augmentor ignition and termination, as applicable, as well as thrust/power demand increase and decrease.

Transient Stability. Transient stability is the condition following a transient in which thrust fluctuations do not exceed a stated margin.

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b. The following definitions apply for turboshaft engines.

Contingency Power. An operating condition at which the engine is capable of operating for at least an incremental time duration of 2.5 minutes. This operating condition is available for use only under one engine inoperative (OEI) conditions.

Delivered Shaft Power. The power delivered at the output shaft(s) as measured by delivered torque and delivered shaft speed.

Emergency Power Capability. The capability to run at 2.5 minutes at emergency power followed by 2.5 minutes at contingency power and 30 minutes at intermediate rated power (IRP).

Intermediate Power. An operating condition at which the engine is capable of operating for at least an incremental time duration of 30 minutes.

Intermediate Test Temperature Setting. An operating condition at which the engine is operated during the durability test to assure an operational capability for at least an incremental time duration of 30 minutes.

Maximum Allowable Speed. As specified for either steady-state or transient conditions, the limit beyond which operation of the engine is not allowed

Maximum Allowable Torque. As specified for either steady-state or transient conditions, the torque limit beyond which operation of the engine is not allowed.

Maximum Continuous Power. An operating condition at which the engine is capable of operating continuously.

Maximum Continuous Temperature Setting. An operating condition at which the engine is operated during the durability test to assure a continuous operational capability.

Maximum Power. An operating condition at which the engine is capable of operating for at least an incremental time duration of 10 minutes.

Maximum Test Temperature Setting. An operating condition at which the engine is operated during the durability test to assure an operational capability for at least an incremental time duration specified in this specification for maximum power.

Rated Delivered Shaft Speed. The speed of the output shaft when the engine is delivering rated torque while operating at a table IIA, IIB, IIC, or III power rating.

6.4 Symbols, subscripts, abbreviations, and acronyms.

The symbols, subscripts, abbreviations and acronyms used in this specification are listed below and should be per SAE AS 681 and SAE AS 755.

AFB	Air Force Base
AGI	Armament Gas Ingestion
AMT	Accelerated Mission Test
AS	Aerospace Standard
ASC	Aeronautical Systems Center
ASTM	American Society for Testing and Material
BIT	Built in Test
BOCLE	Ball-on-Cylinder Lubricity Evaluator
C&E	Controls and Externals

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CDR	Critical Design Review
CG	Center of Gravity
COTS	Commercial Off The Shelf
ECS	Environmental Control System
EFH	Engine Flight Hours
EHM/PHM	Engine Health Monitoring/ Prognostic Health Monitoring
EM	Electromagnetic
EMC	Electromagnetic Compatibility
EMI	Electromagnetic Interference
EMS	Engine Monitoring System
EMT	Elapsed Maintenance Time
ENSIP	Engine Structural Integrity Program
FADEC	Full Authority Digital Engine Control
FFR	Full Flight Release
FOD	Foreign Object Damage
FSII	Fuel System Icing Inhibitor
HCF	High Cycle Fatigue
ICD	Interface Control Document
IFR	Initial Flight Release
IPS	Inlet Particle Separator
IR	Infrared
IRP	Intermediate Rated Power
ISR	Initial Service Release
JTA	Joint Technology Architecture
LCF	Low Cycle Fatigue
LRU	Line Replaceable Unit
MAF	Mission Affecting Failure
MAU	Maintenance Actions Unscheduled
MEFH	Mean Engine Flight Hours
MOPP	Mission Oriented Protective Posture
MTBF	Mean Time Between Failures
MTBMA	Mean Time Between Maintenance Action
NBC	Nuclear, Biological and Chemical
NDI	Non-Destructive Inspection
OCR	Operational Capability Release
OEI	One Engine Inoperative
PBPDG	Performance Based Product Definition Guide

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PDR	Preliminary Design Review
PLA	Power Lever Angle
PPM	Parts Per Million
PTO	Power Takeoff
R&M	Reliability and Maintainability
RAM	Random Access Memory
RAM	Radar Absorbent Material
RCS	Radar Cross Section
RMS	Root Mean Square
ROM	Read Only Memory
RVP	Reid Vapor Pressure
SAE	Society of Automotive Engineers
SFC	Specific Fuel Consumption
SHP	Shaft Horsepower
SI	International System of Units
SLS	Sea Level, Static
STOVL	Short Take-Off and Vertical Landing
TAC	Total Accumulated Cycles
TJ/TF	Turbojet and Turbofan
TLV	Threshold Limit Value
TP/TS	Turboprop and Turboshaft
TRIT	Turbine Rotor Inlet Temperature
USA	United States Army
USN	United States Navy
V/STOL	Vertical/Short Takeoff & Landing
WRA	Weapon Replacement Assembly
WSD	Wear Scar Diameter
UAV	Unmanned Air Vehicle
UCAV	Unmanned Combat Air Vehicle

6.5 Material Safety Data Sheet.

Contracting officers should identify those activities which require copies of completed Material Safety Data Sheets.

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6.6 Subject term (key word) listing.

Bearings
Bird Ingestion
Bleed Air
Commercial Off The Shelf (COTS)
Domestic Object Damage (DOD)
Foreign Object Damage (FOD)
Fuel
Gas
Ice
Idle
Ignition System
Mount
Oil
Propulsion
Sand
Smoke
Spiral Development
Stall
Thrust
Torque
Turbofan
Turbojet
Turboprop
Turboshaft
Windmilling

6.7 Metrication.

The engine will be of metric design per Public Laws 94-168 and 100-418 unless stated otherwise in the contract. The engine will conform to the SI standard described in IEEE/ASTM SI 10, American National Standard for Use of the International System of Units (SI): The Modern Metric System, as listed in the Department of Defense Index of Specifications and Standards (DoDISS). All new metric items (requiring a new part identification) or any new design practice created or designed to replace any customary system of measurement will be integrated into the SI Metric System of Measurement per Public Laws 94-168 and 100-418, and Executive Order 12770. The specification will contain both metric and English units. The metric units will be primary. English units will be provided within parenthesis.

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6.8 Responsible Engineering Office.

The Air Force office responsible for the development and technical maintenance of this specification is ASC/ENF, 2530 LOOP ROAD WEST, WRIGHT-PATTERSON AFB OH 45433-7101. Requests for additional information or assistance can be obtained from this office by writing to the above address or by calling Commercial (937) 255-8604 or DSN 785-8604.

Any information relating to Government contracts must be obtained through the contracting officers. The responsible Navy engineering office is the Naval Air Systems Command, AIR-4.4.7, 22195 Elmer Road, Patuxent River MD 20670-1534; DSN 757-0509, Commercial (301) 757-0509. The responsible Army engineering office is the USA AMCOM, Engineering Directorate, AMSAM-AR-E, Redstone Arsenal AL 35898-5000.

6.9 Changes from previous issue.

Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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APPENDIX A

ENGINES, AIRCRAFT, TURBINE

HANDBOOK FOR

A.1 SCOPE.

This appendix provides rationale, guidance, background, lessons learned and instructions necessary to tailor sections 3 and 4 of the General Engine Specification JSSG-2007A for a specific application. The appendix has been developed and coordinated by technical personnel from the Air Force, Navy, Army, and Aerospace Industries Association (AIA). This Appendix is not a mandatory part of the specification. The information contained herein is intended for guidance only.

A.1.1 Purpose.

This appendix provides information to assist the Using Service in the use of JSSG-2007A.

A.1.2 Use.

This Engine Joint Service Specification Guide (JSSG-2007A), in conjunction with its companion JSSGs, is intended for use by Government and Industry program teams as guidance in developing program-unique specifications. This document should not be placed on contract.

The complete set of JSSGs establish a common framework to be used by Government-Industry Program Teams in the Aviation Sector for developing program-unique requirements documents for Air Systems, Air Vehicles, and major Subsystems. Each JSSG contains a compilation of candidate references, generically stated requirements, verification criteria, and associated rationale, guidance, and lessons learned for program team consideration. The JSSGs identify typical requirements for a variety of aviation roles and missions. By design, the JSSG sample language for “requirements” and “verification criteria” are written as generic templates, with blanks that need to be completed in order to make the requirements meaningful. Program teams need to review the JSSG rationale, guidance, and lessons learned to: (1) determine which requirements are relevant to their application and (2) tailor the requirements to meet program-specific needs.

A.1.3 Format.

A.1.3.1 Requirement and verification identity.

Section 3 and 4 of this appendix parallel sections 3 and 4 of the basic specification; paragraph titles and numbering are in the same sequence. Sections 3 and 4 provide each requirement (section 3) and associated verification (section 4) as stated in the basic specification. Both the requirement and verification have sections for rationale, guidance, and lessons learned.

A.1.3.2 Requirement and verification package.

Sections 3 and 4 of this appendix have been arranged so the requirement and associated verification is a complete package to permit addition to or deletion from the criteria as a single requirement. A requirement is not specified without an associated verification. Considerations for requirements and verification across the acquisition phases including concept refinement, technology development and system development and demonstration have been addressed in this document.

Not all requirements shown in this appendix will apply to every system. Unmanned Air Vehicles (UAV) may not need the same level of design and verification as manned systems thus allowing requirements to be significantly tailored or even deleted from the final contract specification. This should be reflected when using table XXXI to develop the verification matrices.

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A.1.3.2.1 Commercial Off The Shelf (COTS) Applications.

The use of Commercial Off The Shelf (COTS) systems may rely heavily on alternate verification and certification approaches such as those used by the FAA or Joint Aviation Authority (JAA). Every effort should be made to ensure all qualified missions of COTS applications meet military applications. Differences may exist between commercial and military missions and usage. These differences may force re-verifications of some JSSG-2007A requirements reflected in table XXXI. Differences may force the user to use information available other that what was used by the civilian airworthiness authority to meet military requirements. Fleet usage information in addition to the qualification information should be utilized as an aide to evaluate the differences for applicable requirements in table XXXI with moderate/high risk.

A.2 APPLICABLE DOCUMENTS.

A.2.1 General.

The documents listed in this section are referenced in sections 3 and 4 of this appendix. These documents are listed for informational purposes only.

A.2.2 Government documents.

A.2.2.1 Specifications, standards, and handbooks.

The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

INTERNATIONAL STANDARDIZATION AGREEMENTS

NATO STANAG 1135	Interchangeability Chart of Fuels, Lubricants and Associated Products used by the Armed Forces of the North Atlantic Treaty Nations
NATO STANAG 1414	Guidelines to Ensure that Contractors Design and Supply New Equipment Capable of Using Standardized Lubricants
NATO STANAG 3595	Aircraft Fitting for Pressure Replenishment of Gas Turbine Engines with Oil

(Application for copies should be addressed to the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia PA 19111-5094; phone (215) 697-2667.)

DEPARTMENT OF DEFENSE SPECIFICATIONS

AN-E-30	Army-Navy Aeronautical Specification Engines: Aircraft Turbo-Jet General Specification for
JSSG-2001	Air Vehicle
MIL-E-5007	Engine, Aircraft, Turbojet and Turbofan, General Specification for
MIL-E-8593	Engines, Aircraft, Turboshaft and Turboprop, General Specification for
MIL-DTL-12468	Decontaminating Agent, STB
MS14169	Circular Spline and Adapter Details Engine Driven Accessories
MS14184	Nonmetallic Shaft-Coupling Details Engine Driven Accessories
MIL-F-17874	Fuel Systems: Aircraft, Installation and Test of
MIL-PRF-25017	Inhibitor, Corrosion/Lubricity Improver, Fuel Soluble

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MIL-H-25579	Hose Assembly, Tetrafluoroethylene, High Temperature, Medium Pressure
MIL-DTL-25677	Adapter, Cap, and Nozzle, Pressure Lubricating Oil Servicing, Aircraft, General Specification for
MIL-H-27267	Hose, Tetrafluoroethylene, High Temperature, Medium Pressure
MS33787	Wrenching Element, External Spline, Dimensions for
MIL-DTL-38999	Connector, Electrical Circular, Miniature, High Density Quick Disconnect (Bayonet, Threaded and Breech Coupling), Environment Resistant, Removable Crimp and Hermetic Solder Contacts, General Specification for
MIL-PRF-49506	Logistics Management Information
MIL-D-50030	Decontaminating Agent
MIL-P-82522	Propellant, Jet Engine, T-H Dimer, Grade RJ-4
MIL-DTL-83133	Turbine Fuels, Aviation, Kerosene Types, NATO F-34 (JP-8) and NATO F-35
MIL-PRF-83282	Hydraulic Fluid, Fire Resistant, Synthetic Hydrocarbon Base, Aircraft, Metric, NATO Code Number H-537
MIL-PRF-85045	Cables, Fiber Optics, (Metric), General Specification for
MIL-DTL-85470	Inhibitor, Icing, Fuel System, High Flash NATO Code Number S-1745
MIL-PRF-85704	Cleaning Compound, Turbine Engine Gas Path
DOD-L-85734(AS)	Lubricating Oil, Helicopter Transmission System, Synthetic Base
MIL-PRF-87100	Lubricating Oil, Aircraft Turbine Engine, Polyphenyl Ether Base
MIL-P-87107	Propellant, High Density Synthetic Hydrocarbon Type, Grade JP-10
AFGS-87219	Electrical Power System, Aircraft
AFGS-87271	Engines, Unmanned Air Vehicle, Airbreathing Gas Turbine, Expendable

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-461	Control of Electromagnetic Interference Emissions and Susceptibility, Requirements for the
MIL-STD-464	Electromagnetic Environmental Effects Requirements for Systems
MIL-STD-704	Aircraft Electric Power Characteristics
MIL-STD-882	System Safety Program Requirements
MIL-STD-889	Dissimilar Metals
MIL-STD-1472	Human Engineering
MIL-STD-1553	Digital Time Division Command/Response Multiplex Data Bus
MIL-STD-1773	Fiber Optics Mechanization of an Aircraft Internal Time Division Command/Response Multiplex Data Bus
MIL-STD-7179	Finishes, Coatings, and Sealants for the Protection of Aerospace Weapon Systems

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DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-5	Metallic Materials and Elements for Aerospace Vehicle Structures
MIL-HDBK-275	Guide for Selection of Lubricants, Fluids, and Compounds for Use in Flight Vehicles and Components
MIL-HDBK-454	General Guidelines for Electronic Equipment
MIL-HDBK-470	Designing and Developing Maintainable Products and Systems, Volume I and Volume II
MIL-HDBK-516	Airworthiness Certification Handbook
MIL-HDBK-781	Reliability Test Methods, Plans, and Environments for Engineering, Development, Qualification, and Production
MIL-HDBK-784	Guidelines-Design to Minimize Contamination and to Facilitate Decontamination of Military Vehicles and Other Equipment: Interiors and Exteriors
MIL-HDBK-838	Lubrication of Military Equipment
MIL-HDBK-1559	Numbers, Serial, Aircraft Gas Turbine Engine and Engine Module, Assignment of
MIL-HDBK-1568	Materials and Processes for Corrosion Prevention and Control in Aerospace Weapons Systems
MIL-HDBK-1798	Mechanical Equipment and Subsystems Integrity Program
MIL-HDBK-1799	Survivability, Aeronautical Systems (For Combat Mission Effectiveness)
MIL-HDBK-1812	Type Designation, Assignment and Method for Obtaining
MIL-HDBK-2069	Aircraft Survivability
MIL-HDBK-2084	General Requirements for Maintainability of Avionic and Electronic Systems and Equipment
MIL-HDBK-2165	Testability Handbook for Systems and Equipment
MIL-HDBK-46855	Human Engineering Program Process and Procedures
MIL-HDBK-87244	Avionics/Electronics Integrity

QUALIFIED PRODUCT LISTS

QPL-25017	Inhibitor, Corrosion/Lubricity Improver, Fuel Soluble

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or www.dodssp.daps.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia PA 19111-5094; phone (215) 697-2667.)

A.2.2.2 Other Government documents, drawings, and publications.

The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMITTEE FOR AERONAUTICS (NACA)

NACA TN 1855	Recommended Values of Meteorological Factors to be Considered in the Design of Aircraft Ice-Prevention Equipment

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NACA TN 2569	A Summary of Meteorological Conditions Associated with Aircraft Icing and a Proposed Method of Selecting Design Criterions for Ice-Protection Equipment

(Application for copies should be addressed to National Technical Information Service (NTIS), 5825 Port Royal Road, Springfield VA 22161; www.ntis.gov.)

DEPARTMENT OF DEFENSE

AEDC-TR-73-5	Handbook: Uncertainty In Gas Turbine Measurements
AEDC-TR-83-42	Guidelines for Validation of Turbine Engine Status Decks
AFAPL-TR-75-52	Development of Emissions Measurement Techniques for After Burning Turbine Engines
AFAPL-TR-76-64	Aircraft Exhaust Pollution and Its Effect on the US Air Force
AFWAL-TR-83-2079	Weibull Analysis Handbook
ASD-TR-82-5012	Handbook of Military Aircraft Design Normal Load Factor Exceedance Date
JACG Policy	Management of Aviation Critical Safety Items, JACG Guidance Procedures, 28 Aug 2002

(Application for copies should be addressed to National Technical Information Service (NTIS), 5285 Port Royal Rd, Springfield VA 22161; www.ntis.gov.)

DEPARTMENT OF TRANSPORTATION

FAA Advisory Circular No: 20-115	Software Considerations in Airborne Systems and Equipment Certification
FAA Advisory Circular No: 20-135	Propulsion Installation and Propulsion System Component Fire Protection Test Methods, Standards, and Criteria
FAA Advisory Circular No: 33-76	Bird Ingestion Certification Standards
FAA Advisory Circular No: 33-78.1	Turbine Engine Power-Loss and Instability in Extreme Conditions of Rain and Hail
FAA Advisory Circular No: 150/5200-30A	Airport Winter Safety and Operations
FAR Part 25	Airworthiness Standards: Transport Category Airplanes
FAR Part 33	Airworthiness Standards: Aircraft Engines
FAR Part 36	Airworthiness Standards: Noise

(Application for copies should be addressed to Department of Transportation, FAA Aeronautical Center, AML-611, PO Box 25082, Oklahoma City OK 73125-5082; www.dot.gov.)

ENVIRONMENTAL PROTECTION AGENCY

40 CFR 60	Environmental Protection Agency: Standards of Performance for New Stationary Sources

(Application for copies should be addressed to Environmental Protection Agency, Ariel Rios Building, 1200 Pennsylvania Avenue, N.W., Washington DC 20460; www.epa.gov.)

NAVAL AIR SYSTEMS COMMAND

Joint NAVAIR 16-1-540 and Air Force TO 1-1-689	Avionic Cleaning and Corrosion Prevention/Control Technical Manual

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NAVAIRINST 4200.25D	Management of Critical Application Items Including Critical Safety Items

NAVAL AIR WARFARE COLLEGE

NAWCWPNS TP 8076	Ballistic Tests of Structural Concepts to Mitigate Damage and Reduce Fuel Leakage from Aircraft Tan-Duct Walls
NAWCWPNS TP 8235	Specification and Verification Test Procedures To Improve Fuel Ingestion Tolerance of Turbine Engines

NAVAL OPERATION INSTRUCTION

OPNAVINST 4790.2	The Naval Aviation Maintenance Program

(Application for copies should be addressed to Naval Publications & Forms Directorate, Attn: Code 301, 5801 Tabor Ave., Philadelphia PA 19120-5099; www.navicp.navy.mil.)

UNITED STATES ARMY

ADS-37A-PRF	Electromagnetic Environmental Effects (E^3) Performance And Verification Requirements
AR 70-71	Managing the Nuclear, Biological, and Chemical Contamination Survivability of Army Materiel Program
AV-E-8593D	Specification, Engines, Aircraft, Turboshaft and Turboprop, General Specification for

(Application for copies should be addressed to USA AMCOM, Engineering Directorate, AMSAM-AR-E, Redstone Arsenal, AL 35898.)

UNITED STATES AIR FORCE

AFI 32-1002	Snow and Ice Control
AFOSH STD 48-19	Hazardous Noise Program

(Copies may be obtained from the USAF Publications Distribution Office Website: www.afpubs.hq.af.mil.)

A.2.3 Non-Government publications.

The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AIR TRANSPORT ASSOCIATION (ATA)

ATA Spec 2000	E-Business Specification for Materiels Management

(Application for copies should be addressed to Air Transport Association (ATA) of America, Inc.; 1301 Pennsylvania Avenue; NW - Suite 1100; Washington DC 20004-1707; www.air-transport.org.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI Z39.18-1995	Scientific and Technical Reports -- Elements, Organization, and Design

(Application for copies should be addressed to the American National Standards Institute, 11 West 42nd Street, New York NY 10036-8002; www.ansi.org.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D471	Standard test Method for Rubber Property – Effects of Liquids
ASTM D975	Standard Specification for Diesel Fuel Oils

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ASTM D1655	Aviation Turbine Fuels

(Application for copies should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken PA 19428-2959; www.astm.org.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME 68-GT-37	Problems and Solutions for Sand Environment Operation of Helicopter Gas Turbine
ASME 70-GT-96	Environment Problems Affecting Helicopter Engine Operation
ASME 84-GT-262	Aircraft Engine Control Mode Analysis
ASME Y14.24	Types and Applications of Engineering Drawings
ASME Y14.34	Associated Lists
ASME Y14.35	Revision of Engineering Drawings and Associated Documents
ASME Y14.100	Engineering Drawing Practices

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York NY 10017-2392; www.asme.org.)

COORDINATING RESEARCH COUNCIL, INC.

CRC Fuels Handbook (1983)	Handbook of Aviation Fuel Properties CRC Report No. 530

(Applications for copies should be addressed to the Coordinating Research Council, Incorporated; 219 Perimeter Center Parkway; Suite 400; Atlanta GA 30346-1301; www.crcao.com.)

ELECTRONICS INDUSTRIES ASSOCIATION (EIA)

EIA-455-20	FOTP-20 Measurement of Change in Optical Transmittance
EIA-455-21	Matting Durability of Fiber Optic Interconnecting Devices
EIA/IEEE 12207.0	Software Life Cycle Processes
EIA/IEEE 12207.1	Software Life Cycle Processes – Life Cycle Data
EIA/IEEE 12207.2	Software Life Cycle Processes – Implementation Considerations

(Application for copies should be addressed to Electronics Industries Association, 2500 Wilson Boulevard, Arlington VA 22201-3834; www.geia.org.)

EUROPEAN CENTER FOR STANDARDIZATION

EN 228	British Standard Specification for Unleaded Petrol (Gasoline) For Motor Vehicles
EN 590	European Industry Standard (Specification) for Diesel Fuels
EN 925	British Standard Specification For Automotive Diesel Fuel

(Application for copies should be addressed to CEN/ISSS - Rue de Stassart, 36 - 1050 Brussels – Belgium.)

INTERNATIONAL CIVIL AVIATION ORGANIZATION (ICAO)

ICAO Annex 16, Appendices 3 & 5	Instrumentation and Measurement Techniques for Gaseous Emissions From Afterburning Gas Turbine Engines

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:04Z
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(Application for copies should be addressed to International Civil Aviation Organization, Document Sales Unit, 999 University Street, Montreal, Quebec H3C 5H7, Canada; www.icao.org.)

INTERNATIONAL ELECTROTECHNICAL COMMISSION (IEC)

IEC 179	Precision Sound Level Meters
IEC 225	Octave, Half-Octave, and Third-Octave Band Filters Intended for the Analysis of Sounds and Vibration

(Application for copies should be addressed to American National Standards Institute, 11 West 42nd Street, New York NY 10036-8002; www.ansi.org.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 9001	Quality Management Systems – Requirements
ISO 9000-3	Quality Management and Quality Assurance Standards Part 3: Guidelines for the application of ISO 9001 to the development, supply and maintenance of computer software
ISO 10012-1	Measurement management systems – Requirements for measurement processes and measuring equipment

(Application for copies should be addressed to International Organization for Standardization (ISO), 1, rue de Varembé Case postale 56, CH-1211 Geneva 20, Switzerland.)

INTERNATIONAL SOCIETY OF ALLIED WEIGHT ENGINEERS (SAWE)

SAWE RP7	Mass Properties Management and Control for Military Aircraft
SAWE RP9	Weight and Balance Control for Guided Missiles

(Application for copies should be addressed to the International Society of Allied Weight Engineers (SAWE); P.O. Box 60024; Terminal Annex; Los Angeles CA 90060; ATTN: Mr. Robert Johnston, Executive Secretary.)

NATIONAL AEROSPACE STANDARDS COMMITTEE (NA/NAS)

NASM 1515	Fastener Systems for Aerospace Applications
NASM 9006	Recess-Cross, Low Torque Drive, Dimensions and Gage Dimensions for
NASM 23964	Bolt, Self-Retaining, Positive Locking
NASM 33537	Insert, Screw Thread, Helical Coil, Inch Series, Coarse and Fine Thread, Standard Assembly Dimensions for
NASM 33540	Safety Wiring and Cotter Pinning
NASM 33588	Nuts, Self-Locking, Aircraft, Reliability and Maintainability Usage Requirements for
NASM 33750	Recess, High Torque and Conical High Torque, Dimensions of Recess, Gage, and Driver for
NASM 33781	Recess-Torque-Set, Dimensions of Recess, Gage, and Driver for

(Copies of this document may be purchased from the Aerospace Industries Association of America, 1250 Eye Street, N.W., Suite 1200, Washington DC 20005-3924; www.aia-aerospace.org.)

RADIO TECHNICAL COMMISSION FOR AERONAUTICS, (RTCA) INC.

DO-160	Environmental Conditions and Test Procedures for Airborne Equipment

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DO-178	Software Considerations in Airborne Systems and Equipment Certification
DO-254	Design Assurance Guidance for Airborne Electronic Hardware

(Applications for copies should be addressed to RTCA, Inc.; 1140 Connecticut Ave. NW; Suite 1020; Washington DC 20036; www.rtca.org.)

SAE INTERNATIONAL

SAE AIR 713	Determining, Presenting, and Substantiating Turbine Engine Starting and Motoring Characteristics, Guide for
SAE AIR 781	Determining Engine Starter Drive Torque Requirements, Guide for
SAE AIR 947	Engine Erosion Protection
SAE AIR 1087	Aircraft Accessory Drag Torque During Engine Starts
SAE AIR 1419	Inlet Total-Pressure-Distortion Considerations for Gas-Turbine Engines
SAE AIR 1666	Performance Testing of Lubricant Filter Elements Utilized in Aircraft Power and Propulsion Lubrication Systems
SAE AIR 4246	Aluminum Alloy Welding Wire 7.0si - 0.52mg (357)
SAE AMS 2518	Thread Compound, Anti-seize, Graphic-Petrolatum
SAE ARD 50015	A Current Assessment of the Inlet/Engine Temperature Distortion Problem
SAE ARP 492	Aircraft Engine Fuel Pump Cavitation Endurance Test
SAE ARP 949	Turbine Engine Starting System Design Requirements
SAE ARP 994	Recommended Practice for the Design of Tubing Installation for Aerospace Fluid Power Systems
SAE ARP 1179	Aircraft Gas Turbine Engine Exhaust Smoke Measurement
SAE ARP 1256	Continuous Sampling and Measurement of Gaseous Emissions from Aircraft Turbine Engines, Procedures for
SAE ARP 1401	Aircraft Fuel System and Component Icing Test
SAE ARP 1420	Gas Turbine Engine Inlet Flow Distortion Guidelines
SAE ARP 1533	Procedure for the Calculation of Gaseous Emissions from Aircraft Turbine Engines
SAE ARP 1587	Aircraft Gas Turbine Engine Monitoring System Guide
SAE ARP 1797	Aircraft and Aircraft Engine Fuel Pump Low Lubricity Fluid Endurance Test
SAE ARP 1870	Aerospace Systems Electrical Bonding and Grounding for Electromagnetic Compatibility and Safety
SAE AS 568	Aerospace Size Standard for O-Ring Sizes
SAE AS 848	Attachment Points, Engine, Remote Trimming, Main Fuel Control
SAE AS 870	Wrenching Configuration, Double Hexagon (12 points), for Threaded Fasteners
SAE AS 1055	Fire Testing of Flexible Hose, Tube Assemblies, Coils, Fittings and Similar System Components
SAE AS 1606	Gas Turbine Starter (Jet Fuel Starter) Specification

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SAE AS 4273	Fire Testing of Fluid Handling Components for Aircraft Engines
SAE AS 5202	Port or Fitting End, Internal Straight Thread
SAE AS 8879	Screw Threads, Controlled Radius Root with Increased Minor Diameter, General Specification for

(Application for copies should be addressed to SAE International, 400 Commonwealth Drive, Warrendale PA 15096-0001; www.sae.org.)

MISCELLANEOUS

Kaman Report No. R-169	Amount of Dust Recirculated by a Hovering Helicopter, dated 26 Dec 1969
	"Catapult Steam Ingestion Tests of Three Turbofan Engines in the A-7 Aircraft", R. F. Parcells
	"Development of the Lycoming Inertial Particle Separator", H. D. Conners, presented at the Gas Turbine Conference & Products Show, Washington DC, 17-21 Mar 1968
	"Engine/Airframe Interface Considerations for Future Open-Ocean ASW Aircraft", G. Ward, LTV Aerospace Corporation, Proceedings of the 7th Annual National Conference on Environmental Effects on Aircraft and Propulsion Systems at Princeton NJ, 25-27 Sept 1967
	"Evaluation of the Dust Cloud Generated by Helicopter Blade Downwash", Sheridan Rogers, MSA Research, Proceedings of the 7th Annual National Conference on Environmental Effects on Aircraft and Propulsion Systems at Princeton NJ, 25-27 Sept 1967
	"Steam Ingestion by Aircraft Gas Turbine Engines", by Nicholas R. Tomassetti, presented at the 7th Annual National Conference on Environmental Effects on Aircraft and Propulsion Systems at Princeton NJ, 25-27 Sept 1967

(Application for copies should be addressed to National Technical Information Service (NTIS), 5285 Port Royal Rd, Springfield VA 22161; www.ntis.gov; or the appropriate technical society.)

Ministry of Defense Standards

DEF-STAN 00-3/3	Design Guidance for the Transportability of Equipment
DEF-STAN-91-86	Turbine Fuel, Aviation Kerosene, High Flash Type Containing Fuel System Icing Inhibitor
DEF-STAN-91-87	Turbine Fuel, Aviation Kerosene Type Containing Icing Inhibitor
DEF-STAN-91-88	Turbine Fuel, Aviation Kerosene Wide Cut Type Containing Icing Inhibitor
DEF-STAN-91-91	Turbine Fuel, Aviation Kerosene Type Jet A-1

(These United Kingdom standards and others can be viewed online at http://www.dstan.mod.uk/.)

A.2.4 Order of precedence.

In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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A.2.5 Streamlining.

This specification guide has been streamlined. Those documents listed in Appendix A that are required for acquisition have the same status as those referenced directly in section 2 (first-tier documents). All other documents referenced through tiering may be used for guidance and information.

A.3 REQUIREMENTS.

The paragraphs of this specification and all references herein comprise the complete set of engine requirements for the gas turbine engine(s).

REQUIREMENT RATIONALE (A.3)

The Using Service must define the functional requirements for the engine necessary to support the end item mission. This Guide specification is performance based. The intent is to not tell the contractor how to design and verify the requirements, but to state the requirements in terms of the required results with criteria for verifying compliance. A performance specification defines the functional requirements for the item, the environment in which it must operate, and interface and interchangeability characteristics.

REQUIREMENT GUIDANCE (A.3)

The following should be used to tailor the specification paragraph:

All requirements that are classified should be provided in a classified appendix, as part of this specification. Terms, symbols, acronyms, and definitions should be per Section 6.

REQUIREMENT LESSONS LEARNED (A.3)

Throughout this document, specific examples of aircraft engines and their respective manufacturers have been generalized per the following:

a. Specific engine references are listed as Engine A, B, C, etc.
b. Specific engine manufacturers are listed as Engine Company A, B, C, etc.
c. Specific aircraft are listed as Aircraft A, B, C, etc.
d. Specific aircraft manufacturers are listed as Aircraft Company A, B, C, etc.
e. Specific airline companies are listed as Airline Company A, B, C, etc.
f. Specific engine component companies are listed as Engine Component Company A, B, C, etc.

A.4 VERIFICATION.

The verifications established in Section 4 for the requirements specified in Section 3 are intended to result in a progressive in-process verification of design maturity that will be consistent with key milestones of the Government Acquisition Process relating to System Development and Demonstration (Engineering and Manufacturing Development) program schedule. The Verification Matrices (tables XXXI a and b) provide a cross-reference between the requirements and the associated method and timing of the verification. Table XXXIa may be used on programs employing a Spiral Development approach. Table XXXIb may be used for programs not using a Spiral Development approach but adhering to 5000 Series Acquisition System requirements.

All verification activities are required to provide adequate evidence, as determined by the Using Service, that the requirement has been verified. Detected failures or deficiencies during testing constitute non-compliance with the verification paragraph and corrective action and reverification shall be performed to the satisfaction of the Using Service. Verification procedures and success criteria not specified in this section, shall be established by the Contractor and be subject to review and approval or disapproval by the Using Service, prior to the initiation of qualification milestones.

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VERIFICATION RATIONALE (A.4)

All requirements must be verified and results accepted by the Using Service.

VERIFICATION GUIDANCE (A.4)

The qualification program (see 4.12) requires the Contractor to successfully conduct a set of singular or multiple engine and component tests, analyses, demonstrations, or inspections.

Background:

The intention of this header paragraph is to specify that the verifications are conducted for the purpose of progressing through a qualification program, giving the Using Service an indicator of engine and inspections, as specified, at each milestone. Upon the development engine's successful completion of the verification requirements in each milestone, the Using Service should be reasonably assured of the engine's capabilities, and assess continued advancement of the development program into the following milestone. The verifications specified have been thoroughly researched and developed to provide the Using Service with the best indicator of probable engine while keeping qualification costs at a reasonable level. Individual tests or combination of tests in this mission success section are normally conducted at specific conditions.

VERIFICATION LESSONS LEARNED (A.4)

None.

A.3.1 Design and construction.

An open system approach (see 6.3) shall be followed for the engine design where feasible.

REQUIREMENT RATIONALE (A.3.1)

An open systems approach for major systems is mandatory per E1.27 of DoDD 5000.1.

REQUIREMENT GUIDNACE (A.3.1)

Only applies to major system acquisitions as defined by DoDI 5000.2

Open systems emphasize interface management, both external to the air system and also among internal subsystems and components. Open systems approach employs widely used consensus-based interface definitions where appropriate that have high market acceptance. Developer can choose any implementation to meet interface specification. Open system components conform to standard interfaces, even though implementation may be proprietary.

REQUIREMENT LESSONS LEARNED (A.3.1)

Open systems approach is used to ensure that interfaces and components will be available for future production lots by using interface standards that are both widely accepted and have high market acceptance.

A.4.1 Design and construction.

The acceptability of the open design shall be verified by completion of the qualification requirements.

VERIFICATION RATIONALE (A.4.1)

An Open Systems approach for major systems is mandatory per E1.27 of DoDI 5000.2.

VERIFICATION GUIDANCE (A.4.1)

None.

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VERIFICATION LESSONS LEARNED (4.1)

None.

A.3.1.1 Item and interface definition.

A.4.1.1 Item and interface definition.

A.3.1.1.1 Interface and installation.

Engine and air vehicle interfaces shall be as shown in ____. The input and output from the air vehicle to engine shall be defined in ____. The control system interface shall be compatible with and shall receive and process air vehicle thrust or power or torque requests via input signals for the control system. Control system interpretation of power lever angle signals shall be defined in _____.

REQUIREMENT RATIONALE (A.3.1.1.1)

The interfaces between the air vehicle and the engine must be established and controlled to ensure compatibility. The engine and air vehicle contractors must control the interface to ensure the engine will work properly when installed in the aircraft.

REQUIREMENT GUIDANCE (A.3.1.1.1)

The following should be used to tailor the specification paragraph:

The interfaces can be shown in the Interface Control Document (ICD), or for programs that do not have an ICD, the interfaces can be put into the specification paragraph in 3.7.2.1.

For mechanical systems, the maximum allowable loads at the thrust or power lever connections may be specified for the static (1g) axial, shear and overhung moment and for the maximum maneuver loads defined in MIL-HDBK-1783.

Background:

Item diagrams for lubrication system, fuel system, electrical system, and engine control system may be included in the model specification. Diagrams and schematics, if required, should be on the Contract Data Requirements List (CDRL). Computer Aided Design (CAD) engine definitions may be used to meet this requirement. These CAD drawings may be provided in Initial Graphics Exchange System (IGES) format.

The types and application of engineering diagrams and schematics for engine systems must be specified in the Technical Data Package Contract requirements. This can be done by specifying ASME Y14.24 Types and Applications of Engineering Drawings. When drawing types are specified, unless the requirements for each type are given as a requirement in a Statement of Work (SOW), they may be selected from the ASME document.

If CAD files or an IGES format is requested from the contractor, the following guidance should be used.

a. If the contractor designs the engine using modeling practices in CAD, the 2D drawings produced from the CAD file may not be adequate for most purposes. Current industry practices have expanded to 3D modeling.

b. When IGES files are ordered as the only deliverable format, care must be taken by requiring a test copy in a specified media for validation as well as the identification of the pre-processor and assurance of compatibility to the post-processor to be used for the target system.

REQUIREMENT LESSONS LEARNED (A.3.1.1.1)

Past engine model specifications have included diagrams and schematics of the engine. The use of CAD has been used on commercial engine applications with reductions in overall cost and time.

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A.4.1.1.1 Interface and installation.

The requirements of 3.1.1.1 shall be verified by demonstration and inspection.

VERIFICATION RATIONALE (A.4.1.1.1)

The interfaces must be verified to ensure that they are functionally and physically correct.

VERIFICATION GUIDANCE (A.4.1.1.1)

Inspections should be done at PDR and CDR and a demonstration that all interfaces between the air vehicle and engine work properly should be done before first flight.

VERIFICATION LESSONS LEARNED (A.4.1.1.1)

None.

A.3.1.1.2 Installation hardware.

The engine shall be suitable for installation in any air vehicle configuration for multi-engine aircraft with different engine locations. Parts requiring reconfiguration shall be minimized and controls and externals shall not have to be relocated for different engine installations on multi-engine aircraft. Parts that require reconfiguration are ___.

REQUIREMENT RATIONALE (A.3.1.1.2)

Installation hardware necessary to install the engine must be kept to a minimum to decrease direct support labor, decrease direct support inventory requirements and increase aircraft availability.

REQUIREMENT GUIDANCE (A.3.1.1.2)

The contractor should identify the parts that need replacement, minor adjustments or modifications. An example of this is a part that may need to have a cover or blank off plate changed or relocated. The number of parts requiring reconfiguration should be kept to a minimum and major changes such as removing, replacing or relocating control and external components must be avoided.

For engines installed in multi-engine aircraft, the design should be such that the engine can be installed in the left or right side without having to change hardware.

REQUIREMENT LESSONS LEARNED (A.3.1.1.2)

None.

A.4.1.1.2 Installation hardware.

The requirements of 3.1.1.2 shall be verified by: ____.

VERIFICATION RATIONALE (A.4.1.1.2)

Verification is necessary to ensure all hardware necessary to install the engine is defined.

VERIFICATION GUIDANCE (A.4.1.1.2)

Past verification methods have included inspection of the hardware and a demonstration of installing the engine.

VERIFICATION LESSONS LEARNED (A.4.1.1.2)

None.

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A.3.1.1.3 Interface loads.

The maximum allowable loads for all parts that interface with the air vehicle shall be as shown in ____.

REQUIREMENT RATIONALE (A.3.1.1.3)

This information must be shared by the engine and air vehicle contractors to ensure the interface connections are within limits.

REQUIREMENT GUIDANCE (A.3.1.1.3)

The following should be used to tailor the specification paragraph: The interfaces can be shown in the ICD, or for programs that do not have an ICD, the interfaces can be put into the specification paragraph.

The Using Service should ensure the contractor has analyzed the interface loads, such as the inlet flange, the power lever connection, inlet particle separator exhaust and inlet, starter and generator connection, aircraft PTO or starter drive shaft and exhaust interface.

This paragraph should include all interface loads and deflections not specified in Section 3.7 (e.g.; anti-icing or de-icing, fuel, electrical, hydraulic, pneumatic, water injection, wash, and any other items that interface with the air vehicle).

REQUIREMENT LESSONS LEARNED (A.3.1.1.3)

None.

A.4.1.1.3 Interface loads.

The requirements of 3.1.1.3 shall be verified by: ____.

VERIFICATION RATIONALE (A.4.1.1.3)

This requirement is essential for the air vehicle contractor and can be verified by analysis, inspection and tests.

VERIFICATION GUIDANCE (A.4.1.1.3)

The following should be used to tailor the specification paragraph: Analysis should be used to determine the maximum loads that can be tolerated by the engine interfaces and engine and flight tests should be used to ensure the actual loads are within the analytical values.

The air vehicle contractor and the Using Service should inspect the information for completeness and compatibility with the air vehicle.

VERIFICATION LESSONS LEARNED (A.4.1.1.3)

The USN had an engine mount wear problem on an aircraft model in the late 1980's. The contributing cause was unknown stress levels the air vehicle mount system imparted on the engine mounts. Higher than expected levels were discovered in flight tests when both air vehicle and engine sides of the mount system were instrumented. The calculated stresses, from a previously used model simulation, were erroneous and didn't predict this wear problem. This shows that measured values are needed.

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A.3.1.1.4 Mounts.

A.4.1.1.4 Mounts.

A.3.1.1.4.1 Main mounts.
The engine mounts shall retain the engine, including retained fluids and externals, at all flight, takeoff and landing, and ground conditions and shall withstand elastic limit loads of __(a)__ without permanent deformation and ultimate tensile strength loads of __(b)__ without complete fracture. The main mounts shall meet the requirements of 3.4.1.7. The locations of all engine mounts shall be as specified in __(c)__.

REQUIREMENT RATIONALE (A.3.1.1.4.1)

Engine mounts are necessary to attach the engine to the aircraft. Elastic and ultimate tensile strength load limits must be specified to ensure that both engine and aircraft designs are sufficiently strong for normal operations and safe throughout the specified service life. Mounts are required to have sufficient strength to protect the engine against a worst case single attachment point failure to ensure safety of flight (i.e., fuel fire and single engine loss of power) and ground safety (i.e., crashed engines or aircraft) (see LESSONS LEARNED).

REQUIREMENT GUIDANCE (A.3.1.1.4.1)

The following should be used to tailor the specification paragraph except when the air vehicle's engine mounting requirements have been previously established:

(a), (b): The contractor should specify the maximum system limits in units of force and in reference to the engine. The specified values should include, but not be limited to, the effects of the following requirements and specific design characteristics: externally applied forces (e.g., accelerations) of 3.4.1.2.1, gyroscopic moments of 3.4.1.6.10, all air vehicle loads which are supported through the engine structure (if such loads exist) and safety factors of 3.4.1.6.1, cyclic fatigue, engine mass, material strength and mechanics and service life. The contractor should specify the bending moment limits in the axial, vertical, and lateral directions. The engine mounts should meet the applicable requirements of 3.4 (i.e.; durability, strength, damage tolerance, etc.). (NOTE: For competitive engine development programs the Using Service should ensure the engine and aircraft specifications are compatible with the maximum loads of the worst-case engine and air vehicle combination.)

(c): The contractor should specify the locations of the engine mounts to ensure that both engine and aircraft are designed to a common goal. When identifying mount locations in the engine specification, Figure 1 should be used. When identifying mount locations through the installation drawings, the ICD should be used. Redundant mounts should also be indicated on figure 1.

Background:

For engines that utilize thrust vectoring, the vector forces should be determined by the engine contractor. These forces will be based upon a particular engine and air vehicle combination. Because of the higher vertical loads imposed during thrust vectoring, it is essential that the engine mount system sustain the worst case load predictions that the aircraft would obtain. The engine mounting system must be suitable for the thrust vectoring nozzle arrangement and the mount system must be durable enough to withstand the various loads that will be imposed when the nozzle is vectored at selected angles.

Part 33 of the FAR requires that engine mounts have both elastic and ultimate strength integrity.

REQUIREMENT LESSONS LEARNED (A.3.1.1.4.1)

Engine mounts have failed in flight. The right Engine AG on An Aircraft Company D Aircraft AA jetliner fell off shortly after takeoff from O'Hare International Airport on 20 January 1989. According to an eyewitness report, "There was a lot of smoke coming out of one engine, and we saw it leaning, almost falling off, and then it fell off..." (ref.: Washington Post, Jan 21, 1989). Similar incidents involving Aircraft AA were an Airline Company A flight leaving Dallas on January 3, 1986 and Airline Company B flight leaving Philadelphia on December 5, 1987 (ref.: Washington Post, January 22, 1989).

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The single attachment point guidance for multiple engine aircraft will likely result in two forward mounts and one aft mount, with an aft mount failure possibly resulting in partial engine separation (e.g., inelastic mount elongation) and engine shutdown. Guidance for single engine aircraft will likely result in a design with four mounts arranged in a rectangular or diamond shaped pattern, without loss of engine power or change in engine position (33 percent redundancy factor and stresses in the elastic range) after single attachment point failure.

A.4.1.1.4.1 Main mounts.

The requirements of 3.1.1.4.1 shall be verified by: ____.

VERIFICATION RATIONALE (A.4.1.1.4.1)

Elastic and ultimate tensile strength load limits must be verified to ensure that the engine is sufficiently strong for normal operations and safe for single point attachment failures throughout the service life or a damage tolerant design.

VERIFICATION GUIDANCE (A.4.1.1.4.1)

Past verification methods have included analyses and tests.

Background:

To minimize cost prior to preflight qualification, engine mount requirements should be verified by analysis of the worst case engine mount failures and their consequences. Engine mount testing should be conducted after the completion of endurance test cycling and prior to production qualification. See A.3.1.1.4.1 REQUIREMENT GUIDANCE for load criteria that should be included in the verification.

Thrust vectoring nozzles impose new structural loads upon the engine and the air vehicle. These loads must be examined carefully by analysis of test data. Determination as to whether the mount system will withstand these new forces is a factor of mount system strength, durability, and mount system life considerations.

The engine mounting system must be suitable to the thrust vectoring nozzle arrangement and the mount system should be durable enough to withstand the various loads that will be imposed by directing the engine thrust at selected angles.

The load calculations done by the engine contractor during the design phase should be verified during sea level demonstrations and tests.

VERIFICATION LESSONS LEARNED (A.4.1.1.4.1)

The most commonly known failure mode for the engine mounting system was metal fatigue of the mounting bolts.

A.3.1.1.4.2 Ground handling mounts.

The ground handling mounts shall support the engine, including all engine mounted equipment and externals, components, and operating fluids, under the following maximum inertia load conditions, without deformation to the mounts or damage to the engine: _(a)_ acting in combination at the engine center of gravity.

The locations and descriptions for the individual ground handling mounts shall be as specified in _(b)_. The arrangement shall be compatible with _(c)_.

REQUIREMENT RATIONALE (A.3.1.1.4.2)

Ground handling mounts are required to provide a means to lift the engine during the installation and removal from the aircraft and for ground transportation (intra-depot and maintenance transportation and not over-the-road transportation of any kind) and maintenance.

REQUIREMENT GUIDANCE (A.3.1.1.4.2)

The following should be used to tailor the specification paragraph:

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(a): The mounts must have the capability to handle loads of at least 4g axial, 2g lateral, and 3g vertical.

(b): The interfaces can be shown in the ICD, or for programs that do not have an ICD, the interfaces can be put into the specification paragraph.

(c): The Using Service should identify the associated ground handling equipment to be used with the engine (by National Stock Number and nomenclature).

Figure 1 should be formatted and provided in the specification by the Using Service for the contractor to provide the required information.

Background:

The mounts should be designed to be compatible with existing engine transportation and maintenance equipment. Life cycle cost studies, in conjunction with operational requirement studies, should be conducted to determine if new or existing engine handling equipment should be used.

If adapters are necessary, they should be designed and provided by the engine manufacturer as specified in the contract.

REQUIREMENT LESSONS LEARNED (A.3.1.1.4.2)

Ground handling mounts and related support equipment have not always been adequately designed and compatible. Engine damage has occurred due to engine support equipment not containing sufficient shock absorbing capability, thereby transmitting high “g” forces into the engine. Also, engine ground handling mounts have been damaged or have failed because the mounts are not adequately designed for the loads imposed during engine transportation (excluding shipping), maintenance, and installation and removal from the air vehicle.

A.4.1.1.4.2 Ground handling mounts.

The requirements of 3.1.1.4.2 shall be verified by: ____.

VERIFICATION RATIONALE (A.4.1.1.4.2)

The intent of this paragraph is to verify the ground handling mounts provided on the engine are adequate for ground handling, transportation, and maintenance of the engine.

VERIFICATION GUIDANCE (A.4.1.1.4.2)

Past verification methods have included analyses, demonstrations and tests.

The contractor should perform an analysis or demonstration to ensure the ground handling mounts can withstand the loads of 3.1.1.4.2.

The following should also be included in the verification process:

a. On-base ground transportability
b. Engine installation and removal from the air vehicle
c. Engine maintenance tasks
d. Other ground handling tasks peculiar to the engine.

Background:

Tests or analyses should be conducted to load levels sufficient to verify limit load and ultimate load operational requirements and to verify that minimum strength components can meet the load requirements.

VERIFICATION LESSONS LEARNED (A.4.1.1.4.2)

None.

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A.3.1.1.4.3 Engine stiffness.

For turboshaft and turboprop engines, the estimated stiffness of the engine in resisting loads and moments applied at the outboard end of the output shaft, relative to the engine mounting points, shall be specified in ___. The first “free-free” lateral and vertical engine bending modes shall be specified therein.

REQUIREMENT RATIONALE (A.3.1.1.4.3)

Engine stiffness information is used to determine the stress, deflections and loads for an installed engine. The free-free modes and frequencies provide a measure of the dynamic characteristics of the overall engine structure.

Aircraft maneuvers with turboprop and turboshaft engines cause large moments about the propeller shaft. This may increase propeller shaft and engine or gearbox case deflection thereby requiring better structural rigidity.

REQUIREMENT GUIDANCE (A.3.1.1.4.3)

The following should be used to tailor the specification paragraph:

The interfaces can be shown in the ICD, or for programs that do not have an ICD, the interfaces can be put into the specification paragraph.

The loads should include, but not be limited to, the effects of externally applied forces, gyroscopic moments, safety factors, cyclic fatigue, material strength, and service life.

If specified by the air vehicle manufacturer, the engine manufacturer should include installed engine modes if the locations and spring rates of the air vehicle attachment structure are known. This would not impose a significant additional analytical effort for the engine manufacturer; it would merely be a modal rerun with revised boundary conditions. If the installed modes are not provided the air vehicle can still use the free-free information as a sanity or reality check on his independent modal results.

REQUIREMENT LESSONS LEARNED (A.3.1.1.4.3)

The engine manufacturer would want the first engine response mode to be far removed from any known excitation source; e.g., rotor blade passing frequency which for Engine D installation is on the order of 25 Hz. (5 Blades @ 300 RPM).

A.4.1.1.4.3 Engine stiffness.

The requirements of 3.1.1.4.3 shall be verified by: ____.

VERIFICATION RATIONALE (A.4.1.1.4.3)

Verification of engine stiffness is required to ensure compatibility with the air vehicle.

VERIFICATION GUIDANCE (A.4.1.1.4.3)

Past verification methods have included analyses, demonstrations and tests.

Background:

Engine stiffness should be verified prior to preflight qualification since the design may be impacted. See externally-applied forces of 3.4.1.2.1.

VERIFICATION LESSONS LEARNED (A.4.1.1.4.3)

None.

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A.3.1.1.5 Control lever and signals.

REQUIREMENT GUIDANCE (A.3.1.1.5)

When levers are used to perform any of the functions of 3.1.1.5.1 through 3.1.1.5.4, the requirements will be described here. The interface description should include “dead-bands”, hysteresis or reversal of characteristics with continuous demand input.

When other signals are used, 3.1.1.6 Subsystem interfaces will be used to describe the interface signals.

A.4.1.1.5 Control lever and signals.

A.3.1.1.5.1 Power command lever and signal.

The power command lever positions and signals shall be specified in the _____. Engine power response shall be continuous and proportionally constant in relation to engine power lever input.

For multi-power section applications a signal shall be provided to indicate the power section on-off status.

REQUIREMENT RATIONALE (A.3.1.1.5.1)

The power command or lever characteristics of the engine must be specified to establish the control interface relationship.

REQUIREMENT GUIDANCE (A.3.1.1.5.1)

The following should be used to tailor the specification paragraph:

The interfaces can be shown in the ICD, or for programs that do not have an ICD, the interfaces can be put into the specification paragraph.

Background:

Turbofan, turbojet and turboprop engine power demand can be through a power lever. The power lever can be used to control power from off to the maximum power or from the maximum reverse to the maximum power. (This latter arrangement could require a separate fuel shutoff signal (3.1.1.5.4).) The relationship between lever position and power condition should be specified and the accuracy or range should be specified. For example, power lever angle 0 - 3° off, 15 - 18° idle, etc.

Turboshaft engines that may or may not be free turbine engines may use a combination of a power lever and load demand signals (3.1.1.5.2) for power modulation. When levers have been used, multiple functions have been included in the power lever such as fuel shutoff, ground idle (ungoverned power turbine speed) and fly (governed power turbine speed). The relationship between lever position or signal and condition (fuel shutoff, idle, fly, etc.) should be specified.

REQUIREMENT LESSONS LEARNED (A.3.1.1.5.1)

None.

A.4.1.1.5.1 Power command lever and signal.

The requirements of 3.1.1.5.1 shall be verified by: ____.

VERIFICATION RATIONALE (A.4.1.1.5.1)

The power lever signal characteristics must be verified to ensure engine control interface characteristics.

VERIFICATION GUIDANCE (A.4.1.1.5.1)

Past verification methods have included demonstration and tests.

Engine control interface response is verified by various tests and demonstrations conducted throughout the development program.

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VERIFICATION LESSONS LEARNED (A.4.1.1.5.1)

None.

A.3.1.1.5.2 Load demand command and signal.

If required for turboshaft engines, a means shall be provided at the control interface to adjust for rapid power absorber load changes. The control system interface shall provide for compatibility between the air vehicle load demand output signal and the control system. Engine power response shall be continuous and proportionally constant in relation to engine power lever input.

REQUIREMENT RATIONALE (A.3.1.1.5.2)

The intent of this paragraph is to provide for input from the air vehicle to the engine to be used to anticipate load changes and reduce power turbine transient speed droop.

REQUIREMENT GUIDANCE (A.3.1.1.5.2)

If a helicopter requires high maneuverability that is limited by the helicopter rotor characteristics, a load demand signal may be required. This air vehicle signal is inputted to the control system to provide engine acceleration before the rotor droop occurs.

For engines with augmentors, transitioning to augmented thrust may produce an abrupt change in power that may not be acceptable to the Using Service.

REQUIREMENT LESSONS LEARNED (A.3.1.1.5.2)

None.

A.4.1.1.5.2 Load demand command and signal.

The requirements of 3.1.1.5.2 shall be verified by: ____.

VERIFICATION RATIONALE (A.4.1.1.5.2)

The control signals must be verified to ensure that no incompatibility exists.

VERIFICATION GUIDANCE (A.4.1.1.5.2)

Past verification methods have included demonstrations and tests.

VERIFICATION LESSONS LEARNED (A.4.1.1.5.2)

None.

A.3.1.1.5.3 Output speed lever and signal.

If required for turboprop or turboshaft engines, a separate shaft speed lever and signal, if required, shall be provided to adjust and control the output shaft speed. The control system interface shall provide for compatibility between the air vehicle output speed lever output signal and the control system. No abrupt change in output shaft speed shall occur with linear shaft speed demand input.

REQUIREMENT RATIONALE (A.3.1.1.5.3)

The intent of this paragraph is to specify an adjustable and controllable output speed, providing pilot capability to set delivered shaft speed for the flight or to adjust delivered shaft speed for improved fuel economy and noise reduction.

REQUIREMENT GUIDANCE (A.3.1.1.5.3)

Background:

When required, an output speed command lever and signal is utilized on turboshaft and turboprop engines. For mechanical systems, the Using Service should make sure the air vehicle or interface specification requirements are consistent with the mechanical interface requirements of 3.1.1.5.5.

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For helicopters, output speed lever settings are normally used to set ground idle and “fly” conditions, with an air vehicle trim potential to adjust power turbine speed.

REQUIREMENT LESSONS LEARNED (A.3.1.1.5.3)

None.

A.4.1.1.5.3 Output speed lever and signal.

The requirements of 3.1.1.5.3 shall be verified by: ____.

VERIFICATION RATIONALE (A.4.1.1.5.3)

The output speed lever must be verified to ensure that compatibility exists.

VERIFICATION GUIDANCE (A.4.1.1.5.3)

Past verification methods have included demonstrations and tests.

Engine control interface response is verified by various tests and demonstrations conducted throughout the development program.

VERIFICATION LESSONS LEARNED (A.4.1.1.5.3)

None.

A.3.1.1.5.4 Fuel shutoff lever and signal.

The fuel control shall process air vehicle signals as required in the _____ for fuel shutoff.

REQUIREMENT RATIONALE (A.3.1.1.5.4)

Engines may not be capable of shutdown using the cockpit thrust or power demand mechanism, therefore, signals separate from the thrust or power demand mechanism may be needed to shut off fuel to the engine.

REQUIREMENT GUIDANCE (A.3.1.1.5.4)

The following should be used to tailor the specification paragraph: The interfaces can be shown in the ICD, or for programs that do not have an ICD, the interfaces can be put into the specification paragraph.

REQUIREMENT LESSONS LEARNED (A.3.1.1.5.4)

None.

A.4.1.1.5.4 Fuel shutoff lever and signal.

The requirements of 3.1.1.5.4 shall be verified by: ____.

VERIFICATION RATIONALE (A.4.1.1.5.4)

The fuel shutoff signal(s) must be verified for proper processing by the fuel control.

VERIFICATION GUIDANCE (A.4.1.1.5.4)

Past verification methods have included demonstrations and tests.

VERIFICATION LESSONS LEARNED (A.4.1.1.5.4)

None.

A.3.1.1.5.5 Lever torque.

For systems with mechanical interfaces, the torque required to operate the engine thrust or power level or any additional levers necessary to control the engine through their range of travel shall be no greater than _____ throughout all flight loads and conditions. The lever(s) shall not move with the engine operating

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unless external torque is applied. Movement of the lever(s) throughout the operating range shall be free of abrupt changes in actuating torque.

REQUIREMENT RATIONALE (A.3.1.1.5.5)

A maximum torque should be established so that the pilot isn't inconvenienced. The lever movement is required to be free of abrupt changes to assure that the pilot can accurately control the engine.

REQUIREMENT GUIDANCE (A.3.1.1.5.5)

The following should be used to tailor the specification paragraph:

The air vehicle and engine contractors should collectively determine the maximum and minimum lever torque allowed that is acceptable to the pilot. Past programs have limited the maximum torque to 2.8 N-m (25 pound-inches) and minimum torque to 0.56 N-m (5 lb-in).

Background:

In the case of mechanical control interfaces, the torque should be specified in the air vehicle or interface specification if it is different when the engine is not running. Also, the Using Service should ensure the specification or ICD specifies the relationship between power lever torque and power lever angle. The Using Service should make sure the air vehicle or interface specification requirements are consistent with the mechanical interface requirements of 3.1.1.5.5.

The Using Service should make sure that the air vehicle requirements precludes inadvertent lever movement, eases operation for all pilots, and enhances operational safety. The lever is required to remain fixed while the engine is running because there have been instances where the lever moved because of vibration, "g" load, etc.

REQUIREMENT LESSONS LEARNED (A.3.1.1.5.5)

The maximum allowable torque, between Idle and Maximum, historically has been 15 pound-inches since AN-E-30 was developed. A review of Engine K, Q, W, AB, and AH model specifications revealed that the maximum allowable torque was greater than 15 pound-inches. The limiting value was increased to 25 pound-inches by the USAF during the Tri-Service coordination meeting for MIL-E-5007D.

In some earlier programs there was a requirement from the airframer and pilots for a minimum torque to provide a feedback "feel" to the control levers. Engine T in aircraft K allows for zero minimum torque as long as the throttles don't move without external force applied. The minimum torque of 0.56 N-m (5 lb-in) was proposed by the AIA in 1998.

Modern throttle designs as typified in Engine O in Aircraft H and Engines AN and AO in Aircraft AH use electronic control signals thus eliminating the requirement for lever torque.

A.4.1.1.5.5 Lever torque.

The requirements of 3.1.1.5.5 shall be verified by: ____.

VERIFICATION RATIONALE (A.4.1.1.5.5)

The lever torque must be verified to ensure compatibility with the engine control interface.

VERIFICATION GUIDANCE (A.4.1.1.5.5)

Past verification methods have included demonstration and tests.

Engine control interface response is verified by various tests and demonstrations conducted throughout the development program.

VERIFICATION LESSONS LEARNED (A.4.1.1.5.5)

None.

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A.3.1.1.6 Subsystem interfaces.

The engine shall receive and transmit signals as required by the air vehicle and ground support systems. The signals, signal characteristics and physical interface shall be shown herein. Engine subsystems that interface with the air vehicle and ground support system are:

_____ Subsystem (Contractor to specify subsystem):

Input	Parameter	Function

Output	Function	Parameter

Digital message format (data) and timing requirements (transmission rate) communications between the air vehicle and engine control and monitoring computers shall be specified herein.

REQUIREMENT RATIONALE (A.3.1.1.6)

Engine subsystem interfaces need to be defined to ensure the engine will operate properly when installed in the air vehicle. All interfaces, including electrical, engine monitoring system, caution and warning signals, fiber optics and engine health trending data must be identified to ensure that engine and air vehicle systems are compatible.

REQUIREMENT GUIDANCE (A.3.1.1.6)

The following should be used to tailor the specification paragraph:

The interfaces can be shown in the ICD, or for programs that do not have an ICD, the interfaces can be put into the specification paragraph. The contractor should identify the subsystems that interface with the air vehicle or ground support system.

Digital message format (data) and timing requirements (transmission rate) communications between the air vehicle computer and engine control and engine monitoring computers over a bus such as the MIL-STD-1553 bus should be established in cooperation with the air vehicle manufacturers. The agreed upon data format and timing requirements should be identified and controlled through the use of an ICD.

Background:

Instrumentation should be provided to display the power setting and information to determine normal and abnormal engine operation. Engine power settings may be determined by engine speed, exhaust nozzle position, engine pressure ratio, exhaust gas or turbine temperature, output shaft torque or fuel flow. Engine condition parameters may include oil pressure, temperature, and quantity; engine speeds; and various caution and warning lights. The instrumentation requirements may be more stringent for single engine aircraft.

REQUIREMENT LESSONS LEARNED (A.3.1.1.6)

A turbine over temperature warning system was retrofitted into the pilot headset in one fighter aircraft because the pilot’s attention to the environment outside the cockpit prevented them from noticing an over limit condition.

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A.4.1.1.6 Subsystem interfaces.

The requirements of 3.1.1.6 shall be verified by: ____.

VERIFICATION RATIONALE (A.4.1.1.6)

Instrumentation signals must be verified to ensure that the engine performance parameters are adequate and accurate.

VERIFICATION GUIDANCE (A.4.1.1.6)

Past verification methods have included demonstrations and tests.

An analysis should be accomplished to verify the accuracy of instrumentation signals. During development testing, the instrumentation should be installed and tested to the extremes of engine operation.

VERIFICATION LESSONS LEARNED (A.4.1.1.6)

Pilot comments have been valuable for information and types of engine data that may be required.

A.3.1.1.7 Bleed air interface.

The engine shall provide customer bleed air extraction for aircraft use. The capabilities and maximum flow of each bleed port, including the acceleration or start bleed air ports, shall be as shown in ___(a)___. Bleed duct external surface temperatures shall meet the requirements of 3.1.8.2. The bleed port internal pickup points shall be located at positions which have low susceptibility to FOD, and ingestion of sand, dust, ice, moisture, and any other foreign materials contained in the air. Provisions shall be made to prevent high-pressure bleed air from entering lower pressure bleed ports. A failure in the air vehicle bleed system shall result in __(b)____ engine operation. If compressor venting is necessary, the temperature, pressure and airflow inflow and outflow conditions for which venting provisions must be made shall be specified in ___(c)____. The ducting attachment detail shall be shown on the engine interface and installation drawings.

REQUIREMENT RATIONALE (A.3.1.1.7)

Bleed air may be required for various purposes including environmental control, anti-icing, component and accessory cooling, pneumatic operation of components, aircraft boundary layer control, attitude control, and high lift. Bleed air characteristics must be defined so that the systems using engine air bleed provisions can be properly designed.

The need for bleed air during starting or acceleration should be known by the aircraft designer so a compatible system can be developed for the air vehicle and engine interface and integration.

REQUIREMENT GUIDANCE (A.3.1.1.7)

The following should be used to tailor the specification paragraph:

For blanks (a) and (c), the interfaces can be shown in the ICD, or for programs that do not have an ICD, the interfaces can be put into the specification paragraph. The locations, number, and interface dimensions at all customer bleed air ports should be shown on the engine interface and installation drawings. The maximum allowable bleed airflows, pressures, and temperatures for each port, and the compressor stage(s) from which it is extracted should be included in either the ICD or the specification. The maximum permissible continuous flow capability of each bleed air port in percent of total airflow should be specified. Bleed flow, pressure and temperature characteristics versus power setting and flight condition should be included in either the ICD or the specification. Insulation may be used on bleed ducts to keep the surface temperature below the auto-ignition temperature of the flammable fluids used by the engine.

For blank (b), in the event of an air vehicle bleed system failure, some level of engine operation may still be required. The Using Service should specify what level of engine operation, such as fail operational, fail degraded or complete engine shutdown is required.

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Background:

Bleed ducts should be analyzed to determine their potential as ignition sources for flammable fluids to which they may be exposed. Ducts assessed as potential ignition sources should be insulated. 390°C has been used in the past, however the engine bay installation (i.e.; cooling mass flow, cooling air temperatures) and placement of bleed pipes and flammable fluid lines (location of flammable fluid sources) may require a different temperature limit.

The temperature for self-generating ignition is a function of numerous variables and relationships. At near static conditions the surface ignition temperature is approximately 340°C (650°F) and rises to 621°C (1150°F) at a wind velocity of 0.6 m/sec (2 ft/sec). FAA report January 1971, "Crash fire hazard evaluation of jet fuels" indicates that for kerosene a range of surface ignition temperature from 215°C (417°F) with a confined system to 650°C (1200°F) with an open steel plate can occur. Based on all of the above the existing 370°C (700°F) requirement is considered to be a reasonable limiting temperature for un-insulated ducts.

A potential fire hazard exists where bleed air is taken from the engine at high temperature locations. Consideration should be given to using either an engine or air vehicle furnished shutoff device at the bleed source where such hazard conditions can exist. Alternate solutions are the use of insulation on all high temperature bleed lines and adequate ventilation of the engine compartment.

Some aircraft and engine bleed systems require a minimum bleed pressure or energy level for satisfactory operation. Examples are aircraft ECS and engine bleed air start systems. Many helicopters use bleed air for "cross-bleed starting" for two engine aircraft and "buddy starting" for single engine aircraft.

Bleed flow is required for air vehicle uses such as cabin pressurization, air conditioning and anti-icing. These are air vehicle systems and the bleed flow will vary according to air vehicle requirements during flight. It is necessary that no thrust or power demand adjustment be required for the bleed flow modulation. This would impose an additional workload on the pilot when they least need it. Allowing high-pressure air to flow into the low-pressure stages would result in surge, stall, or performance deterioration. Sonic flow will limit the airflow through a given port size. By selecting the area of the port, the maximum airflow can be controlled. This will act as a safety valve to protect the engine against any failure modes down stream of the aircraft bleed system. Some aircraft concepts, such as aircraft with wing blowing or V/STOL attitude control, will need bleed in excess of the normal bleed limit. This is accommodated by increasing the size of the bleed orifice to either increase flow or lower pressure losses.

REQUIREMENT LESSONS LEARNED (A.3.1.1.7)

None.

A.4.1.1.7 Bleed air interface.

The requirements of 3.1.1.7 shall be verified by: ____.

VERIFICATION RATIONALE (A.4.1.1.7)

The bleed air interface information needs to be confirmed.

VERIFICATION GUIDANCE (A.4.1.1.7)

Past verification methods have included demonstrations and tests.

Background:

Bleed air tests are usually conducted in conjunction with engine performance tests at sea level and altitude. The tests should include the air vehicle requirements and the maximum bleed flow specified by the engine contractor. This will eliminate any operability problems if the bleed flow is increased due to air vehicle changes.

Measurement inspection of ports and locations, surface temperature measurements, and engine tests at various bleed rates up to maximum specified flow rates should be performed.

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Low power settings may require a large amount of anti-ice airflow; therefore, the pilot may be instructed to use high power in icing conditions.

VERIFICATION LESSONS LEARNED (A.4.1.1.7)

None.

A.3.1.1.7.1 Customer bleed air contamination.
Engine generated substances contained in customer bleed air shall be below the threshold limit of ____.

REQUIREMENT RATIONALE (A.3.1.1.7.1)

Since aircraft gas turbine engine compressor bleed air is normally used for cabin air conditioning, bleed air must be supplied with minimal contaminants to assure no possibility of a health hazard to the crew members, or to interfere with their ability to perform the mission.

REQUIREMENT GUIDANCE (A.3.1.1.7.1)

The following should be used to tailor the specification paragraph:

Engine generated substances contained in the bleed air should be within the threshold limit values specified below.

Substance	Parts per million
Carbon dioxide	5000.0
Carbon monoxide	50.0
Ethanol	1000.0
Fluorine (as HF)	0.1
Hydrogen peroxide	1.0
Aviation fuels	250.0
Methyl alcohol	200.0
Methyl bromide	20.0
Nitrogen oxides	5.0
Acrolein	0.1
Oil breakdown products (e.g., aldehydes)	1.0
Ozone	0.05

The bleed air should not contain a total of more than 5.0 mg (0.3 lbm x 10^{-6}) of engine generated particles per m^3 (ft^3) of bleed air.

Background:

Where substances other than those listed are contributed to the extracted air by engine operation, the engine manufacturer should report the substances and contamination in parts per million to the Using Service to determine maximum limits. When two or more engine-generated substances are present their combined effect should be determined and reported. In the absence of information to the contrary, the combined effects of the different substances should be considered additive. If cleaning fluids are specified for use during normal engine maintenance, consideration should be given to their effect on bleed air contamination.

The Using Service should check with the aircraft system specification to make sure all engine generated particles and contaminates in the bleed air have requirements that agree with the system specification.

The source of bleed air contamination may be composed primarily of engine lubricating oil and its decomposition products. Contamination becomes a particularly difficult problem on high-pressure ratio

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engines in which air temperatures of 370 to 425°C (700-800°F) are sufficiently high to cause decomposition of the oil. When decomposition products (organic carbonyl and organic hydroperoxide compounds) are present in concentrations as high as 0.5 to 1.0 parts per million (ppm), anyone breathing this air may experience severe irritation, nausea, and vomiting.

REQUIREMENT LESSONS LEARNED (A.3.1.1.7.1)

The Threshold Limit Values (TLVs) for the substances shown in the table were set by the American Conference of Governmental Industrial Hygienists (ACGIH). These values are considered safe levels for continuous exposure of up to eight hours.

Problems have also occurred on engines with air operated components that are contaminated by foreign particles such as blade abradable material. These parts need to be designed to tolerate the contaminants or a filtering system needs to be included.

A.4.1.1.7.1 Customer bleed air contamination.

The requirement of 3.1.1.7.1 shall be verified by: ____.

VERIFICATION RATIONALE (A.4.1.1.7.1)

Bleed air contamination, because of its importance to crew safety, must be measured.

VERIFICATION GUIDANCE (A.4.1.1.7.1)

Past verification methods have included analysis and tests.

Background:

Bleed air contamination levels should be verified on a test engine at idle, maximum continuous, and maximum power settings. Bleed air samples should be collected from all customer bleed ports and from the engine inlet air. Samples should be submitted to a qualified chemistry laboratory for analysis. Bleed air samples should be taken periodically during the course of the endurance testing. Considerations should be made for evaluating the bleed air contamination levels of the altitude qualification engine.

VERIFICATION LESSONS LEARNED (A.4.1.1.7.1)

One oil breakdown product, acrolein, is a highly active product which has shown to be quite difficult to measure accurately because it cannot be separated easily from the other oil breakdown products.

A.3.1.1.8 Drains and fluid collection devices.

Fluid drains, drain characteristics, and collection requirements shall be shown in ___. There shall be no leakage beyond ___ cc/hr of fluids from the engine except at the drains provided for this purpose. The flow rate into all drains and the maximum amount of combustible fluid drainage following shutdown after normal operation shall be as shown in: ____. Drains shall function statically and throughout the operating envelope. Drains for different fluid types shall not be interconnected.

REQUIREMENT RATIONALE (A.3.1.1.8)

Drains and fluid collection devices are necessary to remove flammable fluids from areas conducive to combustion. Drain location is subject to interface definition. An air vehicle fluid collection device or system is necessary to prevent fluid seepage and leakage into potentially hazardous areas of the engine. Most aircraft now require an environmental collection bottle.

REQUIREMENT GUIDANCE (A.3.1.1.8)

The following should be used to tailor the specification paragraph:

The drain characteristics, flow rates and maximum fluid drainage can be shown in the ICD, or for programs that do not have an ICD this information can be put into the specification paragraph.

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The contractor should provide the leakage flow rate. In the past 0.15 cc/hr has been defined for locations other than drains.

Background:

Typically the interface for the engine fluid drain should define the characteristics of the fluid, including the maximum flow rate from failed components. Drains that can be separated into individual lines are useful when it is necessary to isolate leaking components during troubleshooting. Maximum fluid rates should be specified for each individual system (i.e.; fuel, oil, hydraulic) for when the engine is and is not running. Drain lines that cannot be separated for troubleshooting can mask problems with individual component leakage. The total fluid leakage rate may fall within limits, but a single component may be exceeding its own leakage limits.

The survivability and vulnerability community should be consulted to determine restrictions on drain hole sizes.

Actuators and joints have leaked in the past.

REQUIREMENT LESSONS LEARNED (A.3.1.1.8)

None.

A.4.1.1.8 Drains and fluid collection devices.

The requirements of 3.1.1.8 shall be verified by: ____.

VERIFICATION RATIONALE (A.4.1.1.8)

The adequacy of the routing and drain system size must be verified.

VERIFICATION GUIDANCE (A.4.1.1.8)

Past verification methods have included analysis to determine the interface connections and routing of drain tubes on the engine, inspections to look for leaks and tests to measure the fluid flow from the drains.

During engine endurance and altitude testing, the source and amount of fluid drainage should be verified. Consideration should be given to simulating the drain actual pressure in the test cell to ensure the drains will work properly during flight.

VERIFICATION LESSONS LEARNED (A.4.1.1.8)

None.

A.3.1.1.9 Power absorber interface.

For turboshaft and turboprop engines, the allowable range of characteristics of the shaft power absorber at the power absorber to engine interface shall be as shown in: __(a)___. No resonant frequency shall be transmitted to or from the power absorber through the engine.

The output drive shaft shall be capable of continuously absorbing thrust in either direction of an amount not less than __(b)__ percent of the circumferential force acting on the output drive shaft spline at the steady-state torque limit as specified in the engine specification.

The engine shall provide a signal for operation of a torque indicator throughout the operating envelope of the engine. The signal shall be continuous as a function of torque. The specified accuracy shall be maintained without recalibration after the torque sensor is disassembled and reassembled. The accuracy of the torque sensor, from __(c)__ percent rated delivered shaft speed to __(d)__ percent rated delivered shaft speed, shall be:

a. Within plus or minus __(e)__ percent of the torque at maximum continuous rated power from zero delivered shaft torque to the torque at maximum continuous rated power.

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b. Within plus or minus __(f)__ percent of the torque being measured from the torque at maximum continuous rated power to the transient torque limit.

REQUIREMENT RATIONALE (A.3.1.1.9)

Power absorber to engine interface characteristics must be defined for compatibility.

REQUIREMENT GUIDANCE (A.3.1.1.9)

The following should be used to tailor the specification paragraph:

For blank (a), the interface characteristics can be shown in the ICD, or for programs that do not have an ICD, the interface information can be put into the specification paragraph. For engines using engine oil to actuate the propeller, the following should be added:

If engine oil is necessary for actuation of the propeller, the propeller lubrication and hydraulic system should ensure debris from the propeller mechanism does not contaminate the engine oil or cause engine oil starvation due to failure of the propeller.

The characteristics the engine contractor should include but not be limited to: maximum and minimum polar moment of inertia, torsional spring constant, torsional damping coefficient as a function of torsional spring constant, the maximum allowable static and dynamic loads on the engine output drive shaft, direction of rotation of the output drive shaft as viewed from the engine inlet, design of the output drive shaft spline and the power absorber shaft maximum misalignment during steady-state and transient operation.

For blank (b), a typical value is 20 percent.

For blank (c), a typical value is 50 percent.

For blank (d), a typical value is 110 percent

For blanks (e) and (f), a typical value is 2 percent

REQUIREMENT LESSONS LEARNED (A.3.1.1.9)

In a past engine program a propeller failure caused the failure of an engine. Engine oil was used to lubricate and provide hydraulic pressure to the propeller, and debris from the failed propeller mechanism contaminated the engine oil and induced engine failure.

The engine output spline should be outer diameter piloted with crowned splines to reduce the potential for non-synchronous vibration.

The output drive shaft thrust absorption capability impacts engine and aircraft engine nose gearbox design life. Continuous capability provides reduced spline wear.

A.4.1.1.9 Power absorber interface.

The requirements of 3.1.1.9 shall be verified by _____. The maximum allowable misalignment of the power absorber shall be tested.

VERIFICATION RATIONALE (A.4.1.1.9)

The characteristics of the power absorber to engine interface must be verified.

VERIFICATION GUIDANCE (A.4.1.1.9)

Past verification methods have included inspection of the drawings and tests of the engine with the power absorber attached.

If possible, the actual aircraft propeller should be used as the power absorber. Helicopter systems may be difficult to simulate accurately at idle or no load conditions.

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VERIFICATION LESSONS LEARNED (A.4.1.1.9)

Along with spline fatigue life demonstration, misalignment has indicated problems with turboshaft engine output torque measurement accuracy, especially with phase shift systems.

A.3.1.1.10 Power takeoff (PTO).

When PTO pads and drives are provided for driving aircraft accessories, the ratings, clearance envelopes, dimensions and other pertinent information shall be as shown in: ____. The design and configuration of the PTO pads and drives shall be consistent with the requirements of 3.7.16.

REQUIREMENT RATIONALE (A.3.1.1.10)

The PTO is a drive pad therefore its characteristics must be furnished for interface information, similar to 3.7.16.

REQUIREMENT GUIDANCE (A.3.1.1.10)

The following should be used to tailor the specification paragraph:

The interface characteristics can be shown in the ICD, or for programs that do not have an ICD, the interface information can be put into the specification paragraph. Some of the characteristics the engine contractor may want to identify are: pad and connection details, direction of rotation, accessibility and alignment requirements, and customer power extraction limits for each pad individually and in combination.

PTOs are being used more often on modern engine designs in lieu of integral engine gearboxes. The single PTO provides for simpler engine removal and installation.

REQUIREMENT LESSONS LEARNED (A.3.1.1.10)

The extended PTO shaft is affected by more than maneuver loads. In particular, hard carrier landings are detrimental to the shaft.

A.4.1.1.10 Power takeoff (PTO).

The requirements of 3.1.1.10 shall be verified by: ____.

VERIFICATION RATIONALE (A.4.1.1.10)

There is a need to demonstrate the clearance envelopes, dimensions, and accessibility. A test is required to verify the capability of the PTO.

VERIFICATION GUIDANCE (A.4.1.1.10)

Past verification methods have included demonstrations and tests.

Background:

Since the PTO is a driving pad, it must conform to 3.7.16. Also, the starting drive torque is usually applied through the PTO and must be consistent with 3.1.1.11.1.

VERIFICATION LESSONS LEARNED (A.4.1.1.10)

Engine A had a PTO that developed problems during carrier landings.

A.3.1.1.11 Starting drivetrain.

The engine starting torque acceptance capability shall be specified in _____, shall be sufficient to prevent inadvertent starter drivetrain failure with impact loading, and shall meet the requirements of 3.1.1.11.1 and 3.2.2.3.1. Failure of the air vehicle engine starting drivetrain shall not result in loss of the remaining gearbox functions. The starter drive pad characteristics shall be included in table I. The

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direction of rotation when facing the starting pad on the engine shall be specified herein. No resonant frequency shall be transmitted to or from the starting drivetrain through the engine interface.

REQUIREMENT RATIONALE (A.3.1.1.11)

The starting drivetrain requirements must be specified because it is an important interface requirement between the air vehicle and the starter. The engine starter drivetrain must be capable of accepting a load above that required for a normal start to provide protection from impact loads.

Since the engine starter and starting system are not supplied by the engine manufacturer, it is necessary that sufficient information be supplied to permit procurement of a starting system. This requirement provides the moment of inertia of rotating parts, torsional spring constant and starter train backlash information for system dynamic analysis. This requirement is necessary so that the Using Service can thoroughly verify engine systems (i.e.; transmission, starter, rotor, and assembly) and diagnose various problems throughout the life of the engine. It should be noted that redesigns are generally paid for by the Government. If improper designs could be caught early, tremendous cost savings would be achieved. Additionally, if the Government decided to second source a particular component (starter), this requirement is of great significance.

REQUIREMENT GUIDANCE (A.3.1.1.11)

The following should be used to tailor the specification paragraph:

The interfaces can be shown in the ICD, or for programs that do not have an ICD, the interfaces can be put into the specification paragraph.

Air vehicle engine start time requirements should be captured in 3.2.2.3.1. The time required to start the engine will have an impact on the starting drivetrain because short start times may result in larger loads on the drivetrain to get the engine to turn faster in a short amount of time.

The shear section for the starting drivetrain should be at the starter drive spline. If the shear section is at the engine gearbox, the drive shaft can remain attached to the starter and flail around and possibly cause damage to the engine. The maximum effective mass moment of inertia (kilogram-m^2 (slug-ft^2)) of engine rotating parts to be rotated by the starter and the speed ratio between the starter pad and the driven rotor system, the torsional spring constant (N-m per radian (pound-inches per radian)) for the engine starting drive system and starter train total backlash (radians) at the starter drive pad should be provided to the air vehicle contractor to ensure the starting system will be properly sized.

Background:

An engine drivetrain should be designed to accept the impact loads that can occur during start-up (initial valve opening in hydraulic or pneumatic, or engagement of a mechanical starter jaw clutch on electrical or other types). The factor of 3.33 has been found to provide this safety margin. A 15-second start initiation time to minimum starter cutout speed is an adequate minimum since most engines require 30 to 40 seconds to start (which includes the 15-second time period from start initiation to starter cutout). There should be every effort to use existing starter systems for engines of comparable size. The moment of inertia of rotating parts is interface information that is needed in the early part of the development.

It is not practical or desirable to put a shear section within the engine. All starting systems should have a shear section incorporated at the interface between the engine and the starter system. The shear section is a coupling that can be replaced without requiring return of either the starter or engine to the overhaul facility. The general specification existing for air, hydraulic and electrical starters lists basic design requirements to be used as a guide in developing a starter system. These specifications require a shear section but allow the designer to select the value to be compatible with the engine torque requirements.

There is no requirement defining starter direction of rotation. This would be restrictive because of the extra gears, envelope space, etc., that might be needed to comply. Current day starters turn in either direction.

Backlash is the amount by which the width of a tooth space exceeds the thickness of the engaging tooth measured on the pitch circle. Backlash does not adversely affect proper gear function except for lost motion upon reversal of gear rotation. Backlash inevitably occurs because of necessary fabrication

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tolerances on tooth thickness and center distance plus need for clearance to accommodate lubricant and thermal expansion.

REQUIREMENTS LESSONS LEARNED (A.3.1.1.11)

The original safety factor was 2.0 with a 15-second start. This has resulted in a number of engine removals because of major damage to the engine starter drivetrain. The factor was increased to 2.5 then to 3.33. The fact that there have been no reports of drivetrain failures on recently developed engines indicates that the factor is adequate.

A.4.1.1.11 Starting drivetrain.

The requirements of 3.1.1.11 shall be verified by: ____.

VERIFICATION RATIONALE (A.4.1.1.11)

All components of the starting drivetrain must be verified.

VERIFICATION GUIDANCE (A.4.1.1.11)

Past verification methods have included an analysis of the starter, engine and PTO, and tests to measure the loads, start times, moment of inertia, torsional spring constant and backlash.

Background:

Tests should be conducted at extreme cold conditions since this imposes the greatest load on the starter. The procedure for accomplishing this test should be specified in the pre-test data. Prior to FFR, testing should be conducted at the temperature extremes specified and using the fuels and oils as required.

VERIFICATION LESSONS LEARNED (A.4.1.1.11)

None.

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A.3.1.1.11.1 Starting torque and speed.

The required starting torques and drive speeds shall be specified in _____ and shall include engine drag and engine gearbox drag. Figure 2 shall show the effects, singly and in combination, of ambient temperatures, of altitudes from sea level to the maximum of 3.2.2.3.3, and for air starts with starter assist, using the worst case fuels of the specification. Figure 2 shall be presented for each of the following:

a. No customer bleed air extraction, customer power extraction as specified in the model specification.

b. Maximum allowable customer bleed air extraction, customer power extraction as specified in the model specification.

REQUIREMENT RATIONALE (A.3.1.1.11.1)

Since the engine starter and starting system are not supplied by the engine manufacturer, it is necessary that sufficient information be supplied to permit procurement of a starting system.

REQUIREMENT GUIDANCE (A.3.1.1.11.1)

The following should be used to tailor the specification paragraph:

The interfaces can be shown in the ICD, or for programs that do not have an ICD, the interfaces can be put into the specification paragraph.

This is interface information to reduce logistic load. There should be every effort to use existing starter systems for engines of comparable size.

REQUIREMENT LESSONS LEARNED (A.3.1.1.11.1)

None.

A.4.1.1.11.1 Starting torque and speed.

The requirements of 3.1.1.11.1 shall be verified by: ____.

VERIFICATION RATIONALE (A.4.1.1.11.1)

Starting torque and speed requirements must be verified to procure the proper starting system.

VERIFICATION GUIDANCE (A.4.1.1.11.1)

Past verification methods have included analysis, demonstrations and tests.

Background:

The procedure for accomplishing the test should be specified in the pre-test data. The test should be conducted at the temperature extremes of figure 5 and using the fuels and oils of 3.7.3.1 and 3.7.8.1, respectively.

VERIFICATION LESSONS LEARNED (A.4.1.1.11.1)

Extreme cold conditions are the most difficult and taxing for starter systems.

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A.3.1.1.12 Thrust reverser interface.

The thrust reverser shall deploy and stow after receiving the appropriate command signals. Positive mechanical locking shall prevent reverser deployment in the event of system failure. Failure of the locking system shall not cause reverser deployment. No resonant frequency shall be transmitted to or from the thrust reverser interface.

REQUIREMENT RATIONALE (A.3.1.1.12)

The thrust reverser should only deploy and stow after receiving the proper aircraft signals. Thrust reversers that inadvertently deploy, affect aircraft control and safety; and therefore should operate only upon command.

An in flight failure of the thrust reverser locking system should not cause reverser deployment. Such deployment can cause uncontrollable forces on the aircraft and result in a crash.

REQUIREMENT GUIDANCE (A.3.1.1.12)

The reverser mechanism may be required by the air vehicle to provide a signal indicating whenever the reverser is in the unlocked condition. To preclude inadvertent in-flight reverser deployments, some air vehicle device (i.e., a weight on wheels switch) should be included in the system design.

Background:

The reverser should revert to a stowed position during failure modes. The thrust reverser system should be designed so that no single failure or malfunction under any anticipated flight or ground condition results in reverse thrust.

Thrust reversers are used for braking commercial jet transports. Military aircraft generally use other devices. The thrust reverser system should be properly documented for air vehicle information.

REQUIREMENT LESSONS LEARNED (A.3.1.1.12)

Thrust reverser actuation systems that have a single motor connected to the actuators via flex shafts have encountered binding. Actuator systems that use multiple hydraulic cylinders have shown better reliability and are safer than single motor systems.

A.4.1.1.12 Thrust reverser interface.

The requirements of 3.1.1.12 shall be verified by: ____.

VERIFICATION RATIONALE (A.4.1.1.12)

The thrust reverser interface and compatibility information needs to be verified.

VERIFICATION GUIDANCE (A.4.1.1.12)

Past verification methods have included inspection of the interface drawings and demonstration of reverser operation.

The dimension and envelope should be checked using a full-scale mockup. Mechanical locking and unlocking of the actuation system should be demonstrated during engine testing.

VERIFICATION LESSONS LEARNED (A.4.1.1.12)

None.

A.3.1.1.13 Exhaust system interface.

Maximum allowable static (1g) shear, axial and moment loads in all primary axes (x, y, z, as applicable) for any interface connections between the engine and parts of the exhaust system not supplied with the engine shall be specified in ___. The maximum allowable loads at any exhaust interface shall also be

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specified for the maximum allowable maneuver loads as defined in 3.4.1.2.1. No hot exhaust gas leakage is allowed at the interface. No detrimental resonant frequency shall be transmitted to or from the exhaust system through the interface.

REQUIREMENT RATIONALE (A.3.1.1.13)

Load and moment information for exhaust interface connection is necessary for the air vehicle contractor and engine manufacturer to design parts which may interface with the exhaust system.

REQUIREMENT GUIDANCE (A.3.1.1.13)

The following should be used to tailor the specification paragraph:

The interfaces can be shown in the ICD, or for programs that do not have an ICD, the interfaces can be put into the specification paragraph.

The shear, axial, and moment loads shown in the ICD or figure 3 should include loads due to asymmetric aerodynamic pressure and aircraft seal resistance as well as the forces shown in the ICD or figure 16. For vectoring nozzles, all loads due to vectored forces should also be included on figure 3.

Background:

Exhaust attachments should assure adequate sealing of the hot gases and yet provide for ease of assembly and disassembly. A bolted flange is one type that may be used. The other commonly used means of attachment, the V-Band Clamp type requires accurate alignment on assembly to prevent hot gas leakage. This is not practical on large engines.

Aircraft with highly integrated and unique exhaust systems may require separate exhaust system component installation and removal due to configuration layout or limited physical access. These designs may need a more “rapid-disconnect” system of installation and mounting.

REQUIREMENT LESSONS LEARNED (A.3.1.1.13)

Exhaust nozzle loads, installation, mounting, and removal have been significant issues for certain aircraft arrangements. Thrust vectoring nozzles increase interface loads. Vibratory and acoustical effects have been significant on nozzle design and life.

A.4.1.1.13 Exhaust system interface.

The requirements of 3.1.1.13 shall be verified by: ____.

VERIFICATION RATIONALE (A.4.1.1.13)

The exhaust system interface information needs to be verified.

VERIFICATION GUIDANCE (A.4.1.1.13)

Past verification methods have included analysis of the attachment method and loads, and tests.

Background:

The exhaust nozzle is not only subjected to flight maneuver forces but also aerodynamic, vibratory and thermal loads.

VERIFICATION LESSONS LEARNED (A.4.1.1.13)

The Army was concerned on turboshaft Engine C that attachments could possibly be statically loaded to the maximum allowable limit, in which case the connection would be overloaded in a dynamic situation.

A.3.1.1.14 Compatibility - engine and air vehicle.

The interface of the engine with the air vehicle shall allow the engine to perform as required in the specification.

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REQUIREMENT RATIONALE (A.3.1.1.14)

This is a system level requirement that ties all the installation factors together to ensure that all engine and air vehicle interfaces will work correctly.

REQUIREMENT GUIDANCE (A.3.1.1.14)

Background:

The Using Service should ensure that there is a positive clearance (typically not less than one inch) between the air vehicle and the engine per JSSG-2001A requirement 3.3.1.1.1, “Engine compatibility and installation.”

REQUIREMENT LESSONS LEARNED (A.3.1.1.14)

None.

A.4.1.1.14 Compatibility - engine and air vehicle.

The requirement of 3.1.1.14 shall be verified by: ____.

VERIFICATION RATIONALE (A.4.1.1.14)

The requirement needs to be verified by test and demonstration since an analysis may not reveal all possible problems.

VERIFICATION GUIDANCE (A.4.1.1.14)

Past verification methods have included analysis, demonstrations and tests. Analyses may be used for conditions that may not be adequately demonstrated or tested.

This requirement should be verified by test and demonstration throughout the engine development program.

VERIFICATION LESSONS LEARNED (A.4.1.1.14)

None.

A.3.1.1.15 Control and external components list.

The control, external components, and component functional subsystems of the engine, which require qualification testing separate from the engine, shall be listed, see 4.11.2.

REQUIEMENT RATIONALE (A.3.1.1.15)

Listings of all Controls and Externals (C&E) in the model specification is required to enable the contractor and subcontractors to develop the appropriate critical item development specifications required for adequate design, development, verification, and testing of engine components.

REQUIREMENT GUIDANCE (A.3.1.1.15)

Background:

All control and external components should be listed in the specification; examples include fuel pumps, controls, actuators, valves, electrical component, EHM/PHM or BIT hardware.

This list is used in table XXX that requires qualification tests for each control and external.

REQUIREMENT LESSONS LEARNED (A.3.1.1.15)

None.

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A.4.1.1.15 Control and external components list.

The requirement of 3.1.1.15 shall be verified by: ____.

VERIFICATION RATIONALE (A.4.1.1.15)

Control and external components require qualification testing to establish their integrity prior to commencement of engine qualification testing.

VERIFICATION GUIDANCE (A.4.1.1.15)

Past verification methods have included analysis of the components and inspection of the list.

The actual test requirements for the controls and externals listed in 3.1.1.15 are specified in 4.11.2.

VERIFICATION LESSONS LEARNED (A.4.1.1.15)

None.

A.3.1.2 Physical characteristics.

A.4.1.2 Physical characteristics.

A.3.1.2.1 Dry weight of complete engine.

The weight(s) for the engine shall not exceed:

CONFIGURATION	WEIGHT KG (LBS)	CG
complete engine, dry		
engine exhaust system (if a separate system), dry		
residual fluids (draining attitude ___° from horizontal)		
operating fluids		
items part of the engine parts list not mounted on the engine (identify and list each item)		

REQUIREMENT RATIONALE (A.3.1.2.1)

The dry weight of the engine, the residual fluid weight, the weight of fluids within the engine during operation must be established for air system.

REQUIREMENT GUIDANCE (A.3.1.2.1)

The following should be used to tailor the specification paragraph:

The contractor should specify the weights.

The empty weight of the system may be determined per SAWE RP7.

The reason to specify the attitude angle for draining the residual fluids in the engine is that the weight of residual fluids will vary depending on the attitude of the engine when it is drained. For Engine H this is primarily due to the complex shape of the oil tank. This does not imply that the engine needs to be in a certain attitude for maintenance. The engine does not need to be vertical for maintenance actions. The angle is there to provide a baseline value for weight of residual fluids with the understanding that if it is drained at a different attitude, the value may be different.

For cabling connecting off engine mounted components to the engine, the engine manufacturer should provide the weight and the weight per unit length kg/m (lb_m/ft).

If the FADEC is to be mounted on the aircraft and the location hasn't been determined, the engine contractor may specify the electrical harness weight as kg/m (lb_m/ft)

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Background:

Weight should be specified separately for engine and exhaust if the exhaust system is heavy or otherwise unconventional. The associated weights should be identified individually to separate basic turbo machinery from other components.

Examples of engine components not mounted on the engine include life recorders, electronic control units, and vectoring nozzles.

Items which are not included in the aircraft installed weight (i.e.; cover plates, dust covers, etc.) should not be included in the dry weight.

Residual fluids include fuel, oil, hydraulic fluid, etc.

REQUIREMENT LESSONS LEARNED (A.3.1.2.1)

None.

A.4.1.2.1 Dry weight of complete engine.

The requirement of 3.1.2.1 shall be verified by test.

VERIFICATION RATIONALE (A.4.1.2.1)

The dry weight of the complete engine must be established to be consistent with the aircraft requirements. Accurate determination of the weight of residual fluids in the engine is required for aircraft weight and balance calculations. Accurate determination of the total weight of fluids in the operating engine (i.e.,; with a full oil tank, prime oil, and fuel systems, etc.) is required for aircraft weight and balance computations. Determination of the total weight of the engine must be accomplished for accurate air vehicle weight and balance calculations.

VERIFICATION GUIDANCE (A.4.1.2.1)

The following should be used to tailor the specification paragraph:

An estimate of the engine weight should be maintained and updated by the contractor during development.

Each qualification and production engine should be weighed, using the same procedure, to ensure consistency. The Using Service should make sure that weighing equipment has been calibrated within the last six months and complies with the accuracy of 4.12.4.2.

The weight of all residual fluids (fuel, oil, hydraulic, etc.) remaining in the engine after operation and drainage should be determined while the engine is in a single specified attitude of the main rotor axis relative to the level plane. The residual fluid weight should be determined by calculating the difference between the weight of an engine that has been operated and drained and the dry weight of the same engine before being serviced with fluids. Fluids should only be drained through lines or other connections specified for normal maintenance practices.

The weight of fluids in the operating engine should be determined by calculating the difference between the weight of an operational engine and the dry weight of the same engine before servicing with fluids.

VERIFICATION LESSONS LEARNED (A.4.1.2.1)

Weighing each qualification and production engine will provide data to determine engine-to-engine variation.

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A.3.1.2.2 Mass moment of inertia of complete engine.

The maximum effective mass moment of inertia kg-m^2 (lbm - ft^2) of the complete operational engine about three mutually perpendicular axes with the origin at the center of gravity, shall be specified in _____. The maximum effective mass moment of inertia kg-m^2 (lbm - ft^2) about the resultant rotational axis of each engine rotor, together with the direction of rotation when viewed looking forward from the exhaust exit, shall also be specified in _____.

REQUIREMENT RATIONALE (A.3.1.2.2)

The maximum mass moment of inertia of the complete engine about three mutually perpendicular axes is required for aircraft structural design.

REQUIREMENT GUIDANCE (A.3.1.2.2)

For all blanks, the information can be shown in the ICD, or for programs that do not have an ICD, the information can be put into the specification paragraph.

Background:

The mass moment of inertia of the engine about its center of gravity should be based on the summation of the dry weight of the complete engine, and the weight of operating fluids, and should be specified about the engine pitch, roll, and yaw axes. For turbofan and turbojet applications, the maximum effective mass moment of inertia about the rotational axis of each rotor, together with the direction of rotation when viewed from the exhaust end looking forward, should be specified.

For turboshaft applications, the maximum effective mass moment of inertia of each engine rotor system about the resultant rotational axis, together with the effective direction of the inertia and the direction and location of the resultant axis, should be specified. For engines with geared rotor systems, the shaft to which all inertias of each rotor system have been algebraically referred, should be specified. The maximum effective mass moment of inertia of the complete power output system, including the engine reduction gear train (referred to the output shaft speed), should be specified.

REQUIREMENT LESSONS LEARNED (A.3.1.2.2)

None.

A.4.1.2.2 Mass moment of inertia of complete engine.

The requirement of 3.1.2.2 shall be verified by: ____.

VERIFICATION RATIONALE (A.4.1.2.2)

The mass moments of inertia must be accurately determined so that it is available for use in aircraft design considerations.

VERIFICATION GUIDANCE (A.4.1.2.2)

Past verification methods have included analysis and tests.

Background:

The mass moments of inertia should be determined at each of the milestones. The method used should be the most accurate means available, consistent with the size and weight of the engine. The accuracy of the method used should be determined and specified.

The mass moments of inertia for the complete operational engine should be used for determining forces required as a result of rotational inertia and the moments of inertia of the rotors should be used to determine forces required as a result of gyroscopic moments.

VERIFICATION LESSONS LEARNED (A.4.1.2.2)

None.

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A.3.1.3 Materials, processes and parts.

The materials, coatings, and processes used in the design and manufacture of the complete engine shall be chosen, applied and utilized such that the engine shall meet the requirements of this specification.

REQUIREMENT RATIONALE (A.3.1.3)

The materials and processes used during the design and manufacture of the engine must be sufficient to ensure the engine will meet all specification requirements.

REQUIREMENT GUIDANCE (A.3.1.3)

The use of magnesium should be avoided at all engine locations. Silver should not be used without approval from the Using Service. The temperature environment has a major impact on whether silver will be compatible on various locations of the engine. Engine manufacturers have criteria for deciding when and where to use materials in engine designs. The use of reactive materials or potentially reactive materials should be reviewed by the Using Service. Materials that have been used successfully on Air Force or Army engines may not work well on Navy engines due to the different environment.

Background:

Copper should be prevented from coming in contact with fuel or oil during engine operation. Copper acts as a catalyst, adversely affecting the thermal stability and increases the rate of deterioration in these fluids. Dissimilar metals should not contact one another to avoid galvanic corrosion. MIL-STD-889 can be used for guidance when using dissimilar metals. The use of brass and cadmium plating is not allowed in the fuel system due to possible corrosion of parts which contact fuel.

The engine contractor should specify how, when, and where adhesives and sealants will be applied to prevent sealing and adhesive failures. Adhesives and sealants should not be used as a locking feature. The following document provides guidance and requirements for sealing and retaining compounds: MIL-S-22473.

Contractors should avoid the use of proprietary adhesive and sealant compounds. Adhesives and sealants should be in compliance with current federal environmental hazardous materials regulations. Provided there is no current industry or government standard, the Using Service may utilize the following requirements in the main paragraph if sealing or locking compounds are used on threads. The use of retaining, sealing, and locking compounds applied to the threads or bearing surfaces of aircraft threaded components should:

a. Not be used on threaded components except those that are used in a permanent or semi-permanent installation.

b. Be applied to the threads and bearing surfaces before assembly of components.

c. Not be used as a primary locking or retaining method in those applications where the failure of the compound to seal, retain, or lock would endanger the safety of personnel or would render the equipment inoperative or cause its destruction.

d. Not be used in structural and control system applications except as an auxiliary locking or retaining method.

e. Not be used on threaded components for which a torque or preload value is specified for assembly, unless it is feasible to discard and replace these by new components. Threaded fasteners previously treated with such compounds should not be reused. These fasteners should be replaced by new fasteners newly coated.

f. The compound should not accelerate galvanic or stress corrosion of cadmium plating, corrosion resistance steels, aluminum alloys, or titanium alloys in naval aircraft environments.

(a) The compound should not cause degradation of the finishes prescribed in MIL-F-7179 or other approved finish materials.

(b) The compound should not cause degradation or reduce adhesion of tank sealant materials.

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(c) The compound should not accelerate or cause degradation of aircraft fuels, lubricants, and hydraulic fluids, which may be in contact with the compound.

(d) The compound should be inert in the presence of aircraft fuels, lubricant, and hydraulic fluids.

(e) The compound should be used only where no other satisfactory means exist.

Polychloroprene materials (e.g., neoprene) should not be used in engine parts that are in direct contact with fuels or lubricants during engine operation.

Elastomeric material should have unlimited shelf life and must not be allowed to deteriorate, to permit voids to form in the seals and, in turn allow the fluid to leak past the seal. “Unlimited shelf life” for elastomers is equivalent to the term “non-age sensitive” elastomers. SAE AS 568 should be used for guidance for dimensions and tolerances for nonmetallic “O” rings, seals, and packings. “O” rings, seals, and packings should be compatible with fuels and lubricants and neither the seal nor packing or the fuel and lubricant should be affected. Particular attention should be given to “O” rings used for high temperature applications since they may be susceptible to excessive leakage when exposed to low temperatures. An example of use of a non-elastomeric material would be polytetrafluoroethylene “O” ring in transmissions, variable geometry actuators, gearboxes, etc.

The engine contractor should submit a list of all protective treatments. Past experience with coatings and processes should be used when existing materials are proposed. Metallurgists are continually making small changes to materials, coatings, and processes.

Where a metal and nonmetal are coupled, in presence of 3.5% salt water, the galvanic voltage formed should not exceed 0.15 volts without the interface couple being protected from corrosion. The cathodic material should be top coated except where it needs to be bare to perform an engine function. Acid rain equivalent to pH 3.5 using sulfuric acid solutions should be used to qualify the engine components for resistance to rain and condensed humidity (diurnal cycling through the dew point). Processes and materials used to maintain corrosion prevention should be used and verified for their technical capability. The design should be maintained using the Joint NAVAIR 16-1-540 and Air Force TO 1-1-689, except where the design includes unique materials or design detail that cannot be serviced by the manual. In this instance, a system should be maintained with a unique corrosion prevention technical order with the procedures and materials validated and verified to the system requirement prior to approval for use on the system production design hardware.

If used on the engine, nonmetallic hose (MIL-H-27267) and hose assemblies (MIL-H-25579) should be capable of resisting failure due to such items as stress, over-tightening torque, vacuum, temperature, and manufacturing tolerances.

REQUIREMENT LESSONS LEARNED (A.3.1.3)

Magnesium is restricted because it is extremely susceptible to corrosion, especially in a marine environment. Even a small “pin hole” break in the protective coating will allow corrosion to take place under the remainder of the protective coating. On visits (1978-1979) to the NADEPs, it was observed that Engine Q magnesium accessory gearboxes have been scrapped due to extensive corrosion. An Air Force engine with a magnesium gearbox experienced galvanic corrosion due to dissimilar metals being used at the front frame mount hole and tower shaft interface.

Engine AI was originally developed with a magnesium propeller gearbox. Major corrosion problems were incurred with the gearbox (after the coating was scratched during normal maintenance), and the magnesium was later replaced by aluminum. Other engines were developed with a magnesium gearbox and through the use of proper coatings and maintenance procedures have provided good service.

Copper and cadmium can be eliminated from the fuel and oil system design without undue hardship. MIL-STD-889, Appendix A contains some methods of combining dissimilar metals.

In a specific turbojet engine model a compressor case and compressor blades were made of titanium. When the blades failed during excitation, they produced pieces that rubbed against the case and caused

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titanium fires. If material selection for these blades and case excluded titanium, then the fires could have possibly been avoided.

Adhesives and sealants have complicated maintenance and repair (i.e., having proper type available and using correct age control). They should be minimized. An augmentor liner delta P problem was caused by excessive use of room temperature curing silicone based sealant. In the past silicone was used in the fan section of the engine to seal gaps at the blade platforms. This silicone broke off and plugged cooling holes in the afterburner liner.

Hard tubing can usually be substituted for nonmetallic hose when any problems occur.

A.4.1.3 Materials, processes, and parts.

The requirements of 3.1.3 shall be verified by:

VERIFICATION RATIONALE (A.4.1.3)

Material properties must be established by using industry-accepted practices.

VERIFICATION GUIDANCE (A.4.1.3)

Past verification methods have included analysis, demonstrations, and tests.

Background:

Specific allowances used for design should be statistically derived per MIL-HDBK-5 or other applicable documents. Special “allowances” which represent candidate or final solutions should be developed by test per the test plan established under the overall integrity program for the specific application. The material property characterization plan should be referenced. The contractor, in developing this plan, should review existing data on candidate materials and processes and develop a test plan to generate the data to characterize fully the material for the intended application. The test plan should contain a materials listing and identification of properties that should be characterized for each material. Material variability should be examined and means of insuring minimum properties in critical parts/critical safety items should be developed per integrity program guidance. The Using Service should require this plan on the CDRL.

Exposure to extreme temperatures is a good method to evaluate “O” rings, seals, and packings.

Adhesives and sealants should be evaluated for corrosion, engine fluid loss, loosening of fasteners, and integrity of joined structures.

The engine contractor usually verifies all coatings and processes by strip samples or component tests. This requirement should be verified in 4.3.1.3 (Corrosive atmosphere test).

VERIFICATION LESSONS LEARNED (A.4.1.3)

“O” rings have failed at extreme cold temperatures.

Corrosion has been a major problem and has resulted in huge man-hour expenditures for repair. Corrosion can be a slow and subtle process that may take years to appear during actual field use. It may appear only in certain localities or result from a particular mission. In the past, contractors have attempted to satisfy this requirement by material coupon test (e.g., a sample material is immersed in a corrosive solution and a salt spray environment), however the material is not exposed to the thermal environment normally seen in an operating engine. Coupon tests have not been considered acceptable substitutes for the corrosive atmosphere test.

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A.3.1.3.1 Hazardous materials.
The use of any banned material, as listed in the specification, shall not be permitted in the construction, operation, maintenance, storage, transportation or disposal of engine delivered hardware.

The use of any highly restricted materials, as listed in the specification, shall be subject to procuring activity approval prior to use in the construction, operation, maintenance, storage, transportation or disposal of engine delivered hardware.

REQUIREMENT RATIONALE (A.3.1.3.1)

The use of banned or highly restricted materials could pose health or safety risks to personnel. Many banned materials are prohibited by law.

REQUIREMENT GUIDANCE (A.3.1.3.1)

Banned and highly restricted materials should be listed in the specification. They can be listed directly in the requirements paragraph or in a referenced table.

Examples of banned materials are:

Material	Chemical Abstract Service Number	Material	Chemical Abstract Service Number
Asbestos	1332-21-4	Amosite	12172-73-5
Actinolite	77536-66-4	Anthophyllite	77536-66-4
Chrysotile	12001-29-5	Crocidolite	12001-28-4
Tremolite	77536-68-6	2,4-Dichlorophenoxyacetic Acid	94-75-7
2,4,5-Trichlorophenoxyacetic Acid	93-76-5	Iron Carbonyl	13463-40-6
4,4-Methylenedianiline (MDA)	101-77-9	Nickel Carbonyl	13463-39-3
Pentachlorophenol	87-86-5	Picric Acid	88-89-1
Polychlorinated biphenyls (PCBs)	1336-36-3	Tetrachloromethane	56-23-5
1,1,1-Trichloroethane (TCA)	71-55-6	Trichloroethylene (TCE)	79-01-6
Class I Ozone Depleting Compounds			

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Examples of highly-restricted materials are:

Material	Chemical Abstract Service Number	Material	Chemical Abstract Service Number
Anhydrous Ammonia	7664-41-7	Benzene	71-43-2
Beryllium and Compounds	7440-41-7	Cadmium and Compounds	7440-43-9
Chromium Compounds[1]	7440-47-3	Hydrazine	302-01-2
Inorganic Cyanide	57-12-5	Methyl ethyl ketone (MEK)	78-93-3
Mercury (attrit)	7439-97-6	Methanol	67-56-1
Methylene chloride	75-09-2	Methyl isobutyl ketone (MIBK)	108-10-1
Nickel and Compounds[1]	7440-02-0	Perchloroethylene	127-18-4
Nitric Acid[2]	7697-37-2	Phenol	108-95-2
Toluene	108-88-3	Xylene	1330-20-7
o-Xylene	95-47-6	m-Xylene	108-38-3
p-Xylene	106-42-3	Formaldehyde	50-00-0
>70% Hydrofluoric Acid	7664-39-3	Sulfur Dioxide Gas	7446-09-5
Chlorine Gas	7782-50-5	Lead Compounds except in solder	7439-92-1
Class II Ozone Depleting Compounds		Corrosive Materials[2]	
Oxidizers[2]			
Explosive Materials[2]		Radioactive Materials[2]	

Note 1: Except in alloy form

Note 2: IAW DOT Regulation 49 CFR 172

REQUIREMENT LESSONS LEARNED (A.3.1.3.1)

Both engines AN and AO have adopted the banned and highly restricted requirements as shown above.

A.4.1.3.1 Hazardous materials.

The requirements of 3.1.3.1 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.1.3.1)

Verification of the requirements are necessary to ensure health and safety of personnel as well as adherence to law.

VERIFICATION GUIDANCE (A.4.1.3.1)

Verification may be accomplished by inspection, demonstration, and test.

VERIFICATION LESSONS LEARNED (A.4.1.3.1)

None.

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A.3.1.4 Fasteners.
The loss of a fastener shall not affect safety of flight or ability to control the engine.

REQUIREMENT RATIONALE (A.3.1.4)

Threaded fasteners and other connections must be selected to meet strength and standardization requirements for the application.

This requirement assures that the proper bolts are used to reduce the chance of bolt failure or loss (slipping out) at joints in control systems, at single attachments, and where loss of the bolt would affect safety of flight or ability to control the engine.

REQUIREMENT GUIDANCE (A.3.1.4)

NASM1515, Section 400 should be researched for propulsion system fastener requirements.

Requirements for self-retaining bolts are provided in NASM23964 and screw recess' requirements are provided in NASM33750 and NASM9006.

SAE AS 8879, Class 3A or 3B, (rolled) screw threads, have been used in past programs. SAE AS 8879 can be used for guidance on new programs as well. Rolled screw threads provide good grain flow of material and a very strong thread. They also provide a minimum screw thread root radius that reduces stress concentrations. The two screw thread forms help to insure the integrity of fastener systems and reduce the possibility of engine failures.

NASM33540, safety wiring and cotter pinning general practices, and NASM33588, self-locking nuts, prescribe methods to restrain the rotation and loosening of assembled fasteners. To reduce the chance of FOD, safety wire is not permitted for use without Using Service approval.

NASM33537 provides guidance and requirements for helical coils.

Information on external drives for wrenching elements can be found in SAE AS 870 for twelve point drives and MS 33787 for spline drives.

REQUIREMENT LESSONS LEARNED (A.3.1.4)

Silver plating used as an anti-seize agent on bolts in the turbine section reacted with the disk material at elevated temperatures resulting in the initiation of cracks in the disks. Tack welded captive nuts, if not properly aligned, have resulted in failures due to backing out of the fasteners. The use of silver plate on inner threads of bolts to retain torque of a nut at high temperatures encountered in component operation can lead to the brazement of nut to bolt. Consequently, in the 1970's some exhaust nozzle controls had to have their compressor bleed air motor bolts sawed off during overhaul.

Safety wire was once extensively used throughout the engine to secure fasteners. Poor housekeeping resulted in frequent FOD by stray pieces of safety wire.

A USAF Fighter engine used an adhesive as a retention mechanism on a fuel pump drive spline. The adhesive failed to retain the spline resulting in the loss of aircraft. The splines were mechanically staked to prevent this from happening again.

A.4.1.4 Fasteners.

The requirements of 3.1.4 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.1.4)

Analysis and engine testing is usually required to verify the adequacy and characteristics of fasteners.

VERIFICATION GUIDANCE (A.4.1.4)

Analysis should be used prior to the PDR and CDR milestones to define the proper design parameters such as; strength of the material to ensure the component won't come apart or become loose from the attachment point, redundancy to ensure safety of flight and finally proper material selection for the

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expected environment. A thorough analysis of the environment and anticipated usage may preclude problems later in the program.

Engine testing has been used to verify the requirement since a large spectrum of vibration modes will be imparted to the fastening devices during these tests. Shock and "G" loading requirements may require component tests before IFR. Analysis should be used for conditions that can't be ground or flight-tested.

Background:

Analysis of critical fasteners should be completed early in the design phase (PDR). All fasteners should be analyzed by CDR including a review of critical fasteners.

Fastener torque retention, strength, and removal characteristics should be verified in the engine endurance test. An inspection of each engine submitted for qualification testing should be accomplished to evaluate that all fasteners conform to the parts list. The condition of the fasteners at the completion of tests should be used to determine the acceptability of each for the application.

VERIFICATION LESSONS LEARNED (A.4.1.4)

None.

A.3.1.5 Nameplate and product marking.

Equipment, assemblies, modules, and parts shall be marked legibly and such that component removal is not necessary to read the data plate. The type and length of product identification characters on the data plate shall be compatible with the Using Service's maintenance tracking system. The color red shall not be used for marking unless adjacent non-red color marking is easily readable in red light conditions.

REQUIREMENT RATIONALE (A.3.1.5)

All engine components require marking or identification. The maximum number of digits or characters must be limited due to computer limitations used for inventory and logistical control.

REQUIREMENT GUIDANCE (A.3.1.5)

ASME-Y 14.100, ASME-Y 14.24, ASME-Y 14.35M, ASME-14.34M, and MIL-STD-130 should be used for guidance.

The markings should be visible through normal access openings on the engine and aircraft. The Using Services normally only allows color red marking when adjacent non-red color marking is easily readable in red light conditions. Red lettering should not be used for markings since Navy aircraft may be serviced where red lights are the main source of lighting and the warnings and caution notices would not be visible. Red lighting is used at sea so that the personnel can maintain night vision during fleet operations at night.

Engine part numbers should be limited to a total of 15 digits or characters to be consistent with the capability of the Using Service parts life tracking system and the asset management system.

Modern engine programs are moving towards the use of 2D markings. ATA 2000 marking capability can be used for reference.

REQUIREMENT LESSONS LEARNED (A.3.1.5)

Components have had to be removed to examine nameplates and markings during inspection and maintenance due to the poor placement of those markings. The limitation on number of characters in the part number was first imposed by NAVAIR during the LAMPS MK-3 program.

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A.4.1.5 Nameplate and product marking.
The requirement of 3.1.5 shall be evaluated by inspection.

VERIFICATION RATIONALE (A.4.1.5)

Verification of property marking or identification can only be accomplished by inspection.

VERIFICATION GUIDANCE (A.4.1.5)

Verification should be made on completeness, accuracy, and visibility of the information supplied.

VERIFICATION LESSONS LEARNED (A.4.1.5)

None.

A.3.1.5.1 Engine data plate marking.

The engine data plate shall include: manufacturer's identification, engine serial number, purchase order or contract number and engine model designation.

REQUIREMENT RATIONALE (A.3.1.5.1)

All U.S. military property requires proper identification.

REQUIREMENT GUIDANCE (A.3.1.5.1)

Engine serial numbers should be per MIL-HDBK-1559. The Using Service will assign the engine model designation using MIL-HDBK-1812 for guidance.

REQUIREMENT LESSONS LEARNED (A.3.1.5.1)

None.

A.4.1.5.1 Engine data plate marking.

The requirement of 3.1.5.1 shall be verified by inspection.

VERIFICATION RATIONALE (A.4.1.5.1)

The requirement can only be verified by inspection.

VERIFICATION GUIDANCE (A.4.1.5.1)

None.

VERIFICATION LESSONS LEARNED (A.4.1.5.1)

None.

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A.3.1.5.2 Critical parts/critical safety items identification and tracking.

The engine life critical parts/critical safety items and subassemblies shall be identified by serial numbers marked legibly on the part. The parts shall have multiple, non-wearing reference surfaces and have a space designated for marking the number of cycles and time accumulated between each overhaul period.

Engine critical parts shall be identified and managed through a Critical Item Management Plan.

REQUIREMENT RATIONALE (A.3.1.5.2)

Identification of life critical parts/critical safety items is necessary for adequate tracking during the maintenance and overhaul process.

REQUIREMENT GUIDANCE (A.3.1.5.2)

The critical parts' or assemblies' serial numbers should be legibly and visibly marked on their surfaces. Provisions should be made during the design process for adequate identification of life limited fracture and fatigue critical parts/critical safety items, and for components that have established time between overhauls (TBOs).

Guidance on Critical Item Management Plans may be found in JSSG-2001, requirement 3.3.10.3, Critical Item Management Plan; NAVAIRINST 4200.25D; and JACG Guidance Procedures on Management of Aviation Critical Safety Items, dated 28 Aug 02.

REQUIREMENT LESSONS LEARNED (A.3.1.5.2)

None.

A.4.1.5.2 Critical parts/critical safety items identification and tracking.

The requirement of 3.1.5.2 shall be verified by inspection.

VERIFICATION RATIONALE (A.4.1.5.2)

The requirement must be verified by inspection.

VERIFICATION GUIDANCE (A.4.1.5.2)

Background:

Testing of every part of the engine attempts to simulate the real world environment of thermal cycling, mechanical wear, chemical cleaning, and other deteriorating effects the engine will encounter. Parts should be marked prior to test, and verification of durability of identification should be conducted upon completion of component and engine tests.

Guidance on Critical Item Management Plans may be found in JSSG-2001, 4.3.10.3, Critical Item Management Plan Verification; NAVAIRINST 4200.25D; and JACG Guidance Procedures on Management of Aviation Critical Safety Items, dated 28 Aug 02.

VERIFICATION LESSONS LEARNED (A.4.1.5.2)

In the past, parts identification has worn off in use.

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A.3.1.6 Transportability.

The engine and individual components shall be suitable for transportation as follows:

Features shall be provided to permit installation on and use of appropriate static and mobile ground equipment.

REQUIREMENT RATIONALE (A.3.1.6)

The engine, including individually shipped components, must be compatible with the logistic system equipment for ease of transportation.

REQUIREMENT GUIDANCE (A.3.1.6)

The following should be used to tailor the specification paragraph.

The main engine mounts should be used to support the engine during transportation. The engine should be restrained by the ground handling or main mounts for engine shipment. The engine should be suitable for transportation from factory to aircraft, delivery to transport aircraft, and delivery to overhaul facilities. The engine should be capable of withstanding the externally applied forces and vibrational loads induced during transportation without impact to structural life or engine performance. Land shipment of the engine should be by spring or air ride trucks, or by railroad, as determined by the Using Service. The method of protecting the engine should be specified.

This guidance also applies to individually shipped modules and components, including LRUs.

Background:

Engine transportability should give consideration to total logistic mobility. Turbine engine bearings, seals, mounts, and joint hard coats are subject to damage during transportation and can adversely affect both performance and durability. Damage during engine transportation has been a concern in the propulsion community for many years. The newer turbine engines, primarily dual rotor systems (some have three rotors) have compounded this concern because there are more bearings, seals, etc. Adding to the problem, the newer engines have tighter clearances designed to lower margins, and they possess more sophisticated coatings (some brittle) and material systems than in the past. Small engines have been shipped in reusable or throwaway foam, metal, or wood containers. A life cycle cost analysis should be conducted to determine the most economical method of shipping engines and whether peculiar shipping stands or isolators should be developed to ship engines between the engine and the air vehicle manufacturer and between the field and overhaul facility. The Army standard aircraft maintenance (engine transportation) trailer (NSN 1750-01-086-1653) conforms to data list 4920-EG-081.

The use of air ride semi-trailers has been recommended and used for highway shipment to minimize shipping concerns. However, there are major drawbacks to restricting movement of military engines by air ride transport. Some of these drawbacks are discussed below:

Air ride is more costly than spring ride truck transport.

Air ride is not always available, particularly outside of the Continental United States (CONUS); this results in an adverse effect on readiness when delays are caused by air ride truck availability.

Inspection of air ride equipment to assure it is in working order usually is not conducted.

Air ride characteristics may be worse than other suspension types depending on the position of the engine on the trailer bed and level of load applied.

The development and use of shock isolating adapters are preferred and recommended to alleviate the above concerns.

Defense Standard 00-3/3, “Design Guidance for the Transportability of Equipment”, dated 27 May 1985, is also available for reference at http://www.dstan.mod.uk/.

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REQUIREMENT LESSONS LEARNED (A.3.1.6)

Use of main mounts during shipment has reduced engine weight since additional ground handling mounts were not required. However, Engines A, C and D needed a ground-handling mount so they were compatible with the existing ground support equipment.

A.4.1.6 Transportability.
The requirements of 3.1.6 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.1.6)

Transportability needs to be verified to ensure the engine will not be damaged during shipment.

VERIFICATION GUIDANCE (A.4.1.6)

Past verification methods have included analyses and demonstrations.

If the verification method includes testing, consideration should be given to:

a. Methods of engine, module, and LRU protection

b. An instrumented package drop test

c. Air transportation.

MIL-STD-810 can be used for tailoring guidance on shock and vibration levels and methods of testing.

VERIFICATION LESSONS LEARNED (A.4.1.6)

Aircraft AI, European Distribution Aircraft, provides no active damping between the engine and the aircraft; therefore, the inherent damping characteristics of the aircraft and the anticipated landing, takeoff, and load spectrums should be verified.

A.3.1.7 Interchangeability.
All parts and modules having the same manufacturer's part number shall be functionally and dimensionally interchangeable. Matched parts or selective fits shall be permitted upon Using Service approval. Complete engines shall be interchangeable in multi-engine aircraft (i.e., right side to left side).

REQUIREMENT RATIONALE (A.3.1.7)

It is essential that parts with the same part number be interchangeable to reduce logistic support requirements, minimize maintenance and repair problems, and assure that engine performance and operability are not compromised.

REQUIREMENT GUIDANCE (A.3.1.7)

Background:

All parts having the same manufacturer's part number should be functionally and physically interchangeable and replaceable with each other with no effect on installation and performance, except that matched parts or selective fits may be permitted where necessary to meet other design requirements. The use of matched parts and selective fits should be held to a minimum. The replaceable units should be functionally and physically interchangeable without selective fittings. A list of non-interchangeable, matched, and selected fit parts should be identified and submitted to the Using Service during the engine development program to focus attention on these exceptions and force early, improved design changes. Due to logistical concerns, the Using Service needs to be aware of configurations (i.e., bleed and access port) peculiar to left and right side engine configurations which would preclude interchangeability of right and left engines and modules.

REQUIREMENT LESSONS LEARNED (A.3.1.7)

Functional and physical interchangeability of components and parts is a desired design goal. There have been many instances where engines designed with matched split cases had high scrapping costs. When unusual damage occurs to one-half of the case and the other half is not damaged, both halves must be

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condemned. Maintenance of matched sets throughout distribution and overhaul cycles is extremely costly.

Even though some parts manufactured by an alternate vendor are built to the same blueprint, they don't always function the same as the original part. On an augmented turbofan, an augmentor flame holder was manufactured by an alternate source vendor to the same blueprint tolerance as the original flame holder. However, the engine experienced flame front flashback when the alternate parts were used in the swirl-stabilized augmentor.

A.4.1.7 Interchangeability.
The requirements of 3.1.7 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.1.7)

Verification of the functional and physical interchangeability of engine parts is required.

VERIFICATION GUIDANCE (A.4.1.7)

Past verification methods have included inspections, analyses, demonstrations, and tests which can be piggybacked onto existing verification methods.

Background:

Interchangeability of engine parts can be verified during the course of routine assembly, maintenance, and testing. Documentation of this verification is recommended to insure complete coverage of all parts. Consideration should be given to parts that have been slightly modified to ensure that parts that are functionally not suitable cannot be installed on the engine. If an upgraded engine is developed, parts may be tested to be compatible with the existing fielded engine to reduce transition costs of fielding the upgraded engine. If this is done, you need to make sure that the new parts will meet the most rigorous requirements of either of the engine specifications.

VERIFICATION LESSONS LEARNED (A.4.1.7)

A quality assurance test was done during the 150-hour fighter accelerated mission test on Engine J to ensure interchangeability between Engine J with Engine J1 parts installed. This test resulted in component assurance qualification of Engine J1 parts in the Engine J model.

A.3.1.8 Safety.

A.4.1.8 Safety.

A.3.1.8.1 Flammable fluid systems.

All exterior lines, fittings and components which contain fuel shall be fire resistant (for at least five-minutes at a minimum flame temperature of 1093°C (2000°F)). The lubrication oil system exterior lines, fittings and components and hydraulic fluid system exterior lines, fittings and components shall be fireproof (for at least 15-minutes at a minimum flame temperature of 1093°C (2000°F)). During exposure to the above conditions, the lines and components shall be conveying fluids under the worst possible operational combination of fluid parameters (i.e.; flow rate, pressure and temperature) encountered throughout the complete environmental conditions and operating envelope of the engine.

REQUIREMENT RATIONALE (A.3.1.8.1)

Exterior lines, fittings, and components carrying flammable fluids must operate within a fire environment for a specified period of time. This requirement is to ensure that these parts will not contribute to the fire until corrective action can be taken or the crew can escape.

REQUIREMENT GUIDANCE (A.3.1.8.1)

The Using Service should determine if the aircraft fuel systems have air vehicle supplied shutoff valves at points prior to delivery to the engine. If the air vehicle does not, then the engine fuel system should be

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fireproofed. This means that if the air vehicle does not have a fuel shutoff valve, then the requirement is fireproof for 15-minutes instead of fire resistant for 5-minutes.

Modern air vehicles, such as Aircraft AH, have a higher degree of engine/aircraft control interaction level that could require additional requirements for continued operation in the presence of a 2000°F fire.

REQUIREMENT LESSONS LEARNED (A.3.1.8.1)

The most adverse conditions have occurred at idle when flow rates are low and heat removal capability is minimized.

A.4.1.8.1 Flammable fluid systems.

The requirements of 3.1.8.1 shall be verified as follows: _____.

VERIFICATION RATIONALE (A.4.1.8.1)

Verification of the fire resistance and fireproof requirements is essential for safety.

VERIFICATION GUIDANCE (A.4.1.8.1)

Past verification methods have included analyses, inspection and tests.

FAA Advisory Circular (AC) No: 20-135 has guidance on how to conduct a fire test.

The following procedure may be used as a verification method:

“Testing of the exterior lines, fittings, and components containing flammable fluids shall be accomplished by feeding the fluids at the minimum flow rate, highest fluid temperature, and highest system operating pressure encountered during engine idle power or windmilling conditions. The unit being tested shall be directly exposed to the flame. Component orientation and direction and location of flame impingement during testing shall be subject to Using Service approval. A visual inspection for leakage shall be made during and after the test. Sufficient instrumentation shall be included to continuously record critical parameters. Remotely controlled color video and stop-action photographic test facility equipment shall be provided to record and monitor all fire tests. The entire test shall be taped for test substantiation. Voice input on videotape shall be provided for clarification of test times and conditions. Thermocouple(s) shall be provided at the interface between flame pattern and surface to monitor and record flame temperature during testing to verify that flame temperature does not fall below pre-test burner calibration limits. To provide for accurate readings, thermocouple(s) shall be mounted directly within the flame pattern. SAE AS 1055 shall be used for conducting the fire test. SAE AS 4273 shall be used for fire testing components.

The test shall be considered successful, if there is no leakage during and five minutes after removal of the flame. Analysis may be conducted via computer modeling to verify the requirement with approval from the Using Service.”

VERIFICATION LESSONS LEARNED (A.4.1.8.1)

Component seals, covers, and fittings are primary areas of concern for leakage. Component orientation with respect to the flame pattern during test should simulate realistic engine compartment fire conditions.

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A.3.1.8.2 Fire prevention.

Where engine surface area temperatures are sufficient to ignite flammable fluid that could impinge on the engine, these areas shall be specifically identified in _____. If a fire shield is required, it shall be identified in _____.

REQUIREMENT RATIONALE (A.3.1.8.2)

The purpose of this requirement is to prevent external leakage of combustible liquids from reaching the parts of the engine hot enough to ignite them.

REQUIREMENT GUIDANCE (A.3.1.8.2)

The information can be shown in the ICD, or for programs that do not have an ICD, the information can be put into the specification paragraph.

Fire shields have been used for fire prevention in past designs.

REQUIREMENT LESSONS LEARNED (A.3.1.8.2)

Openings between the engine bay and the engine variable exhaust nozzle (VEN) cavity allowed fuel from an engine leak to contact the hot nozzle surface resulting in an engine bay fire problem on a Navy fighter aircraft. The ignition source was removed by adding boots to the VEN actuators to seal the openings between the engine bay and the VEN cavity.

A.4.1.8.2 Fire prevention.

The requirements of 3.1.8.2 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.1.8.2)

Verification of this requirement is essential for safety.

VERIFICATION GUIDANCE (A.4.1.8.2)

Past verification methods have included analyses.

VERIFICATION LESSONS LEARNED (A.4.1.8.2)

None.

A.3.1.8.3 Explosion-proof.

All electrical components (except igniter plug electrodes and devices intended to ignite fuel air mixtures) shall not ignite any explosive mixture surrounding the equipment.

REQUIREMENT RATIONALE (A.3.1.8.3)

All electrical components must be explosion proof, since an explosive atmosphere may exist near the component.

REQUIREMENT GUIDANCE (A.3.1.8.3)

Background:

Electrical system components should be designed to be non-spark producing or should be sealed sufficiently to prevent entrance of combustible materials.

REQUIREMENT LESSONS LEARNED (A.3.1.8.3)

During the development of the USAF Turbine Engine Monitoring System (TEMS), it was determined some of the connectors would be mated and unmated in a hazardous environment with power applied between the mating halves. Any spark of sufficient energy from this connector could create an explosion. It was

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determined that electrical connectors should not allow a spark to propagate outside of its shell of enclosure into the hazardous environment.

A.4.1.8.3 Explosion-proof.

The requirements of 3.1.8.3 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.1.8.3)

The explosion-proof properties of the electrical components must be fully verified.

VERIFICATION GUIDANCE (A.4.1.8.3)

MIL-STD-810 contains an explosion-proof procedure that can be used for guidance. If an explosion-proof electrical connector is used, then verification by similarity may be considered in lieu of component testing prior to initial flight release

Background:

All electrical components, including electrical connectors, should be subjected to explosion proof testing per MIL-STD-810 or Radio Technical Commission for Aeronautics, Environmental Conditions and Test Procedures for Airborne Equipment (RTCA/DO-160). Hermetically sealed components and passive devices such as thermocouples and chip detectors with no make and break contacts do not need to be tested.

The following tailoring is recommended for DO-160 or MIL-STD-810:

> "During the test, components shall have maximum input voltage applied to them and shall be operated continuously at their maximum loads. During each altitude condition, all make and break contacts shall be operated at least ten times. An over voltage power supply transient shall be applied to the components at least four times during each altitude condition. At least four of these power supply transients shall be applied during operation of make and break contacts. Power supply transients shall consist of the application of an over voltage for the times specified in MIL-STD-704. The four points selected for performance of an over voltage shall include the following 180, 160, 140, and 125 volts AC RMS line to neutral or 50, 43, 37, and 31.5 volts dc. Ignition components or systems shall be operated continuously. Electrodes of spark igniters shall be mounted in such a manner that the explosive vapor in the test chamber shall not be contacted. Electrically self-sufficient ignition systems shall be exempted from the application of power supply transients. Failure criteria shall be as defined in MIL-STD-810, Method 511, "Explosive Atmosphere". Test altitudes shall be every 10,000 above 20,000 ft up to the absolute altitude and every 5000-ft below 20,000 ft. The mixture shall be verified to be an explosive mixture at each altitude during the test procedure."

VERIFICATION LESSONS LEARNED (A.4.1.8.3)

None.

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A.3.1.8.4 Combustible fluid drains.

Provisions shall be made to prevent entrapment of fuel after each false start and for preventing excess combustible fluids from entering the combustion areas after shutdown with the engine in a range of positions from _(a)_ degrees nose up to _(a)_ degrees nose down. For turboshaft engines, no pilot or maintenance action shall be required to clear the engine of fuel after each false start or following engine shutdown.

For engine applications in V/STOL, STOVL aircraft requiring engine attitude changes, the drain limits of 3.1.1.8 shall not be exceeded for combustible fluids after engine shutdown with the engine in a range of positions from __(b)___ nose up to __(b)___ nose down. Provisions shall also be made for clearing all vent areas and other pockets or compartments where combustible fluids may collect during or subsequent to operation of the engine. The engine shall be designed so that combustible fluids cannot enter the combustion area when the engine is in, or turned to, a vertical position for maintenance.

The fluid drain system shall be configured to collect and route fuel and oil as specified in the __(c)____.

REQUIREMENT RATIONALE (A.3.1.8.4)

This requirement assures that combustible fluids do not remain in the combustion area after a false start. This reduces the possibility of a “hot” start on the next attempt. The various engine positions are specified to allow for different engine mounting positions for different engine applications. The no fuel leakage requirements from the engine in the vertical position are to eliminate the possible spillage of fluid on maintenance personnel or in the working area.

REQUIREMENT GUIDANCE (A.3.1.8.4)

The contractor should specify the range of engine positions where safe and reliable operation of the combustible drains will occur. For blank (a), values of 15 degrees nose up and 20 degrees nose down have been used on other applications.

The fluid drain system should be compatible with the aircraft fluid collection system. For blank (b), the contractor should provide the values for the V/STOL application.

For blank (c), the interfaces can be shown in the ICD, or for programs that do not have an ICD, the interfaces can be put into the specification paragraph.

REQUIREMENT LESSONS LEARNED (A.3.1.8.4)

None.

A.4.1.8.4 Combustible fluid drains.

The requirements of 3.1.8.4 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.1.8.4)

Combustible fluid drain requirements must be verified.

VERIFICATION GUIDANCE (A.4.1.8.4)

Past verification methods have included analysis, inspections and tests. The test should be performed as part of the attitude test.

VERIFICATION LESSONS LEARNED (A.4.1.8.4)

Engine testing has identified leakage rates, from drains, in the test cell.

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A.3.1.8.5 Air and gas leakage.

There shall be no air and gas leakage of sufficient temperature and concentrated impingement to present a safety hazard or affect installation requirements or adversely affect aircraft nacelle equipment.

REQUIREMENT RATIONALE (A.3.1.8.5)

The leakage requirement provides interface information and limits leakage to a tolerable level.

REQUIREMENT GUIDANCE (A.3.1.8.5)

Leakage of air and gas can interfere with engine bay cooling and cause safety problems.

REQUIREMENT LESSONS LEARNED (A.3.1.8.5)

Hot gas leakage has caused engine control failures in engine tests. Reverse flow of hot gases into helicopter engine bays or nacelles has caused failure of electrical and fluid lines and components, precipitating fires.

A.4.1.8.5 Air and gas leakage.

The requirements of 3.1.8.5 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.1.8.5)

Air and gas leakage must be verified to ensure no injury to personnel nor damage to nacelle equipment will occur.

VERIFICATION GUIDANCE (A.4.1.8.5)

Past verification methods have included analysis and tests.

Typically a performance analysis is performed to account for air leakage. Data uncertainty associated with airflow measurements makes it difficult to determine the leakage flow using testing as the verification method.

VERIFICATION LESSONS LEARNED (A.4.1.8.5)

None.

A.3.1.8.6 Ground safety.

Warning notices shall be provided, where applicable, for high voltage ignition systems and other high voltage electrical sources, radioactive devices, explosive devices, and high-powered optical sources.

REQUIREMENT RATIONALE (A.3.1.8.6)

Standard and uniform safety considerations must be used for engine safety.

REQUIREMENT GUIDANCE (A.3.1.8.6)

Ground maintenance personnel use warning notices to avoid safety problems during engine bay area maintenance procedures. For engines that may require service while under red light conditions, the Using Service should ensure that all warning notices can be read under red light conditions.

REQUIREMENT LESSONS LEARNED (A.3.1.8.6)

Warning notices have given maintenance personnel needed warnings that they need to increase safety precautions.

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A.4.1.8.6 Ground safety.

The requirements of 3.1.8.6 shall be verified by analysis and inspection.

VERIFICATION RATIONALE (A.4.1.8.6)

Ground safety warning notices can only be verified by analysis and inspection.

VERIFICATION GUIDANCE (A.4.1.8.6)

An analysis is needed to determine if and where notices and warnings are needed, and an inspection that the notices are properly located.

VERIFICATION LESSONS LEARNED (A.4.1.8.6)

None.

A.3.1.8.7 Jet wake.

Jet wake temperature and velocity profiles at sea level static standard day conditions when operating at idle, intermediate, and maximum shall be as shown on figure 4. Areas hazardous to personnel and equipment shall be appropriately defined and marked on figure 4.

REQUIREMENT RATIONALE (A.3.1.8.7)

Jet wake diagrams aid the aircraft manufacturer in initial aircraft design and in development of safe ground and shipboard operating procedures.

It is important to ensure that the engine and air vehicle is compatible with ground or shipboard equipment such as the Jet Blast Deflector (JBD).

REQUIREMENT GUIDANCE (A.3.1.8.7)

Background:

The jet wake should not become a hazard to equipment and personnel.

On shipboard operations where space is at a premium, the hazardous zones must be minimized. The heat and vibration loads on flight deck equipment and areas where the jet wake will impinge should be within the design limits of the equipment.

Issues regarding Jet Blast Deflector compatibility are addressed in JSSG-2001A (section 3.4.8).

REQUIREMENT LESSONS LEARNED (A.3.1.8.7)

It is important that the engine and air vehicle combination is compatible with existing equipment, and that the safety of ground personnel is not compromised. Some aircraft systems experienced jet wake impingement on portions of the air vehicle at certain flight conditions that affected air vehicle structural integrity. Jet wakes have caused personnel injury and weapons and ordnance and equipment damage.

A.4.1.8.7 Jet wake.

The requirements of 3.1.8.7 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.1.8.7)

Temperature and velocity characteristics must be verified for aircraft designers, ground equipment designers, and for development of uninstalled engine operating procedures.

VERIFICATION GUIDANCE (A.4.1.8.7)

Past verification methods have included analyses and tests. This test could be included with the noise test.

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For vectoring nozzles, the full range of thrust vectoring angles including partially modulated conditions should be tested.

A sea level static test is satisfactory for uninstalled engine data. Engineering judgments can be made about flight conditions from computer models.

VERIFICATION LESSONS LEARNED (A.4.1.8.7)

None.

A.3.1.9 Design control.

A.4.1.9 Design control.

A.3.1.9.1 Parts list.

The parts list for the engine including software version that successfully completes the designated engine milestone verifications shall constitute the approved parts list. Changes to the approved parts list shall require approval by the Using Service prior to parts incorporation. Prior to the initiation of IFR, FFR, ISR, and OCR, the contractor shall submit a proposed parts list, which shall include hardware and software configuration. Hardware configuration shall be identified by part number and associated drawings. Software configuration shall be identified by a specific source code listing and associated logic diagrams. The parts list for the engine configuration that successfully completes the IFR, FFR, ISR, and OCR verifications shall be the approved parts list for the respective engine model. The parts list shall be updated, as required, by changes incorporated in the parts list approved by the Using Service.

REQUIREMENT RATIONALE (A.3.1.9.1)

A parts list from engines that successfully completed qualification tests identifies the configuration released for flight and provides a baseline to measure improvements.

REQUIREMENT GUIDANCE (A.3.1.9.1)

Modern engine development programs using a performance-based approach or COTS may not require the engine manufacturer to provide a parts list. Commercial practice utilized during a COTS approach may not require approval from the services before changes are incorporated.

REQUIREMENT LESSONS LEARNED (A.3.1.9.1)

None.

A.4.1.9.1 Parts list.

The requirements of 3.1.9.1 shall be verified by inspection.

VERIFICATION RATIONALE (A.4.1.9.1)

The parts list must be verified to ensure an accurate description of the engine.

VERIFICATION GUIDANCE (A.4.1.9.1)

The Using Service should ensure the proper parts list is documented, and that the production parts list reflects the parts that have been qualified.

VERIFICATION LESSONS LEARNED (A.4.1.9.1)

None.

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A.3.1.9.2 Assembly of components and parts.

Equipment, parts, and components that are not structurally or functionally interchangeable with other parts on the same engine shall not be physically interchangeable unless approved by the Using Service. Parts and components shall be designed such that it is impossible to install them incorrectly (e.g.; backwards, upside down, reversed in an assembly, or installed in the wrong location in an assembly.) Connections located in close proximity to each other shall be made physically non-interchangeable.

REQUIREMENT RATIONALE (A.3.1.9.2)

System safety of the engine must not be compromised by the improper assembly or installation of equipment, parts, and component assemblies.

REQUIREMENT GUIDANCE (A.3.1.9.2)

Background:

Nonmetallic “O” ring seals and packings should remain on the part onto which they are installed when the associated mating part is not installed, and the associated mating part should be designed to prevent cutting or other damage to the “O” ring seals and packings.

REQUIREMENT LESSONS LEARNED (A.3.1.9.2)

Past experience has shown that catastrophic damage could result when the design of a hydraulic filter allows easy installation onto a connection intended for a lube filter. Lube or hydraulic system failure may result which in turn could cause engine failure. Another typical situation would be if a plug were installed in reverse causing reversed diode polarity, resulting in failed diodes or other polarity sensitive devices.

A.4.1.9.2 Assembly of components and parts.

The requirement of 3.1.9.2 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.1.9.2)

The non-interchangeability of parts must be verified.

VERIFICATION GUIDANCE (A.4.1.9.2)

Past verification methods have included demonstration.

VERIFICATION LESSONS LEARNED (A.4.1.9.2)

None.

A.3.1.9.3 Changes in vendors or fabrication process.

Changes in any vendor, fabrication process, or fabrication source for any component or part shall be subject to all requirements specified herein.

REQUIREMENT RATIONALE (A.3.1.9.3)

The Using Service must be assured that changes in vendors or fabrication processes will not result in the use of inferior parts.

REQUIREMENT GUIDANCE (A.3.1.9.3)

Background:

The contractor should prepare and submit a list of those parts, components, and assemblies that require substantiation tests to qualify an alternative vendor source or process. The specific test(s) required to qualify parts as engine parts should be defined and submitted with the list. The fabrication source and process of selected vendor components will be included in this list. The contractor should be responsible for insuring that all parts, components, and assemblies on the substantiation list comply with the qualified

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fabrication source and process, and that any changes to those sources or processes are effectively controlled.

The contractor should be responsible for performance of the substantiation test to establish satisfactory alternate vendors or fabrication sources or processes. A fabrication source is defined as the prime physical source producing the part, component, or assembly. Changes of fabrication location, such as to another plant of an individual vendor, should be construed as a change of fabrication source.

The Critical Item Management (CIM) Desktop Guide is a useful reference that implements the policies in NAVAIRINST 4200.25D by providing guidance to Navy personnel who participate in engineering reviews of new sources for procurement, repair or overhaul of naval aviation replenishment spare parts that meet the definition of critical safety items (CSI's) defined within.

Commercial practice utilized during a CLS or COTS approach may not require approval from the services of change in vendors or fabrication process.

REQUIREMENT LESSONS LEARNED (A.3.1.9.3)

Change in vendors or processes was a source of inferior parts being introduced into fleet equipment in the past.

A.4.1.9.3 Changes in vendors or fabrication process.

The requirements of 3.1.9.3 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.1.9.3)

The Using Service must monitor all changes in vendors or fabrication processes to ensure acceptable parts for the engine.

VERIFICATION GUIDANCE (A.4.1.9.3)

Past verification methods have included inspection of the part numbers to determine if a vendor or fabrication process change has been made, and testing of the component to ensure it meets all requirements. A change in vendor or process that effects the tests shown in table XXXI a or b may indicate a need to reevaluate the results of these tests.

VERIFICATION LESSONS LEARNED (A.4.1.9.3)

The Navy performed performance testing of alternate vendor Engine L combustor cans at NAWCAD Trenton’s Hot Gas Facility to verify combustor can airflow and combustor can fit checks were performed at NADEP Norfolk to verify form, fit, and function of alternate vendor parts that had vast improvements over the previous design parts such as structural improvements and Thermal Barrier Coatings in the late 1980’s.

A.3.1.9.4 Standardization.

Standardization principles, standard parts, materials, processes, tools, subsystems, and components shall be used to the maximum extent possible without compromise in design, performance, operability, or economic life of the engine. All parts, materials, and processes, whether or not identified as a Government, industry, or contractor standard shall be qualified for the intended use as a part of the qualification specified herein. Items already in the Government inventory shall be used to the maximum extent possible where suitable for the intended purpose. Variation in similar components or parts shall be held to the absolute minimum. Proprietary designs shall be kept to a minimum. Under conditions wherein economics of production conflict with standardization objectives, the Using Service shall be requested to select the component desired for use. Military and industry standards and parts developed for engines in air vehicles shall be used unless they are unsuitable for the intended purpose. Open Systems (see 3.1 and 6.3 for definition) shall be used to the maximum extent possible.

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REQUIREMENT RATIONALE (A.3.1.9.4)

Standardization is needed to reduce the number of parts in the logistic system, the number of man-hours required to maintain the system, and avoid significant increase in man-hours required to track assets in the system.

REQUIREMENT GUIDANCE (A.3.1.9.4)

A parts management process for acquisition parts management should be in place.

Background:

This requirement helps to assure that, to the maximum extent possible, standardization will be incorporated in the design of gas turbine engines even when economics of production conflict with standardization objectives but not to compromise engine design or performance.

REQUIREMENT LESSONS LEARNED (A.3.1.9.4)

In cast parts, contractors have requested changes in design to lower rejection rates. This has caused reduction in standardization and an increase in ILS tracking efforts but has decreased the cost of parts.

A.4.1.9.4 Standardization.

The requirements of 3.1.9.4 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.1.9.4)

Standardization principles, standard parts, materials, processes, tools, subsystems, and components must be verified.

VERIFICATION GUIDANCE (A.4.1.9.4)

Past verification methods have included inspections and demonstrations.

Verification of standardization is an ongoing process during development and into production.

VERIFICATION LESSONS LEARNED (A.4.1.9.4)

None.

A.3.2 Performance and operability.

A.4.2 Performance and operability.

A.3.2.1 Performance characteristics.

A.4.2.1 Performance characteristics.

A.3.2.1.1 Steady-state performance.

The minimum engine steady-state performance requirements shall be defined by the steady-state performance computer program (a) dated (a) , within the range of effects specified in 3.2.1.3. The specification computer program shall calculate minimum performance for both new engines and engines deteriorated for the time period specified in 3.2.1.4. Cycle calculations for the minimum engine computer program shall be based on an average engine with thrust or power and fuel flow related output adjusted to minimum engine performance as follows: (b) . The relationship between minimum and average performance shall be specified throughout the operating envelope of 3.2.2.1. Tables IIa, IIb, IIc, and table III shall provide reference performance points from the performance computer program for new and deteriorated engines (c) . In the event of conflict between the tables and the performance computer program, (d) shall take precedence.

For turboprop and turboshaft engines, the first stage turbine rotor inlet temperature for each rating established at (e) conditions shall be constant, for all atmospheric and Mach number conditions except when limited by fuel flow, compressor aerodynamics, or torque limits.

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REQUIREMENT RATIONALE (A.3.2.1.1)

All engines must meet or exceed minimum performance to ensure the capability of the air system to accomplish established missions. Performance computer programs provide a means of assessing performance without actual testing of engine hardware. Engine performance information is of critical importance in the development of the aircraft mission capability, and influences air vehicle inlet design, takeoff, landing, and various other operational characteristics to be generated and assessed by the air vehicle contractor. Engine performance under various conditions and loads (i.e.; altitude, Mach number, sea level, varying temperatures, bleed air extraction, power extraction, attitudes, power settings, etc.) also needs to be specified to assess air vehicle performance under these conditions.

Engine performance characteristics must be standardized to allow easy comparison of engine models and for assessing engine capability. Without a common baseline it is difficult to conduct a one-to-one comparison of proposals from different engine manufacturers. Performance must be stated as the minimum engine to assure that the resultant air vehicle performance estimates will be achievable.

REQUIREMENT GUIDANCE (A.3.2.1.1)

The following should be used to tailor the specification paragraph:

(a): Insert the computer program number and date of its release.

(b): Minimum engine is defined in Section 6. Requirement 3.2.1.1 assumes this definition is implemented in the steady-state performance deck in the following manner: the engine component performance and control schedules modeled in the cycle deck represent an “average new” engine. “Minimum” performance is determined by applying derates or multipliers to the thrust or power and fuel flow related output. These multipliers are applied to the output after the cycle is balanced. They account for control schedule tolerances and engine-to-engine variability in component performance due to manufacturing variations, but do not account for deterioration effects. “Deteriorated” performance is determined by derating component efficiencies, flow parameters, etc. by means of adders or multipliers applied internal to the cycle balance calculations. Combining the two gives “deteriorated minimum” performance, which must be consistent with the performance retention requirements of 3.2.1.4.

(c): Tables IIa, IIb, and IIc are reserved to provide performance points at sea level static for specified atmospheric temperatures. Table III provides performance at selected altitude, Mach number, inlet temperature, and installation factor conditions.

The contractor should provide the conditions for the tables, in conjunction with air vehicle requirements, for Using Service approval. Some conditions that should be considered are: inlet air distortion, relative humidity, ram recovery, customer air bleed, customer power extraction, nozzle effects, parasitic flows, engine anti-icing, and variable geometry, etc., as applicable, for standard and nonstandard days throughout the complete environmental conditions and operating envelope.

While the performance tables must necessarily reflect specific values for the environmental and installation factors, ultimately the “specification” performance requirements are determined by the cycle deck, and any thrust or power and fuel flow output by the deck within a “reasonable” range of environmental and installation factors as defined in 3.2.1.3 are “specification” requirements according to 3.2.1. This convention allows for a definition of “specification” performance even if expected installation factors change during the development program. It also gives the using service two options when determining specification compliance: (1) to adjust qualification test data to the table conditions, or (2) to compare the test data directly to the cycle deck run at the as-tested conditions. For more information on verification, refer to the guidance for 4.2.1.1

MIL-HDBK-310 should be consulted to choose appropriate frequency of occurrence days, or for special requirements, the Using Service can choose a specific temperature. Previously, MIL-STD-210 A or C has been used in the following manner: The USN traditionally uses cold day 51°C (-60°F), standard day 15°C (59°F), tropical day 32. 1C (89.8°F), and hot day 49°C (120°F) for tables IIa, IIb, and IIc temperatures, respectively. Recent USAF practice has been to use the MIL-STD-210C as follows: “hot day” is the 1% risk, worldwide air environment, 5.3.1.1.2, p. 35 of MIL-STD-210C; “cold day” is the 20% risk, worldwide air environment, table XX, p.90 of MIL-STD-210C. The Army in tables IIa, IIb, and IIc uses Sea Level

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static standard day and Sea Level static hot day 50°C (122°F). An important Army table II point for helicopters is 1.2 km (3937 ft) 35°C (95°F).

Any detrimental performance effects caused by an integral IPS (i.e., the IPS is provided by the engine manufacturer) should be modeled in the performance computer program and should be included in tables IIa, IIb, IIc, and III. If the IPS is not part of the engine, the computer model should be able to accept inlet losses associated with an air vehicle IPS to predict installed performance.

(d): The Using Service should indicate whether the computer program or the tables would take precedence. The following factors should be considered when deciding on whether the computer program or the performance tables should take precedence for establishing engine performance.

Pro the computer program:

The computer program simulation would produce continuous thrust or power, airflow and fuel flow values throughout the range of operation. There would not be “specification” points in the simulation that produce engine performance inconsistent with surrounding conditions (either as part of a power hook or as part of a Mach number sweep).

Because the performance is continuous, air vehicle mission analysis and test data analysis would be more accurate.

Pro the tables:

Non-mission essential points produced by the simulation would not become design or verification issues between the Using Service, air vehicle contractor, and the engine contractor.

The tables are easy to understand and always available. There is no concern about which computer system was used, nor how the data was generated (i.e.; whether the proper installation factors, ambient temperature, and power lever angle were used).

(e): 1.2 kilometers, 35°C has been used.

Background:

The performance requirements should be integrated with other engine design requirements such as operability, durability, and reliability to produce a satisfactory air system.

Minimum steady-state engine performance should be established early in the development program. Minimum performance should be obtained without exceeding limits.

The relationship between average and minimum performance levels for thrust or power and fuel flow (specific fuel consumption) throughout the flight envelope should be approximately 2-3% for turbojet and turbofan engines and 2-6% for smaller turboshaft and turboprop engines. The relationship used should consider engine class, maturity, production history, and database.

For more information concerning methodology for generation of performance margin consult ASME Paper 84-GT-262 by H. Brown and J.A. Elgin.

The usual definition of minimum engine is -2 sigma. The (-) sign refers to the lowest thrust or highest SFC observed in the engine population at any given operating condition. While the Air Force has historically used -2 sigma for most performance requirements, -3 sigma is used for certain situations which are considered to be safety critical for the application (e.g., takeoff thrust in a backup control mode). It is important to keep in mind that the term “minimum” refers to the statistical distribution of a population of engines. The “minimum specification” engine is best thought of as a fictitious engine which is simultaneously -2 sigma in both thrust and SFC at all flight conditions. In the actual population of engines, it is highly unlikely that any single specific engine will be both -2 sigma in thrust and simultaneously -2 sigma in SFC at any given flight condition, regardless of whether the engine is fan speed or turbine temperature limited at that condition. Also, the engine which is -2 sigma in either thrust or SFC at one flight condition will probably not be the -2 sigma engine in that parameter at some other flight condition. When performing a mission analysis, consideration must be given to whether one wants to simulate a specific engine chosen from the population, or a statistical minimum which is both -2 sigma in thrust and -2 sigma in SFC at all flight conditions (a “minimum specification” engine), or some other

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choice. Using a minimum specification engine for mission analysis ensures that the worst case thrust is used for the thrust-limited portions of the mission (take-off, climb, etc.) and the worst case SFC is used for the mission legs which are driven by fuel burn. Furthermore, by having the minimum specification performance model take precedence over the performance tables (see 3.2.1.1), it is hoped that the engine manufacturer will avoid discontinuities in thrust at specific table points, and that the specification model will then be useful for mission analyses and test data analyses which require consistent data at points not in the tables, including use of the model as the “grading scale” for the performance qualification test.

The performance tables specified should define the steady-state values for thrust or power, airflow, fuel flow, rotor speeds, torque and temperatures critical to overall air system performance. The performance parameters should be presented in tables IIa, IIb, IIc, and III format for sea level and altitude conditions and should be chosen to:

a. Assure sufficient engine thrust or power at crucial mission points such as takeoff, climb, or transonic acceleration.

b. Assure that fuel usage during climb, loiter and cruise is consistent with the overall air system design and mission.

Tables IIa, IIb, IIc, and III reflect the performance requirements that the engine contractor must meet. Seemingly small performance table changes can greatly impact the engine or control design. Selection of performance points which cover the operating extremes and critical mission points should be done with care because they are very powerful drivers on the engine design.

One performance table should present sea level static uninstalled performance, which should form the basis for the sea level acceptance tests for production engines. The air vehicle performance specification requirements are reflected in the engine specification via the table III characteristics. Table III engine performance is required to allow the air vehicle to meet its system performance levels. Table III should include combinations of flight points, installation factors, alternate front frames, and exhausts, if applicable. To link the two, the lowest corrected airflow in table III should also be included in the power hook in tables IIa, IIb, and IIc. This provides a sea level performance point that is equivalent in corrected airflow (engine cycle match point) to an altitude cruise point.

There should be no derates applied between these tables, i.e., a minimum engine at SLS condition should also represent a minimum engine at the altitude performance points.

Performance presentation via computer programs, plots, data and reports should be consistent. The computer program performance defines tables IIa, IIb, IIc, and III for establishing minimum steady-state engine performance (including thrust or power, fuel flow, SFC). Hardware and software differences (e.g.; word length, compiler, and operating system) among computers can cause differences in performance output from otherwise identical programs. SAE AS 681 requires that the variation in the cycle deck output values between computer systems be less than 0.25% and Using Service experience has been well within that requirement.

Engines that require airflow and exhaust flow for supplemental lift systems will need special consideration in the performance program and tables. If the ducting and nozzles are to be engine contractor furnished equipment, thrust, duct pressure losses and nozzle characteristics should be defined in the specification and computer program. If the supplemental lift system is furnished by the air vehicle contractor, the mass flow, pressure and temperature characteristics should be specified for the engine and air vehicle interface and in the engine performance computer program. When the performance is obtained by means of augmentation, it should be so designated. If liquid injection is provided, the liquid composition, flows, and limitations in use should be specified.

REQUIREMENT LESSONS LEARNED (A.3.2.1.1)

Performance curves were used in the past to clarify performance requirements to Generals and Admirals (G and A curves), and for operators in the test cells before the advent of computer programs. Older computer programs produced varied output depending on the sequence of the input parameters. Some of these programs had to be reprogrammed to adapt to Using Service computers. Secondary and tertiary

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airflow was only estimated in the past, since airflow instability had a pronounced affect. The Using Services no longer require the G and A curves because of limited use and cost of preparation and government verification.

Engine Company A uses 2 sigma for Engines J and V.

A.4.2.1.1 Steady-state performance.

The requirements of 3.2.1.1 shall be verified by:

VERIFICATION RATIONALE (A.4.2.1.1)

Engine steady-state performance must be verified.

VERIFICATION GUIDANCE (A.4.2.1.1)

Past verification methods have included analyses and tests, including endurance and altitude tests.

Past programs have used the following; “The engine performance shall be at least that specified in the steady-state performance computer program. The ratio of measured performance [at altitude] to measured performance at sea level shall be within $\pm$3 percent of the ratio of specified altitude performance to specified performance at sea level.”

For turbofan and turbojet engines, some parameters used to verify performance include net thrust, fuel flow, rotor speeds, inlet airflow, turbine gas path, or blade metal temperature. For turboshaft and turboprop engines the parameters should include shaft horsepower, torque, airflow, fuel flow, and measured turbine gas path or blade metal temperature. This verification should be satisfied by the utilization of a combination of tests listed in the engine milestones (i.e.; altitude, sea level, steady-state, transient, Accelerated Mission Test (AMT), corrosion, etc.). The air vehicle contractor should provide the installation bleed air and horsepower extraction values.

The steady-state performance and computer program should be verified initially by analysis and component rig test, and subsequently by engine testing. Testing should be conducted at extreme conditions and loads (i.e.; altitude, sea level, varying temperatures and pressures, bleed air extraction, power extraction, attitudes, power settings, etc.).

Background:

A sea level static acceptance test of the altitude performance qualification engine should be performed in the contractor’s production type test cell and repeated in a government altitude test cell. Altitude testing is usually performed in a government test facility. The engine should be the production configuration with nominal production control schedules. The test condition should result in the measurement of steady-state performance at stabilized engine conditions. Engine bleed air and power extraction should be taken from the engine if called for by tables IIa, IIb, IIc, and III.

Instrumentation should be capable of measuring thrust and power, fuel flow, exhaust gas temperature, and airflow, as well as internal pressure and temperatures. The specific instrumentation requirements and accuracies should be defined in the test plans. Control parameters such as rotor speeds and variable geometry position should also be measured.

Customer bleed air verifications should be accomplished at sea level and at altitude cruise condition, from idle to maximum power, for inlet air temperatures over the range of sea level hot and cold atmospheric temperatures of figure 5 (a minimum of five test temperatures). The test should be conducted in such a manner as to demonstrate the amount of customer bleed air available over and above that required for engine systems, such as for engine anti-icing systems, and to establish the relationship between bleed air extraction and engine performance loss.

Measurement of scale force should be the primary method of determining thrust for tables IIa, IIb, IIc, and III verification. Other methods of determining thrust, such as the use of nozzle coefficients, should be used to confirm the thrust as measured by the scale force method. The data reduction program should be approved by the Using Service and should be fully approved prior to the start of testing. Test cell performance correction factors should be verified by back-to-back indoor and outdoor testing or a calculation procedure approved by the Using Service. If installation of test instrumentation results in

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significant performance losses, the engine performance results may be adjusted with the concurrence of the Using Service. Once the official tables IIa, IIb, IIc, and III verification begins no further adjustments of the engine, its components, or fuel control should be allowed unless specifically authorized by the Using Service. Throughout tables IIa, IIb, IIc, and III verification testing, the measured performance should not be adjusted for engine degradation.

When conducting altitude performance tests, collecting data for status deck development and verification, including temperature and pressure lapse rates and installation effects (e.g.; air bleed and horsepower extraction, inlet distortion) should be considered. The objective is to characterize the engine components across the full range of their maps, including Reynolds number, clearance, Mach number, and other effects. This may require steady-state power hooks at several flight conditions other than those listed in tables IIa, IIb, IIc and table III. Additional guidance may be found in AEDC-TR-83-42.

The points in tables IIa, IIb, IIc, and III are performance points of the steady-state performance computer program and may not represent all the points required for demonstration or testing.

Present USN philosophy for verifying the engine performance is not based on a direct comparison of the as tested performance corrected to the table II and III conditions and then compared to the table II and III values. Instead, the minimum engine performance computer program is run at the same flight conditions, fuel type, bleed and horsepower extraction and internal instrumentation as the actual engine was tested at for each individual data point and comparison is made between the specification performance computer program and the as tested data to determine if the engine meets its guaranteed performance.

VERIFICATION LESSONS LEARNED (A.4.2.1.1)

Experience has shown that in an attempt to achieve a high level of performance, engines have been built up so tight that heavy rubs and excessive deterioration resulted. In the past, some contractors delivered high performing engines for performance tests and low performing (cold running) engines for durability tests. It has proven difficult to measure secondary and tertiary air for exhaust nozzles.

A.3.2.1.2 Transient performance.

The status or average engine transient performance requirements shall be modeled by the transient performance computer program (a) dated (a), per the conditions of 3.2.1.3. Engine thrust or power transient times from the transient computer program shall be specified in 3.2.2.7, as required therein.

REQUIREMENT RATIONALE (A.3.2.1.2)

All engines must meet or exceed minimum performance to ensure the capability of the air system to accomplish established missions. Performance computer programs provide a means of assessing performance without actual testing of engine hardware. Engine performance information is of critical importance in the development of the air vehicle mission capability, and influences air vehicle inlet design, takeoff, landing, and various other operational characteristics to be generated and assessed by the air vehicle contractor. Engine performance under various conditions and loads (i.e.; altitude, various Mach numbers, sea level, varying temperatures, bleed air extraction, power extraction, attitudes, power settings, etc.) also needs to be specified to assess air vehicle performance under these conditions.

The transient computer program provides a means to verify engine or fuel control design changes. This computer model of the engine is needed in solving transient engine problems involving engine stall margin, transient time, and combustion operation.

Transient engine performance is needed to calculate engine and air vehicle flight behavior such as takeoff, landing, glide path control, speed brake sizing, engine thrust steering, augmentor ignition, transient spool downs, air starting, etc.

REQUIREMENT GUIDANCE (A.3.2.1.2)

The following should be used to tailor the specification paragraph.

(a): Insert the computer program number and date of its release.

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One or two separate cycle decks should be specified or the transient deck may be integrated into the steady-state deck listed in 3.2.1.1 above.

The transient performance computer program must present pressure, temperature, and mass flow rates of bleed air from idle to maximum power. These presentations include the following conditions: operating altitudes, air inlet temperatures, flight speeds, and the effects of bleed air extraction on engine performance. The effects of air bleed for compressor surge protection during starting and acceleration need to be modeled in the program."

Background:

Status transient programs can model specific test engines. The status performance computer program should be used to define the margin between the status model and the minimum engine performance requirements. For identified deficiencies in status to minimum margin, a plan should be presented defining the engine design or configuration changes necessary to achieve adequate production margin. The average engine should reflect the predicted nominal engine characteristics, taken from contractor database and rig tests, until actual engine performance data can be used from the FETT. After the FETT, the average or status performance computer programs should reflect actual engine test data. Minimum performance and average transient performance may, at the option of the Using Service, be specified in two separate cycle decks or may be incorporated in one. If separate from the minimum engine transient performance computer program, the status or average transient performance computer program should be defined in the statement of work and required for submittal in the CDRL. The advantage of having two separate cycle decks is that the minimum deck will remain constant, and only the average or status deck will need to be updated.

Starting, acceleration, and deceleration operation should be closely checked since these represent conditions during which spikes (temperature, speed, pressure, vibrations, fuel flow, etc.) may occur.

REQUIREMENT LESSONS LEARNED (A.3.2.1.2)

Adequate planning and design attention during initial development is required to understand compressor stall characteristics, including those in the starting and non-idle regions. In one engine development program, the initial design did not have a start and acceleration bleed system or the associated control functions or logic. During the course of development, it was discovered that such a system was required to meet the engine specification. This resulted in flight restrictions during initial testing, schedule slippage, and engine design modification.

In a recent Navy program a "unified " cycle deck was created which incorporated the minimum and average steady-state performance and the average transient performance.

A.4.2.1.2 Transient performance.

The requirements of 3.2.1.2 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.2.1.2)

The test engine transient performance must be compared to the transient computer program to ensure the model accurately predicts the engine transient capability.

VERIFICATION GUIDANCE (A.4.2.1.2)

Past verification methods have included analyses and tests.

The transient performance and computer program should be verified initially by analysis and component rig test and subsequently by engine testing. Testing should be conducted at extreme conditions and loads (i.e.; altitude, Mach number, sea level, varying temperatures, bleed air extraction, power extraction, attitudes, power settings, etc.).

Background: See 4.2.1.1.

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VERIFICATION LESSONS LEARNED (A.4.2.1.2)

In the past, transient computer programs have not accurately simulated engine operability nor been capable of predicting surge or stall problems.

A.3.2.1.3 Performance computer program.

The specification performance computer program shall be prepared per the practices of SAE AS 681. In addition, the program shall be capable of representing the effects of the following parameters on the engine cycle performance, singly and in any combination:

a. Fuel lower heating value at constant pressure of (a) BTU/lbm, and otherwise conforming to the fuel specified in 3.7.3.1.

b. Fuel specific gravity of (b).

c. Standard and non-standard atmospheric conditions of (c).

d. Inlet pressure recovery of (d).

e. Exhaust pipe and jet nozzle, fan exit duct and fan nozzle, and engine front and rear frame features specified in (e).

f. Customer bleed air extraction (f).

g. Customer power extraction (g).

h. Using the engine control system specified in (h).

i. Humidity (i).

j. For turboshaft/turboprop engines, the shaft torque absorber characteristics specified in 3.1.1.9 (j).

k. Stall margin and operating line consistent with requirements when matched with vehicle flight exhaust and distortion requirements as specified in (k).

l. Inlet distortion of (l)

REQUIREMENT RATIONALE (A.3.2.1.3)

The computer programs define engine performance throughout the complete operating envelope. The format of the programs must be specified to ensure software compatibility.

REQUIREMENT GUIDANCE (A.3.2.1.3)

The following should be used to tailor the specification paragraph:

(a): The minimum specification Lower Heating Values for JP-4, JP-5, and JP-8 should be 42.8 MJ/kg (18,400 Btu/lbm), 42.6 MJ/kg (18,300 Btu/lbm) and 42.8 MJ/kg (18,400) Btu/lbm respectively. These LHVs are normally used in the Performance Tables since these represent the worst LHVs for the particular fuels. For alternate fuels, the fuel usage should be corrected for the worst LHV specified by the Using Service. The performance computer program should be capable of providing accurate performance estimates over a range of fuel heating values. Typical ranges for various fuels are presented in the CRC Handbook of Aviation Fuel Properties (1983), Figure 28.

(b): The performance computer program should be capable of providing accurate performance estimates over a range of fuel specific gravity. Typical density versus temperature characteristics for aviation fuels are presented in the CRC Handbook of Aviation Fuel Properties (1983), figure 3. The minimum and maximum fuel specific gravities per MIL-DTL-5624 and MIL-DTL-83133 are as follow:

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Fuel	JP-4	JP-5	JP-8
min	0.751	0.788	0.775
max	0.802	0.845	0.840

(c): U.S. Standard Atmosphere 1976 is a universally agreed upon standard of temperatures and pressures for a range of altitudes sufficient for aircraft operations. Geopotential altitude is designated since it produces a consistent definition of engine altitude capability. The 1976 Standard Atmosphere edition is the same as the 1962 edition for altitudes below 50km (164,050 ft), but has new values for higher altitudes. Non-standard atmospheres are defined in MIL-HDBK-310. The definition of non-standard atmospheres has evolved over time. In the past, we have referenced MIL-STD-210A tropical, hot, cold, and polar days. Recent USAF practice has been to use the MIL-STD-210C as follows: "hot day" is the 1% risk, worldwide air environment, requirement 5.3.1.1.2, p. 35 of MIL-STD-210C; "cold day" is the 20% risk, worldwide air environment, table XX, p.90 of MIL-STD-210C.

(d): Where possible, estimates of the actual inlet pressure recovery should be used, as well as the "Military Standard" recoveries, which are defined by the equations below:

Pt1(ref)/PtO = 1.0, from 0.0 to 1.0 Mach Number.

Pt1(ref)/PtO = $1.0 - 0.075(M - 1)^{1.35}$ from 1.0 to 5.0 Mach Number.

Pt1(ref)/PtO = $800/(M^4 + 935)$, above 5.0 Mach Number.

Pt1(ref) = referenced total pressure at the propulsion system entrance or external/internal interface.

PtO = free stream total pressure.

M = flight Mach Number.

Inlet pressure recovery is dependent on the inlet design and methods of shock recovery. Every inlet design has a different characteristic pressure recovery relationship. The "Military Standard" relationship is normally selected as a standard baseline, and has been used in turbojet and turbofan specifications as far back as historical records exist. In practice, ground testing is usually accomplished with bellmouth inlets that reproduce the mil standard recovery. The engine scheduling, performance and testing where possible should be based on actual engine inlet recovery.

(e): An exhaust system should conform aerodynamically with the exhaust nozzle of the engine installation and interface drawing and have the thermal and pressure growth properties of the material(s) as specified thereon. The exhaust system characteristics should be based on steady-state operating conditions of the primary nozzle, primary plug, fan exit duct, and fan nozzle (including the pylon shape and other surface areas affecting performance), and should include all other details affecting performance. These exhaust system performance characteristics should be presented in the steady-state and transient computer programs of 3.2.1.1 and 3.2.1.2. If referee nozzles are used, then the referee nozzle performance should be substantiated and included in the steady-state and transient performance computer programs when generating performance. The use of a referee nozzle other than a straight circular duct should be coordinated with the Using Service. Substantiating performance of referee nozzles is necessary for engine performance bookkeeping when changing from the referee to flight nozzle. Thrust (and SFC) with vectoring nozzles should be based on resultant axial thrust (i.e., zero vector angle). Thrust and (SFC) for fixed geometry non-axially aligned nozzles should be based on resultant gross thrust minus ram drag. Thrust (and SFC) of multiple nozzle (e.g.; V/STOL, STOVL) arrangements should specify the thrust or force components of each nozzle.

Engine cycle performance should include the effects of specified front and rear face devices, engine supplied frames, coatings, and features (if applicable).

(f): Where possible, estimates of actual customer bleed demands should be used. Zero customer bleed air extraction is normally specified in the Performance Tables since it is a baseline from which performance effects can be obtained.

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(g): Where possible, estimates of actual customer power extraction demands should be used. Zero customer power extraction is normally specified in the Performance Tables since it is a baseline from which performance effects can be obtained.

(h): The control system defined in the engine specification should be used. Control system tolerance on such items as variable geometry schedules, stability bleed schedules, etc. can affect performance.

(i): The performance computer program should be capable of providing accurate performance estimates over a range of humidity levels. A dry atmosphere is normally specified in the Performance Tables since it is a baseline from which performance effects can be obtained.

(j): For turboprop and turboshaft engines, it is necessary to know the mechanical and performance characteristics and limitations of the output shaft torque absorber. This is particularly needed for matching rotational speeds, observing torque and power and speed limitations, and designing the engine control for transient operation. It is not possible to exactly match the turboshafts' or turboprops' inertia and dynamic characteristics with a dynamometer or water brake (Engine Company A and B).

(k): Engine performance should be determined using the operating line and stall margin needed for the operational engine. This includes matching with the vehicle inlet and exhaust systems and engine surge control bleed ducting when present. Zero inlet pressure and temperature air distortion is chosen as the standard, since air distortion has an effect on performance. Zero distortion is a baseline from which performance effects can be specified with various degrees of inlet air distortion. Performance credit or debit should be allowed for facility, but not engine (Engine Company A) FOD screens or bellmouths.

(l): The program should be capable of modeling the effects of steady-state inlet total pressure distortion on engine performance.

The following requirement may be included in the specification paragraph:

"The computer program shall be prepared per the practices of SAE AS 681, except as modified herein.

a. Program Requirements. The performance program shall be a thermodynamic cycle simulation in which component identity is maintained; e.g.; the compressor, turbine, and combustor must each be identifiable as entities in the model logic as required to obtain and maintain an accurate simulation. The computer program shall be submitted in source language compatible with the computer specified.

b. Program Capabilities. The program shall be capable of operating throughout the complete environmental conditions and operating envelope. Compilation shall not be necessary for each different run. Capability for determination of installation effects shall be included in the computer program. Effects of distortion, relative humidity, test specific instrumentation, ram recovery, customer air bleed, customer power extraction, nozzle effects, parasitic flows, engine anti-icing, windmilling, surge bleed ducting, coating installation, and variable geometry, etc., shall be included as applicable. Installation effects which are interactive with engine characteristics shall be modeled as dummy subroutines for Using Service inclusions and shall be called within the appropriate parameter convergence logic of the engine model. The ability to thermodynamically match with complex exhaust and suppressor system models shall also be included. The distortion limits of stall-free and surge-free operation shall be included in the performance computer program. The program shall provide the ability to input component performance modifiers. The computer program shall include the engine inlet and exhaust systems as discrete modules. The performance computer program shall be capable of being integrated into air vehicle installation programs to determine installed performance.

c. Documentation Requirements. A user's manual and source program card image, disk, CD ROM, or other suitable media shall be provided.

d. User's Manual. In addition to those items specified in SAE AS 681, the user's manual shall contain a general description of the simulation techniques, general overall model flow chart and a clear explanation of the calculation process and related assumptions for all engine components. The user's manual shall include a tabulation of all parameter limits and reference to all engine limits described in the model specification; e.g.; measured temperature, speed. Additionally, graphs of all empirical functions used, reference values for normalized parameters, an index of subroutines including their inputs, outputs and functions, and a listing of all test case program inputs and the corresponding required outputs should

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be included in manual or on the delivered media (disk, etc.) All input parameters, output parameters, and parameters used in calculations shall be identified along with corresponding units for each parameter. The manual shall describe the relationship between average and minimum engine performance and how it is calculated in the computer program. The user's manual for Government users should include component maps, principal factors, nomenclature, and basis for internal performance.

e. Source Program. The Using Service computer program shall be in source form, and shall include all program subroutines with comment card images to identify subroutines and their functions. Sufficient comment card images shall appear in the program code to elucidate the calculation process. The first card images on the delivered media shall be comment cards indicating the contractor, engine type designation, program number, and date. The format of the card image on the delivered media shall be as specified by the Using Service.

f. Inputs and Outputs. The program output listing shall be organized so that the input data shall print separately from the output. The program output listing shall show the engine designation, computer program number, and date. On classified computer programs, provision shall be made in the program for a line on the output sheet which can print declassification and authority. The program shall have features to print internal station properties such as physical flow, total pressure, and total temperature at all internal station locations. Output is to be labeled as to performance basis (i.e.; average, minimum, new, deteriorated, specification, status, etc.) All program inputs must be independently variable and the programs shall be capable of sequentially accepting multiple numerical changes to the computer inputs. In addition to the inputs listed in SAE AS 681, the program shall provide inputs of fuel heating value and specific gravity. The program shall be capable of calculating all the required output parameters of SAE AS 681, with input options of net thrust or power in addition to the options of rating code and power lever angle or power code, as specified in SAE AS 681. In addition to the required output parameters of SAE AS 681, the program shall output the gas temperature at the first stage turbine rotor inlet of each turbine and the measured gas or metal temperatures at the measurement plane (contractor shall parenthetically insert the measurement plane station identification here). For turboshaft and turboprop engines, the program outputs should include exhaust gas velocity and exhaust gas swirl angle. The program shall have the capability of accepting inputs and providing outputs of all temperatures in the unit of degree Celsius (C). The program shall have the capability of accepting inputs and providing outputs in the U.S. Customary System of Units and, additionally, in the International System of Units (SI) per IEEE/ASTM SI 10. SI units shall be made selectively available. In the event of conflict between SI and US Customary Values (due to improper conversion), the US Customary Value shall prevail.

All aero thermodynamic cycle parameters used in component performance determination and cycle calculations should be selectively available by means of a user input. All program output characteristics should present status levels of performance with the exception of the guaranteed performance parameters which will be characterized as both status and minimum performance.

Expanded input and output capability should be required to allow the performance prediction to be tailored for special analysis. For example, the model may be used to support field service verifications (including deterioration effects) or to verify the effect of proposed design changes on status and minimum engine performance.

Expanded inputs should include adders and scalars used to modify the performance of each of the engine components, such as: component efficiencies, burner pressure loss, duct pressure losses, intermediate case pressure loss, nozzle discharge coefficient, leakages and cooling flows, and turbine flow parameters, etc. Similarly the ability to adjust engine control schedules should be provided through expanded input.

Expanded output should include all control and aero thermodynamic cycle parameters used in component performance verification and engine power management, such as: component efficiencies, fan and compressor surge margins, turbine flow parameters, and nozzle areas.

For U.S. Army Aviation and Missile Command applications; performance computer program(s), the computer processor platform, it's location, computer operating system, and program source code format or language should be specified. The Air Force prefers not to declare an "official computer" because a variety of systems are used.

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Background:

At the initiation of the engine program, the engine designer has designed an “average” engine, in terms of the component performance maps and the thermodynamic cycle calculations he has used, and has created a cycle deck which predicts the overall performance of that engine. The Using Service then “grades” that model for realism and thermodynamic consistency, apply estimated derate factors to the thrust or power and fuel flow from this model, and thereby derives the minimum specification requirements (this is a process of negotiation between the engine contractor and the Using Service). The initially estimated derate factors represent an assumed variability between the average of the population of engines (still to be built) and the projected -2σ engine. Thus, the initially projected -2σ engine is the minimum specification requirement. The minimum specification requirements never change until renegotiated with the Using Service. The average status engine and the projected -2σ engine will change with time as we undergo engine testing and design iterations.

The average engine should initially reflect the predicted nominal engine characteristics, taken from contractor database and rig tests, until actual engine performance data can be used from the first engine to test (FETT). After the FETT the average or status performance computer programs should reflect actual engine test data. Status “average engine” should reflect a true arithmetic average of sufficient accepted production engine samples.

Note the term “margin” is reserved for comparisons to the minimum specification requirements, while the difference between the average engine and the projected -2σ engine is called the “variability”. Initially the variability and the margin are assumed to be identical. This margin is usually approximately 2-3% for turbojet and turbofan engines and 2-6% for smaller turboshaft and turboprop engines. Like the average status engine, the variability and the margin will each change with time. We want the projected -2σ engine to always be above the minimum specification requirements. For identified deficiencies in status to minimum margin, a plan should be presented that defines the engine design and configuration changes necessary to achieve adequate production margin.

The status performance computer program should be used to define the margin between the status model and the minimum engine performance requirements.

The capability for determining installation effects should be included in the computer program. Installed engine performance should include effects of distortion, relative humidity, test specific instrumentation, ram pressure recovery, customer air bleed, customer power extraction, nozzle effects, parasitic flows, engine anti-icing, windmilling, variable geometry, and internal engine instrumentation, etc. as applicable.

Installed performance is required by the airframe contractor to predict or determine aircraft performance. As a result, the performance computer program of 3.2.1.3 should be capable of including engine installation effects (e.g.; inlet distortion, humidity, inlet pressure recovery, customer bleed and power extraction, alternate exhaust characteristics, etc.) and be capable of integration with an airframe installation computer code.

REQUIREMENT LESSONS LEARNED (A.3.2.1.3)

A requirement for a computer program first appeared in MIL-E-5007B in January 1959. It has since been continually improved, based upon experience, and use of the programs. Since 1960, aircraft manufacturers have provided computer programs along with engine model specification performance graphs. The last vehicle application based on specification performance curves and correction factors, rather than the customer computer program, was the Navy Aircraft B application circa 1968. After that, the customer computer program became the official basis of air vehicle performance.

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A.4.2.1.3 Performance computer program.

The requirements of 3.2.1.3 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.2.1.3)

The use of actual engine and components data is the only acceptable method of verifying the computer programs.

VERIFICATION GUIDANCE (A.4.2.1.3)

Past verification methods have included analyses, inspection and tests.

Substantiating data analysis for performance and computer program verification should be submitted to the Using Service. The engine transient computer program output should be within one percent (1%) of the steady-state performance computer program output at the performance points. The contractor should demonstrate the successful operation of the computer program(s) using all input mode options at all performance points on the specified computer. Failure of the computer program to produce valid output within the engine operating envelope should be cause for the computer program(s) to be judged as unacceptable. The test cases for each input mode option should cover the operating envelope and power range of the engine. The contractor should arrange the test case inputs in such a manner as to demonstrate the capability of the computer program(s) to accurately calculate engine data regardless of the order of the inputs or the option modes being exercised. Each test case should be verified to ensure that the computer program(s) is operating properly and the data produced for each test case represents the performance of the engine including program options, limits, cutbacks, and error messages. More than one item can be verified with each point.

The contractor's verification of the computer program(s) should include a comparison of the engine performance steady-state data with the figures and tables in the model specification and user's manuals for consistency and accuracy. Engine test data should be used to verify the computer program predictions of performance. If there are significant discrepancies between the engine test data and the computer program, a status computer program should be generated, upon Using Service approval, to represent current engine performance levels. A sufficient number of test cases, as determined by the Using Service, should be used for verification to test and analyze all areas of the operating envelope.

Background:

Verification of the minimum engine model is important in that the minimum performance quotations of thrust or power, specific fuel consumption, and limiting temperature should be a reasonable representation of a $\pm$2 or 3 sigma production engine. It is intended that the component performance levels are known or assumed at the time the specification is developed.

This verification requires the consideration of several factors:

The quoted component performance characteristics are realistic and attainable.

The computer representations of the component aero thermodynamic characteristics are of adequate fidelity.

The laws of continuity, momentum, and energy conservation and the second law of thermodynamics are applied in a valid manner.

VERIFICATION LESSONS LEARNED (A.4.2.1.3)

In the past, some computer programs had problems predicting engine performance at the extremes of the operating envelope. Performance programs have not always reflected all the hardware changes that could affect performance.

Unrealistic component performance predictions, errors in bookkeeping of loss terms, and programming mistakes have often been discovered in the computer models, and in some cases have resulted in unattainable performance quotes.

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On a past vehicle application (Navy Aircraft I, 1975), several candidate engines had very detailed transient computer programs available. When exercised, these engines had significantly varying transient performance. When the Government RFP was issued, all engine vendors changed their transient performance to agree with the Government specification (some up and some down).

A.3.2.1.4 Performance retention.

Unless otherwise specified, during and at the completion of one cold parts life, the engine thrust or power shall not fall more than (a) below the minimum engine thrust or power requirements specified in 3.2.1.1 and 3.2.1.2. at (b) temperature. During and at the completion of the specified hot section life, the engine thrust or power shall be no less than (c) percent of the actual thrust or power at the beginning of the hot section life at (b) temperature. The specific fuel consumption (SFC) shall be no more than (d) percent of the actual SFC at the same thrust or power at the beginning of the hot section life at (b) temperature.

REQUIREMENT RATIONALE (A.3.2.1.4)

This requirement sets limits on power and specific fuel consumption (SFC) degradation that is needed to ensure an adequate performance life of the engine for the Using Service.

REQUIREMENT GUIDANCE (A.3.2.1.4)

The following should be used to tailor the specification paragraph:

(a): A value from 0 to 5%.

(b): For constant temperature machines insert “constant turbine (station no. ___)” with the engine station number in parentheses. For machines whose turbine temperatures vary (i.e., control modes based on rotor speed or pressure ratio), the words “at ___ temperature” should be deleted unless the engine performance is based on a specific turbine temperature. In this case, the EPR or rotor speed should be included in the blank.

(c): This value is usually bounded by the cold section life performance retention requirement but expected to be in the range of 95% to 100%.

(d): A value from 100% to 105%.

The engine should retain at least 95 percent of the initial power, and SFC should not increase to more than 105 percent of the initial SFC when operated for any hot parts life specified and per the mission and mission mix of table XI and the ambient temperatures of figure 5. The amount of performance degradation during each hot section life should be specified to preclude large performance drops. It is assumed an average engine would be used for the accelerated life test and an average performance degradation of 3 to 5 percent would be expected (3-5 percent depends upon what flight point, control scheme, and sigma engine). This requirement limits the performance drop during a hot section life test so that a poor-performing engine (e.g., high turbine temperature and high thrust) would not have a large performance drop of 7 to 10 percent. The engine should complete the hot parts life interval without replacement of parts specified in table X. Turbine temperature should not exceed a limit consistent with hot section life limits. The contractor should specify the station number of the turbine rotor inlet.

Background:

Performance retention is characterized by power, turbine, or exhaust gas temperature, and SFC retention at all SLS specification performance points. Performance retention requirements should be established in such a way that any engine should be able to deteriorate (at the established rate) to the maximum allowable turbine temperature for operational use and always provide the specified percent of rated power for the required number of engine operating hours between scheduled refurbishments.

Deterioration characteristics based on engine manufacturer field data will include the environmental and ingestion effects experienced by individual postproduction engines. This level of deterioration should be higher than that obtained in qualification testing in “clean air”. Conversely, deterioration in non-production test engines may be high because of tight clearances and thermal mismatch of materials. Field deterioration for land based helicopters is significantly higher than on qualification tests due to actual

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environmental and ingestion effects. Comparison and development of field deterioration data, performance retention requirements, and performance test recalibration requirements needs to consider the environmental and ingestion effects of each engine class or application.

REQUIREMENT LESSONS LEARNED (A.3.2.1.4)

Hot day takeoff, hot day low altitude penetration, and supersonic operation have been areas where past air systems have failed to maintain adequate system performance, and therefore, have driven engines back to overhaul prematurely. Thrust available from deteriorated engines at key operating points, such as catapult launch and wave off on a tropical and hot day can affect flight safety. Thrust available for vertical takeoff, hover, and landing for helicopters, V/STOL, and STOVL aircraft is important for flight safety and mission accomplishment. Performance deterioration is a principal cause of engines being pulled for maintenance on helicopters. Water washing was sometimes used in the field for performance recovery.

Some recent CFE air vehicle programs have had performance retention warranties with vehicle performance guarantees based on performance of deteriorated or “warranty” engines. This was because of concern for deterioration rates and engine-to-engine variation expected in service.

A.4.2.1.4 Performance retention.

The requirements of 3.2.1.4 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.2.1.4)

Demonstration of performance retention is needed to ensure compliance with the specification and provides information on the expected performance life of the engine for the intended application.

VERIFICATION GUIDANCE (A.4.2.1.4)

Past verification methods have included analyses and tests. If tests are used, the thrust or power deterioration is usually measured at a constant temperature or at the same power setting, dependent upon the control mode. The SFC deterioration should be determined using a power hook from low thrust or power to full thrust or power. The power hook should go from the lowest part power performance point of table III to sea level static intermediate or maximum continuous thrust or power. Ideally performance deterioration needs to be determined at altitude and sea level, but altitude endurance testing may be excluded due to costs. If only sea level endurance testing is utilized, it is very important to include the part power table III performance point in the SFC power hook to ensure the engine will meet the SFC requirement at that condition. Computer simulations and predictions of performance deterioration rates cannot be used solely for verification of engine performance. Deterioration modeling can be useful for future testing and is used to update the deterioration model once the simulation results have been verified through engine testing.

Background:

The Using Service will determine the cycles of testing and the equivalent engine flight hours. Test cycles and time can be determined from table XI and the hot section life.

Scale force measurement is the preferred method for determining power for turbofan and turbojet applications. The test setup, instrumentation, and data reduction procedures should be acceptable to the Using Service and should be approved prior to the test start. Test calibrations should be held to a minimum but will not count toward endurance time.

VERIFICATION LESSONS LEARNED (A.4.2.1.4)

Past SLS endurance testing (AMT or Accelerated Simulated Mission Endurance Test (ASMET)) on high performance turbofan engines revealed that SFC and operating line deterioration cannot be determined by SLS endurance test alone. High ram temperature and pressure gradients and upper left-hand corner cold temperatures appear to impose significant additional engine deterioration on modern tight clearance

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engines. Therefore, an AMT that includes sea level, high Mach, and high altitude conditions should be accomplished since it is more representative of expected operational usage.

Deterioration in the field is also affected by environmental, design, and operating factors such as sand, rain, ice, thermal mismatch of materials, design clearances, and hard landings on carrier decks. New engines in test development can have problems with initial design clearances. One engine on its initial runs lost 5% performance in the first 40 hours because of clearances that were too tight. Engines in turboshaft engine service can also lose 15-20% power in 500-1000 hours due to engine design and environmental factors, such as operation in a high sand environment.

A.3.2.2 Operating characteristics and limits.

All engine steady-state and transient operating limits (maximum, minimum) shall be specified in table IV and figure 6. The specified limits shall be predicated on the most critical tolerances of the engine.

REQUIREMENT RATIONALE (A.3.2.2)

Engine operating characteristics and limits must be specified for the Using Service to know the full range of engine capabilities.

The limits are the limiting conditions within which the engine must meet all specified steady-state and transient performance characteristics.

REQUIREMENT GUIDANCE (A.3.2.2)

Background:

The engine and control system has a tolerance band, for each parameter with a limit, which should be taken into account when the limits are specified.

REQUIREMENT LESSONS LEARNED (A.3.2.2)

None.

A.4.2.2 Operating characteristics and limits.

The requirements of 3.2.2 shall be verified by:

VERIFICATION RATIONALE (A.4.2.2)

All engine operating characteristics and limits must be verified by analysis, demonstrations, and tests.

VERIFICATION GUIDANCE (A.4.2.2)

Past verification methods have included analyses and tests.

Test results should be analyzed for worst case tolerance in the engine and the control system.

VERIFICATION LESSONS LEARNED (A.4.2.2)

The full impact of some operating limits is not discovered until the engine has been in service for a number of years.

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A.3.2.2.1 Operating envelope.

The engine shall meet all the requirements of the specification throughout the complete operating envelope without exceeding any limits. The engine operating limits shall be specified for the environment of figure 5, per the format shown on figures 6 and 7 and in tables IIa, IIb, IIc, and III. If applicable, the thrust augmentation-operating envelope shall be included on figures 6 and 7.

REQUIREMENT RATIONALE (A.3.2.2.1)

This requirement defines the operating envelope within which the engine must meet its functional, performance, and durability requirements (aero thermodynamic and mechanical limitations).

REQUIREMENT GUIDANCE (A.3.2.2.1)

The following should be used to tailor the specification paragraph:

If the referenced figures are insufficient to describe the operating envelope, either the Using Service or contractor can add a table.

The specified limits should be predicated on the most critical parameters and characteristics of the engine. The absolute altitude of the engine and the range of Mach numbers applicable at standard day, cold, tropical, and hot atmospheric conditions should be specified on figure 7. Figure 6 should represent the operating envelope for pressure and temperature. Both transient and steady-state operations should be specified. The engine operating envelopes should meet or exceed the envelope requirements of all current and anticipated aircraft applications for the engine.

Background:

Design requirements are set by envelope extremes, such as the maximum inlet pressure, which helps define the combustor case maximum pressure requirements and blade aero-elastic requirements for blade vibration. Low inlet pressure and temperatures impose design requirements on the combustor and augmentor performance parameters.

Altitude and Mach number extremes will impact the cooling requirements of the engine lubrication system. Bearing loads need to be quantified throughout the engine envelope. In summary, most parts of the engine are, in some way, impacted by the engine envelope requirements.

REQUIREMENT LESSONS LEARNED (A.3.2.2.1)

In-service engines have operated outside of the operating envelope. The aircraft envelope is normally within the engine envelope.

A.4.2.2.1 Operating envelope.

The requirements of 3.2.2.1 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.2.2.1)

Verification that the engine operates successfully throughout the specified operating envelope must be accomplished.

VERIFICATION GUIDANCE (A.4.2.2.1)

Past verification methods have included analyses and tests.

Engine testing should be conducted to demonstrate that the engine can meet the functional, performance, and durability requirements of this specification throughout the operating envelope of figure 7. All the sea level and altitude tests specified should be used to demonstrate satisfactory operation throughout the operating envelope. Test demonstration figures should show both the operating envelope and demonstration points. The Using Service and contractor should negotiate the test points, and the testing should be accomplished in an altitude test facility.

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Characteristics of the engine which should be verified around the envelope include: steady-state and transient performance, engine stability, starting, internal stresses and temperatures, augmentor operation, lubrication system operation, control system operation, and inlet distortion tolerance.

VERIFICATION LESSONS LEARNED (A.4.2.2.1)

Thorough testing of the engine around the envelope extremes in an altitude test facility has been a cost effective method of reducing the possibility of encountering engine problems during flight testing and deployment of the weapon system. Early identification of engine problems and demonstration of corrective changes around the engine envelope has been accomplished with this testing. Experience has shown that augmentors on afterburning engines should be tested for stable combustion at high altitude, low air speed as well as at maximum Mach number conditions. Maximum inlet pressure conditions on the exhaust nozzles have caused buckling of nozzle components. This test is, therefore, of particular importance to engines with variable geometry exhaust nozzles. There have been cases where oil hiding has been observed in the engine gearbox at high inlet pressure. Therefore, the behavior of lubrication and fuel systems should be carefully monitored at flight envelope extremes and during altitude starts.

A.3.2.2.2 Operating attitude and conditions.

The engine operating attitude limits shall be shown on figure 8. The engine shall meet the requirements of the specification when operating in the unshaded area of figure 8, and operate continuously at least (a) seconds in the shaded areas of figure 8. Operation in the shaded area shall not degrade engine performance or cause any damage. The engine shall start, stop, and be stowed in any of the attitudes shown in the unshaded area of figure 8. Engine stowing capability outside of the unshaded area of figure 8 shall be specified herein. The engine shall function satisfactorily for at least (b) seconds in negative g and for at least (c) seconds in zero g conditions. The engine shall meet all requirements of this specification with unlimited occurrences of negative and zero g flight defined above.

REQUIREMENT RATIONALE (A.3.2.2.2)

The engine is required to operate throughout the attitudes imposed by the aircraft, such as takeoff, climb, inverted flight, air combat maneuvers, stowage, and terrain following. Time duration of these factors is a significant consideration to the engine design.

REQUIREMENT GUIDANCE (A.3.2.2.2)

The values used should be consistent with the requirement in the air vehicle specification.

For engines that don't have an air vehicle specification available, the following values have been used on past programs:

(a): A value of at least 30 seconds.

(b): A value of at least 60 seconds.

(c): A value of at least 30 seconds.

For helicopter applications, past programs have used figure 8 clear limits of not less than 40 degrees nose up with a left inclination of 15 degrees and a right inclination of 15 degrees, 0 degrees nose up with a left inclination of 20 degrees and a right inclination of 20 degrees, and 40 degrees nose down with a left inclination of 15 degrees and a right inclination of 15 degrees. Corresponding figure 8 shaded limits have been not less than 90 degrees nose up with a left inclination of 15 degrees and a right inclination of 15 degrees, 0 degrees nose up with a left inclination of 45 degrees and a right inclination of 45 degrees, and 60 degrees nose down with a left inclination of 25 degrees and a right inclination of 25 degrees.

For V/STOL aircraft such as the Aircraft F, requiring engine attitude changes, the starting and stopping attitude limits should be not less than 105 degrees nose up, 20 degrees nose down, and 30 degrees roll to each side. Above 6 km (3.7 miles), continuous operation is not required at nose up attitudes greater than 45 degrees.

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REQUIREMENT LESSONS LEARNED (A.3.2.2.2)

Extended flight operation in zero or negative “g” conditions resulted in oil system malfunctions, such as oil starvation, bearing sump flooding, gearbox flooding, oil foaming problems, and seal leaks. Special provisions, such as an auxiliary lube system may be necessary in the lubrication system for extended inverted flight operation.

Stowing for Engine J was limited to attitudes of zero to 20 degrees in pitch, or zero to 5 degrees in roll. An Engine C stowage requirement permitted 20 degrees roll and 40 degrees down to 40 degrees up for pitch.

A.4.2.2.2 Operating attitude and conditions.

The requirements of 3.2.2.2 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.2.2.2)

Verification is required to demonstrate satisfactory engine functional capability under a variety of attitude and maneuvering conditions.

VERIFICATION GUIDANCE (A.4.2.2.2)

Past verification methods have included analyses and tests.

The engine should be subjected to an attitude test to meet the requirements of 3.2.2.2. A lube system simulator may be used in lieu of an attitude test with the concurrence of the Using Service. Engine capability to operate for the time specified in 3.2.2.2 for both negative “g” and zero “g” conditions should be verified by analysis or by a rig test of the lubrication oil system. The engine should be started and operated at intermediate thrust or power demand position, for at least thirty minutes, at each of the test points shown in the unshaded area on figure 8. The engine should also be operated at intermediate thrust or power demand position for at least 30 seconds at each of the test points shown in the shaded area of figure 8. This test should be considered completed when the engine starts satisfactorily, remains within all operating limits, and there is no evidence of mechanical damage.

The qualification test program should have a strong foundation of component tests, especially in the oil system component area. Oil tanks, oil pumps, scavenge pumps, gearboxes, deareators, and bearing compartments should all be component tested in various simulated flight conditions. It may be possible to test complete small engines at various attitudes.

VERIFICATION LESSONS LEARNED (A.4.2.2.2)

None.

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A.3.2.2.3 Starting.

The engine shall satisfactorily make ground and air starts, and shall restart under the conditions, limitations, and procedures, specified in this section. The engine shall start with inlet air at any temperature between and including the extreme hot and cold atmospheres of figure 5, with fuel inlet temperatures as specified in 3.7.3.2.2. Starts shall be made with the primary fuel(s) of 3.7.3.1.1 and oil(s) of 3.7.8.1, unless specified otherwise. A start shall be considered satisfactory when the engine rotor is accelerated from either rest or wind milling speed to idle, when the minimum assist torque is provided by a starter or ram air, as specified in 3.1.1.11, and the engine stays within the engine starting and operating limits of figure 6. Starting time shall be measured from the initiation of the starting sequence to the attainment of a stabilized uncorrected engine idle rotor speed.

REQUIREMENT RATIONALE (A.3.2.2.3)

The conditions under which an engine can be successfully started and the definition of a successful start need to be specified.

REQUIREMENTS GUIDANCE (A.3.2.2.3)

Background:

It has been recognized that with a starter and starting system of constant capability, start times will be longer at both high and low inlet air temperatures than at standard day temperature. Low temperature starts take longer because of the increased viscosity of the oil and high temperature starts take longer due to the high temperature and low-density air. The engine contractor should specify the effect of degraded equipment. The starts may be acceptable on a new engine, but may not be acceptable on a time worn service engine, or if a poorly performing APU, battery, or starter is used.

REQUIREMENTS LESSONS LEARNED (A.3.2.2.3)

None.

A.4.2.2.3 Starting.

The requirements of 3.2.2.3 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.2.2.3)

The ability to start the engine using the specified procedure with minimum starter torque under various temperature and load conditions must be demonstrated.

VERIFICATION GUIDANCE (A.4.2.2.3)

Past verification methods have included analysis, demonstrations and tests.

Background:

Adequate testing should be accomplished to show that the starting system could meet all starting requirements. As a minimum, start testing should be conducted at the cold, standard, and hot ambient temperatures specified for both the ground and air start environments. Start system performance is highly dependent upon aircraft accessory drive loads and power extraction, and these loads must be simulated or analytically accounted for in all engine testing. Sufficient data that adequately defines both the fired and unfired torque characteristics of the engine as a function of ambient temperature should be collected. Analysis can then be made to determine the starting characteristics of the engine at conditions other than those that were tested.

VERIFICATION LESSONS LEARNED (A.4.2.2.3)

Previous altitude and ground starting tests did not use any load or power extraction. The aircraft contractor criticized the testing since it did not give realistic results. Analytically accounting for the power extraction does not give the confidence of an actual test. Starting information is included in the pilot's handbook and should be as accurate as possible.

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A.3.2.2.3.1 Ground starts.
The engine shall start at the specified ambient temperatures and altitudes, in times equal to or less than those shown on figure 9a. The engine shall be capable of (a) consecutive start cycles, with a maximum time interval of (b) seconds between the completion of one cycle and the initiation of the next cycle. The wind condition envelopes for ground starting shall be shown on figure 9b.

Special starting requirements apply as follows:

a. Hot temperature soak start. The engine shall start, after a (c) hour hot soak at the temperatures specified on figure 5, using (d) fuel, and with inlet air, fuel, and oil at the maximum hot ambient temperatures shown on figure 5. Following shutdown, the engine shall start, after a resoak period of (e) minutes at the temperatures specified on figure 5, using (d) fuel, and with inlet air, fuel, and oil at the maximum hot ambient temperature shown on figure 5.

b. Cold temperature soak start. The engine shall start, using (f) fuel and (g) oil, after a cold soak for a period of (h) hours with inlet air, fuel, and oil at the minimum cold ambient temperature shown on figure 5.

REQUIREMENT RATIONALE (A.3.2.2.3.1)

Worldwide, all-weather operation requires the use of specified fuels and oil under extreme hot and cold ambient temperature conditions.

REQUIREMENT GUIDANCE (A.3.2.2.3.1)

The following should be used for tailoring the specification paragraph:

(a): The engine should be capable of an unlimited number of start cycles, or if an electric starter is an engine component, then a starter duty cycle needs to be specified.

(b): A value between 0 and 60 seconds.

(c): The time required to thermally stabilize the engine and its contained operating fluids.

(d): The fuel used should be the worst case primary fuel.

(e): A value between 15 and 30 minutes.

(f): The fuel used should be the worst case primary fuel.

(g): The oil should be either MIL-PRF-23699 for maximum cold ambient temperature at -40°C and above; and MIL-PRF-7808 for below -40°C. MIL-PRF-23699 can be used below -40°C if the contractor chooses to use that oil in that temperature environment."

(h): The time required to thermally stabilize the engine and its contained operating fluids.

The location in the engine used for the thermal measurement should be identified.

Background:

Figure 9a should be constructed by the Using Service and should be consistent with the ambient temperature requirements of the engine as required by 3.2.2.1 and figure 5. The Using Service should indicate on figure 9a the maximum hot and cold ambient temperatures (consistent with figure 5) and indicate the hot soak, hot resoak, and cold soak temperatures vs. times vs. altitudes.

The contractor should provide curves on figure 9a specifying the engine starting times for the complete ambient temperature range at incremental altitudes up to the maximum ground-starting altitude of 3.2.2.3.3. The Using Service should be aware that according to SAE AS 1606, starters should be capable of at least five consecutive start cycles or attempts. However, the engine should not be limited in the number of consecutive starts.

REQUIREMENT LESSONS LEARNED (A.3.2.2.3.1)

Tail winds have adversely affected the starting performance of engines, particularly the high bypass ratio turbofans. Since cold weather conditions can affect compression and combustion interaction, the cold

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soak requirement has proven difficult for engine contractors to meet. Combustor stability and sub-idle (start) stall margin have seen deficiencies that have led to operational starting problems. During a rapid restart attempt, the unburned fuel can collect and cause torching and excess turbine temperatures.

A.4.2.2.3.1 Ground starts.
The requirements of 3.2.2.3.1 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.2.2.3.1)

Engine capability to be started on the ground under the required environmental extremes must be demonstrated.

VERIFICATION GUIDANCE (A.4.2.2.3.1)

Past verification methods have included analysis and tests.

The following procedure has been used on past programs:

"(a) High temperature test. The engine, serviced with the oil specified, shall be subjected to a hot soak period of at least ten hours duration at an ambient temperature equal to the sea level hot ground soak atmospheric temperature of figure 5. The 10-hour soak period shall be started after the temperature measurement point specified in the model specification has reached the hot ground soak temperature. After the soak period, and when supplied with fuel and inlet air both at the sea level hot atmospheric temperature of figure 5, a start shall be made with the power lever in the idle position. Immediately after the engine has reached idle speed, an acceleration to intermediate thrust or power shall be accomplished by moving the engine power lever within 0.5 second from idle to intermediate power lever position. The engine shall then be returned to idle and shutdown. The above procedure, including the soak period, shall be repeated.

If more than one fuel or oil is specified, the complete test above shall be repeated using these fuels or oils.

The test will be considered to be satisfactorily completed when, in the judgment of the Using Service, the above two successive starts have been satisfactorily accomplished within the time limits specified on figure 9a; the engine has demonstrated its ability to accelerate to intermediate thrust without exceeding any engine starting or operating limits and there were no fuel or oil leaks.

(b) Low temperature test. The engine, serviced with the oil specified, shall be subjected to a cold soak period of at least ten hours duration at an ambient temperature equal to the sea level cold atmospheric temperature of figure 5. The 10-hour soak period shall be started after the temperature measurement point specified in the model specification has reached the cold atmospheric temperature. At the end of the 10-hour soak period, the electrical connectors shall be disconnected and reconnected to verify the requirements of 3.7.4.3. In addition, the oil reservoir filler cap and the other servicing features shall be functionally checked to demonstrate their proper functioning under cold soak conditions.

After the soak period, and when supplied with fuel and inlet air both at the sea level cold atmospheric temperature of figure 5, a start shall be made with the power lever in the idle position. Immediately after the engine has reached idle speed, an acceleration to intermediate thrust shall be accomplished by moving the engine power lever within 0.5 second from idle to intermediate power lever position. The engine shall then be returned to idle and shutdown. The above procedure, including the soak period, shall be repeated twice for a total of three times per fuel or oil.

If more than one fuel or oil is specified, the complete test above shall be repeated using these fuels or oils. The ambient soak, fuel, inlet air and oil temperatures shall be subject to the limitations of 3.7.3.2.2, 3.7.3.2.3 and 3.7.8.1.1, respectively. The test will be considered to be satisfactorily completed when the above three successive starts have been satisfactorily accomplished within the time limits specified on figure 9a; the engine has demonstrated its ability to accelerate to intermediate thrust or power without exceeding any engine starting or operating limits; there were no fuel or oil leaks; functional checks of electrical connectors and servicing features have revealed no damage or difficulties during operation."

In the above paragraphs the temperature measurement point was usually the mid-engine bearing cavity.

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Background:

The high and low temperature testing, if required, should be scheduled near the end of the qualification program so that the engine configuration tested is as close to the final production configuration as possible. Climatic testing is best accomplished in an environmental facility because it is not subject to the variability of the natural environment.

VERIFICATION LESSONS LEARNED (A.4.2.2.3.1)

High and low temperature testing has identified shortcomings in designs that were not possible to find through analyses or were overlooked in the design process. For example, O-rings were found to leak badly at low temperatures because they lost their resiliency and elasticity. Gearboxes have contracted at a faster rate than the gears therein, causing severe pitch line interference and almost immediate bearing failure. The initial start at an extreme low temperature will cause the oil pressure to “spike”. This rapid increase in pressure has occasionally ruptured the filter casing and cracked oil tubes.

A.3.2.2.3.2 Air starts.

Windmilling, spooldown, and starter assisted airstart envelopes and maximum start times, with customer extractions shall be defined in (a). Air starts within figure 10 shall be considered acceptable when the engine achieves idle speed within the air start times specified and no starting or operating limits are exceeded. The engine shall be capable of consecutive start cycles, with a maximum time interval of (b) seconds between the completion of one cycle and the initiation of the next cycle. Starting time shall be measured from the initiation of the starting sequence to the attainment of (c) percent of stabilized uncorrected engine idle rotor speed. During air starts the engine rotor load shall be at least (d) percent of the total maximum accessory drive load specified for the drives shown in table I.

For turboprop engines that are shut down in flight on patrol aircraft, the engine shall start at the post loiter altitude start envelope of figure 7 after being inoperative for at least (e) hours.

REQUIREMENT RATIONALE (A.3.2.2.3.2)

The engine starting characteristics must be sufficiently defined so that an adequate starting system can be provided. Restart time(s) are needed to meet operational requirements.

REQUIREMENT GUIDANCE (A.3.2.2.3.2)

The following should be used to tailor the specification paragraph:

(a): The envelopes should be defined in the ICD or for programs that do not have an ICD the envelopes should be included in the specification on figure 10.

(b): A value from 0 to 60 seconds.

(c): A value from 95% to 100%.

(d): The contractor should fill in the blank. The recommended value is from 5 to 10 percent.

(e): The loiter period should be specified in the ICD or for programs that do not have an ICD, the interface information can be put into the specification paragraph.

Background:

Air starting is accomplished in flight either by windmill, spooldown, or starter assist. Figure 10 should be formatted by the Using Service and filled in by the contractor, and should include altitude, ambient temperature, bleed air, power extraction, and start time for air starts. The ambient temperatures in the figure should include the temperature extremes shown on figure 5. Aircraft accessory drive loads and power extraction should be included in starting torque requirements. There may be aircraft electrical loads imposed during an air start attempt that the pilot may be unable to uncouple from the engine that would hinder an engine start. Torque corrections for altitude should be defined on a figure. The calculation methods of AIR 1087 should be used to calculate aircraft accessory drag torque during an engine start. For purposes of timing, the start should commence when torque is applied to the starter pad and is complete when idle speed is reached. Voltage drops associated with electrical starts should have

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no effect on other engine systems. Additional guidance is available in AIR 713, AIR 781, and ARP 949. Starts should be satisfactory from either a cold engine or a hot engine condition and from any time interval after shutdown. The engine-starting envelope should be defined and consistent with the aircraft operational requirements. Spooldown start characteristics should be investigated throughout the engine speed range because experience has shown some engines are stall prone in certain rotor speed ranges during spooldown starts.

REQUIREMENT LESSONS LEARNED (A.3.2.2.3.2)

Cold weather conditions can affect compression and combustion functions within the engine.

A.4.2.2.3.2 Air starts.

The requirements of 3.2.2.3.2 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.2.2.3.2)

Air start requirements must be fully verified since this information will be included in the aircraft flight manual.

VERIFICATION GUIDANCE (A.4.2.2.3.2)

Past verification methods have included analysis and tests.

VERIFICATION LESSONS LEARNED (A.4.2.2.3.2)

Stabilization requirements may affect results. These requirements should be defined considering normal and worst case scenarios.

A.3.2.2.3.3 Starting limits.

The starting altitude for ground starts shall be up to (a) km, ((a) ft). The maximum turbine temperature for ground and air starts shall be specified (b). Differences in the altitude starting limits for “hot” and “cold” engines shall be specified (b). Limits with maximum customer bleed air and maximum customer power extraction shall be specified (b). For turboprop applications, the post loiter altitude start limit shall be specified (b).

REQUIREMENT RATIONALE (A.3.2.2.3.3)

Field use of the engine requires information on the ground starting limitations at the extreme high and low temperatures specified. The pilot also requires information on air start limitations.

REQUIREMENT GUIDANCE (A.3.2.2.3.3)

The following should be used to tailor the specification paragraph:

(a): For turbofan, turbojet and turboprop engines: 4.5 km (14,764 ft). For turboshaft engines: 5 km (16,404 ft.).

(b): This information should be put in the ICD and for programs that do not have an ICD put the information here.

The air start altitude and airspeed limits for single or twin-engine aircraft should be as high as possible at an airspeed as low as possible. This helps the pilot preserve altitude during air start attempts. For cargo or patrol aircraft the altitude and airspeed requirements may not be as critical since these aircraft usually have four engines. Turbofan and turbojet engines should be capable of starting at 8 kilometers (26,247 ft).

Background:

The type of fuel and oil determines the extreme temperatures for starting on the ground. The use of customer bleed air and power extraction during air starts is more realistic. These will degrade the limitations for air start.

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REQUIREMENT LESSONS LEARNED (A.3.2.2.3.3)

The air vehicle contractor has wanted to see air start limitations with customer bleed and power extraction consistent with air vehicle usage. Some engines require a starter assist during air starts.

A.4.2.2.3.3 Starting limits.

The requirements of 3.2.2.3.3 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.2.2.3.3)

Starting limits need to be verified.

VERIFICATION GUIDANCE (A.4.2.2.3.3)

Past verification methods have included analysis and tests.

Background:

Tests should be conducted at a test facility that is capable of simulating exact conditions for air and ground starts. The facility needs a capability of operating at -60°F air temperature and the maximum sea level airflow of the engine. For air starts it must be capable of maintaining a constant pressure and temperature at the engine inlet while the airflow is increased during engine acceleration. These tests should be conducted using the worst case fuel and oil combination (i.e., for cold temperatures JP-5 and MIL-PRF-7808 oil).

The test should require air starts and restarts with and without starter assist and at the worst case starting conditions.

VERIFICATION LESSONS LEARNED (A.4.2.2.3.3)

In the past, contractors only supplied starting limits for a “bare” engine, there was no customer bleed air or power extraction. The air vehicle contractors had to develop their own starting limits.

A.3.2.2.3.4 Starting procedure.

The starting procedure shall be simple and shall not require critical timing. The engine shall be capable of starting with the thrust or power demand at idle or above and, after initiation of the starting sequence, being accelerated immediately to any power request. This shall be accomplished within specified engine starting limits and without stall.

REQUIREMENT RATIONALE (A.3.2.2.3.4)

Actions required for ground and air starting must be established to ensure aircraft and engine compatibility.

REQUIREMENT GUIDANCE (A.3.2.2.3.4)

Background:

The air vehicle specification or ICD should keep the starting procedure simple for the pilot. Starts should not require simultaneous manual operation procedures, actuation of switches, levers, or combinations thereof. If a special operating procedure or ground equipment is necessary to obtain satisfactory starts after normal shutdown, the requirement should be specified.

REQUIREMENT LESSONS LEARNED (A.3.2.2.3.4)

Complex starting procedures have been responsible for aircraft loss and an increase in pilot workload.

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A.4.2.2.3.4 Starting procedure.

The requirements of 3.2.2.3.4 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.2.2.3.4)

Starting procedures need to be verified to ensure proper engine operation.

VERIFICATION GUIDANCE (A.4.2.2.3.4)

Past verification methods have included analysis, demonstrations, and tests. The tests have been performed as part of the altitude test.

Background:

Engine starting procedures should be defined prior to and used during engine ground starting verification. Ground test starting systems used for verification should reflect actual start systems as closely as possible to provide valid qualification results.

VERIFICATION LESSONS LEARNED (A.4.2.2.3.4)

None.

A.3.2.2.3.5 Automatic relight.

The engine shall incorporate an automatic rapid relight system that shall immediately reestablish engine operation (without any pilot action) after a main combustor flameout or augmentor blowout has occurred. The automatic relight system shall detect any flameout that may have occurred and initiate a sequence for automatic recovery anywhere in the automatic relight operating envelope shown on figure 7.

REQUIREMENT RATIONALE (A.3.2.2.3.5)

An automatic relight capability is essential in a single engine aircraft and unmanned air vehicles as well as being desirable in other aircraft during critical phases of flight, such as takeoff. The automatic relight capability improves system operating reliability and reduces pilot workload.

REQUIREMENT GUIDANCE (A.3.2.2.3.5)

The following should be used to tailor the specification paragraph:

The contractor should fill in figure 7 showing the limitations of the automatic relight system in the engine-operating envelope.

Background:

The flameout or blowout detection and recovery capabilities of the automatic relight system should be based on aircraft mission requirements, and if required by the aircraft mission, the engine should incorporate one of the following automatic relight systems: 1) an engine initiated system that relights the engine 2) a continuous ignition system that provides ignition via constant igniter firing; 3) a passive ignition system, through coating of the combustor liner, that maintains sufficient combustor temperature to allow automatic ignition.

The system should consider the rotor speed at which relight occurs to avoid stalls and stagnation from low power flameout events. Automatic relight lockout provisions may be required in stall sensitive regions of the engine speed range to prevent stalls.

The system should be analyzed to determine if:

(a) The system is sensitive enough to the measured parameters to generate the required signal before spool-down occurs.

(b) The system will respond quickly enough to prevent rotor spool-down after a blowout or flameout signal is generated.

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(c) The ignition remains on long enough to ensure that stabilized combustion has occurred.

The automatic relight capability shown on figure 7 should be equal to or better than the airstart envelop.

Continuous ignition or rapid relight ignition is useful in preventing or overcoming the flameouts that result from:

(a) ingestion of exhaust products from gun or rocket firings

(b) distortion of the airflow at the engine inlet face during aircraft maneuvers, particularly at high altitudes

(c) ingestion of ice

(d) adverse weather conditions

REQUIREMENT LESSONS LEARNED (A.3.2.2.3.5)

In rotary wing aircraft automatic rapid relight is required to prevent excessive gas generator rotor speed deterioration. Excessive speed deterioration and subsequent relight can cause a “tail kick” torque spike.

A.4.2.2.3.5 Automatic relight.

The requirements of 3.2.2.3.5 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.2.2.3.5)

Automatic relighting procedures must be fully verified during qualification.

VERIFICATION GUIDANCE (A.4.2.2.3.5)

Past verification methods have included demonstrations and tests.

VERIFICATION LESSONS LEARNED (A.4.2.2.3.5)

None.

A.3.2.2.4 Stopping.

Upon receiving a signal from the air vehicle for stopping, the engine shall terminate fuel flow in any operating condition.

Stopping of the engine from any power setting (including augmentor) or at any rate shall not:

(a) result in immediate or subsequent exceedance of any engine limits

(b) adversely impact engine durability, structural integrity, or operational capability

(c) cause unacceptable rotor bowing or rotor seizure, and

(d) delay a satisfactory start or augmentor light.

The engine shall not experience any post-shutdown fires. No damage to the engine shall result from shutting off the fuel supply by the foregoing means or from shutting off the fuel supply to the engine inlet connection during any engine operating condition.

Provisions for stopping the engine by means other than a completely-mechanical system shall be subject to approval of the Using Service.

REQUIREMENT RATIONALE (A.3.2.2.4)

A simple method must be provided to stop the engine for operational and safety considerations.

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REQUIREMENT GUIDANCE (A.3.2.2.4)

All methods for stopping the engine, including emergency provisions, should be specified. Shutdown procedures should not cause rotor seizure or excessive oil leakage on subsequent starts. If the provisions for normal stopping include systems other than mechanical, an additional completely mechanical emergency system for shutting off all fuel flow to the engine should be supplied. For turboprop or turboshaft engines, adjusting the load of the shaft power absorber should not be required when shutting down the engine.

Consideration should be given to whether the air vehicle has a fuel shutoff valve.

If an air vehicle shutoff valve is used, the engine interface should be specified including the volume of fuel that can flow after shutoff and consideration should be made in the engine overspeed and containment verification.

Cool down periods at idle before shutdown have been beneficial in preventing undesirable effects of a high power shutdown.

REQUIREMENT LESSONS LEARNED (A.3.2.2.4)

Hot shutdowns (termination of fuel when the engine is above a stabilized idle condition) have caused rotor seizure and bowed rotors. This resulted in delaying the subsequent start.

A.4.2.2.4 Stopping.

The requirements of 3.2.2.4 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.2.2.4)

Engine stopping and shutdown procedures should be fully verified.

VERIFICATION GUIDANCE (A.4.2.2.4)

Past verification methods have included analyses and tests.

Engine shutdowns during ground testing should be accomplished per established procedures. Engine shutdowns should also be accomplished during the altitude testing to verify the adequacy of altitude shutdown procedures at a wide range of temperature conditions.

Analysis may be needed for verifying stopping and subsequent restarting from high power settings.

VERIFICATION LESSONS LEARNED (A.4.2.2.4)

Multiengine patrol aircraft have routinely shut down engines in flight to conserve fuel.

A.3.2.2.5 Low power conditions.

A.4.2.2.5 Low power conditions.

A.3.2.2.5.1 Idle thrust or power - [ground or flight idle].

When the engine incorporates provisions for an additional idle power setting (e.g., ground idle), the performance shall be included in table II, and the means of accomplishment shall be specified herein or in the ICD.

REQUIREMENT RATIONALE (A.3.2.2.5.1)

Idle power levels must be consistent with aircraft limitations and requirements and allow the engine to maintain steady-state, stall-free operation. Keeping the power as low as possible also lowers the wear on the aircraft brakes and reduces the possibility of aircraft skids on wet or icy pavements.

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REQUIREMENT GUIDANCE (A.3.2.2.5.1)

The following should be used to tailor the specification paragraph:

In determining idle power, consideration must be made of the following:

a. landing and taxi speed requirements

b. bleed pressure to assure minimum aircraft environmental control system performance

c. engine stability and acceleration times when doing engine transients

d. minimum horsepower extraction

Maximum idle power levels and minimum gearbox speeds and bleed pressures should be defined in this specification. The steady-state computer program should be used to define actual thrust, bleed pressure, and rotor speeds.

It is imperative that once the engine cycle has been selected, the relationship between combustor burner pressure and power in the region near idle should be accurately characterized. Combustion blowout limits should be defined near the idle region to prevent flameout during decelerations.

Ground and flight idle speeds must be compatible with power requirements and still provide adequate stall margin for off-idle and bodie accelerations. Attention must be given to the tradeoff between Idle thrust, ECS bleed pressure, and compressor off-idle stability margin. Idle speed should be limited to minimize power while taxiing so the pilot does not need to constantly apply the brakes or use collective to adjust the rotor load.

REQUIREMENT LESSONS LEARNED (A.3.2.2.5.1)

Some engines reduced ground idle power by opening the variable exhaust nozzles while taxiing. Opening the nozzles in-flight had the favorable effect of increasing rotor speeds and transient responses at low power settings.

Early computer simulations contained significant errors during early development phases because of the inaccuracies of component maps at near idle operation and inaccurate definitions of idle power. This led to costly system redesigns in aircraft ECS and braking systems or in the engine control system. Minimum bleed pressure at altitude can affect the bleed takeoff locations(s), heat exchanger sizing, and overall ECS design. Arbitrary idle control schedules in early program phases have resulted in errors in calculation for mission fuel, payload range, airplane sizing, and engine or cycle comparisons. This results from credit being given for fuel burned and distance gained on idle descent mission segments when idle is not well defined at altitude. Definition of idle power can also affect operation and stability of secondary flow systems and flows used for exhaust or aircraft cooling. Off-idle and core stalls have plagued engines where the inherent compressor stall margin at the idle power selected was not sufficient to meet acceleration time requirements. In this case, it would be prudent to select a higher power setting for idle and identify this change to the air vehicle contractor as soon as possible to minimize the overall system impact.

A.4.2.2.5.1 Idle thrust or power - [ground or flight idle].

The requirements of 3.2.2.5.1 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.2.2.5.1)

The idle power must be verified to ensure that it does not exceed the specified values and that it will be compatible with aircraft requirements.

VERIFICATION GUIDANCE (A.4.2.2.5.1)

Past verification methods have included analyses and tests.

Ground and flight idle power as a function of inlet temperature, altitude, and flight Mach should be verified in sea level and altitude facilities. Use of a measured power from a scale force load cell for thrust and

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from a torque meter for shaft horsepower are the preferred methods. Calculation of power is less accurate in most cases.

Measurement of stall margin in the idle region is highly desirable to ensure engine acceleration capability.

VERIFICATION LESSONS LEARNED (A.4.2.2.5.1)

For turbofan and turbojet engines, experience has shown that stability problems have been encountered on numerous engines in the idle speed range.

A.3.2.2.5.2 No-load condition.

For turboprop and turboshaft engines, the no-load condition shall be an engine operating condition of zero delivered output shaft torque while maintaining constant steady-state governed output shaft speed with a specified power absorber and operating at altitudes up to _____ km (___ft). Engine gas generator speed at no-load shall be specified in the performance computer program of 3.2.1.3.

The engine(s) and torque absorber system shall be capable of transitioning between load and no-load conditions without unacceptable mismatches in rotor speeds or torques defined in 3.2.2.7, and without exceeding gas generator operating limits.

REQUIREMENT RATIONALE (A.3.2.2.5.2)

No-load conditions must be consistent with aircraft limitations and prevent output shaft overspeed when power absorber torque is zero. Transitions to and from no-load are important in control system design and flight dynamics.

REQUIREMENT GUIDANCE (A.3.2.2.5.2)

The following should be used to tailor the specification paragraph:

Typical altitudes are from sea level to 4.5 km (15,000 ft) for helicopters and sea level to 9.0 km (30,000 ft.) for fixed wing aircraft.

REQUIREMENT LESSONS LEARNED (A.3.2.2.5.2)

Helicopters have often encountered no-load conditions during combat and terrain following or masking maneuvers when the engine is decoupled from the main rotor system. Transitions to and from no-load have resulted in helicopter rotor speed and rotor thrust droop, shaft overspeed, over torque, exceedance of engine operating limits, and unacceptable flight dynamics.

A.4.2.2.5.2 No-load condition.

The requirements of 3.2.2.5.2 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.2.2.5.2)

The no-load condition must be verified to ensure that the engine does not exceed the specified values of operating limits and that the engine is compatible with the aircraft requirements. Transitions to and from no-load must be verified on a transient basis for performance and control characteristics. Operation at no-load is a power setting lower than idling with a coupled rotor system.

VERIFICATION GUIDANCE (A.4.2.2.5.2)

Past verification methods have included analyses and tests.

Transitions to and from no-load must be verified and tested on a transient basis for performance and control characteristics. Past Army programs have considered no-load as 6.78 N-m (5 ft-lb) or less. It is unlikely that you can completely replicate principal application so the power absorber used in the test should replicate key characteristics of the principal application.

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VERIFICATION LESSONS LEARNED (A.4.2.2.5.2)

None.

A.3.2.2.6 Stability.

Under steady-state operating conditions, engine thrust or power fluctuations shall not exceed (a) percent of the intermediate or maximum continuous between idle and intermediate or maximum continuous conditions, or (b) percent of the thrust or power available at the power level position and operating condition, whichever is less. During operation above intermediate up to maximum augmentor, fluctuations shall not exceed (c) percent of the thrust or power available at that condition. During steady-state operating conditions, the period between major discrete frequency fluctuations, as defined below, (see 3.7.11 Guidance Background) shall not be less than (d) seconds. Main combustor and augmentor blowout margins shall be sufficient to protect against blowouts or flameouts during rapid throttle movements under worst case conditions, and engine control and main burner and augmentor stackups. Bleed air extraction shall not require thrust or power demand modulation to maintain engine stability and limits. The maximum allowable variation shall never be less than (e) thrust or power.

REQUIREMENT RATIONALE (A.3.2.2.6)

A stable operational and stall and surge-free engine is important to the operation of the aircraft. Stall or surge can result in loss of aircraft and personnel. Stable operations within all temperature and rotor speed limits during power transients is necessary to ensure mechanical integrity and parts durability and allow pilots to maintain attention on other aspects of flight.

REQUIREMENT GUIDANCE (A.3.2.2.6)

The following should be used to tailor the specification paragraph:

The contractor should propose the values for (a)-(e). The values for (a)-(d) should be as low as possible to avoid problems during approach for landing and to avoid problems during in-flight refueling.

The following values were used in previous specifications:

(a): A value of 1%.

(b): A value of 5%.

(c): A value of 1%.

(d): A value of 5 seconds.

(e): The value should be between 25 and 50 lbs. thrust and TBD for power.

SAE ARD 50015, Section 3, provides guidance toward developing a methodology for temperature stability assessments. SAE ARP 1420 identifies the methodology for pressure stability assessments. A stability audit CDRL should be prepared requiring the engine contractor to submit updated stability audits to the Using Service at each milestone, or more often as necessary.

REQUIREMENT LESSONS LEARNED (A.3.2.2.6)

Non-recoverable stalls have been encountered in some engines. Investigations involving engine flow paths, control systems, and afterburner operating conditions have been required to establish redesign and operating limitations necessary for satisfactory operation. The problems have reduced the operational utility of the aircraft. Engine stalls have been encountered due to external influences such as inlet distortion during aircraft maneuvers, ingestion of aircraft gun gas, and improper design of aircraft bleed systems that allowed high pressure bleed air to flow into the compressor. It has been found that determining idle and bodie stall margins and burner blowout margins early in the development process is necessary for a successful and timely program. Burner blowout margin problems and off idle and bodie stall problems have been encountered late in the development cycle; these have impacted early production and operational engines. Problems were encountered because of a lack of understanding (database) of the stack up of engine-to-engine and control system variations.

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A.4.2.2.6 Stability.

The requirements of 3.2.2.6 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.2.2.6)

Stability audits and tests need to be conducted during the engine development program to provide assurance that there is adequate stall margin available for the intended aircraft application.

VERIFICATION GUIDANCE (A.4.2.2.6)

The following should be used for tailoring the specification paragraph:

Past verification methods have included analyses and tests. The Using Service and contractor, with air vehicle contractor input, should determine the most critical flight conditions, and these should become the conditions for the stability tests and audits. Tests should be performed in each control mode and with control mode transfer.

Stability audits and tests should be performed to identify marginal stability areas within the engine and flight envelope, and the results should delineate the steady-state and transient stall-free envelopes. The stability audit methodology should be presented in appropriate documentation such as the Interface Control Document and should be per ARP 1420. The stability audit is a stack-up of destabilizing events that may raise the operating line or lower the surge line of the fan or compressor. The audit is statistical in nature, and contains both non-random and random components. The random components represent events that may or may not occur simultaneously with the non-random components, and do not represent precision error. The root-sum-square of the random components is added to the sum of the non-random components to produce a worst case (-3 sigma) stack-up. The objective is that the -3 sigma engine will have positive remaining stability margin at each of the audit points. Note that the Air Force typically uses a -2 sigma criterion for performance requirements but -3 sigma for operability requirements. Special situations such as safety critical points, failure mode accommodation, or field maintenance issues may necessitate unique requirements. The stability audit should identify the critical conditions for stability tests. Stability tests should include verification of the effects of inlet airflow distortion, customer air bleed, power extraction, exhaust nozzle backpressure, and thrust reverser installation factors.

Engine and component tests should be conducted to verify engine stability and ensure stall-free operation prior to aircraft operational tests, to ensure stable operation and to verify margins.

Stability tests should be conducted at:

(a) critical stability margin and flight conditions

(b) high altitude, low Mach conditions

(c) sea level takeoff - static operation

(d) cruise operation

(e) envelope extremes

(f) conditions examining extreme engine or aircraft effects on engine stability

(g) control tolerances and engine-to-engine variations

VERIFICATION LESSONS LEARNED (A.4.2.2.6)

Improper test planning and execution has led to improper conclusions regarding engine stability margins. Raised operating line tests such as in-flow bleed and fuel pulsing afford some measure of assuring adequate surge margin to account for engine-to-engine variations and control tolerances.

With modern electronic engine controls, fuel pulsing can be accomplished with no special equipment other than a "bread-board" computer to interface with the control. In-bleed, on the other hand, requires external air compressors, as well as very specialized test hardware to route the air into the compressor. In some cases, however, compressor in bleed may be the only way to drive the compressor to stall.

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In-flow bleed ovalized/distorted the compressor case on a core test. In-flow bleed worked well on a full-up sea level test on Engine B. Fuel spiking worked well on Engine B, using a breadboard. For a later test on a new Engine B, a fuel spike cart was used. In-flow bleeding has been successfully done on Engine O at AEDC and Engine P at NAPC during altitude testing.

A NASA fiber optic combustor viewing system used during alternate fuels combustor ignition and blowout testing proved more accurate in determining initiation of combustor ignition and blowout than using thermocouples to detect the combustor temperature change in the NAWCAD Trenton Hot Gas Facility during the mid-eighties timeframe.

A.3.2.2.7 Transients.
The engine shall satisfactorily perform any thrust or power transient throughout the operating envelope while meeting the requirements of the specification. Thrust or power requests in any sequence and at any rate for both primary and backup control modes shall not result in exceeding any engine operating limit (including overspeed and overtemperature), result in unstable operation of the engine, or cause any mechanical damage.

For engines operating in backup control modes, thrust or power transient times shall be within (a) of the primary control mode.

For engines operating in primary control mode, thrust or power transient times in all backup control modes shall be within (a) of the primary control mode. Thrust or power transient times for installed engine conditions, including the effects of customer bleed air, customer power extraction, anti-icing bleed, inlet distortion, and the exhaust nozzle, singly or in any combination, for non-standard days throughout the complete environmental conditions and operating envelope, shall not exceed (b) percent of the standard day thrust or power transient times.

For thrust or power lever movements of (c) degrees/second, the transient times shall be the slowest thrust or power transient times for standard day conditions with no customer power or bleed air extractions and no engine anti-icing bleed air operation but with all other engine system bleed air requirements (e.g.; acceleration bleed air, cooling bleed air).

The times required for the engine to accomplish (d) % of the thrust or power change, and the total time required to accomplish the full transient and reach transient stability shall be specified below.

For a turboshaft engine in a helicopter application, the times required for the engine to accomplish (d) % of the power change and the total time required to accomplish the full transient and reach transient stability when the power absorber load is applied or removed per figure 11 shall be specified below.

Continuous sinusoidal throttle transients with peak-to-peak thrust or power change demands of up to (e) % with a frequency up to (f) cycle per second shall not result in a divergent sinusoidal thrust or power response. Thrust or power response phase lag shall be no more than (g) degrees and thrust or power peaks shall be at least (h) % of demand. Symmetrical throttle demands shall not result in divergent symmetrical throttle responses.

REQUIREMENT RATIONALE (A.3.2.2.7)

Thrust or power transient characteristics must be defined to meet aircraft mission performance requirements during critical phases of aircraft operation (i.e.; takeoff, wave off, go-around and combat aircraft acceleration). Power transient acceleration and deceleration times are critical because of their impact on engine life and operability (i.e.; compressor stall, combustor stability). Stable operation within all temperature and rotor speed limits during power transients is necessary to ensure mechanical integrity and parts durability and allow pilots to maintain attention on other aspects of flight.

REQUIREMENT GUIDANCE (A.3.2.2.7)

Today's military engines use a variety of control mode configurations. These range from electrical primary/hydromechanical backup to fully functioning Full Authority Digital Engine Controls (FADEC) with no backup. If a FADEC system has no backup mode, it should contain sufficient redundancy to meet safety of flight requirements. For systems containing primary and backup modes, it is very important to define transient requirements for all modes.

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The following should be used to tailor the specification paragraph:

(a) The contractor should provide the value for all backup control modes.

(b) The value should not exceed 125%.

(c) The power lever rate should be equivalent to a 0.5 second or less movement from idle to maximum. This rate would be used for all acceleration and deceleration transients, including small part power transients to ensure that the slew rate is faster than the engine control-metering device to keep the control on the intended maximum schedule.

(d) The value should be from 95% to 98%.

(e) A value of +5%.

(f) A value of 1 cycle/second.

(g) The value should not exceed 135.

(h) The value should be provided by the contractor.

The following may be used as guidance to define transient requirements in specific, high interest areas of the engine-operating envelope:

A: FOR TURBOFANS AND TURBOJETS:

The standard day engine transient times for turbofans and turbojets shall be as follows:

From sea level up to 10,000 feet and 0 to 150 knots (0.0 to 0.25 Mach number) indicated airspeed (IAS):

a. From idle to intermediate thrust available (a) seconds. From intermediate to idle thrust available, (a) seconds.

b. From idle to maximum thrust available, (b) seconds. From maximum to idle thrust available, (b) seconds.

c. From idle to maximum reverse thrust available, (c) seconds. From maximum reverse to idle thrust available, (c) seconds.

d. From 30 percent intermediate to intermediate thrust available, (d) seconds. From intermediate thrust to 30 percent intermediate thrust available, (d) seconds.

e. From 10 percent intermediate to 20 percent intermediate thrust available, (e) seconds. Power lever movement shall not exceed 125 percent of the power lever movement required to obtain a corresponding steady-state change.

f. From 20 percent intermediate to 10 percent intermediate thrust available, (f) seconds. Power lever movement shall not exceed 125 percent of the power lever movement required to obtain a corresponding steady-state change.

g. From 20 percent intermediate to 30 percent intermediate thrust available, (g) seconds. Power lever movement shall not exceed 125 percent of the power lever movement required to obtain a corresponding steady-state change.

h. From 30 percent intermediate to 20 percent intermediate thrust available, (h) seconds. Power lever movement shall not exceed 125 percent of the power lever movement required to obtain a corresponding steady-state change.

i. The thrust versus time from idle and 10, 20, 30, 40 and 50 percent intermediate to intermediate thrust and to maximum thrust shall be presented in the specification as shown in (i).

j. The thrust versus time from idle and 10, 20, 30, 40 and 50 percent maximum to maximum thrust shall be presented in the specification as shown in (j).

From 10,000 feet up to the absolute altitude:

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k. From idle to intermediate thrust available, (k) seconds. From intermediate to idle thrust available, (k) seconds.

l. From idle to maximum thrust available, (l) seconds. From maximum to idle thrust available, (l) seconds.

For all operating conditions (throughout the complete operating envelope):

m. From intermediate to maximum thrust available (m) seconds. From maximum to intermediate thrust available (m) seconds.

n. From intermediate to maximum reverse thrust available (n) seconds. From maximum reverse to intermediate thrust available (n) seconds.

o. From maximum to maximum reverse thrust available (o) seconds. From maximum reverse to maximum thrust (o) seconds.

B: FOR TURBOPROP/FIXED WING:

The standard day engine power (turboprop) transient times shall be as follows:

Sea level up to 10,000 feet and 0 to 150 knots (0.0 to 0.25 Mach number) indicated airspeed (IAS):

a. No load to maximum continuous or intermediate power available, (aa) seconds. From maximum continuous or intermediate power to no load (aa) seconds.

b. From idle to maximum power available (bb) seconds. From maximum to idle power available, (bb) seconds.

c. From maximum continuous or intermediate power to full reverse power available in (cc) seconds. From full reverse power to maximum continuous or intermediate power available in (cc) seconds.

d. From 30 percent maximum continuous or intermediate to maximum continuous or intermediate power available (dd) seconds. From maximum continuous intermediate power to 30 percent maximum continuous or intermediate power available, (dd) seconds.

e. From 10 percent maximum continuous or intermediate to 20 percent maximum continuous or intermediate thrust available, (ee) seconds. Power lever movement shall not exceed 125 percent of the power lever movement required to obtain a corresponding steady-state change.

f. From 20 percent maximum continuous or intermediate to 10 percent maximum continuous or intermediate thrust available, (ff) seconds. Power lever movement shall not exceed 125 percent of the power lever movement required to obtain a corresponding steady-state change.

g. From 20 percent maximum continuous or intermediate to 30 percent maximum continuous or intermediate thrust available, (gg) seconds. Power lever movement shall not exceed 125 percent of the power lever movement required to obtain a corresponding steady-state change.

h. From 30 percent maximum continuous or intermediate to 20 percent maximum continuous or intermediate thrust available, (hh) seconds. Power lever movement shall not exceed 125 percent of the power lever movement required to obtain a corresponding steady-state change.

i. The power versus time from idle and 10, 20, 30, 40 and 50 percent maximum continuous or intermediate to maximum continuous or intermediate power and to maximum power shall be presented in the model specification as shown in (ii).

From 10,000 feet to absolute altitude:

j. From idle to maximum continuous or intermediate power available, (jj) seconds. From maximum continuous or intermediate to idle power available, (jj) seconds.

For all operating conditions:

k. From maximum power to full reverse power available, (kk) seconds. From full reverse power to maximum power available, (kk) seconds.

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l. From full reverse power to maximum continuous or intermediate power available, (ll) seconds. From maximum continuous or intermediate power to full reverse power available (ll) seconds.

m. From maximum power to maximum continuous or intermediate power available (mm) seconds. From maximum continuous or intermediate to maximum power available (mm) seconds.

C: FOR TURBOSHAFT/HELICOPTER:

For turboshaft engines in helicopter applications, with the power absorber load applied as shown on figure 11, the following requirements apply:

From Sea Level to 3 km (9843 ft), and Mach 0.0 to 0.2:

a. From idle to maximum power, (nn) seconds.

b. From no load to maximum power, (oo) seconds.

c. From maximum power to no load, (pp) seconds.

From 3 km (9843 ft) to maximum altitude, and Mach 0.0 to 0.2:

d. From no load to maximum power, (qq) seconds.

e. From maximum power to no load, (rr) seconds.

D: FOR TURBOFANS AND TURBOJETS (To be used with “A” above):

The following transient criteria are recommended for turbojet and turbofan (TJ/TF) fighter engines. The times should be negotiated among the engine contractor, air vehicle contractor, and the Using Service for each specific program. Transient times may be expected to vary depending on the class of engine (i.e.; bypass ratio, augmentation), and its mission (fighter, bomber, transport, and trainer). For turbofan and turbojet engines, the flight envelope is relatively large with a wide range in transient times. The transient times should be specified graphically on a flight envelope to clearly define the capability. The Using Service may opt to leave the blanks as is, and then the contractor should complete the blanks.

(a): The value should not exceed 5, 5.

(b): The value should not exceed 7, 7.

(c): The value should not exceed 5, 5.

(d): The value should not exceed 4, 4.

(e)-(h): 0.5 to 1.0

(i): Contractor should provide in a table or figure.

(j): Contractor should provide in a table or figure.

(k): The value should not exceed 12, 12.

(l): The value should not exceed 14, 14.

(m) - (o): The value should not exceed 2, 2.

E: FOR TURBOPROP IN FIXED WING APPLICATION (To be used with “B” above):

The following power transient times are recommended for turboprop (TP) engines. The times should be negotiated among the engine contractor, air vehicle contractor, and the Using Service for each specific program. The Using Service may opt to leave blanks as is, and then the contractor should complete the blanks. For this application, removal of load to the output shaft should not compromise the transient responses in any sequence and at any rate.

(aa): The value should not exceed 5, 5.

(bb): The value should not exceed 7, 7.

(cc): The value should not exceed 3,3 for engines that do not have free turbines. To keep the propeller from an over-speed condition on free turbine engines, the engine power may have to be

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reduced to idle, then accelerated up to the reverse power position. The transient times required to do this should be consistent with other transients specified in this paragraph (i.e.; subparagraph bb., above).

(dd): The value should not exceed 4, 4.

(ee)-(hh): The value should not exceed 0.5.

(ii): Contractor should provide in a table or figure

(jj): The value should not exceed 12, 12.

(kk): The value should not exceed 2,2 for engines that do not have free turbines. In order to keep the propeller from overspeeding on free turbine engines, the engine power may have to be reduced to idle, then accelerated up to the reverse power position. The transient times required to do this should be consistent with other transients specified in this paragraph (i.e.; subparagraph bb., above).

(ll): The value should not exceed 2,2 for engines that do not have free turbines. To keep the propeller from overspeeding on free turbine engines, the engine power may have to be reduced to idle, then accelerated up to the reverse power position. The transient times required to do this should be consistent with other transients specified in this paragraph (i.e.; subparagraph bb., above).

(mm): The value should not exceed 2, 2.

F: FOR TURBOSHAFT/HELICOPTER (To be used with “C” above):

The following times are recommended for turboshaft engines in helicopter applications.

(nn): The value should not exceed 7.

(oo): The value should not exceed 3.

(pp): The value should not exceed 3 for attack/scout, 5 for cargo/utility.

(qq): The value should not exceed 8.

(rr): The value should not exceed 8.

Note that helicopter transient to and from no-load are at constant commanded output shaft speed. Transients to and from idle use the minimum self-sustaining engine and output shaft speeds for idle.

The tolerance band on thrust transients for all engines delivered to the Using Service should be +0 and -___ seconds. The minus time (in seconds) should be negotiated between the contractors and Using Service. Quicker acceleration times are desirable as long as they do not excessively use up available surge margin, cause engine compatibility problems with the air vehicle, and reduce engine life. All production engines should have faster transient times than those specified herein, because the specification transient times are based on the poorest performing engine. A very rapid acceleration time is not desirable since this could cause additional problems.

Transient characteristics from initial PLA movement up to 95% or 98%, depending on engine size, thrust change should be shown on figure 22. The transient characteristics between 95% or 98%, and 100% of the final transient stabilized thrust (transient stability) should be shown on figure 22. Transient characteristics from transient stability up to steady-state stability should be shown on figure 22. Transient figures permit a better pictorial presentation of the critical transient area from initiation of the transient to final stabilization.

Component maps should be sufficiently accurate in the idle and sub-idle range to characterize ground and airstart transient times to within 10 percent. Actual engine idle to intermediate and maximum transient times should be accurate to within 5 percent of the transient performance computer program. These accuracies may only be achieved near the end of the development program after component and engine tests have been performed.

Power loss should be defined for a cold engine transient from idle to intermediate and idle to maximum power. The cold engine transient time should be consistent with SLS idle to intermediate and idle to maximum power times. This requirement represents the USAF “tiger start” and is the first start of the day

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when the idle dwell time does not exceed 90 seconds. This scenario represents alert status/base escape requirements.

Special system needs should be considered when transient requirements are established. For example, some applications have a need for expanded inlet airflow distortion limits in order to provide for the desired maneuvering capability.

Engine stall margin available for distortion may be increased at the expense of transient acceleration time if system requirements allow. System requirements for customer power or bleed air capability should also be considered and balanced with distortion and acceleration capabilities. Some systems may require increased idle speeds (ground or flight) which will alter transient time requirements.

REQUIREMENT LESSONS LEARNED (A.3.2.2.7)

Some operational engines have experienced hung accelerations at high altitude, low airspeed regions of the operating envelope.

In the process of reducing transient times considerable internal stall margin is consumed. This resulted in operability problems in many fighter aircraft.

Crisp and repeatable engine response is needed during aerial refueling and aircraft carrier approaches.

A.4.2.2.7 Transients.

The requirement of 3.2.2.7 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.2.2.7)

Transient characteristics must be verified to ensure that the mission requirements of the aircraft can be met.

VERIFICATION GUIDANCE (A.4.2.2.7)

Past verification methods have included analyses and tests.

Transient verification should be performed on all qualification engines, especially during the altitude test. In addition to thrust or power, the engine measured turbine temperature, rotor speeds, fuel flow, and compressor discharge pressure, as well as other pertinent parameters, should be recorded during each transient. Testing, simulations, or analyses should be performed for all control modes.

It should be noted that when the engine is operating in backup mode, full aircraft mission capability may be degraded.

VERIFICATION LESSONS LEARNED (A.4.2.2.7)

On Engine A, when a modified turbine temperature probe design was tested, a power overshoot characteristic occurred. This allowed the engine to meet the specification requirement in terms of time from idle to 95 percent intermediate, but the thrust subsequently dipped below 95 percent before finally trimming to a stable value.

Turboshaft engine transient compliance is difficult to directly determine since the test cell load absorber characteristics usually do not reflect the aircraft rotor system characteristics defined in 3.1.1.9. In one program a reference load path was defined and the test cell Transient Acceleration Dynamometer Loading Curve was developed to determine adequate system transients, compliance with transient time requirements and compliance with the power turbine speed fluctuation requirement. This Curve was based on a representative flight simulation maneuver demanding maximum power from a low power level where the initial power demand was 55% torque per second. The curve is straight line from 0% to 55% torque during the first second and a third order polynomial curve from 55% to 100% during the last two seconds. (Complete details are in ATCOM AMSAT-R-EPT Memorandum Subject: Determination of the Engine D Transient Acceleration Dynamometer Loading Curve, dated May 17, 1994.)

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The turboshaft engine time to power characteristic needs to be judged on its time to power, not time to core speed, pressure ratio, etc. On turboshaft applications, the only real concern to the aircraft is the torque and speed coming out of the shaft; all other items are secondary.

A.3.2.2.7.1 Overshoot and undershoot.

The engine thrust or fan, core, rotor speed overshoot following any acceleration demand shall not exceed (a) percent of the final stabilized steady-state thrust or speed value. The engine thrust or fan, core, rotor speed undershoot following any deceleration demand shall not exceed (b) percent of the final stabilized steady-state thrust or speed value.

For turboshaft engines, during the transient to the stabilized value, output shaft speed and torque shall not exceed the limits of table IV.

REQUIREMENT RATIONALE (A.3.2.2.7.1)

Overshoot and undershoot need to be specified to assess control limiting functions (overspeed and overtemperature), surge margin loss and aircraft system performance. Undershoot is especially important during landing or aerial refueling.

REQUIREMENT GUIDANCE (A.3.2.2.7.1)

The following should be used to tailor the specification paragraph:

(a) and (b): The value should be 2%-3%.

Background:

This requirement is most critical between 95% and 100% power. The overshoot or undershoot should not exceed the recommended values to prevent using up surge margin and excessive engine life. The amount of temperature overshoot or undershoot generally increases with engine bypass ratio, and a direct correlation may be established for a given engine between the amount of overshoot or undershoot and the time at idle prior to acceleration or the time at intermediate maximum prior to deceleration. This paragraph should be tailored to address EGT overshoot or undershoot depending on the engine control mode. Note that the engine should be capable of limiting the overshoot or undershoot.

Helicopter control systems typically try to govern output shaft speed during power transients. Observing rotor system torque limits is typically left to pilot action. The engine control has anticipators to increase transient response, lower overshoots, and lower speed and performance droop.

REQUIREMENT LESSONS LEARNED (A.3.2.2.7.1)

Several Air Force engines have been found to experience turbine temperature overshoot after a burst from idle to maximum. The overshoot was caused primarily by a large loss in High-Pressure Turbine (HPT) efficiency, due to open blade tip clearances caused by the differing thermal responses of the turbine disk, blades, and case. The typical amount of temperature overshoot was approximately 50 degrees.

In one engine, the turbine temperature overshoot resulted in a thrust droop whenever the engine turbine temperature limit was encountered (e.g., a new engine on a hot day or a deteriorated engine on any day). The maximum temperature overshoot and thrust droop occurred 45 to 90 seconds after the throttle movement. Engine rotor temperatures were found to be a big driver on the amount of turbine temperature overshoot.

A cold rotor resulted in a large overshoot - running time at idle from a “cold” shutdown condition warmed the rotor and decreased the overshoot. Conversely, running time at idle on a “hot” rotor cooled the rotor and increased the overshoot.

The “worst case” overshoot condition occurred on a burst to maximum power with less than 5 minutes at idle after a shutdown period of 4 hours or more. Aircraft on alert can suffer overtemperature or low takeoff thrust since no stabilization time is possible.

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Naval aircraft can also experience over temperatures and low thrust on catapult launch on a tropical or hot day. Thrust undershoots and overshoots and lengthy stabilization times have resulted from hysteresis in the control system. In attempts to compensate for low thrust and overtemperature on a hot day takeoff or launch the Using Service has had to implement turbine “fix kits”.

A.4.2.2.7.1 Overshoot and undershoot.

The requirement of 3.2.2.7.1 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.2.2.7.1)

Thrust or power overshoot must be verified to prevent potential performance losses at critical points, minimize engine stalls, and preserve engine life.

VERIFICATION GUIDANCE (A.4.2.2.7.1)

Past verification methods have included analyses and tests.

Thrust or power overshoot problems should be exposed during various sea level and altitude qualification performance tests. Testing with and without customer bleed air and power extraction should be mandatory. Verification should be performed for all control modes.

VERIFICATION LESSONS LEARNED (A.4.2.2.7.1)

Thrust or power overshoot or undershoot has not occurred on all engines or particular models. It can be very elusive to find and may occur only on a deteriorated engine.

A.3.2.2.7.2 Droop.

During any acceleration, no thrust or power droop below (a) percent of the demanded thrust or power value shall occur between the attainment of (b) percent thrust or power change and achievement of transient stability as defined by 3.2.2.7. In addition, there shall be no more than a (c) percent thrust or power droop from the point of transient stability up to attainment of steady-state stability defined in 3.2.2.6.

For turboshaft engines in helicopter applications, the output shaft speed shall not deviate more than the following for the table XLVII power transients at the specified ambient conditions. This shall be verified through computer modeling using either a full, linearized rotor model or a lumped inertia model (attached to the power turbine) derived from the full, linearized rotor model. The actual value for 100% aircraft torque shall be determined using 100% engine output shaft speed and the value of the power required for each engine as specified by the Using Service through coordination with the air vehicle manufacturer.

Table XLVII points:

(d11) The value shall not exceed __(d)__.

(d12) The value shall not exceed __(e)__.

(d13) The value shall not exceed __(f)__.

(d14) The value shall not exceed __(g)__.

(d15) The value shall not exceed __(h)__.

(d16) The value shall not exceed __(i)__.

(d18) The value shall not exceed __(i)__.

(d21) The value shall not exceed __(g)__.

(d22) The value shall not exceed __(h)__.

(d23) The value shall not exceed __(h)__.

(d24) The value shall not exceed __(i)__.

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(d25) The value shall not exceed __(i)__.

(d26) The value shall not exceed __(i)__.

(d28) The value shall not exceed __(i)__.

(d31) The value shall not exceed __(j)__.

(d32) The value shall not exceed __(k)__.

(d33) The value shall not exceed __(l)___.

(d34) The value shall not exceed __(m)__.

(d35) The value shall not exceed __(n)___.

(d36) The value shall not exceed __(n)___.

(d38) The value shall not exceed __(n)__.

(d41) The value shall not exceed __(l)___.

(d42) The value shall not exceed __(m)__.

(d43) The value shall not exceed __(n)__.

(d44) The value shall not exceed __(n)___.

(d45) The value shall not exceed __(n)___.

(d46) The value shall not exceed __(n)__.

(d47) The value shall not exceed __(n)___.

REQUIREMENT RATIONALE (A.3.2.2.7.2)

During a change in power level there is a rapid change in gas temperature. All the parts of the engine do not respond thermally in the same manner to the temperature change. These differential rates of thermal expansion between cases, disks, and other parts can result in a power or thrust droop. This can be a critical condition during and immediately after catapult launch or hot day takeoff from high altitude airfields. It can also be noteworthy for V/STOL aircraft.

For helicopters, rotor speed falloff or “rotor droop” is an important consideration for low level flight dynamics, designing the engine control, and aircraft interface. Thrust loss occurs when the pilot commands a sudden increase in blade pitch (rotor power required) and the rotor slows down while waiting for engine power to increase. When engine power increases, it may exceed the rotor system torque limit (at the lower speed) and thereby lengthen the time to accelerate and achieve thrust. If the rotor speed or thrust droop becomes too large, generators can come off-line, sink rates can become excessive, and the pilot’s ability to recover from extreme droop is reduced.

REQUIREMENT GUIDANCE (A.3.2.2.7.2)

The following should be used to tailor the specification paragraph:

(a): A value between 95% and 98%.

(b): A value between 95% and 98%.

(c): A value between 0% and 2%.

(d): The value should not exceed 12%.

(e) The value should not exceed 6%.

(f) The value should not exceed 4%.

(g) The value should not exceed 3%.

(h) The value should not exceed 2%.

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(i) The value should not exceed 1%.

(j) The value should not exceed -8%.

(k) The value should not exceed -4%.

(l) The value should not exceed -3%.

(m) The value should not exceed -2%.

(n) The value should not exceed -1%.

Background:

The engine thrust droop should not exceed 2% to maintain air vehicle performance demands. The so-called “thrust droop” phenomenon that has been encountered on many engines is due to transient thermal response mismatch between the massive turbine rotor disk and the turbine case. A cold rotor (i.e., one at or near idle for less than about 15 minutes) expands more slowly than the case during acceleration, resulting in transiently open clearances and high turbine temperatures. On engines with automatic temperature limiting controls, the control will cut back fuel flow to hold the temperature limit, resulting in the “thrust droop”. On engines with no automatic temperature limiting, thrust is maintained but exhaust gas temperature (EGT) may exceed limits, resulting in aborted takeoffs or requires maintenance actions. Thus this paragraph should be tailored to address thrust droop or EGT overshoot depending on the engine control mode. Note also that automatic turbine clearance control devices may be utilized to counteract the problem in severe situations; however, they have an associated weight penalty on the engine.

For helicopter applications, consideration of air vehicle system effects should be made. The transient maximum time to power requirements of 3.2.2.7 may be decreased to meet the table XLVII droop requirements, however, the engine manufacturer becomes specification bound to the new transient requirements for qualification, verification and delivery. The values of table XLVII will be specified by the Using Service. The full, linearized rotor model must be included in the procurement or RFP to provided for torsional stability analysis and should therefore be available for the droop analysis.

Helicopter engines may have backup control modes. Operation in a backup mode may affect the droop characteristics of the engine.

REQUIREMENT LESSONS LEARNED (A.3.2.2.7.2)

In one instance of thrust droop, the problem was reduced or eliminated by the addition of an active turbine clearance control system. In another instance, the engine added “droop compensation” in the control followed by “extended life” turbine kits.

Transient thrust droop has often been overlooked by the air vehicle and engine contractors in the vehicle proposal phase. The assumption was often made that the thrust droop could be compensated for and thereby neglected in the proposal phase. Experience shows many cold engines encounter some amount of thrust droop when operating on a temperature limit. An exception to the cold engine droop phenomena was the small engine class used in helicopters and light transports. Since the engines were physically small, parts and components heated up rapidly and cold engine thrust droop was less obvious.

Thrust droop on helicopters resulted from decreasing rotor speed with sudden changes in blade pitch.

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A.4.2.2.7.2 Droop.

The requirements of 3.2.2.7.2 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.2.2.7.2)

Power or thrust droop must be verified to prevent potential performance losses at critical points, minimize engine stalls, and preserve engine life.

VERIFICATION GUIDANCE (A.4.2.2.7.2)

Past verification methods have included analyses, demonstrations, and engine tests, and rotorcraft flight test.

Power or thrust droop problems should be exposed during various sea level and altitude qualification performance tests. Testing with and without customer bleed air and power extraction should be mandatory. In theory, the thrust droop may occur whenever a cold engine of sufficient size is operating on a temperature limit. Since a deteriorated engine is more often on a temperature limit, the thrust loss may be more apparent. The droop can also vary with engine-to-engine variation.

There can be excessive cost and time required in providing a power absorber that accurately simulates a helicopter rotor drive system, only the transient time requirements will be demonstrated during engine testing. It should be agreed upon by the Using Service and the prime contractor that the output shaft speed droop requirement be demonstrated both through computer modeling of the engine and aircraft rotor drive system, and through flight test of the engine and aircraft system.

Verification should be performed for all control modes.

VERIFICATION LESSONS LEARNED (A.4.2.2.7.2)

Power or thrust droop has not occurred on all engines or particular models. It can be very elusive to find and may occur only on a deteriorated engine.

A.3.2.2.8 Windmilling.

The engine shall be capable of continuous windmilling for ___ hours, without damage to the engine, and without affecting air restart and operational capability.

The following information shall be specified (for an uninstalled engine):

a. The limits of windmilling operation, including curves showing windmilling rotor speed and windmilling drag vs. flight Mach number.

b. The oil consumption rate during windmilling operation.

c. In the event of lubrication failure, the time duration and limits of windmilling operation after depletion of oil supply. This is not required for turboshaft applications.

d. The power extraction available during windmilling operation as shown on figure 12a. This is not required for turboshaft applications.

e. The customer bleed air extraction available during windmilling operation as shown on figure 12b. This is not required for turboshaft applications.

REQUIREMENT RATIONALE (A.3.2.2.8)

The windmilling capability of the engine should be compatible with the intended air system usage and with all air vehicle systems that use bleed air and power extraction.

Installed windmilling performance is needed to determine aircraft get-home capability and engine out trim for cruise, takeoff, or landing. Installed windmilling performance and capability can vary significantly from traditional specification curves with high loss or buried engine installations.

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REQUIREMENT GUIDANCE (A.3.2.2.8)

The following should be used to tailor the specification paragraph:

The time interval required should be for a period equal to the longest mission. This requirement is for a multi-engine aircraft.

REQUIREMENT LESSONS LEARNED (A.3.2.2.8)

Multiengine aircraft have used windmilling techniques to conserve fuel during cruising and to conserve brakes while taxiing.

In past programs, acceptable minimum times for subparagraph (c), above, have been one hour at full speed (USAF fighter program), and one hour at 30% speed (USN fighter program). Rotor axial loading during windmilling is a force opposite to normal loading that has caused oil seals to leak.

A.4.2.2.8 Windmilling.

The requirements of 3.2.2.8 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.2.2.8)

Windmilling should be verified to ensure that oil leakage and other related problems would not occur in flight.

VERIFICATION GUIDANCE (A.4.2.2.8)

Past verification methods have included analyses and tests.

During altitude testing, a functional test run of windmilling drag versus flight Mach number and windmilling oil consumption should be accomplished. Estimates of windmilling time after oil supply depletion and the power extraction capability should be determined from these tests. Verification of windmilling should be done in the air starts verification test.

Windmilling data (oil consumption, rotor speeds, etc.) should be analyzed to verify windmilling capability.

An assessment should be made of windmilling performance installed in the aircraft using the performance computer program(s) of 3.2.1.3. This is necessary for high loss or buried type installations that affect system flow balance and pressure ratio across the engine.

VERIFICATION LESSONS LEARNED (A.4.2.2.8)

None.

A.3.2.2.9 Reverse thrust.

For turbofan and turbojet engines, the operating envelope, limitations, and description for thrust reverser operation shall be specified herein. The percentage of total engine thrust used for reverse thrust shall be ___.

REQUIREMENT RATIONALE (A.3.2.2.9)

The reverser may be required to slow the aircraft after touch down, aircraft maneuvering, and aircraft braking and backing.

REQUIREMENT GUIDANCE (A.3.2.2.9)

For turboprop and turboshaft engines replace the specification paragraph text with “Not Applicable”.

The contractor should provide the percentage of total engine thrust used for reverse thrust.

The following requirement may be included in the specification paragraph:

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"The reverser system shall be capable of operating from the fully stowed position to the full deployed position in no more than (a) seconds and from the fully deployed position to the fully stowed position in no more than (b) seconds at all power settings and throughout the reverser operating envelope."

The following should be used to tailor the preceding paragraph:

(a): The value should not exceed 2 and is defined in ICD.

(b): The value should not exceed 5 and is defined in ICD.

The mechanism and thrust reverser-operating envelope should be defined by the contractor together with the level of performance of reverse thrust, deployment time, and stowing time. Reverser performance should be included in the steady-state and transient performance computer programs. Thrust reversers should be designed to avoid re-ingestion of exhaust products and foreign object damage. Air vehicle backing requirements, if any, should be included in the thrust reverser design. The amount of reverse thrust should be specified in the ICD.

Background:

Reverse thrust systems are mainly used to brake large turbofan or jet powered aircraft after landing. Military fighter and attack aircraft tend to use wheel brakes, arresting gear systems, or drag parachute systems for braking. Thrust reversers have the disadvantage of providing a weight penalty to the aircraft.

REQUIREMENT LESSONS LEARNED (A.3.2.2.9)

Past problems have included re-ingestion of hot exhaust products, foreign object damage due to exhaust gas impingement on runways, and engine stalls due to transient operation.

A.4.2.2.9 Reverse thrust.

The requirements of 3.2.2.9 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.2.2.9)

The performance and operating characteristics of the thrust reverser must be verified.

VERIFICATION GUIDANCE (A.4.2.2.9)

For turboprop and turboshaft engines, replace the specification paragraph text with "Not Applicable".

Past verification methods have included analyses and tests.

If tests are used, they should include component and engine tests. Component tests may include model tests to establish the aerodynamic characteristics, and mechanical tests should demonstrate functional characteristics of the thrust reverser structure.

Engine reverser tests should include operation throughout the full range of engine power settings and thrust reverser positions. Full range transients from full forward thrust to full reverse and full reverse to full forward thrust should be accomplished to demonstrate transient requirements and satisfactory operation.

VERIFICATION LESSONS LEARNED (A.4.2.2.9)

Experience has shown that strain gages on fan and turbine blades are very useful in defining blade stress levels during reverser operation. This type of testing is useful to establish operating limitations relative to re-ingestion of reverser exhaust gas into the engine inlet.

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A.3.2.2.10 Steady-state and transient airflow.

The engine maximum allowable steady-state and transient airflow limits shall be shown in table IV. The transient airflow limits specified shall include augmentor ignition and termination, as applicable, as well as thrust or power demand increase and decrease. The steady-state airflow of any engine shall not vary by more than (a) percent of the steady-state intermediate airflow computed by the steady-state computer program of 3.2.1.1 and shown in tables IIa, IIb, IIc, and III. The transient airflow of any engine shall not vary by more than (b) percent of the transient airflow computed by the transient computer program of 3.2.1.2.

Transient airflow shall not deviate from the maximum steady-state airflow by more than (c) percent and shall be controlled to assure compatible engine inlet operation. The engine steady-state and transient airflow limits shall include engine deterioration, control tolerances (including temperature lags and transients), engine-to-engine variations and shall apply to the nonstandard atmospheres of this specification as shown on figure 5.

The effects of customer air bleed and power extraction shall be identified.

REQUIREMENT RATIONALE (A.3.2.2.10)

Airflow values and tolerances must be established to meet power requirements and to ensure inlet compatibility. Steady-state operational tolerances must be determined and controlled so that the lower and upper tolerance airflow limits still allow stable engine operation. Transient airflow deviation from maximum steady-state airflow must be established to ensure stall-free operation.

REQUIREMENT GUIDANCE (A.3.2.2.10)

The following should be used to tailor the specification paragraph:

(a)-(c): The value should not exceed 3%

The contractor normally provides the maximum and minimum airflow limits, including the deviation during transient operation to be specified in table IV.

This requirement is primarily applicable to turbofan and turbojet engines.

Background:

Airflow limits are a fallout of trade studies that optimize installed performance based on mission requirements while maintaining compatible inlet and engine operation during steady-state and transient operation. Both maximum and minimum airflow limits must be considered to assure stall-free operation. The engine control system must maintain upper and lower airflow limits both during steady-state and transient operation.

Engine-to-engine airflow variations must be considered within the desired limits. Steady-state and transient airflow values for any specific flight condition and bleed and power extraction levels are required in the computer programs of 3.2.1.1 and 3.2.1.2.

REQUIREMENT LESSONS LEARNED (A.3.2.2.10)

Gusts, maneuvers, and abrupt altitude changes can cause transient airflow to exceed maximum steady-state limits and lead to engine stall. Low inlet mass flow (near idle and minimum airflow limits) has caused planar pulse distortion generated in the inlet resulting in engine stall. This has classically been associated with supersonic operation where it is known as buzz; however, it has also existed in the subsonic region of the envelope at levels severe enough to cause engine stall. If the inlet has been properly characterized at low mass flows, the lower airflow limit can be properly scheduled in the engine

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control to maintain compatible inlet and engine operation. If the engine control system is limited in its capability, inlet redesign may be necessary.

A.4.2.2.10 Steady-state and transient airflow.

The requirements of 3.2.2.10 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.2.2.10)

Verification of airflow is required to ensure compatibility with the aircraft inlet.

VERIFICATION GUIDANCE (A.4.2.2.10)

Past verification methods have included analyses and tests.

Accurate verification of airflow limits and requirements can be obtained only through controlled test conditions. Altitude wind tunnel testing with adequate instrumentation and instrumented flight-testing are methods that may be used for verification. There are several methods of measuring airflow, such as venturis, bellmouths, through fan and compressor pumping characteristics and exhaust nozzle flow coefficients. It is desirable to confirm the airflow values by measurement with at least two different techniques.

VERIFICATION LESSONS LEARNED (A.4.2.2.10)

Experience has shown that airflow limits must be maintained throughout the entire engine population. One engine program experienced difficulty because control system stackup and engine-to-engine variation caused some engines to exceed the required airflow limit; for stable operation, an airflow topping device was added to the control system to maintain stable engine operation throughout the engine envelope during cold weather operation.

A.3.2.2.11 Inlet airflow distortion.

The engine shall not surge, stall, flameout, or incur any damage with the steady-state or time variant inlet distortion (pressure, temperature, or any combination of both) shown on figure 13. The aerodynamic interface plane shall be defined as (a). Engine stability and performance assessments shall use the methodology and inlet distortion descriptors defined in (b) for total pressure distortion. Temperature distortion methodology and the methodology for combination of both pressure and temperature distortion shall be specified.

REQUIREMENT RATIONALE (A.3.2.2.11)

Aircraft inlet distortion is an important consideration in the design of gas turbine engines. The distortion limits provide interface information for making engines and aircraft compatible.

REQUIREMENT GUIDANCE (A.3.2.2.11)

The following should be used to tailor the specification paragraph:

(a): The air vehicle and engine contractors should define the aerodynamic interface plane. If no agreement can be made between the contractors, the Using Service should make the decision.

(b): The contractor should specify the method used to assess inlet total pressure distortion. ARP 1420 has been used for many engine programs.

The most severe distortion conditions should be identified on figure 13. Normally 7 patterns are specified, but the number may be minimized to reduce testing.

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When selecting patterns, the primary objectives are:

(1) to include the most severe patterns (i.e., the heaviest distortion and greatest engine sensitivity)

(2) to include patterns associated with critical mission points (i.e., combat maneuver points and stability margin pinch points).

A secondary objective is to include patterns of each type, if the aircraft inlet produces them: tip radial, hub radial, 1/rev circumferential, and multiple-per-rev circumferential. In lieu of specific patterns, a plot of distortion handling capability may be used that shows the maximum circumferential and radial distortion, separately or in combination, that the engine can tolerate.

As an option, the following may be transferred verbatim into the specification paragraph to supplement the requirement:

> "An engine stability and performance assessment for total pressure and total temperature distortion shall be performed using the methodology and inlet distortion descriptors defined below:
>
> a. A distortion descriptor methodology indicating the maximum distortion limits at which the engine shall operate without stall, rotating stall, and surge, throughout the complete environmental conditions and operating envelope, due to steady-state and time-variant inlet air total pressure and temperature variation, shall be defined in the specification. Where exhaust nozzle backpressure effects on the fan and gas generator affects the tolerance of the engine to inlet air pressure variation, the effect shall be specified in the specification. Engine stability and performance assessments shall be performed using the methodology and inlet distortion descriptor elements defined in ARP 1420. This information shall be provided with the specification.
>
> b. The engine contractor, using the distortion descriptor methodology shall provide approximately five sets of pressure distortion patterns (screens), that, with Using Service approval, reflect the most severe conditions permitted within the limits of the distortion descriptor specified for the engine. Approximately three of these distortion patterns (screens), exceeding the most severe inlet spatial and planar pressure conditions expected in the weapon system installation, shall be used for IFR testing. All distortion patterns (screens) shall be used during ISR and OCR engine testing. One of the patterns (screens), if required by the Using Service, shall be used for the Acceptance Test surge and stall margin test. For each set of inlet flow distortion data specified, the engine interface operating conditions shall be defined, as applicable, in terms of Mach number, altitude, power setting, customer bleed air, customer power extraction, etc. Measurements of the engine inlet total pressure, temperature, and flow variation shall be made at the engine inlet (aerodynamic interface plane), which shall be defined in the specification. All inlet instrumentation utilized in measuring airflow, pressures and temperatures, the arrangement, location, response, and instrumentation accuracies shall be consistent with the guidelines of ARP 1420 and defined in the specification. For each set of specified inlet flow distortion data, the total airflow, average total pressure recovery and pressure and temperature for each individual probe shall be specified. The available surge margin, maximum distortion limits, and the differences between the contractor's methodology and ARP 1420 shall be presented in the specification."

SAE ARP 1420 provides generalized parameters for inlet airflow total pressure distortion. The distortion limits should be defined in terms of both spatial and planar content. When the inlet flow descriptions and key inlet operating conditions are established, a table should be constructed by the contractor to specify the engine requirements.

Consideration should be given to:

a. Mach number and altitude

b. inlet angle of attack and yaw

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c. inlet flow margins
d. inlet flow descriptions (steady-state and dynamic pressures and temperatures)
e. planar wave description (frequency and amplitude)
f. steady-state and dynamic flow disturbances
g. planar pulse distortion under low inlet mass flow conditions
h. cross wind effect during ground operation
i. control tolerances and engine-to-engine variations
j. compression system bleed and vane schedule
k. shift in compressor operating line
l. steam ingestion for carrier based turbojet, turbofan and turboprop engines

Background:

An in depth discussion of inlet total pressure distortion for gas turbine engines is contained in SAE AIR 1419 and provides the technical background and development of the guidelines of SAE ARP 1420. A current assessment of the inlet and engine temperature distortion problem is contained in SAE ARD 50015. The aircraft engine industry is most knowledgeable about steady-state total pressure distortion. Pressure distortion is the most prevalent disturbance, easiest to measure, and can be duplicated by screens. Time variant pressure distortion requires more complex measurement system and elaborate equipment to duplicate in the laboratory and test cell. Engine performance degradation, including power loss due to combustor blowout and compressor instability, has been attributed to engine inlet total temperature distortion. It can be a time-variant spatial temperature distortion or a temperature ramp. The armament gas ingestion requirement (3.3.2.6) is a form of time variant distortion. Steam ingestion (3.3.2.7) is a combined form of pressure and temperature distortion.

REQUIREMENT LESSONS LEARNED (A.3.2.2.11)

Early definition of the inlet distortion limits has proven to be necessary for successful integration of the engine into the aircraft and should be provided by the engine manufacturer as part of the tradeoffs that must be made during the design and development process. Early model tests of the inlet provide assurance in the validity of the distortion limits. Inlet distortion has resulted in premature fan blade failure and has also been shown to have an impact on steady-state and transient augmentor operations. Reported inlet distortion causes include aircraft maneuvers, afterburner operation on fan engines, ingestion of gases from armament firing, ingestion of steam from catapult launch systems, water ingestion, bird ingestion, ingestion of engine exhaust gases from thrust reverser systems, and ingestion of engine exhaust during helicopter and V/STOL aircraft operations. Fighter aircraft were reported to be most susceptible to inlet distortion in the upper left-hand corner of the aircraft flight envelope. Aircraft that fly with speeds in excess of Mach 1 have experienced engine surge caused by aircraft inlet shock waves.

A.4.2.2.11 Inlet airflow distortion.

The requirements of 3.2.2.11 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.2.2.11)

Verification is needed to ensure the engine can tolerate the specified levels of inlet distortion.

VERIFICATION GUIDANCE (A.4.2.2.11)

Past verification methods have included analyses and tests.

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Inlet pressure distortion tolerance can be verified by several techniques: distortion screens or distortion simulators in engine test facilities, and actual aircraft inlets in wind tunnel facilities. If screens are used, the distortion patterns should be selected on the basis of peak instantaneous dynamic distortion level from inlet tests. A minimum of five screens should be tested. For turbofan engines, distortion sensitivity is determined by throttling the engine to surge at constant rotor speed behind each screen. The process is repeated at several levels of rotor speeds for each screen. Consideration should be given to testing classical patterns (e.g., pure circumferential and pure radial content) as well as flight patterns for the purposes of methodology development and verification. Fan distortion transfer characteristics are determined by measuring distortion in front of the fan and the compressor. Accurate measurement of distortion transfer requires adequate instrumentation and screen rotation. By rotating the screen to (usually 3) different orientations, the distorted flow field is more accurately mapped. Care should be taken that the engine match point does not shift during the test. This may be accomplished by use of a remotely controlled screen rotator (so the engine does not have to be shut down to rotate the screen) and by locking the engine control settings. Also, the effect of fan operating line on distortion transfer may be significant, so it is recommended that distortion transfer data be collected at a high (near surge) operating line as well as the normal operating line.

Inlet temperature distortion tolerance can be verified by one of the following techniques in a test chamber: a controlled rocket charge in front of the engine, a hydrogen fueled distortion generator, or a hot air injection system. Flight tests are also used to verify engine distortion by firing rockets in flight. The effects of customer bleed air, power extraction, and exhaust nozzle backpressure on engine tolerance to inlet airflow distortion should be determined. Tests should be accomplished to verify the engine capability to accept distortion generated by air vehicle inlets.

A complete aircraft inlet can be used during the engine development for distortion verification, but it does not ensure total engine and air vehicle integration. The test must simulate the most severe cases of distortion at the flight condition and maneuver at which they occur. The attachment of the aircraft inlet to an engine in a test chamber may not simulate the worse case condition.

The results of the verification should be incorporated into the transient performance computer program for a better understanding of the effects of distortion.

Consideration should be given to running endurance testing with a realistic, steady-state distortion screen.

The effects of crosswinds on inlet distortion are generally uncharacterized on military programs because inlet designs on military applications tend to be significantly different in geometry. Any available data on the effects of crosswind distortion patterns generated by the inlet geometry should be thoroughly analyzed to help minimize stall potential.

VERIFICATION LESSONS LEARNED (A.4.2.2.11)

Aircraft inlet tests have not always simulated the worse case pressure distortion for the installed engine. The tests used scale models that did not have engines installed. These tests did not always produce the correct Strouhal Number or the Reynolds Number that are experienced in the full-scale aircraft with the engine installed. In spite of these drawbacks, distortion screens, calibrated for the proper distortion level, have proven to be effective, economical tools for engine distortion testing. Other tools for generating inlet distortion were a planar wave generator or a distortion valve. Distortion transfer coefficient is extremely sensitive to the accurate mapping of the distorted flow field.

Experience has shown that the distortion transfer coefficient may be in error by as much as 50% if screen rotations are not used, even with the full complement of 40 probes per SAE ARP 1420. In some cases instrumentation is less than the recommended levels due to inoperative probes or missing rakes. In one test program, the engine manufacturer elected not to remove the engine gearbox to install rakes on the lower half of the compressor face, leaving a gap of greater than 100 degrees unmapped. The performance computer program was also used for analytical distortion verification.

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The earliest cases where armament gas ingestion caused significant engine surge problems were during the 1950s on the Aircraft O and P programs. During the late 1960's, Aircraft N encountered steam ingestion induced surges during catapult launch on aircraft carrier. A 1979 study documented the effects of gas ingestion from a nose mounted cannon on Aircraft Q.

Inlet distortion from crosswinds have caused engine stalls on development programs. These stalls were not predicted due to a lack of available inlet model distortion data.

A.3.2.2.11.1 Pressure and temperature rate of change.

The engine shall withstand the following rate-of-change of pressure and temperature, separately or in combination, without surge, stall, flameout, or damage:

Parameters	Rate-of-change	Duration
Engine inlet pressure	(a) kPa/sec (psi/sec)	(b) sec
Engine inlet temperature	(c) °C/sec (°F/sec)	(d) sec

REQUIREMENT RATIONALE (A.3.2.2.11.1)

Rate of pressure and temperature changes must be specified since large changes in pressure and temperature may occur at the engine face due to ingestion of turbine engine exhaust, gun gas, or nuclear and nonnuclear explosions.

REQUIREMENT GUIDANCE (A.3.2.2.11.1)

The following should be used to tailor the specification paragraph:

(a) - (d): The contractor should provide the values.

The maximum rates of pressure and temperature change required for the engine should be determined from examination of the intended mission of the aircraft. Rate of change of pressure may be set by nuclear or conventional weapons effects or climb/dive requirements on the aircraft. The effects of rocket/gun gas ingestion may determine temperature rate of change. Steam ingestion can also cause high rates of change in temperature of greater than 1000 deg F/sec.

Background:

Temporary losses in stall margin result when pressure and temperature changes cause substantial variations in density of the air entering the engine. Engine exhaust from minimum interval takeoff operation of tactical and strategic aircraft can reduce the surge margin of engines.

Past turboshaft requirements have been engine inlet pressure rate of change 48.3 kPa/sec (7 psi/sec), 0.40 sec duration and inlet temperature rate of change 1000°C/sec (1800°F/sec), 0.15 sec duration.

REQUIREMENT LESSONS LEARNED (A.3.2.2.11.1)

Fighter and attack aircraft in the past have experienced compressor stall when exposed to rapid temperature changes caused by rocket or gun gas ingestion.

A.4.2.2.11.1 Pressure and temperature rate of change.

The requirement of 3.2.2.11.1 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.2.2.11.1)

Verification must be conducted to ensure the engine meets the requirements.

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VERIFICATION GUIDANCE (A.4.2.2.11.1)

Past verification methods have included analyses and tests.

Analytical predictions should be verified by testing, if technically and economically feasible, using instrumented component rigs, core engines, and full engines, as appropriate. Pressure distortion testing should be accomplished using an air jet distortion generator or distortion screens in front of the engine face. Use of a shock tunnel to generate appropriate full-face overpressure transients should be considered. High response pressure probes mounted in an approved array should be placed directly behind the distortion device to verify the pressure values presented to the inlet of the compression component being tested. Temperature measurement sensors should be mounted in an array of rakes to validate the temperature changes and rates being experienced by the engine. The inlet temperature distortion tests can be done in flight test aircraft.

VERIFICATION LESSONS LEARNED (A.4.2.2.11.1)

The Defense Nuclear Agency has developed a test technique to satisfactorily demonstrate the adequacy of turbomachinery against the stated overpressure requirement.

Tests on Engine AA have shown that turbomachinery can be aerodynamically and mechanically sensitive to relatively small increases in full-face overpressure that results from nuclear and non-nuclear explosions. These tests showed that the engine failed before any part of the air vehicle failed. Therefore, engine survivability from a nuclear or non-nuclear explosion is a serious concern.

A.3.2.2.11.2. Engine Recoverability.

The engine shall recover from any stalls encountered during steady-state operation within, or any transient exceedence of, the operating envelope of 3.2.2.2, and/or with transient exceedence of inlet pressure and temperature distortion limits as defined in 3.2.2.11 and 3.2.2.11.1 without pilot action. After the destabilizing influence(s) have been removed, the engine shall meet the requirements of this specification, and shall automatically recover to a safe, steady-state operating condition in a maximum of ___ seconds longer than the sum of the transient times and start times as defined in 3.2.2.7 and 3.2.2.3.

REQUIREMENT RATIONALE (A.3.2.2.11.2)

The engine needs to be capable of recovery in the unlikely event that it does stall. Operational engines have experienced engine stalls both in their operating envelope and by temporarily exceeding some operational limit. The presence of engine recoverability capability enhances the safety and reliability of weapon systems, particularly single engine applications.

REQUIREMENT GUIDANCE (A.3.2.2.11.2)

The following should be used to tailor the specification paragraph:

The engine must recover once the cause is removed, with the minimum time being the combined time to start and transient to the pre-stall power setting. Some applications may not require a specified recovery time and this portion of the requirement may be tailored out.

REQUIREMENT LESSONS LEARNED (A.3.2.2.11.2)

None.

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A.4.2.2.11.2 Engine Recoverability

The requirements of 3.2.2.11.2 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.2.2.11.2)

Verification is needed to ensure the engine can recover from stalls once the destabilizing influence has been removed.

VERIFICATION GUIDANCE (A.4.2.2.11.2)

Past verification methods have included analyses, engine ground level and altitude tests.

VERIFICATION LESSONS LEARNED (A.4.2.2.11.2)

None.

A.3.2.2.12 Gas path and measurement plane temperature limits.

The first stage high or low pressure turbine rotor inlet gas temperature limit(s) shall be _____.

The maximum allowable steady-state gas temperature shall be shown in table IV. The location of the measurement plane (including station identification number) shall be shown in table II. No engine shall operate above the maximum allowable measuring plane turbine temperature. When a direct measuring metal temperature sensing system is provided at the measurement plane, the maximum allowable steady-state metal temperature(s) shall be shown in table IV. The maximum allowable transient measured temperature during starting and acceleration shall also be specified herein or shown in table IV.

REQUIREMENT RATIONALE (A.3.2.2.12)

The temperature of the hot gas in a gas turbine is important to reliability, endurance, and performance and limits must be established.

REQUIREMENT GUIDANCE (A.3.2.2.12)

The following should be used to tailor the specification paragraph:

The contractor should provide the temperature limit and the location in the engine where it occurs.

The following requirement may be included in the specification paragraph blank above:

“Maximum allowable operating gas temperature(s): The maximum allowable operating steady-state and transient gas temperature(s) at the first stage high or low pressure turbine rotor inlet(s) shall be specified for the range of engine inlet air temperatures defined by figure 6. No engine shall operate above the maximum allowable first stage turbine rotor inlet temperature. Additionally, the maximum allowable operating transient gas temperature(s) during starting and acceleration shall be specified.”

If short period ratings (less than 30 minutes) are used, the maximum allowable temperature will exceed the highest value for continuous operation. The Army requires the engine to be capable of operating at the maximum allowable temperature for 12 seconds during accelerations above idle. A maximum allowable starting temperature that can be tolerated by the engine for 12 seconds should also be specified.

Some Navy engines have required emergency thrust ratings above intermediate power. These engines exceeded the normal maximum operating temperatures at intermediate. The maximum allowable temperature limit(s) should allow for droop compensation of engines operating at the highest steady-state

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temperature. This condition may last for one to three minutes per application. When feasible, the maximum allowable temperature limit(s) and material selection should allow for near-term thrust or power growth or “throttle push”.

Prior to IFR, FFR, ISR and OCR, the correlation between measured and gas temperature at the measurement plane should be updated for review by the Using Service. Since the temperature may vary during development, and changes in the temperature affect engine life, the Using Service should monitor it during testing and be aware of changes the contractor makes.

REQUIREMENT LESSONS LEARNED (A.3.2.2.12)

None.

A.4.2.2.12 Gas-path and measurement plane temperature limits.

The requirements of 3.2.2.12 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.2.2.12)

All engine limits must be verified.

VERIFICATION GUIDANCE (A.4.2.2.12)

Past verification methods have included analyses and tests.

VERIFICATION LESSONS LEARNED (A.4.2.2.12)

The full impact of operating limits is not discovered until the engine has been in service for a number of years.

A.3.2.2.13 Surface temperature and heat rejection.

The maximum operating surface temperatures and heat rejection rates of the engine shall be shown on figure 14a. The engine surface emissivity shall be presented as shown on figure 14b. Accessory pad loading, compressor bleed air conditions, oil system cooling requirements, and air and gas leakage from engine case flanges and split lines shall be used in establishing engine heat rejection rates. The conditions surrounding the engine, the engine power condition, and the oil system temperatures for which the surface temperatures are applicable shall be shown. For components and externals on the surface of the engine, the specified component and externals surface temperature and heat generation curves may differ from the engine temperature as shown on figure 14a.

REQUIREMENT RATIONALE (A.3.2.2.13)

The surface temperature and heat rejection must be controlled to ensure compatibility between the engine and the aircraft for safe and durable operation.

REQUIREMENT GUIDANCE (A.3.2.2.13)

The following should be used to tailor the specification paragraph:

The heat rejection and surface temperature data presented should be for the conditions listed below. If heat rejection rates during flight and ground operating conditions other than those listed below are critical or limiting, that data should be specified:

FOR TURBOJET, TURBOFAN AND TURBOPROP ENGINES:

a. Maximum power, Sea Level static, hot atmosphere of figure 5.

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b. Maximum power, Sea Level, maximum stagnation inlet air temperature for hot atmosphere of figure 5.

c. Maximum power, 36,089 ft or maximum altitude of figure 5, or maximum stagnation inlet air temperature, whichever is higher.

d. Maximum power, highest altitude of figure 5, hot atmosphere of figure 5, subsonic level flight.

e. Low power, Sea Level, low speed loiter condition, tropical or hot atmosphere of figure 5.

f. Idle power, Sea Level static, hot atmosphere of figure 5, ground operation.

FOR TURBOSHAFT ENGINES:

a. Maximum power, sea level static, 55°C (131°F).

b. Maximum power, 1.2 km (3937 ft), 45°C (113°F) ambient, maximum stagnation inlet air temperature.

c. Maximum power, highest altitude of figure 5, maximum stagnation inlet air temperature for the hot atmosphere of figure 5.

d. Contingency power, sea level static, 55°C (131°F).

The conditions to be specified in subparagraphs a - c are generally considered the most severe heat rejection conditions.

As a minimum, the following factors should be considered in the analysis: horsepower extraction, compressor bleed air conditions, anti-ice operation, oil system cooling requirements, engine skin temperature and radiation properties, and air/gas leakage from flange joints and split lines.

Skin temperature has an influence on the adjacent air vehicle structure and significantly affects the weight and cost of that structure and the engine bay cooling requirements.

This paragraph provides interface data necessary for the air vehicle contractor to determine the amount of heat that must be dissipated by the air vehicle from the engine skin. The surface temperatures and emissivity are necessary to determine the radiant heat transfer to the inner skin of the air vehicle engine compartment.

Multiple flight points are required to survey bay cooling flows and thermal soak back. Aircraft with largely composite structure are more difficult to cool and more vulnerable to engine surface heat and temperature.

REQUIREMENT LESSONS LEARNED (A.3.2.2.13)

In the past, there were errors made by the requirement to specify component and externals surface temperatures. Components are items furnished by the engine manufacturer and component surface temperatures are the engine manufacturer’s responsibility.

Accessories are items mounted on the engine and not furnished by the engine manufacturer. Accessory surface temperatures are not addressed in this specification.

In the Aircraft I program, Engine V components are primarily fuel actuation driven, which needs to be considered in fire safety assessments. Engine V’s Systems Safety Work Group recommended for this engine that surface temperatures at sea level static, with no airflow, needed to be in the 450°F to 500°F range to prevent hydraulic or fuel leak auto-ignition.

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A.4.2.2.13 Surface temperature and heat rejection.

The requirements of 3.2.2.13 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.2.2.13)

Surface temperatures and heat rejection rates must be verified.

VERIFICATION GUIDANCE (A.4.2.2.13)

Past verification methods have included analyses and tests.

The required verification should include, but not be limited to the considerations listed in 3.2.2.13. The determination of whether to use analysis or test should be based on the accuracy and relative simplicity of each method.

The engine should be tested at the highest inlet temperature it expects to encounter in the flight envelope. All other temperatures, including surface temperatures, are proportional to the inlet air temperature. Heat rejection also is a function of temperature and pressure.

This should also be in air vehicle verification as aircraft tests.

VERIFICATION LESSONS LEARNED (A.4.2.2.13)

None.

A.3.2.2.13.1 Controls and external components limiting temperatures.

Engine components mounted on the engine shall not exceed their allowable temperatures when surrounded by still air under the following conditions:

A tabulation of the maximum ambient air temperature surrounding each component, the limiting surface temperatures for all components, the time period(s) applicable, and the limiting cooling fuel inlet temperature for fuel cooled components, for the most severe engine condition(s), shall be specified in table V. The specific points of measurement, where the ambient air temperature and surface temperature with respect to the three coordinate axes of the component are maximum, shall be specified. The engine components requiring special cooling and ventilation shall be listed in table V. Length of operation of these components without special cooling shall be defined in table V.

Off-engine components shall not exceed their allowable temperatures when surrounded by still air at _ degrees with no special cooling such as forced ventilation or refrigeration. The specific points of measurement, where the ambient air temperature and surface temperature with respect to the three coordinate axes of the component are maximum, shall be specified.

REQUIREMENT RATIONALE (A.3.2.2.13.1)

The Using Service needs to ensure that the engine components will not exceed their limiting temperatures under the most adverse conditions of heat generation and rejection.

REQUIREMENT GUIDANCE (A.3.2.2.13.1)

The contractor should provide the environmental conditions that cause the components to encounter their limiting temperatures. Some of the conditions that have been used on other programs are:

a. Continuous operation (including high and low power settings) with ambient air at the maximum stagnation temperature.

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b. Flight shutdown from the most adverse condition and continued soaking with ambient air at maximum stagnation temperature.

c. Ground shutdown with ambient air at the sea level hot resoak temperature conditions of figure 5 with no special cooling. For turboshaft engines, ground shutdown with ambient air at 55°C (131°F) with no special cooling. The value is to be determined by analysis in the intended installation.

The contractor should establish the maximum allowable temperature that the components can withstand and list these components and corresponding limits in the engine model specification so that adequate cooling schemes can be provided.

REQUIREMENT LESSONS LEARNED (A.3.2.2.13.1)

Non-unidirectional flow in the engine compartments has resulted in overheating of components. Overheated fuel, which was intended for cooling electronic components, damaged electronic controls, resulting in an engine overspeed and turbine failure.

A.4.2.2.13.1 Controls and external components limiting temperatures.

The requirement of 3.2.2.13.1 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.2.2.13.1)

Component limiting temperatures must be verified to ensure that they are not exceeded.

VERIFICATION GUIDANCE (A.4.2.2.13.1)

Past verification methods have included analyses and tests.

Applicable cooling data should be obtained for the conditions tested for all individually cooled components and externals and for each section of the engine case (compressor case, combustor case, turbine case, and exhaust duct). Sufficient data should be obtained to ensure that the engine and all critical components are within the specified temperature limits.

An instrumented engine in a simulated or actual engine bay should be tested with maximum allowable fuel and ambient air temperatures. Data should be taken at critical component locations at the power and airflow conditions that produce the highest component temperatures. Cooling data should be obtained for the conditions tested for all individually cooled components/externals and for each section of the engine case (compressor case, combustor case, turbine case, and exhaust duct). Sufficient data should be obtained to ensure that all critical components are within specified temperature limits.

VERIFICATION LESSONS LEARNED (A.4.2.2.13.1)

None.

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A.3.3 Environmental conditions.

A.4.3 Environmental conditions.

A.3.3.1 Atmospheric conditions.

A.4.3.1 Atmospheric conditions.

A.3.3.1.1 Humidity.

The engine shall meet all requirements of this specification when subjected to _____% humidity at _____ °C (°F).

REQUIREMENT RATIONALE (A.3.3.1.1)

Humidity must be required because it may physically and chemically deteriorate engine material. Humidity can lead to: (1) swelling of materials due to moisture absorption, (2) loss of physical strength, (3) changes in mechanical properties, (4) degradation of electrical and thermal properties in insulation materials, (5) electrical shorts due to condensation, (6) binding of moving parts due to corrosion or fouling of lubricants, and (7) deterioration of electrical components.

REQUIREMENT GUIDANCE (A.3.3.1.1)

A value of at least 95% humidity at 32°C (90°F) should be used to tailor the specification paragraph.

Background:

This requirement is intended to assure that certain components will not be affected by exposure to high humidity conditions, such as in tropical areas.

The 95% humidity requirement is based on MIL-STD-810 requirements.

REQUIREMENT LESSONS LEARNED (A.3.3.1.1)

Fundamental design problems have resulted because of the long-term effects of humidity related corrosion of metals due to oxidation and of effects on electrical and electronic components.

A.4.3.1.1 Humidity.

The requirements of 3.3.1.1 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.3.1.1)

The deteriorating effects of humidity must be determined. Components must be tested and the materials of construction analyzed to verify their ability to withstand high humidity conditions.

VERIFICATION GUIDANCE (A.4.3.1.1)

Past verification methods have included analyses, demonstrations, and tests.

The following procedure may be used as a verification method:

> "The requirements of 3.3.1.1 shall be verified by test as follows: Components shall be subjected to a humidity test per MIL-STD-810, Method 507 (Humidity), figure 507.4-1 (Aggravated temperature-humidity cycle) for a test duration of 15 cycles (360 hours). The components shall be subjected to pre- and post-test calibrations. Operational verification for brief periods during the test phase is allowed subject to Using Service approval. At the end of the tenth cycle (not less than 240 hours), electrical and electronic components shall be dried without forced

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convection for 12 hours in air at a temperature not greater than 21°C (70°F) and a relative humidity not less than 50 percent. At the end of this period, the component shall be subjected to four power supply over voltage transients and transient spike voltages. The component shall then complete the humidity test. After the post-test calibration, the component shall be disassembled and inspected. There shall be no corrosion or other defects present that will affect the function or structural strength of the component. Additional success criteria shall be per MIL-STD-810."

MIL-STD-810 cannot be invoked on a blanket basis, but must be tailored to suit a particular operating environment.

Background:

A full-up engine test to satisfy the humidity test requirement cannot be accomplished in the engine corrosion test. The engine corrosion test does not contain the humidity requirements of 3.3.1.1. Operation of the component during the corrosion test deters humidity buildup within the component, and would diminish the ability of the facility to maintain a specified humidity level. Therefore, engine components should be subjected to a humidity test per MIL-STD-810.

VERIFICATION LESSONS LEARNED (A.4.3.1.1)

Past engines have had flow path coatings on compressor cases that swelled when exposed to humidity. When the engine was started there was interference in the blade tip clearances for the compressor blades. This caused some blade tip corner losses or extreme rubbing in the flow path coating. This would not have been discovered until production, if the humidity test had not been run during development.

A.3.3.1.2 Fungus.

The engine shall meet all specification requirements when subjected to fungus.

REQUIREMENT RATIONALE (A.3.3.1.2)

Fungus requirements must be specified because it can create problems such as metal corrosion, grease hardening, reduction of insulation effectiveness, and bridging of conductors causing electrical shorts and arcing. These problems are particularly severe in tropical and Naval environments.

REQUIREMENT GUIDANCE (A.3.3.1.2)

Background:

New engine designs are now including porous ceramic material layers on hot section or augmentor parts. These materials are cooled with compressor or fan bleed or discharge air. The air flows through the “pores” in this material. Any entrained particles larger than the pore sizes, will be deposited in this material. This can degrade durability and operability.

Fungus may be resisted by the selection of non-nutrient materials and the use of special fungus resistant coating on nutrient materials, as well as other methods, as approved by the Using Service.

Adding a requirement that selection of all non-nutrient materials be mandatory is a very difficult requirement because this would preclude the use of polymer composite materials. Some polymer composite materials in gas turbine engines contain fillers that are fungus nutrient materials.

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REQUIREMENT LESSONS LEARNED (A.3.3.1.2)

Some of the typical fungus-related problems experienced in Southeast Asia:

Microorganisms digest organic materials as a normal metabolic process thus degrading the substrate, reducing the surface tension, and increasing moisture penetration.

Enzymes and organic acids produced during metabolism diffuse out of the cells and onto the materials and cause metal corrosion, glass etching, hardening of grease, and other physical and chemical changes to the materials, such as insulation resistance and arcing.

The physical presence of microorganisms produces living bridges across components which have resulted in electrical failures.

A.4.3.1.2 Fungus.

The requirements of 3.3.1.2 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.3.1.2)

The engine must be resistant to fungus growth and fungus related problems. Fungus resistance must be verified by satisfactory completion of a component fungus test or analysis.

VERIFICATION GUIDANCE (A.4.3.1.2)

Past verification methods have included analyses, demonstrations, and tests.

The following procedure may be used as a verification method: “The requirements of 3.3.1.2 shall be verified by test as follows: "Components shall be subjected to a fungus test per MIL-STD-810, Method 508, or evidence must be provided in a fungus analysis report which indicates all materials used are fungus inert per MIL-HDBK-454, Guideline 4, or have been previously tested per MIL-STD-810.”

RTCA-DO-160C, Section 13, Fungus Resistance may also be used to tailor the specification.

Background:

A full up engine test to satisfy the fungus test requirement cannot be easily accomplished, therefore engine components should be subjected to a fungus test per MIL-STD-810, Method 508, which is the recognized standard military fungus test. The contractor should review each engine component for the materials used that comprise all parts and determine those materials that are fungus resistant. MIL-HDBK-454, Guidance 4, should be used for electrical components.

VERIFICATION LESSONS LEARNED (A.4.3.1.2)

Past engine materials have contained fungus food (e.g., carbon particles in flow path coatings) that was found to promote fungus growth during testing for fungus.

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A.3.3.1.3 Corrosive atmosphere.

The engine shall meet all requirements of this specification during and after exposure to the corrosive atmosphere of table VI for (a) corrosion cycles. The engine shall not deteriorate more than (b) percent in thrust or power, or gain (c) percent in specific fuel consumption. Engine parts shall show no corrosion which impairs function, component integrity, or prescribed maintenance. Corrosion protected parts shall show no corrosion effects after cleaning and stripping of corrosion protection coatings.

REQUIREMENT RATIONALE (A.3.3.1.3)

This requirement is specified because during engine operation in the proximity of the sea and salt air environment, engine materials can be susceptible to corrosive attack. The air and gas path areas of the engine can be particularly susceptible to corrosive attack. Also, if external engine hardware, such as fasteners and clamps, corrode in service, drilling and cutting bolts to maintain or repair engines result in high maintenance costs.

REQUIREMENT GUIDANCE (A.3.3.1.3)

This requirement is applicable for aircraft deployed and operating in a corrosive environment, such as on aircraft carriers, at or around seacoast bases, and in some desert environments such as in the Middle East, where sand composition contains a relatively large concentration of salt. If the weapon system will not operate in such environments, then this paragraph and 4.3.1.3 should be labeled “Not Applicable.”

The following should be used to tailor the specification paragraph:

(a): A value of at least 25 cycles should be used unless a reduced value is defined based on a system exposure analysis.

(b): The number should not exceed 5 percent.

(c): The number should not exceed 5 percent.

Background:

The number of test cycles is directly related to the hot section life. In some engines, the compressor materials may require the test cycles to be doubled to the cold section life. One hundred cycles were used for engine A and H since these engines compressor cases and seals were susceptible to corrosion. Twenty-five cycles would be the minimum number of cycles to verify corrosion resistance for 1000 operational hours per Gas Turbine Materials Conference Proceedings in Oct 72, paper, "Service Corrosion Problems Produced in the Test Cell", by W. W. Wagner of the Naval Air Propulsion Test Center, Trenton, NJ. Further information on the condition of engine Q1 engine hardware post-test condition and test results are discussed in, Propulsion Technology & Project Engineering Department NAPTC-PE-5 report, "Full Scale Corrosion Test of the J79-GE-8 Engine", final report NAVAIR Work Unit Assignment No. NAPTC-337, prepared by J. E Newhart and W. W. Wagner, and approved by Commander, USN Director PT&PE Department F.B. Stumcke, JR. Feb 73.

Engine corrosion will vary with environment and exposure time. Critical corrosion parameters are time at temperature and salt concentration. Materials may continue to corrode after exposure to salt laden air. There are many environments that are conducive to corrosion. Operation in proximity to salt water or salt laden air is one of the biggest causes of corrosion. The burning of fossil fuels causes air pollution and an acidic environment. Exposure time will depend on aircraft installation, mission, and physical location of the aircraft. Naval aircraft, particularly helicopters, are very susceptible to corrosion.

The corrosion of parts in the engine is a large contributor to increased maintenance and aircraft downtime. Some parts require special protective coatings (see 3.1.3, Materials, processes and parts) to make them corrosion resistant. For example, no corrosion resistant material could match the high

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strength to weight ratio of certain steel alloys. Therefore, a corrosion resistant coating needs to be used. These coatings must withstand extreme temperatures and thermal stresses as well as environmental effects. Refer to MIL-HDBK-1568 for selection of materials and processes for corrosion prevention and control.

The state-of-the-art of combustors is increasing the required hot section inspection to 1000 hours and longer. As a result, engine inspection periods are longer with less possibility of uncovering a corrosion problem in the engine. Therefore, there is a need for ensuring engine parts do not corrode for longer periods of time.

Corrosion may take place while the engine is operating or when the aircraft is parked on the ground in a static condition. Cold corrosion continues long after the engine is removed from the corrosive environment. Hot corrosion is recognized as a form of high temperature oxidation caused by sodium sulfate from sea water, condensing on turbine blades and vanes. Hot corrosion or sulphidation will only take place under the proper condition of temperature (760 - 980°C (1400 - 1800°F)) and hot gas pressure. These are metal temperatures, gas path temperatures will be higher. Approximation can be made for the metal temperatures, but care must be used since some hardware is air cooled. The engine parts should be resistant to both types of corrosion.

The Using Service should look into the aviation 3M database for current estimates of annual costs of corrosion induced maintenance actions to get a better feel for the severity of corrosion on engines of various applications.

REQUIREMENT LESSONS LEARNED (A.3.3.1.3)

A COMNAVAIRLANT message in 1974 stated: “exorbitant maintenance man-hrs, 2.5 million documented in FY73, on corrosion control and are barely holding our own against deterioration effects.” A COMNAVAIRPAC message in 1974 also discussed the “major problem” of structural corrosion and cited huge man-hour expenditures on various aircraft for corrosion control and repair. This message also urged that the “best anticorrosion materials, processes and methods” should be “specified in new hardware procurement contracts.” Two specific engine corrosion problems are on file: (1) Engine K first stage fan turbine blade failed due to FOD at a crack which propagated rapidly under wet salt conditions and (2) Engine AH compressor second through thirteenth stage wheels were being scrapped at an increasing rate at overhaul because of corrosion pits in or near the dovetails. Also, Engine AH had a history of turbine sulphidation problems and the Engine AI had magnesium gearbox corrosion problems in the 60’s.

A point paper by LCDR M. Johns, NAVAIR, dated 8/10/90, stated that squadron HS-10 removed three USN Aircraft C, Engine C engines for low power. The cumulative operating hours were between 650 and 857 hours. Examination of disassembled engines revealed 1st stage GGT stage 1 blade tip and LE damage caused by hot corrosion.

ASME 70-GT-96 provided information on salt concentrations at various altitudes above salt water. J.C. Arribat presented information such as concentrations of 0.03 g/m^3 at 8 meters above sea water, with a total ingestion of 900 grams/hour for a helicopter was to be expected.

Application of the standard corrosion test to Engine D would require first stage gas generator turbine blade tip caps, which are an expensive feature. Analysis of the Army operating environment resulted in tailoring the requirement to fifty cycles based on 150 hours of test operation.

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A.4.3.1.3 Corrosive atmosphere.

The requirements of 3.3.1.3 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.3.1.3)

The corrosion resistant properties of the engine must be verified.

VERIFICATION GUIDANCE (A.4.3.1.3)

If requirement 3.3.1.3 is labeled “Not Applicable”, then this paragraph should be labeled “Not Applicable”.

Past verification methods have included inspections, analyses, demonstrations, and tests.

The following engine test may be used as a verification method: ”The test may be conducted using a new or newly overhauled engine, operated per table VI and figure 23. Turboshaft engines will be subjected to two hours of sand ingestion testing per 4.3.2.4 prior to the start of this corrosion susceptibility test. Prior to starting the corrosion susceptibility test, the engine shall be disassembled sufficiently to inspect the surface condition of all parts normally exposed to atmospheric conditions. Detailed photographic coverage of these parts shall be taken. Upon reassembly and after an initial calibration, including data required to establish compliance with transients of 3.2.2.7, the engine shall be subjected to the cycles of 3.3.1.3 for 24-hours each per table VI. Anti-icing and customer bleed air should be extracted for ten-minutes of the table VI three-hour phase 1 operating cycle. Should engine performance during the test deteriorate more than the amount specified in 3.3.1.3 from that determined during the initial calibration, water washing shall be accomplished per 3.7.12. Water wash shall be used only to correct for performance loss. External water wash is not permitted. If performance cannot be recovered after water washing, the engine shall be disassembled and inspected to determine the effect of the corrosion testing on performance loss. If the unrecoverable performance loss is determined to be caused by a problem not related to corrosion, the engine shall be repaired and reassembled and the test continued. During the test, the engine shall be subjected to internal inspections after every tenth cycle to detect any evidence of corrosion or progression of corrosion of internal parts. Additional inspection may be conducted with approval of the Using Service.

After completion of the corrosion susceptibility test, the cleaning procedure of 3.7.12 shall be accomplished prior to recalibration of the engine. During recalibration, the applicable transient performance specified in 3.2.2.7 shall be demonstrated. Following recalibration, the engine shall be disassembled and inspected for evidence of corrosion. Detailed photographs shall be taken of all parts which show evidence of corrosion. Metallurgical analyses that completely characterize the types of corrosion found and test specimen evidence shall be available for review.

The corrosion susceptibility test will be considered to be satisfactorily completed when upon recalibration: (1) the engine non-recoverable performance deterioration at steady-state measured temperatures or Engine Pressure Ratio (EPR) at intermediate or maximum continuous and above does not exceed the deterioration specified in 3.3.1.3, for the same value of measured temperature or EPR obtained during the initial calibration, (2) the engine exhibits not more than the increase in specific fuel consumption specified in 3.3.1.3, for the same values of measured temperature or EPR obtained during the initial calibration at idle and above, (3) the transients conducted are per the specification requirements, and (4) the extent of corrosion evident at test completion satisfies the following criteria for all engine parts:

All internal parts exposed to gas path air, upon cleaning, shall show no impairment of their function due to corrosion. Minor corrosive attack is acceptable only when the part design criteria (e.g., fatigue resistance) are not affected.

All corrosion-protected parts shall indicate no effects of corrosion upon cleaning and stripping of the protective schemes. Recoating of parts designed for recoating shall restore such parts to as new condition.

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All other parts shall show no corrosion that affects component integrity or prescribed maintenance procedures.

Background:

An initial verification by analysis can be made of the materials used in the engine to determine if any are prone to corrosion. The engine contractor may want to demonstrate the corrosion properties of a material by the use of a “strip test”. It has been found that hot corrosion can occur in a wide range of temperatures 760 - 980°C (1400 - 1800°F). Therefore the engine should be operated over a wide range of turbine temperatures. This is to ensure all possible engine parts are exposed to the hot corrosion temperature range. A mission type of engine operation could be substituted for the generic temperature test, if the range of temperatures are covered. A mission-oriented test is designed to determine structural durability while a corrosion test is designed to determine the corrosion resistance properties of the engine under various environmental conditions in the shortest possible time. Conducting the test according to table VI means conducting the test 24 hours a day, seven days a week. This is not always possible, because facility or maintenance problems can disrupt the test. The test schedule must be flexible to allow for any disruptions.

The external air velocity was not specified in the previous test, only the salt concentration. A high air velocity across the external surface of the engine would deposit more salt than a low velocity air stream for the same period of time. A static engine may only be exposed to the outside winds. Therefore, a velocity of 4.6 - 6.1 m/sec (15 - 20 ft/sec) was chosen to represent an average wind velocity. On some corrosion tests conducted at NAWCADTRN, the engine contractor requested the engine to be water washed if there was more than a 24 hour break in the test. This does not represent field operation of the engine. Fresh water is a valuable commodity on board aircraft carriers. It is rare to use it for water washing engines and external water wash is rarely practiced even on land bases. The test should simulate actual use of the engine so water wash should only be used to restore engine performance.

Dilution of the salt concentration is permitted to give better distribution and atomization of the salt spray when testing large engines with high airflow. Dilution has caused problems with the sodium monitor and the salt sampling instrumentation. On small engine tests, the dilution actually improved the performance of the engine. The 40:1 dilution is an arbitrary figure, which is a compromise for the spray nozzle equipment used at NAWC AD TRENTON. Generating small droplets at very low flows is another challenge. Small droplet size is needed to effectively atomize the salt solution. The 25-micron droplet size is the smallest droplet size DoD equipment will generate. Smaller size droplets would be desirable.

Turboshaft engine operation can include significant time in sand and dust environments. To simulate this effect, the corrosion susceptibility test is preceded by two hours of sand ingestion per 4.3.2.4.”

VERIFICATION LESSONS LEARNED (A.4.3.1.3)

The corrosion test developed in the mid-1960’s for the general engine specification has uncovered numerous corrosion problems on such engines as Engine AB, Engine AD, and Preliminary Flight Rated Engine C. These tests have resulted in material changes to improve the life of engine parts in service.

NADEP Alameda cited reduction gearbox corrosion on engines stored in improper storage containers.

A corrosion test on engine H was conducted at NAWC-AD Trenton in 1997 to determine the susceptibility of the engine design to the effects of a corrosive atmosphere and to qualify it for use in a maritime environment. The test was to include 12 cycles consisting of salt water ingestion, dynamic soaks and static soaks to simulate approximately 1000 hours of operation in the maritime environment.

Though this test was planned from the beginning of engine H development program, the TEAM was challenged on the benefits of the test from both the contractor and Program Officials. The contractor cited the design similarities between engine H and engine AH which has many years of service in the

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Navy therefore negating the need to perform the test. Program Officials were concerned about the costs and benefits associated with this test. The TEAM pressed Program Officials and contractor personnel in what they believed the benefits of conducting the test would provide to the Program and the Navy. Eventually a test program was agreed to consisting of 12 cycles vice the originally planned 25 cycles.

Approximately one third of the way through the test the engine gas generator rotor was found to be seized following an attempted start. Borescope inspection of the compressor did not reveal the cause of the failure and a decision was made to disassemble the engine.

Disassembly revealed heavy swelling, blistering, delamination and other corrosive type attack of the aluminum-graphite compressor inner band abradable seal material. The deterioration of this material caused interference between the compressor vanes and the knife edge seals on the compressor disks and the subsequent seizure.

Redesign efforts were conducted to replace the seal material with one having improved corrosion resistant properties. A coupon test program was conducted at NAWC-AD Patuxent River to evaluate several alternate materials. The results of the test program confirmed the failure mechanism and the limitations with the aluminum-graphite coating. A replacement material was identified which was determined to be a superior material from a corrosion resistance standpoint. The new material also successfully completed the engine corrosion test at Trenton. The combined results of the coupon testing and engine corrosion testing allowed for the necessary qualification efforts, including performance testing, to replace the current with the new material as the compressor inner band seal material and introduce it into production engines. The change was not available for flight test engine incorporation however, these engines are being reworked on an opportunistic basis (during engine repair).

Aircraft F No. 10 Incident - MCAS New River: During a two month operational test in 1998 for aircraft F, an incident occurred where an engine failed to start. It was discovered that the gas generator rotor was seized and attempts to free the rotor were unsuccessful. The engine was removed and sent back to Engine Company D for investigation.

Disassembly of the engine revealed swelling and blistering of the compressor inner band seal material similar to what was seen during the corrosion test and is believed that this caused the compressor rotor to seize. Aircraft No. 10 had conducted several over water flights (low level hover deployment of troops and gear) just prior to the incident.

Review of the maintenance records indicate that a fresh water rinse was attempted after a flight (standard procedure for over water operations) but was aborted due to an unrelated problem with the aircraft. It was discovered that fresh water was introduced into the engine just prior to the aircraft incident which did not allow it to be properly dried out. The engine sat for approximately 2-3 days in this condition. It is well documented that failure to properly dry an engine following fresh water rinses can lead to excessive corrosion attack.

A.3.3.1.4 Icing conditions.

The engine shall meet the requirements of this specification while operating in the meteorological conditions shown on figures 15a and 15b and table VII, with not more than (a) percent total loss in thrust or power and (b) percent total increase in specific fuel consumption at all operating conditions. After removal of the icing conditions, the engine shall perform the same as prior to encountering the icing conditions. No damage to the engine shall be allowed as a result of operating in the icing environment.

REQUIREMENT RATIONALE (A.3.3.1.4)

Service aircraft are intended to operate under all types of environmental conditions, therefore the engines must be capable of operating under conditions conducive to icing.

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REQUIREMENT GUIDANCE (A.3.3.1.4)

The following should be used to tailor the specification paragraph:

(a) and (b): The number inserted should not be more than five percent except engines with IPS should not be more than 9 percent.

RTCA-DO-160C, Section 24 may also be used to tailor the specification paragraph.

Background:

Stratiform clouds occur at altitudes from 0.9 km (3,000 ft) to 1.8 km (6,000 ft). These clouds have a low to moderate liquid water content (LWC) and are characterized by figure 15a. Cumuliform clouds occur from 1.2 km (4,000 ft) to 7.3 km (24,000 ft) and have a LWC moderate to high. The characteristics of icing conditions in these clouds are depicted by figure 15b. Icing encounters in cumuliform clouds tend to be of relative short duration (about one minute). The icing conditions of this requirement are fairly consistent with encounters in cumuliform clouds, but the engine is required to operate for a much greater period than would be expected for these types of encounters.

Civilian aviation figures are located in Appendix C of FAR Part 25. Data sources for figures 15a and 15b are from NACA technical notes 1855 and 2569 published in 1949 and 1951, respectively. These data have been generally accepted as the aircraft and engine industry standard.

REQUIREMENT LESSONS LEARNED (A.3.3.1.4)

Some contractors have interpreted that 5.0 percent power loss should only be applied to the intermediate thrust or power demand setting since this is the highest available thrust condition. The contractor's reasoning is that at any part power the throttle could be moved forward if the anti-icing or icing condition caused a thrust loss at the original throttle setting. In the case of a turboprop or turboshaft engine that runs at constant speed, the loss in power is automatically compensated by an increase in turbine temperature. These engines are not always run at maximum power. Therefore, they have excess power to make up for small loses in power.

The USN Engine C Model Specification permitted a loss of 8.5 percent delivered shaft horsepower available for conditions of figures 15a and 15b and a 5.5 percent loss relative to table VII at operating conditions above 50 percent maximum continuous power setting. The specification also provides a lengthy discussion relative to an actual power restriction and a time restriction on the engine, depending on the original power.

The Engine J Model Specification permitted a "10 percent total loss in thrust commanded" and a 10-percent increase in fuel consumption above 50 percent maximum continuous power setting.

The Engine D is projecting a 9% power loss and a 9% increase in SFC. These performance losses are greater than other engines because they incorporate integral particle separators therefore having greater surface area to anti-ice.

Navy Research Laboratory Report 9025 dated 12/30/86 concludes that some Navy aircraft have "shown worrisome susceptibility to ice damage" and others possess "a continuing history of icing related mishaps." The report also cites a National Transportation Safety Board finding of 100 icing related accidents and 28 fatalities per annum in the general aviation community.

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A.4.3.1.4 Icing conditions.

The requirements of 3.3.1.4 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.3.1.4)

Since aircraft are required to operate in icing conditions, there is a need to verify engine performance when operating in icing conditions.

VERIFICATION GUIDANCE (A.4.3.1.4)

Past verification methods have included inspections, analysis and tests.

The following procedure may be used as a verification method.

> "a. This part shall consist of two runs at each of several engine thrust or power settings under each of the conditions in part 1 of table VII. The engine thrust settings shall be for turbofan/jet engines: idle, 25, 50, 75 and 100 percent intermediate thrust or power settings shall be for turboshaft or turboprop: idle, 25, 50, 75 and 100 percent maximum continuous, intermediate, and maximum. At each icing condition and at each thrust or power setting, the engine shall be operated for a period of not less than (a) minutes. During each period, at intervals after ice buildup, the engine shall be rapidly accelerated to intermediate thrust or maximum power to demonstrate acceleration response.
>
> b. This part shall consist of a (b) minute run at idle with no throttle movement, followed by an acceleration to intermediate thrust or maximum power at the end of the period. During this run the engine shall be operated under the conditions listed in part 2 of table VII. At each of the test thrust or power demand settings above, thrust or power shall be measured to verify performance loss, without any power lever movement.
>
> If the engine incorporates an anti-icing system, the above tests shall be performed using the anti-icing system to demonstrate the requirements of 3.3.1.4."

The following should be used to tailor the preceding guidance paragraph:

(a) and (b): The contractor should establish the time based on mission analysis.

If altitude testing is desired as part of the test plan, the following is recommended to replace the sea level test in the specification paragraph:

Runway hold and takeoff. This run shall be accomplished at an altitude of _____ m (_____ ft) or less at zero flight speed under table VII icing conditions. Run for _____ minutes at idle power with maximum customer air bleed and then accelerate to and maintain maximum _____ power for a minimum of _____ minutes.

Low altitude climb. This run shall be accomplished at an altitude of _____ m (_____ ft) at a flight speed equivalent to _____ Mach number, _____ ambient temperature. Run for _____ minutes at intermediate power setting with icing conditions established at figure(s) 15 conditions for the first _____ followed by _____ minutes at figure(s) 15 conditions.

Low altitude descent. This run shall be accomplished at an altitude of _____ m (_____ ft) at a flight speed equivalent to _____ Mach number, _____ ambient temperature. Run for _____ minutes at flight idle power setting icing conditions established at figure(s) 15 conditions.

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High altitude climb. This run shall be accomplished at an altitude of _____ m (_____ ft) at a flight speed equivalent to _____ Mach number, _____ ambient temperature. Run for _____ minutes at maximum continuous power setting with icing conditions established at figure(s) 15 conditions.

High altitude descent. This run shall be accomplished at an altitude of _____ m (_____ ft) at a flight speed equivalent to _____ Mach number, _____ ambient temperature. Run for _____ minutes at idle power setting with icing conditions established at figure(s) 15 conditions.

Low altitude loiter. This run shall be accomplished at an altitude of _____ m (_____ ft) at a flight speed equivalent to _____ Mach number, _____ ambient temperature. Run for _____ minutes at maximum continuous power setting with icing conditions established at figure(s) 15 conditions.

High altitude loiter. This run shall be accomplished at an altitude of _____ m (_____ ft) at a flight speed equivalent to _____ Mach number, _____ ambient temperature. Run for _____ minutes at cruise power setting with icing conditions established at figure(s) 15 conditions.

Low altitude landing loiter. This run shall be accomplished at an altitude of _____ m (_____ ft) at a flight speed equivalent to _____ Mach number, _____ ambient temperature. Run for _____ minutes at idle power setting with icing conditions established at figure(s) 15 conditions followed by an acceleration to intermediate/ maximum continuous.

The Using Service, engine contractor, and air vehicle contractor should negotiate the values for the blanks for flight conditions, thrust or power setting, and duration under which the engine should be tested. The values should be based on the anticipated aircraft missions.

Background:

Official anti-icing qualification tests are conducted at sea level conditions in an icing tunnel where all icing conditions can be controlled. Some icing tests are conducted with the aircraft flying behind a tanker aircraft. The tanker sprays water at an altitude where the water will freeze. However, testing under these conditions is difficult to monitor and control.

For testing in an icing tunnel the following icing conditions should be used to demonstrate compliance with the requirements of 3.3.1.4. For this test, the engine should be operated under the free air conditions listed in table VII. For each test run, the liquid water content and droplet size, should be measured at a distance within five feet of the engine inlet face and still within the engine inlet duct. The liquid water content measured at this station should be corrected to the free air conditions as specified in table VII. This meteorological data should be recorded at suitable intervals during each test run. The method and procedure for collecting and determining the water droplet size and liquid content should be specified. The engine performance level (i.e., thrust or parameters correlated to thrust), speed and vibration should be continuously recorded during the test and high speed photographic coverage of the engine inlet should be provided. The baseline for determining engine performance loss should be established by operating the engine with no customer bleed air or power extraction and under the inlet temperature conditions of tables IIa, IIb, and IIc with air between 80 and 100 percent relative humidity and zero liquid water content. The thrust or power loss and specific fuel consumption increase should be determined by comparison of engine performance when operating at the icing conditions defined in table VII with the aforementioned baseline values.

For engines that use oil as the anti-icing fluid, the oil temperature should be maintained at the minimum operating oil temperature specified in the specification, or less, during all runs except at idle condition where the oil should be maintained at the same temperature as the engine inlet air.

Engines that use compressor bleed air for anti-icing may have a variable capacity for anti-icing capability. At high power, the engine speed and pressure ratio are higher than at low power. This results in higher velocity and temperature air for anti-icing. Therefore more heat is delivered to the anti-ice surface at high

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power. This assumes a fixed flow area and a simple on/off valve to control the anti-ice air. Some engine contractors use a thermostatic valve which controls the anti-ice airflow. At high power it reduces the flow, while at idle it is wide open. With time this type of valve tends to corrode and stick in the closed position.

VERIFICATION LESSONS LEARNED (A.4.3.1.4)

Early engine requirements called for altitude test in addition to sea level tests. The requirement for altitude tests was dropped, since it was felt that the most potentially serious icing problem of a jet airplane occurs at the ground and very low altitudes.

Environmental Icing Tests for Turboshaft and Turboprops with Non-standard Inlets:

In all cases, the test should be performed with the representative inlet or flow path that is intended for the application. However, there will be cases where the air vehicle inlet is unavailable or unsuitable for testing. This may be a result of either the status of the development program for the air vehicle, or test facility limitations of power as a result of absorber and shafting arrangements. In these instances a suitable “S” duct will be required to conduct the test.

The S duct must be adequately designed to prevent icing (of the S duct) while providing the desired icing conditions at the engine inlet. The duct must first be verified for uniform temperature and pressure distribution at the engine face. Droplet size and Liquid Water Content (LWC) will normally be set and measured at the entrance to the S duct, as this is the final location that the conditions can be adjusted and controlled effectively. More elaborate calibrations have documented the conditions at the engine face plane.

The capture area of the duct and the water droplet distribution at the engine face have been concerns in the past. Water droplets impacting the S duct surfaces have a tendency to influence the amount of water delivered to the outer portions of the engine inlet (greater or less than desired, several arguments develop). If you heat the S duct sufficiently the impacting water droplets could be evaporated. Though this will likely result in a non-uniform temperature distribution. It could also be suggested that the non-uniform water distribution would occur in an installed condition. Though the capture area then becomes an issue, and the inlet must be sized correctly to have a meaningful test.

A straight bellmouth inlet installation with a front mounted cantilevered dynamometer has proved to be troublesome in the past. Vertical struts housing water, lube and electrical services and instrumentation for the dynamometer were shown to adversely influence the distribution of water droplets. The accumulation of ice on duct walls has been correlated to coincide with disturbed flow around the struts. This was proven conclusively, by observing the location of ice formation changing when the struts were rotated.

It should be noted that for engines with a power or free turbine the amount of inlet duct heating required will likely change with power, as will the local icing environment as the inlet Mach number changes with engine airflow. The local overheating of the inlet air near the engine inlet air temperature sensor is especially critical, and non-intrusive instrumentation should be considered for monitoring conditions in this area.

If the engine incorporates an anti-icing system for the aircraft inlet duct, this system may or may not adequately support the facility installation. Consideration must be given to augmenting or bleeding off the engine provided resource. In either case, the system design restrictions and exit conditions must be observed

In 1959 the Navy conducted its first icing tests at NAPTC, Trenton, NJ. Aircraft V powered by Engine AF experienced a number of accidents when flying through rain or icing conditions. The test found that ice blockage of the PT2 sensor affected the fuel control.

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In 1961 the Navy tested Engine AD to determine its suitability for all weather application. On this engine only the inlet struts were anti-iced. The bullet nose was not supplied with anti-ice air. An alternator and constant speed drive was located in the bullet nose. The heat given off by these units was supposed to anti-ice the bullet nose. The tests showed that sufficient heat was not produced to prevent ice formation.

In the test of Engine AE a thrust loss greater than the allowable 5% occurred at the lower engine speeds. The loss was as high as 23% after 5 minutes in icing conditions.

In 1966 testing of Engine M in icing conditions resulted in modification to Aircraft W and Aircraft X aircraft flight manuals. The pilot was warned to use high power when entering icing conditions.

Tests of Engine Q found that modification to the anti-icing system was necessary to supply sufficient heat to the inlet guide vanes of the engine.

In 1968, Engine L was the first turbofan engine tested under icing conditions at NAPTC. The tests revealed inadequate anti-icing for the inlet guide vanes and the inlet total pressure sensor.

Engine J was flight tested in Aircraft I that flew behind a tanker that sprayed water on the engine inlet duct. Ice accumulated on the inlet duct lip at the same flight conditions where field experience showed ice ingestion problems. When the aircraft pulled away from the tanker, ice shed off and caused engine damage that represented field experience.

A.3.3.2 Ingestion capability (hazard resistance).

A.4.3.2 Ingestion capability (hazard resistance).

A.3.3.2.1 Bird ingestion.

The engine shall continue to operate and perform during and after the ingestion of birds as specified in table VIII.

REQUIREMENT RATIONALE (A.3.3.2.1)

Engines must be capable of ingesting birds encountered during missions without significant power loss, deterioration, or safety implications. The total weapon system mission environment must be studied to examine the probability of bird strike occurrence, bird sizing criteria, flocking densities, mission routing, training, etc., to determine the design criteria for bird ingestion capability requirements for an engine.

REQUIREMENT GUIDANCE (A.3.3.2.1)

The following should be used to tailor table VIII:

In the event specific air system bird strike criteria has not been established for the engine, the following birds vs. inlet area criteria should be used.

The inlet area to be used should be the aircraft inlet. This area should then be documented in the interface control document as a limitation or constraint.

The shape and orientation of the inlet ducting should be considered in establishing bird ingestion requirements. Engines that utilize inlet ducts, which would not permit direct impact on the front face of the engine, may be allowed to qualify assuming that birds will strike the walls of the inlet duct prior to being ingested into the engine. Impact on the duct walls will have an effect on the size and energy level of the ingested birds. Engines with a unique orientation can also affect the size and quantities of birds impacting the front face. For example, an engine with an inlet flush mounted on the side of the aircraft fuselage would have fewer potential bird impacts than one oriented normal to the direction of flight. If an engine is qualified to either of these standards then the interface control document must state that an inlet system with a specific geometry must be used with the engine.

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The number of birds to be ingested should be based on inlet area as follows: one 100 gm (3.5 oz) bird per 300 cm^2 (46.5 in^2) of inlet area plus any fraction larger than 50 percent thereof, up to a maximum of 16 birds; one 1 kg (2.2 lb.) bird per 1500 cm^2 (232.5 in^2) of inlet area plus any fraction larger than 50 percent thereof; one 2 kg (4.4 lb.), regardless of the size of the inlet, provided the inlet is large enough to admit a 2 kg (4.4 lb.) bird.

The 100 gm (3.5 oz) birds should be ingested at random intervals and be randomly dispersed over the inlet area. Birds 1 kg (2.2 lb.) and larger should be directed at critical areas of the engine face.

The bird velocity and engine power setting for each condition should be as described below:

Birds weighing 100 gm (3.5 oz) (a maximum of sixteen at a time) and birds weighing 1 kg (2.2 lb.) (one at a time) ingested at a bird velocity equal to the takeoff flight speed, with the engine at intermediate thrust for TJ/TFs or maximum power for TP/TSs.

Birds weighing 100 gm (3.5 oz) (a maximum of sixteen at a time) and birds weighing 1 kg (2.2 lb.) (one at a time) ingested at a bird velocity equal to the cruise flight speed with the engine at cruise power setting.

Birds weighing 100 gm (3.5 oz) (a maximum of sixteen at a time) and birds weighing 1 kg (2.2 lb.) (one at a time) ingested at a bird velocity equal to the descent flight speed with the engine at descent power.

For aircraft that have a low level, high speed mission requirement: birds weighing 100 gm (3.5 oz) (a maximum of sixteen at a time) and birds weighing 1 kg (2.2 lb.) (one at a time) ingested at a bird velocity equal to the aircraft maximum sea level speed and the engine power setting required to achieve that speed.

Birds weighing 2 kg (4.4 lb.) ingested at a bird velocity equal to the aircraft takeoff speed or low level operational airspeed, whichever is more severe, with engine power equal to that required for the flight condition.

For 100 gm (3.5 oz) birds, the engine should sustain performance 95 percent or greater of the initial thrust or power, and all damage to the blades and vanes should be blendable (within repair limits) with flight line type tooling. The 1 kg (2.2 lb.) bird ingestion may cause some damage; however, it should not result in immediate engine shutdown, and post ingestion thrust or power levels should be 75 percent or greater of the initial thrust or power at the operating condition.

The 2 kg (4.4lb.) ingestion should not cause an engine failure which results in damage to the aircraft or adjacent engines. No bird ingestion should prevent the engine from being safely shutdown.

Performance recovery times will vary as a function of the bird size, number of birds, and size of the engine. The performance recovery time after ingestion of the 100 gm (3.5 oz) bird(s) should occur in 5 seconds or less after the final volley of birds has been ingested. The performance recovery time after ingestion of the 1 kg (2.2 lb.) bird(s) should occur in 5 seconds or less for small engines, 5-10 seconds for moderate sized turbofans or turbojets, and up to 5-15 seconds for large by-pass turbofans.

Turboshaft engines should recover within 5 seconds after an ingestion event with no less than 95 percent of the power prior to the ingestion event, and without exceeding any engine control limits.

Background:

Bird strike durability is a necessary safety design criteria to be used in all engines. Analyses should be conducted to determine the sensitivities to blade design, blade to stator spacing design, control system operation, and stator design for structure performance, and engine control. Bird strike tolerance can be enhanced by providing large axial clearances between blade and stators at the front of the engine.

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In 2001, an AIA bird ingestion task group issued a set of recommendations concerning the management of large flocking birds. The group identified a growing threat posed by populations of flocking bird species over 1.8 kg (4 lb.). To counter this threat, the task group, with FAA and JAA participation, drafted a new requirement for engines to demonstrate an ability to withstand encounters with large flocking birds (birds weighing between 1.14 kg. (2.5 lb.) and 3.65 kg. (8.0 lb.). Note that this new requirement affects engines with inlet areas of 2.50 sq. meters (3875 sq. inches) and greater. As of early 2002 this requirement had not become law. Draft copies of this requirement (CFR 14 Part 33, Paragraph 33.76) are available from the FAA Engine & Propeller Directorate Standards Staff in Burlington, MA.

FAA Advisory Circular 33-76.1A contains guidance for compliance to ingestion requirements including: conduct of ingestion tests, selection of target locations, evaluation of post test engine performance and assessment of test deviations.

REQUIREMENT LESSONS LEARNED (A.3.3.2.1)

Since helicopter takeoff flight speed differs from fixed wing, the following was used for Engine D:

A bird of 50 to 100 grams (0.11 to 0.22 lb.) ingested at a bird velocity equal to Mach number of 0.1 and with the engine at maximum rated power set by rated Measured Gas Temperature (MGT).

A bird of 50 to 100 grams (0.11 to 0.22 lb.) ingested at a bird velocity equal to Mach number of 0.3 with the engine at maximum continuous power set by rated MGT.

A bird of 50 to 100 grams (0.11 to 0.22 lb.) ingested at a bird velocity equal to a Mach number of 0.2 with the engine at 25 percent of the maximum continuous rated power.

Military studies have indicated that the engine may receive over 60 percent of the total aircraft strikes, and most strikes occur below 1.8 km (6000 feet) at takeoff, landing, and low level penetration speeds.

A 2002 Washington Post article stated, “Since the Navy Safety Center began keeping bird strike records in 1980, about 20,000 bird strikes have been recorded, resulting in two deaths, 25 aircraft destroyed and more than $310 million in damage, according to the center.”

A GAO/NSIAD-89-127 report dated July 1989 states that from 1983 to 1987, military aircraft have collided with birds over 16,000 times. Many of these collisions caused only minor damage; however, the services lost six crew members, incurred $318 million in damages, and lost nine aircraft. During this period the Air Force lost six aircraft, the Navy lost two aircraft, and the Army lost one aircraft. A review of nine military jet engines developed since the early 1970’s showed that the Services lessened the military specification requirements in engine model specifications for the sizes and the numbers of medium birds used in testing engines.

Recent studies (USAF/ASC & AIA) of bird ingestion data indicate that older specifications did not accurately reflect the sizes and the number of birds actually ingested. Prior to the publication of this specification, engine military specifications required medium birds sizes for ingestion tests of 1.5 pounds for the Air Force and 2 pounds for the Navy.

DOT Report DOT/FAA/CT-84/13 by Gary Frings, dated 9/84, states bird ingestion is a rare but probable event. For every one million aircraft operating hours, 230 bird ingestions (of all weights) will occur, on average. An average bird weighs 26 ounces. Most likely weight of birds in the areas of runways is 11 ounces. A small percentage (<3%) of ingestion events involved birds weighing 61 ounces. Ten percent of events involved ingestion of multiple birds into a single engine. One-half of one percent of events involved multiple engine ingestions. Five percent of all bird ingestions resulted in engine failure.

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A.4.3.2.1 Bird ingestion.

The requirements of 3.3.2.1 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.3.2.1)

A test with a complete engine is necessary to ensure all interactive effects of a bird strike are properly accounted for in the engine design. Analysis and component test (fan and compressor rigs) have been successfully used to predict loading and deflection criteria leading up to full-scale engine tests.

VERIFICATION GUIDANCE (A.4.3.2.1)

Past verification methods have included analyses, demonstrations and tests.

If the inlet shape was considered in the size and energy level of bird fragments ingested, then the verification test should be conducted with an aircraft inlet in place.

The contractor should specify in the pretest data the critical target area for bird ingestion. Test target area is subject to approval by the Using Service.

Background:

A development test program consisting of analysis and component test should be established to provide confidence in the design prior to formal engine ingestion tests. The number of birds, bird sizes, engine RPM's, bird velocities, performance criteria and amount of damage for the test engine should be based on the parameters of 3.3.2.1. Analysis should indicate the location for critical bird strikes at the front face of the engine and pass/fail criteria should be included in the test plan. The birds should be ingested in a random sequence and dispersed over the inlet area to simulate an encounter with a flock. Synthetic "birds" have not been used by the military services to date, but their use should not be precluded in future programs if sufficient information and justification is presented to the Using Service.

VERIFICATION LESSONS LEARNED (A.4.3.2.1)

Severe out-of-balance conditions creating high vibrations, along with rotating to static structural contact, have been encountered following bird ingestion events.

A.3.3.2.2 Foreign object damage (FOD).

The engine shall have foreign object damage requirements specified per _____.

REQUIREMENT RATIONALE (A.3.3.2.2)

The engine inlet airflow velocities create conditions where loose foreign objects may be ingested into the engine resulting in gas path damage. The engine must tolerate the ingested objects up to a specified level.

REQUIREMENT GUIDANCE (A.3.3.2.2)

MIL-HDBK-1783 should be used to tailor the requirement.

REQUIREMENT LESSONS LEARNED (A.3.3.2.2)

Blades insufficiently designed for FOD considerations have caused In Flight Shut Downs (IFSDs) and costly damage to the engine. FOD tolerance can be enhanced by use of more damage tolerant materials and a redistribution of the blade mass.

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Foreign object damage has cost the Navy over $15M per fiscal year based on a conservative estimate in one source reference. Common culprits are “housekeeping” items, such as nuts, bolts, safety wire, screwdrivers, etc. This paragraph is aimed at this type of FOD problem (environmental factors such as ice, sand, water and birds, are covered in other specification paragraphs).

A.4.3.2.2 Foreign object damage (FOD).

The requirements of 3.3.2.2 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.3.2.2)

Analysis, demonstration, and test are required to ensure that the fan and compressor airfoils can meet the operational requirement of 3.3.2.2 and to establish accept or reject criteria for damage that is detected during flight line inspections.

VERIFICATION GUIDANCE (A.4.3.2.2)

Past verification methods have included analysis, demonstration, and test.

MIL-HDBK-1783 should be used for further verification guidance.

VERIFICATION LESSONS LEARNED (A.4.3.2.2)

Past engine programs have shown that FOD ingestion and subsequent engine repair is a problem aboard aircraft carriers, because it requires engine module rework or extra maintenance to blend out damage.

A.3.3.2.3 Ice ingestion.

The engine shall operate and perform per table IX, during and after ingestion of hailstones and sheet ice at the take-off, cruise, and descent aircraft speeds. The engine shall not be damaged beyond field repair capability after ingesting the hailstones and ice.

REQUIREMENT RATIONALE (A.3.3.2.3)

Sufficient structural capability is needed to tolerate ingestion of environmentally generated ice (hailstones), ice shed from the air vehicle, and engine inlet generated ice.

Particles or chunks of ice can be dislodged or break off of inlet duct components such as cowl lips, boundary layer bleed wedges, inlet accessory covers, and variable inlet spikes, and cause compressor damage.

REQUIREMENT GUIDANCE (A.3.3.2.3)

The following should be used to tailor table IX:

For inlet capture area of 0.065 m^2 (100 in^2) the engine should be capable of ingesting one 25 mm (1.0 in) diameter hailstone. For each additional 0.065 m^2 (100 in^2) increase of the initial capture area (0.065 m^2 (100 in^2)), supplement the first hailstone with one 25 mm (1.0 in) and one 50 mm (2.0 in) diameter hailstone. The engine should be capable of ingesting pieces of ice of various sizes and shapes including spears, slabs and sheets. Past engine programs have used 5-8 and 7-9 pieces with one piece weighing at least 0.34 kg (0.75 pounds). For turbofan engines the amount of ice has ranged from 2.3 to 4.1 kg (5 to 9 pounds) total ingested weight.

Within 5 seconds after an ice ingestion event, the engine thrust or power should be at least 95 percent of the thrust or power immediately prior to the event. The hailstones and sheet ice should be between 0.80 g/cm^3 (50 $lb./ft^3$) and 0.90 g/cm^3 (56 $lb./ft^3$) specific gravity. Hailstones should be ingested at typical

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takeoff, cruise, and descent conditions. Sheet ice should be ingested at typical takeoff and cruise condition.

Background:

All-weather aircraft engines should be designed to withstand potential hail conditions. An engine should also have some capability to ingest, without major damage, chunks or sheets of ice which may dislodge or break off of inlet duct components, such as cowl lips, inlet ramps or doors, inlet accessory covers, and air vehicle surfaces. See FAA Advisory Circular 33-78.1 section 2 for additional background information on ice ingestion design factors.

REQUIREMENT LESSONS LEARNED (A.3.3.2.3)

Past engine programs have used these ice ingestion requirements:

Airflow kg/s (pps)	No. of Pieces	Weight Kg (lb.)	Velocity km/hr (ft/sec)	Miscellaneous Information
112 (246)	7-9	2.3 (5)	54 (50)	no piece weighing greater than 0.34 kg (0.75) lb.
118 (260)	7-9	2.3 (5)	54 (50)	no piece weighing greater than 0.34 kg (0.75 lb.)
161 (355)	5-8	4.1 (9)	not specified	no piece less than 0.4 kg (1 lb.), one 0.9 kg (2 lb.) spear 114 cm (45 inch) long

Engines that have inlet guide vanes and lack adequate anti-icing or deicing provisions have had operational restrictions and increased maintenance workloads when exposed to icing conditions. The problems were eliminated by providing adequate anti-icing systems that reduced inlet case ice accumulation and resultant fan blade damage. Whereas the damage from inlet case ice shedding is a maintenance problem (blending and replacement of fan blades), ice shed from the air vehicle or environmentally generated ice can be a safety of flight concern if sufficient structural capability in the engine design is not provided.

Numerous incidents of axial-flow compressor damage caused by ice ingestion are on record. Some of these resulted in complete engine failure and disintegration of the engine.

The Engine L, on the Navy's Aircraft B, passed its Full Scale Development (FSD) icing condition requirement by similarity. However, during fleet operations, engines experienced damage from ice ingestion events. The damage occurred when ice, formed on the inlet lip and duct walls during flight, was dislodged and ingested during descent in warm weather and during hard carrier landings.

Naval Research Laboratory Report 9025 dated 30 December 1986, states that about 525 mishaps related to ice ingestion have occurred on U.S. Navy aircraft from 1964 through 1984. It mentions that the Engine J (Aircraft I) often sustained compressor blade damage that was substantial or required an overhaul. Engine J had not been tested for ice ingestion during qualification tests so the effects of ice ingestion were unknown when the engine entered service. It became apparent in field deployment that ice ingestion was a concern and this may have been determined earlier if an ice ingestion test had been conducted during development of the engine.

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A.4.3.2.3 Ice ingestion.

The requirements of 3.3.2.3 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.3.2.3)

Engine ice ingestion verification is needed to ensure satisfactory engine performance is maintained in icing conditions throughout the aircraft flight envelope.

VERIFICATION GUIDANCE (A.4.3.2.3)

Past verification methods have included analysis, demonstration, and test.

The test procedure should require the engine to run for at least five minutes following ice ingestion, before it is shut down for inspection. During the ingestion test, high speed photographic coverage should be taken. Sufficient instrumentation for monitoring the structure of the engine should be included in the test engine.

The procedures to be used for introduction of the hailstones and sheet ice at the engine inlet and the engine power settings and speed at which the hailstones and sheet ice are to be ingested should be specified in the pretest data. The temperature of the ice should be between -2.2 to 0°C (28 - 32°F). The test procedure for sheet ice ingestion should require the most severe ice velocities representing ice shedding off the aircraft inlet lip. FAA Advisory Circular 33-78.1 section 5 has some guidance for verification of the ice ingestion capability of engines for commercial supersonic aircraft.

VERIFICATION LESSONS LEARNED (A.4.3.2.3)

After undergoing icing tests behind a tanker, Aircraft I with Engine J sustained severe damage from ice when the aircraft descended to a warmer altitude and inlet ice was ingested by the engine.

For turboshaft engines the most severe ice ingestion velocities may occur at low velocities reflecting the ingestion of ice shed from the inlet.

A.3.3.2.4 Sand and dust ingestion.

The engine shall meet all requirements of the specification during and after the sand and dust ingestion event specified herein. The engine shall ingest air containing sand and dust particles in a concentration of (a) mg sand/m3. The engine shall ingest the specified coarse and fine contaminant distribution for (b) and (c) hours, respectively. The engine shall operate at intermediate thrust for TJ/TFs or maximum continuous power for TP/TSs with the specified concentration of sand and dust particles, with no greater than (d) percent loss in thrust or power, and (e) percent gain in specific fuel consumption (SFC). Helicopter engines shall ingest the 0-80 micron (0 - 3.15 x 10-3 in.) fine sand and dust of table XXXV in a concentration of 53 mg/m3 (3.3 x 10-6 lb/ft3) air for 54 hours and 0-1000-micron coarse sand and dust in table XXXV for 50 hours and inspection shall reveal no impending failure.

REQUIREMENT RATIONALE (A.3.3.2.4)

The operation of aircraft in sand and dust environments can result in serious erosion damage to engine parts. Sand and dust particles are highly abrasive and tend to erode the thin metal tips and trailing edges of gas turbine engine compressor blades and vanes.

Sand and dust ingestion is also needed to determine the effect on surface coatings.

Engine power loss, surge margin loss, and specific fuel consumption (or fuel usage) gain may adversely affect system safety.

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REQUIREMENT GUIDANCE (A.3.3.2.4)

Table XXXV should be used to tailor the specification paragraph.

Background:

This requirement is based upon a severe but realistic potential service ground environment. It is recognized, however, that the time the engine is subjected to the concentration of sand and dust particles is quite dependent upon the particular engine and air vehicle contribution.

The coarse and fine sand and dust particle size is representative of field deployment where pilots will train in the United States desert areas and be sent to the Middle East desert as in Operation Desert Shield and Desert Storm.

Large particles are more likely to cause erosion on airfoils (blades and stators) while small particles are more likely to block cooling holes in the turbine and cause corrosion. The SiO_2 in both particle size distributions are likely to melt in the combustor discharge gases and be deposited on first stage turbine nozzle vanes. A verification should be made to determine the effect of different sand compositions on the combustor and HPT at elevated temperatures.

Notes on sand and dust concentration in guidance:

The thrust or power loss above should be verified at constant turbine temperature for all engine classes. If verifying at constant commanded power setting, the engine thrust or power loss will be less. This is because engines commanding constant fan speed tend to increase thrust as components deteriorate rather than decrease. Fuel usage or SFC loss should be verified at constant thrust or power output.

The ratio of thrust or power loss to SFC loss varies with engine cycle. The ratio of shaft power loss to SFC loss is as high as 3:1 for typical helicopter engines. The relative losses can be verified by calculating performance with changes in compressor efficiency.

The Air Force ATF sand and dust requirement is the MIL-E-87231 and MIL-E-5007D and MIL-E-8593A sand and dust contaminant requirement at a “53 mg/m^3 ($3.3x10^{-6}$ lb/ft^3) of air concentration” with a two hour test allowing a 10-percent loss of thrust and a 10-percent increase in SFC. Recent Army helicopter engine specifications since 1971 have all used “C-Spec”(MIL-E-5007C) sand with a concentration of 53 mg/m^3 (3.3x10-6 lb/ft3). The Engine D specification added an additional 54-hr sand test with 0-80 micron (0 - 3.12 x 10^{-3} in) AC Fine air cleaner test dust. The Engine H specification uses a 0-200 micron (0 - 7.87 x 10^{-3} in) sand (AZ Road Dust Coarse) at a “53 mg/m^3 ($3.3x10^{-6}$ lb/ft^3) of air” concentration with a 20-hour test allowing a 10-percent loss of power and a 10-percent increase in SFC.

The deterioration factors for power and SFC loss may be approximately the same for higher pressure ratio engines. Helicopter engines, with lower pressure ratios, experience power loss more rapidly than SFC loss (at constant T41). The factors may be more equal for higher pressure ratio engines.

Coarse sand produces blade erosion and a near term loss of compressor efficiency. Fine sand causes plugging of cooling holes, deposits, and long term engine distress. Fine sand tests should be conducted to determine the engine’s vulnerability to air hole plugging and nature of deposits. Deposits formed also depend upon the composition of the sand.

Most of the arid regions of the world correspond to “C-Spec” sand in size except regions in the Middle East. Saudi sand, in general, is close to “C-Spec” in size, very sharp particle shape, with high alumina content. When mixed with water (i.e., water wash), Saudi sand may form cement, which, when heated, may form porcelain. Saudi fine sand has been encountered as high as 4,572 m (15,000 feet). Very fine sand is found in parts of Israel and some other nearby regions.

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The performance loss due to coarse sand ingestion depends upon the total quantity of sand ingested, particle size, and engine power setting. Total quantity of sand depends upon engine airflow rate, sand concentration in the air and duration of the test. (53 mg/m^3 ($3.3x10^{-6}$ lb/ft^3) for 50-hr. may be equivalent to 530 mg/m^3 ($3.3x10^{-5}$ lb/ft^3) for 5 hr.). Particle size ingested varies rapidly with hover height, for clouds created by rotor downwash. Particle size is largest near the surface and decreases rapidly with height. Power setting affects the effectiveness of the IPS. The IPS is most effective at high power settings where high airflows result in high velocities. Filtration is needed to be effective at all power settings or with the finer particles. A helicopter on the ground at low engine power settings could conceivably experience more sand damage than when hovering a few feet above the ground at high power. A comprehensive sand test should, therefore, consider sand concentration level, composition, particle size in the bed, particle size in the cloud, hover height, time duration, and power setting. The present sand test (constant concentration, time, and power setting regime) forms a good benchmark for engine vulnerability to sand erosion, but may not be indicative of actual field experience.

USAF bases and commercial airports are permitted to sand icy taxiways and runways with sand particles up to 0.475 cm (0.187”). If prolonged operation in these environments is anticipated, the FOD potentials of this size sand need to be considered. AFI 32-1002 and FAA Advisory Circular 150/5200-30A should be consulted for specific information on allowable sand dimensions and quantities.

While this requirement paragraph is oriented around thrust or power and SFC performance, it is stated that the engine has to meet all requirements of the specification. This includes life and performance effectiveness of special technology materials and features on the engine inlet or front face which may be vulnerable to sand erosion and FOD.

REQUIREMENT LESSONS LEARNED (A.3.3.2.4)

The operation of aircraft in sand and dust environments has resulted in serious erosion damage to engine parts. Sand and dust particles are highly abrasive and tend to erode the thin metal tips and trailing edges of gas turbine engine compressor blades and vanes. Helicopter operations are most significantly impacted by the sand and dust problem, although fan blade leading edge damage has been experienced on Engine L in Aircraft B as a result of operation “near sandy, dry desert-like areas such as NAS Miramar”. One source pointed out that the “accelerated replacement of erosion damaged helicopter turbines in Southeast Asia was estimated to cost the U.S. Government about $150 million a year” (during the Vietnam war period). The finer particles of sand and dust have been known to clog turbine cooling passages and cause engine failures. Sand and dust has been encountered several hundred miles out at sea as well as over land masses.

Sand has been determined to be more detrimental during foreign operation due to the differences found in the sand composition. A lower melting point has been observed in foreign sand test samples which allows the sand to melt and form a glaze coating on the turbine airfoils. Spalling of the coating and base metal then occurs.

Due to the unexpected turbine blade failure and short life encountered in engines with gas temperatures in excess of 1093°C (2000°F), special consideration in blade design should be taken into account. The soil analysis of Saudi Arabia and other Middle Eastern countries reveals that very high amounts of sulfur, calcium, and magnesium exist. A study has shown that an HPT blade failure occurred because of excessive sulphidation. At high temperatures, a flux or coating of calcium sulfate and magnesium sulfate attaches to the blade, attacks the base metal, and corrodes it, leading to failure.

USAF Aircraft A bases have raised questions about operating on taxiways and runways with sand particle sizes specified in FAA and Air Force Instructions. Subsequent analysis revealed that Aircraft A engines are vulnerable to stage 1 fan FOD when operating on sanded areas. Sand and dust ingestion and subsequent performance is a primary cause of engine removal on helicopters.

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Operation in volcanic ash can result in erosion of compressor and turbine blades, blockage of turbine blade cooling holes and passages, contaminated oil systems, etc. Requirements regarding tolerance of this material are not specified because operation in a volcanic ash environment is rare. The general practice is to specify cleaning and/or inspection requirements following operation in volcanic ash.

References:

Particle distribution: “Development of the Lycoming Inertial Particle Separator”, H.D. Conners, presented at the Gas Turbine Conference & Products Show, Wash. DC, 17-21 Mar 1968.

Particle densities: ASME 70-GT-96, J.C. Arribat, ASME Gas Turbine Conference, Brussels, Belgium, 24-28 May 1970.

Particle distribution: SAE AIR 947, Issue 1, 2/71.

Blade damage including blade and vane erosion, secondary airflow deposits resulting in power reduction and stall margin loss. ASME 68-GT-37, G.C. Rapp and S.H. Rosenthal, presented at Gas Turbine Conference & Products Show, Wash. DC, 17-21 Mar 1968.

Particle distribution: “Evaluation of the Dust Cloud Generated by Helicopter Blade Downwash”, Sheridan Rogers, MSA Research, Proceedings of the 7th Annual National Conference on Environmental Effects on Aircraft and Propulsion Systems, 25-27 Sep 67. Single helicopter dust concentration reached 16.2 mg/cu ft, (572 mg/m^3 (35.7 x 10^{-6} $lb./ft^3$)) takeoff and approach reach 40 mg/cu ft (1,413 mg/m^3 (88.2 x 10^{-6} lb/ft^3)). Two helicopter levels of 64 mg/cu ft (2260.1 mg/m^3 (141.1 x 10^{-6} lb/ft^3)) were reached.

Particle distribution: Kaman Report No. R-169, “Amount of Dust Recirculated by a Hovering Helicopter”, dated 26 Dec. 1969. Densities ranged from .29 gm/cu ft to 26 gm/cu ft.

A.4.3.2.4 Sand and dust ingestion.

The requirements of 3.3.2.4 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.3.2.4)

The effect of sand ingestion on the performance, bleed air quality, and internal air cooling system must be verified.

VERIFICATION GUIDANCE (A.4.3.2.4)

Past verification methods have included analysis, inspection, and test. ISO/FDIS 12103-1 can be used for guidance on Arizona test dust.

The contractor should propose a sand ingestion test to verify the operation and performance of the engine at worst case power levels derived from the mission profile.

This test could be combined with the AMT. If combined, the sand test should occur after post-AMT recalibration and endurance test completion, due to performance degradations of the engine.

The following procedure may be used as a verification method.

For fixed wing and VSTOL engines:

> “During the engine test, the coarse sand and dust shall be ingested first, with the fine particle sand and dust ingested afterward. An engine disassembly and inspection shall be conducted between the coarse and fine sand tests as specified by the Using Service.

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The engine shall be tested at Intermediate thrust, with sand and dust ingested at the concentration levels and for the length of time specified in 3.3.2.4. During each hour of operation, at least one deceleration to idle and acceleration to maximum augmentation shall be made, with power lever movements of 0.5 seconds or less. If an anti-icing system is provided, ten periods of one minute operation of the anti-icing system shall be performed during the first test hour. During the entire test, maximum customer bleed air shall be extracted from the engine. The customer bleed air shall be continually filtered and the total deposits measured and recorded. Following the post-test performance check, the engine shall be disassembled to determine the extent of erosion, and the degree to which the contaminant may have entered critical areas in the engine. The test will be considered satisfactorily completed when the criteria of 3.3.2.4 have been met and the teardown inspection reveals no failure or evidence of impending failure."

For helicopter engines:

"The engine shall be tested with a 0 to 80 micron (0 - 3.15 x 10^{-3} in.) sand and dust ingested at the concentration levels specified in 3.3.2.4. The engine shall be tested for 9 hours at maximum, 27 hours at intermediate, and 18 hours at maximum continuous, for a total of 54 hours with not less than 27 starts. The test cycle shall be 10 minutes at maximum, 20 minutes at maximum continuous and 30 minutes at intermediate. All ratings shall be initially set to measured gas temperature associated with rated rotor inlet temperature. After test initiation, the engine shall be run at a constant gas generator speed unless the measured gas temperature associated with the T4.1 deterioration limit is reached in which case the measured gas temperature will be held constant. This test shall be terminated and the engine shall be removed and disassembled if power deterioration based on measured gas temperature is excessive or if engine failure or impending failure is evident. The engine shall be tested with the IPS if it is an integral part of the engine design. During each hour of operation, at least two deceleration to idle and acceleration to maximum continuous speed shall be made, with power lever movements of 0.5 seconds or less. A calibration test shall be performed at the beginning, at 25 hours and end of the test run. The engine shall be inspected at 25 hours by borescope and other visual techniques that do not require disassembly of the engine. Upon completion of the test run the engine shall be disassembled in such a way that the contamination displacement is minimized. The engine shall be disassembled to determine the extent of sand erosion, and the degree to which sand may have entered critical areas in the engine. Each major rotating and stationary component subject to the effects of sand ingestion shall be weighed at engine build, disassembly and after cleaning. The test will be considered satisfactorily completed when 54 hours of testing have been completed and the teardown inspection reveals no failure or evidence of impending failure."

"The engine shall be tested with the coarse sand at maximum continuous rated measured temperature with sand and dust ingested at the concentration levels and the length of time specified in 3.3.2.4 The engine shall be tested with the IPS if it is an integral part of the engine design. During each hour of operation, at least one deceleration to idle and acceleration to maximum continuous rated measured temperature shall be made, with power lever movements of 0.5 seconds or less. If an anti-icing system is provided, ten periods of one minute operation of the anti-icing system shall be performed during the first hour of each five hour cycle. The engine shall be shut down and cooled at least 12 hours following each five hours of sand ingestion. During the entire test, maximum customer bleed air shall be extracted from the engine. The customer bleed air shall be continually filtered, and the total deposits measured and recorded. If an engine internal washing system is provided, it shall be demonstrated once during each shut down. Components such as air-oil coolers with exposure to inlet sand and dust conditions shall be considered inlets for this test but a rig test may be performed to satisfy the requirements herein. Following the post-test performance check, the engine shall be disassembled to determine the extent of sand erosion, and the degree to which sand may have entered critical areas in the engine. The test will be considered satisfactorily completed when the criteria of 3.3.2.4 have been met and the teardown inspection reveals no failure or evidence of impending failure."

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Background:

The recommended text decreases the operational time in the extreme sand and dust environment from ten hours to two hours for turbofan and turbojet engines. Engine contractors have been unwilling in the past to guarantee their engines for ten hours (helicopter subjected to the severe Vietnam sand and dust environment typically used inlet filtration systems). The time requirement will have to be negotiated with each engine contractor in specific future specification negotiations based upon the intended usage in regions of the world where sand will be a concern.

The sand concentration should be calculated with customer bleed air extraction. The anti-icing switch should be activated five times during each hour of sand ingestion at equally spaced intervals. The test should be conducted with a thrust bed and load cell measurement of thrust in lieu of calculating thrust by EPR. Disassembly and inspection between the coarse and fine sand tests should be conducted for 45.4 kg/s (100 lb./sec) airflow or smaller engines.

VERIFICATION LESSONS LEARNED (A.4.3.2.4)

The Engine V sand and dust test did not use the recommended sand and dust mixture due to commercial unavailability of the mixture. The specification for fine sand calls for a particle size distribution which cannot be obtained commercially. Specifically, calcite and gypsum could not be obtained with a particle size distribution to match the specified particle size distribution. Tables XXXVIa and b show the closest particle size distributions which the Engine V sand and dust test team could find along with the required size distribution.

During the desert storm experience, many Engine B engines were shown to have a glassified buildup on the cooling holes of the turbine blades. Upon a more detailed examination of the buildup, it was found that the calcium oxide sand had been getting into the cooling flows of the engine. As compared to the Arizona type sand which is a courser mixture of silicon dioxides, the Saudi sand is more akin to talcum powder in texture. It has a significantly different mode of attachment to turbine hardware than the Arizona type sand. Marine experience with Engine P shows clogging due to fixed particle size and geometry. Whereas, with the Saudi sand, the particles are fused into a glassy coating which accumulates to eventually shutoff the holes. Manufacturers should use lessons learned to mitigate these effects in new turbine designs.

A.3.3.2.5 Atmospheric liquid water ingestion.

The engine shall start and operate satisfactorily with (a) percent of the total airflow weight in the form of water (liquid and vapor), with (b) percent of the liquid water entering the inlet through a segment equivalent to (c) of the inlet area.

REQUIREMENT RATIONALE (A.3.3.2.5)

Large amounts of water in the engine inlet affect engine performance, resulting in stall, rotor speed rollback, main or augmentor combustor blowout. Two sources of water are rainstorms and aircraft nose wheel spray generated when passing through standing water on runways during takeoff and landing.

REQUIREMENT GUIDANCE (A.3.3.2.5)

The following should be used to tailor the specification paragraph:

(a): A value of at least 5 percent.

(b): A value of at least 50 percent.

(c): One third has been used historically. The contractor may alternatively propose the worst case situation for water ingestion area.

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Background:

Ingestion requirements which concentrate half of the liquid water in one-third of the inlet area are an attempt to simulate the concentration that often occurs because of installation effects.

Consistent with the requirement for all-weather aircraft, the engine must be capable of operating in severe rain conditions without flaming out or stalling. There is also a potential for water to be ingested during takeoff and landing situations where standing water exists on the runway surface.

The 5.0 percent water-to-air mixture tolerance requirement for engines is somewhat arbitrary, although MIL-HDBK-310 supports it with data for the highest recorded rainfall in a one minute period at sea level (3.12 cm/min (1.23 in/min.) (6.4 percent water-to-air mixture)). At 6,005 m (19,700 ft), the highest recorded rainfall rate for one minute was 4.3 cm/min (1.7 in/min.) (17.2 percent water-to-air mixture). The water-to-air ratios are reduced significantly, however, when the 0.5 or 0.1 percent extremes are considered in MIL-HDBK-310. For example, at 6,005 m (19,700 ft) a water-to-air mixture of 2.6 percent will be exceeded on an average of only 0.1 percent of the time in the rainiest area of the world during the rainiest month.

Although the probability of exceeding a 5.0 percent water-to-air mixture on a worldwide basis is extremely low, a finite probability does exist. The extreme values in MIL-HDBK-310 represent those recorded officially at weather stations throughout the world; it is likely that higher rainfall rates have occurred at locations where weather stations do not exist. Another interesting point is that, “It is unlikely, however, that one minute precipitation rates from storms elsewhere will greatly exceed rates measured in the United States since it is the world center of intensely precipitating thunderstorms.”

Tabulated rainfall information was obtained from MIL-HDBK-310 for the worldwide extreme and 0.1 percent extremes between sea level and 20 km (65,616 ft) altitude. The worldwide extreme water-to-air mixture is 10.5 percent at 13,990 m (45,900 ft) and 1.9 percent for the 0.1 percent extreme. See FAA Advisory Circular 33-78.1, Section 2, for additional guidance on design factors that can diminish water ingestion.

In 1988, the Aerospace Industries Association (AIA) initiated a study of turbine engine power-loss and instability that were attributed to operating in inclement weather. This study concluded that the FAA requirements in place at that time did not adequately address the flight safety threat posed by operating in an extreme rain or hail environment (defined as an extremely remote probability of encounter ($1x10^{-8}$)). Investigations have revealed that ambient rain or hail concentrations can be amplified significantly through the turbine engine core at high flight speeds and low engine power conditions. Rain or hail through the turbine engine core may degrade compressor stability, combustor flameout margin, and fuel control run down margin. Ingestion of extreme quantities of rain or hail through the engine core may ultimately produce a number of engine anomalies, including surge, power loss, and engine flameout. Consequently, new FAA regulations have been promulgated to address this threat. The FAA requirement (CFR 14, Part 33, paragraph 33.78) typically yield higher required water-to-air mixtures than the 5% value stated in (a) above. This may be due to the use of the previously mentioned amplification factors. FAA Advisory Circular 33-78.1 contains guidance regarding: Rain and Hail Concentration Amplification Effects, Design Factors, Critical Point Analysis (determination of operating conditions within the engine flight envelope at which an engine’s operability margin is reduced to a minimum level) and Compliance Methods.

REQUIREMENT LESSONS LEARNED (A.3.3.2.5)

Several aircraft have encountered engine water ingestion problems in ground operation during takeoff or landing through water. Problems were reduced through the use of nose wheel chine tires or nose wheel deflectors.

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Incidents have been recorded where engines were damaged or aircraft were lost because of water ingestion. Fuselage runoff water supposedly drained into the inlet of Engine AF which led to the loss of several Aircraft V. In 1972, Aircraft Y experienced a dual engine stall (Engine L) due to "severe precipitation" at 5,029 m (16,500 ft). Another Aircraft Y experienced engine damage from runway water ingestion. Water ingestion was suspected as the possible cause of an Engine L stall in an Aircraft B on 11 Sept. 1976 (runway water).

In a paper entitled "Engine/Airframe Interface Considerations for Future Open-Ocean ASW Aircraft", G. Ward, LTV Aerospace Corp., Proceedings of the 7th Annual National Conference on Environmental Effects on Aircraft and Propulsion Systems, 25-27 Sep 1967, the presenter recommended that military aircraft engines be designed to operate satisfactorily with ingested water levels up to 5% of total airflow.

A.4.3.2.5 Atmospheric liquid water ingestion.

The requirements of 3.3.2.5 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.3.2.5)

The effects of water ingestion must be determined.

VERIFICATION GUIDANCE (A.4.3.2.5)

Past verification methods have included analysis, inspection, and test.

The contractor should propose a water ingestion versus power level schedule to verify the engine performance under conditions of the worst case water ingestion.

The following procedure may be used as a verification method.

"The engine shall start and operate satisfactorily from ground idle to maximum power with water ingested at varying rates of (a) to (b) percent with no airbleed, (c) to (d) percent with air bleed, and with a minimum of (e) percent (up to the limit of the engine) of total inlet mass flow in the form of water entering the engine inlet through a segment equivalent to one third the inlet area. Idle-max-idle throttle transients shall be conducted at each test condition. The engine shall be operated at each condition for five minutes. During the test, the effects of the water ingestion on engine performance shall be noted and recorded.

Following the performance check, the engine shall be disassembled sufficiently for inspection. This test shall be considered to be satisfactorily completed when adequate clearances were maintained, no damaging or detrimental rubbing occurred during the test, the performance has not deteriorated, and the gas flow path parts show no damage."

The following should be used to tailor the preceding guidance paragraph:

(a): A value of 2 percent.

(b): A value of 3 percent.

(c): A value of 3.5 percent.

(d): A value of 5 percent.

(e): A value of 5 percent.

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Background:

The water ingestion test should be conducted at sea level static conditions.

Flight tests to verify the effects of the engine installation on water ingestion should be conducted behind an aircraft water spray test vehicle. Tanker flight tests may be conducted at the same time as tanker icing tests; however, spray nozzle configurations may differ for water and ice. FAA AC 33-78.1, Section 4, contains guidance for conducting an engine test to demonstrate compliance with the requirement.

VERIFICATION LESSONS LEARNED (A.4.3.2.5)

None.

A.3.3.2.6 Armament gas ingestion.

The engine shall meet all the requirements of the specification during and after Armament Gas Ingestion (AGI), except as follows.

If any performance loss occurs during AGI, the engine shall recover within (a) seconds and without pilot intervention, to the thrust or power available immediately prior to the AGI event. In addition, when an AGI signal is present, engine thrust or power shall never fall below (b) percent of the thrust or power available immediately prior to the AGI.

REQUIREMENT RATIONALE (A.3.3.2.6)

AGI may cause a rapid rate of pressure and temperature change and attendant airflow distortions to the engine of combat aircraft. The engine's tolerance to such ingestion should be determined to avoid compressor stall, surge, engine flameout, excessive fan or compressor blade stresses, or other mechanical damage.

REQUIREMENT GUIDANCE (A.3.3.2.6)

The engine and air vehicle contractors should determine the values for (a) and (b) based on mission effectiveness studies. The values used are subject to approval by the Using Service.

In the past the following has been used to tailor the specification paragraph:

(a): A value of at least 5 -10 seconds.

(b): A value of at least 93 percent.

If a performance loss occurs, the engine should recover full performance in 5 -10 seconds depending on the type of gas ingested and length of gas ingestion. This performance loss is assumed to occur without an AGI signal. When an AGI signal is provided by the aircraft, the engine should lose no more than 7% thrust or power for 5 -10 seconds depending on the type of gas ingested and length of gas ingestion. The engine should recover to full performance within 5 seconds following AGI signal cancellation.

The engine operating condition(s) and armament gas ingestion condition(s) and characteristic(s) should be specified by the engine contractor using information provided by the air vehicle contractor. The conditions and characteristics should include: engine power setting, pressure altitude, Mach number, gas ingestion duration, mass flow rate, propellant type and missile position.

For turboshaft engines, the engine operating and armament gas ingestion conditions specified by the contractor should follow the guidelines below:

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The engine operating conditions should be at the maximum continuous power setting, zero Mach number and armament gas conditions should be with rocket firing duration of 0.1 second with the rocket positioned 4.6 meters (15.1-ft) from the engine inlet face with the rocket exhaust perpendicular to the engine face and on the engine center line.

Characteristics for demonstration should be as follow:

a.	Altitude	1.2 kilometers (4,000 ft)
	Rocket mass flow rate	13.6 kilograms per second (30 pps)
	Propellant characteristics	Non-aluminized double base
b.	Same as a., above, except propellant characteristics will be aluminized composite	
c.	Altitude	Sea level
	Rocket mass flow rate	36.3 kilograms per second (80 pps)
	Propellant characteristics	Non-aluminized double base
d.	Same as c., above, except propellant characteristics will be aluminized composite	

In the case of gun gas ingestion, define the critical conditions from the aircraft's design and mission, and for each of these conditions define the engine power setting, altitude, Mach number, maximum firing duration, and quantity of gas ingested.

In establishing rocket gas ingestion conditions, consider the type of rocket, rocket mass flow rates, firing durations, rocket plume characteristics versus altitude and Mach number, and rocket flight paths for the various loading configurations.

Background:

All engine control system accommodations for AGI should be specified when utilized in the design. Such control accommodations should be capable of being actuated by the armament firing trigger system in the aircraft cockpit and should be a normally off type design. All corresponding aircraft requirements (e.g., signals and interface) should also be specified. Further, the timing of the air vehicle signal to the ingestion event should be considered and integrated into the air vehicle and engine control algorithms.

Avoiding armament gas ingestion is desired and most successful air vehicle and engine integrations have adopted this philosophy. Engine contractors should work closely with the weapon system contractor during the preliminary design phase, and efforts should be directed at avoiding and minimizing armament gas ingestion.

REQUIREMENT LESSONS LEARNED (A.3.3.2.6)

During the Aircraft Q development program, gun gas ingestion from the PAUSE-8 mm gun caused Engine K to experience compressor stalls resulting in engine flameout or overtemperature in some instances. Extensive testing of different baffles and gas stripper configurations was required to develop a satisfactory solution. Periodic water washing of engines to remove gun gas combustion products was necessary for field operations.

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In the past, weapon systems such as the SPARROW and SIDEWINDER missiles had engine operating conditions, AGI conditions, and characteristics specified by contractors as follow:

Engine power setting - Intermediate. This represents the power setting at which most air-to-air rockets would be fired. Idle represents power setting for air-to-ground rockets.

Altitudes of 11 km (36,089 ft) and 2.3 km (7,500 ft) representing the maximum altitudes where either guided or unguided rockets could be fired.

Mach number - maximum and minimum aircraft flight speed where a rocket could be fired.

Rocket mass flow rates of 13.6 and 36.3 kg/sec (30 and 80 lb/sec) 13.6 kg/sec (30 lb/sec) was the highest flow rocket in the 1972/73 time period and 80 lb./sec was the flow rate for a rocket under consideration.

Rocket firing duration of 2.0 seconds - this represents the maximum time interval during which the engine inlet is exposed to the rocket exhaust plume (i.e., maximum time interval during which the engine can ingest rocket gases as the aircraft travels through the rocket plume).

Rocket position of 4.5 m (15 ft) - selected because this position was considered a realistic position.

The Using Service provided propellant types.

A.4.3.2.6 Armament gas ingestion (AGI).

The requirements of 3.3.2.6 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.3.2.6)

The impact of armament gas ingestion on engine performance must be determined.

VERIFICATION GUIDANCE (A.4.3.2.6)

Past verification methods have included analysis and testing.

It is recommended that tests should be conducted on the engine during ground and flight tests.

VERIFICATION LESSONS LEARNED (A.4.3.2.6)

Estimates for various engine inlet temperature rates range from 550°C (1000°F) to 5500°C (10,000°F) per second for an armament gas ingestion period of one-tenth second.

Helicopter drive train transients can be caused by AGI therefore ground tests should be conducted to determine transient magnitude.

Temperature changes due to ingestion of rocket exhaust or gun gas can be simulated by discharging the device mounted in an appropriate location ahead of the engine in a sea level facility.

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A.3.3.2.7 Steam ingestion.

The engine shall meet all specification requirements without stall, surge, flameout, mechanical/structural damage, or permanent performance loss, when the total inlet mass flow is at least (a) steam, and has a minimum temperature ramp (increase) of (b) °C/sec ((b) °F/sec) with a (c) °C ((c) °F) temperature rise in a region less than (d) percent of the engine inlet flow area. After a steam ingestion event, the engine shall meet all requirements of the specification.

REQUIREMENT RATIONALE (A.3.3.2.7)

Steam from catapult launching causes an adverse environment for aircraft engine operation. Steam ingested by gas turbine engines can cause compressor surge by the distortion effect on the compressor inlet temperature and specific humidity, combined with the effect of ingesting water.

REQUIREMENT GUIDANCE (A.3.3.2.7)

This requirement and the verification of 4.3.2.7 should be labeled “Not Applicable” for weapon systems not utilizing steam catapults.

For carrier-based aircraft, the following should be used to tailor the specification paragraph:

(a): Two percent has been used previously.

(b): A maximum value of 2840°C/sec (5150°F/sec).

(c): A value of at least 100°C (212°F) .

(d): A value no greater than 25 percent.

Background:

The results of tests indicate that during actual catapult the free leakage steam follows a flow pattern that concentrates the steam ingestion in 25% of the inlet on some aircraft. On aircraft with serpentine inlets, the inlet flow patterns may concentrate the steam ingestion into an area greater than 25% of the total inlet area or may cause concentration in several smaller areas less than 25%. When the steam ingested is not concentrated the engine is capable of ingesting over 3 times the quantity of steam without any compressor surge occurring.

REQUIREMENT LESSONS LEARNED (A.3.3.2.7)

In 1966 during the initial land based catapult testing of Aircraft N powered by turbofan Engine L, it experienced compressor surge by steam ingestion. This is a single engine aircraft with a low chin inlet, and it was capable of ingesting a large quantity of steam. The older degraded catapults leaked more steam than the new ones. An improvement to the steam catapult serving arrangement was incorporated on all the carriers which solved the problem for the Aircraft N steam ingestion problem. Testing experience from Aircraft N with Engine L revealed a temperature ramp of 2840°C/sec (5,150°F/sec) and a total temperature rise of 100°C (212°F). Reference: “Catapult Steam Ingestion Tests of Three Turbofan Engines in the A-7 Aircraft, W. B. Mallett and R. F. Parcells.

In 1988, Aircraft I with Engine J started to experience engine stalls on catapult launches. This is a twin engine aircraft and there were times when both engines would stall with fireballs exiting the exhaust.

Data taken on Engine J during catapult shots recorded 100°C (212°F) temperature rises within 0.2 seconds. No exact data on temperature range and temperature rises where determined because low response thermocouples were used.

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A.4.3.2.7 Steam ingestion.

The requirements of 3.3.2.7 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.3.2.7)

The Using Service must determine if the engine can operate satisfactorily while ingesting steam.

VERIFICATION GUIDANCE (A.4.3.2.7)

Past verification methods have included analysis and test.

Steam should be introduced into the engine inlet as 0.5, 1.0, 1.5, and 2.0% of the total mass flow. Initial temperature of the steam should be 245°C (475°F) and gradually raised to obtain a total temperature rise of 100°C (212°F) in the 25% region of the inlet that received the steam.

This verification and the requirement of 3.3.2.7 should be labeled “Not Applicable” for weapon systems not utilizing steam catapults.

Background:

Considerations for referencing other paragraphs or aircraft tests:

The Using Service should determine if the results of 4.3.2.5 Atmospheric Liquid Water Ingestion and 4.3.2.6 AGI can be substituted for this test. Another alternative would be a flight test with a catapult that can vary the steam leakage. This test should be done on the engine during flight tests.

VERIFICATION LESSONS LEARNED (A.4.3.2.7)

Steam ingestion is a problem that is a function of aircraft inlet design, compressor stall margin, and steam leakage or condition of the catapult. The first incident was reported in 1966, but after the catapult service was improved, there were no incidents reported until 1988.

The only documented laboratory steam ingestion test was by Engine Company B on an aircraft in 1966. This was presented at the Seventh Annual National Conference on Environmental Effects on Aircraft and Propulsion Systems at Princeton, New Jersey on September 25-27, 1967. Its title is “Steam Ingestion by Aircraft Gas Turbine Engines,” by Nicholas R. Tomassetti.

A.3.3.3 Electromagnetic environmental effects (E^3).

The engine shall meet the electromagnetic environmental effects requirements specified in this section.

REQUIREMENT RATIONALE (A.3.3.3)

An integrated design approach to E^3 is needed to achieve compatible operation of the engine and compatible operation of the engine with the aircraft and its electromagnetic environment. Electromagnetic fields are emitted from friendly and hostile ships, aircraft, missiles, satellites, and ground transmitters as well as from high altitude nuclear detonations and lightning. The engine must be protected from these intentional or non-intentional electromagnetic sources. Electromagnetic compatibility between the electrical and electronic systems and subsystems of aircraft, aircraft carrier deck, transient electromagnetic fields, and lightning electromagnetic environments is essential for proper and safe operation of the engine. Electrical and electronic equipment performance may be degraded, or the equipment may malfunction, as a result of internally generated interference or susceptibility to external fields and voltages.

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REQUIREMENT GUIDANCE (A.3.3.3)

Background:

An E^3 program is needed to provide maximum confidence that the engine will operate in the overall electromagnetic environment of the air vehicle. See MIL-STD-464 for additional guidance. Compliance with these requirements alone will not guarantee operation on the air vehicle or with the external environment. A parallel effort is required by the air vehicle or integration contractor to demonstrate that the stress levels specified for the engine electronics are consistent with the air vehicle electromagnetic environment. Through the systems engineering process, requirements are allocated to the engine subsystem through transfer functions which translate external environmental factors (lightning, EMP, and RF) to engine level stresses based on the air vehicle architecture. If there are any conflicts, it may require tailoring of the MIL-STD-461 limits or special considerations in the air vehicle installation.

REQUIREMENT LESSONS LEARNED (A.3.3.3)

Past use of these E^3 requirements has resulted in successful operation of engines with the aircraft in many varied electromagnetic environments.

A.4.3.3 Electromagnetic environmental effects (E^3).

N/A.

A.3.3.3.1 Electromagnetic interference (EMI).

All engine electrical and electronic equipment and subsystems shall meet their performance requirements when subjected to the susceptibility requirements of (a) requirement (b) and shall comply with the emission requirements of (a) requirement (c) . These requirements shall be met for engine operation in all environmental conditions within the operating envelope for all control system operating modes.

REQUIREMENT RATIONALE (A.3.3.3.1)

The engine will be subjected to various electromagnetic environments from sources within the aircraft as well as those external to the aircraft which may degrade or interrupt engine performance. The engine may also be a source of interference to other equipment on the aircraft.

REQUIREMENT GUIDANCE (A.3.3.3.1)

The Using Service should review the requirements of the latest version of MIL-STD-461 for applicability to unique EMI environments. MIL-STD-461 should be tailored as specified in the paragraph about Tailoring of requirements of MIL-STD-461 for the specific engine application.

The following should be used to tailor the specification paragraph:

(a): MIL-STD-461 or an equivalent commercial standard such as RTCA-DO-160.

(b): If MIL-STD-461 is used, then the Using Service should specify the specific design requirements of MIL-STD-461 that are applicable. The applicable susceptibility requirements of MIL-STD-461 for all engines are CS101, CS114, CS115, CS116 and RS103. If the engine is intended for Army use, then RS101 is also required. If commercial standards are used, specify the particular applicable sections.

(c): If MIL-STD-461 is used, then the Using Service should specify the specific design requirements of MIL-STD-461 that are applicable. The applicable emission requirements of MIL-STD-461 for all engines are CE102 and RE102. If the engine is intended to be installed on Navy aircraft with Anti-Submarine Warfare capability or intended for Army use, CE101 and RE101 are also required. If commercial standards are used, specify the particular applicable sections.

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Background:

MIL-STD-461 has been coordinated and approved by the Tri-Services. MIL-STD-461 may not adequately address certain unique environments for all possible applications, therefore MIL-STD-461 may need tailoring as specified in the paragraph entitled, Tailoring of requirements of MIL-STD-461.

For engines that have safety critical (full authority) electronics, MIL-STD-461 defaults to design requirements for CS114 and RS103 which represent hardening levels that have worked successfully in most aircraft applications. Included are concerns with external environments such as ground based and carrier deck radars and communications equipment. These two requirements may require some tailoring of the levels after review of the intended operational environment.

Two particular areas that may require tailoring are for the indirect effects of lightning for composite air vehicles and electromagnetic pulse (EMP) requirements. CS115 and CS116 specify transient requirements that are adequate for most applications. However, the low frequency content of lightning waveforms together with the conductivity characteristics of composites can result in substantial currents and voltages internal to the air vehicle. The lightning, EMP and external RF environment areas, in particular, require coordination with air vehicle contractors to determine the flow-down of internal stress levels (MIL-STD-461) from the external lightning, EMP, and RF environments imposed upon the entire aircraft system. MIL-STD-464 contains additional information on how the external aircraft environments can be transferred into equipment level specifications, such as this engine specification. Also, RTCA-DO-160 provides general waveform and amplitudes for use in composite air vehicles for commercial aircraft applications, which may be applicable for military use. If the engine wiring harnesses are intended to be installed on the aircraft in a relatively electromagnetic protected area, then it is expected that the levels specified in MIL-STD-461 are usually adequate without tailoring. If the harnesses are to be installed in an open area, such as in a nonconductive nacelle, then the limits may need to be tailored. A possible alternative to imposing higher susceptibility levels (may exceed test capability) is to require that the engine components meet the requirement with the overall shields removed from the wiring harness. The use of overall shields in the actual installation then provides additional protection.

REQUIREMENTS LESSONS LEARNED (A.3.3.3.1)

Past use of E^3 requirements has resulted in successful operation of aircraft and engines in many varied electromagnetic environments. There have been EMI problems encountered with some engine installations in the past, but in those cases, the EMI requirements were either inadequate for the particular application or waivers were given to the requirements. Application of effective EMI requirements and continual monitoring of the design development by the Using Service has resulted in compatible operation of engines on most aircraft.

There is a documented case where ground support equipment was affected on the aircraft carrier deck as Engine K was being trimmed on Aircraft T.

A.4.3.3.1 Electromagnetic interference (EMI).

The requirements of 3.3.3.1 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.3.3.1)

Verification using standardized electromagnetic interference methods is necessary to ensure the engine meets all E^3 interference requirements and system safety requirements.

VERIFICATION GUIDANCE (A.4.3.3.1)

Past verification methods have included analysis and tests.

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MIL-STD-461 should be specified, if MIL-STD-461 is specified in 3.3.3.1. If a commercial document is specified in 3.3.3.1, then specify the appropriate commercial test standard. RTCA-DO-160 contains both the requirements and test procedures in the same document.

For EMI, the requirements of MIL-STD-461 should be verified with the testing techniques and testing methods established in MILSTD461. The contractor should prove by test that the engine electrical and electronic systems and subsystems will operate satisfactorily within an EMI environment, and within safe limits, as established in the contract. MIL-STD-461 establishes general techniques for use in the measurement and determination of the electromagnetic emission and susceptibility characteristics of electronic, electrical, and electromechanical equipment and subsystems.

The air vehicle contractor should flow-down the external environment requirements to the engine contractor at time of engine and air vehicle development. The carrier environment will change during development. A software regression test can be conducted to determine limit exceedances taking into account worst case noise combinational effects. The software test can be run at representative flight conditions and thrust fluctuation can be calculated to determine whether the system meets its EMI susceptibility requirements. This testing should be repeated with any subsequent software package prior to its release to flight test or production. This regression testing should be done to ensure that the control system would meet its EMI susceptibility requirements throughout the system life cycle. The EMI software regression test should be focused on the signal conditioning and signal processing code for the engine and air vehicle sensors and communication links that are connected to the engine control unit.

Background:

Each EMI design requirement in MIL-STD-461 has a companion test procedure. Demonstration tests per MIL-STD-461 should be conducted on test articles which simulate that of the flight configurations. Interconnected wiring, physical arrangement, and mounting methods should fully emulate the aircraft installation.

Hardware and software changes could adversely affect EMI hardness. Anytime such a change is made, consideration needs to be given to its impacts on EMI hardness. The contractor should propose the type and level of re-verification in the Engineering Change Proposal (ECP).

VERIFICATION LESSONS LEARNED (A.4.3.3.1)

There were problems found on Engine C Digital Electronic Control Unit (DECU), during EMI testing performed at government facilities.

Engine C had an EMI problem with its electronic engine control erroneously tripping a fuel flow rate limiter. Troubleshooting revealed the rate limiting software needed modification. The contractor's ability to adequately conduct the necessary EMI software tests was a concern.

A.3.3.3.2 Intrasystem electromagnetic compatibility (EMC).

The engine, installed in the air vehicle, shall meet the applicable requirements of: _____.

REQUIREMENT RATIONALE (A.3.3.3.2)

The engine contractor must design into the engine sufficient hardening to allow the engine to be able to be installed in the intended aircraft and be electromagnetically compatible with the aircraft.

REQUIREMENT GUIDANCE (A.3.3.3.2)

The following should be used to tailor the specification paragraph:

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The contractor should propose to meet the requirements of 5.2 of MIL-STD-464. There is no known commercial equivalent document. The Tri-Service coordinated EMC standard, is MIL-STD-464 and must now be used on all joint programs.

Background:

Normally there are design and test requirements in MIL-STD-461 applicable for each of the external environments, but they need modification using an overall E-cubed system level approach for the particular system application using MIL-STD-464. MIL-STD-464 includes the overall system E^3 requirements and MIL-STD-461 includes the component level design and test requirements.

The Using Service must tailor the applicable requirements from the indicated document, based on the intended aircraft.

Any specific additional hardening measures that should be accomplished by the integrator, should be identified by the engine contractor and specified in the corresponding ICD.

A proactive approach must be taken to ensure that the new engine is not designed in a “vacuum” and that considerations of the intended platform are made.

REQUIREMENT LESSONS LEARNED (A.3.3.3.2)

Experience has shown that compliance with electromagnetic interference (EMI) standards, alone, is not always sufficient to ensure EMC, especially for programs that have allowed EMI waivers for some of the equipment already on the aircraft or aircraft that have unique emissions or susceptibility characteristics.

A.4.3.3.2 Intrasystem electromagnetic compatibility (EMC).

The requirement of 3.3.3.2 shall be verified by analysis and test on the engine as installed on the aircraft.

VERIFICATION RATIONALE (A.4.3.3.2)

The engine must be tested on the intended platform to ensure the requirements of 3.3.3.2 have been met. If this testing is not done by the engine contractor, the engine contractor should support the integration contractor in the conduct of the test.

VERIFICATION GUIDANCE (A.4.3.3.2)

The engine should meet the EMC requirements.

VERIFICATION LESSONS LEARNED (A.4.3.3.2)

None.

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A.3.3.3.3 Intersystem electromagnetic compatibility (EMC).

The engine as installed in the air vehicle shall meet the requirements of the specification when exposed to the following external electromagnetic environment:

REQUIREMENT RATIONALE (A.3.3.3.3)

The engine must be able to operate in the electromagnetic environment external to the aircraft. The engine contractor must design into the engine sufficient hardening to allow the engine to be installed in the intended aircraft and operated in the intended electromagnetic environment of the aircraft.

REQUIREMENT GUIDANCE (A.3.3.3.3)

The following should be used to tailor the specification paragraph:

The contractor should propose using or tailoring one of the tables in 5.3 of MIL-STD-464. The requirement should take into account the actual aircraft installation characteristics which may modify the total air vehicle system requirements so that the engine requirements can be properly tailored.

The intersystem EMC requirement is a total air vehicle requirement which must be specifically tailored for each engine application. It is possible for the total environment to be imposed on the engine manufacturer. An example would be an engine manufacturer that was providing the engine and nacelle to a transport type of air vehicle. In other applications the environment of MIL-STD-464 would be reduced by the shielding provided by the air vehicle.

Any specific additional hardening measures that must be accomplished by the integrator should be identified by the engine contractor and specified in the corresponding ICD. The contractor should specify the applicable requirements from the indicated documents, based on the intended aircraft for the engine.

REQUIREMENT LESSONS LEARNED (A.3.3.3.3)

Based on experience with Aircraft C engines, when the engines are not designed to operate in the intended operational environment of the aircraft, numerous problems are often found, many with flight safety consequences. The solutions to these problems are often time consuming and expensive. The experience with these aircraft have shown that the integrator cannot sufficiently protect a system that does not have sufficient hardening already designed into it.

A.4.3.3.3 Intersystem electromagnetic compatibility (EMC).

The requirement of 3.3.3.3 shall be verified by analysis or test.

VERIFICATION RATIONALE (A.4.3.3.3)

The engine must be tested on the intended platform (or in the intended installed environment) to ensure the requirements of 3.3.3.3 have been met.

VERIFICATION GUIDANCE (A.4.3.3.3)

In addition to air vehicle level ground tests, this requirement can be verified by means other than testing on the aircraft. One example would be an engine in a nacelle which could be tested as a unit without the aircraft. Another example is where aircraft electromagnetic coupling data is used to demonstrate a margin between the aircraft coupling data and the engine EMI data to allow for the earliest possible testing even prior to full-up engine nacelle testing to provide for innovative component pre-testing that might find problems early on.

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The Using Service should provide the specific contractor support requirements for the testing. Typically, this testing is performed by a Government test activity based on a test plan generated by that activity. The contractor provides the technical expertise on the engine and its operations and assists in the troubleshooting of any problems encountered during the testing.

VERIFICATION LESSONS LEARNED (A.4.3.3.3)

None.

A.3.3.3.4 Lightning.

The engine shall meet the applicable lightning requirements of: _____.

REQUIREMENT RATIONALE (A.3.3.3.4)

The lightning requirement is needed for any engine but the actual lightning environment is dependent on the installation.

REQUIREMENT GUIDANCE (A.3.3.3.4)

The following should be used to tailor the specification paragraph:

The contractor should propose to meet the requirements of MIL-STD-464, requirement 5.4. The lightning environment defined in MIL-STD-464 may need to be tailored depending upon the actual aircraft installation characteristics. The requirements would be totally different for a nacelle installation as opposed to a situation in which the engine is installed more deeply in the air vehicle. Once the shielding level of the aircraft has been determined, RTCA-DO-160 should be used to establish the waveforms and amplitudes to which the engine electrical components will be subjected.

REQUIREMENT LESSONS LEARNED (A.3.3.3.4)

None.

A.4.3.3.4 Lightning.

The requirement of 3.3.3.4 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.3.3.4)

Lightning verification is required.

VERIFICATION GUIDANCE (A.4.3.3.4)

The following should be used to tailor the specification paragraph:

Past verification methods have included analysis and test. The contractor should propose the appropriate verification methods. The SAE and EUROCAE are in the process of preparing an international lightning test standard which should be utilized when published. See the final draft of AC/AMJ 20-TBD. Online access to this document as of November 2003 was: **http://www.aea.net/Tech_Pubs/RegDocuments/WG-330_-_AC20-TBD_Lightning_Environment.pdf**. The testing standard is EUROCAE WG 31/SAE AE4L document APR-TBD (in preparation) "Aircraft Lightning Testing Standard".

VERIFICATION LESSONS LEARNED (A.4.3.3.4)

None.

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A.3.4 Integrity.

A.4.4 Integrity.

A.3.4.1 Structural integrity.

A.4.4.1 Structural integrity.

A.3.4.1.1 Design service life.

The engine shall have a design service life specified per _____.

REQUIREMENT RATIONALE (A.3.4.1.1)

The service life must be established since it is one of the primary design goals and is needed as a criteria for quantifying structural integrity and to achieve logistic and economic effectiveness.

REQUIREMENT GUIDANCE (A.3.4.1.1)

MIL-HDBK-1783 should be used to tailor the specification paragraph.

REQUIREMENT LESSONS LEARNED (A.3.4.1.1)

Structural life requirements are the most difficult primary design goals to fulfill. This was the case on many past engine development programs due to a lack of adequate usage parameter definitions. Although requirements for mission profiles, mission mix, and flight hours have often been defined accurately, the important usage parameters that govern cyclic life (major throttle cycles other than the start-stop excursion, time at or above Intermediate power, dwell times, etc.) have not been accurately defined. As a result, operational data has revealed usage parameters not accounted for in design and has resulted in significant reduction in life limits for critical parts/critical safety items and the associated need for redesign and spare parts. Therefore, it is important that realistic design usage information be identified at the outset of the development program for use in design, analysis, and test. The information in table XXXVIII presents design service life requirements that have been used in the past.

In the past, time to cracking did not necessarily become the life limit for a hot part; therefore, the design of hot section parts have been based on verification of wear, LCF, creep, stress rupture, oxidation and erosion, and sulphidation.

Cooled turbine vanes were designed such that cooling air would outflow in the presence of cracking or other distress that extends through the thickness. Dual compartment positive outflow design has been needed to minimize erosion rates subsequent to the occurrence of thermal mechanical fatigue cracks thereby maximizing total usable life of the airfoil.

Turbine vanes have also been designed for positive retention so that vane segments would not fall into the gas flow path and cause secondary damage subsequent to total burn-through or severance of an airfoil.

Predictions of bearing performance and life have not always been reliable, especially those bearings whose operating conditions exceed recent engine experience or whose designs are unusual. For example, bearing performance analytical models and limited life tests did not have the capability to predict or reveal roller dynamic instability which occurred in some high DN engine roller bearings. More extensive life testing may have uncovered this potentially catastrophic bearing failure mode.

Air Force engine related mishap data from 1976-1987 shows a high incidence of failure or malfunction of engine components and externals. During that time period, engine component and external failures or malfunctions accounted for no less than 35% of the total engine related mishaps.

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A.4.4.1.1 Design service life.

The requirements of 3.4.1.1 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.4.1.1)

Design service life requirements must be verified to ensure the desired levels of damage tolerance, durability, functional capability, operability, performance, reliability, and strength are attained.

VERIFICATION GUIDANCE (A.4.4.1.1)

Past verification methods have included analyses and tests.

MIL-HDBK-1783 should be used for further verification guidance.

VERIFICATION LESSONS LEARNED (A.4.4.1.1)

Actual service life can vary from the design service life. Verification that the service missions are or are not the same as the design service missions should be completed within the first five years after ISR or before the first parts reach 25% of the life limit.

Insufficient qualification testing (methods and duration) has resulted in the fielding of engine components that were not capable of meeting the desired operational life.

A.3.4.1.2 Design usage.

The engine structure shall be capable of withstanding the design usage specified per ____.

REQUIREMENT RATIONALE (A.3.4.1.2)

One of the major shortcomings in past air vehicle and engine development programs has been inadequate definition and verification of the operational usage parameters critical to the durability of engine components. Although requirements for mission profiles, mission mix, and flight hours have often been defined accurately, the important usage parameters that govern cyclic life (major throttle cycles other than the start-stop excursion, partial throttle cycles, time at or above Intermediate power, dwell times, etc.) have not been accurately defined as part of design information on many systems. As a result, operational data has revealed usage parameters not accounted for in design and has resulted in significant reduction in life limits for critical parts/critical safety items and the associated need for redesign and spare parts. Therefore, it is important that realistic design usage information be identified at Milestone B, the initiation of System Demonstration and Development (SDD) for use in design, analysis, and test of the engine.

REQUIREMENT GUIDANCE (A.3.4.1.2)

MIL-HDBK-1783 should be used to tailor the specification paragraph.

REQUIREMENT LESSONS LEARNED (A.3.4.1.2)

One of the major shortcomings in past engine development programs has been inadequate definition of the design usage parameters critical to engine durability. Although requirements for mission profiles, mission mix, and flight hours have often been defined accurately, the important usage parameters that govern cyclic life (major throttle cycles other than the start-stop excursion, time at or above intermediate power, dwell times, etc.) have not been accurately defined and other type cycles have been ignored because of the belief that there were non-damaging. As a result, operational data has revealed usage parameters not accounted for in design and has resulted in significant reduction in life limits for critical parts and the associated need for redesign and spare parts. Therefore, it is important that realistic design

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usage information be identified at the outset of the development program for use in design, analysis, and test.

Past program-specific usage parameters are specified in table XXXIX.

A.4.4.1.2 Design usage.

The requirements of 3.4.1.2 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.4.1.2)

Requirement 3.4.1.2 usage requirements will entail multiple mission profiles each with separate speed, altitude and throttle excursions as a function of time. It is not always practical to analyze and test each of these discrete profiles with appropriate mission mix throughout the various development tasks but where possible, this should be done. Therefore, a minimum number of design duty cycles must be derived early in the development phase (as early as contract award) for use in all subsequent analysis and test tasks. These design duty cycles may be provided by the Using Service as part of the Request for Proposal (RFP).

VERIFICATION GUIDANCE (A.4.4.1.2)

Past verification methods have included analyses and simulator tests.

MIL-HDBK-1783 should be used for further verification guidance.

VERIFICATION LESSONS LEARNED (A.4.4.1.2)

None.

A.3.4.1.2.1 Externally applied forces.

The engine shall function satisfactorily under externally applied forces specified per _____.

REQUIREMENT RATIONALE (A.3.4.1.2.1)

Engines must be designed to withstand the externally applied and the induced forces which may result from flight maneuvers, landings, and takeoffs. The components must be designed to withstand the relatively infrequent, nonrepetitive shocks, or transient vibrations induced by the external forces applied to the vehicle in which they are mounted.

REQUIREMENT GUIDANCE (A.3.4.1.2.1)

MIL-HDBK-1783 should be used to tailor the specification paragraph.

REQUIREMENT LESSONS LEARNED (A.3.4.1.2.1)

Aircraft such as the N, W, X, AB, AC, and AD have all recorded load factors up to 7g's, and Aircraft A's go up to 9g's. Aircraft maneuver requirements have increased so that particular weapon systems must now be capable of 8 to 10 g's.

Engine J's model specification used a maneuver load spectrum which showed various values of positive and negative "g's" and cycles per 1000 hours for each "g" value. The maximum static load requirement was reduced to 1.4 times the landing loads rather than the 1.5 times requirement.

The forces induced by acceleration can cause: permanent deformations and fractures that disable or destroy equipment, broken fasteners or mounting hardware that cause equipment to become projectiles, short or open circuits on electronic boards, changes in inductances and capacitances, relays to open or

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close, actuators and other mechanisms to bind, seals to leak and pressure and flow regulators to charge valves.

A.4.4.1.2.1 Externally applied forces.

The requirement of 3.4.1.2.1 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.4.1.2.1)

The ability of the engine to withstand the external forces to which it may be subjected due to flight maneuvers, landings, and takeoffs must be verified.

VERIFICATION GUIDANCE (A.4.4.1.2.1)

Past verification methods have included analyses and component and engine tests.

MIL-HDBK-1783 should be used for further verification guidance.

VERIFICATION LESSONS LEARNED (A.4.4.1.2.1)

None.

A.3.4.1.3 Material characterization.

Material properties shall be based per _____:

REQUIREMENT RATIONALE (A.3.4.1.3)

This requirement is needed to ensure adequate material capability by limiting the statistical variation of the properties to conventional aerospace design practices. Material structural properties must be quantified during detail design so that materials selection and operating stress levels can be established which provide a high degree of confidence that operational requirements will be met. Early generation of sufficient data for use in preliminary and detail design is emphasized since downstream surprises relative to structural properties will have a significant impact on redesign, substantiation and replacement needs and weapon system availability.

REQUIREMENT GUIDANCE (A.3.4.1.3)

MIL-HDBK-1783 should be used to tailor the specification paragraph.

REQUIREMENT LESSONS LEARNED (A.3.4.1.3)

None.

A.4.4.1.3 Material characterization.

The requirement of 3.4.1.3 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.4.1.3)

Material properties must be verified to ensure appropriate selection of materials.

VERIFICATION GUIDANCE (A.4.4.1.3)

Past verification methods have included documentation for existing materials that have been characterized and component tested.

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MIL-HDBK-1783 should be used for further verification guidance.

VERIFICATION LESSONS LEARNED (A.4.4.1.3)

Many durability problems can be traced to the selection of unsuitable materials. The need to reduce engine weight has forced gas turbine engine contractors to make compromises in the selection of materials or higher risk designs. Characterization and choice of materials should be closely monitored by the Using Service. Material selection should be reviewed under a risk assessment or management plan.

A.3.4.1.4 Parts classification.

All engine parts shall be classified as fracture critical, durability critical, and durability non critical. Fracture critical parts/critical safety items shall be further classified as safety and mission critical per _____.

REQUIREMENT RATIONALE (A.3.4.1.4)

Parts, components and expendables must be classified to assure the appropriate design requirements are applied.

REQUIREMENT GUIDANCE (A.3.4.1.4)

MIL-HDBK-1783 should be used to tailor the specification paragraph.

REQUIREMENT LESSONS LEARNED (A.3.4.1.4)

It has been difficult to determine, in advance, engine parts or components whose failure would have resulted in secondary failure which would have lead to loss of aircraft or essential mission capability. Therefore, continued verification has been required as the subsystem was being defined. An example is the high pressure hydraulic or pneumatic pressure vessel which, by itself, may not be safety-of-flight critical, but if located on or close to primary air vehicle structure could precipitate a failure of the air vehicle if it were to fail.

The number of critical parts/critical safety items can be significantly larger for an engine model used in a single engine configuration as opposed to a dual engine configuration. For example, the number of critical parts/critical safety items for Engine B in Aircraft A and R is 45 and 70, respectively, excluding externals.

A.4.4.1.4 Parts classification.

The requirement of 3.4.1.4 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.4.1.4)

Verification of classification is necessary to ensure the appropriate design requirements are applied.

VERIFICATION GUIDANCE (A.4.4.1.4)

Past verification methods have included analyses.

MIL-HDBK-1783 should be used for further verification guidance.

VERIFICATION LESSONS LEARNED (A.4.4.1.4)

None.

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A.3.4.1.5 Durability.

The engine shall be durable for the design service life when subjected to the usage of 3.4.1.2 and the requirements specified per _____.

REQUIREMENT RATIONALE (A.3.4.1.5)

Durability requirements must be applied since engine durability is a primary design requirement. Durability requirements must be applied to minimize cracking or other structural or material degradation which could result in functional impairment or excessive in-service maintenance problems.

REQUIREMENT GUIDANCE (A.3.4.1.5)

MIL-HDBK-1783 should be used to tailor the specification paragraph.

REQUIREMENT LESSONS LEARNED (A.3.4.1.5)

Increased turbine engine performance requirements have resulted in higher thrust-to-weight ratios. This trend has led to higher stresses in engine components which in turn has forced development of specialized high-strength alloys and new manufacturing processes and techniques. As a result, Air Force engine systems have experienced early structural cracking and often contain components that have to be replaced one or more times during the operational life of the engine. Experience with past engine systems highlights the fact that a disciplined approach for design, analysis, test, and improvement of engine components is needed so that in-service maintenance and component replacement needs over the operational life of the engine will be minimized.

Engine W had a compressor blade flutter problem (high cycle fatigue) which was discovered after the engine had successfully passed qualification testing and after engine production started. Structural failures during Engine B's qualification testing threatened the existence of the weapon system program and raised serious questions about military engine procurement techniques. Fatigue failures in Engine Y have caused loss of life and aircraft. Although the Navy Aircraft B has two engines, Engine L problems resulted in loss of life and aircraft. Cracking of expensive turbine blades in Engine L produced serious maintenance support problems through excessive replacement rates during engine overhaul.

A.4.4.1.5 Durability.

The requirements of 3.4.1.5 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.4.1.5)

Attainment of durable parts is achieved through application of detail requirements in the functional areas of design, materials selection, manufacturing control, and inspection. The strength and life analysis report is one of the most important reports submitted by the engine contractor. Special inspections conducted periodically during the durability test programs are essential to preclude part failure and loss of the development engine.

Definitive criteria are needed to judge the success of qualification tests. Problems will likely be discovered during teardown inspection of these engines prior to attaining the required test durations. Therefore, criteria is needed to establish course of action in the event problems occur and to define follow on actions (i.e.; remaining test duration, problem cause, corrective action, operational implications, qualification tests for redesign, etc.).

VERIFICATION GUIDANCE (A.4.4.1.5)

Past verification methods have included strength and life analysis and part, component, and engine tests.

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MIL-HDBK-1783 should be used for further verification guidance.

VERIFICATION LESSONS LEARNED (A.4.4.1.5)

Complete failure of test engines has occurred in past development programs due to undetected damage growth. Many of these failures could have been prevented by use of detail inspection requirements (methods and intervals) developed through the durability and damage tolerance control plans and experience obtained from earlier development testing (e.g., previously observed deterioration and distress).

A.3.4.1.5.1 High cycle fatigue (HCF) life.

Engine parts shall have the high cycle fatigue life specified per _____.

REQUIREMENT RATIONALE (A.3.4.1.5.1)

Experience has shown that engine structural components operating under combined steady and vibratory stress conditions must be designed to ensure resistance to HCF cracking.

REQUIREMENT GUIDANCE (A.3.4.1.5.1)

MIL-HDBK-1783 should be used to tailor the specification paragraph.

REQUIREMENT LESSONS LEARNED (A.3.4.1.5.1)

USAF experience on Engine A and B programs has shown that titanium airfoils can be very sensitive to vibratory stresses. Failures of fan and compressor airfoils, resulting from vibratory stresses, have caused domestic object damage and catastrophic failure. Controlling this problem has required significant additional effort by field units. Testing performed as part of these investigations has shown that the fatigue strength continues to decline past 3 X 10^7 cycles. Based on the results of this testing, the life requirement of titanium materials should be set at 10^9 cycles. USAF experience and materials data for typical blade stresses for titanium, nickel and single crystal blades in the fan, HPC and HPT has shown that 10-15 ksi peak-peak to be the maximum vibratory stress desirable to meet the HCF life requirements.

A.4.4.1.5.1 High cycle fatigue (HCF) life.

The requirements of 3.4.1.5.1 shall be verified by _____:

VERIFICATION RATIONALE (A.4.4.1.5.1)

The HCF life must be verified to avoid HCF failures.

VERIFICATION GUIDANCE (A.4.4.1.5.1)

Past verification methods have included analyses and tests.

MIL-HDBK-1783 should be used for further verification guidance.

VERIFICATION LESSONS LEARNED (A.4.4.1.5.1)

None.

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A.3.4.1.5.2 Low cycle fatigue (LCF) life.

All engine parts shall have their minimum LCF lives specified per _____.

REQUIREMENT RATIONALE (A.3.4.1.5.2)

Low Cycle Fatigue is one of the most severe and costly problems encountered in service. This requirement is an attempt to reduce or eliminate LCF failures.

REQUIREMENT GUIDANCE (A.3.4.1.5.2)

MIL-HDBK-1783 should be used to tailor the specification paragraph.

REQUIREMENT LESSONS LEARNED (A.3.4.1.5.2)

Most LCF problems did not surface until a number of years after qualification and acceptance by the Using Service.

The Navy has over the years had a number of low cycle fatigue problems due to drastic, unanticipated reductions in predicted lives which have resulted in aircraft groundings and required large scale programs to recover from the events. Most notable were Engines L, M, V, and Z. Although there were a number of incidents of parts initiating cracks, to the best of our knowledge, none of these events resulted in catastrophic in-flight failures. The cause of these problems was attributed primarily to lack of updated missions and mixes. Updated lives reflecting the updated actual missions and mixes were drastically lower than the previous lives or limits.

The visibility of life management increased significantly following two separate in-service engine failures. In October 1991, an Engine J experienced an uncontained failure of a high pressure turbine forward cooling plate (1787 hours vs. 1900 hour field limit); five months later, another Engine J experienced a catastrophic failure of a first stage fan disk (1950 hours vs. 4700 hour field limit).

A.4.4.1.5.2 Low cycle fatigue (LCF) life.

The requirement of 3.4.1.5.2 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.4.1.5.2)

The LCF lives must be verified to preclude the occurrence of part failures.

VERIFICATION GUIDANCE (A.4.4.1.5.2)

Past verification methods have included analyses, component and engine tests.

MIL-HDBK-1783 should be used for further verification guidance.

VERIFICATION LESSONS LEARNED (A.4.4.1.5.2)

Most LCF tests on the complete engine were not conducted for the LCF life of the parts. This would be a very long and expensive program, even when it is accelerated. The lack of statistical significance of one data point (a single engine) justifies running several identical parts in spin pits. An increase in sample size provides an increase in the statistical significance. Disks, which are expensive to replace, have been tested in a spin pit with as many as five samples. Spin pit testing has certain disadvantages. Specifically, it only simulates the centrifugal loading, and does not simulate the vibratory, aero-elastic, thermal or loads from adjacent disks. Of those loads, simulating the thermal stresses imposed on disks is the most difficult.

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A.3.4.1.5.3 Creep.

All engine parts shall not creep to the extent that the engine does not have the requirements specified per_____.

REQUIREMENT RATIONALE (A.3.4.1.5.3)

Growth must be minimized on static and rotating parts to ensure that acceptable engine operation and maintenance is not impaired, and part replacement is not required.

REQUIREMENT GUIDANCE (A.3.4.1.5.3)

MIL-HDBK-1783 should be used to tailor the specification requirement.

REQUIREMENT LESSONS LEARNED (A.3.4.1.5.3)

None.

A.4.4.1.5.3 Creep.

The requirements of 3.4.1.5.3 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.4.1.5.3)

Creep and stress rupture verification is required to ensure critical parts/critical safety items can operate satisfactorily for the design service life when exposed to the design usage. Final verification of creep and stress rupture capability must include extended operation of the full engine.

VERIFICATION GUIDANCE (A.4.4.1.5.3)

Past verification methods have included analyses and tests.

MIL-HDBK-1783 should be used for further verification guidance.

VERIFICATION LESSONS LEARNED (A.4.4.1.5.3)

None.

A.3.4.1.6 Strength.

The engine shall have strength requirements specified per _____.

REQUIREMENT RATIONALE (A.3.4.1.6)

Limit and ultimate loading must be addressed since the engine may be exposed to limit and ultimate loading conditions during operation.

REQUIREMENT GUIDANCE (A.3.4.1.6)

MIL-HDBK-1783 should be used to tailor the specification paragraph.

REQUIREMENT LESSONS LEARNED (A.3.4.1.6)

None.

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A.4.4.1.6 Strength.

The requirements of 3.4.1.6 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.4.1.6)

Verification is required to verify that the engine and its components can meet strength requirements.

VERIFICATION GUIDANCE (A.4.4.1.6)

Past verification methods have included structural analysis and part, component, and engine tests.

MIL-HDBK-1783 should be used for further verification guidance.

VERIFICATION LESSONS LEARNED (A.4.4.1.6)

Heat exchanger integrity problems have been addressed with past verification methods which included component cycling, proof pressure, and vibration tests. If the heat exchanger assembly incorporated a bypass valve, regulator, or indicating feature, appropriate tests were conducted to demonstrate proper functioning. The cycling test and pressure test were conducted as follows:

a. Flow, Pressure, and Temperature Cycling Test. The heat exchanger shall be subjected to a flow, pressure, and temperature cycling test for one design life. The number of cycles or hours shall be as specified in 3.4.1.2. The fluid pressure and temperature cycling shall be mission oriented and be derived from the engine mission duty cycles contained in the specification. An appropriate number of cycles shall include the maximum delta pressure that will occur between fluid passages, i.e., highest fuel pressure with lowest oil pressure and lowest fuel pressure with highest oil pressure. Ambient temperature shall be varied during the test if it is a parameter that degrades cooler life. At the completion of the test there shall be no leakage or permanent deformation.

b. Heat Exchanger Proof Pressure. Upon successful completion of the flow, pressure, and temperature cycling test, the same heat exchanger should be subjected to a proof pressure test. Each fluid side of the heat exchanger shall be individually subjected to proof pressure for at least one or two time(s) and held two minutes for each pressure application. During the application of pressure to one side the other element shall be empty and at atmospheric pressure. There shall be no evidence of leakage into the dry side. Following this test, both sides of the heat exchanger shall be simultaneously subjected to their maximum working pressures for at least one or two time(s) and held two minutes for each pressure application. At test completion there shall be no evidence of leakage or permanent deformation."

A.3.4.1.6.1 Factors of safety.

Factors of safety shall be applied to design usage induced loads to establish limit and ultimate conditions specified per _____.

REQUIREMENT RATIONALE (A.3.4.1.6.1)

Sufficient factors of safety must be determined to assure adequate safety margins exist in designs. Pressurized vessels, cases and components require internal and external load safety margins to preclude failure (e.g., burst and hazardous venting conditions). Also, pressurized vessels, cases and components must be protected from degradation caused by combustion processes or erosion. The engine pressure vessels and gas pressure loaded components must be capable of withstanding the combined operating ultimate loads without catastrophic failure to assure that the engine has been designed with appropriate safety margins, and that it can operate satisfactorily for the required design usage and service life.

REQUIREMENT GUIDANCE (A.3.4.1.6.1)

MIL-HDBK-1783 should be used to tailor the specification paragraph.

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REQUIREMENT LESSONS LEARNED (A.3.4.1.6.1)

Catastrophic failures of cast parts and pressure vessels have occurred due to porosity and poor manufacturing processes. These parts were designed with a 1.5 factor of safety for ultimate load conditions. A more graceful (i.e., noncatastrophic) failure would have occurred if the hardware had been designed with a 2.0 factor of safety.

Experience has shown that if a case is designed to a factor of safety of two, problems of rupture, LCF and burn through are reduced. The Engine AJ's burner case was initially designed close to yield and exhibited problems in service due to LCF and burn through. After the engine was redesigned to eliminate these problems, the factor of safety on rupture was checked and found to be approximately two. On Engine AK, a factor of safety of two was mandated. During testing of this engine it was found to be capable of tolerating a burn through without rupture. The Engine A program required a factor of safety of two. Subsequently, Engine A has experienced several burn throughs without rupture.

A.4.4.1.6.1 Factors of safety.

The requirements of 3.4.1.6.1 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.4.1.6.1)

Factors of safety must be verified to assure safety margin.

VERIFICATION GUIDANCE (A.4.4.1.6.1)

Past verification methods have included analyses and component and engine tests.

MIL-HDBK-1783 should be used for further verification guidance.

VERIFICATION LESSONS LEARNED (A.4.4.1.6.1)

None.

A.3.4.1.6.2 Blade and disk deflection.

The engine shall have blade and disk deflection requirements specified per _____.

REQUIREMENT RATIONALE (A.3.4.1.6.2)

Sufficient rigidity must be provided so that the engine can operate to the limit loads and repeated loads that occur within the flight envelope without detrimental damage.

REQUIREMENT GUIDANCE (A.3.4.1.6.2)

MIL-HDBK-1783 should be used to tailor the specification paragraph.

REQUIREMENT LESSONS LEARNED (A.3.4.1.6.2)

Rotor shifts have caused blades to contact vanes and led to subsequent blade failure.

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A.4.4.1.6.2 Blade and disk deflection.

The requirements of 3.4.1.6.2 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.4.1.6.2)

Verification that blade and disk deflection does not result in contact with any static parts of the engine other than seals and shrouds must be accomplished. Blade and disk rigidity must be verified to ensure flight safety against blade failures and titanium fires.

VERIFICATION GUIDANCE (A.4.4.1.6.2)

Past verification methods have included analyses and tests.

MIL-HDBK-1783 should be used for further verification guidance.

VERIFICATION LESSONS LEARNED (A.4.4.1.6.2)

X-ray photography has been used to determine that seals and clearances are effective under all operational loads.

A.3.4.1.6.3 Containment.

The engine shall have containment requirements specified per _____.

REQUIREMENT RATIONALE (A.3.4.1.6.3)

For safety of personnel and equipment, it is imperative to contain all rotor system failures within the engine.

REQUIREMENT GUIDANCE (A.3.4.1.6.3)

MIL-HDBK-1783 should be used to tailor the specification paragraph.

REQUIREMENT LESSONS LEARNED (A.3.4.1.6.3)

The consequences of engine component failures increase significantly when the failed parts are not contained. For example, there have been many cases of uncontained blades that penetrated the air vehicle and severed air vehicle fuel tanks and lines, hydraulic lines, etc., resulting in loss of the air vehicle. Excessive weight penalties have occurred for containment provisions for larger mass items such as disks and seals; therefore, past designs have provided containment for only blades. Uncontained rotor failures result in numerous air vehicle losses, mission aborts, and fatalities.

Experience has led to current designs that allow uncontained fan blades. In association with these designs, critical external engine components are kept off of the fan case to the maximum extent possible. Those components that must be placed on or in the fan case are grouped together in as few circumferential locations as possible, limiting the probability that an uncontained failure will impact a critical component.

The decision to contain fan blades should take into consideration total system survivability. While for a multi-engine air vehicle full containment may maximize the probability of safe recovery of the air vehicle following a failure, it might not be the best approach for single engine air vehicle. For example, containment of a fan blade within the engine virtually assures massive damage to the engine and near complete loss of thrust. In a single engine air vehicle this scenario assures loss of the air vehicle unless a suitable airfield is available very nearby. In contrast, allowing a fan blade to exit the case and be contained by air vehicle shielding increases the chance that internal secondary damage will be limited to an extent that allows the engine to continue to operate at a reduced power and permit safe recovery.

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This has been shown in eight cases of uncontained fan blade failures where single engine air vehicles were able to get home.

The turboshaft engine cylindrical exhaust pipe containment requirement identifies a potential exhaust configuration and attempts to preclude engine unilateral transfer of containment responsibilities to the air vehicle.

A.4.4.1.6.3 Containment.

The requirements of 3.4.1.6.3 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.4.1.6.3)

Verification of engine and component containment capability is necessary to avoid the occurrence of rotating component failures.

VERIFICATION GUIDANCE (A.4.4.1.6.3)

Past verification methods have included analyses and tests.

MIL-HDBK-1783 should be used for further verification guidance.

VERIFICATION LESSONS LEARNED (A.4.4.1.6.3)

Early use of containment criteria and analysis can avoid a redesign.

A.3.4.1.6.4 Blade out.

The engine shall have requirements for blade out specified per _____.

REQUIREMENT RATIONALE (A.3.4.1.6.4)

The engine must possess adequate structural integrity after blade loss so that a stable time period exists without uncontained catastrophic destruction to allow time for pilot awareness and appropriate action. In addition to the requirement for containment of the failed blade, the secondary failure modes that would result in catastrophic failure must be avoided.

REQUIREMENT GUIDANCE (A.3.4.1.6.4)

MIL-HDBK-1783 should be used to tailor the specification paragraph.

REQUIREMENT LESSONS LEARNED (A.3.4.1.6.4)

Blade failures have caused uncontained fire and catastrophic rotor failure in compressors that use titanium for blades and cases. An imbalance condition caused by loss of blades can lead to bearing and rotor support failure. Interactive blade and disk vibration modes have resulted in the failure of a blade retainer and subsequent loss of a fan rotor assembly.

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A.4.4.1.6.4 Blade out.

The requirements of 3.4.1.6.4 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.4.1.6.4)

Verification is required to ensure that blade failure does not cause the engine to experience uncontained fire; catastrophic rotor, bearing, support or mount failures; overspeed conditions; leakage of flammable fluids or loss of ability to shut down the engine.

VERIFICATION GUIDANCE (A.4.4.1.6.4)

Past verification methods have included analyses and tests.

MIL-HDBK-1783 should be used for further verification guidance.

VERIFICATION LESSONS LEARNED (A.4.4.1.6.4)

Verification by engine test during development is very rare. As part of FAA certification, Engine AA was required to demonstrate compliance with blade out requirements via engine test.

A.3.4.1.6.5 Overspeed and overtemperature.

The engine shall have overspeed and overtemperature requirements specified per _____.

REQUIREMENT RATIONALE (A.3.4.1.6.5)

These requirements are needed to provide an operational margin for rotor structural integrity while allowing continued use.

REQUIREMENT GUIDANCE (A.3.4.1.6.5)

MIL-HDBK-1783 should be used to tailor the specification paragraph.

REQUIREMENT LESSONS LEARNED (A.3.4.1.6.5)

Past experience indicates that the engine can overspeed or exceed design temperature due to control system malfunctions or other engine operating anomalies.

A.4.4.1.6.5 Overspeed and overtemperature.

The requirements of 3.4.1.6.5 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.4.1.6.5)

Overspeed and overtemperature of the engine must be verified to ensure structural integrity of rotating parts.

VERIFICATION GUIDANCE (A.4.4.1.6.5)

Past verification methods have included analyses and tests.

MIL-HDBK-1783 should be used for further verification guidance.

VERIFICATION LESSONS LEARNED (A.4.4.1.6.5)

None.

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A.3.4.1.6.6 Disk burst speed.

The minimum loaded disk burst speed of the complete disk assembly shall have the requirement specified per _____.

REQUIREMENT RATIONALE (A.3.4.1.6.6)

This requirement assures adequate margin against the risk of a disk burst in service.

REQUIREMENT GUIDANCE (A.3.4.1.6.6)

MIL-HDBK-1783 should be used to tailor the specification paragraph.

REQUIREMENT LESSONS LEARNED (A.3.4.1.6.6)

The FAA uses 120 percent of maximum allowable steady-state rotor speed. The USN and USAF have allowed minimum disk burst speeds of 115-117 of maximum allowable steady-state rotor speed. For titanium fan disks, the USAF has required minimum disk burst speeds of 130 percent of maximum allowable steady-state rotor speed.

A.4.4.1.6.6 Disk burst speed.

The requirements of 3.4.1.6.6 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.4.1.6.6)

The required disk burst speed must be verified to prevent the occurrence of a catastrophic failure.

VERIFICATION GUIDANCE (A.4.4.1.6.6)

Past verification methods have included analyses and tests.

MIL-HDBK-1783 should be used for further verification guidance.

VERIFICATION LESSONS LEARNED (A.4.4.1.6.6)

None.

A.3.4.1.6.7 Output shaft torque limits.

For turboprop and turboshaft engines the maximum allowable steady-state delivered shaft torque (mechanical) limit shall have the requirements specified per _____.

REQUIREMENT RATIONALE (A.3.4.1.6.7)

A limit is required to provide a margin of torque to prevent catastrophic component failure.

REQUIREMENT GUIDANCE (A.4.4.1.6.6)

MIL-HDBK-1783 should be used to tailor the specification paragraph.

REQUIREMENT LESSONS LEARNED (A.4.4.1.6.6)

Past experience indicates that the engine output shaft torque can increase because of a malfunction or other anomalies. Some engine contractors have asked for a deviation. They felt they have a good control of output shaft torque and torque limits could be lower. Other systems (helicopters) with torque limits depend upon pilot action for limit observance.

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A.4.4.1.6.7 Output shaft torque limits.

The requirements of 3.4.1.6.7 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.4.1.6.7)

Verification is needed to assure the engine will not be degraded by shaft torque operation up to the limit.

VERIFICATION GUIDANCE (A.4.4.1.6.7)

Past verification methods have included analyses and tests.

MIL-HDBK-1783 should be used for further verification guidance.

VERIFICATION LESSONS LEARNED (A.4.4.1.6.7)

None.

A.3.4.1.6.8 Output shaft speed limits.

For turboprop and turboshaft engines the maximum allowable steady-state delivered shaft speed (mechanical) limit shall have the requirements specified per _____.

REQUIREMENT RATIONALE (A.3.4.1.6.8)

An output shaft speed limit is required to provide a margin of speed to prevent catastrophic failure.

REQUIREMENT GUIDANCE (A.3.4.1.6.8)

MIL-HDBK-1783 should be used to tailor the specification paragraph.

REQUIREMENT LESSONS LEARNED (A.3.4.1.6.8)

Past experience indicates that the engine output shaft speed can increase because of a malfunction or other anomalies. Engine contractors have asked for a deviation on the speed requirement since they believe the newer control systems will prevent any overspeed.

A.4.4.1.6.8 Output shaft speed limits.

The requirements of 3.4.1.6.8 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.4.1.6.8)

Verification is needed to prevent catastrophic component failure.

VERIFICATION GUIDANCE (A.4.4.1.6.8)

Past verification methods have included analyses and tests.

MIL-HDBK-1783 should be used for further verification guidance.

VERIFICATION LESSONS LEARNED (A.4.4.1.6.8)

The verification was conducted by analysis or similarity on some engines in the past.

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A.3.4.1.6.9 Bearing load.

The engine bearings shall react the loads applied during engine operation and meet the design service life of 3.4.1.1. The engine bearings shall react the loads applied during flight when the engine is not operating (i.e., shutdown and windmilling). The engine bearings shall have any applicable load requirements specified per _____.

REQUIREMENT RATIONALE (A.3.4.1.6.9)

The thrust bearings must meet their design service life after exposure to all loading conditions when the engine is operated throughout the operating envelope.

REQUIREMENT GUIDANCE (A.3.4.1.6.9)

MIL-HDBK-1783 should be used to tailor the specification paragraph.

REQUIREMENT LESSONS LEARNED (A.3.4.1.6.9)

None.

A.4.4.1.6.9 Bearing load.

The requirement of 3.4.1.6.9 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.4.1.6.9)

The engine pressure balance system must be verified to ensure compliance with the requirement of 3.4.1.6.9.

VERIFICATION GUIDANCE (A.4.4.1.6.9)

Past verification methods have included analysis and test.

MIL-HDBK-1783 should be used for further verification guidance.

VERIFICATION LESSONS LEARNED (A.4.4.1.6.9)

None.

A.3.4.1.6.10 Gyroscopic moments.

The engine shall have gyroscopic moment requirements specified per _____.

REQUIREMENT RATIONALE (A.3.4.1.6.10)

Engine flight loads are increased due to rotations and accelerations that occur during aircraft rolling, pitching, and yawing maneuvers. The engine must be designed to resist these loads at the limiting conditions.

REQUIREMENT GUIDANCE (A.3.4.1.6.10)

MIL-HDBK-1783 should be used to tailor the specification paragraph.

REQUIREMENT LESSONS LEARNED (A.3.4.1.6.10)

In 1998, spin tests were conducted on a Navy aircraft. Maximum yaw rates achieved were 130°/sec for upright, and 102°/sec for inverted.

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A USAF study showed that a bearing load could be increased by as much as 15 times by a 3.5 rad/sec aircraft turn rate. The force on an individual rotor blade could be three times the magnitude of the corresponding aerodynamic force. Gyroscopic forces are cyclic in nature, thus tending to reduce the cyclic life of rotor blades. The vector sum of pitch and yaw rates is very important from the standpoint of the gyroscopic moments produced on the engine.

The fighter and attack aircraft and the lighter weight engines with tighter clearances will make the gyro requirement even more essential. Simulation data from an Aircraft AB produced yaw rates of less than 0.5 rad/sec and pitch rates less than 1 rad/sec. Intentional departure stall maneuvers have been used as a “last chance” evasive action against SAM’s. This action results in high yaw rates and gyro loads. Yaw rates of 2.7 rad/sec were demonstrated years ago, in flight, during accelerated departure stall maneuvers, with Aircraft N. Yaw rates of 3.1 and 3.2 rad/sec were measured on Aircraft R and AB. A recent mishap involving the Aircraft I was reported as yawing in excess of 200 degrees per second (3.5 rad/sec). The aircraft had violently departed controlled flight due to asymmetrical positioning of the leading edge flaps. Testing Engine X on the gyro rig showed significant mechanical interference occurring at 2 rad/sec. The manufacturer was able to correct the problem without major redesign.

A.4.4.1.6.10 Gyroscopic moments.

The requirements of 3.4.1.6.10 shall be verified by:

VERIFICATION RATIONALE (A.4.4.1.6.10)

Aircraft are exposed to gyroscopic moments during normal operation and the ability of their engines to withstand those conditions must be verified.

VERIFICATION GUIDANCE (A.4.4.1.6.10)

Past verification methods have included analysis and test.

MIL-HDBK-1783 should be used for further verification guidance.

VERIFICATION LESSONS LEARNED (A.4.4.1.6.10)

None.

A.3.4.1.7 Damage tolerance.

Fracture and safety and mission critical engine parts shall have damage tolerance requirements specified per _____.

REQUIREMENT RATIONALE (A.3.4.1.7)

The requirement protects fracture, safety, and mission critical parts/critical safety items from potentially degrading effects of handling damage, and material, manufacturing, and processing anomalies which could result in premature engine failures and loss of aircraft. This process ensures proper material choices, control of operating stress levels, use of fracture-resistant design concepts, manufacturing and process controls, and the use of reliable inspection methods during production and in-service maintenance. Attainment of engine damage tolerance is achieved through application of detail requirements in the functional areas of design, materials selections, manufacturing control, and inspections.

REQUIREMENT GUIDANCE (A.3.4.1.7)

MIL-HDBK-1783 should be used to tailor the specification paragraph.

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REQUIREMENT LESSONS LEARNED (A.3.4.1.7)

There have been numerous Class A incidents on Air Force aircraft and engines due to structural failures caused by material defects, manufacturing defects, or fatigue induced cracks. These defects grew in size due to repetitive cycles of maneuvers or throttle excursions until such time as the residual strength of the component became less than the applied load and failure occurred. Causes have been: (1) use of high strength low fracture toughness materials, (2) improper detail designs resulting in high stress levels and structural discontinuities, and (3) lack of adequate quality control requirements (both in production and depot maintenance). Also, past review of commercial engine experience reveals uncontained failures of blades, disks, and spacers due to structural cracking. When compared against the total number of parts and flying hours, these occurrences are low. However, the demonstrated consequences of failure on Air Force systems has been high in terms of loss of aircraft and crew members. Many of the incidents could have been avoided by proper material selection, control of stress levels, use of fracture resistant design concepts, manufacturing and process controls and by use of reliable inspection methods during production and in-service maintenance.

Recent examples of optimized part designs following the ENSIP and DTD guidelines include the Engines A, B, B1, O, and X designs. These design configurations have shown that damage tolerance requirements can be met with small or modest increases in overall engine weight, will have little impact on engine performance, and will provide greatly improved engine durability while significantly reducing weapon system life cycle cost.

Most of the tasks to be contained in the Damage Tolerance Control Plan have been accomplished by engine manufacturers in past development and production programs. However, the durability and damage tolerance requirements established here impose tighter controls and more interface involvement between the functional areas.

Damage tolerance requirements for composite parts have been derived from the work done for the Aircraft Structural Integrity Program (ASIP).

A.4.4.1.7 Damage tolerance.

The requirements of 3.4.1.7 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.4.1.7)

Damage tolerance verification is needed to support damage tolerance design concepts, trade studies, material selection, maintenance requirements, and cost and weight impacts and verify allowable stress levels and chemical and thermal environment spectra.

VERIFICATION GUIDANCE (A.4.4.1.7)

Past verification methods have included analyses, tests, process documentation, and quality control during production and assembly.

MIL-HDBK-1783 should be used for further verification guidance.

VERIFICATION LESSONS LEARNED (A.4.4.1.7)

Methods for verification of damage tolerance of composites are derived from the work done for the Aircraft Structural Integrity Program (ASIP).

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A.3.4.1.7.1 Residual strength.

All damage tolerant designed parts and components shall have residual strength requirements specified per _____.

REQUIREMENT RATIONALE (A.3.4.1.7.1)

The load-carrying capability of fracture, safety, and mission critical parts/critical safety items, with "damage" present, must remain above some minimum value during part design service lives.

REQUIREMENT GUIDANCE (A.3.4.1.7.1)

MIL-HDBK-1783 should be used to tailor the specification paragraph.

REQUIREMENT LESSONS LEARNED (A.3.4.1.7.1)

None.

A.4.4.1.7.1 Residual strength.

The requirements of 3.4.1.7.1 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.4.1.7.1)

Verification of residual strength is required to ensure less than critical size flaws will not grow and cause failure due to the application of the required residual strength load.

VERIFICATION GUIDANCE (A.4.4.1.7.1)

Past verification methods have included analyses and tests.

MIL-HDBK-1783 should be used for further verification guidance.

VERIFICATION LESSONS LEARNED (A.4.4.1.7.1)

None.

A.3.4.1.7.2 Initial production and in-service flaw size.

All fracture critical parts/critical safety items shall have initial flaw sizes specified per _____.

REQUIREMENT RATIONALE (A.3.4.1.7.2)

This requirement is necessary to establish the probable flaw size that can exist in a part after manufacture or after a depot, intermediate or base level inspection. Damage tolerance, as applied in ENSIP, assumes the presence of a flaw in each fracture, safety, and mission critical part, at the highest stressed location with an orientation most unfavorable with respect to the stress field. Tolerance to these initial flaws must be designed into the part. In-service inspection flaw sizes must be specified to establish part life limitations, and the maintenance capability requirements.

REQUIREMENT GUIDANCE (A.3.4.1.7.2)

MIL-HDBK-1783 should be used to tailor the specification paragraph.

REQUIREMENT LESSONS LEARNED (A.3.4.1.7.2)

Experience has identified preferred processing methods or key processing parameters as indicated below:

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Fluorescent Penetrant Inspection. The preferred process utilizes a high sensitivity post emulsified penetrant with a hydrophilic (water soluble) emulsifier. A non-aqueous or water soluble developer is preferred over the dry powder or wet suspendable type of developer. Surface preparation is very important and should include an etch to remove smeared metal, dirt, combustion products, etc. In process FPI preceded by a heavy etch that removes considerable surface material 0.254 - 0.508 mm (.001 - .002 inches) is recommended for each fracture critical rotating component. Etch of the finished component prior to FPI is also recommended for each fracture critical component. However, the selection of an etching media for the finished component must be thoroughly evaluated to assure no detrimental life effects are caused when adequate surface material removal is achieved 0.0254 - 0.0508 mm (.0001 - .0002 inches).

Fluorescent Penetrant Inspection

Principle:	Liquid penetrant is drawn into surface flaws by capillary action. Flaws are revealed by a dye.
Applications:	Surface cracks, laps, porosity, shrinkage areas, laminations, etc., that are open to surface.
Advantages:	Inexpensive, portable, and sensitive. Not dependent on magnetic or electrical properties of a material.
Disadvantages:	Limited to flaws on the surface of the part. Not usable on porous and rough surfaces.

Eddy Current Inspection - The preferred process uses automated scanning and automated data recording, and special fixturing when necessary.

Eddy Current Inspection

Principle:	Coil induces a current in a metal then detects the resulting current that fluctuates with a change in electrical property of the part.
Applications:	Surface cracks, laps, porosity, shrinkage areas, laminations, etc., that are open to surface. Detects variations in the metal and microstructure of parts. Lower frequency probes can be used for subsurface flaw detection.
Advantages:	Very sensitive. Not necessary to contact part.
Disadvantages:	Often too sensitive to unimportant properties such as conductivity variations. Difficult to apply on ferromagnetic alloys.

Ultrasonic Inspection - The preferred process uses automated scanning and may use more than one mode (i.e.; longitudinal, shear, surface, lamb, etc.) where appropriate.

Ultrasonic Inspection

Principle:	Sound waves are transmitted through material and reflected by flaws.
Applications:	Subsurface flaws such as cracks, laminations, and bonds with principle plane perpendicular to sound source.
Advantages:	Capable of detecting flaws that exist deep in a material. The capability to produce images of the flaw exists.
Disadvantages:	Flaws that are parallel to source of sound are undetectable. Inspection must be performed by trained personnel. Angle beam methods can sometimes be used to detect flaws not oriented parallel to the scanning surface. Most ultrasonic testing (UT) methods require a fluid couplant (with the exception of air coupled transducers and laser UT).

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Radiographic Inspection - The preferred process requires selection of proper kilovoltage and exposure geometry.

Radiographic Inspection

Principle:	X-rays and gamma rays are sent through the metal and strike a film. The existence of flaws is seen as dark shadows on the film.
Applications:	Subsurface flaws in castings and weldments with the principle plane of flaw parallel to radiation beam.
Advantages:	Flaws are detectable at any depth.
Disadvantages:	Flaws perpendicular to radiation beam are undetectable. Inspection may be hazardous and must be performed by trained personnel. Inspection is expensive.

Magnetic Particle Inspection - The preferred process provides that insecure adequate magnetic field strengths are introduced in the part and at critical locations in the part.

Magnetic Particle Inspection

Principle:	Discontinuities distort an applied magnetic field causing leakage fields that attract iron powder.
Applications:	Cracks, inclusions, and other discontinuities on or near the surface of parts.
Advantages:	Inexpensive. Suitable for extremely large objects.
Disadvantages:	Limited by depth of flaw and coatings. Personnel performing inspections must be trained to interpret results.

A.4.4.1.7.2 Initial production and in-service flaw size.

The requirements of 3.4.1.7.2 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.4.1.7.2)

Verification of the initial flaw sizes for production and in-service is necessary to ensure that flaw sizes greater than the sizes specified in 3.4.1.7.2, do not exist in finished or inspected parts.

Damage tolerance of many parts is achieved by providing minimum flaw growth intervals based on initial flaw sizes. The initial flaw size values selected for design only have significance when production and in-service NDI capability is confirmed by the demonstration programs.

VERIFICATION GUIDANCE (A.4.4.1.7.2)

Past verification methods have included analyses, inspections, demonstrations, and tests.

MIL-HDBK-1783 should be used for further verification guidance.

VERIFICATION LESSONS LEARNED (A.4.4.1.7.2)

In past programs, inspection capabilities have been quoted, for generic areas and processes with no real basis in reality. For example, an eddy current capability was quoted as 0.254 x 0.508 mm (.010 x .020 inch) for Titanium broach slots. However, eddy current couldn't detect this or any reasonable flaw size within 2.54 mm (0.100 inches) of the edge of the broach slot due to the geometry signal. Today detection of flaws in edges are possible with certain probes, but the inspection time is three times slower than an inspection which does not include the edge. This was only discovered with an NDI demonstration program.

Reliability assessment methodologies can provide very different POD results based on the same data sets. Proposed (DRAFT) MIL-STD-1823 provides a reliability assessment procedure for establishing POD and should be used for all POD and process quality control assessments. In order to standardize POD calculations, software developed under contract with the University of Dayton Research Institute has been programmed based on this MIL-STD and should be used for POD calculations.

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A.3.4.1.7.3 Flaw growth and inspection intervals.

The initial flaw size as specified in 3.4.1.7.2 shall not grow to critical size and cause failure of the part due to the application of the required residual strength load within the inspection intervals as specified per __.

REQUIREMENT RATIONALE (A.3.4.1.7.3)

The flaw growth interval (i.e., safety limit) must be specified to ensure the assumed initial flaw will not grow to critical size and fail due to application of the required residual strength load.

Also, the design objective is to eliminate the need for in-service inspections. However, the weight penalty incurred to achieve a safety-limit or damage growth interval sufficiently large to preclude the need for in-service inspections may be prohibitive on some components. Therefore, in-service inspections will be allowed on some parts subject to justification.

These requirements are intended to provide the minimum information necessary to show that basic maintenance functions have been considered in the design of the engine.

REQUIREMENT GUIDANCE (A.3.4.1.7.3)

MIL-HDBK-1783 should be used to tailor the specification paragraph.

REQUIREMENT LESSONS LEARNED (A.3.4.1.7.3)

Since average fracture properties have been used in analysis, parts made from materials with scatter factors greater than two have failed prior to their inspection interval. Thus, for materials with large scatter factors (i.e., greater than two), factors of safety greater than two, on residual life, should be considered.

A.4.4.1.7.3 Flaw growth and inspection intervals.

The requirements of 3.4.1.7.3 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.4.1.7.3)

Verification of inspection intervals is required to ensure the flaw growth interval of 3.4.1.7.3 is of sufficient duration to preclude failure between inspections.

VERIFICATION GUIDANCE (A.4.4.1.7.3)

Past verification methods have included analyses, inspections, and tests.

MIL-HDBK-1783 should be used for further verification guidance.

VERIFICATION LESSONS LEARNED (A.4.4.1.7.3)

None.

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A.3.4.1.8 Vibration and dynamic response.

The engine shall have vibration and dynamic response requirements specified per _____.

REQUIREMENT RATIONALE (A.3.4.1.8)

Absence of damaging vibration and dynamic response is essential to maintain the desired mechanical and structural integrity of the engine. This requirement is intended to apply to rotating equipment and to other critical components in a vibratory environment on the air vehicle. Determining the frequency and relative amplitude of various sources (forcing functions) will provide design information on frequencies to be avoided in components as well as information to determine fatigue resistance and vibration associated wear problems.

REQUIREMENT GUIDANCE (A.3.4.1.8)

MIL-HDBK-1783 should be used to tailor the specification paragraph.

REQUIREMENT LESSONS LEARNED (A.3.4.1.8)

Some of the sources of vibration, which have been identified and studied by researchers are; (1) asymmetric stiffness properties of shafts or bearing supports, (2) internal friction in shafts or other rotating parts, (3) aerodynamic or gas flow excitation, (4) excitation from fluids trapped within rotor shafts, (5) rubbing friction between radial surfaces or rotor and stator housings, (6) excitations from other rotating components in the same engine structure, and (7) excitation from gears at the gear tooth mesh frequencies.

Small turboshaft engines are more susceptible to rotor vibration problems. An increase in engine performance has been obtained through higher shaft speeds. This has heightened the importance of rotor dynamics considerations in the development of the small engine.

Rotor vibration experienced on multishaft engines has resulted directly from rotor imbalances existing in the numerous critical speeds inherent in lightweight, high speed engines.

Campbell frequency diagrams have been required to assure frequency mistuning margins against vibration and fatigue failures. This method has not always predicted blade flutter. The damage caused by blades failing in vibration fatigue has caused catastrophic results in life and aircraft. When determining the total stress of an engine part, contractors typically allow 25% - 50% of yield stress for stress due to vibrational loads. The more conservative values contractors have used are between 30-35%.

Air seal and spacer failures were experienced by an engine. As part of the investigation it was learned that transient engine operational cavity temperatures caused acoustic excitations and mechanical resonances to coincide and caused damaging vibrations. Another engine's front frame experienced cracking during development testing that was attributed to the fan blade tip exceeding supersonic speed. The shock wave off the blade tip provided the stimulus to excite natural frequencies of the frame strut panels. Since Mach number is related to temperature, the fan speed at which the blades went supersonic varied with inlet temperature conditions. The stresses in the struts peaked when the fan blade tip reached supersonic speeds and coincided with the natural frequency of the strut panels.

Past gear resonance testing was performed at 130% of normal operating speed. This would equate to 121% maximum allowable steady-state speed or 116% maximum transient speed.

Blade failures have been experienced on several occasions due to lack of margin between a resonance or critical condition and a sustained power operating condition.

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A.4.4.1.8 Vibration and dynamic response.

The requirements of 3.4.1.8 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.4.1.8)

Analysis is required to establish equipment and subsystem rigidity and to verify equipment and subsystem design to minimize the potential vibration and dynamic response problems. Testing is needed to verify the vibration and dynamic response characteristics of the engine. Verification is required to ensure compliance with the requirements of 3.4.1.8.

VERIFICATION GUIDANCE (A.4.4.1.8)

Past verification methods have included analyses and tests.

MIL-HDBK-1783 should be used for further verification guidance.

VERIFICATION LESSONS LEARNED (A.4.4.1.8)

Engine contractors have said that it is difficult to build an engine with a maximum imbalanced rotor for vibration tests. They have said it is difficult to predict the total imbalance of the rotor when the engine is completely assembled.

The engine rotors are supported by bearings mounted on relatively stiff supports. The vibration loads are transmitted to the engine via the bearing supports. These loads and deflections can be reduced by soft-mounting the bearings through “squirrel cages” or squeeze-film bearings.

Vibration problems appear to occur more frequently in rotors with a cantilevered disk. The overhung rotor configuration is characterized by a disk having a shaft bearing only on one side. There has been a long history of dynamic related problems associated with overhung rotors.

Small turboshaft and turboprop engines with their high speeds and front drive are more susceptible to vibration problems. The vibration analysis of blades was usually made prior to engine buildup to preclude the possibility of resonance problems. Experimental testing was always feasible unless the hardware was non existent or unavailable.

Aerodynamic blade flutter or vibration was not uncovered by test more by lack of environmental definition than from blade characteristics. Flutter is a function of pressure and temperature which makes its prediction a very complex problem. On the compressor map, many areas of flutter can be depicted (e.g.; subsonic and supersonic stalled flutter, subsonic and supersonic unstalled flutter, choke flutter, etc.). The areas are so numerous that it seems impossible to avoid flutter. The Campbell diagram shows how close some rotor speeds come to resonant vibration with certain harmonics of rotor speed. This can all change with the ambient environment.

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A.3.4.1.8.1 Vibration limits.

The engine shall have the vibration limits specified per _____.

REQUIREMENT RATIONALE (A.3.4.1.8.1)

The intent of this paragraph is to specify the vibration limits which are utilized as a diagnostic tool and monitoring engine health. Vibration limits are also needed to establish baseline performance of developmental engines to determine when design changes are warranted, for uninstalled engine production and depot limits, and installed steady-state, transients and maneuver limits.

REQUIREMENT GUIDANCE (A.3.4.1.8.1)

MIL-HDBK-1783 should be used to tailor the specification paragraph.

REQUIREMENT LESSONS LEARNED (A.3.4.1.8.1)

Vibration monitoring has been used during development testing, production acceptance, flight testing, field troubleshooting, etc. Field experience with helicopters and turboprops has shown that no vibration pickups are necessary in day to day field use. Few, if any, problems would have been identified if they had been present that were not obvious from other deterministic methods. A clear case for reduced LCC or readiness must be presented before specifying routine continuous vibration measurement for these engines.

A.4.4.1.8.1 Vibration limits.

The requirements of 3.4.1.8.1 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.4.1.8.1)

Verification is required to ensure compliance with the requirements of 3.4.1.8.1.

VERIFICATION GUIDANCE (A.4.4.1.8.1)

Past verification methods have included analysis and test.

See 4.4.1.8 VERIFICATION GUIDANCE.

VERIFICATION LESSONS LEARNED (A.4.4.1.8.1)

See 4.4.1.8 VERIFICATION LESSONS LEARNED.

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A.3.4.1.8.2 Critical speeds.

The engine shall be free of resonance conditions at all shaft rotational speeds that would cause the engine not to meet all the requirements of the specification. Critical speeds existing below the engine operating range shall be at least (a) percent below idle. Critical speeds existing above the maximum operating speed shall be at least (b) percent above the maximum allowable transient shaft rotational speed. The natural frequencies of the mounting system with the engine installed shall be at least (c) percent below idle shaft rotational speed(s) in all damaging modes of vibration which can be excited by the rotor imbalances.

REQUIREMENT RATIONALE (A.3.4.1.8.2)

Resonance conditions should be avoided so that amplified response and structural failures do not occur. Margin is required between engine speeds and resonance speeds due to the variation that can occur in speeds due to Mach number, deterioration, ambient conditions, and combinations, thereof.

REQUIREMENT GUIDANCE (A.3.4.1.8.2)

The following should be used to tailor the specification paragraph:

(a) - (c): 20 percent.

Background:

It is recommended that a margin of at least 20 percent be specified for damaging resonance conditions that exist above maximum allowable transient shaft operating speed or below idle speed. Damping and appropriate balancing should be provided so that any critical speed existing below maximum operating speed will be traversed safely with smooth engine operation including the blade out conditions of 3.4.1.6.4. It is also recommended that the natural frequencies of the mounting system with the engine installed should be 20 percent below idle rotor speeds in all detrimental modes of vibration.

REQUIREMENT LESSONS LEARNED (A.3.4.1.8.2)

Steady-state engine operation at a critical speed has quickly consumed the high cycle fatigue life of parts due to vibratory resonance.

A.4.4.1.8.2 Critical speeds.

The requirements of 3.4.1.8.2 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.4.1.8.2)

Verification is required to ensure compliance with the requirements of 3.4.1.8.2.

VERIFICATION GUIDANCE (A.4.4.1.8.2)

Past verification methods have included analysis and test.

See A.4.4.1.8 VERIFICATION GUIDANCE.

VERIFICATION LESSONS LEARNED (A.4.4.1.8.2)

See A.4.4.1.8 VERIFICATION LESSONS LEARNED.

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A.3.4.1.9 External surface foreign object damage.

The engine external surface, protruding from the air vehicle, shall meet all requirements of this specification, without repair, after FOD that produces damage equivalent to a stress concentration factor (Kf) of _____.

REQUIREMENT RATIONALE (A.3.4.1.9)

The outer surfaces of thrust vectoring nozzles, which are external of the air vehicle surface, may be susceptible to hits by birds and ice while vectoring takes place. This requirement may not be applicable to TS and TP engines if none of their surfaces protrude beyond the air vehicle surfaces.

REQUIREMENT GUIDANCE (A.3.4.1.9)

The following should be used to tailor the specification paragraph:

The engine should be more resistant to FOD, bird, ice, etc. than aircraft structure (except for canopy) as it is needed for flight operation.

REQUIREMENT LESSONS LEARNED (A.3.4.1.9)

None.

A.4.4.1.9 External surface foreign object damage.

The requirement of 3.4.1.9 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.4.1.9)

The capability of any engine part to withstand external FOD damage must be verified.

VERIFICATION GUIDANCE (A.4.4.1.9)

Past verification methods have included analysis and test.

VERIFICATION LESSONS LEARNED (A.4.4.1.9)

None.

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A.3.4.2 Mechanical equipment and subsystems integrity.
The engine's mechanical equipment and subsystems shall meet the integrity requirements specified per _____.

REQUIREMENT RATIONALE (A.3.4.2)

None.

REQUIREMENTS GUIDANCE (A.3.4.2)

MIL-HDBK-1798 should also be used as guidance for tailoring the requirements.

REQUIREMENT LESSONS LEARNED (A.3.4.2)

None.

A.4.4.2 Mechanical equipment and subsystems integrity.

The requirements of 3.4.2 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.4.2)

Verification is required to ensure compliance with requirements of 3.4.2.

VERIFICATION GUIDANCE (A.4.4.2)

Past verification methods have included analysis, inspection, demonstration, and test.

VERIFICATION LESSONS LEARNED (A.4.4.2)

None.

A.3.4.3 Avionics and electronic integrity.
The engine shall meet the electronic system integrity requirements specified per _____.

REQUIREMENT RATIONALE (A.3.4.3)

None.

REQUIREMENTS GUIDANCE (A.3.4.3)

MIL-HDBK-87244 should be used to tailor the requirement paragraph.

REQUIREMENT LESSONS LEARNED (A.3.4.3)

None.

A.4.4.3 Avionics and electronic integrity.

The requirements of 3.4.3 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.4.3)

Verification is required to assure compliance with requirements of 3.4.3.

VERIFICATION GUIDANCE (A.4.4.3)

Past verification methods have included analysis, inspection, demonstration, and test.

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VERIFICATION LESSONS LEARNED (A.4.4.3)

None.

A.3.4.4 Software integrity.

The software integrity shall be obtained through the use of ___.

REQUIREMENT RATIONALE (A.3.4.4)

A Software Development Integrity Program (SDIP) is necessary to assure a high quality software product.

REQUIREMENT GUIDANCE (A.3.4.4)

Commercial documents that are used to maintain software integrity should be referenced in the specification paragraph.

If required by the weapon system contractor, the equipment should identify the operational configuration of the equipment and software combination.

"Quality Management and Quality Assurance Standards Part 3: Guidelines for the application of ISO 9001 to the development, supply and maintenance of software" is a DOD adopted standard for software integrity which may be used for guidance.

Additional guidance may be found in DO-178B.

REQUIREMENT LESSONS LEARNED (A.3.4.4)

Due to its increased complexity and amount of code, Engine O successfully used SDIP during its development.

A.4.4.4 Software integrity.

The requirements of 3.4.4 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.4.4)

The design of software programs for the engine computer resource must be verified to ensure that all requirements have been met.

VERIFICATION GUIDANCE (A.4.4.4)

Past verification methods have included an inspection of the contractor's process, closed and open loop demonstrations, analysis and tests. The contractor may also use a documented process control.

Background:

Verification of software design should be accomplished through the contractor's quality assurance program. Plans, procedures, and test results should be reviewed by the Using Service. The Higher Order Language (HOL) and structure may be verified by inspection. Real-time processing should be verified by analysis and demonstration and the Built-in-Test (BIT) by test.

VERIFICATION LESSONS LEARNED (A.4.4.4)

None.

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A.3.5 Reliability and maintainability (R&M).

A.4.5 Reliability and maintainability (R&M).

A.3.5.1 Reliability.

A.4.5.1 Reliability.

A.3.5.1.1 Reliability quantitative requirements.

The engine design reliability shall be capable of meeting engine or air system availability goals for the mission and mission mix profile of table XI. The engine reliability shall be as defined in table XIII.

REQUIREMENT RATIONALE (A.3.5.1.1)

The requirement for quantitative reliability must be specified for safety, mission capability, and cost reasons. The designer must take reliability considerations into account, or these three key factors will be adversely impacted.

REQUIREMENT GUIDANCE (A.3.5.1.1)

The following should be used to tailor the specification paragraph:

The Using Service should provide a formatted table for the contractor to complete.

Background:

A reliability program has been used to attain reliability requirements. The plan should include program objectives, management and technical resources, plans, procedures, schedules, and controls for achieving reliability requirements. OPNAVINST 4790.2 may be used for tailoring guidance for terminology.

R&M program plans should take into consideration the following:

a. Identification of the life cycle, mission, and environmental profiles which represent in service usage.

b. Detailed description of hardware configuration, equipment, and system operation under test.

c. Performance parameters to be monitored during the test.

d. Provision of reliability and maintainability test data and reports to the Using Service.

Effort should be taken to include demonstrations and tests for verification of reliability and maintainability.

The contractor should establish a reliability program and procedures to identify and control Reliability Critical Items, items whose failure would critically affect the weapon system safety, availability, ability to meet mission objectives, or cause extensive or expensive repair. Reliability Critical Items should be identified by a FMECA containing item functional traceability to the end item, probability of failure, and criticality of failure

The Using Service may consider data provided by the contractor from other programs (e.g., operational data for a specific component) for determining initial reliability. After converting reliability data to the equivalent engine flight hours, the subassembly reliability should be established by analysis of the data using appropriate techniques as those described in MIL-HDBK-781 or AFWAL-TR-83-2079.

MIL-HDBK-781 may be used as tailoring guidance for prediction and modeling of engine and component reliabilities.

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REQUIREMENT LESSONS LEARNED (A.3.5.1.1)

MIL-STD-882 has been used as tailoring guidance for establishing the hazard categories for failures and their end item effects.

The reliability program should (1) improve operational readiness and mission success of the major end item, (2) reduce item demand for maintenance manpower and logistic support, and (3) provide essential management information.

Insufficient reliability growth testing during development has resulted in failures during operational usage and have required Component Improvement Programs for corrective action. Insufficient reliability requirements have resulted in increased maintenance and operations and support cost.

Contractors have included some engine caused events in the non-engine caused category for In Flight Shutdowns (IFSD) such as: in flight stall events (within aircraft and engine flight envelopes), engine over temperature (within flight and operating envelopes), engine over temperature beyond limits during air starts (within airstart envelope requirements), engine variable geometry cycling, and engine low oil pressure readings (known not to be A/C system caused). This biased the engine caused IFSD rate to be lower than actual. Contractors have not listed nor shown shop visit rate and maintenance events that are not engine caused. This did not let the Using Service determine if some should be engine caused.

A.4.5.1.1 Reliability quantitative requirements.

The requirements of 3.5.1.1 and table XIII shall be verified by: _____.

VERIFICATION RATIONALE (A.4.5.1.1)

Reliability tests are necessary to verify that the quantitative reliability requirements are satisfied.

VERIFICATION GUIDANCE (A.4.5.1.1)

Past verification methods have included analyses, demonstration, and test.

The engine contractor should specify the planned tests and analyses to verify reliability requirements in table XLII. The Using Service should provide the table format for the contractor to complete.

Reliability verifications must be designed to assess reliability with the low sample sizes and operation times available prior to engine qualification. Reliability verification plans should be robust in planning alternatives for items that fail in testing. Weibull analysis provides statistical tools for handling small sample sizes and operation times. AFWAL-TR-83-2079, Weibull Analysis Handbook provides guidance for Weibull analysis.

Background:

Data gathered on a component or subassembly should be considered pertinent to reliability determination only when a severity factor(s) can be established for the test. To find events per equivalent engine flight hours (EFH), multiply test events per test hour by the corresponding severity factor.

Test time which accumulated during tests for which the severity level cannot be established for failure modes; e.g., test which exceed the required operational envelope should not be considered pertinent to reliability determination. An example of non-pertinent data for reliability determination is a pump test operated at overspeed conditions, elevated temperatures, and with excessive fluid contamination.

MIL-HDBK-781, 5.4.1 through 5.4.2.5 may be used as tailoring guidance for Reliability Qualification Testing.

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VERIFICATION LESSONS LEARNED (A.4.5.1.1)

Considerable cost has been saved by using the severity level concept to allow the use of all pertinent reliability data to establish reliability levels. If all testing were conducted at a severity level equivalent to actual operation, the test cost would be prohibitive for a comparable equivalent test time.

A.3.5.2 Maintainability.

A.4.5.2 Maintainability.

A.3.5.2.1 Maintainability quantitative requirements.

The engine design maintainability shall be capable of meeting engine or air system availability goals for the mission and mission mix profile of table XI. Maintainability requirements shall be specified in tables XIV and XV.

The diagnostic capabilities necessary to meet engine maintainability requirements shall be specified herein.

REQUIREMENT RATIONALE (A.3.5.2.1)

Quantitative requirements are essential to assess engine achievement of the maintainability requirements.

REQUIREMENT GUIDANCE (A.3.5.2.1)

The following should be used as guidance to tailor the specification paragraph:

The values for the maintainability requirements of table XIV should be derived from the system maintainability allocations and should be based on the planned air system utilization rate and mission mix. Table XIV, items 8 - 14 should consider the following:

a. Percent of success that diagnostic testing can isolate faulty WRA/LRU engine components should be at least __(a)__ percent and should be within __(b)__ minutes.

b. Percent of success which diagnostic tests correctly isolate faulty replaceable parts within WRA/LRU engine components should be __(c)__ percent.

c. Percent of success which diagnostic tests correctly diagnose fault by retest should be ___(d) percent.

d. Percent of false alarms from diagnostics system should be __(e)__ percent.

e. Percent of WRA/LRU fault detection should be __(f)__ percent.

f. Percent of WRA/LRU component fault detection should be __(g)__ percent.

g. Mean flight hour between false alarm rate for the diagnostics system should be __(h)__ hours.

The following values have been used to tailor the previous paragraph:

(a): A value from 70% to 75%.

(b): A maximum value of 15 minutes (this includes time for hookup of all needed diagnostic equipment, external and internal to the aircraft, and to perform all diagnostic testing).

(c): A value from 90% to 95%.

(d): A value from 95% to 100%.

(e): A value less than 10%.

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(f): A value greater than 90%.

(g): A value greater than 75%.

(h): The contractor should provide the number of hours.

Background:

A maintainability program has been used to identify, apportion, and control maintainability requirements for engine design and development to achieve the operational readiness, maintenance manpower, and life cycle cost objectives of the engine program.

MIL-HDBK-470 or an equivalent commercial standard may be used as tailoring guidance for establishing a maintainability program.

The Maintainability program should establish program structure and requirements; a maintainability program plan; identify monitoring and control programs for subcontractors and vendors; identify a data collection analysis and corrective action system; perform maintainability monitoring, allocations, predictions and analysis; prepare inputs for the Logistics Support Analysis; and perform maintainability testing and demonstration.

The R&M terms defined in OPNAVINST 4790.2 should be used in this requirement. Maintainability index should be kept low, so that support costs will be kept low and mission capability will be improved. Special attention should be given to keeping task times low at the organizational level because of their direct bearing on aircraft turnaround time and thus on mission capability. The maintainability index, MMH/EFH, is a very useful parameter to control cost, since cost is a function of total maintenance time. The maintainability index is a versatile parameter, since it can be varied by changing task frequency that is affected by reliability, or task times. Engine replacement time and WRA/LRU and LRU replacement time are key maintainability considerations at the organizational maintenance level and increase mission capability by allowing aircraft to be turned around more quickly between sorties. Rapid, reliable and accurate diagnostic capability is needed to reduce maintenance time, thus reducing maintenance costs. The need for diagnostics should be identified starting with known failure modes as identified via reliability considerations. Diagnostics should be able to isolate failures down to the WRA/LRU level within 30 minutes for at least 95% of all known failures. For ultimate maintainability, all known failures should be isolated to the faulty WRA/LRU within a reasonable amount of time. Special attention should be given to making diagnostic techniques reliable. Attempts should be made to monitor the conditions that exist when a failure occurs. Intermediate shops and, if necessary, depot facilities should be equipped to fault isolate using techniques which simulate flight conditions.

All maintenance events, scheduled and unscheduled should be completed at the lowest maintenance level feasible and limited to the least number of actions possible.

A Level of Repair Analysis (LORA) should be performed to determine the proper level of maintenance for a repair action as shown in MIL-PRF-49506.

A Reliability Centered Maintenance (RCM) analysis is required in conjunction with the Level of Repair Analysis to determine or change the number of scheduled maintenance events.

REQUIREMENT LESSONS LEARNED (A.3.5.2.1)

The Army made significant maintainability improvements in Engine C by paying attention to maintainability considerations. Many improvements are now requirements in the specification. Some of these considerations were modular construction, ease in removal for repair, and no fix time before overhaul.

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Significant reductions in “cannot duplicate rate” of electronic components on Engine B program were achieved when Combined Environmental Reliability Testing (CERT) was used to fault isolate problems. CERT is commonly called “shake and bake” testing and involves subjecting electronic components to realistic temperature and vibration environments. It is intended to identify potential problems and associated fixes, early in development.

A.4.5.2.1 Maintainability quantitative requirements.

The requirements of 3.5.2.1 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.5.2.1)

Quantitative requirements for maintainability and maintainability indices must be verified to confirm that the engine meets maintainability requirements.

VERIFICATION GUIDANCE (A.4.5.2.1)

Past verification methods have included analysis, demonstration, and test.

The following may be used to tailor the specification paragraph.

a. Total maintenance man-hour/EFH should be determined via demonstration and analysis, and should be based on the following:

Frequency of failure based on the most up to date reliability predictions, results of development testing, and like item field data where available.

Maintainability demonstration task times as determined utilizing Using Service personnel and procedures for task times that are demonstrated. The number and skill level of personnel should be consistent with current Using Service maintenance policy.

Maintainability predictions for task times that are not demonstrated.

a. Mean time of repair should be measured via demonstration.
b. Engine replacement time should be measured via demonstration.
c. WRA/LRU replacement times should be measured using established Using Service procedures.
d. Diagnostics:

   WRA/LRU fault isolation of installed engine components should be verified by demonstration. This demonstration should be based on failure modes that can be simulated. At least one failure mode should be selected from each WRA/LRU that comprises at least the top 80 percent of the total WRA/LRU removal rate. Also, the top 30 failure mode drivers should be included in the demonstration.

   Percent of success which diagnostic tests correctly isolates faulty parts within WRA/LRU engine components should be verified by demonstration. The requirements of table XV should be verified during the demonstration.

   Percent of success in which diagnostic tests correctly diagnose fault by retest should be verified by analysis of data gathered during development, documenting the number of times that the diagnostic testing properly isolated problems or gave false readings.

The Using Service should provide guidance to the Contractor as to the relationship between system life cycle phases and the maintainability verification. These phases are: Concept Refinement, Technology Development, System Development & Demonstration (SDD), Production and Deployment (including Modifications and Low Rate Initial Production), and Operations and Support. Coordination of the maintainability verification with other required demonstrations should be accomplished whenever possible to avoid unnecessary duplication of effort. The use of mockups, Engineering Development Models or

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development hardware, in the early phases of SDD, should be allowed for maintainability demonstrations since production hardware will not be available.

Maintainability requirements will be assessed for compliance with the specification at the time of the initial service release. However, it is important that the contractor conduct incremental maintainability demonstrations on prototype hardware throughout the development program to assure that the ultimate design will be capable of meeting maintainability requirements. The results of incremental maintainability demonstrations will be reviewed throughout the development program, especially at the time of design reviews.

Maintenance demonstrations should be conducted under the conditions that are expected in the operational environment. Expected weather conditions should be considered if a large percentage of the maintenance tasks will be performed under adverse conditions. Where portions of the maintenance actions are expected to occur under different environmental conditions, multiple maintainability demonstrations should be conducted. The effects of varying task times should be weighed depending on the proportionate period of time that each of the varying environments is expected to be encountered.

Engine replacement time should be demonstrated using either an installation mockup or an aircraft. For a new centerline engine, a development mockup may be used to verify WRA/LRU accessibility.

As a minimum, failure modes observed during development should be analyzed during the diagnostics and effectiveness demonstration. Specific demonstration procedures should be established based on the automatic test equipment system under consideration.

The maintainability index, MMH/EFH, is a function of task time frequency. The most up to date reliability demonstrations, accounting for measured failure rates, provide the most accurate failure frequencies. Real or simulated failures must be used to verify the diagnostic and fault isolation capabilities.

The reliability centered maintenance analysis is meant to be an objective view of the maintenance requirements such as preventative maintenance, scheduled on-condition tasks, scheduled rework tasks, scheduled discard tasks, safe life limits, economic life limits, scheduled failure finding tasks, and age exploration.

VERIFICATION LESSONS LEARNED (A.4.5.2.1)

In those instances in which maintainability considerations were not demonstrated, operational maintainability and mission capability suffered. The Engine C program proved to benefit significantly in this area because of the effects of conducting maintainability demonstrations. During this program the Army required three demonstrations for maintainability. Each demonstration was an improvement on the previous one. They also required a full time human engineering engineer present at all time.

Past demonstrations of maintainability have used unrealistic procedures. A separate ratchet, socket, and extension was used for each nut run up on all threaded studs during replacement of all controls and externals. The nuts were preplaced in all sockets before the demonstration began. This resulted in unrealistic maintenance times.

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A.3.5.2.1.1 Excluded maintenance functions.

All excluded maintenance and repair functions shall be listed in table XVI.

REQUIREMENT RATIONALE (A.3.5.2.1.1)

The Contractor needs to specify all maintenance functions that are excluded from the maintenance index calculation.

REQUIREMENT GUIDANCE (A.3.5.2.1.1)

The contractor should provide the listing in table XVI. If the Using Service prefers to have the list provided separately from the specification, then make sure the list of excluded maintenance and repair functions is in the CDRL. This paragraph should then be labeled “Not Applicable”.

The following maintenance and repair functions should be excluded from the maintenance index calculation and added to the list:

a. Technical Order and Time Compliance Technical Order compliance when related to upgraded engine performance or other changes in specification requirements.

b. Removal and installation of non-engine supplied equipment to perform engine maintenance (e.g.; access panels, parts installed on the engine or nacelle by the weapon system Contractor). Excluded maintenance functions for non-engine supplied equipment to access engine components should be reviewed on a case-by-case basis to prevent shifting the maintenance accountability burden to the air vehicle side or losing accountability altogether.

c. Engine fuel servicing.

d. Engine removal and replacement in the aircraft except when required due to: chargeable engine failures or malfunctions, engine design that requires removal to perform routine maintenance on such items as ignitors, fuel filters, oil filters, etc.

e. All maintenance tasks resulting from failures or discrepancies that are not chargeable to the engine.

f. All maintenance tasks required to position ground equipment, and move the aircraft.

REQUIREMENT LESSONS LEARNED (A.3.5.2.1.1)

Some engine maintenance requirements were specified in the contract, such as for Engine C1, rather than the engine model specification.

A.4.5.2.1.1 Excluded maintenance functions.

The requirements of 3.5.2.1.1 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.5.2.1.1)

Analysis and demonstrations of the excluded maintenance functions are needed to determine if any pertinent engine maintenance was excluded.

VERIFICATION GUIDANCE (A.4.5.2.1.1)

Past verification methods have included analysis and demonstration.

Logistic and maintenance personnel of the Using Service should review the excluded maintenance functions.

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VERIFICATION LESSONS LEARNED (A.4.5.2.1.1)

None.

A.3.5.2.2 Maintainability qualitative requirements.

The engine shall meet the qualitative maintainability requirements designated in table XVII.

REQUIREMENT RATIONALE (A.3.5.2.2)

When only quantitative requirements are specified and qualitative requirements are not considered, maintainability is adversely affected. Qualitative requirements are needed also.

REQUIREMENT GUIDANCE (A.3.5.2.2)

The following should be used to tailor table XVII:

MIL-HDBK-470, MIL-HDBK-471, and MIL-STD-1472, or equivalent commercial standards may be used as tailoring guidance.

The maintainability data developed from the maintainability program should be used as source data for the MIL-PRF-49506, Support Analysis Summaries.

The following list below may be used to develop the requirements for table XVII:

a. The contractor should design the engine to minimize required skills, tools, and support equipment.

b. Parts of the engine requiring routine service, checking, adjustment, or frequent replacement should be made readily accessible without removal of other parts or components, and without disassembly of the engine.

c. Provisions for access to the location of any WRA/LRU should be based on reliability predictions, i.e., the least reliable WRA/LRU on the engine should have the best accessibility.

d. WRA/LRUs should be no more than one deep. Removal of the WRA/LRU should not require the removal of unaffected WRA/LRUs.

e. Shimming or measuring of WRA/LRUs should not be required for WRA/LRU replacement.

f. No adjustments, balancing, matching, or rigging should be required to install or replace WRA/LRUs on an assembled engine.

g. Plumbing or electrical removal should not be required for WRA/LRU access or removal.

h. Captive fasteners should be used for WRA/LRU mounts and connections. No fasteners smaller than 6-mm (1/4-inch) diameter should be used for WRA/LRU mounts or for organizational level maintenance. Individual WRA/LRUs should have the same style and size fastener for mounting and the same style and size fasteners for mating.

i. Positive securing clamps, nuts, bolts, connectors, and keyed interfaces should be provided to prevent misalignment and improper installation of WRA/LRUs, plumbing, and cables. The design should permit removal without exceeding capacity or capability of required tools after the engine has been in service.

j. Fluid carrying systems subject to disassembly for WRA/LRU or engine replacement should be designed to minimize system drainage or spillage on the maintainer.

k. Features should be provided to prevent functionally different components with the same or similar design from being improperly installed.

l. On wing fan trim balance capability should be provided when applicable.

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m. Safety wire, cotter pins, and bent tab washers should not be used in securing connectors, fasteners, etc. at the intermediate and organizational levels and should be kept to a minimum at the depot level. Safety wire and cotter pins should not be used for turboshaft engines at any maintenance level.

n. The engine design should allow centrally located engine to air vehicle disconnects, subject to survivability and vulnerability requirements.

o. The design should not require engine-to-engine, engine to aircraft, or accessory and installation rigging. System alignment or complicated operational checks should not be required.

p. The design should allow for adequate access, space, and tool clearance around fasteners and connectors to provide: adequate clearance for component removal and installation at all levels of assembly; and to provide proper seating of tools and uniform torque application.

q. WRA/LRU modules and any SRU should be interchangeable with like items without match balancing.

r. The design should provide provisions for rapid repair such as external tubing repair versus whole line replacement.

s. Placement of engine furnished electronic equipment in the air vehicle rather than on the engine should be considered to avoid the hostile environment posed by the engine.

t. Sensor check or replacement should not require engine rig or trim. Any sensor that requires more than 20 minutes to replace while the engine is installed in the aircraft should not be mission critical.

u. Aircraft leveling should not be required for installed engine servicing or quantity status reading.

v. Fluid level quantity indicators, if part of the engine design, should be clearly visible for accurate reading when using red or white flashlight as approved by the Using Service or natural daylight illumination sources.

w. Connecting devices should be selected to prevent improper installation while minimizing the number of different parts. MIL-HDBK-965 or an equivalent commercial practice can be used for guidance.

x. No routine or scheduled maintenance should be needed for field washing, cleaning, or drying of engine components. However, the engine should incorporate provisions for water wash compressor cleaning if required.

y. The design should consider including provisions for the incorporation and use of pyrometers to check temperatures on all turbine blades during depot level testing. This should be evaluated during the initial design reviews.

z. Engine operation after installation in the aircraft should not be required except for leak check and control.

aa. Disconnect points for external plumbing should coincide with module or SRU flange and disconnect locations.

bb. The engine hydraulic system should include dedicated fittings conveniently located on the engine for each access and attachment of ground test equipment when installed in the aircraft.

cc. The color red should not be used on the engine for any identification means without an adjacent symbol readily visible under red-light conditions.

dd. Components subject to wear, requiring repair or rework to restore dimensioned features, should provide a dimensional baseline datum that is not destroyed or degraded in service or during repair or rework.

ee. The engine configuration should permit installation in the aircraft with as-received components and connectors mounted on the engine.

ff. Tubing installation should use ARP 994 as a guide. Plumbing and tubing lines should be identified by labels, markings, color coding, or a combination of these methods.

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REQUIREMENT LESSONS LEARNED (A.3.5.2.2)

Previous engine designs that violated the provisions of the above list experienced increased support costs and reduced mission capability.

In the past, many engines sustained FOD when nuts, bolts, and fasteners were left on the ramp. Therefore, captive fasteners are a desirable maintenance feature. Engines D and O have demonstrated that safety wire and cotter pins are not required.

A.4.5.2.2 Maintainability qualitative requirements.

The requirements of 3.5.2.2 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.5.2.2)

The qualitative maintainability requirements must be verified to assess the engine design in meeting maintainability goals.

VERIFICATION GUIDANCE (A.4.5.2.2)

Past verification methods have included analysis, demonstration, and inspection.

Qualitative maintainability should be demonstrated in an operational environment or on an installed engine mockup. For each of the requirements listed in table XVII, a verification method should be noted. Most of the requirements can be verified by demonstration and inspection, while a few may also include some amount of engine operation or test as part of the demonstration (e.g., fan balancing should be verified by demonstration of the balancing task, and operating the engine to demonstrate the task has been properly performed).

VERIFICATION LESSONS LEARNED (A.4.5.2.2)

When qualitative maintainability requirements were not adequately analyzed early during development, operational support costs went up and mission capability was reduced.

A.3.5.2.3 Maintenance.

A.4.5.2.3 Maintenance.

A.3.5.2.3.1 Modules.

The structural components that define each engine module, and the length of time to remove and replace each module, shall be specified in table XVIII. All modules shall be capable of removal and replacement at the ____ maintenance level.

REQUIREMENT RATIONALE (A.3.5.2.3.1)

The modular concept is needed for easy replacement of major engine sections and to reduce or eliminate the requirement to send the complete engine back to the depot for maintenance.

REQUIREMENT GUIDANCE (A.3.5.2.3.1)

The following should be used to tailor the specification paragraph:

The Using Service should provide the maintenance level.

Disconnects for all external plumbing and wiring associated with a module should be located as near the front and back planes of the module as possible.

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Background:

Replacing major engine sections at the organizational or intermediate maintenance level can either reduce or eliminate the requirement to send the complete engine back to the depot for maintenance when an internal engine repair is necessary. This approach will tend to improve the availability of the engine (and weapons system) for service use and reduce labor time and cost at the field and intermediate maintenance levels.

Modules should be completely interchangeable among engines. This interchangeability should be possible for any combination of modules at any time during their required life. Complete engines should be interchangeable in multi-engine aircraft (i.e., right side to left side). Typical engine modules are fan, compressor, turbine, augmentor and nozzle, etc.

REQUIREMENT LESSONS LEARNED (A.3.5.2.3.1)

In the past, an internal engine part failure often required that the complete engine be shipped back to the depot to effect the repair. This tied up two sets of hardware: all the satisfactory components of the faulty engine and another set of duplicate components in the replacement engine shipped into service. The modular concept has reduced the need for a second set of satisfactory duplicate components.

A.4.5.2.3.1 Modules.

The requirements of 3.5.2.3.1 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.5.2.3.1)

The maintenance level and time required for modular removal and replacement must be verified.

VERIFICATION GUIDANCE (A.4.5.2.3.1)

Past verification methods have included analysis and demonstration.

Background:

Performing demonstrations on engine mock-ups or zero time engines will not result in representative remove and replacement times because interface wear between modules which results in alignment difficulties will not exist.

Maintenance personnel of the Using Service should perform the verification.

VERIFICATION LESSONS LEARNED (A.4.5.2.3.1)

Past engine programs have shown that field engines have longer module replacement times than the contractor showed on development engines due to field environment effects.

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A.3.5.2.3.2 Maintenance, inspection, and repair cycle.

The maintenance, inspection, and repair periods for the total engine and each module shall be specified in tables XIXa-c, together with the estimated time in man-hours required to perform these functions.

REQUIREMENT RATIONALE (A.3.5.2.3.2)

The Contractor needs to provide the information necessary to show that basic maintenance, inspection, and repair functions have been considered in the design of the engine.

REQUIREMENT GUIDANCE (A.4.5.2.3.1)

The following should be used to tailor tables XIXa-c of the specification:

Table XIXa should contain the time to remove and replace modules, assuming engine uninstalled and including the time to remove and refit other modules and externals as necessary.

Table XIXb should contain the removal and replacement times of externally mounted engine components with no allowance being made for aircraft access.

Table XIXc should contain the module inspection period:

Background:

The maintenance and repair cycles of the engine components should coincide or be some multiple of each other and to the aircraft maintenance cycle to eliminate duplicate and frequent engine removals.

REQUIREMENT LESSONS LEARNED (A.4.5.2.3.1)

See tables XLIII and XLIV for values, based on USN and USAF experience for current engine types.

A.4.5.2.3.2 Maintenance, inspection, and repair cycle.

The requirements of 3.5.2.3.2 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.5.2.3.2)

Maintenance, inspection, and repair cycles need to be performed to verify requirements.

VERIFICATION GUIDANCE (A.4.5.2.3.2)

Past verification methods have included analysis and demonstration.

VERIFICATION LESSONS LEARNED (A.4.5.2.3.2)

None.

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A.3.5.2.3.2.1 Maintenance inspection techniques.

Engine design shall permit maximum use of non-destructive inspection techniques and multi-purpose test and inspection equipment. Inspection provisions, including access envelopes, shall be shown on the engine configuration and envelope figure.

REQUIREMENT RATIONALE (A.3.5.2.3.2.1)

An effective maintenance philosophy requires the detection of a potential failure condition. To be cost effective the inspection techniques should not require disassembling the engine. The borescope inspection method is the most common non-destructive inspection technique used, so a sufficient number, size, and location of access ports needs to be stated so an adequate internal inspection of the engine can be accomplished.

REQUIREMENT GUIDANCE (A.3.5.2.3.2.1)

The following should be used to tailor the specification paragraph:

The information can be shown in the ICD, or for programs that do not have an ICD, the information can be put into the specification paragraph.

Provisions for 360-degree inspection of the installed engine should be made for the fan, compressor, combustor, and turbine sections of the engine. A positive means of slowly rotating the rotor system should be provided to facilitate inspection. The access port size should be a minimum of (a) mm ((a) inch). The same tool should be used for removal and replacement of port covers, plugs, and associated fasteners. The location of access ports should, as a minimum, permit inspection of the following locations: (b) . Access ports should be accessible without removing other components

The following should be used to tailor the preceding paragraph.

(a): Borescope ports should be large enough to accept the largest borescope used by the maintenance group. The Using Service should indicate the borescope size. A typical size for large engines is 10-mm (0.4-inches). For small engines, the borescope access port size should be a minimum of 5-mm (0.2-inches).

(b): The contractor should propose the location of access ports or the areas to be seen through inspection. Typical areas of interest are:

a. Fan
b. Compressor inlet guide vane leading edge.
c. Compressor inlet guide vane trailing edge and compressor first stage rotor leading edge.
d. Compressor last stage rotor trailing edge and combustor inlet.
e. Combustor liner and fuel nozzle faces.
f. Combustor outlet and turbine first stage vane leading edge.
g. HPT first stage vane trailing edge and first stage rotor leading edge.
h. HPT last stage rotor trailing edge and low pressure turbine first stage vane leading edge.
i. LPT first stage vane trailing edge and first stage rotor leading edge.
j. Power turbine first stage vane trailing edge and first stage rotor leading edge.

The number of access ports should be kept to a minimum. They produce areas of high stress that eventually produce cracks in cases. Access ports are a source of leakage and maintenance discrepancies.

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REQUIREMENT LESSONS LEARNED (A.3.5.2.3.2.1)

In the past, some engines had a limited number of access ports. Therefore, complete 360-degree coverage of non-rotating parts was impossible. This prevented detection of impending failures.

A.4.5.2.3.2.1 Maintenance inspection techniques.

The requirements of 3.5.2.3.2.1 should be verified by: _____.

VERIFICATION RATIONALE (A.4.5.2.3.2.1)

Maintenance inspection techniques must be demonstrated by the Using Service personnel using procedures and tools and equipment available in the field.

VERIFICATION GUIDANCE (A.4.5.2.3.2.1)

Past verification methods have included inspection and demonstration on an installed engine.

Background:

Each inspection method developed for the engine should be employed during routine inspections of development engines. Any deficiencies that are discovered should receive design attention as early as possible so that improvements can be made prior to engine flight operation.

VERIFICATION LESSONS LEARNED (A.4.5.2.3.2.1)

Past demonstrations by the contractor used procedures and tools that were never used in the field.

A.3.5.2.3.3 Tools.

Required tooling shall be held to a minimum. The engine shall be capable of being maintained with standard hand tools, as identified in table XX. Where provisions for standard hand tools are not feasible, the design should provide, wherever possible and cost effective, for special tools and test equipment that are available and in use on other in-service engines, as identified in table XX.

REQUIREMENT RATIONALE (A.3.5.2.3.3)

The use of standard hand tools is needed to minimize the cost of tooling for the engine. Required tooling must be kept to the absolute minimum for economy and ease of maintenance.

REQUIREMENT GUIDANCE (A.3.5.2.3.3)

The following may be used to tailor the specification paragraph.

The contractor or Using Service should identify the tools in the table. The table should include a tool set number if available, and at which maintenance level the tool set will be used. If no tool set number has been assigned, then a description of the tools should be provided. The description should identify the type of tools (i.e.; wrenches, sockets, gear pullers, etc.)

The Army uses tool set number SC-5180-99-CL-A07 for standard tools, and tool set number SC-4920-99-CL-A92 for supplemental tools.

Any requirement for new special tools, fixtures, and test equipment should be subject to specific approval of the Using Service.

Cadmium plated tools should be avoided and require the specific approval of the Using Service.

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Background:

Cadmium plated tools may be prohibited if titanium materials exist in the engine. However, the Air Force has not encountered a detrimental problem when cadmium tools are used in titanium at temperature of 316°C (600°F) or less. If titanium is used above 316°C (600°F) with cadmium, liquid metal embrittlement may occur.

REQUIREMENT LESSONS LEARNED (A.3.5.2.3.3)

Non-standard tools have been quite expensive, and logistics and provisioning problems have occurred in service when an inadequate number of each special tool was purchased to support maintenance and repair efforts. An engine designed to be maintained with standard tools has been simpler for personnel to maintain. It has taken less time to train personnel to maintain an engine using standard tools rather than special tools.

NADEP, Norfolk, found in the early 70's that "cracking of stressed titanium can occur due to contact with solid metals such as cadmium". The NADEP "banned the use of cadmium plated tools while working on titanium hardware" and coated the tools with black iron oxide.

The Air Force has discovered that iron oxide may cause corrosion in titanium.

A.4.5.2.3.3 Tools.

The requirements of 3.5.2.3.3 should be verified by: _____.

VERIFICATION RATIONALE (A.4.5.2.3.3)

The use of hand tools can only be verified by inspection and demonstration.

VERIFICATION GUIDANCE (A.4.5.2.3.3)

Past verification methods have included inspection and demonstration.

VERIFICATION LESSONS LEARNED (A.4.5.2.3.3)

None.

A.3.5.2.4 Battle damage repair.

The engine design shall support the battle damage repair requirements of the air vehicle. Battle damage repairs shall not take longer than _(a)_ hours and shall be of sufficient strength to sustain at least _(b)_ flight hours of operation.

REQUIREMENT RATIONALE (A.3.5.2.4)

The primary purpose of combat damage repair is to allow aircraft to continue flying combat missions. A secondary purpose and absolute necessity is to allow ferrying of the damaged aircraft to rear areas for more lasting repairs.

REQUIREMENT GUIDANCE (A.3.5.2.4)

The following should be used to tailor the specification paragraph:

(a) and (b): The contractor should provide the values for repair time and hours of operation, and the values should be consistent with the air vehicle requirements. If the air vehicle has not been defined at this time, then values from past programs should be used. The Using Service may delete this requirement and use the other requirements of section 3.5 to cover battle damage repair.

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Combat damage repair design techniques should utilize tools and materials found at organizational and intermediate levels.

Background:

Survivability is measured by the capability of the aircraft to accomplish its mission in a hostile environment. The ability of the subsystems to retain mission essential performance following a hit is a major contribution to survivability. The ideal repair resolution is to fly the airplane with existing engine damage if possible. Where damage would jeopardize flying safety or mission accomplishment, repairs must be authorized that can be completed within a short time with tools, men, material, and spares on hand.

REQUIREMENT LESSONS LEARNED (A.3.5.2.4)

Aircraft operating in high threat, high intensity conflicts have received frequent damage from combat. War games conducted with computer models have indicated that during the early days of a conflict many aircraft will be returning from combat damaged.

A.4.5.2.4 Battle damage repair.

The requirements of 3.5.2.4 should be verified by: _____.

VERIFICATION RATIONALE (A.4.5.2.4)

Battle damage repair techniques must be verified.

VERIFICATION GUIDANCE (A.4.5.2.4)

Past verification methods have included analysis and demonstration.

Background:

Specimens from the engine component static ballistic tests may be used to develop battle damage repairs. Verification testing can be conducted in conjunction with the engine Live Fire Test (LFT) program. LFT programs address specific engine vulnerability issues and incorporated the combination of controlled damage tests as well as ballistic testing on an operating engine. Emergency repair methods, materials and restrictions for noncatastrophic battle damage should be analyzed. The analysis should verify the effects of repairs on engine operability, reliability and maintainability (e.g.; performance, structural integrity, oil and fuel consumption, etc.).

Structural repair or patching of advance engine materials, especially composites, must be demonstrated since the database may be very limited. Repair of cooling liners in augmentors and tailpipes especially needs attention to enable puncture repair and preserve cooling path.

VERIFICATION LESSONS LEARNED (A.4.5.2.4)

None.

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A.3.5.3 Human performance and human engineering.

The engine shall be designed to be maintained by at least the central ninety percent (90%) of the maintainer population (5th percentile female stature through 95th percentile male stature) wearing a full complement of personal protective equipment, under all environmental conditions (including day and night), defined in the specification and all operational scenarios at operating bases and deployed locations.

REQUIREMENT RATIONALE (A.3.5.3)

The application of human engineering is necessary in engine systems, subsystems, equipment, and facilities to ensure usability.

REQUIREMENT GUIDANCE (A.3.5.3)

The following should be used to tailor the specification paragraph:

MIL-STD-1472 and MIL-HDBK-46855 or equivalent commercial standards may be used as tailoring guidance. The Using Service should select the applicable requirement of MIL-STD-1472.

Background:

Human engineering should be applied in order to achieve mission success through integration of the human into the system, subsystem, equipment, and facility, and achieve effectiveness, simplicity, efficiency, reliability, safety of system operation, training, and maintenance, and foster design standardization within and among engines.

REQUIREMENT LESSONS LEARNED (A.3.5.3)

None.

A.4.5.3 Human performance and human engineering.

The requirements of 3.5.3 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.5.3)

Verification of human engineering is necessary to achieve required performance by operations and maintenance personnel.

VERIFICATION GUIDANCE (A.4.5.3)

Past verification methods have included demonstration and analysis.

Background:

During the maintenance demonstration the contractor should use the minimum and maximum physical requirements for the personnel. They should be clothed for battlefield conditions and extreme low temperatures.

VERIFICATION LESSONS LEARNED (A.4.5.3)

Maintenance personnel dressed in chemical protective clothing with arctic gear for cold protection had problems performing maintenance task.

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A.3.6 Combat survivability.

The engine shall apply a combat survivable design concept that includes state of the art survivability technology (vulnerability and susceptibility) which insures survival in the intended threat environment.

REQUIREMENT RATIONALE (A.3.6)

Combat survivability must be required. An aircraft engine will affect, and in many cases dominate, critical survivability characteristics of the aircraft systems.

REQUIREMENT GUIDANCE (A.3.6)

The engine may have a major impact on air system survivability.

DoDI 5000.2 requires that a total system approach be used to satisfy the survivability requirements. The directive should be consulted for further guidance. DoD Instruction 5000.2 contains the following air system requirement:

E 7.8 Survivability. For systems with missions that might require exposure to combat threats, the PM [program manager] shall address personnel survivability issues including protection against fratricide, detection, and instantaneous, cumulative, and residual nuclear, biological, and chemical effects; the integrity of the crew compartment; and provisions for rapid egress when the system is severely damaged or destroyed. The PM shall address special equipment or gear needed to sustain crew operations in the operational environment.

Section 3.6 should be classified when tailored for a particular air system. A classified appendix with the same numeration, paragraph titles, and sequence of Section 3.6 should be attached to or referenced by the RFP or model specification. In the event a classified appendix is utilized, the paragraph numbers and titles in Section 3.6 should be retained but type “See Classified Appendix __” in the text portion of each paragraph.

If the engine is intended for a single aircraft or class of aircraft, the engine should be optimized for the threat environment described in the System Threat Report (STR). The STR may result in survivability allocations to the engine from the aircraft. Engine requirements allocated from an air vehicle system specification should be included in the engine model specification.

If the engine is partly or wholly intended for a general class of aircraft with combat missions (e.g.; patrol or anti-submarine warfare, combat reconnaissance, combat transport, strike fighters or attack, assault helicopter, or other), the engine should be optimized for the general threat environment described in the threat assessment for those missions or classes of combat.

Background:

Critical engine survivability characteristics should be specified for intended aircraft systems mission performance requirements, which might include observable reduction, vulnerability, degree of benign failure, fail-safe feature or tolerance to other threat effects such as fuel ingestion or blast pressure transients. The Joint Services agree that engine performance related to threat weapons capabilities must be classified, although the degree of classification may vary.

REQUIREMENT LESSONS LEARNED (A.3.6)

Observable and ballistic test experience in Aircraft F (Engine H), Aircraft H (Engine O), and other engine programs, demonstrate that the critical engine survivability characteristics are capable of considerable design optimization, provided that appropriate design and performance requirements are specified. Development tests and verifications (DT&V) need to contribute to the design process and should be

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conducted as early as possible in development. These tests and verifications should be directed toward specific design adaptation prior to IFR.

A.4.6 Combat survivability.

The engine combat survivability design features that include state of the art survivability technology (vulnerability and susceptibility) to insure survival in the intended threat environment shall be demonstrated.

VERIFICATION RATIONALE (A.4.6)

Considerable DT&V is required to help define the achievable engine survivability characteristics and their contribution to aircraft system mission effectiveness.

Optimizing the engine in combination with the total aircraft system and knowledgeable allocation of critical survivability characteristics between air vehicle and engine will produce the most effective combat system. Survivability DT&V is required to provide information to designers seeking to incorporate survivability requirements with the lowest penalties.

To ensure that ongoing DT&V is accomplished, progress toward achievement of critical survivability characteristics must be measured and reported at appropriate engine design milestones. Survivability DT&V is also required to determine that specification requirements have been met and to measure the actual survivability characteristics that have been achieved.

VERIFICATION GUIDANCE (A.4.6)

Background:

The individual survivability verification requirements for each critical survivability characteristic or requirement should be specific and detailed in the subparagraphs below, and should be based on engine missions or mission types. The description of the overall verification procedure and critical milestones, as applicable, should be specified in this section. Tests required in the survivability program should reference specific paragraphs in MIL-HDBK-2069 or MIL-HDBK-1799.

VERIFICATION LESSONS LEARNED (A.4.6)

A considerable database of engine survivability verifications has been available from individual services and repositories such as the Survivability and Vulnerability Information Analysis Center (SURVIAC), which is sponsored in part by the joint technical coordinating group on aircraft survivability (JTCG/AS), and is located at Wright-Patterson Air Force Base in Ohio.

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A.3.6.1 Susceptibility.

A.4.6.1 Susceptibility.

A.3.6.1.1 Noise.

The near and far field noise levels for an uninstalled engine, vectored or non-vectored, operating at sea level static shall not exceed the levels of figures 17a, 17b, and 17c.

REQUIREMENT RATIONALE (A.3.6.1.1)

Engine acoustic signature noise represents a health hazard to personnel, an annoyance to civilian communities, and an observable that can be used to detect combat and search and rescue aircraft.

REQUIREMENT GUIDANCE (A.3.6.1.1)

The following should be used to tailor the specification paragraph: The contractor should provide levels in figures 17a, 17b, and 17c. The figures should include the full range of vectored positions, if applicable, including those for take-off and landing. Uninstalled engine noise requirements may involve significant engine testing. Installed noise testing may be more valuable in verifying the air system overall noise level.

Background:

Noise reduction, while desirable, should remain secondary to performance requirements. No noise limits are defined here since such limits could place constraints on necessary performance. Engine noise limits should be defined on a program-by-program basis. Combat aircraft are usually not required to meet any standards, unless imposed for detection reduction purposes. Transport-type aircraft should be required to meet flight noise requirements set forth in the Federal Aviation Regulations (FAR) Part 36. Transport aircraft are exempt from FAR 36 only if it is determined that compliance would inhibit mission success.

Engine noise limits should include consideration of crew environment, aircraft structural requirements, maintenance, and ground operation. Noise limits should be established to preclude health hazards to personnel. The government should specify personnel noise exposure limits per AFOSH STD 48-19. Aural signature must be controlled so that noise is not detectable by threat weapons.

REQUIREMENT LESSONS LEARNED (A.3.6.1.1)

Air vehicle structural response to engine acoustic energy has led to premature failure of aircraft exhaust assemblies and adjacent areas, some of which necessitated mission abort and precautionary landings. . For more design and verification guidance concerning structural integrity see MIL-HDBK-1783.

Aircraft engine noise has caused hearing loss. Noise from aircraft engines being tested or operated in the vicinity of populated areas produced complaints from the public and in some cases, resulted in restricted operation.

With the current flow-down of requirements, it seems that the noise requirements will start at the overall system level. (Some noise can be controlled by operational requirements.) The requirements will then flow-down to the A/C system. For military A/C (especially highly integrated configurations) some of the requirements can be handled by A/C configuration/design tradeoffs. The A/C requirements are flowed to the "Propulsion System" which may have requirements such as noise suppression (liners/blockers/nearly choked inlets etc.) From here requirements are flowed down to the engine. The engine/gas generator requirements are at the end of a long trail. It also should be remembered that for some configurations the exhaust nozzle is part of the airframe (Aircraft X, etc.) In this situation the engine is accepted with a reference nozzle, and any noise requirements for the engine are with a non-representative exhaust system.

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Once the requirements have gotten down to the Tier III, there should be a place to capture the requirements and methods to verify that the engine is meeting those requirements.

A.4.6.1.1 Noise.

The requirements of 3.6.1.1 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.6.1.1)

Test measurement of engine noise may be required to demonstrate the actual engine does not exceed specification limits.

VERIFICATION GUIDANCE (A.4.6.1.1)

Past verification methods have included analysis and test.

The blanks in the paragraphs below must be filled in; the Survivability and Vulnerability (S&V) community should provide recommended numbers.

The following procedure may be considered as a verification method:

"A noise survey shall substantiate that limits will not be exceeded. The engine shall be mounted on an outdoor test stand with a minimum clearance of seven feet between the lowest part of the engine and the ground. Microphones shall be located in relatively flat terrain, free of excessive ground absorption characteristics. There shall be no obstructions that significantly influence the engine noise field.

The weather shall be free of precipitation with relative humidity between 30 and 60 percent, ambient temperature between 0°C (32°F) and 30°C (86°F), wind velocity less than 9.66 kilometers per hour (6 miles per hour) and no temperature inversions or anomalous wind conditions. No post-test engine recalibration shall be required. The signal level shall be at least 10 dB greater than the background noise level in each third octave band in the frequency range of interest.

a. Data presentation. The engine noise level data shall be presented as follows and per the format shown on the referenced figures:

Equal overall sound pressure level contours (dB reference of 0.0002 microbars) as shown on figures 17a and 17b.

One-third octave frequency spectra from a center frequency of 20 Hz to 10 kHz at the positions A through G specified on figure 17a shall be tabulated. All near field sound pressure levels shall be based on measurements at head level where personnel are located during engine maintenance or other operation. In addition, one-third octave band sound pressure levels from a center frequency of 20 Hz to 10 kHz in the form of uncorrected data (e.g.; uncorrected for weather, terrain, etc.) shall be tabulated for all table IIa, IIb, and IIc performance points and for 19 positions. The 19 positions shall be based on measurements located 76.2 meters (250 feet) radially from the center of the engine's exhaust plane at ten degrees increments starting at zero degrees directly in front of the engine and ending at 180 degrees directly behind the engine.

Equal sound pressure level contours plotted for octave bands with center frequencies of 250, 500, and 1000 Hz shall be shown on figure 17a for the maximum and maximum continuous power of table IIa, IIb, and IIc only.

Equal perceived noise level (tone-corrected) contours in 5 PNdB increments shall be shown on figure 17b for all table IIa, IIb, and IIc performance points.

Three narrow band spectrum plots from 20 Hz to 10 kHz for the inlet and exhaust noise shall be shown in figures. These narrow band spectrum plots shall be presented in a spectral density format (e.g., sound pressure spectrum level) which normalizes the data to account for changing analysis bandwidth.

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(Position and power setting for the narrow band data shall be determined for an examination of the third-octave spectra).

The total acoustic power (dB ref 10-12 watts) generated at the maximum power condition.

There shall be sufficient data in the report to enable prediction of the engine noise signatures at any power setting. The required data shall include rotor speed, number of blades, hub to tip ratio, diameter, discharge total temperature, pressure ratio, exit velocity, exhaust mass flow, and ambient temperature and pressure.

a. Measurement system. The acoustical measurement system shall consist of approved equipment equivalent to the following:

A microphone system with frequency response compatible with measurement and analysis system accuracy as stated in c. below.

Tripods or similar microphone mountings that minimize interference with the sound being measured.

Recording and reproducing equipment with characteristics, frequency response, and dynamic range compatible with the response and accuracy requirements of c. below.

Acoustic calibrators using sine wave or broadband of known sound pressure level. If broadband noise is used, the signal shall be described in terms of its average and maximum RMS value for a non-overload signal level.

Analysis equipment with the response and accuracy requirements of d. below.

a. Sensing, recording and reproducing equipment. The sound produced by the engine shall be recorded in such a way that the complete data history is retained. A magnetic tape recorder is acceptable.

The characteristics of the system must comply with recommendations given in International Electromechanical Commission (IEC) 179 with regard to the microphone and amplifier characteristics.

The response of the complete system to a sensibly plane progressive sinusoidal wave of constant amplitude must lie within the tolerance limits specified in IEC 179 over the frequency range of interest.

The equipment must be acoustically calibrated using facilities for acoustic free field calibration and electronically calibrated as stated in paragraph d., below.

a. Analysis equipment. A frequency analysis of the acoustical signal shall be performed using one-third octave filters complying with the recommendations given in IEC 225.

The analyzer-indicating device must be analog, digital, or a combination of both. The preferred sequence of signal processing is:

(a) Squaring the one-third octave filter outputs.

(b) Averaging of integrating and linear to logarithmic conversion.

The indicating device must have a minimum crest factor capacity of three and shall measure, within a tolerance of +1.0 dB, the true root-mean-square (rms) level of the signal in each one-third octave band. If other than a true rms device is utilized, it must be calibrated for nonsinusoidal signals. The calibration must provide means for converting the output levels to true rms values.

The amplitude resolution of the analyzer must be at least 0.25 dB.

Each output level from the analyzer must be accurate within $\pm$1.0 dB with respect to the input signal, after all systematic errors have been eliminated. The total systematic errors for each of the output levels must not exceed $\pm$3 dB. For continuous filter systems, the systematic correction between adjacent one-third octave channels may not exceed 4 dB.

The dynamic range of the analyzer must be at least 55 dB in terms of the difference between full-scale output level and the maximum noise level of the analyzer equipment.

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The complete electronic system must be subjected to a frequency and amplitude electrical calibration by the use of sinusoidal or broadband signals at frequencies covering the range of interest of known amplitudes covering the range of signal level furnished by the microphone.

If broadband signals are used, they must be described in terms of their average and maximum rms values for a non-overload signal level.

Narrowband analysis shall be conducted with a maximum bandwidth of 5 Hz in the 20 to 500 Hz frequency range, 20 Hz in the 500 to 5000 Hz frequency range, and 100 Hz in the 5000 to 10000 Hz frequency range.

a. Noise measurement procedures. In order to ensure uniform practices relative to acoustic testing of engines, the following procedures shall be required:

Immediately prior to and after each test, a recorded acoustic calibration of the system shall be made in the field with an acoustic calibrator to check system sensitivity and provide an acoustic reference level for the analysis of the sound level data.

For the purpose of minimizing equipment or operator error, field calibrations shall be supplemented with the use of an insert voltage device to place a known signal at the input of the microphone, just prior to and after recording engine acoustic test data.

The ambient noise, including both acoustical background and electrical noise of the measurement system, shall be recorded and determined in the test area with the system gain set at levels which will be used for aircraft engine noise measurements.

a. Reporting and correcting measured data. Data representing physical measurements or corrections to measured data shall be recorded in permanent form and included in the final report. Estimates must be made of the individual errors inherent in each of the operations employed in obtaining the final data.

Measured and corrected sound pressure levels must be presented in one-third octave band levels obtained with equipment conforming to the standards described in previous paragraphs.

The type of equipment used for measurement and analysis of all acoustic, meteorological, and engine performance data shall be reported.

The following atmospheric environmental data shall be measured at representative microphone positions.

(a) Air temperature in degree Celsius (Fahrenheit) and relative humidity in percent.

(b) Maximum, minimum, and average wind velocity in knots and wind direction relative to engine centerline.

(c) Atmospheric pressure in kPa (psi).

(d) Comments on local topography, ground cover, and events that might interfere with sound recordings shall be reported.

All data shall be corrected to the following:

(a) Sea level pressure of 101.325 Kpa (2116 psf).

(b) Ambient temperature of 25°C (77°F) (ISA + 10°C).

(c) Relative humidity of 70 percent.

(d) Zero wind.”

On an engine with vectoring exhaust nozzles, the noise survey shall include all vectored positions from the centerline in increments of 5 degrees or less, as determined by the Using Service. This may require a change in the engine mounting on the outdoor test stand.

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VERIFICATION LESSONS LEARNED (A.4.6.1.1)

None.

A.3.6.1.2 IR signature.

The engine shall not exceed the IR signature levels contained in table XXI while meeting the requirements specified herein at any point during the engine life.

REQUIREMENT RATIONALE (A.3.6.1.2)

Combat survivability depends on keeping IR levels below which the enemy can detect and utilize, or use to target or defeat (kill) the aircraft.

REQUIREMENT GUIDANCE (A.3.6.1.2)

The following should be used to tailor the specification paragraph:

The purpose of 3.6.1.2 is to set the requirements on the propulsion system IR signature. The requirements should be tied to the Air Vehicle Specification. From the air vehicle requirement, the contractor budgets the propulsion system contribution. If the engine is provided under a separate contract as GFE, then the engine system signature specification from the government to the engine system contractor is required. The government issued specification should be consistent with the total air system specification. The IR levels should be derived for engine operating conditions, specified in the ICD, to ensure that the signature is properly specified. The IR signature requirements are usually specified in a table format.

If the engine is not required to meet IR signature requirements, this paragraph should be labeled “Not Applicable” or deleted.

Table format:

The IR signature should be specified for various azimuth and elevation angles. Table XXI provides general IR signature requirements table example for an installed engine. Not all the entries in the table may be required for a specific case; the required angles should be derived based on the mission, threat and tradeoffs between ease of verification with fewer angles and complete coverage with more angles. The number of tables required will depend on what parameters are important in the specific air system mission.

Specification parameters:

i) Azimuth and Elevation angles: Not all the entries in the table may be required for a specific platform, the required angles should be derived based on the mission, threat, and tradeoffs between ease of verification with fewer angles and complete coverage with more angles. For example, a very high flying vehicle may only require elevation angles of 5, 0, -5, -10, -20, -45 degrees.

ii) IR spectral bands: The band over which the IR spectral intensities (watts per steradian-micrometer) are computed as a function of the aircraft mission and threat. For example, a high flying vehicle may require a few bands in the 3 - 12 micron range (e.g., 3.0 - 5.0 and 8.0 - 12.0), while a low-flying vehicle may require more bands in the 1 - 15 micron range (e.g. 1.0 - 3.0, 3.0 - 5.0 and 8.0 - 12.0). The spectral resolution recommended is 20 cm^{-1}.

iii) Operational parameters: The operational parameters are dependent on the air vehicle mission and postulated threat. Parameters that are subject to change and will affect the IR signature must be specified in detail for each table. Examples of important operational parameters are:

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1) Altitude: affects engine and aerodynamic heating signature.

2) Throttle position: affects engine signature, and aerodynamic heating signatures.

3) Engine Material properties: affects IR emissions and reflections.

iv) All environmental parameters that are subject to change and will affect the IR signature must be specified in detail for each table. Examples of important operational parameters are:

1) Ambient temperature: affects atmospheric transmission, engine and aerodynamic signatures.

2) Ambient pressure: affects engine signature and atmospheric transmission.

3) Sun position: affects solar reflections.

4) Humidity, aerosol content, ozone: affects atmospheric transmission.

5) Cloud cover, earth surface temperature and emissivity: affects reflections and background level.

6) Visibility: affects atmospheric transmission.

Note: Temperature pressures, choosing a “standard” atmosphere often specifies humidity, and aerosol content. These atmospheric models were developed as average values over a period in certain types of regions. They probably will not reflect actual conditions in a specific locality on a particular day, as would be encountered in a test or in area mission. The Staff Meteorologists can help with these specific conditions. Section 3.2.1 defines the atmospheric conditions for the engine for temperature and pressure.

REQUIREMENT LESSONS LEARNED (A.3.6.1.2)

Having a common table format (including common conditions) with the vehicle is necessary to understand properly and make decisions regarding the IR impact to the vehicle.

Air vehicle mission determines which angles should be used. A more comprehensive specification is recommended for engines for multi-purpose applications due to the expense of obtaining the data later.

A.4.6.1.2 IR signature.

The requirements of 3.6.1.2 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.6.1.2)

The purpose of section 4.6.1.2 is to verify that the engine has not exceeded the limits set in 3.6.1.2. There are readily available instruments to measure the IR signature, but there are several problems that prevent direct verification that the IR signature limits have not been exceeded. The measured data must be processed, often extensively, to enable the desired comparison. In the past, this process has taken several months to complete. Therefore this section must not only specify how to make the measurements, but also to specify a straight-forward methodology for processing the measured data in a timely and accurate manner.

A database of propulsion system signature measurements is required for computer code validation and calibration.

IR characteristics must be measured for air vehicle integration, engine and air vehicle design tradeoffs, and combat tactics development.

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VERIFICATION GUIDANCE (A.4.6.1.2)

Past verification methods have included inspection, analysis, simulation, and test. Flight tests are needed to obtain platform signature. Analysis is needed to determine the engine contribution to the air system signature level.

The verification tests should be set up to match the operating conditions of the requirements as closely as possible. Ground test should be considered because of the ability to get more accurate, repeatable results, and to save valuable resources. Another alternative is to utilize ground or flight test measurements at the obtainable operating conditions to calibrate a computer code. The computer code then will be utilized with a set of validated input conditions to match the specification conditions exactly.

If the engine is not required to meet any observable requirements, this paragraph should be labeled “Not Applicable”.

VERIFICATION LESSONS LEARNED (A.4.6.1.2)

Because of the problems involved in IR measurements, IR signature modeling can be a key to successful verification. An IR signature prediction program, coupled with a detailed, validated input set, should facilitate the verification process. Rather than designing the verification tests specifically for every requirement value, a subset of specially selected tests can be accomplished to validate the inputs to an IR signature prediction program which could then be used to generate the values to compare directly with the requirements. This greatly simplifies the processing of the measured data, and saves test resources.

An additional reason for not using source signature as the signature limits is because of the difficulty in correcting measured signature to zero range (source). In several parts of the IR spectrum, IR transmission is zero or near zero. Source signature is obtained by dividing the measured value by the IR transmission. If the IR transmission is zero, the source signature is undefined.

Errors in measuring IR signature have been made because of background interference from the sun. This made it necessary to determine IR signature at night.

A.3.6.1.3 Smoke.

The engine shall not emit smoke trails or smoke puffs defined by ___, during any transient or steady-state operation when using any primary or alternate fuel.

REQUIREMENT RATIONALE (A.3.6.1.3)

Visible or detectable exhaust smoke trails or smoke puffs are observables and can be used to detect combat aircraft.

REQUIREMENT GUIDANCE (A.3.6.1.3)

The following should be used to tailor the specification paragraph:

Figure 25 should be used for guidance to determine the acceptable level of exhaust smoke. If a method other than figure 25 can’t be defined, then figure 25 should be included in the specification.

A smoke trail is a steady stream of carbon particulate matter emitted by the engine that is visible, and a smoke puff is a small whiff or cloud of carbon particulate matter emitted by the engine that is visible.

Background:

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The smoke parameter “nd”, should be employed where “d” is the diameter of the engine and “n” is the maximum number of engine exhaust streams through which an observer could possibly sight. The invisibility criterion graph is used in AFAPL-TR-76-64.

REQUIREMENT LESSONS LEARNED (A.3.6.1.3)

Smoke is a function of combustor primary zone pressure and is highest at intermediate, and minimal at maximum since augmentor pressure is significantly lower than intermediate primary zone pressure.

For mixed flow fan engines, the smoke number is usually much lower than for other engines. However, this does not necessarily mean that plume visibility is less since plume diameter is also a factor. The fuel control and combustor components and system interaction, or lack of interaction, has been the primary source of smoke and smoke puffing. Statistics have shown that for aircraft with visibly high smoking engines, a tenfold increase in combat aircraft losses have occurred.

A.4.6.1.3 Smoke.

The requirements of 3.6.1.3 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.6.1.3)

Smoke emission must be verified to provide values for visible signature suppression.

VERIFICATION GUIDANCE (A.4.6.1.3)

Past verification methods have included analyses, demonstrations, and tests. SAE ARP 1179 should be used as guidance.

Background:

Smoke level should be determined for at least four engine power settings including intermediate, for example, idle, 40 percent intermediate, 75 percent intermediate, and intermediate. For turboshaft engines idle, 25% maximum continuous, 75% maximum continuous, maximum continuous, intermediate and maximum should be used. Prior to sampling, the engine should be run long enough to become fully stabilized at each power setting. No post-test performance check is required. Visual or photographic means should be used to determine if smoke puffs are present during throttle transients. Fuel used should be the primary fuel or fuel per table XXIV.

VERIFICATION LESSONS LEARNED (A.4.6.1.3)

Post qualification redesign has been needed to correct visible smoke conditions which began apparently due to some subtle change in the manufacturing process of combustion systems. Smoke measurements using the principle of ARP 1179 have occasionally led to the erroneous conclusion that the exhaust gas of an engine is invisible. Ambient conditions (e.g., humidity) and viewing angle have produced a visible smoke trail from an engine considered smoke free. A good understanding of this method is needed to prevent these erroneous conclusions.

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A.3.6.1.4 Gaseous emissions.

The engine shall not produce byproduct exhaust emissions greater than the levels specified on figures 18a and 18b when using any primary or alternate fuel.

REQUIREMENT RATIONALE (A.3.6.1.4)

Emissions (carbon monoxide (CO), total hydrocarbons (C_xH_y), and oxides of nitrogen (NO_X)) represent hazards to personnel and the environment. These emissions including carbon dioxide (CO_2) and water are also observable at some conditions.

REQUIREMENT GUIDANCE (A.3.6.1.4)

The following should be used to tailor the specification paragraph:

The contractor should specify the emission limits.

Background:

Guidance is provided under the Federal Clean Air Act of 1990, as amended, for aircraft emission standards (sec. 231) conformity (sec. 176) and stationary source permitting requirements (sec. 504).

Aircraft engines mounted in test cells/ stands for testing are considered stationary sources and must comply with state and local air regulations.

Currently, there are no engine certification, exhaust emissions standards for gaseous or particulate mass emissions from military aircraft engines. However, this data in the form of emissions indexes (pounds of pollutant per thousand pounds of fuel burned) is needed in order to prepare state and local emissions inventory estimates and to comply with state and local inventory reporting requirements associated with the Federal Clean Air Act of 1990. Gaseous and particulate mass emissions inventories are used to compare air emission impacts associated with replacing older aircraft and redeploying squadrons, and are mandated under the conformity provisions of the Act.

Particulate mass emissions are fast becoming a major concern area and military engine designers should pay close attention to all technical and legislative issues as they develop.

Emission limits should be in compliance with the Federal Clean Air Act Amendments of 1990 as follows:

a. CO and HC levels must be below levels that result in an idle combustion efficiency of 99.5 percent for engines with an idle pressure ratio above 3:1.

b. CO and HC levels must be below levels that result in a combustion efficiency of 99 percent for engines with an idle pressure ratio below 3:1.

c. NO_X levels must be less than 50 percent of the uncontrolled level of figure 18a.

d. For engines using water injection, NO_X levels must be less than 25 percent of the uncontrolled level of figure 18a.

The contractor should supply the emission limits in the format of figures 18b and tabulation of pollutant emission index versus thrust or power setting.

REQUIREMENT LESSONS LEARNED (A.3.6.1.4)

Absence of gaseous and particulate mass emissions data from production engines has delayed the introduction of new aircraft into the fleet and relocation of squadrons pending data collection. The lack of this data in the past has been addressed by substituting emissions indexes from similar engines. This

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solution usually causes increased community scrutiny and costly legal delays associated with introduction and redeployment of military aircraft.

A.4.6.1.4 Gaseous emissions.

The requirements of 3.6.1.4 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.6.1.4)

Gaseous emission levels must be demonstrated to show conformance with the Clean Air Act of 1990 and limit the possibility of future threats.

VERIFICATION GUIDANCE (A.4.6.1.4)

Past verification methods have included analysis and test.

Multiple representative engines, if possible, should be tested at several power settings (i.e.; Idle, 100%, and at least two mid-power settings). Engines equipped with augmentors should also be tested at the augmentor settings.

Additional verification guidance may be found in the following:

SAE ARP 1256 and SAE ARP 1533 for test and data reduction.

a. Appendices 3 and 5 to International Civil Aviation Organization (ICAO) Annex 16, Environmental Protection, Volume II, Aircraft Engine Emissions, Second Edition, July 1993.

b. An acceptable alternative to testing every engine is described in Appendix 6 to International Civil Aviation Organization (ICAO) Annex 16, Environmental Protection, Volume II, Aircraft Engine Emissions, Second Edition, July 1993.

c. Regardless of the procedure, measurements should be made at various power settings including maximum, intermediate thrust and idle where efficiency is lowest and total hydrocarbons is highest.

For measuring gaseous emissions from engines with augmentors the following may be considered for guidance:

(1) ICAO Annex 16, Appendix 5 (Instrumentation and Measurement Techniques for Gaseous Emissions From Afterburning Gas Turbine Engines), and

(2) Procedures outlined in Air Force Report AFAPL-TR-75-52 (Development of Emissions Measurement Techniques for After Burning Turbine Engines).

The fuel used during verification engine testing should be per table XXIV.

VERIFICATION LESSONS LEARNED (A.4.6.1.4)

Past engine programs have successfully tested gaseous emissions in engine altitude test cells during performance and operability tests.

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A.3.6.1.5 Fuel streaming and vapor puffing.

The engine shall not emit visible, unburned fuel (fuel streams or vapor puffs) as defined by ___, at any power setting at or above flight idle or during any transient or steady-state operation when using any primary or alternate fuel.

REQUIREMENT RATIONALE (A.3.6.1.5)

Fuel streaming and vapor puffing is a fuel control and augmentor problem that, if visible, is an observable and can be used to detect combat aircraft.

REQUIREMENT GUIDANCE (A.3.6.1.5)

The following should be used to tailor the specification paragraph:

The Using Service and contractor determine what method should be used to determine fuel streaming at the time of contract award. If no method is available, then change the requirement to read as follows: "The engine shall not emit visible unburned fuel (fuel streams or vapor puffs) at any power setting at or above flight idle or during any transient or steady-state operation when using any primary or alternate fuel."

This requirement applies primarily, but not restricted, to augmented aircraft and may be waived, if not applicable, at the discretion of the Using Service. If this requirement is not used, put "Not Applicable" following the paragraph number.

Background:

Vapor puffing can occur during the activation of augmentation but is usually more evident during throttle chops out of augmentor operation. Special attention should be given to the upper left-hand corner of the flight envelope where vapor puffing is more likely to occur. Vapor puffing is more frequent with JP-5 and JP-8 fuels that are less volatile than JP-4 fuel.

REQUIREMENT LESSONS LEARNED (A.3.6.1.5)

There have been significant lessons learned in this area and consulting with experts should be done. Survivability experts can be found at NAVAIR Code 4.1.8 and ASC/ENA (USAF).

A.4.6.1.5 Fuel streaming and vapor puffing.

The requirements of 3.6.1.5 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.6.1.5)

Fuel streaming and vapor puffing should be verified to provide values for visible signature suppression.

VERIFICATION GUIDANCE (A.4.6.1.5)

Past verification methods have included test and analysis.

The following should be used to tailor the specification paragraph:

The contractor should specify the method or procedure to be used. If this verification paragraph is not to be used, label the paragraph number "Not Applicable".

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Background:

An analysis using an infrared radiation transmissometer for measuring the level of hydrocarbons (unburned fuel) in the flow path should be used during the tests for gaseous emissions. It can be difficult to determine the level of unburned fuel that represents a visible stream or vapor puff. Visual or photographic inspections, during flight test, are also a feasible means of detecting fuel streaming and vapor puffing.

VERIFICATION LESSONS LEARNED (A.4.6.1.5)

While not common, vapor puff testing, using an IR transmissometer, has been performed on the Engine A at NAWCAD, Trenton NJ.

A.3.6.1.6 Water vapor contrails.

The engine shall not emit water vapor contrails during any transient or steady-state operation when using any primary or alternate fuel.

REQUIREMENT RATIONALE (A.3.6.1.6)

Contrails are observables and can be used to visually detect low observable designed aircraft at extremely long ranges.

REQUIREMENT GUIDANCE (A.3.6.1.6)

The following should be used to tailor the specification paragraph:

This requirement applies to low observable aircraft and may be waived if not applicable at the discretion of the Using Service. If this requirement is not used, label the paragraph number “Not Applicable”. This requirement is normally not applicable to turboshaft (helicopters) engines since normal operating altitudes are too low to generate contrails. Requirements should focus on mission scenarios where contrails are unacceptable.

REQUIREMENT LESSONS LEARNED (A.3.6.1.6)

Contrails have been visible as low as 14,000 feet.

A.4.6.1.6 Water vapor contrails.

The requirements of 3.6.1.6 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.6.1.6)

The engine’s ability to suppress water vapor contrails must be verified.

VERIFICATION GUIDANCE (A.4.6.1.6)

Past verification methods have included analyses and tests.

The following should be used to tailor the specification paragraph:

The contractor should specify the test method or procedure to be used. If this verification paragraph is not to be used, label the paragraph number “Not Applicable”.

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VERIFICATION LESSONS LEARNED (A.4.6.1.6)

Water vapor contrails have not been successfully created nor suppressed during engine testing in test cells.

A.3.6.1.7 Radar cross-section.

The engine shall not exceed the Radar Cross Section (RCS) signature requirements of tables XXII and XXIII. This requirement is not applicable to turboprop engines with no externally visible rotating parts.

REQUIREMENT RATIONALE (A.3.6.1.7)

The above paragraph provides the RCS requirement for the engine. To meet the mission requirements of the air vehicle, the signature of the engine may need to be specified.

REQUIREMENT GUIDANCE (A.3.6.1.7)

The following should be used to tailor the specification paragraph:

The purpose of 3.6.1.7 is to set the requirements on the propulsion system RCS levels. The requirements should be tied to the Air Vehicle Specification. If the engine is provided under a separate contract as GFE, then the engine system signature specification from the government to the engine system contractor is required. The government-issued specification should be consistent with the total weapon system specification. The RCS levels should be derived for engine operating conditions, specified in the ICD, to ensure that the signature is properly specified.

If the engine is not required to meet any observable requirements, this paragraph should be labeled “Not Applicable”.

Table format:

The azimuth sectors of most interest are nose, tail, and broadside. The sectors may be further subdivided into elevation sectors or additional azimuth sectors. The number of angular sectors selected depends on the platform shape as well as the mission requirements.

The threats to the vehicle may operate over a wide range of frequencies and at different polarization. A separate RCS requirement table for different polarization and frequency bands may be required.

The entries in the table can be specified for the required polarization (i.e., one table per polarization) and for a group of N discrete frequencies (i.e., N tables required) or as a statistical average over certain band of frequencies. Discrete frequencies are typically chosen as the mid band frequency for all applicable bands but more than one frequency may be used in each band. Statistical frequency averages are selected over the band of interest (e.g.; C-band, X-band, etc.) and include more information about the behavior of the target as a function of frequency.

Specifications within the defined angular sectors are typically provided in terms of some angular statistical averaging process of the raw data at discrete frequencies or the average of the raw data over multiple frequencies. Requirements may be given in terms of estimates of the first and second moments of the sector data. Examples of an estimate of the first moment are median, arithmetic mean, or a geometric mean. Geometric mean (or averaging in decibels) is a bias estimator of the first moment in that the estimate is always less to or equal than the actual mean. One particular estimator is not always best for all cases. Table XXII provides an example of an estimate of the first moment at a given frequency and polarization. Estimates of the second moments can be specified in conjunction with the first. Examples include Pcum 90, Pcum 95, or standard deviation. An alternative to an estimate of the second moment is a spike width criterion within a sector. Table XXIII provides an example of an estimate of the second

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moment at a given frequency and polarization. The number of tables required depends on the number of frequencies, polarization, and statistical measures used.

REQUIREMENT LESSONS LEARNED (A.3.6.1.7)

Radar guided threats have demonstrated high kill probability against aircraft with high RCS signature.

Make sure the units actually represent the physics of the situation. There have been significant lessons learned in this area and consulting with experts should be done. Survivability experts may be found at NAVAIR Code 4.1.8 and ASC/ENA (USAF).

A.4.6.1.7 Radar cross-section.

The requirements of 3.6.1.7 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.6.1.7)

RCS characteristics must be measured for air vehicle integration, engine and air vehicle design tradeoffs and combat tactics development. Verification of the air vehicle RCS is needed to ensure that the RCS falls within specified signature requirements.

VERIFICATION GUIDANCE (A.4.6.1.7)

Past verification methods have included analysis, static testing, and dynamic testing. Dynamic testing is usually not appropriate for engine specification compliance. Static RCS tests must be conducted at a facility that has a low RCS background at the angle, polarization and frequencies of interest.

VERIFICATION LESSONS LEARNED (A.4.6.1.7)

RCS measurement on engines and engines with aircraft inlets installed have shown that math models do not adequately predict actual measurements from hardware. There have been significant lessons learned in this area and consulting with experts should be done. Survivability experts may be found at NAVAIR Code 4.1.8 and ASC/ENA (USAF).

A.3.6.1.8 Radar absorbent materials and coatings.

If Radar Absorbent Materials (RAM) or radar absorbent coatings are used, the ingestion requirements of 3.3.1.3 and 3.3.2.4 combined shall not deteriorate the RCS signature level by more than __ dB.

REQUIREMENT RATIONALE (A.3.6.1.8)

Sand and salt ingestion can change the surface characteristics of some radar absorbent materials and coatings and the engine RCS signature levels.

REQUIREMENT GUIDANCE (A.3.6.1.8)

The following should be used to tailor the specification paragraph:

The maximum allowable change in RCS signature due to environmental effects should be specified and should be compatible with the air system specification.

REQUIREMENT LESSONS LEARNED (A.3.6.1.8)

Sand ingestion has shown that erosion of coating can change RCS signatures.

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A.4.6.1.8 Radar absorbent materials and coatings.

The requirements of 3.6.1.8 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.6.1.8)

The change in RCS signature levels can be determined from the engines used in the salt corrosion and sand ingestion tests. These engines must be fitted with all the RAM and RAM coatings specified for production. This will show the cumulative effect of sand and salt environments on changing RCS signature levels.

VERIFICATION GUIDANCE (A.4.6.1.8)

If a test is used, the test should be with engines fitted with all the equipment and devices specified for production, and that have satisfactorily completed the sand ingestion and corrosion tests of 4.3.2.4 and 4.3.1.3, respectively.

VERIFICATION LESSONS LEARNED (A.4.6.1.8)

Analysis and computer model simulation techniques have not been developed to adequately substitute for engine test results on effects of sand ingestion and corrosion on RCS signatures.

A.3.6.2 Vulnerability.

A.4.6.2 Vulnerability.

A.3.6.2.1 Ballistic weapons.

The engine, including externally mounted components, shall tolerate ballistic weapon threats as follows: _____.

REQUIREMENT RATIONALE (A.3.6.2.1)

The engine must be capable of withstanding a combat hostile environment defined for the weapon system.

REQUIREMENT GUIDANCE (A.3.6.2.1)

The following should be used to tailor the specification paragraph:

The Using Service should establish the ballistic requirements. Ballistic requirements should be defined based on the required performance after ballistic damage has occurred (i.e., for smaller threats, (up through 14.5 mm API) the requirement would be: “The engine shall be capable of producing (x) % power for (y) minutes after damage.” The power level and duration should be established by the Using Service. For larger threats, (include 23 mm and 30 mm API) the requirement would be “For all ballistic damage, excluding a direct hit on a disk or mainshaft, the engine shall not suffer an uncontained, instantaneous catastrophic failure. The engine shall be capable of being shut down following ballistic damage.”)

The ballistic requirements should include such things as threat weapon, kill type, penetration condition (velocity, direction, fragment weight), engine operating condition, effect on engine operation, etc. Fuel ingestion tolerance, loss of lubrication power and duration requirements, lube level sensing or lube loss rate sensing, and catastrophic failure containment should be specified in this paragraph or included by cross-reference to other paragraphs in this specification.

The engine design should incorporate enhancement features that reduce vulnerability. Such features include component redundancy, component location, component shielding, suppression of active and

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passive damage and minimizing vulnerable components, and battle damage repair. These survivability enhancement features should be incorporated during the early stages of development. Engine requirements allocated from an air vehicle system specification should be included in the engine model specification.

Consideration of engine components toughness, ductility, and rupture resistance is vital to combat damage resilience. Attention to lubrication, fuel and control systems has a large effect on propulsion system overall survivability. Survivability biased control algorithms, to provide proper engine control after combat damage, incorporated into FADEC engine controls should be considered.

The location and packaging of engine components and externals, to minimize a vulnerable area, should also consider the engine's maintainability requirements. Accessibility and removal of externals should be considered when positioning these parts.

REQUIREMENT LESSONS LEARNED (A.3.6.2.1)

On one particular aircraft, combat damage analysis, all hits received on returned damaged aircraft were plotted on a single outline of the aircraft for the purpose of determining the likelihood of hitting various parts of the aircraft. As an accidental result researchers noticed that one particular rectangular area in the belly of the fuselage toward the aft edge of the wings showed no hits on returned aircraft. On further examination they discovered that the blank area on the diagram corresponded with a bay on the aircraft which contained fuel pumps and fuel controls. It took dozens of hits during combat sorties and a considerable number of lost aircraft and crews to discover a correctable design flaw.

Some external components, controls, and externals are redundant to the engine and have "back up" redundant features. Redundant components, controls, and externals have prevented penetration of the main engine by absorbing the impact energy of incoming ballistic threats. Engine J has experienced this and allowed the engine to continue operation for the next mission, after replacement of the external component or part.

In the mid-1980's Engine J altitude test had an incident where a Digital Electronic Control (DEC) was "over-temperatured" and damaged when an insufficiently torqued combustor igniter plug became loose. This caused "hot gases" from the combustor to impinge upon the DEC chassis and cause the DEC to become inoperable. The engine transferred control to the hydromechanical back-up control and continued to operate satisfactorily to allow for a safe landing. If the back-up control had been a redundant "DEC channel", inside the same control chassis, engine control would have been lost and probably the aircraft also (i.e., this was a single-engine application). This emphasizes the importance of designing a secondary channel DEC, if it is designated as the back-up control, in a separate box and location from the first.

A.4.6.2.1 Ballistic weapons.

The requirements of 3.6.2.1 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.6.2.1)

Ballistic damage effects are complex and dynamic and must be verified through testing, unless applicable experimental results are available.

VERIFICATION GUIDANCE (A.4.6.2.1)

As an option, the paragraph may require tests to be conducted per the survivability program and plan as called out by MIL-HDBK-2069 or MIL-HDBK-1799.

The following should be used to tailor the specification paragraph:

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Past verification methods have included a combination of analyses, transfer of data from other programs, and ballistic testing of replicas, components, systems, major assemblies, and running engines under conditions that simulate realistic combat and Live Fire Test and Verification (LFT&V).

The verification should be structured to provide design information at times supportive of program design milestones, verify achievement of specification requirements, and verify the actual engine ballistic hardness achieved at a measure of expected combat performance. The test verification should be tailored to support intended present, or future use of the engine in systems covered under the Live Fire Test Law (Title 10 USC 2366) and as discussed by DODI Instruction 5000.2, "Operation of the Defense Acquisition System" signed, 12 May 2003 in E5.4 T&E Planning and Table E3.T2, entitled "Regulatory Information Requirements".

Background:

Tests to provide design information should also be conducted where the experimental base is lacking. Live Fire Test and Verification (LFT&V) is a process of analyses, data gathering, and live fire test that begins as early as possible in the acquisition cycle preferable during demonstration and validation to gather design data and reduce risk, and proceeds during SDD and ISR to verify achievement of specification compliance and to measure expected combat performance under realistic conditions. During SDD it is desirable to test before the end of IFR to allow enough lead-time for redesign and again after the hardware configuration has been established. After hardware configuration (after IFR) has been established, engines dedicated for live fire testing should be selected from those having completed development or flight testing to maximize use of development hardware.

VERIFICATION LESSONS LEARNED (A.4.6.2.1)

Prior ballistic testing of aircraft and engines has provided valuable design feedback to development programs resulting in improved combat effective systems.

Experience has shown that a combination of ballistic test data and vulnerability analysis is the most accurate and economical way to determine engine ballistic tolerance. Engines that have been tested include the Aircraft F and Aircraft H.

A.3.6.2.1.1 Static structure.

No case, frame or gas pressure loaded non-rotating component, except airfoils, shall collapse or rupture when subject to its maximum operating pressure and a round hole of ____ diameter placed anywhere on the structure.

REQUIREMENT RATIONALE (A.3.6.2.1.1)

It is not practical to build engine cases and static structures with the capability to repel bullets and warhead fragments; in fact, increasing demands for weight reduction dictate construction that is increasingly vulnerable to penetration. Engines will seldom fail from mere penetration and the resulting overboard leakage. However, it is critical that the engine response to penetration be graceful; i.e, it should not cease operation or release parts, even internally, which could augment or cascade the damage. In addition, the engine should remain firmly in its mountings, remain responsive to power demands (although degraded) and if required, be capable of being shutdown by the aircrew.

"Static structure" includes internal non-rotating parts, in addition to external cases. Engine failure could result from removal of load-carrying material, stress concentration, or unintended airflow through the hole. The requirement is not intended to apply to bearings, lube and fuel lines, or holes aligned such that the entire load path is severed. Cases should have enough ductility or thickness to tolerate without pressure-loading caused rupture, buckling, or crack propagation, holes that nominally produce elastic stress concentration factors of 3.0.

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REQUIREMENT GUIDANCE (A.3.6.2.1.1)

The following should be used to tailor the specification paragraph:

The hole diameter should not be less than that corresponding to a 14.5 mm armor-piercing round, which makes a minimum 15.2 mm (0.6 inch) diameter hole, if not yawed to the direction of flight. The 23-mm API makes a minimum 25.4-mm (1-inch) diameter hole. Warhead fragments and high-explosive projectiles can make holes generally up to the equivalent of 25.4-mm (1-inch) diameter. In general, the hole should be assumed to have its axis perpendicular to the surface it penetrates. The current projectile threat is 30 mm.

Background:

Provided that it is within the allowances of the structural integrity requirements, stationary airfoils should be made of ductile materials that will allow the airfoil to be bent 90° through the airfoil chord axis at the root with a mean bend radius of five times the nominal airfoil root maximum thickness (disregarding any root fillet) without separating or fracturing. Naval Air Warfare Center China Lake has provided the input for this requirement.

REQUIREMENT LESSONS LEARNED (A.3.6.2.1.1)

Testing at NAWCWD China Lake has indicated that engines generally continue to produce acceptable performance despite multiple hits by ballistic threats, if the original damage is not amplified by brittleness in the structure which allows release of engine structural debris into the flow path. Most engines tested to date use vane and blade materials sufficiently ductile that, although ballistic threats and foreign objects mangle and destroy the aerodynamic function, the damaged airfoils usually do not release and cause cascading failures.

A.4.6.2.1.1 Static structure.

The requirements of 3.6.2.1.1 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.6.2.1.1)

Analysis and demonstration are required to determine the vulnerability of a pressure-loaded component when it is pierced.

VERIFICATION GUIDANCE (A.4.6.2.1.1)

Past verification methods have included analyses, demonstration, and test.

The following should be used to tailor the specification paragraph:

Analyses may be presented to satisfy this requirement, except where confidence in analytical tools is lacking, in which case, ballistic firings, or static tests of parts with drilled holes, should be required as negotiated with the Using Service.

Background:

Tests should be per 4.6.2.1.

VERIFICATION LESSONS LEARNED (A.4.6.2.1.1)

There has been experience in testing and in analysis to establish tolerance of engines to ballistic damage.

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A.3.6.2.2 Nuclear weapons effects.

The engine shall have the following operational capabilities and balanced hardness, in a nuclear environment that includes blast and gust, dust, thermal, transient radiation effects, and electromagnetic pulse:

REQUIREMENT RATIONALE (A.3.6.2.2)

Future aeronautical vehicles may have to operate in a strategic or tactical nuclear warfare environment. Engines must be optimized to permit mission completion within that environment.

REQUIREMENT GUIDANCE (A.3.6.2.2)

The following should be used to tailor the specification paragraph:

The engine operating capabilities and balanced hardness for a particular nuclear environment must be specified.

Nuclear requirements for an engine should be allocated from requirements derived from the air vehicle mission threat encounter conditions or from generic air vehicle mission threat encounter conditions for the engine intended class or type of aircraft. All potential missions in which exposure to enemy or friendly Nuclear Weapon Effects is expected must be considered including all expected yield and detonation conditions. For example, an air vehicle which has a very high altitude mission profile or which may have to provide close air support in a tactical nuclear environment would very likely be exposed to gamma missions or disruptions of electronic signals. Aircraft for which base escape is an issue or must support tactical operations at low altitude would have to deal with nuclear detonation generated dust. Nearly all aircraft expected to operate in a nuclear environment will have to deal with some degree of blast, gust, flash and thermal effects. All aircraft with missions that are mission essential to the Using Service will have to resist electromagnetic pulse. Specifications for engines should match expected air vehicle requirements and reduce or eliminate single point weaknesses. The Using Service must ensure that the ICD includes the air vehicle mission threat encounter conditions and air vehicle degree of protection provided, for each user's mission requirements and environments. The most stringent users mission requirements should dictate the final requirement, if practical.

Nuclear contamination includes only radioactive particles and induced gamma activity resulting from nuclear detonation. It does not include primary effects such as blast, initial radiation, thermal radiation, and electromagnetic pulse.

REQUIREMENT LESSONS LEARNED (A.3.6.2.2)

Nuclear hardening programs on many past and present aircraft indicate that careful tailoring of the requirement is crucial. Analysis concerning mission threat encounter, mission completion, and parametric trade studies which calculate the cost or benefits and performance or benefits in relation to mission effectiveness are an absolute must for specification requirement derivation. Dust and ash raised by nuclear weapons will have an extremely deleterious effect on engines equivalent to volcanic ash, if ingested. Engines intended for such mission encounters must be designed to withstand the environment.

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A.4.6.2.2 Nuclear weapons effects.

The requirement of 3.6.2.2 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.6.2.2)

Verification is required to accurately determine the blast and gust, dust, thermal, transient radiation, and EMP characteristics of the engine.

VERIFICATION GUIDANCE (A.4.6.2.2)

Past verification methods have included the analyses, demonstrations, and tests.

VERIFICATION LESSONS LEARNED (A.4.6.2.2)

None.

A.3.6.2.3 Chemical and biological agent effects.

The engine shall survive contamination from chemical and biological agents, and decontamination without losing mission capability as follows: (a) . The engine shall be capable of being decontaminated as defined in FM 3-5, NBC decontamination, to reduce the hazard to a level where personnel operating under Mission Orientated Protective Posture (MOPP) level 0 suffer no physical impairment or injury within (b) of completing the procedures and decontamination agents defined by (c) . Engine materials shall be compatible with the decontamination agents. The engine shall be capable of being operated and maintained by personnel in nuclear, biological and chemical (NBC) MOPP-4 protective ensembles.

REQUIREMENT RATIONALE (A.3.6.2.3)

Chemical and biological weapons are widely available and have been used in recent warfare so the engine must be capable of sustaining combat in a chemical and biological environment.

REQUIREMENT GUIDANCE (A.3.6.2.3)

The following should be used to tailor the specification paragraph:

(a): The Using Service should specify engine mission capabilities in a chemical and biological contamination environment.

(b): Thirty minutes has been used as the requirement on other air vehicles.

(c): The contractor should specify the decontamination procedures and agents when standard government chemical-biological ensembles are being worn by the decontamination personnel.

Decontamination should be per Army Regulation (AR) 70-71.

The engine should withstand the chemical and biological environments defined by the applicable System Threat Report (STR).

The characteristics of NBC Contamination Survivability (NBCCS) in addition to decontaminatability addressed in the requirement are:

a. Hardness. The engine should be capable of full, unimpaired operation after exposure to NBC contamination and the procedures and agents required to decontaminate the system. Surfaces should be constructed of materials that are resistant to NBC agents and decontaminants. The contractor should be responsible for providing data proving that selected materials are resistant to NBC agent and decontaminants. They can acquire the required data through either the Chemical-Biological Information

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Analysis Center or through material testing as required. The engine design should minimize entrapment of agents and decontaminants. Design guidance is available from MIL-HDBK-784.

Compatibility. The engine should be capable of being operated, maintained, and resupplied by persons operating at the MOPP Level 4. Equipment should be designed to minimize performance degradation under NBC conditions.

Background:

Much of the information regarding chemical warfare requirements is classified. The materials to which the engine may be exposed during decontamination operations are as follows:

a. MIL-D-50030 - used to neutralize chemical warfare agents on surfaces.

b. MIL-DTL-12468 - used to neutralize chemical warfare agents on surfaces.

c. Ethylene Oxide - used to neutralize bacteriological agents.

d. Water - incorporated as a matrixing-dissolving fluid and as steam.

e. Isopropyl Alcohol - used as a solvent to wash away contaminants.

f. Additional types of decontamination fluids may be used as the threat changes. The possibilities should be researched when the specification is prepared.

All three services should utilize the U.S. Army Nuclear and Chemical Agency guidance in letter DAMO-SW, subject: Department of Army Approved Quantitative Contamination Survivability Criteria, dated 24 Oct 91.

REQUIREMENT LESSONS LEARNED (A.3.6.2.3)

NBC contamination can have direct effects upon equipment items by degrading critical properties (e.g.; physical, chemical, mechanical, electrical, optical, or thermal) of the material's construction. In addition, materials or procedures used to decontaminate equipment items may also reduce component and system capabilities. Indirect effects may arise if it is not possible to decontaminate to levels that are not hazardous to unprotected personnel. Decontamination of a surface may be hindered because materials absorb the contaminant or the contaminant becomes trapped in cracks or crevices. The heat associated with normal engine operation in the air vehicle should substantially reduce the degree of residual contamination.

A.4.6.2.3 Chemical and biological agent effects.

The requirements of 3.6.2.3 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.6.2.3)

NBC decontamination must be verified.

VERIFICATION GUIDANCE (A.4.6.2.3)

Past verification methods have included analysis and demonstrations.

Material compatibility should be shown by coupon tests or existing data, as applicable. Prior to any exposure to agents or decontaminants, the critical functions and properties of the engine should be demonstrated to measure engine performance and material properties after the decontamination.

Background:

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Decontamination procedures may be conducted concurrent with the maintainability demonstration of 4.5.2.1. Long-term effects should be determined by the compatibility tests. A test should be conducted to evaluate engine compliance to the AR 70-71, or equivalent, requirement. All chemical agent resistant coatings should be verified by documentation.

VERIFICATION LESSONS LEARNED (A.4.6.2.3)

None.

A.3.6.2.4 Fuel ingestion effects.

A.4.6.2.4 Fuel ingestion effects.

A.3.6.2.4.1 Fuel ingestion - steady-flow.

The engine shall be capable of withstanding the effects of steady flow fuel ingestion via the inlet duct with the fuel and plume characteristics (a) and initial flow rates as follows. The engine shall not suffer more than (b) % thrust loss for ingestant flow rates up to (c) kg/hr (PPH) for 30 seconds, and shall be capable of operating for 30 minutes at the residual thrust level, at (d) flight condition. The swirl displacement angles that would be experienced by entrained fuel in core and bypass flow paths shall be described by the contractor for intermediate and flight idle throttle settings, and the windmill state, at (d) flight condition.

REQUIREMENT RATIONALE (A.3.6.2.4.1)

Engines vary in their ability to tolerate the ingestion of steady fuel leakage through a wound in duct walls, their tolerance subject to functions of engine size and bypass ratio. For small engines, and low bypass ratios, wounds resulting from small fire can cause engine case burn throughs with subsequent damage to flight-critical air vehicle-mounted components and loss of the aircraft. Swirl angles are required to predict the locations of engine case burn throughs and where air vehicle thermal protection may be required.

REQUIREMENT GUIDANCE (A.3.6.2.4.1)

The following should be used to tailor the specification paragraph:

The plume (a) is most easily and accurately described as that which results from the flow of fuel out of a typical wound in the nominal inlet duct that is a nominal distance forward of the engine face; the flow being driven by the expected head and ullage pressure at the flight condition (d). It is conceivable that a fuel tank with an entrance wound can experience ram pressure that will force fuel through an exit wound into the inlet duct.

Instead of requiring no performance degradation, a loss of (b) % of the normally available net thrust is allowed for the ingestant mass flows specified. (A (b) value in the range of 0 to 50% is suggested; higher losses imply that the engine is unstable, unless the loss results from deep throttling.) The flow rate (c) is selected to correspond to the threat(s) described in the vulnerability allotment for the aircraft, or is selected by the Using Service according to its needs. The flight condition (d) is one or more of the conditions at which fuel ingestion vulnerability is of concern, such as the nominal ordnance delivery speed, altitude, and throttle condition for an attack mission.

The flight idle and windmilling conditions are needed due to a possible burn-through when there is steady fuel ingestion as when the engine flames out due to a hit, then windmills and is restarted at flight idle.

Background:

Given predicted air vehicle leakage of fuel into the engine inlet, the engine contractor should define the swirl displacement angles for core and fan flow paths that would be experienced by entrained fuel.

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Predictions of burn through zones are straightforward when the swirl displacements of both the fan-duct-entrained ingestant and core-captured ingestant are known for various throttle settings. For further information contact NAWCWPNS, China Lake for fuel ingestion recommendations.

Engine and air vehicle vulnerability to inlet fuel ingestion effects and adverse engine reaction to fuel ingestion can be minimized by contractor design methods. The air vehicle developer bears the responsibility of protecting the engine from fuel leakage to the extent possible by eliminating common tank and duct walls, and where this cannot be done, providing the lowest possible fuel leakage rates upon penetration of the common wall by specified threats. This paragraph requires the engine contractor to specify techniques to increase the engine's tolerance to fuel ingestion caused by combat damage, tank failures, and aerial refueling accidents.

REQUIREMENT LESSONS LEARNED (A.3.6.2.4.1)

Analyses and demonstrations have indicated that engine control activity affected by sufficiently intelligent control systems can mitigate the effects of inlet fuel ingestion. Swirl angles are easily measured by injecting dye at the engine face and observing its track through the engine.

The Joint Live Fire (JLF) program has completed numerous realistic tests to assess the vulnerability of U.S. and foreign aircraft to inlet fuel ingestion. Experimental determinations of swirl displacements and burn through zones during JLFT have led to recommendations for selective application of thermal protection materials.

A.4.6.2.4.1 Fuel ingestion - steady-flow.

The requirements of 3.6.2.4.1 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.6.2.4.1)

The engine's ability to tolerate fuel ingestion must be verified.

VERIFICATION GUIDANCE (A.4.6.2.4.1)

Past verification methods have included test, unless analysis acceptable to the Using Service indicates the requirement could be met with large margins.

Fuel should be ingested at flow rates in steps of (a) with cooling periods of (b) between injections. Each fuel ingestion condition should be conducted with the engine operating at (c). The engine should be inspected for damage at periodic intervals. The fuel injection mechanism should simulate air vehicle fuel system behavior and characteristics.

The following should be used to tailor the preceding paragraph:

(a), (b), (c): The fuel flow rate step increase and cooling period, and engine operating requirements during the test, should be established by the Survivability experts within the Using Service.

This is a potentially destructive test, in that structural overheating and thermal distortions will likely occur. No other reliable method of determining actual tolerance has been developed. It is important that the injection mechanism allow the ingestant flow rate to adjust to engine reactions to demonstrate effectiveness of the mitigation techniques.

One result of steady-flow fuel ingestion is burn through of the engine cases upon ignition of the ingested fuel. It is important to be able to predict and test the locations of these burn through zones so that thermal protection materials may be applied at only the vulnerable locations in the engine compartment (the materials are heavy and expensive). Testing may be the only viable means for accurately verifying the tolerance of an engine to fuel ingestion.

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Fuel ingestion requirements and testing should be conducted in conjunction with aircraft vulnerability testing. Engine fuel ingestion requirements should be based on the expected fuel leakage rates from the common wall fuel cells when subject to anticipated ballistic threats. Inlet-fuel cell wall testing should be conducted early in the development program to verify predicted leakage rate levels. Engine ingestion threshold levels should be defined by test. If the engine is unable to ingest flow rates comparable to the inlet leakage rates, the air vehicle manufacturer should investigate techniques available to minimize the fuel leakage rate due to ballistic damage. Information about techniques and materials to reduce fuel cell leakage rates can be found in NAWCWPNS TP 8076 titled “Ballistic Tests of Structural Concepts to Mitigate Damage and Reduce Fuel Leakage from Aircraft Fan-Duct Walls.” Additional information regarding fuel ingestion testing techniques can be found in NAWCWPNS TP 8235 titled “Specification and Verification Test Procedures To Improve Fuel Ingestion Tolerance of Turbine Engines”.

For further information regarding fuel ingestion, contact NAVAIR Code 4.1.8.

Swirl angles are easily measured by injecting dye at the engine face and observing its track through the engine. Some compressor and turbine design codes output swirl displacement, but these may be inaccurate or not capable of accounting for swirl in the combustor, fan duct, or exhaust ducting, so testing should be done.

VERIFICATION LESSONS LEARNED (A.4.6.2.4.1)

Testing of the Engines B, P, J, and certain foreign engines indicate that air vehicle and engine vulnerability to fuel ingestion can be significantly reduced with minor design changes (with minimum penalty in weight, cost, or reliability). In many instances, these weaknesses could have been identified early in the engine development cycle and fixed at minimal cost to the program. On Engine J, high-pressure compressor blade failures caused by fuel ingestion resulted in titanium fires. Design changes were implemented late during engine development at extremely high cost.

The Joint Live Fire Program sponsored by OSD (OT&E) (Live Fire Test) has suggested injection techniques that replicate aircraft fuel system behavior.

A.3.6.2.4.1.1 Fuel ingestion - cooling air contamination.

Contamination of engine internal cooling air from fuel ingestion shall not degrade engine life nor engine parts life below that specified in 3.4.1.1.

REQUIREMENT RATIONALE (A.3.6.2.4.1.1)

Turbine or bearing coolant air extracted from the compressor, if contaminated by ingested fuel, can accelerate engine failure by releasing heat at locations where cooling is needed. Contaminated cooling air can cause failure of turbine blades by releasing heat where the system was designed to transfer heat away from the turbine section. The desirability of keeping fuel out of cockpit conditioning air and other engine bleed air destinations is obvious.

REQUIREMENT GUIDANCE (A.3.6.2.4.1.1)

The following should be used to tailor the specification paragraph:

This requirement may be waived if the Using Service determines that inlet fuel ingestion is not a threat.

Background:

Bleed air extraction locations should be determined very early in the design stage. Changes during qualification testing could be very difficult.

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REQUIREMENT LESSONS LEARNED (A.3.6.2.4.1.1)

None.

A.4.6.2.4.1.1 Fuel ingestion - cooling air contamination.

The requirements of 3.6.2.4.1.1 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.6.2.4.1.1)

Cooling air contamination from fuel ingestion must be verified.

VERIFICATION GUIDANCE (A.4.6.2.4.1.1)

Past verification methods have included inspection and analysis.

Compliance can be determined by reviewing the engine design.

Background:

The cooling air requirements of 3.6.2.4.1.1 should be verified by inspection of the engine drawings.

VERIFICATION LESSONS LEARNED (A.4.6.2.4.1.1)

Ingested fuel has been generally concentrated at the fan case and core case walls by centrifuging in the fan and compressor. This has been clearly responsible for more rapid thermal erosion and failure at turbine nozzle outboard ends than at inboard ends, as seen in ingestion tolerance testing of engines B and J. In some tests of engines that extract coolant air at the core case, as opposed to the rotor hub, the failure of the turbines was suspected to have been accelerated by contamination of the coolant air by ingested fuel. Drawing uncontaminated cooling air from the hub, instead of the fan and compressor blade tip, (where the ingested fuel tends to concentrate) has allowed the cooling air system to perform as designed even with fuel ingestion present. Engines Q and A are examples of designs that successfully extracted required bleed air from the hub, and passed the JLFT.

A.3.6.2.4.2 Fuel ingestion - transient (“quick dump”).

The engine shall withstand ingestion of transient fuel with no more than _(a)__% permanent thrust loss under operating conditions _(b)_. The fuel shall be injected from the duct wall into the center of the inlet duct at a flow rate sufficient to carburet to twice the stoichiometric ratio the complete inlet air column for a length corresponding to the amount of air contained between engine face and exhaust nozzle throat. Engine instability and total loss of thrust is permitted during the event, but normal engine function shall be reestablished within __(c)__ seconds of the ingestion event with no pilot input. Normal throttle agility may be degraded, but the thrust must be capable of modulation down to flight idle with gentle throttle input. The engine shall withstand ingestion of transient “quick dump” fuel that may occur during aerial refueling operations.

REQUIREMENT RATIONALE (A.3.6.2.4.2)

Impact of small arms fire on tank and duct common walls can carburet inlet duct air in what is known as “quick-dump” fuel ingestion. The event happens so fast that engine control activity probably cannot be initiated quickly enough to provide mitigation. However, when the stoichiometric amount of fuel has been mixed in the inlet duct, the potential for damage is maximized.

REQUIREMENT GUIDANCE (A.3.6.2.4.2)

The following should be used to tailor the specification paragraph:

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The allowable thrust loss (a) should be in the range 0 to 50%, depending on the needs of the Using Service. More than 50% loss suggests severe engine damage and very short life expectancy. No thrust loss suggests complete invulnerability to quick-dump events, a desirable ultimate capability, and one that is feasible, if the Using Service is willing to trade the benefits of reduced vulnerability for an increase in weight. The operating condition (b) should be the flight condition and throttle setting at which vulnerability is of greatest concern. The most vulnerable flight condition for quick-dump events is in the lower right corner of the flight envelope (high subsonic, low altitude), cold day, intermediate throttle. This is the condition at which the engine internal pressures are greatest, and at which the greatest violence can be expected during the quick-dump. Although the flight envelope may extend well past Mach 1.0 at sea level, most engines' control systems limit engine pressures to those which are attained at speeds of Mach 0.8 or lower at sea level. Engine A, for example, begins P_3-limiting at Mach 0.67 at sea level standard day. So, the worst-case test condition is not difficult to generate in a test facility. The duration for which the engine is required to operate after the ingestion event should be stated with the operating condition. The recovery time (c) depends on the needs of the Using Service; modern engines should recover within 5 seconds, if they recover at all. One minute of stable operation after the injection should establish the outcome.

If the fuel spilled and ingested during aerial refueling is more than the amount specified above the contractor should specify the amount expected to be ingested during aerial refueling.

Background:

The values specified above should be selected based on the operating envelope and control scheme. The greatest potential for engine damage occurs at the highest air density condition in the compression machinery. The most recent correlations between threat types and quick-dump parameters, and the latest thinking on engine-affected mitigation, using the current technology is necessary.

REQUIREMENT LESSONS LEARNED (A.3.6.2.4.2)

None.

A.4.6.2.4.2 Fuel ingestion - transient (“quick dump”).

The requirement of 3.6.2.4.2 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.6.2.4.2)

Verification that the engine can meet the “quick-dump” fuel ingestion requirements is necessary.

VERIFICATION GUIDANCE (A.4.6.2.4.2)

Past verification methods have included analysis and test. In the event of failure to demonstrate compliance, retest should not be necessary if the contractor executes design changes that in the opinion of the Using Service would allow satisfactory completion.

Background:

An inlet air temperature and pressure of 15°C (59°F) and 154.4 kPa (22.4 psia) correspond to flight at Mach 0.8 sea level, cold day, that is a typical stores-delivery condition where threat impact is likely, and it is a condition at which engine pressures (and damage potential) are near maximum values.

VERIFICATION LESSONS LEARNED (A.4.6.2.4.2)

Most engines exhibit damage modes and severity that suggest that minor improvements could significantly reduce vulnerability to quick-dump fuel ingestion. The Joint Live Fire Program tested Engines B, J, and P recently for fuel ingestion tolerance; and Engines M, L, R, and certain foreign

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engines have been tested under other sponsorship. The Engine B test report is available from the office of WL/FIVS, Wright-Patterson AFB. Engine J report was written at NWC, China Lake. In general, these engines exhibit damage which could be precluded by minor redesign and strengthen certain critical parts/critical safety items such as IGV actuation linkages and afterburner liners. Engines B and J recovered very quickly from the ingestion event and continued to operate stably with minor thrust degradation for substantial periods. Modern engines already have control logic to provide good surge recovery and relight if necessary. Advanced control systems being developed will have some quick-dump fuel ingestion capability, although thrust may be degraded.

A.3.7 Subsystems.

A.4.7 Subsystems.

A.3.7.1 Anti-icing and de-icing system.

Within ____ seconds following termination of the anti-icing or de-icing system, engine performance shall return to the level prior to system actuation.

The anti-icing or de-icing system shall allow (no/no detrimental) accumulation of ice on any engine part while operating under the icing condition specified in 3.3.1.4. Operation of the anti-icing or de-icing system throughout the operating envelope shall not damage the engine. The performance and operability effects of the anti-icing or de-icing system operation in a non-icing environment shall be specified. The anti-icing or de-icing system shall provide a signal for indicating that the anti-icing or de-icing system is operating. The fail-safe mode for the ice protection system shall allow safe return to base in icing conditions and non-icing conditions without safety of flight damage to the engine. If an inlet particle separator (IPS) is used, ice formation on the IPS shall not be permitted. Limited ice accumulation shall be permitted in the IPS scavenge system if it does not affect IPS operation.

REQUIREMENT RATIONALE (A.3.7.1)

An anti-icing or de-icing system must be required since accumulation of inlet ice affects safety of flight, can cause loss of assets, and can considerably reduce mission capability. Inlet ice can produce stalls, loss of thrust, rapid increase in turbine temperatures, and compressor blade damage due to ice ingestion.

REQUIREMENT GUIDANCE (A.3.7.1)

The following should be used to tailor the specification paragraph:

A value up to and including five seconds.

The contractor should indicate whether or not the system will allow no ice accumulation or no detrimental ice accumulation.

The Using Service should ensure that the affected paragraphs correctly identify that either an anti-icing or de-icing system is used. An anti-icing system prevents ice from forming; therefore there should be no ice build up. A de-icing system sheds ice that has already formed, therefore there should be no detrimental ice build up or damage when the ice sheds once the system has been activated.

Background:

Icing conditions typically occur below freezing temperatures in clouds, precipitation, fog, or sea spray but can also occur at ambient temperatures well above freezing due to condensation icing. This is due to the decrease in static pressure in the inlet area by the engine suction effect or by air intake vortex generation, which causes a corresponding temperature drop. When the temperature falls below freezing, either in the air stream or on the component surface, ice starts to accumulate on the sides of the inlet, inlet components, on the inlet struts, IGVs, and the first stage compressor blades from any liquid water

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components in the air stream (either above freezing or supercooled). Solid hydrometers can also accumulate in air induction components, possibly reform into ice, and be shed into the engine.

The critical periods for aircraft are during ground idle and low power landings when there may be insufficient anti-icing or de-icing bleed air and during takeoffs when the inlet temperature drop is the greatest.

Anti-icing or de-icing systems should be designed to minimize the amount of bleed air or other source of energy (electrical, mechanical, and chemical), avoid thermal stress and fatigue of the engine anti-iced structure and not cause engine operational problems.

Anti-icing or de-icing should be provided at the front of the engine for anti-icing or de-icing a customer supplied inlet dome or accessory section. Details of the connection should be shown on the engine configuration and envelope figure. If compressor bleed air is used for this purpose, it should be considered part of the quantity specified in 3.1.1.7.

For electrical anti-icing or de-icing systems, the engine should be capable of simultaneous operation of the anti-icing or de-icing system and all other engine electrical systems.

A means of actuating the anti-icing or de-icing system automatically is preferred. Manual override should be provided if the system is automatically actuated.

Turbofan and turbojet engines operate at higher air speeds (Mach number) than turboshaft and turboprop engines. The operation at higher speeds results in a higher inlet temperature (T_2). An anti-icing or de-icing system in a fail-on position may produce bleed air temperatures up to 1240°F (671°C) or higher. The combination of higher inlet temperatures (T_2) and constant bleed airflow may induce fan or strut integrity problems due to thermal damage. An anti-icing or de-icing system in a fail-off position and the aircraft encountering an icing environment may result in loss of the aircraft due to engine failure from ice buildup on the leading edge of the fan frame struts and bullet nose. Consequently, the contractor should provide a system that allows safe return to base without damage to engine regardless of the design [fail-on or fail-off].

Anti-icing and de-icing systems are complicated by the use of relatively low temperature materials in the frames and inlet structures for which there are few substitutes available to meet other weapon system requirements. Normal candidates are engine bleed air, electrical heating, pneumatic boot or pulse deicers, and hot oil or fluid recirculation. Failed open valves could lead to failure of these parts unless other actions are taken, such as power level reductions, etc. Meanwhile, failed closed valves could lead to unacceptable ice buildup, if the system is in use at the time. The broader wording recommended allows for the design of fault accommodation to be approached on a system level. It is quite possible that the valves (if indeed a hot air system is employed) need to be designed so that they fail in different ways depending on their state at the time of failure. Also, for systems that don't use hot air, there still has to be consideration of preferred failure modes.

If an IPS is required, the separating efficiency design should consider removal of controlled and passive de-ice system ice shards that are shed from engine, air induction or air vehicle components such that the requirements of 3.3.1.4 are met. If the engine or inlet IPS is a "running wet" design, the runback of liquid water should not refreeze on any subcomponents during any phase of flight. Consideration should be given to condensation icing, vortex icing, ice crystal, snow, the potential for accumulation of ice pellets, water or slush plug formation, and ramp, guide vane and fence design.

REQUIREMENT LESSONS LEARNED (A.3.7.1)

On a hot air anti-icing or de-icing system, when anti-icing or de-icing was required for ground idle operation, design considerations were made to determine which stage of compression was to be used for adequate heat transfer to the anti-iced parts. Engine anti-ice systems were prone to thermal distress in

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thin sheet metal structures, such as inlet guide vanes. The usual design problem encountered in hot air anti-icing or de-icing systems was to provide adequate anti-icing or de-icing capability when the engine was operating at low approach power conditions without overheating the system under takeoff conditions.

An engine has experienced a significant adverse impact on the anti-ice valve because the actual bleed air temperature was higher than what was predicted by the original design analysis.

Moisture buildup and contamination from cleaning procedures and environment have caused valve malfunctions. Valve orientation and design should minimize moisture buildup and tolerate contamination as much as possible.

Some engines utilize modulating valves to minimize the amount of anti-ice flow at high power. One engine switched to a modulating valve design because inlet struts were cracking due to thermal stresses from the anti-ice flow at high power in flight test engines.

Engine M has had problems preventing anti-ice buildup under mid and low power conditions. However, in cases where the pilot had to throttle up to 75% power or higher for anti-icing purposes while on icy runways it has been difficult or impossible to stay on the runway. Also, due to ice buildup, the engine has experienced flameouts on the deck because of the constrictions in the engine airflow.

Both Aircraft G and K have experienced a problem of engines building up with ice which can break free and destroy the engines.

Aircraft F's Engine H and engine inlet are anti-iced, but other components (spinner/rotor root fairing) in front of the engine inlet are de-iced. Thus, while in icing conditions, Engine H will have a cyclical ice ingestion event roughly every minute or so. The design goal is ensure that the aircraft particle separator will remove a considerable portion of those shed ice shards, and that Engine H will not have its fatigue life substantially altered due to the ingestion events of unseparated ice shards.

A.4.7.1 Anti-icing and de-icing system.

The requirement of 3.7.1 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.7.1)

The anti-icing or de-icing system performance must be verified in an icing environment. The performance degradation caused by the ice accretion process is very complex and varied such that a combination of development methods must be used to provide a robust functional system to the fleet.

VERIFICATION GUIDANCE (A.4.7.1)

Past verification methods have included analyses, demonstrations, and tests.

In general, as computational methods evolve, they may become an acceptable verification procedure, and the required verification efforts may shift from physical testing to computational optimization of designs or design options, both at the subcomponent, component, and subsystem levels.

Prior to an engine test, the anti-ice system should be bench tested to characterize the temperature, pressure, and flow of the anti-ice system at different power settings.

Flight conditions that should be considered in establishing icing test requirements are listed in 4.3.1.4

A combination of the following methods may also be used:

a. empirical correlation of experimental data or historical design data

b. computer simulations using flow field, trajectory, heat transfer, and accretion codes

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c. ground-based icing facility testing on the component and subsystem levels

d. ground-based component and subsystem functional, reliability, and fit evaluations

e. ground-based full system mockup functional testing

f. ground-based climatic lab testing

g. in-flight artificial spray tanker testing

h. natural icing survey flight testing

i. cold weather ground/flight operations evaluations

Background:

Engine performance in icing and non-icing conditions should show compliance with power loss and fuel consumption requirements.

During testing, power, rotor speeds, and vibrations should be continuously recorded. High speed photographic coverage of the inlet and engine face should be provided.

VERIFICATION LESSONS LEARNED (A.4.7.1)

In most tests conducted in the past, the most severe test condition usually was running the engine at ground idle followed by acceleration to high power. Ice accumulated at ground idle power and was ingested when the engine accelerated to high power conditions. More recent designs have modulating or mixing bleed air valves that place the most severe test point where mixing cutoffs lie at intermediate freezing temperatures and engine flow rates.

Engines incorporating IPS have complicated flow paths requiring use of bleed air and oil cooling for anti-icing or de-icing protection. Protecting the entire IPS, including the scavenge system was difficult and adversely affected the engine design.

A.3.7.2 Control system.

The control system shall be fault tolerant and automatically control the engine to meet all specification requirements.

Regions of control limiting functions, as a function of altitude, mach number, thrust and power setting for standard, hot, and cold day atmospheres of figure 5, shall be shown in the format of figure 19 in _____. The limiting values used to establish the various regions shall be specified in _____.

For turboshaft applications, the engine control system shall provide for load sharing and torque matching for multiple turboshaft engines.

REQUIREMENT RATIONALE (A.3.7.2)

The control system architecture and all interfaces to the engine control system, engine and integrated control modes, and related functional engine capability require a description that shows compliance with engine operational, environmental, and mission requirements.

There are regions in the envelope, for example low altitude and high Mach number, where engine pressures may exceed the maximum design pressure of the compressor case. Therefore, engine limiting must provide protection in this region of the flight envelope.

REQUIREMENT GUIDANCE (A.3.7.2)

The following should be used to tailor the specification paragraph:

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The information can be shown in the ICD, or for programs that do not have an ICD, the interfaces can be put into the specification on figure 19.

Engines that have an overtemperature device, which prevents exceedance of the starting gas temperature limit, may have provisions to deactivate this device and provide a non-resetable indication of deactivation.

Background:

The engine control system should optimize engine performance and operability regardless of harsh atmospheric conditions. Implementation of closed loop control logic should accurately schedule fuel during engine starting, acceleration, deceleration, steady-state operation, and augmentor sequencing through engine shutoff. If practical, augmentor fuel scheduling should be closed loop using augmentor fuel flow ratio units parameters such as (W_f/P_3) to control fuel scheduling. Limits should be imposed on critical engine parameters to avoid exceeding aerodynamic, thermodynamic, and structural engine limits to ensure engine protection, safety, and durability. Engine limit considerations should consist of stall margin protection, pressure, temperature, torque, engine speed, and blowout protection during rapid decelerations. In addition, the controller should provide flameout detection and corrective action, stall and surge detection and recovery, automatic trim and engine deterioration compensation, fault detection and isolation, and should automatically clear hung starts and prevent hot starts.

Modulation of thrust from cutoff to maximum power by means of unrestricted thrust and power demand excursions should reduce pilot workload, increase combat maneuverability, and mission effectiveness.

Engine control modes should be implemented to reduce pilot workload by providing linear correlation between power demand and power output. Integrated control modes should be considered to improve aircraft operation such as carrier approach, landing, wave-off, aerial refueling, loiter, combat, and autorotation power recovery.

Dual redundant FADEC systems, by definition, should provide main engine, augmentor and nozzle control. The dual units or channels can both be active or one can be active and the other provide passive control. Both units and channels should be fault tolerant and should provide automatic control unit switching in the event of a single failure without engine degradation. The mission reliability of the engine will drive the design of the control system. Therefore, the requirement does not preclude the use of a triplex system or other designs that provide fault tolerant systems and enhance fault accommodation. There should not be any single point safety critical or mission critical failures, software or hardware, in redundant electronic controls. Trade studies should be made to ensure that the redundant systems are more cost efficient and reliable than other control system designs.

At design reviews or other technical interface meetings, descriptions of the control system should identify the functional capability of each subcomponent including the FADEC unit(s), electrical harness, sensors, and the actuation, fuel management, and ignition systems. The system should be described as a function of power demand to identify engine variable geometry modulation and fuel scheduling during engine starting, steady-state and transient operation, augmentor sequencing, and shutdown. Additional control system input parameters from the air vehicle such as gas ingestion, FOD, icing, weight on wheels, distortion index, and Mach number may also be needed.

The control system input and output signals can be electrical, mechanical, hydraulic, or pneumatic. The input signals may include aircraft Mach number, altitude, armament reset, idle exhaust nozzle reset, etc. The output signals may include engine parameters for cockpit display or engine condition monitoring equipment (for fault detection and isolation to the control system component weapon replacement assembly or LRU level) and other interrogation systems external to the engine.

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For complex data systems it may be desirable to separate this section into subsections to clarify what interfaces are required. Possible headings are Discrete Inputs, Scaled Inputs, Digital Inputs, Discrete Outputs, Scaled Outputs, and Digital Outputs.

The digital data bus, MIL-STD-1553, MIL-STD-1773, or others should provide the integration of the engine air vehicle controls and transmit data from the engine control to the engine monitoring system. The type and amount of data transmitted should be in the format required by the engine monitoring system. The design of the control system, hardware and software, dedicated to conditioning and transmitting data to the engine monitoring system should be partitioned from normal control functions to eliminate fault propagation.

The contractor should complete figure 19. There are times when the engine contractor must limit pressure and temperature of the engine in a given region of the flight envelope. This is done for structural or aerodynamic reasons. Therefore, a pressure limiter is used in this region of the flight envelope. At supersonic flight speed, airflow through the aircraft inlet duct needs to be controlled between an upper and lower limit to prevent supersonic flow anomalies such as “inlet buzz”. At sea level, intermediate airflow may be reduced to lower turbine temperature.

Overtemperature and overspeed are limited under normal operating conditions. These limiting functions usually lose all authority under abnormal conditions and cannot prevent overspeed and overtemperature.

The engine must not sustain damage due to inadequate limiting of critical engine parameters.

The control system should be designed to accept and condition electrical power from engine driven alternators and power from the aircraft power bus. This provides preflight back-up capability and control system interrogation of built-in test (BIT) to determine “go” or “no go” control status without starting the engine.

The electronic control should allow for external reprogramming to minimize the cost associated with a field software retrofit.

REQUIREMENT LESSONS LEARNED (A.3.7.2)

Increased complexity of engines has placed higher demands on control system functions. Full Authority Digital Electronic Controls (FADECs) have proved to be highly flexible and were quickly modified to solve unexpected design problems during development stages.

In a recent NAVY engine program, the following separately wired input and output signals to the engine control system were required:

Input Signals	Output Signals
Thrust or power demand	Turbine exit temperature
Speed lockup discrete	Core speed
APC throttle command	Fuel flow
1553 data bus (redundant)	1553 data bus (redundant)
	PLA position

Existing engines, such as the Engine C, have had operational problems when they are coupled into a common gearbox. Past specification did not adequately cover load share requirements.

In past engine programs, FADEC systems have controlled the following variables: main fuel flow to combustor, augmentor fuel flow, ignition, variable geometry, exhaust nozzle position, temperature, pressure, fuel thermal bypass flow, and engine anti-icing or deicing.

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Engine A's control system uses PS3 signal inputs in the high altitude, low airspeed portion of the engine operating envelope to limit augmentor fuel flow to avoid rich stability augmentor light off problems.

A.4.7.2 Control system.

The requirements of 3.7.2 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.7.2)

The control system must be verified to assure that it controls the engine as required.

VERIFICATION GUIDANCE (A.4.7.2)

Past verification methods have included analysis, demonstrations, and tests.

Background:

Real time dynamic engine simulations should be used to verify FADEC hardware and software operability and performance to allow early detection and correction of problems prior to actual engine test. This test should simulate sensor and actuator functional and engine operability capabilities and provide verification of control laws and logic during steady-state and transient conditions. Engine limits and operational logic such as automatic trim, engine deterioration compensation, fault detection, isolation and accommodation, control switching, limiting of speed, temperature and pressure, the prevention of flameouts, hot and hung starts should be verified.

Control system and engine testing should ensure critical engine parameter limiting throughout the entire operating envelope at all ambient conditions.

The integration of the FADEC system, including the fuel delivery system, should be bench tested and verified in a closed loop system interfaced to a real time dynamic engine simulator to demonstrate total engine control capability. Thrust and power should be modulated throughout the test to verify steady-state and transient performance between startup and maximum augmentation at sea level, altitude, and selected Mach number conditions for all ambient operating conditions. The test demonstration and analysis should ensure proper engine operation without engine damage.

Engine control modes and integrated control modes, if applicable, should be tested by a flight simulator to verify aircraft and engine performance and operability including loiter, takeoff and landing, aerial refueling, autorotation recovery, and other control modes required by the Using Service. This requires accurate modeling of helicopter dynamic systems.

Aircraft nacelle and engine bay environmental conditions and engine skin temperatures subject the control units to high temperature at high Mach number conditions. Component testing does not address these conditions. To properly verify the control system operability, structural and thermal design, and cooling effectiveness, the control units should be subjected to simulated nacelle environmental atmospheric and aircraft flight conditions. This could be accomplished by a Combined Environmental Reliability Test (CERT) which is the simultaneous application of temperature, vibration, humidity and altitude cycling per defined aircraft mission profiles. This test should be structured to replicate control failures caused by design flaws or inadequate component specification requirements, and will assist in projecting the reliability MTBF at maturity (1×10^6 engine flight hours). The number of test hours and test units should be sufficiently large to assure that design weaknesses are weeded out. A typical CERT program for an electronic control unit would test five to ten units for a total of 50,000 to 100,000 unit hours.

The structural integrity analyses required in 3.4 should state which engine structures require regions that limit engine functions. The performance presentation of 3.2.1 should show how the control affects engine performance when it is limiting engine functions. The altitude test requirements of 4.11.1.2 should be

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used to demonstrate the regions of control limiting functions, which should also be demonstrated during the component tests of 4.11.2.

VERIFICATION LESSONS LEARNED (A.4.7.2)

Real time dynamic simulations have identified control system problems and fixes prior to engine test. Fixes incorporated during engine test were accomplished through the data bus or an engine connector on-line.

Timing of the CERT is critical to its value, the earlier the better. On an Air Force program, the CERT was accomplished late and revealed problems that had already been experienced in the field. The CERT should be done before full production.

Engine contractors claim it was very difficult to overspeed or overtemperature an engine with an electronic control. However, during a test conducted at NAWCADTRN, the production configuration FADEC on a Navy engine allowed the engine to exceed its speed limits.

A.3.7.2.1 Control system performance.

The control system fault accommodation shall detect failures that shall result in fail-operational, fail-degraded, and fail-safe engine operation, during and after control system malfunction(s) or failure(s). The engine control system shall provide fully modulated thrust or power response, and prevent exceedance of aerothermodynamic and mechanical limits when subjected to unrestricted thrust or power demand inputs. Thrust or power signal demand versus engine thrust or power output shall be defined in (a) and free of abrupt changes between ground idle and intermediate thrust or maximum power. A step change in thrust at augmentor light off should not exceed (b). The relationship between control demand and power output shall be shown on figure 20 and in the ICD if available. For thrust reversers, the relationships between reverse thrust and thrust demand input signals shall be defined in (a) and free of abrupt changes.

For turboshaft and turboprop engines, the system shall allow the selection of any output shaft speed between (c) and (c) percent, inclusive, at all powers equal to and greater than no load. For turboshaft engines, the system shall have the capability to automatically match the output shaft torque of both engines of a twin engine installation within (d) percent of the torque available from a single engine at maximum continuous power.

For US Army turboshaft multi-engine applications; engine power loss shall be sensed and power automatically increased on the other engine(s) to provide power required up to contingency power (one engine inoperative) capability. The control system shall automatically enable an emergency power capability from a contingency power condition when the engine is operating at contingency $T_{4.5}$ and gas generator speed and rotor droop is detected.

REQUIREMENT RATIONALE (A.3.7.2.1)

Control system fault accommodation must be addressed via the level of degradation (fail operational, fail degraded, fail-safe), and the identification of which system is affected (i.e.; electrical, hydraulic, etc.) should be included.

Modulation of engine thrust or power from start up through maximum power and shutdown by means of unrestricted throttle movement must be linear between power input and thrust or power output.

REQUIREMENT GUIDANCE (A.3.7.2.1)

The following should be used to tailor the specification paragraph:

(a): This information should be placed in the ICD, for programs without an ICD put the information here.

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(b): As agreed to with the Using Service - may vary with altitude

(c): Typical values for this requirement on several engine programs are from 70 to 110 percent for turboprop and turboshaft engines.

(d): A maximum value of five (5) percent.

Turboprop engines may also be required to automatically match the output shaft torque of another engine if both engines are connected to a common drive system (i.e., an Aircraft F application). A turboprop may also be required to automatically increase power in a twin engine application if the other engine suddenly loses power.

For the fault accommodation requirement, the engine electrical and electronic system should be a first fail operational design except for failures in the augmentor ignition system. Augmentor operation should not be inhibited by the control system as a result of any single electrical failure in the augmentor system.

Any interfaces transmitting Integrated Flight and Propulsion Control data to and from the aircraft should have a fail operational capability for any first failures of the communication system. The engine control should not propagate any internal failures to the aircraft.

Background:

The requirement of fail-operational should provide full engine operational capability after any first failure or control system malfunction. Switching logic should require transfer to the redundant unit. Critical control unit malfunctions such as switching circuitry and logic and the impact on the main engine, augmentor, and nozzle control should be described at design reviews and technical meetings. Single, dual, and triple failures should be verified to determine the impact on engine performance. The contractor should provide the above information in the FMECA.

The US Army has nap-of-the-earth operational requirements and the pilot should not have to monitor engine instruments. The pilot should be advised of the engine power loss and the other engine should provide power to the contingency or emergency capability.

A nonintegrated control system places limitations on the pilot's ability to make unrestricted throttle movements due to the possibility of engine stall. The FADEC system integrated with the air vehicle control system should allow unrestricted movement of the engine throttles throughout the flight envelope. The FADEC response to a throttle transient should increase or decrease fuel flow at a rate appropriate to the aircraft operating condition. In the event of engine stall or rapid increase in engine turbine temperature the FADEC should take recovery action or delay the transient until it is safe to continue. Unrestricted power input should require integration of flight and propulsion controls that provide automatic flight and engine envelope limitations to effectively increase combat maneuverability and reduce pilot workload.

Closed loop control modes should provide: accurate and responsive control of thrust and linear correlation between power demand and power output, repeatable starting, precise fuel scheduling, control of critical engine parameters, overspeed, blowout, and stall protection. Closed loop control eliminates adjustments for FADEC trim for engine tolerance or engine deterioration and should provide improved engine operability by providing corrective actions in the event of flameout, hung, and hot starts.

With closed loop control (feedback control), some output quantity is measured and then compared with a desired value, and the resulting error is used to correct the systems output. Without feedback, it is often difficult to build a controller that will be suitably accurate under all conditions.

A major advantage of closed loop control is that the error in the controlled quantity is substantially less sensitive to system-parameter variations than an open-loop control, and therefore precise knowledge of

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system characteristics is not required in order to achieve precise control. This is very important when the system is a large high-powered device that is naturally subject to substantial variations.

Integrated control modes should be implemented to optimize aircraft performance and operational capability (i.e., integration with the FADEC and aircraft inlet control should provide the capability to eliminate or significantly reduce airflow instability (buzz) by modulating ramp and bleed door positions with engine corrected airflow to allow an increase in the stability corridor). In addition, reducing SFC while maintaining the same level of thrust at cruise conditions should be considered.

Augmentor thrust step limits should be compatible with engine and aircraft operation. Abrupt changes occurring during initiation and termination of augmentor sequencing should be minimal but sufficient for pilot awareness. Step changes in thrust should be minimal when initiating and terminating augmentor operation. Thrust step change should be small enough not to disorient the pilot, but sufficient enough to let the pilot know the augmentor is lit. It also assures a smooth transition that will prevent a pressure spike from traveling up the fan duct and causing a low pressure compressor or fan stall.

REQUIREMENT LESSONS LEARNED (A.3.7.2.1)

In past engines, abrupt thrust changes occurring in the system have resulted in stalls and flameouts. Augmentor induced stall, caused by hard augmentor lights, have caused fan stall in turbofan engines, which have resulted in shutdown.

Large thrust increases at augmentor ignition have resulted in pilot disorientation, death, and loss of aircraft. Conversely, pilots have, at times, not realized that the augmentor was lit, resulting in reduced augmentor life and increased fuel consumption.

During some aerial refueling exercises, aircraft at intermediate thrust settings experienced abrupt thrust changes during augmentor initiation. This caused unacceptable closing rates between the tanker and the aircraft.

Engine and air vehicle integrated control laws and modes have been verified and demonstrated by most engine and air vehicle manufacturers. These control modes have improved SFC during cruise conditions.

Both Aircraft B1 and I have had performance problems at entry into service because of nonlinear control system performance.

A.4.7.2.1 Control system performance.

The requirements of 3.7.2.1 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.7.2.1)

Control system performance must be verified to ensure satisfactory engine operation under all expected flight conditions.

VERIFICATION GUIDANCE (A.4.7.2.1)

Past verification methods have included analysis, demonstration, and test.

Background:

Control System Testing:

Fail-operational, fail-degraded, and fail-safe operation should be tested and a FMECA should identify the impact of FADEC, sensor and effector failures, or malfunctions on engine operation. Automatic engine and flight control limiting, as an integrated system should be tested to verify unrestricted power

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modulation. Linear correlation between power demand and thrust output should be verified. Control laws and logic should be tested and implemented to eliminate pressure spikes and reduce abrupt thrust step changes during augmentor sequencing.

Engine Testing:

Engine testing should consist of sea level static and selected flight conditions throughout the envelope demonstrating performance and operability and should demonstrate fail-operational, fail degraded, and fail-safe engine operation. The engine should conform to the same parts list as the endurance test engine, starting in IFR. Control system performance should be demonstrated and verified in conjunction with the engine tests required in 4.2. Engine starting, steady-state, and transient operation should demonstrate consistent engine performance at the same simulated flight conditions.

During the high and low temperature qualification test, consistent starting times should be demonstrated at each test condition and the control should prevent hot and hung starts.

Fault detection, control unit switching (transfer) and fault accommodation should be demonstrated to assure smooth engine control transfer with no engine degradation. Control system faults should be simulated covering control system inner and outer loops. Hard over, soft, and out of range sensor failures should be simulated and compared to the results of the FMECA. The fail-degraded and fail-safe test should consist of those failures providing the highest and lowest levels of thrust. Fail-safe operation should demonstrate the capability to shutdown with no engine damage and minimum engine performance.

Qualification testing of the engine control system should include all integrated air vehicle signals, (which requires the air vehicle flight control interaction algorithms). Historically, some of this testing has been left to the air vehicle contractor to complete.

VERIFICATION LESSONS LEARNED (A.4.7.2.1)

Successful engine performance and operability is dependent on the thoroughness of FADEC system test and verification. Schedule changes, logic changes and hardware fixes should be incorporated and verified prior to engine test. After minor adjustment impact on performance and operation have been fully determined through test and simulation, minor adjustments could be made on-line (i.e.; starting schedules, while the engine is running, which significantly reduces down time and cost). For critical engine tests, experience has shown that the control system should be the intended production configuration to provide valid test results.

FADEC development programs have demonstrated full operational capability after any first sensor and actuator failure and fault accommodation after the second and third failures which resulted in performance degradation less than 10 percent.

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A.3.7.2.1.1 Backup control.

If required by the Using Service, a backup control in conjunction with a redundant control system shall be required for single engine aircraft unless sufficient statistical evidence on the proposed system shows acceptable levels of risk and the backup control shall be considered part of the control system. During transfer from the primary mode to the backup mode and during operation, the backup control shall provide, as a minimum, a safe return to base. The capabilities and limitations of the engine when operating with the backup control shall be specified. Transfer from the primary system to the backup control and back to the primary system shall be _(a)_ at any power setting without engine instability or change to the control system input. Transfer from primary control to backup control shall be within _(b)_ seconds after system malfunction or failure. Thrust or power demand input signal relationships shall be similar to those of the primary control system, but cut-off, idle and intermediate thrust or power signal positions shall be the same as for the primary mode. The backup control shall prevent the engine from exceeding any limits.

REQUIREMENT RATIONALE (3.7.2.1.1)

Backup control must be required wherever Electromagnetic Interference (EMI) or Electromagnetic Pulse (EMP) threat may cause failure of the primary electronic control system. Control capabilities and limitations, when operating in the backup control mode, must be established to ensure compatibility with mission and safety requirements.

The requirement to transfer between FADEC and backup control without changing the power input to match power improves overall safety and ease of operation. The requirement results in a control system where power in the backup control and FADEC modes are similar at the same power setting. However this requirement may be impossible to implement in a simple backup control, particularly for turboshaft engine applications. Therefore, “failed-fixed” transfer modes can be considered as an alternative, particularly for helicopter applications.

REQUIREMENT GUIDANCE (3.7.2.1.1)

The following should be used to tailor the specification paragraph:

(a): Either “automatic” or “manual” should be inserted depending upon results of system trade studies for cost, complexity, and weight. In single engine applications the backup control should use some method to avoid possible single point failures in the fuel delivery system.

(b): Typical transfer times are 0.5 seconds or less. Depending on the criticality of the control function, transfer times may be in the millisecond range.

The backup requirements could be met via redundancy in the electrical system.

Background:

A backup to the control system should be required if the loss of an engine affects flight safety, the aircraft will be subjected to a combat environment in which survivability becomes a large factor, or when mission requirements dictate that power on all engines be maintained. Single engine aircraft not using a backup control of dissimilar technology should be justified to the Using Service.

The contractor should specify the description of the backup control, and the capabilities and limitations of the engine when operating on the backup control. The backup control capability, compatibility, and limitations that require description should include but not be limited to compatibility with the FADEC, pilot notification, PLA modulation from idle to any thrust or power setting, and flight envelope (temperature and pressure) limitations.

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Dual, multi, or single engine aircraft should require redundant FADEC systems to provide fail-operational capability and increased mission reliability which cannot be accomplished with single strand control systems. Either FADEC unit can act as a primary control with the standby unit providing active or passive engine control. A dissimilar backup control should be considered for applications where either friendly or hostile forces (e.g., submarine patrol aircraft that are bombarded with EMI or EMP from surface ships), generate EMI or EMP. The backup control design of dissimilar technology provides protection against common mode failures and possible aircraft loss, if the air vehicle critical systems are hardened and the FADEC is not. The entire control system architecture and internal design should provide enhanced survivability by reducing the vulnerability of the system.

Verification should be conducted to establish backup control functions and design architecture. The control system (backup control and FADEC) requires compatibility to effectively perform switching functions. Upon primary system failure, it is preferred that switching be automatic with an indication to the pilot. The transfer should occur without stall or flameout, “failed-fixed” transfer schemes can be considered. System checkout should be conducted prior to flight to assure the system is working at full capability. The backup control may be required to accommodate gun and rocket gas ingestion. To enhance combat survivability, a reduction of control system vulnerability should be provided. This can be accomplished by effectively separating the primary control units if dual, to minimize the probability of kill of both controllers by a single hit. If the FADEC channel(s) are engine mounted, they should not be placed by the backup control to increase ballistic survivability. Composite air vehicle structures provide little protection against the electromagnetic environment, friendly or unfriendly, increasing the vulnerability of the control system.

For turbofan and turbojet applications the backup control should have the following to ensure safe return to base capability:

a. It should be possible to modulate power between Idle and 80 - 100 percent of the intermediate power available on a standard day from sea level to 10-km (32,800 feet) altitude. Multi-engine applications may be able to sustain safe flight with less than 80 - 100 percent. The low end to the power range (ground idle) should provide low enough forward velocity on the ground so that the brakes can stop the aircraft. The top of the power range should provide sufficient power to complete a takeoff maneuver safely on a hot day. The 10-km (32,800-foot) altitude is high enough to clear any mountain.

b. It should be possible to prevent engine operation beyond established limits. There should be no overspeed or overtemperature beyond the stated transient limits and no combustor, augmentor or compressor instability during or following a power demand change.

For turboshaft applications, the backup control should:

a. Allow engine power modulation from flight idle to maximum continuous power available on a standard day from sea level to 10-km (32,800 feet) altitude.

b. Provide overspeed limiting

c. Consider provision of power turbine speed governing.

REQUIREMENT LESSONS LEARNED (3.7.2.1.1)

Experience has shown that it is particularly difficult to design and incorporate a backup control for application on an existing engine design due to space limitations, interference problems and the inability to detect a primary failure and accomplish a transfer. The backup control system must be sufficiently sophisticated so that it does not require undue attention of the pilot during starting and operation.

The automatic transfer requirement has been usually matched with an altitude bias in the backup system. This type of system can be found on various engines such as O and P. Engine P1’s backup control has no altitude bias and can only allow transfer without matching at high power at sea level. The pilot must be very careful not to cause a surge during normal throttle handling.

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Hydromechanical fuel controls have provided backup control in the manual mode and have required excessive pilot workload to stay within engine limits.

For turboshaft applications, different backup control implementations have been tried. Engine D for Aircraft E utilizes a very simple fuel valve modulated by a DC motor, controlled by the aircraft flight control computer. This provides some integration benefits compared with letting the pilot operate the DC motor with conventional beeper switches. Other backup controls have been more complicated, but not necessarily more controllable.

Engine N has a single-channel FADEC with a rudimentary hydromechanical back-up control. While the backup control has an altitude-compensated fuel flow to power lever relationship, the pilot must be careful to avoid compressor surges during acceleration and must monitor all instrumentation to prevent engine limit exceedances.

In addition, the unique requirements of helicopter applications usually means that the power lever is set at maximum during flight, allowing the FADEC to modulate engine power anywhere from flight idle to the maximum allowed by the PLA setting in order to match engine power to the rotor load demand. This means that when a failure in the primary system is detected, automatic transfer to the backup control could result in a fuel increase to the maximum value due to the PLA being set to maximum. This transfer problem has usually been solved by use of “fail-fixed” systems where the fuel valve is held at its last position upon detecting and confirming a primary control failure. This allows the pilot time to assess the failure and take any appropriate actions (such as placing a hand on the PLA in order to retard or advance the throttle) prior to making the switch to the backup control.

A.4.7.2.1.1 Backup control.

The requirements of 3.7.2.1.1 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.7.2.1.1)

Backup control operation must be demonstrated.

VERIFICATION GUIDANCE (A.4.7.2.1.1)

Past verification methods have included engine and component analysis, demonstration, and test. Development tests will normally be done prior to CDR. Component tests, including systems rig tests should be done prior to IFR. The engine tests shown in tables XXXIa and b are done as part of the engine altitude and sea level testing to ensure the backup control works properly throughout the operating envelope. The flight-testing shown in the tables are not necessarily dedicated to backup control verification, but would include backup control testing as a normal course of the flight test.

Background:

The backup control should be analyzed and tested for single point failure modes.

VERIFICATION LESSONS LEARNED (A.4.7.2.1.1)

Past engine tests have shown some problems in transfer to the backup control from the primary control. These problems were fixed prior to first flight.

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A.3.7.2.2 Control system adjustments.

No adjustment of the engine control system shall be permitted except as follows: _____.

REQUIREMENT RATIONALE (A.3.7.2.2)

Control designs requiring mechanical adjustments should not be allowed. Control adjustments result in increased maintenance workload, engine ground running, and potential for misadjustment.

REQUIREMENT GUIDANCE (A.3.7.2.2)

The following should be used to tailor the specification requirement:

The contractor should identify what mechanical adjustments are needed such as rigging of helicopter load demand input signals.

No adjustments should be required to install control system components on an assembled engine. There should not be any field trims. The engine control system should not require adjustments when operating on primary or alternate fuels.

In the event the Using Service permits external adjustment and a remote trimmer is required, the following should replace the specification paragraph:

Adjustment features shall accommodate SAE AS 848 and shall be self-locking under all operating conditions. All other adjustments shall be protected to prevent tampering. Once adjusted, no further adjustment shall be necessary to provide within limits operation throughout the complete environmental conditions and operating envelope of the engine. All adjustment locations shall be shown on the engine installation and interface drawing provided in the ICD or for programs that do not have an ICD, the drawings shall be included in the specification.

Background:

State-of-the-art fuel controls may not require adjustments. All recent designs are digital electronic. Digital electronic systems, using closed loops, can eliminate adjustments for idle, maximum speed, fuel type, temperature, and pressure. In the past, hydromechanical designs were not able to meet this requirement due to the use of open loop control mode design.

FADEC systems should not need adjustments. FADECs can provide automatic trim, compensate for engine deterioration, and may not require fuel specific gravity adjustments.

If allowed by the Using Service, external adjustments to the controls should be limited to adjustments that can be made with the system installed on the engine and with reference to the operating characteristics on the ground. The adjustments should be clearly marked, accessible, and adjustable with the engine running. Once adjusted, no further adjustment should be necessary for within limits operation. External adjustments may be necessary to achieve peak engine performance and economy. Adjustments are made on the ground with the engine in the aircraft or test stand and while the engine is running.

Incorrect control settings often reduce operational capability and cause extensive engine damage. Self-locking adjustments are required to ensure that vibration, maneuver loads, etc., and will not change control settings while the engine is running. The fail-safe characteristic of the self-locking requirement exists to reduce risks associated with other methods of securing adjustments. Protection against tampering is required to prevent unauthorized adjustments.

Adjustments may be permitted because of reasons peculiar to a specific design, only during the development program, or only at overhaul. A variety of engine parameters may be considered for control system operation and adjustment. Some examples are described below:

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a. Fuel grade adjustment. No adjustment should be required when operating with the primary fuels specified in the specification.

b. Idle speed adjustment. Idle rotor speed should be adjustable within a range of +5 percent of the specified idle speed. This adjustment should be independent of and should not affect the maximum speed setting.

c. Maximum speed adjustment. Maximum rotor speed should be adjustable within a range of +5 percent of the intermediate value. If the engine has more than one rotor and each rotor speed is adjustable, the specification should specify the adjustment ranges and how the adjustments are made.

d. Measured temperature adjustment. Maximum measured temperature should be adjustable within a range of +5 percent of the intermediate value.

On modern digital electronic controls, these and many other adjustments are made through reprogramming. Reprogramming digital electronic engine controls is strictly controlled through the Engineering Change Process (ECP).

REQUIREMENT LESSONS LEARNED (A.3.7.2.2)

A common result of incorrect control settings has been combustor and turbine damage. Engine life can be significantly reduced, and the resulting damage can be extensive and expensive to repair.

The worst case example of an adjustable fuel control required the time between maintenance actions to be less than 30 hours. This control was replaced by a digital system requiring no adjustments.

Joseph J. Curran and Allan M. Levine, of Engine Company A, wrote a paper entitled, “Development of a ‘No Adjustment’ Turboshaft Engine Control System”.

A.4.7.2.2 Control system adjustments.

The requirements of 3.7.2.2 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.7.2.2)

Engine and control response to external adjustments must be verified.

VERIFICATION GUIDANCE (A.4.7.2.2)

Past verification methods have included engine and component test and demonstration.

Background:

The verification of self-locking adjustments should be accomplished during engine testing. Adjustment positions should be marked prior to testing and checked at the end of the test prior to recalibration. The existence of tamper protected adjustments should be accomplished by component examination.

Verification of reprogramming changes is usually accomplished by system level and engine testing.

VERIFICATION LESSONS LEARNED (A.4.7.2.2)

None.

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A.3.7.2.3 Overspeed protection system.

The control system shall detect all main engine rotor overspeed conditions and take corrective action to prevent overspeed limits from being exceeded.

For turboshaft and turboprop applications, the measurement of the power rotor system speeds shall be sensed directly with no intermediate mechanical devices such as gears, flexible shafts, or clutches and the rotor speeds at which the overspeed protection occurs shall be specified.

REQUIREMENT RATIONALE (A.3.7.2.3)

There is a need for an overspeed protection system for safety purposes. Overspeeds can result in damage and failure of rotating components such as fans, compressors and turbines.

REQUIREMENT GUIDANCE (A.3.7.2.3)

Overspeed limits should be established by the engine contractor and set so as to prevent damage to rotating components.

Background:

Loss of power turbine load may result in rotor overspeed. Propeller blades and helicopter rotor blades should not be stressed by the power turbine rotor going overspeed. Overstressed blades and power turbines have caused the loss of aircraft. The safety elements required of the overspeed device requires a system that directly senses rotor system rotational speed.

Small engines used for turboshaft and turboprop applications have engine rotors that can withstand as much as 150% overspeed. Yet the propeller or helicopter blade system that the engine drives may only be substantiated up to 120% overspeed. It is important to interface with the total system to determine the minimum overspeed that the total system can withstand. Some overspeed devices not only use a speed level trip point actuation, but also use a speed rate trip point actuation to limit the rotor speed.

Helicopter overspeed systems are particularly difficult to implement since deliberate overspeeds of the main rotor are common during autorotations, and modern FADEC control laws can deliberately overspeed the power turbine during power recovery from autorotations to minimize helicopter rotor droop. Because of intentional operation in the overspeed regime, it is often difficult to design an overspeed system which can detect a real overspeed condition early enough, to prevent it from being destructive, without cutting fuel flow completely. This may make it difficult to relight the engine, particularly if operating in the backup mode. This problem will become more common as low-inertia, high-temperature power turbines are designed to meet the goals of light-weight engines, and as FADEC control laws become more responsive to helicopter load demand changes.

In many cases, only speed rate anticipation can be used to distinguish an engine failure early enough so that a min flow overspeed response is adequate to arrest the power turbine overspeed before it reaches a destructive level. While speed rate anticipation is a possible solution, experience has shown that it is often intentionally disabled after fielding since it is very susceptible to noise and false overspeed trips.

REQUIREMENT LESSONS LEARNED (A.3.7.2.3)

Engine overspeed incidents have resulted in structural failure of rotating components. Overspeed incidents require unscheduled maintenance, ranging from a complete borescope to engine replacement.

An engine overspeed incident occurred in a Navy test cell with a FADEC on the engine. A fuel-metering valve was stuck full open and the engine control system had no protection device to stop the engine from overspeeding. This occurred with a fully operational FADEC.

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A.4.7.2.3 Overspeed protection system.

The requirements of 3.7.2.3 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.7.2.3)

The rotor speed at which overspeed protection occurs, needs to be demonstrated on an engine test.

VERIFICATION GUIDANCE (A.4.7.2.3)

Past verification methods have included analysis, demonstration, and test.

If a test is needed, the following procedure should be considered:

For turboshaft and turboprop applications the engine shall be set up in a test cell with a shaft power absorber that can be instantaneously decoupled from the engine. The power turbine output shaft speed shall be adjusted to the highest speed with the engine operating at intermediate power setting. After reaching a stabilized condition, the shaft power absorber shall be instantaneously disconnected from the engine. This test shall be considered satisfactorily completed if the power turbine overspeed logic prevents destructive overspeed of the power turbine and no destructive overspeed of engine rotors.

VERIFICATION LESSONS LEARNED (A.4.7.2.3)

Past specifications identified this test as the engine overspeed control system test.

A.3.7.3 Fuel system.

A.4.7.3 Fuel system.

A.3.7.3.1 Fuels.

The engine shall meet all applicable requirements when supplied with the fuels specified per _____ .

REQUIREMENT RATIONALE (A.3.7.3.1)

Fuels must be identified so that the engine can be designed to operate and perform using the selected fuels.

REQUIREMENT GUIDANCE (A.3.7.3.1)

Consider the following proposed options for manned and un-manned air vehicles where world-wide logistics support (refueling) will be a primary operating concern:

The following should be used to tailor the primary, alternate, and restricted fuel specification paragraphs that follow in this section: table XXIV contains three options for choice of fuels. The Using Service should select one option and delete the others from table XXIV.

TS-1 or JP-4 fuels only were available at specific remote locations during the first few weeks of Operation Enduring Freedom; hence, these fuels are listed in the primary fuels option 1 in table XXIVa.

The options in table XXIV are listed in order of preference. The information below explains the rationale for choosing options:

Option 1 (table XXIVa)

Requiring all of the fuels listed in table XXIVa to be “Primary Fuels” affords the following advantages:

a. The engine/aircraft will be capable of unrestricted operation on any common aviation turbine fuel found worldwide greatly simplifying the logistics necessary to support emergency operations.

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b. Operations out of commercial airfields are simplified since the aviation jet fuel normally supplied will be usable.

c. Foreign Military Sales (FMS) attractiveness is increased since several foreign Military Organizations continue to operate on JP-4 fuel.

d. Recent operational experience has required the use of kerosene-based fuels such as Russian TS-1 and other foreign manufactured fuels. Designing US Military aircraft to operate satisfactorily on such fuels would preclude major logistics problems and/or significant delays in urgent operations.

e. The inclusion of Wide-Cut fuels means that engine performance will not be compromised if fuel or crude oil supply shortages require military operations to be converted to the more easily refined Wide-Cut fuels.

f. There is no need to have an alternate or restricted fuel entry since the engine will operate without restrictions on any of the aviation turbine fuels commonly used worldwide.

g. This option is expected to be the most expensive since it requires a more robust fuel system design supported by extensive analysis as well as component and engine verification.

Option 2 (table XXIVb)

Exclude Wide-Cut fuels (i.e., JP-4 and Jet B) from the Primary Fuels category. Moving them to a "Restricted fuels" category could simplify the airframe and engine design since the high volatility of this type of fuel will not have to be accommodated. Moving Wide-Cut fuels will negate advantages c, e, and f, above. The current thinking among fuel logisticians is that now that the USAF and Army have converted to JP-8 it will be very unlikely that the AF and Army will reconvert to JP-4 even in an extremely tight oil supply scenario. If the Wide-Cut fuels are placed in a restricted category, the engine manufacturer must identify the operating restrictions that will be placed on the engine (and aircraft) such as high fuel temperature, when operating on a Wide-Cut fuel. There may be a severe restriction on operational capability when operating at low temperatures when JP-8 is used. JP-4, with the high volatility, has provided good low temperature engine starting capabilities below -40°F.

Option 3 (table XXIVc)

Exclude the fuels without additives (much less desirable), Fuel System Icing Inhibitor (FSII) and Corrosion Inhibitor/Lubricity Improver (CI/LI) and move into an alternate or restricted fuels category depending on whether the lack of additives affects long term durability or operational performance respectively.

Background:

Research has shown that fuel thermal additives have the ability to increase the stability of existing fuels. Advanced systems are showing potential needs to be able to handle increased thermal loads and these types of additives may be of use. The logistics impacts (injection equipment and carrier operations) of using these additives must be considered.

For manned and unmanned aircraft operating from fixed facilities where logistics concerns are less important, designers can consider the following specialty fuels:

a. High thermal stability / high heat sink fuels:

- NATO F-37 (JP-8+100, MIL-DTL-83133): This fuel offers higher heat sink capability and reduced coke deposit formation in hot section components. F-37 is normally produced by injecting the +100 additive into the fuel (F-34, JP-8) while loading the aircraft refueling vehicles. Since the +100 additive is incompatible with its fuel supply and delivery infrastructure especially on its ships, the Navy does not authorize the use of F-37 except in certain shore-based aircraft.

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- JP-7 (MIL-T-38219B): This fuel was developed for the aircraft J and has very limited availability. It is also significantly more expensive than F-37 or standard kerosene jet fuels.

- JPTS: Developed for and used by the aircraft AI this fuel is also expensive and available in a limited number of locations.

b. High density fuels:

- JP-10 (MIL-P-87107): This fuel offers higher volumetric energy density than kerosene based fuels; i.e., approximately 141,500 BTUs per gallon versus 120,400 for JP-5. (Weight density is slightly lower at 18,100 BTUs / lb versus 18,300 BTUs / lb for JP-5.) JP-10 is also relatively expensive; it costs three to four times as much per gallon as the regular military and commercial jet fuels.

- Other High Density Fuels: Several additional high density fuels have been used over the years, including RJ-4 (MIL-F-82522) which had long-term stability problems; i.e, formed gums which caused problems with sticking fuel system valves in missiles. RJ-5 proved too viscose at low temperature for operational/readiness requirements.

For unmanned aircraft operating in remote locations, designers should consider the use of common ground products as primary fuels. Unfortunately the properties of these products vary according to the location and time of year.

Some other specifications to consider:

a. Diesel Fuels: ASTM D975, EN 590, and A-A-52557

b. Automotive Gasolines: ASTM D4814, EN 228

REQUIREMENT LESSONS LEARNED (A.3.7.3.1)

During the first few months of Desert Shield, Navy and Marine Corps aircraft were often required to use commercial Jet A1 without additives since these materials and the required injection facilities were not available in country. Future conflicts could require even longer periods of commercial jet fuel use if additive and fuel supplies are effected. For this reason it is recommended commercial jet fuels be included among the "Primary Fuels".

During the mid-1970's, the Navy experienced problems with Engine L's fuel controls and pumps that were related to the use of low lubricity fuels (hydrotreated JP-5 being delivered in the Mediterranean). These engine components were modified to make them less sensitive to low lubricity fuels. At the same time corrosion inhibitor and lubricity improver additives were added to JP-5 to prevent similar (potential, unidentified) problems from surfacing in other engines and aircraft components. It is prudent to develop and qualify equipment such that it is not dependent on this additive since commercial fuels do not contain it.

The minimum use level for FSII varies by Service (0.03 vol. % for USN and USMC, 0.07 vol. % for USAF and USA and NATO). Several studies of the quality of aviation jet fuel delivered to aircraft by Navy ships

has shown that a percentage of the fuel (<2%) contains less than 0.07 vol. % FSII. In addition, commercial turbine fuels do not contain FSII. Engine fuel systems should be designed for unrestricted operation on fuel without FSII.

Antioxidant additives are included in Military fuels to insure stability of the fuel during long-term storage (>6 months). Static Dissipater Additive (SDA) is injected in JP-4 and JP-8 fuels to improve static charge dissipation during fuel handling operations. No evidence has ever been found that these additives affect engine performance or operation.

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Design, development, and qualification testing of the engine should be performed primarily on Jet A without additives since the engine manufacturer will have to design the engine for operation on the worst case fuel.

It is good practice to avoid MIL-G-3056 gasoline due to handling and availability problems.

A.4.7.3.1 Fuels.

The requirements of 3.7.3.1 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.7.3.1)

The specific type(s) of fuel, including the types of additives, if any, must be specified in order to be designated for use in tests throughout the specification.

VERIFICATION GUIDANCE (A.4.7.3.1)

Past verification methods have included test as described in the verification paragraphs below.

Background:

Choosing the relevant fuel properties and testing the worst case extremes, rather than all fuels, can reduce the scope of the verification test program. For example, combustor hot section durability should be qualified based on fuel that burns with highest luminosity (e.g.; JP-5, JP-8, or specially prepared worst-case fuel) and not all fuels, and operability should be qualified using fuels that cover the range of selected properties relevant to the tests to be conducted.

Advanced systems require fuels to be used for applications other than just combustion. Examples are thermal management (fuel cooling of electronic subsystems), hydraulics (actuating exhaust nozzle design, integrated flight controls) and flying qualities (aircraft CG management). These applications may require a more complex system design and rigorous verification methods resulting in increased development cost.

VERIFICATION LESSONS LEARNED (A.4.7.3.1)

None.

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A.3.7.3.1.1 Primary fuel.

The primary fuels shall be as specified in table XXIV. The engine shall meet the requirements of the specification when using any of the fuels conforming to and having any of the variations in characteristics permitted by the specifications listed in table XXIV. The engine shall meet all requirements of the specification after transition between any primary fuels with no engine damage. The altitude relight envelop for each of the three major fuel types (High Flash Kerosene, Kerosene and Wide-Cut) shall be identified.

REQUIREMENT RATIONALE (A.3.7.3.1.1)

Utilization of established fuel specifications insures wide availability and constant supply.

REQUIREMENT GUIDANCE (A.3.7.3.1.1)

Primary fuels are the fuels on which the engine is designed to operate continuously without restrictions. The following should be used to tailor the specification paragraph: table XXIV should reflect the option chosen for the engine application. Refer to guidance of 3.7.3.1. Use of engine company A's +100 thermal stability additive is prohibited on most Navy aircraft since shipboard fuel handling systems are not designed to handle fuel containing the +100 additive.

REQUIREMENT LESSONS LEARNED (A.3.7.3.1.1)

None.

A.4.7.3.1.1 Primary fuel.

Unless noted otherwise within the specification, engines shall be verified using the worst-case fuel(s) below:

VERIFICATION RATIONALE (A.4.7.3.1.1)

Conducting verifications with the appropriate specific fuel insures satisfactory operation with the fuels listed.

VERIFICATION GUIDANCE (A.4.7.3.1.1)

The following should be used to tailor the specification paragraph:

The Using Service should select one or more of the following:

a. MIL-DTL-5624 Grade JP-5 without FSII and lubricity improver ASTM D1655 Jet A or Jet A-1
b. MIL-DTL-83133 Grade JP-8 without FSII and lubricity improver or Jet A-1
c. MIL-DTL-5624 Grade JP-5
d. MIL-DTL-83133 Grade JP-8 .

Background:

The following provides background for selecting a test fuel:

A Special Test (ST) fuel is preferred since JP-8 and JP-5 ST fuel is specially formulated to be the worst-case fuel in terms of engine durability, emissions, cold temperature start, and altitude relight characteristics. Note: The use of Wide-Cut primary and restricted fuels can be critically limiting factors in hot fuel tests because of their high volatility; therefore, high temperature operation and performance tests in Section 3.2 should be specifically written to use either JET B or JP-4 fuel. ST fuels are usually locally developed to match the requirements of the design.

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JP-5's general volatility characteristics make it slightly worse than Jet A or JP-8 in terms of engine durability, emissions, cold temperature start, and altitude relight characteristics.

ASTM D1655 Jet A and Jet A-1 are commercial jet fuels without additives which are readily available throughout the US.

Fuels containing FSII and lubricity improver are listed at the bottom since the presence of these additives may mask design problems during testing.

Utilization of established fuel specifications ensures wide availability and constant supply.

Use of JP-5 and JP-8, for example, can be expected to result in higher combustor liner temperature and lower life because of their reduced hydrogen content relative to JP-4.

VERIFICATION LESSONS LEARNED (A.4.7.3.1.1)

Cold day ground start and altitude relight capabilities have been observed to vary significantly with fuel types. In general, JP-8 and JP-5 fuels have reduced capabilities in these areas compared to JP-4 because of reduced fuel atomization and volatility.

JP-4, JP-5, or JP-8 have potentially differing durability impacts.

A.3.7.3.1.2 Alternate fuel.

The engine shall be capable of starting, and operating satisfactorily when using the alternate fuels. The operating limits, thrust outputs, and thrust transients specified herein shall not be adversely affected when using alternate fuels. There shall be no adverse effect on aircraft mission(s). Those external adjustments permitted in 3.7.2.2 shall be allowed in order to meet these requirements. Alternate fuels shall be the commercial jet fuels specified in table XXIV. Any limitations on the engine when using alternate fuels shall be specified herein.

REQUIREMENT RATIONALE (A.3.7.3.1.2)

Alternate fuels must be specified so the Using Service knows which fuels can be used when primary fuels are not available.

REQUIREMENT GUIDANCE (A.3.7.3.1.2)

An alternate fuel is one on which the aircraft can be flown without operational restrictions but which can have long term durability or maintainability impact if used for continuous operation (multiple flights). Alternate fuels are used only on an occasional or intermittent basis. The following should be used to tailor the specification paragraph: table XXIV should reflect the option chosen for the engine application. Refer to guidance of 3.7.3.1.

If no alternate fuel is required in the Option selected, delete the text in the paragraph and insert "Not Applicable".

Background:

Characteristics of NATO and commercial fuels (such as NATO F-35 and Jet A or Jet A1) should be taken into consideration during design of the engine. Variations in light off capability and lubricity have been problems in the past when fuel changes were required.

Commercial jet fuels differ only slightly from the fuels prescribed in 3.7.3.1 and are suitable for most flight operations. The option for designating alternate fuels permits the Using Service to tailor for specific

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conditions as well as accommodate the use of shale and coal derived fuels that deviate from specification fuels.

REQUIREMENT LESSONS LEARNED (A.3.7.3.1.2)

The conversion from JP-4 to Jet A-1, JP-8, or JP-5 fuel may increase aircraft maintenance and degrade systems safety due to increased fuel leaks. This is due to the high swell properties of JP-4 and the permanent set taken by the seals (especially fluorosilicone) after prolonged periods of operations with JP-4 fuel. The introduction of Jet A-1, JP-8, or JP-5 that have low swell properties will result in seal shrinkage and the leakage of seals. These problems are further aggravated by transition from high ambient ground temperature to low flight temperatures. The conversion from JP-4 to other kerosene based fuels has resulted in a limited availability of JP-4.

A.4.7.3.1.2 Alternate fuel.

The requirements of 3.7.3.1.2 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.7.3.1.2)

The effect of alternate fuels on engine operation must be verified.

VERIFICATION GUIDANCE (A.4.7.3.1.2)

Past verification methods have included test.

The following should be used to tailor the specification paragraph:

If 3.7.3.1.2 is not required, delete the text in the paragraph and insert “Not Applicable”.

Background:

Alternate fuels testing is required only when option 3 of table XXIV is selected. Alternate fuels are essentially identical to primary fuels with the exception that they do not contain additives.

Durability tests should be conducted with the fuels of 4.7.3.1.1.

Some fuels shown in 4.7.3.1.1 contain FSII and lubricity improver additives; therefore the use of these fuels in AMT will result in higher durability.

VERIFICATION LESSONS LEARNED (A.4.7.3.1.2)

None.

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A.3.7.3.1.3 Restricted fuel.

The restricted fuel(s) the engine shall be capable of using shall be specified in table XXIV. Any limitations (fuel temperature limits, altitude, etc.) on the engine when using restricted fuels shall be specified herein.

REQUIREMENT RATIONALE (A.3.7.3.1.3)

Situations may occur where the engine has to use a specific fuel on an extremely limited basis. Use of a restricted fuel may have operational limitations associated with it.

REQUIREMENT GUIDANCE (A.3.7.3.1.3)

A restricted fuel is one which imposes operational restrictions on the aircraft when used.

The following should be used to tailor the specification paragraph: table XXIV should reflect the option chosen for the engine application. Refer to guidance of 3.7.3.1.

If no restricted fuel is required in the Option selected, delete the text in the paragraph and insert “Not Applicable”.

REQUIREMENT LESSONS LEARNED (A.3.7.3.1.3)

None.

A.4.7.3.1.3 Restricted fuel.

The requirements of 3.7.3.1.3 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.7.3.1.3)

Specific tests must be conducted to define the operating restrictions imposed by the use of “restricted fuels”.

VERIFICATION GUIDANCE (A.4.7.3.1.3)

Past verification methods have included test.

This paragraph is only necessary if restricted fuels are included in 3.7.3.1; otherwise this paragraph should be labeled “The use of restricted fuel is not required.”

Background:

Wide-Cut fuels can effect hot fuel operations and altitude relight envelopes. The specific test paragraphs where these tests are established should state that the fuel to be used would be JP-4 or Jet B.

VERIFICATION LESSONS LEARNED (A.4.7.3.1.3)

None.

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A.3.7.3.1.4 Emergency fuel.

The engine shall function satisfactorily for a time period of at least six hours from sea level to 35,000-ft altitude, at least throughout a range from idle to 90 percent of maximum thrust and at no greater than 120 percent of the specification rated or estimated specific fuel consumption when using automotive and/or marine diesel fuels (A-A-52557, MIL-PRF-16884, and ASTM D4814). Those external adjustments permitted in 3.7.2.2 shall be allowed in order to meet this requirement. If applicable, operating limitations, special inspection or maintenance actions required as a result of using this fuel shall be specified in the model specification. If applicable, emergency fuel shall be specified herein. If applicable, emergency fuel shall be specified herein.

REQUIREMENT RATIONALE (A.3.7.3.1.4)

This requirement identifies fuels which can be used in the engine but which may cause significant damage to the engine or other systems under extended use.

REQUIREMENT GUIDANCE (A.3.7.3.1.4)

An emergency fuel is one which significantly limits the overhaul life of the engine and imposes operational restrictions on the aircraft as well.

The following should be used to tailor the specification paragraph:

The contractor should specify all limitations when emergency fuel is used. The limitations should include but not be limited to altitude, airspeed, thrust or power, SFC, hot parts life reduction or other special maintenance and inspections and the period the engine is using the emergency fuel. Some emergency fuels that have been used are VV-F-800 (superseded by A-A-52557), MIL-F-16884 (superseded by MIL-PRF-16884), and MIL-G-3056 (superseded by ASTM D4814). It is good practice to avoid MIL-G-3056 gasoline (superseded by ASTM D4814) due to handling problems. Diesel fuels should conform to NATO Code F-54, OCONUS DF-2, with a sulfur mass of 0.30%, and a kinematic viscosity of at least 6.0 centistoke (64.6 ft^2/s) (9.5 centistokes (102.3 ft^2/s) desired) at 20°C (68°F).

Background:

The minimum operating time period, altitude and power ranges and maximum specific fuel consumption limit for which the engine should function satisfactorily when using the selected emergency fuels should be established by the Using Service based on mission requirements or a mission operational trade study. If applicable, operating limitations, special inspection or maintenance actions required, as a result of using an emergency fuel should be described in the specification.

Automotive diesel fuels should be considered as an emergency fuel due to their wide availability.

Automotive-type fuels generally have low lubricity characteristics that may adversely affect the life of fuel system components.

REQUIREMENT LESSONS LEARNED (A.3.7.3.1.4)

None.

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A.4.7.3.1.4 Emergency fuel.

The requirements of 3.7.3.1.4 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.7.3.1.4)

This verification is to ensure the engine's capability to accomplish typical missions with emergency fuel.

VERIFICATION GUIDANCE (A.4.7.3.1.4)

Past verification methods have included test.

Background:

Verification should determine all significant engine performance limitations using designated emergency fuels. The engine may require overhaul after completing any of the typically stated sorties.

Function and operation of the engine should be tested for compliance with design requirements, using the endurance test procedures as closely as possible, except as modified below:

a. The test should consist of at least 10 hours of engine operation.

b. Separate tests should be conducted for each fuel specified.

c. Engine calibration using primary fuel and recalibration using emergency fuel should be conducted to identify performance deterioration levels as defined in the pretest data.

d. Engine disassembly and inspection should be limited to the extent necessary to perform inspections of the hot section, including fuel nozzles.

e. This test will be considered satisfactorily completed, when the engine continues to meet the requirements specified and there are no part failures or impending failures which might compromise safety of flight.

Tertiary butyl disulfide should be added to the diesel fuel, if necessary, to obtain the required sulfur content.

VERIFICATION LESSONS LEARNED (A.4.7.3.1.4)

None.

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A.3.7.3.2 Fuel system performance.

A.4.7.3.2 Fuel system performance.

A.3.7.3.2.1 Fuel contamination.

The engine shall meet specification requirements when supplied with fuel contaminated in an amount up to the extent specified in table XXVa, and when supplied with (a) liters ((a) gallons) of fuel, contaminated per table XXVb, with the filter in full bypass. Engines for land based rotorcraft and engines with unassisted (suction) fuel systems shall meet specification requirements with fuel contaminated in amounts specified in table XXVc, with fuel filter not in bypass mode.

REQUIREMENT RATIONALE (A.3.7.3.2.1)

Contamination may be introduced into the fuel during maintenance, storage, or transfer systems from the atmosphere, through the aircraft vent systems or due to deterioration of materials used in the aircraft fuel systems. Engine development and fuel systems design requires consideration of contaminated fuel during long term continuous usage or usage for short periods of highly contaminated fuel.

The results of an Army survey support the table XXVa requirements. Land based rotorcraft operate in dirty air and fuel environment and need to withstand higher fuel contamination. Also, helicopters typically have unassisted fuel systems and less total filtration.

REQUIREMENT GUIDANCE (A.3.7.3.2.1)

The following should be used to tailor the specification paragraph:

(a): The Using Service should require a volume of fuel equivalent to at least that required to complete one mission.

The Using Service should specify the appropriate table for the application. Fixed wing turboprop aircraft that will be operated in and out of remote fixed base operating sites should use table XXVc levels of contamination.

The engine should be capable of meeting the specification requirements in the presence of air vehicle fuel tank inerting substances. Some substances used in the past are Halon, nitrogen and a foam material.

Background:

Engines should be tolerant of contaminated fuel. Engine filtration systems may be necessary to eliminate or reduce contamination levels for servo systems, pumps, nozzle systems and actuators. Tables XXVa and XXVb should be used for design and test purposes when specific aircraft system interface data has not been defined or is unavailable. The fuel system should be able to handle all of the fuel used by the engine during the mission that uses the most fuel with the filter in full bypass. The intent is to ensure that the aircraft can get home after the filter has gone into bypass. The engine should operate satisfactorily with the filter in bypass.

REQUIREMENT LESSONS LEARNED (A.3.7.3.2.1)

One engine was initially designed to accommodate or pass through fuel with contaminate particles up to the size passed through the refueling system. Experience has shown, however, that large particles, such as bits of safety wire which were introduced to critical parts/critical safety items of the engine fuel system, resulted in stuck fuel nozzle, a torched turbine vane, and a turbine failure caused by response to a one-per-rev pulse.

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Fuel contamination was a significant problem on land based helicopters in Desert Storm.

Fuel has been contaminated up to the levels contained in table XXVc at Army forward area refueling sites and in Marine environments during hot refueling.

A.4.7.3.2.1 Fuel contamination.

The requirements of 3.7.3.2.1 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.7.3.2.1)

The ability of the fuel system to handle contaminated fuel must be verified.

VERIFICATION GUIDANCE (A.4.7.3.2.1)

Past verification methods have included analysis to size the filter and look at wetted surface clearances and engine and component testing.

Background:

Test assemblies containing components normally in contact with fuel should be supplied with contaminated fuel downstream from the fluid tank with at least the amount of contaminant specified in table XXVa and XXVb. The solid contaminant should not be recirculated. During the test the fuel filters, if furnished for the engine fuel system, should be serviced as recommended by the engine manufacturer, but at intervals representing a volume of fuel equivalent to not less than that required in 3.7.3.2.1.

VERIFICATION LESSONS LEARNED (A.4.7.3.2.1)

Past verification methods have included missionized testing as described in 4.11.2.1. For economy of test reasons, table XXVa continuous duty testing has typically been limited to 300 hours for USAF engines.

A.3.7.3.2.2 Fuel system performance with external assistance.

The fuel system shall supply the required amount of fuel at the required pressure and temperatures for operation, including starting and augmentation (if applicable), with the following conditions at the fuel inlet connection:

Normal Operation:

a. Fuel temperature: From a minimum of (a) °C, ((a) °F) when using worst case allowable viscosity fuels of 3.7.3.1.1, to a maximum of no less than (b) °C, ((b) °F).

b. Fuel pressure: From a minimum of (c) kPa, ((c) psi) above the true vapor pressure of the fuel to a maximum of (d) kPa, ((d) psig) with a vapor/liquid volume ratio of (e) as defined at worst case operating conditions.

c. Fuel pressure surges: The fuel system shall accept air vehicle induced inlet pressure surges up to (f) kPa, ((f) psig) occurring at a maximum rate of change of up to (g) kPa, ((g) psi) per second. Engine induced pressure surges at the fuel inlet flange shall not exceed (h) kPa, ((h) psi) above the steady-state fuel inlet pressure.

REQUIREMENT RATIONALE (A.3.7.3.2.2)

This requirement provides information to the air vehicle contractor so that the aircraft system (e.g.; boost pumps, filters, lines, heat exchangers, valves, etc.) can be properly sized. The requirements of this paragraph tend to lead to an optimized sizing of the overall aircraft fuel pumping system. Specifying the

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minimum and maximum fuel pressures is to ensure that the air vehicle contractor provides a reasonable fuel pressure to obtain a vapor/liquid ratio of zero.

REQUIREMENT GUIDANCE (A.3.7.3.2.2)

The following should be used to tailor the specification paragraph:

(a): The Using Service should specify the coldest temperature of figure 5

(b): The maximum hot soak temperature of figure 5

(c), (d): The contractor should provide the design parameter values to define the system capabilities and limitations.

(e): The contractor should provide the design value, but 0 (zero) is desired.

(f) - (h): The contractor should provide the design parameter values to define the system capabilities and limitations.

Background:

Past engine programs have used the following values:

(c): 34 kPa (5 psi)

(d): 345 kPa (50 psig)

(e): 0

(f): 689 kPa (100 psig)

(g): 13.79 mPa (2,000 psi)

(h): 689 kPa (100 psi)

The minimum fuel temperature should be found on figure 5 for the sea level cold atmosphere. This represents the coldest temperature the fuel will be at in service when the engine is initially started. The maximum fuel temperature should be that which would permit the air vehicle contractor to use the fuel as a cooling medium. For high altitude, hot fuel conditions this requirement will be more difficult to achieve. Minimum fuel pressure value is necessary for aircraft designs in general to meet almost all operating conditions. Systems have been designed to operate at a minimum pressure of 14 to 21 kPa (2 to 3 psi) above the True Vapor Pressure (TVP). Engine D's design requirement includes one psi above TVP.

The maximum fuel pressure must be coordinated between the aircraft and engine manufacturers to assess the impact of high-pressure operation on system strength and weight. This pressure represents the no flow pressures from aircraft tank mounted boost pumps and takes into consideration the fuel feed requirements, aircraft weight and line size tradeoffs. Fuel systems have exceeded the 345 kPa (50 psig) required in previous specifications. The value should not be considered a state-of-the-art limitation. Previous operating pressures of 414 kPa (60 psi) and proof pressures of 827 kPa (120 psi) were consistent with the aircraft fuel system structural design requirements. Ideally the air vehicle contractor attempts to design the aircraft system to the lowest pressure that will meet the engine's V/L = 0 requirement. Engine AA's fuel system operates up to a maximum inlet gage pressure of 517 kPa (75 psig) and surge peaks up to 1724 kPa (250 psi). The aircraft induced pressure variations have been compensated for by the pump inlet structure design.

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A minimum pressure of 34 kPa (5 psi) above the true vapor pressure of the fuel should be maintained to assure the fuel (mainly JP-4) will not boil or the system may become vapor locked at elevated temperatures.

This paragraph defines the fuel properties at the fuel inlet interface. From a structural integrity aspect the fuel inlet should be designed to withstand the startup surges of the aircraft system or the emergency shutdown induced surges.

REQUIREMENT LESSONS LEARNED (A.3.7.3.2.2)

The earlier design requirements that the fuel supplied to the engine at no less than 103 kPa (15 psia) was criticized as unrealistic. This design condition, in some applications, required the design and installation of oversize fuel boost pumps and did not take into account the margin of pressure above the vapor pressure of 5 psi plus the true vapor pressure of the fuel (a design condition that is feasible within present state-of-the-art pump design). Operating pressures at the engine interface of 517 kPa (75 psi) have been designed with surge pressures up to 1724 kPa (250 psi), with resulting higher weight to provide structural integrity.

A study of Aircraft R's Life Cycle Costs (LCC) related to this subject showed that by operating the engine with 93°C (200°F) fuel rather than 57°C (135°F) fuel (an old limit), the aircraft ram air coolers would not be required at a savings of $200M. Trade studies of this type should be performed on all new engine and aircraft programs. Fuel at temperatures of 149 - 177°C (300 - 350°F) may begin to suffer stability problems, causing deposits and rapid nozzle performance deterioration. Other disadvantages are rising fuel temperatures at low fuel quantities and operation of the ECS during ground maintenance.

High fuel pressure is associated with pump shaft seal leakage and if the designer is unaware of the high pressures, the pump may be improperly designed.

Engine AL's flight-test engines for Aircraft H, allowed fuel inlet temperatures up to 235°F (based on JP-4 and fuel pressurization). However due to significant aircraft system and engine heat loads and the use of fuels as the heat sink medium, an integrated fuel thermal management approach was adopted. Critical limiting parameters were sensed and a modulating fuel bypass control was incorporated to allow maximum heat rejection without compromising system limits. Protection was provided for fuel nozzle temperature (coking), fuel inlet temperature (vaporization), engine hydraulic fluid and lube oil temperature, as well as specific aircraft system temperatures.

A.4.7.3.2.2 Fuel system performance with external assistance.

The requirements of 3.7.3.2.2 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.7.3.2.2)

The adequacy of the fuel system must be verified by test observations and experience gained during qualification.

VERIFICATION GUIDANCE (A.4.7.3.2.2)

Past verification methods have included analyses, demonstrations and tests including component tests, low and high temperature engine tests, alternate fuel and emergency fuel tests.

Background:

During the endurance and altitude tests the fuel system is required to operate at the pressure and temperature extremes specified for the engine. Observations of the fuel system function and performance throughout these tests should be recorded and included in the test reports for use in

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verifying the fuel system. Verification should be obtained by the fuel pump altitude proof test, the component tests, the low and high temperature engine tests, and the alternate and emergency fuel tests.

VERIFICATION LESSONS LEARNED (A.4.7.3.2.2)

None.

A.3.7.3.2.3 Fuel system performance with no external assistance.

The fuel system shall supply the required amount of fuel at the required pressure and temperatures for operation, including starting and augmentation (if applicable), with the following conditions at the fuel inlet connection:

a. Fuel Temperature: From a minimum of (a) °C ((a) °F) when using worst case primary fuel or (b) centistoke, whichever is worse, to a maximum of no less than (c) °C ((c) °F).

b. Fuel Pressures: From a minimum of (d) above the true vapor pressure of the fuel to a maximum of (e).

c. Vapor/liquid volume ratio from zero to (f).

d. For turboshaft engines, vapor shall be introduced continuously as vapor only (bubbles) equivalent to 300 mm (11.8 in.) of fuel line of the same area as the engine fuel inlet followed by liquid fuel only equivalent to 300 mm (11.8 in.) of fuel line of the same area as the engine fuel inlet.

e. If required, quantity of air to be ingested (g).

REQUIREMENT RATIONALE (A.3.7.3.2.3)

The fuel system must supply fuel at the required pressures and temperatures when it is designed as a suction system.

REQUIREMENT GUIDANCE (A.3.7.3.2.3)

The following should be considered for tailoring the specification paragraph:

(a): The Using Service should specify the coldest temperature of figure 5.

(b): The centistoke value based on the temperature in (a).

(c): The maximum hot soak temperature of figure 5.

(d): Minimum fuel pressure of 1 psi above TVP.

(e): Maximum fuel pressure not less than 50-psig.

(f): V/L not less than 1.0.

(g): Define any requirements for air ingestion.

Background:

For aircraft with a suction fuel system, the fuel system must provide the required amount of fuel up to the highest altitude required in 3.2.2.1, the ambient temperatures of figure 5, the externally applied forces of 3.4.1.2.1 and the ground and air start requirements of 3.2.2.3. The maximum fuel temperature should include the effects of a wing tank being heat soaked by the sun in conjunction with a low fuel condition in the aircraft when the fuel in used as a cooling medium (heat sink).

Suction feed systems are desired on helicopters and other low altitude mission aircraft to provide improved crash worthiness and increased safety against ballistics that may result in fuel leakage and a fire.

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REQUIREMENT LESSONS LEARNED (A.3.7.3.2.3)

Engines E, G, and K were examples of engines designed with a suction fuel system. Most suction systems in Army experience have not worked well and have continuing problems with flameouts and operability due to air vehicle fuel system layout, refueling configuration, and fuel aeration (bubbles).

A.4.7.3.2.3 Fuel system performance with no external assistance.

The requirements of 3.7.3.2.3 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.7.3.2.3)

The adequacy of the fuel system to function as a suction feed, either normally or in emergency situations, must be verified by specific component tests.

VERIFICATION GUIDANCE (A.4.7.3.2.3)

Past verification methods have included analysis and test.

The following Fuel System Suction Test, Fuel Pump Altitude Test, and Fuel Pump Cavitation procedures may be used as a verification method:

Fuel System Suction Test. Prior to completion of FFR testing, a fuel system suction test shall be conducted. This test shall include a simulation of temperature soak-back on engine starting, and operation at the extremes of fuel and air temperature and altitude environment of the engine fuel system. Fuel system performance parameters shall be demonstrated with the designated primary fuel that is most critical for that parameter. Performance parameters that are critical due to vapor and air evolution shall be demonstrated with JP-4, if JP-4 is designated as a primary or alternate fuel. The test shall be conducted within the guidelines established in (4.11.1.2, Altitude Test). Demonstration test points shall include: (1) sea level, (a) fuel temperature, and (b) surrounding air temperature; (2) 1.2 km (3937 ft), (c) fuel temperature, and (d) surrounding air temperature; and (3) 3.0 km (9842 ft), (e) fuel temperature, and (f) surrounding air temperature. Soak-back testing shall be accomplished on the first two test conditions. For soak-back testing, the engine shall be operated for 30 minutes before shutdown. A restart shall be attempted after soak periods of 15 and 30 minutes."

The following should be used to tailor the above paragraph:

(a): 68°C (154°F)

(b): 70°C (158°F)

(c): 68°C (154°F)

(d): 70°C (158°F)

(e): 55°C (131°F)

(f): 38°C (100°F)

All components shall then be completely disassembled and inspected for indications of failure or excessive wear.

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The following Fuel Pump Altitude Test and Fuel Pump Cavitation procedures have been used as verification methods:

For IFR:

FOR TURBOFAN and TURBOJET APPLICATIONS:

Fuel Pump Altitude Test. The portion of the engine fuel system from the fuel inlet of the engine to the engine fuel pump inlet and augmentation pump inlet, if applicable, shall be simulated in the test assembly. This shall include lines, fittings, filters, and other items, as applicable, between the engine fuel inlet and fuel pump inlet, as well as any elements of the fuel system that might have an effect on the pump such as an interstage fuel heater. All independent and separately replaceable fuel pumps shall be operated for five hours at flow and pressure conditions corresponding to those required by design maximum engine performance at 30,000 feet altitude conditions. Performance parameters that are critical due to vapor and air evolution shall be demonstrated with JP-4, if JP-4 is designated as a primary or alternate fuel. The fuel pump shall be operated with pressure on the fuel tank maintained at 4.37 psia (30,000 ft. altitude) and an engine fuel inlet temperature of 43°C (110°F). The fuel vapor/liquid ratio at the engine fuel inlet shall be maintained at not less than 0.45. Except as stated above, the test procedure shall be per ARP 492. During this test the fuel pump dry lift capability defined in 3.7.3.2.5 shall also be verified.

The fuel pump altitude test shall be considered to be satisfactorily completed when, in the judgment of the Using Service, the pump performance during the course of testing and the performance deterioration, determined from calibration runs, conform to the requirements established by the applicable design specification and do not adversely affect engine performance; the pump dry lift capability as defined in 3.7.3.2.5 has been demonstrated; and the component teardown inspection does not disclose parts failure or impending failures.

For ISR:

Fuel Pump Cavitation. For the fuel pump cavitation test, the portion of the engine fuel system from the fuel inlet of the engine to the engine fuel pump inlet and augmentation pump inlet, if applicable, shall be simulated in the test assembly. This shall include lines, fittings, filters, and other items, as applicable, between the engine fuel inlet and fuel pump inlet, as well as any elements of the engine fuel system downstream of the pump that might have an effect on the pump. Prior to the start of this test, the system shall have had fluid passed through it at intermediate power engine fuel flow for two hours contaminated with at least twice the amount of contaminant specified in tables XXVa and XXVb. Unweathered, clean fluid may be used to conduct the test. Fuel pump cavitation performance shall be demonstrated with the designated primary fuel that is most critical for vapor and air evolution. The primary and secondary pumps shall be operated for 47 hours at intermediate rated speed and at the maximum flow and discharge pressure required by the engine at sea level standard altitude absolute condition and ram pressure ratio of 1.15. Pressure on the fuel tank will be maintained at 20 inches Hg absolute during the cavitation test. An additional restriction will be introduced ahead of the engine fuel inlet to provide the required vapor/liquid ratio. The fluid vapor/liquid ratio at the engine inlet shall be maintained at not less than 0.45 and the fluid temperature shall be at least 43°C (110°F). Augmentation pumps shall be operated at the above conditions at least 12 hours, continuously. Except as stated above the test procedure shall be per ARP 492. All components shall then be completely disassembled and inspected for indications of failure or excessive wear. (See 4.11.2.1.6 for success criteria.)

For IFR:

FOR TURBOSHAFT and TURBOPROP APPLICATIONS:

Fuel Pump Altitude Test. The portion of the engine fuel system from the fuel inlet of the engine to the engine fuel pump inlet shall be simulated in the test assembly. This shall include lines, fittings, filters and other items, as applicable, between the engine fuel inlet and fuel pump inlet, as well as any elements of the fuel system downstream of the pump that might have an effect on the pump. An additional restriction

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will be introduced ahead of the engine fuel inlet to provide the required vapor/liquid ratio. All independent and separately replaceable fuel pumps shall be operated for:

a. One hour at the flow, pressure, and speed corresponding to those required by design maximum engine performance at sea level, 35°C (95°F) conditions with a fuel temperature of at least 57°C (135°F) and a fuel inlet pressure not more than the minimum specified in 3.7.3.2.3, followed by one hour at the flow, pressure, and speed corresponding to those required by design idle engine performance at sea level, 35°C (95°F) conditions with a fuel temperature of at least 57°C (135°F) and a fuel inlet pressure as established by the line restriction determined above.

b. One hour at the flow, pressure, and speed corresponding to those required by design maximum engine performance at 7.0 km (22966 ft), -10°C (14°F) conditions with a fuel temperature of at least 57°C (135°F) and a fuel inlet pressure not more than the minimum specified in 3.7.3.2.3, followed by one hour at the flow, pressure, and speed corresponding to those required by design idle engine performance at 7.0 km (22966 ft), -10°C (14°F) conditions with a fuel temperature of at least 57°C (135°F) and a fuel inlet pressure as established by the line restriction determined above.

c. One hour at the flow, pressure, and speed corresponding to those required by design maximum engine performance at 9.0 km (29528 ft), -25°C (-13°F) conditions with a fuel temperature of at least 57°C (135°F) and a fuel inlet pressure not more than the minimum specified in 3.7.3.2.3, followed by one hour at the flow, pressure, and speed corresponding to those required by design idle engine performance at 9.0 km (29528 ft), -25°C (-13°F) conditions with a fuel temperature of at least 57°C (135°F) and a fuel inlet pressure as established by the line restriction determined above.

The fuel shall be per MIL-T-5624, grade JP-4. Except as stated above, the test procedure shall be per ARP492. During this test the fuel pump dry lift capability defined in 3.7.3.2.5 shall also be verified. The fuel pump altitude test shall be considered to be satisfactorily completed when, in the judgment of the Using Service, the pump performance during the course of testing and the performance deterioration, determined from calibration runs, conform to the requirements established by the applicable design specification and do not adversely affect engine performance; the pump dry lift capability as defined in 3.7.3.2.5 has been demonstrated; and the component teardown inspection does not disclose parts failure or impending failures.

For ISR:

Fuel Pump Cavitation. The portion of the engine fuel system from the fuel inlet of the engine to the engine fuel pump inlet and augmentation pump inlet, if applicable, shall be simulated in the test assembly. This shall include lines, fittings, filters, and other items, as applicable, between the engine fuel inlet and fuel pump inlet, as well as any elements of the engine fuel system downstream of the pump that might have an effect on the pump. Prior to the start of this test, the system shall have had fluid passed through it at maximum continuous power engine fuel flow for five hours contaminated with at least as much contaminant specified in tables XXVa and XXVb. Clean fluid may be used to conduct the test. Each test cycle shall consist of 45 minutes at the maximum speed and flow required by the engine at the altitude corresponding to the fuel tank pressure followed by 15 minutes at the minimum idle speed and flow required at the same altitude. The test shall consist of two parts described below.

(1) For part one, the fuel shall conform to MIL-DTL-5624, grade JP-4; or MIL-DTL-83133, grade JP-8. A restriction shall be introduced ahead of the engine fuel inlet to provide a fuel pressure at the engine fuel inlet during the maximum speed and flow portion of each cycle of not more than 7.0 kPa (1.0 psi) above the true vapor pressure of the fuel or not more than the fuel pressure which corresponds to a fuel vapor to liquid (V/L) ratio of 1.0, whichever is the higher pressure. The restriction shall be maintained during the idle speed and flow portion of each cycle. Part one shall consist of:

a. 20 cycles with pressure on the fuel tank maintained at 101 kPa (14.6 psi) and with engine fuel inlet temperature of 71°C (160°F).

b. 20 cycles with pressure on the fuel tank maintained at 94.2 kPa (13.7 psi) and with engine fuel inlet temperature of 71°C (160°F).

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c. 20 cycles with pressure on the fuel tank maintained at 64.2 kPa (9.3 psi) and with engine fuel inlet temperature of 71°C (160°F).

d. 20 cycles with pressure on the fuel tank maintained at 94 kPa (13.6 psi) and with engine fuel inlet temperature of 57°C (135°F).

During the last 10 of the cycles below (e, f, g), the vapor shall be introduced continuously as vapor only (bubbles) equivalent to 300 mm (12 in.) of fuel line of the same area as the engine fuel inlet followed by liquid fuel only equivalent to 300 mm (12 in.) of fuel line of the same area as the engine fuel inlet.

a. 20 cycles with pressure on the fuel tank maintained at 78.8 kPa (11.4 psi) and with engine fuel inlet temperature of 57°C (135°F).

b. 20 cycles with pressure on the fuel tank maintained at 48.3 kPa (7.0 psi) and with engine fuel inlet temperature of 57°C (135°F).

c. 20 cycles with pressure on the fuel tank maintained at 82.0 kPa (11.9 psi) and with engine fuel inlet temperature of 57°C (135°F).

During the last 10 of the cycles below (h, i, j, k, l), the vapor shall be introduced continuously as vapor only (bubbles) equivalent to 300 mm (12 in.) of fuel line of the same area as the engine fuel inlet followed by liquid fuel only equivalent to 300 mm (12 in.) of fuel line of the same area as the engine fuel inlet.

a. 20 cycles with pressure on the fuel tank maintained at 67.0 kPa (9.7 psi) and with engine fuel inlet temperature of 43°C (109°F).

b. 20 cycles with pressure on the fuel tank maintained at 37.3 kPa (5.4 psi) and with engine fuel inlet temperature of 43°C (109°F).

c. 20 cycles with pressure on the fuel tank maintained at 76.0 kPa (11.0 psi) and with engine fuel inlet temperature of 29°C (84°F).

d. 20 cycles with pressure on the fuel tank maintained at 61.5 kPa (8.9 psi) and with engine fuel inlet temperature of 29°C (84°F).

e. 20 cycles with pressure on the fuel tank maintained at 37.3 kPa (5.4 psi) and with engine fuel inlet temperature of 29°C (84°F).

(2) For part two, the fuel shall conform to MIL-T-83133, grade JP-8. A restriction shall be introduced ahead of the engine fuel inlet to provide a fuel pressure at the engine fuel inlet during the maximum speed and flow portion of each cycle of not more than 35 percent of the fuel tank pressure or 15.0 kPa (2.2 psi) absolute, whichever is the higher pressure. This restriction shall be maintained during the idle speed and flow portion of each cycle. The fuel temperature at the engine fuel inlet shall be maintained to provide a fuel viscosity of not less than 12 centistoke. Part two shall consist of:

a. 20 cycles with pressure on the fuel tank maintained at 100 kPa (14.5 psi) absolute.

b. 20 cycles with pressure on the fuel tank maintained at 60 kPa (8.7 psi) absolute.

c. 20 cycles with pressure on the fuel tank maintained at 30 kPa (4.4 psi) absolute.

Except as stated above the test procedure shall be per ARP 492.

(See 4.11.2.1.6 for success criteria.)

Background:

Bench testing should be used to demonstrate the capability of the fuel system to operate without external assistance at the worse case conditions.

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A fuel pump test should be run with the vapor/liquid (V/L) ratio at the pump inlet from 0 to the section 3 limit. Additionally, a fuel pump cavitation test should be conducted on the portion of the fuel system from the engine fuel inlet to the primary and augmentation pumps (as applicable), with the V/L simulating aircraft fuel system conditions at the maximum fuel flow rates and the lowest inlet fuel pressure for that condition.

VERIFICATION LESSONS LEARNED (A.4.7.3.2.3)

Engine C was the first engine specifically designed for a suction fuel system. The problems experienced on Aircrafts D and E indicated that the design must consider actual phases of liquid separated from fuel phases in addition to high V/L ratios. Engine D was the first attempt to include this type of vapor and fuel separation (bubbles). The resulting fuel system design will handle large bubbles and has significant dry lift capability. These capabilities are not in any other current Army engine.

A.3.7.3.2.4 Fuel system performance under conditions of excessive fuel vapor.

When the vapor to liquid ratio at the fuel inlet is greater than _____, it shall not be necessary for the fuel system to supply fuel for engine operation. However, after the conditions of 3.7.3.2.2 or 3.7.3.2.3 have been reestablished, it shall be possible to accomplish normal starts and operate satisfactorily thereafter.

REQUIREMENT RATIONALE (A.3.7.3.2.4)

Following conditions of excessive fuel vapors, the engine must operate satisfactorily after re-establishment of normal operating conditions has occurred.

REQUIREMENT GUIDANCE (A.3.7.3.2.4)

The following should be used to tailor the specification paragraph:

For JP-5, JP-8 and JP-4 fuels the V/L ratio should be 1.0. The V/L requirement in this paragraph must be coordinated with the V/L requirements in 3.7.3.2.2 and 3.7.3.2.3.

Vapor relief provisions should be incorporated to vent excess vapor during pump priming.

Background:

When the maximum V/L ratio of 3.7.3.2.2 or 3.7.3.2.3 is exceeded, the fuel system must clear the excess vapor or air from the system when the inlet conditions of 3.7.3.2.2 or 3.7.3.2.3 have been reestablished, and the engine must be able to be restarted and operate satisfactorily.

In systems using fuel as an air vehicle system cooling medium, fuel pressurization may be required continuously to prevent total system failure due to vaporization in the aircraft plumbing and heat exchangers, particularly with JP-4 and when actual failure scenarios are considered. In general, such systems will require a “prime reliable” aircraft boost system.

Without venting, a severe vapor lock problem could occur in the fuel system quickly if the aircraft incurred a boost pump failure above 9.1-km (30,000 feet) (for a constant aircraft fuel system pressure drop, V/L ratio increases rapidly as altitude increases). Venting the fuel system as the altitude decreases below 9.1-km (30,000 feet) would be expected to permit the engine to start and operate per 3.7.3.2.2 and 3.7.3.2.3.

REQUIREMENT LESSONS LEARNED (A.3.7.3.2.4)

Excessive fuel system vapor/liquid ratios have resulted in lean combustor flameouts on trainer aircraft.

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A.4.7.3.2.4 Fuel system performance under conditions of excessive fuel vapor.

The requirements of 3.7.3.2.4 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.7.3.2.4)

The requirements of this paragraph must be verified by test.

VERIFICATION GUIDANCE (A.4.7.3.2.4)

Past verification methods have included test.

VERIFICATION LESSONS LEARNED (A.4.7.3.2.4)

None.

A.3.7.3.2.5 Fuel pump priming.

The fuel pump shall be self priming and be capable of a dry lift of at least (a) meter, ((a) feet) at a minimum absolute pump inlet pressure of (b) kPa, ((b) psia), using the fuels of 3.7.3.1 with the RVP not to exceed (c) kPa, ((c) psi), and a fuel temperature not to exceed (d) °C ((d) °F).

REQUIREMENT RATIONALE (A.3.7.3.2.5)

The fuel pump must be capable of priming itself and starting the engine with air in the fuel lines.

REQUIREMENT GUIDANCE (A.3.7.3.2.5)

The following should be used to tailor the specification paragraph:

(a): The vertical distance from the fuel tank to the inlet.

(b) - (d): The contractor should provide the numbers.

Background:

The requirements of 3.7.3.2.5 need not apply to operation with emergency fuel.

For fixed wing aircraft, a self-priming or suction capability is needed to provide limited starting and operating capability with no external assistance (boost pressure) from the air vehicle fuel system. A gravity feed aircraft fuel system with sufficient pressure head would negate this requirement.

The fuel pump should be capable of priming itself when subjected to a dry lift of the vertical distance from the fuel tank to the inlet or of at least 0.3 meter (1 foot) at an absolute pump inlet pressure of 30.4 kPa (9 in Hg) with the Reid Vapor Pressure not to exceed 21 kPa (3 psi), and at a fuel temperature not to exceed 43°C (110°F). For turboshaft engines, a 0.46-meter (1.5-foot) dry lift is recommended at these conditions.

The Army allows initial priming of the fuel system by an externally supplied and driven pump for cases where the engine is high above the fuel tank. For large vertical distances an external means may be necessary to supply fuel to the engine.

The fuel pump may have to be vented if air or vapor created after the failure of the aircraft boost pump cannot be pumped through the fuel system.

The requirements of this paragraph need not apply to operation with emergency fuel.

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Engine fuel lines between the inlet and the fuel pump should perform satisfactorily and should not collapse at standard sea level conditions when the minimum fuel pressure is applied.

REQUIREMENT LESSONS LEARNED (A.3.7.3.2.5)

None.

A.4.7.3.2.5 Fuel pump priming.

The requirements of 3.7.3.2.5 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.7.3.2.5)

The capability of the fuel pump to prime itself must be verified.

VERIFICATION GUIDANCE (A.4.7.3.2.5)

Past verification methods have included engine and component tests.

Fuel pump priming may be verified during the fuel pump altitude and cavitation tests.

VERIFICATION LESSONS LEARNED (A.4.7.3.2.5)

None.

A.3.7.3.2.6 Fuel lubricity.

The engine shall operate satisfactorily when using fuel with a lubricity value equivalent to a 0.75 - 0.90-mm (0.030 - 0.035 inch) Wear Scar Diameter (WSD) as defined by ASTM D5001.

REQUIREMENT RATIONALE (A.3.7.3.2.6)

The engine must operate satisfactorily with a low lubricity fuel.

REQUIREMENT GUIDANCE (A.3.7.3.2.6)

The Ball-On-Cylinder Lubricity Evaluator (BOCLE) measures Wear Scar Diameter.

Background:

The requirement paragraph applies to commercial fuels with no lubricity additives. Military specification fuels have a lubricity value lower than 0.42 mm.

Current and future fuels are being derived from lower quality crude oil sources to improve fuel availability. These crude oils require higher levels of refining which are detrimental to fuel lubricity. Poor lubricity fuels have BOCLE WSD diameters as high as 0.90 mm (0.035 inch). This value was determined to be a realistic value through extensive research at United Technology Hamilton Standards Test Facility. Pump failures during accelerated mission tests and bench type endurance runs have revealed a variety of failure modes.

REQUIREMENT LESSONS LEARNED (A.3.7.3.2.6)

This requirement resulted from fuel component problems on several engines (Engine M's fuel control, Engine L's fuel pump, etc.) caused by low lubricity fuel acquired from several refineries. Adding a corrosion inhibitor to the fuel increases the lubricity of the fuel. Hydromechanical controls had numerous problems with low lubricity fuels.

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A.4.7.3.2.6 Fuel lubricity.

The requirement of 3.7.3.2.6 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.7.3.2.6)

Engine operation with a low lubricity fuel must be verified.

VERIFICATION GUIDANCE (A.4.7.3.2.6)

Past verification methods have included analysis, demonstration and test. Component tests using a simulated endurance cycle have also been used. SAE ARP 1797 provides guidance for test facility setup. A fuel's lubricating characteristics are measured by ASTM D5001, Standard Test Method for Measurement of Lubricity of Aviation Turbine Fuels by the Ball-on-Cylinder Lubricity Evaluator (BOCLE).

The following procedure may be used as a verification method:

Fuel systems assemblies or components normally requiring fuel for self-lubrication shall undergo a simulated mission operational test for at least 100 hours. If necessary, the cycle shall be modified to give a minimum of 50 throttle excursions per hour. The fuel shall be per MIL-DTL-5624, grade JP-5 degraded to provide the lubricity of 3.7.3.2.6.

Background:

Engine fuel system components which have moving parts (such as pumps, valves, and regulators) should be subjected to a low lubricity fuel test. The test should be per SAE ARP 1797 except that the components shall be subjected to twenty cycles of five hours of operating time followed by five hours of non-operating time. The fuel system components should be subjected to an acceptance test prior to and after the low lubricity test. Performance should conform to new part standards.

VERIFICATION LESSONS LEARNED (A.4.7.3.2.6)

None.

A.3.7.3.2.7 Fuel system performance with water saturated fuel.

The engine shall meet the requirements of this specification when supplied with fuel saturated with water up to ____ at worst case fuel pressure and temperature conditions.

REQUIREMENT RATIONALE (A.3.7.3.2.7)

Prevention of ice buildup in the fuel filters and screens is necessary for acceptable system safety.

REQUIREMENT GUIDANCE (A.3.7.3.2.7)

The Using Service may provide the value based on the anticipated usage or past field experience. SAE ARP 1401 can be used for guidance with the exception of testing at -18°C (0°F) in lieu of the temperatures cited in the ARP. This requirement is applicable when operation with fuel without anti-icing inhibitor (FSII) is required or Service experience shows that delivered fuel does not comply with FSII minimum requirements. (See 3.7.3.1 Lessons Learned) An alternative to this requirement is to require anti-icing additive to be added in the field to fuel without FSII or to limit operational capability.

Background:

When no fuel anti-icing additive is used, a fuel anti-icing heat exchanger type system should be incorporated. If an air-fuel heat exchanger is used, the contractor should specify provisions required for the cooling air. The fuel anti-icing heat exchangers should be installed upstream of the fuel filter and fuel

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control. It should be a pilot manually selected type, incorporating a warning feature for cockpit indication during icing conditions. A means should be provided to indicate when the fuel anti-icing system is in operation. Operational requirements should determine to a large extent whether a heat exchanger system is used. A fuel system icing inhibitor should be incorporated into locally available fuel. The inhibitor chemical composition, use, and handling should conform to MIL-DTL-85470. When the fuel system design does not incorporate an anti-icing heat exchanger and an alternate commercial fuel (such as Jet A1) is being used, engine operation should be limited to areas of the flight envelope that preclude freezing of moisture in the fuel system. Methods of fuel system anti-icing should perform so as not to affect the fuel system components or engine operation and should provide complete fuel icing protection.

If a fuel heater is required for the fuel system to perform with water saturated fuel, the form, fit, and functional interfaces should be described.

REQUIREMENT LESSONS LEARNED (A.3.7.3.2.7)

Army helicopters have not experienced fuel icing problems in the field, since they use FSII in the fuel and consequently fuel de-icing using fuel system heaters is not a requirement on Army engines.

Fuel anti-icing additives have the beneficial side effect of reducing microbiological contamination in the fuel tank(s).

A.4.7.3.2.7 Fuel system performance with water saturated fuel.

The requirements of 3.7.3.2.7 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.7.3.2.7)

The capability of the fuel system to perform at low temperatures with water saturated fuel must be verified.

VERIFICATION GUIDANCE (A.4.7.3.2.7)

Past verification methods have included analysis and test.

The following procedure may be considered as a verification method:

The engine should be supplied with water saturated fuel to demonstrate that icing does not adversely affect engine operation. The fuel inlet temperatures should be set at -18°C (0°F) and the minimum temperature specified in the specification. Fuel should be specified by the Using Service (either MIL-DTL-5624 [grade JP-5], or MIL-DTL-83133 [grade JP-8]) and should not contain a fuel system icing inhibitor. At each inlet temperature, fuel which has been initially saturated with water at 27°C (80°F) and having 0.75 cc of free water per gallon per SAE ARP 1401 and MIL-F-17874, should be supplied to the engine. The engine should be operated for the length of the design mission plus one in-flight refueling if the aircraft is so equipped or for the three (3) hours (longest ferry mission), whichever is longer at each fuel inlet temperature using the AMT cycle. This test should be considered satisfactorily completed when there has been no deterioration in the steady-state performance or loss of the capability to perform a transient during or after the test. It is recommended that the anti-ice/de-ice system be activated during the test to evaluate any effects on it.

VERIFICATION LESSONS LEARNED (A.4.7.3.2.7)

It is not practical to carry out a fuel icing test with the inlet temperature controlled to -3.2 to 1.2°C for a period of 3 hours due to the difficulty of controlling the bulk fuel temperature to such tight limits. Navy experience is that the effects of iced fuel between -1°C and -10°C are very similar. A test with saturated fuel at a temperature above 0°C should be carried out to assess the effects of free water.

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A.3.7.3.2.8 Fuel filter.

Fuel filtration on the engine shall be provided, as needed, for the engine to meet the requirements of the specification. The fuel filter shall consist of (a) filter(s), and shall be made of a material that does not cause fuel contamination due to media migration. The filtration ratings of the filter elements, contaminated per table XXVa, shall be per table XXVI.

The filter assembly shall incorporate a pressure relief bypass and shall prevent the discharge of filter contaminant through the bypass. The filter assembly shall have (b) indications of both impending and actual filter bypass. The indication shall remain activated until the filter is removed for inspection. The filter shall be removable without spills or draining of the fuel lines. Fuel pressure surges during operation within the limits specified in 3.7.3.2.2 shall not cause false activation of the impending or actual bypass indicators.

REQUIREMENT RATIONALE (A.3.7.3.2.8)

Fuel filtration is necessary to ensure successful engine operation in a contaminated fuel environment.

REQUIREMENT GUIDANCE (A.3.7.3.2.8)

The following should be used to tailor the specification paragraph:

(a) The contractor should specify if the filter is cleanable by including either “cleanable” or “noncleanable” in the requirement.

(b) The contractor should complete the blank with “remote”, “visual” or “remote and visual”. A remote indication would be needed if cockpit warning or EMS notification is required. If only a pop out button is required, then “visual” should be used. The fuel filter impending bypass indicator with pressure activated pop-out buttons should be manually reset, internally requiring disassembly of the filter. Bypass indicators with electronic signals may not require manual resetting to clear the flag if the impending bypass signal is retained as a maintenance action indication.

The following should be used to tailor table XXVI.

Filter capacity from impending bypass to bypass mode should be sufficient to allow completion of at least two combat missions.

Background:

Fuel filter requirements should be tailored to include the following system design and envelope requirements: fuel flow rates, capacity, mesh size, media, fuel viscosity, and scheduled servicing intervals (removals). Desirable features to be considered in filter selection are: visual impending and actual bypass indication (pop out button), remote impending and actual bypass indication (to cockpit or monitoring system), internal or external reset for indicator, and bowl removal shutoff valve.

Redundant indications for filter impending bypass may be unnecessary, depending on the system design. Some modern engines have bypass indications automatically passed to the engine and aircraft health monitoring systems.

The type of element material must be compatible with the fuel to prevent adverse reactions. The capacity of the filter element must be specified since it will determine the frequency of element removal and cleaning.

The fuel filtration may be supplemental or accomplished by air vehicle filters. Fuel capacity should be based on anticipated maintenance procedures and practices. The manufacturer must size the filter using the dirty fuel specified in conjunction with the amount of fuel used during the mission that consumes the

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most fuel. The definition of dirty fuel is provided for standardization. The pressure relief bypass is necessary to prevent system damage in the event of excessively high pressure due to a plugged filter element. The pressure relief feature must not allow debris on the element to be washed through the filter housing to prevent contamination of the whole fuel system. The filter assembly will provide visual indication of impending and actual bypass when the differential pressure across the element exceeds specified values. Visual indication of impending and actual bypass alerts the ground crew that the filter element must be changed. The remote indication provides bypass information to the monitoring system or cockpit.

The manual reset requirement is necessary to prevent maintenance personnel from resetting the indicator without removing, cleaning, or replacing the filter element, which would defeat the purpose of the warning indicator.

REQUIREMENT LESSONS LEARNED (A.4.7.3.2.7)

A fuel strainer located between the augmentor fuel-oil cooler and the augmentor fuel control was originally positioned horizontally. When serviced, contaminants, which had collected near the strainer lip, would contaminate the augmentor fuel control and require its removal and return to depot for repair. By repositioning the fuel strainer vertically, servicing could be accomplished with minimal chance of contaminating the augmentor fuel control.

A.4.7.3.2.8 Fuel filter.

The requirements of 3.7.3.2.8 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.7.3.2.8)

The fuel filter requirements must be verified to ensure the filter will remove the required level of contaminants.

VERIFICATION GUIDANCE (A.4.7.3.2.8)

Past verification methods have included component missionized testing. Fuel filter durability and maintenance features should be demonstrated during the engine endurance test.

VERIFICATION LESSONS LEARNED (A.4.7.3.2.8)

None.

A.3.7.3.2.9 Fuel flow limit.

The maximum and minimum fuel flow shall be specified in _____.

REQUIREMENT RATIONALE (A.3.7.3.2.9)

The air vehicle contractor must specify all limits so the Using Service knows the full range of the engine capabilities and for use. Fuel flow limits affect fuel system design and performance limits for maximum power and idle.

REQUIREMENT GUIDANCE (A.3.7.3.2.9)

The following should be used to tailor the specification requirement:

The fuel flow limits should be in the ICD and for programs that do not have an ICD put the fuel flow limits in table IV of the specification. Many applications specify fuel flow limits in both the ICD and specification.

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Background:

This requirement is an interface requirement so that the air vehicle fuel tubes and pipes can be sized properly. Also, in the event that an ICD is not available, this is where the ICD information would be available. The control system tolerances should be taken into account when the limits are specified. Minimum fuel flow often defines flight idle power at high altitude.

REQUIREMENT LESSONS LEARNED (A.3.7.3.2.9)

None.

A.4.7.3.2.9 Fuel flow limit.

The requirements of 3.7.3.2.9 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.7.3.2.9)

All engine limits must be verified by test.

VERIFICATION GUIDANCE (A.4.7.3.2.9)

Past verification methods have included analysis and component and engine tests.

Background:

Test results should be adjusted for worst case tolerance in the engine and the control system.

VERIFICATION LESSONS LEARNED (A.4.7.3.2.9)

None.

A.3.7.4 Electrical system.

A.4.7.4 Electrical system.

A.3.7.4.1 Electrical power.

After reaching compressor speeds specified herein, engine electrical power shall be uninterruptible and supplied by the (engine or air vehicle) as specified herein. Engine electrical shall have (b) indications equipment shall be capable of accepting externally supplied air vehicle electrical power as a (primary or alternate) power source as specified herein.

Transfer between primary and alternate power sources shall not interrupt engine operation.

REQUIREMENT RATIONALE (A.3.7.4.1)

The electrical power source needs to be specified. Air vehicle electrical alternate power may be required to ensure continued engine operation in the event of engine electrical power system failure or interruption. This makes the engine more reliable. It also makes the engine more survivable in the event of external damage.

REQUIREMENT GUIDANCE (A.3.7.4.1)

The following may be used to tailor the specification paragraph:

The primary and alternate power sources and the compressor speed should be provided here. The contractor should specify whether the engine or air vehicle will provide the primary and alternate electrical power and compressor speeds in the ICD.

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MIL-STD-704 or an equivalent commercial standard may be used as guidance to define the alternate power.

In the past, uninterruptible engine supplied electrical power has been accomplished through redundant systems.

If air vehicle electrical power is specified, the interface information should be included in the ICD. For programs that do not have an ICD, the interface information can be put in the specification paragraph. If air vehicle power is specified, the engine contractor should ensure the source is uninterruptible, and there are no single point failure sources, and the Using Service should ensure the air vehicle program office is aware of the requirement for the air vehicle to provide the power. Air vehicle power supply requirements need to be defined at the start of the design process to insure compatibility.

When the engine provides the primary or alternate power, interface power definition in the specification is not required, but when the air vehicle provides primary or alternate power, interface power definition is required. If the air vehicle provides primary power, then the air vehicle must also provide alternate power. When the air vehicle provides both primary and alternate power, the engine and air vehicle contractors should also ensure that the air vehicle primary and alternate power sources and wiring or harnesses minimize the use of common resources to avoid a single point failure mode.

Background:

The electrical power for the engine control system, engine ignition system, and any other engine electrical accessory should provide for safe operation at all engine speeds throughout the operating envelope of the engine.

An external electrical power failure should not result in loss of electrical power to the engine or the engine control system.

Loss of air vehicle electrical power should not result in loss of engine electrical power and should not cause the engine control system to be unavailable (off-line) for any operating condition other than during an engine start.

Unmonitored electrical power characteristics have resulted in unpredicted operation of the devices being powered. Undervoltage or overvoltage conditions can adversely affect electrical circuitry function, performance and reliability.

REQUIREMENT LESSONS LEARNED (A.3.7.4.1)

Numerous false fault flags occurred on an engine health monitoring system due to an interrupted power input.

Engine E has experienced FADEC faults caused in part by power interruptions during starting and bus switching during ground operations.

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A.4.7.4.1 Electrical power.

The requirements of 3.7.4.1 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.7.4.1)

Electrical power requirements must be verified.

VERIFICATION GUIDANCE (A.4.7.4.1)

Past verification methods have included analysis, demonstration, and test.

VERIFICATION LESSONS LEARNED (A.4.7.4.1)

None.

A.3.7.4.1.1 Generator.

If an engine generator is required it shall provide power to satisfy the full electrical load of the engine. The generator performance shall not degrade during and after continuous operation at a speed equivalent to _____ percent of the maximum allowable steady-state engine speed under full electrical load. The generator performance shall not degrade following the overspeed of 3.4.1.6.5, with no electrical load. The generator housing shall completely contain all damage if a mechanical failure should occur when operating at or below maximum transient rotor speed.

REQUIREMENT RATIONALE (A.3.7.4.1.1)

Electrical power is required over the entire operating speed range of the engine. Containment of all generator damage within the housing is a safety requirement to prevent damage to the adjacent accessories or possible catastrophic engine failure due to severing of fuel or oil lines or major control functions.

REQUIREMENT GUIDANCE (A.3.7.4.1.1)

The following should be used to tailor the specification paragraph:

The contractor should fill in the blank. The generator overspeed capability should be compatible with the maximum engine overspeed value.

The generator should be capable of supplying the full electrical load from the lowest idle rotor speed to the maximum transient engine rotor speed.

Background:

Because of possible control system degradation the generator must have sufficient design margin to provide electrical power up to the maximum allowable transient rotor speed. Continuous operation of the generator at the maximum overspeeds condition without load ensures the generator integrity is equivalent to the main rotor requirements.

Some modern engines utilize an integral starter/generator system provided by the airframe manufacturer. Its power requirements should be documented in the ICD.

REQUIREMENT LESSONS LEARNED (A.3.7.4.1.1)

This requirement was waived for Engine P in aircraft Z. Engine Company A's negotiated specifications, for Engines C and J, in aircraft I and C and revised the 115 percent speed maximum speed requirement upward to 135 and 125 percent, respectively, and eliminated the generator containment requirement.

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Engine C's specification included a statement that the minimum design burst speed was equivalent to 190 percent of the engine maximum allowable transient overspeed. The contractor also tested the units to 200 percent speed without failure.

A.4.7.4.1.1 Generator.

The requirements of 3.7.4.1.1 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.7.4.1.1)

The generator specification performance requirements must be verified.

VERIFICATION GUIDANCE (A.4.7.4.1.1)

Past verification methods have included analysis, demonstration, and test.

The following procedure may be used as a verification method:

In addition to any other tests required by this specification, the following shall be conducted.

a. Overspeed. The generator shall be tested per the overspeed requirement of 3.4.1.6.5, with no electrical load. At the completion of the test there shall be no evidence of mechanical or electrical damage or impending failure.

b. Load test. The generator should be operated at a speed corresponding to the percent of the maximum allowable rotor speed under full rated electrical load for __(a)__ hour(s). During this test, the generator shall be subjected to the maximum component limiting temperature. At the completion of the test there shall be no evidence of mechanical or electrical damage or impending failure.

c. Containment. The generator shall be tested at the maximum transient rotor speed in a manner to cause a mechanical failure of the rotor system. All damage shall be contained within the generator housing.

The following should be used to tailor subparagraph b., above.

(a): Past programs have used one hour.

VERIFICATION LESSONS LEARNED (A.4.7.4.1.1)

None.

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A.3.7.4.2 Alternate electrical power.

The engine shall automatically accept alternate power when the source is either air vehicle generated power or external (ground) power or a combination thereof as defined in (a).

Engine performance and operability shall not be degraded with electrical power transients of: (b).

REQUIREMENT RATIONALE (A.3.7.4.2)

An alternate electrical power source is required to ensure continued engine operation to provide a safety of flight redundancy.

The engine alternate power source electrical characteristics must be defined to ensure the air vehicle and engine requirements will be compatible.

The engine electrical requirements must be specified so that the aircraft power source will supply the engine with the proper alternate electrical power.

REQUIREMENT GUIDANCE (A.3.7.4.2)

The following may be considered to tailor the specification paragraph:

(a): This definition can be shown in the ICD, or for programs that do not have an ICD, the definition can be put into the specification paragraph.

(b): The contractor should provide a description of the alternate electrical power and the type of electrical transients that can be tolerated by the engine without degradation of performance or operability. These requirements must be consistent with the air vehicle design requirements and coordinated in the ICD. This needs to be accomplished early in the design phase. MIL-STD-704 or an equivalent commercial standard can be used for guidance. Either a tailored version of the military standard or commercial document can be referenced in the blanks, or the applicable portions or subparagraphs of the military standard or commercial document should be transferred into the requirement paragraph.

Background:

Electrical power interrupts exceeding the requirements of MIL-STD-704 should not permanently or temporarily affect engine performance or operability including any transients.

The characteristics of the power interrupts need to be clearly defined at the start of the design process.

The engine manufacturer should specify the form and quantity of electrical power required. The engine should accomplish satisfactory air starts and should operate safely at all rotor speeds when it is utilizing air vehicle electrical backup power.

REQUIREMENT LESSONS LEARNED (A.3.7.4.2)

There have been cases in which an engine, equipped with a supervisory electronic control powered by the engine mounted generator, encountered chronic generator connector failures. This caused the engine to be susceptible to augmentor anomalies and stall. If an alternate electrical power source had been available, these anomalies could have been averted.

Engine C1’s specification states that some control functions are lost and no ignition capability is available when the engine is operating on an external power source.

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A.4.7.4.2 Alternate electrical power.

The requirements of 3.7.4.2 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.7.4.2)

Operation with alternate electrical power must be verified.

VERIFICATION GUIDANCE (A.4.7.4.2)

Past verification methods have included analysis, demonstration and test.

The following procedure may be used as a verification method:

a. Operate the engine at intermediate or maximum continuous power with full customer bleed air and power extraction until thrust or power is stabilized, then switch off the electrical power to the control system. The engine shall automatically switch to the alternate electrical power and stabilize at intermediate or maximum continuous power. Decelerate to part power and switch back to the electrical power. The engine shall not exceed any of its limits nor exhibit performance or operability degradation during the test.

b. Interrupt the electrical power to the control system at the most critical point in acceleration and a deceleration. During the time the electrical power is off and during the switching, the engine shall not exceed any of its limits nor exhibit performance or operability degradation. The engine shall complete the transient satisfactorily.

c. With the engine operating at idle and also maximum augmentation, switch off the normally supplied external electrical power and check for limitations or loss of functions.

d. With the engine off, the external electrical ground power on, and the electrical power off, start and accelerate to intermediate or maximum continuous power. Operate at this power until the engine is stabilized, decelerate to idle and stabilize then shutdown. The engine shall not exceed any of its limits nor exhibit performance or operability degradation during the test."

Background:

Transfer to and from the air vehicle electrical power backup should be conducted at critical operating conditions. Evaluation of the operating characteristics of the engine when external electrical power is removed can be done on a bench or systems rig test before the engine verification test is conducted. If these tests are used, the variable geometry and other similar control functions need to be fully verified to ensure no engine performance or operability problems are encountered.

VERIFICATION LESSONS LEARNED (A.4.7.4.2)

In past programs this capability has been verified during endurance testing.

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A.3.7.4.3 Electrical connectors and cables.

Electrical connectors shall be specified herein. Engine interface connectors shall be specified in _____. At the sea level cold atmospheric temperature of figure 5, it shall be possible to flex electrical cable and conductors during routine maintenance without damage to these items and to connect or disconnect electrical connectors using normal maintenance procedures.

REQUIREMENT RATIONALE (A.3.7.4.3)

The functional, environmental, durability and maintainability requirements for the electrical connectors and cables must be established to ensure suitable operation and life. Physically interchangeable connectors in close proximity can and do get mixed which may render the system inoperative.

REQUIREMENT GUIDANCE (A.3.7.4.3)

The following should be used to tailor the specification paragraph:

Interface connectors can be shown in the ICD or for programs that do not have an ICD, the connectors can be put into the specification paragraph.

MIL-DTL-38999, Series III and Series IV, or an equivalent commercial standard for self-locking, scoop-proof connectors should be used as tailoring guidance.

Electrical connectors and cables should have sufficient spare contacts in the connectors and wires in the cables to permit future growth and should be repairable.

The contractor should minimize the number of electrical connections between the air vehicle and engine to help meet the maintainability requirements of section 3.5.

Background:

The cable environment should include heat, vibration, spilled fluids, sand, salt spray, dust, and other engine bay environmental conditions. The minimum ambient temperature should be the sea level cold atmospheric temperature. Self-locking connectors should be considered for all engine applications. Spare contacts and wires should be considered for future growth.

Past turboshaft programs have had three electrical connections to the air vehicle.

REQUIREMENT LESSONS LEARNED (A.3.7.4.3)

Connectors that were physically the same but connected to different systems were, on occasion, inadvertently mixed causing an engine malfunction. Some electrical insulation became embrittled at low temperatures causing electrical shorts leading to engine system failures.

A certain attack helicopter experienced at least two in-flight dual engine shutdowns caused by water intrusion into the wiring harness.

Moisture intrusion has been a connector problem on Engine C.

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A.4.7.4.3 Electrical connectors and cables.

The requirements of 3.7.4.3 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.7.4.3)

The electrical connector and cable requirements must be verified by component test, demonstration, and inspection.

VERIFICATION GUIDANCE (A.4.7.4.3)

Past verification methods have included analysis, demonstration, inspection, and test.

Background:

Demonstration of removal and installation of connectors and cables should be done on an engine at the specified ambient temperature of 3.7.4.3. Non-interchangeability of connectors should be verified by engine inspection.

VERIFICATION LESSONS LEARNED (A.4.7.4.3)

None.

A.3.7.4.4 Electrical bonding.

The engine shall have a clearly defined electrical bonding interface with the air vehicle which shall be identified in the __. All electrical and electronic components on the engine shall have a clearly defined electrical bonding interface to the engine structure that shall be identified in the design documentation including drawings.

REQUIREMENT RATIONALE (A.3.7.4.4)

Good electrical bonding practices have long been recognized as a key element of successful system design. Since electrical bonding involves obtaining good electrical contact between metallic surfaces while corrosion control often tries to avoid electrical continuity between dissimilar materials, it is necessary to ensure that both disciplines are properly considered. Good electrical bonding is required for the following reasons:

a. Electrical continuity across external mechanical interfaces on electrical and electronic equipment, both within the equipment and between the equipment and system structure for control of EMI effects. Systems generally include ground planes to form equipotential surfaces for circuitry. If voltage potentials appear between electronics enclosures and the ground plane due to internal circuitry operation, the enclosure will radiate interference. Similarly, electromagnetic fields will induce voltage potentials between poorly bonded enclosures and the ground plane. These potentials are imposed as common-mode signals on all circuitry referenced to the enclosure. The same two effects will occur for poorly bonded shield terminations.

b. Electrical continuity for wire and cable shield terminations. Shields are provided on cable to prevent radiated interference and to protect the system for externally generated electrical fields. For a shield to be effective, the “faraday shield” must be maintained at the shield termination through the connector to the structure of the equipment.

c. Prevention of voltages hazardous to personnel, fuel, and electronics during lightning strikes. Without proper bonding, lightning interaction with systems can produce voltages which can shock personnel, ignite fuel through arcing and sparking, and upset or damage electronics.

d. Assurance that voltage drops for power current returns using structure meet power quality requirements. It is essential that system electrical and electronic equipment is provided with adequate

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voltage levels from prime power sources for proper operation. Bonding provisions help control voltage drops in power current return and fault paths.

e. Prevention of fuel ignition hazards from electrical fault currents. Electrical fault conditions must not introduce potential fuel or fire hazards due to arcing or sparking from melted or vaporized structural material.

f. Prevention of shock hazards from electrical faults. The system design must protect personnel from shock hazards.

REQUIREMENT GUIDANCE (A.3.7.4.4)

The following should be used to tailor the specification paragraph:

The engine to air vehicle bonding interfaces can be shown in the ICD, or for programs that do not have an ICD, the interfaces can be put into the specification paragraph or on figure 1.

The role of bonding is essentially to control voltage by providing low-impedance paths for current flow. Unconventional joints should receive special attention to ensure their adequacy, particularly conductive joints in fuel vapor areas. SAE ARP 1870 provides details on electrical bonding concepts for aerospace systems and examples of bonding techniques.

Special attention should be given to the interdependent relationship between electrical bonding and corrosion control especially when dissimilar materials are used. Design techniques for effective corrosion protection, such as the use of material finishes that are not electrically conductive, can result in lack of bonding. Conversely, obtaining a good electrical bond can lead to potential corrosion problems, if the bonding is not properly implemented. Alternate methods of electrical bonding may be achieved by the use of jumper cables or bonding straps across mating surfaces. Detailed corrosion requirements for systems are imposed by other documents, such as MIL-HDBK-1568 for airborne systems.

While specific bonding levels required to obtain required performance are system dependent, 2.5 milliohms has long been recognized as an indication of a good bond across a metallic interface, particularly aluminum. There is no technical evidence that this number must be strictly met to avoid problems. However, higher numbers tend to indicate that a quality assurance problem may be present and bonding may be degrading or not under proper control. Higher values may be more appropriate for other metals such as stainless steel or titanium. Also, composite materials will exhibit much higher levels and imposed requirements should be consistent with those materials.

Controls need to be implemented in shield termination paths at connector assemblies. A realistic value would be 2.5 milliohms maximum for any particular joint.

Power quality standards, such as MIL-STD-704 for aircraft, control the supply voltage for utilization equipment within specified limits. The voltage is maintained at a monitoring location termed the “point of regulation" with allocations for allowable voltage drops beyond this point to the input of the utilization equipment. These drops must be controlled through wire conductor type and size selection and current return path design. Most aircraft use structure as the return path for power currents. Bonding provisions must be incorporated to control the impedance of this path.

Bonding for prevention of fuel ignition hazards from electrical fault currents need to address areas on the system where flammable vapors may be present, installed electrical equipment (such as fuel pumps), the electrical paths that fault currents will follow, the available level of fault current, and the bonding value necessary for the implemented design architecture to prevent arcing, sparking and hot spots.

Provisions must be included to prevent hazardous voltages from appearing on any electrically conductive assembly. These voltages could result from sources such as broken components in assemblies allowing “hot" wiring to contact the housing or from electrical referencing of a circuit to the housing. MIL-HDBK-454, Requirement 1, controls voltage for shocks to 30 volts rms. and DC.

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REQUIREMENT LESSONS LEARNED (A.3.7.4.4)

Numerous instances of the need for good bonding have been demonstrated. Bonding improvements or corrections have solved many system problems including precipitation static in UHF radios, susceptibility of electronics to external electromagnetic fields, radiation of interference into antenna-connected receivers, and lightning vulnerabilities.

The actual need for certain bonding in a particular application is not easily ascertained. It is dependent on various items such as the shielding topology, type of circuit interfaces, and the use of the enclosure as a ground reference for circuits and filters. For example, a subsystem which is wholly contained (all enclosures and cable interfaces in a continuous unbroken shield) typically does not necessarily require bonding for RF potential control. External currents will remain outside the shield and internal currents will remain inside. This configuration is rare. The increasing use of differential interface circuits makes bonding less critical since there is better rejection of common-mode noise. However, there is often a trade-off between interface design and the amount of wiring and number of connector pins since differential interfaces require the use of two wires and pins for each signal.

Maintaining required voltage levels from electrical system power sources at utilization equipment on metallic aircraft has not been a problem since the current return paths have low impedance. With increasing use of composites, the need for separate wire returns or implementation of a ground plane becomes a consideration.

The bonding resistance should be based on the maximum short circuit currents expected from the configuration in order to prevent shock hazards.

On a current fighter engine, it was discovered that anodizing a connector surface, for corrosion control, resulted in unacceptable levels of electrical bonding. EMI testing of this connection resulted in the control system transferring into backup mode.

A.4.7.4.4 Electrical bonding.

The requirements of 3.7.4.4 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.7.4.4)

Verification of protection measures for electrical bonding is necessary to ensure that adequate controls are implemented.

VERIFICATION GUIDANCE (A.4.7.4.4)

Past verification methods have included analyses, inspections and component, engine, and installed engine(s) tests. Multiple engines should be tested to account for manufacturing variability on electrical bonding characteristics. Electrical bonding measurements should be performed prior to and following corrosion testing.

The electrical bonding area involves a number of different concerns. The particular verification methodology needs to be tailored for the bonding control being assessed. Many elements require more than one form of verification. When bonding values in the several milliohm range or less is required, testing is a necessity. When much higher values are adequate, inspection of surface finishes and mounting techniques supplemented by analysis can be acceptable. Analysis is an element of assessing structural voltage drops for power returns, fuel ignition hazards and personnel shock.

Detailed corrosion control requirements for systems are imposed by other documents. For airborne systems, MIL-HDBK-1568 provides requirements. Demonstration is required that appropriate manufacturing processes are in place to address corrosion concerns.

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The requirement for electrical continuity across external mechanical interfaces on electrical and electronic equipment is normally verified during the development of the equipment. The equipment to structure interface is normally verified at the system level. A measurement is made from an enclosure surface to the next major assembly. For example, in an installation with an enclosure mounted in a tray, separate measurements would be applicable from the enclosure to the tray and from the tray to structure. The measurement is normally performed with a DC resistance meter. Where RF currents are of concern, the AC impedance should be maintained as high in frequency as possible. The impedance will normally remain low for enclosures that are hard-mounted to structure. However, for enclosure installations that use bonding straps, such as shock mounts, the impedance of bonding straps will be significant due to the inductance of the strap. Standard practice is adding one strap across each shock mount (with a minimum of four) to achieve < 2.5 milliohms.

On most military aircraft, aircraft structure is used as the current return for electrical power. The controls on bonding between structural members, the resistance of structure, and electrical current levels need to be considered. For aircraft and other systems that use wired returns, the resistance of the wire is the primary consideration and bonding is not a significant player. The location of the point of regulation for the power system also plays a role.

Verification of electrical bonding measures for design against electrical shock is primarily achieved by demonstrating that voltage faults to electrically conductive surfaces will not result in hazardous voltages on the surface. These types of faults should normally trip circuit protection equipment.

VERIFICATION LESSONS LEARNED (A.4.7.4.4)

The adequacy of much electrical bonding can be evaluated through DC resistance measurements and inspection. AC measurements should be performed where RF currents are of concern; however, they require more sophisticated instrumentation. DC measurements have proven to provide a good indication of the quality of a bond.

With metallic systems, voltage drops through structure are typically very low. Much higher levels are possible with graphite and epoxy structure.

Powerline filtering arrangements in electronics, which isolate the powerline neutral from chassis, can result in hazardous voltages on the enclosure if the frame ground is disconnected. Typically, filters will be present on both the high side and the return that will have capacitance to the chassis. If the chassis is floating with respect to earth ground, the capacitors act as a voltage divider for AC waveforms with half the AC voltage present on the case with respect to earth. The value of the capacitors determines the amount of current that may flow. The capacitance should be limited to the values specified in MIL-STD-461.

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A.3.7.4.5 Ground isolation.

The engine to aircraft interface shall not use aircraft structure for power or signal return.

REQUIREMENT RATIONALE (A.3.7.4.5)

This requirement ensures current return paths are established to control ground loops and minimize electromagnetic coupling. Provisions should be taken to ensure that there is sufficient isolation resistance between grounds.

REQUIREMENT GUIDANCE (A.3.7.4.5)

The following may be used as tailoring guidance:

The isolation between power returns and signal returns and structure should be at least 1 mega ohm at DC.

Background:

This requirement is necessary to prevent grounding of components and parts in a series circuit such that static discharge or lightning strike to one component or part will not go through another on its way to being grounded out. A series circuit for grounding may cause unnecessary loss of both components and parts because of a faulty ground route taking place.

REQUIREMENT LESSONS LEARNED (A.3.7.4.5)

None.

A.4.7.4.5 Ground isolation.

The requirements of 3.7.4.5 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.7.4.5)

Ground isolation must be verified to ensure electromagnetic coupling is minimized and ground loops prevented.

VERIFICATION GUIDANCE (A.4.7.4.5)

Past verification methods have included inspection and test.

Background:

Inspection of engineering drawings showing the isolation between component and structure may satisfy the verification criteria. The contractor should submit verification methods in the E^3 Control Plan.

VERIFICATION LESSONS LEARNED (A.4.7.4.5)

None.

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A.3.7.4.6 Potting compounds.

The use of potting compounds shall not be permitted except where specifically approved by the Using Service. When approved, the potting compounds and their application(s) shall be described.

REQUIREMENT RATIONALE (A.3.7.4.6)

Potting materials are required in some electrical and electronic components to protect the internal parts from environmental damage (e.g.; temperature, vibration, corrosion, etc.).

REQUIREMENT GUIDANCE (A.3.7.4.6)

A proper follow-on maintenance concept on whether the item containing the potting compound is repairable or replaceable needs to be addressed during development.

Background:

The potting materials and processes used in electrical components for environmental protection should not deteriorate the chemical, physical, and electrical properties of the components. Potting materials in repairable components should be easily removable and replaceable. Water-absorbing potting compounds should not be used for electrical soldering structural supports. For additional guidance refer to AFGS-87219.

REQUIREMENT LESSONS LEARNED (A.3.7.4.6)

Potting compounds can significantly influence the reliability and maintainability of an electrical component. In one case, epoxy used to stabilize the windings in a wetted stepper motor was eventually found to create corrosion of the motor wires after the epoxy was exposed to fuel. In another component that required frequent maintenance, the potting compound had to be removed by technicians using dental picks. This was extremely costly and time consuming.

A.4.7.4.6 Potting compounds.

The requirements of 3.7.4.6 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.7.4.6)

Verification is needed to ensure the potting compound will not degrade or cause deterioration of the electrical component in the engine operating environment and to ensure that the removal of the compound for repair does not further damage the internal components.

VERIFICATION GUIDANCE (A.4.7.4.6)

Past verification methods have included demonstration, inspection, and test.

The following should be used to tailor the specification paragraph:

Removable potting compounds should be removed and replaced following the engine and component endurance test. The components should then be functionally tested to ensure that no damage exists as a result of potting compound removal and replacement.

Background:

The verification should be conducted on the engine endurance test that has the greatest number of thermal cycles. This ensures that the compound will be thermally exercised so that the ability to resist cracking may be known.

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VERIFICATION LESSONS LEARNED (A.4.7.4.6)

There have been problems that result from the use of potting compounds that cannot be found during engine and bench testing. These problems have shown up after hundreds of thousands of fleet hours of operation, and may result in serious support problems, possible fleet grounding and large redesign and retrofit activities.

A.3.7.5 Ignition system(s).

The engine ignition system(s) shall ignite the combustor and if applicable the augmentor. An external means shall be provided to deactivate the engine ignition system(s), and it shall be described in _____.

The engine ignition system(s) shall ignite the combustor(s) under carbon and water fouling conditions.

a. Main ignition system. The main engine ignition system shall be:

Each separate igniter and circuit shall release sufficient energy for all ground and air starting requirements.

a. Augmentor ignition system. If applicable, the augmentor ignition system shall be:

REQUIREMENT RATIONALE (A.3.7.5)

The main ignition system must be capable of supplying sufficient energy for starting and relighting the engine under all environmental and operating envelope conditions. Two separate igniter systems are needed for redundancy in a single engine application.

REQUIREMENT GUIDANCE (A.3.7.5)

The following should be used to tailor the specification paragraph:

The interfaces can be shown in the ICD, or for programs that do not have an ICD, the interfaces can be put into the specification paragraph.

For single engine applications the main ignition system should be redundant and provide a minimum of two separate igniters and two independent exciter output circuits.

The specification paragraph should, as a minimum, include the following information: igniter type and method of operation, such as continuous or intermittent spark, or if the spark is dependent upon thrust or power setting.

Background:

Each igniter and associated exciter output circuit should be independent to ensure proper operation with any other exciter output open circuited or short circuited. In areas of the flight envelope where there is a high risk of a combustor blowout, one of the igniter systems should be capable of continuous operation to permit successive starting attempts without delay. Continuous duty operation is desirable as a preventive measure against flameout. The ignition system should have provisions to allow its external activation and termination that will provide safety during maintenance by eliminating a lethal shock hazard.

The contractor should describe the augmentor ignition system (igniter plug, fuel slug from main combustor, etc.). Augmentor ignition is controlled by the engine control system and additional requirements are contained in 3.7.2.

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REQUIREMENT LESSONS LEARNED (A.3.7.5)

Experience has shown that continuous ignition requires the use of a low energy system to provide long igniter life. The combination of one high and one low energy ignition will usually suffice. Capacitive discharge has proven to be superior to inductive discharge. Past experience has shown that a minimum energy level, discharged in 10 to 100 microseconds at a frequency of at least one spark per second is essential to satisfactory operation.

Augmentor ignition requires the application of correct fuel-air ratio and air velocity in the vicinity of the igniter. Experience has shown that a pilot burner may be required to meter airflow and fuel flow in combination with low air velocity adjacent to the igniter to ensure reliable augmentor ignition throughout the flight envelope.

A.4.7.5 Ignition system(s).

The requirement of 3.7.5 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.7.5)

The ignition system(s) must be verified.

VERIFICATION GUIDANCE (A.4.7.5)

Past verification methods have included tests. The tests should include fouling tests.

The contractor should propose a test procedure and schedule.

The following may be considered when tailoring the specification paragraph:

The self-contained ignition system or component test assemblies should be tested per the following 4-hour schedule(s). For the purpose of these tests the minimum and maximum voltages and frequencies should correspond to those extreme conditions permitted on the engine for satisfactory functioning of the ignition system. The ignition system should be tested with the same number of igniters as used on the engine. The igniters should be installed in a chamber. The chamber pressure and airflow should be regulated to simulate the internal engine pressures and flows from minimum windmilling pressure and flow to the maximum pressures and flows the igniters will see in the engine flight envelope. For the high temperature test, the complete ignition system should be placed in a suitable chamber to simulate, prior to initiating the test, conditions of 95°C (203°F) or the maximum component limiting temperature of 3.2.2.13.1.

Procedures:

a. High temperature endurance. The ignition system should be operated for 300 hours of cycling per appropriate schedule outlined below and at the maximum component limiting temperature given in 3.2.2.13.1. A 2-hour shutdown should follow each cycle at the temperature conditions corresponding to the requirements of 3.2.2.13.1(c). At the conclusion of testing, checks should be made of insulation resistance, over-voltage capability, igniter output energy, and spark rate.

b. Room temperature endurance. The ignition system should be operated at an ambient temperature between 16°C (60°F) and 38°C (100°F) for 100 hours of cycling per the appropriate schedule outlined below.

c. Low temperature endurance. The ignition system should be tested for 24 hours at an ambient temperature equal to the sea level cold atmospheric temperature of figure 5. The system should be operated for 12 hours of cycling, per the appropriate schedule outlined below, followed by a 10-hour minimum inoperative soaking period, and a final 12 hours of cycling.

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Schedules:

a. Continuous duty ignition system. The continuous duty start and augmentor ignition systems should be cycled simultaneously according to the schedules shown in tables XLVa and XLVb.

For the start ignition system, each cycle should be as shown in table XLVa.

For a continuous duty augmentor ignition system, each cycle should be as shown in table XLVb.

a. Intermittent ignition system. Intermittent ignition systems should be cycled according to the schedule shown in table XLVc.

All ignition system tests should be conducted on test assemblies that have the same serial number, consisting of groups of related components arranged and interconnected to simulate their normal relationship and function on the engine. No adjustments should be made subsequent to initial calibration.

Ignition system environmental tests should be per 4.11.2.1.5, 4.7.5.1.1, and 4.7.5.1.2. Test assemblies should have the same serial number, consisting of groups of related components arranged and interconnected to simulate their normal relationship and function on the engine. No adjustments should be made subsequent to initial calibration.

VERIFICATION LESSONS LEARNED (A.4.7.5)

None.

A.3.7.5.1 Ignition system fouling.

A.4.7.5.1 Ignition system fouling.

A.3.7.5.1.1 Carbon fouling.

The spark igniters of the ignition system shall provide sparking performance with spark gaps completely covered or bridged with a material simulating carbon fouling. With the minimum power input of section 3.7.5, and the carbon fouling conditions, the sparking rate shall not be less than the minimum design value.

REQUIREMENT RATIONALE (A.3.7.5.1.1)

The ignition system must be capable of igniting while fouled with carbon.

REQUIREMENT GUIDANCE (A.3.7.5.1.1)

Graphite Petrolatum Compound (SAE AMS 2518) may be used as tailoring guidance for the material simulating carbon fouling.

REQUIREMENT LESSONS LEARNED (A.3.7.5.1.1)

None.

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A.4.7.5.1.1 Carbon fouling.

The requirement of 3.7.5.1.1 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.7.5.1.1)

The ignition system assembly spark igniters must be tested to verify igniter performance under carbon fouling conditions.

VERIFICATION GUIDANCE (A.4.7.5.1.1)

Past verification methods have included test.

VERIFICATION LESSONS LEARNED (A.4.7.5.1.1)

None.

A.3.7.5.1.2 Water fouling.

The spark igniters of the ignition system shall provide sparking performance when thoroughly drenched with water or frosted over. With the minimum power input of section 3.7.5, and the water fouling conditions, the sparking rate shall not be less than the minimum design value.

REQUIREMENT RATIONALE (A.3.7.5.1.2)

Engine flameout in extreme environmental conditions may introduce moisture or water to the spark igniters. Spark igniters under these conditions must be capable of providing minimum performance necessary to start the engine consistently.

REQUIREMENT GUIDANCE (A.3.7.5.1.2)

None.

REQUIREMENT LESSONS LEARNED (A.3.7.5.1.2)

None.

A.4.7.5.1.2 Water fouling.

The requirement of 3.7.5.1.2 shall be verified by:

VERIFICATION RATIONALE (A.4.7.5.1.2)

Igniter performance under water fouling conditions must be verified.

VERIFICATION GUIDANCE (A.4.7.5.1.2)

Past verification methods have included test.

Background:

The spark igniters should be positioned in a manner simulating the mounted position in the engine. The minimum power or voltage input should be supplied per section 3.7.5. The spark igniters of the ignition system test assembly should be thoroughly drenched with water to simulate extreme atmospheric conditions.

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VERIFICATION LESSONS LEARNED (A.4.7.5.1.2)

None.

A.3.7.6 Engine health monitoring system (EHMS).

The EHMS as defined in the _____ shall provide, monitor, and coordinate the recorded appropriate parameters for maintenance, performance trending, parameter limit exceedance, and life usage. The malfunction of any EHMS hardware or software shall not affect engine performance or operability.

REQUIREMENT RATIONALE (A.3.7.6)

An EHMS is required to monitor and record parameters for Operation and Support (O&S) cost reduction, operational readiness improvements, and to monitor complete mission recording for mission analysis updates and life limit updates.

REQUIREMENT GUIDANCE (A.3.7.6)

The following should be used to tailor the specification paragraph:

The interfaces can be shown in the ICD, or for programs that do not have an ICD, the interfaces can be put into the specification paragraph.

Modern engine health monitoring systems may include prognostic capabilities where failures are predicted based on sensed changes in engine operating conditions.

The EHMS should be compatible with existing engine maintenance systems including the entire logistics footprint; e.g.; forms and records, supply, interactive manuals, etc.

The engine health monitoring system should be completely functional following failure of any other subsystem of the engine, including engine control and electrical systems. This may help troubleshoot in-flight engine anomalies that result in an engine shutdown.

Some of the parameters that should be monitored are rotor speeds, gas path pressures, and temperatures, accelerometers for vibration measurement, FADEC signals, aircraft parameters (velocity, angle of attack, yaw, missile launch, pressure altitude, etc.) and discrete events (anti-ice valve on or off, weight on wheels switch on or off, in-flight refueling probe out or in, etc.). Tables XLVIa and XLVIb should be the tables where the monitored parameters and their accuracies will be shown. Two tables are recommended, one each for the engine and aircraft. The contractor should fill in the table. Tables XLVIa and XLVIb should be coordinated with logistics and life management personnel to establish data format, parameters, frequency, and requirements (low cycle fatigue cycles, full thermal cycles, multiple categories of partial thermal cycles, multiple categories of minor thermal cycles, etc.). The engine maintenance concept (i.e., two or three level) needs to be chosen before the EHMS is designed so the contractor can properly design the system. Additional guidance can be found in SAE ARP 1587.

Due to the rapidly-changing technologies involved in EHMS, data acquisition, memory, and processing, it is necessary to have a set of firm functional requirements for the system. The functional requirements should be traceable to mission accomplishment, supply or maintenance management requirements. Data acquisition should be minimized and parameters should support fault detection and isolation, life usage or scheduled maintenance.

Continuous mission recording is necessary to achieve high fidelity critical part life limits during service. As initial life limits are established from design missions, the mission recording capability of the EHMS is necessary to confirm that the design missions are being flown, or if not, the true operational missions. The missions should be reviewed no less frequently than every five years as operational mission requirements, capabilities, aircraft configurations, and tactics change.

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Some fielded helicopters and turboprops do not require vibration monitoring.

Background:

The EHMS should be an integral part of the engine maintenance system and should be compatible with the aircraft data collection, ground support systems, and the maintenance support concept. The EHMS data may or may not be retrieved through the aircraft system. All aircraft to engine EHMS interfaces should be described in the interface section 3.1.1 of this specification. The contractor should describe the equipment included as part of the engine monitoring system (i.e.; engine data collection box and sensors and, if applicable, the aircraft data collection system and ground support system, data transfer system [removable memory card, computer port]).

The EHMS should include installed airborne equipment, ground equipment, flight line support equipment, intermediate level support equipment, JOAP, etc., and depot support equipment, technical publications, visual and manual inspections and personnel required to accomplish the complete task. Obviously, collecting data is a small part of the systems approach required to develop an effective EHMS. Some of the details to be considered in the design are the sensors, signal conditioners, data processor, data storage, crash survivability, data exchange rate(s), engine or air vehicle mounting, ground based processing station, and programming language.

Numerous parameters are to be recorded continuously from the beginning to the end of the mission to identify the actual missions flown in service. The list should include aircraft parameters (altitude, mach no, angle of attack, ‘g’, direction, etc.), engine parameters (rotor speeds, temperatures) and discrete events (weight on wheels, anti-ice on or off, etc.).

There has been considerable proliferation of EHMS ground support equipment. It is generally recognized that the airborne electronic engine monitoring equipment will be somewhat application unique, but the ground equipment required to extract, transmit, process, and distribute the data can be adaptable to the application.

EHMS program software algorithms and instrument range, system accuracy, time response, sample rates, and electrical characteristics for each parameter should be provided to the Using Service via the Contract Data Requirement List (CDRL) or as an appendix to this specification.

REQUIREMENT LESSONS LEARNED (A.3.7.6)

The USAF has had numerous bad experiences with EHMS that did nothing more than collect data. The data that were collected could not be reduced into useful information, and some of the more important parameters were not collected. Care needs to be exercised when selecting the data to be recorded.

Aircraft I and Z engine monitoring system designs were frozen at an early stage of development because they were incorporated into mission computers. This has provided zero growth capability for the EHMS software in these aircraft. More importantly, it has prevented operational lessons learned from being incorporated into these aircraft. On Aircraft I, the MIL-STD-1553 data bus used for EHMS parameters does not have a fast enough data rate to support real time computations of engine health based on state-of-the-art engine modeling techniques. This severely limits the future growth capability of the In-Flight Engine Condition Monitoring System (IECMS) on this aircraft.

Engine O on aircraft H and engine AN on aircraft AH utilize sophisticated EHM systems that include prognostics, automated data downloads and digital technical order subsystems.

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A.4.7.6 Engine health monitoring system (EHMS).

The requirement of 3.7.6 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.7.6)

EHMS drawings and electrical diagrams and the engine maintenance concept need to be reviewed to ensure the EHMS design will comply with the requirements.

VERIFICATION GUIDANCE (A.4.7.6)

Past verification methods have included inspection, analysis, demonstration, and test.

The following should be used to tailor the specification paragraph:

The accuracies of the EHMS sensors should be as good as the control system sensors, otherwise incorrect life usage may be calculated, or the EHMS may not capture critical events, as when components exceed their operational limits. The accuracies of the parameters in tables XLVIa and XLVIb should be verified by independent engine test, and where appropriate, by the calculated relationship between the actual measured parameter and the output signal. The above calculations should be made available to the Using Service.

The accuracies of the algorithms used to calculate parts life should be verified by engine sea level and altitude tests as well as flight test. Mission data recording should also be verified during the engine AMT, other endurance testing and flight-testing.

Background:

The monitoring system should be installed on all engines that are tested during development, to ensure all problems have been identified and solved prior to the official qualification testing.

VERIFICATION LESSONS LEARNED (A.4.7.6)

None.

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A.3.7.6.1 EHMS fault detection and isolation.

The EHMS fault detection and isolation capability shall be described herein. The EHMS functions required to achieve this level of detection and isolation faults shall be described herein. The onboard engine diagnostics shall provide fault detection and isolation, to the faulty component level, to meet the reliability and maintainability requirements as specified in section 3.5. Onboard engine diagnostics system false alarms shall be considered as failures and shall be used in the reliability and maintainability calculations per section 3.5. The mission critical equipment and functions of the onboard diagnostic system, which monitor mission and safety critical parameters, shall be defined herein. The onboard engine diagnostics system shall not cause a failure of any other mission or safety critical system.

REQUIREMENT RATIONALE (A.3.7.6.1)

The EHMS fault detection and isolation capabilities and the equipment and maintenance efforts necessary to accomplish these functions need to be specified.

REQUIREMENT GUIDANCE (A.3.7.6.1)

The following should be used to tailor the specification paragraph:

Past specifications have required a 90-100% detection and isolation of all faults as the specification capability for fault detection and isolation. Ground based systems have required 95-100% detection and isolation of all faults.

Background:

The contractors list and description of the functions to detect and isolate faults may include: the onboard engine diagnostics, inspections, results from processed EHMS data, and troubleshooting (automated, computer assisted and manual).

For higher percentage coverage of fault detection and isolation, more maintenance functions should be listed. That is, if 90 percent coverage is specified, then the list may only have onboard engine diagnostics. If 100 percent coverage is required, then the list may have all of the features described above.

As the Using Services reduce their number of personnel and affordable systems (including maintenance functions throughout operational support) continue to drive future requirements, the importance of the EHMS will increase. The goal is to have the onboard system identify any faulty component and have maintenance personnel only conduct minor troubleshooting tasks on the ground.

A high percentage of the fault detection and isolation should be accomplished through automated processes (both in flight and on the ground) to save maintenance work hours. Verifying the problem in the field through the EHMS can also save depot rework hours and other supply costs by reducing the number of “cannot duplicate” occurrences. The EHMS should be compatible with existing engine maintenance systems; Core Automated Maintenance System (CAMS), and Comprehensive Engine Management System (CEMS) for the USAF; Aircraft Engine Management System (AEMS), Naval Aviation Logistics Data Analysis (NALDA), Aviation Maintenance Material Management System (Aviation-3M), and Engine Component Tracking System (ECOMTRAK) for the USN.

REQUIREMENT LESSONS LEARNED (A.3.7.6.1)

None.

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A.4.7.6.1 EHMS fault detection and isolation.

The requirement of 3.7.6.1 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.7.6.1)

The EHMS must be functionally verified to ensure its usefulness as maintenance, performance trending, and warranty tool will be fulfilled.

VERIFICATION GUIDANCE (A.4.7.6.1)

Past verification methods have included inspection, analysis, demonstration, and test.

Background:

Testing should be performed throughout the development cycle and should include qualification testing, test cell testing, and flight-testing. The flight-testing should verify the effectiveness of the entire system including airborne equipment, ground equipment, and data output product usage. Flexibility of the EHMS during flight test should be maintained so that changes can be made to the system to ensure it is a useful maintenance tool.

VERIFICATION LESSONS LEARNED (A.4.7.6.1)

Many systems have been thoroughly tested from a functional standpoint and adequate hardware testing has also been performed. However, the software area has been neglected, resulting in problems after the EHMS has been fielded. These problems were due to “bugs” that were not discovered during functional testing but surfaced when the EHMS became operational. These types of problems occurred specifically on Engine AK’s Engine Cycle Counter (ECC) and Aircraft AE’s Turbine Engine Monitoring System (TEMS). Thorough verification of the software is necessary to prevent this from happening.

USAF fighter engines have experienced instances where the EMS limits were originally set too tight. This resulted in numerous false indications and can not duplicate (CND). Eventual expansion of the limits eliminated the false calls.

Past experience has shown that during flight test, necessary and desired changes to the EMS do occur in every system. Reasons for some of these changes are as follows:

a. Some of the EHMS functions originally chosen do not adequately aid maintenance.

b. Excessive and useless data is being gathered.

c. More information about the engine is available resulting in additional required functions being added to the EHMS.

d. Several trend points are originally programmed with the “best” point being chosen based on flight test results. Due to these known changes in the EHMS, it is recommended that the EHMS be as flexible as possible during flight-tests. This can be implemented via “flexible” software that can be reprogrammed quickly during flight test. The final EHMS production software can then take advantage of flight test results and can be made “permanent”.

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A.3.7.6.2 Onboard engine diagnostic functions.

The EHMS shall acquire the following data: _____.

REQUIREMENT RATIONALE (A.3.7.6.2)

A listing of the diagnostic function performance requirements needs to be specified to ensure the functions are consistent with the overall monitoring system requirements.

REQUIREMENT GUIDANCE (A.3.7.6.2)

The following should be used to tailor the specification paragraph:

There needs to be coordination on integrated diagnostic requirements between the air vehicle manufacturer and the engine manufacturer.

The Using Service should ensure the EHMS data being recorded can be downloaded and analyzed. The ground support system, engine or air vehicle should be identified as the primary entity responsible for ensuring the data will be maintained and analyzed.

The onboard engine diagnostics system should perform the following functions:

a. Acquire data to determine engine performance status and performance trends, and mission profile data (altitude, Mach number, alpha, beta, etc.).

b. Acquire data for parts life tracking.

c. Acquire limit exceedance data (a) seconds prior to the event until (b) seconds after.

d. Acquire pilot initiated data records.

For carrier based aircraft, the onboard engine diagnostics system should perform:

a. Takeoff thrust or power check and provide a signal to the air vehicle. The thrust or power should be within (c) percent of the actual net thrust or power. The EHMS should be able to determine engine mission capability and provide a signal to the air vehicle.

Past specifications have used the following values to tailor the preceding paragraph:

(a): A value of at least 15 seconds.

(b): A value of at least 15 seconds.

(c): For turbofan and turbojet applications: a maximum value of 4 percent or the accuracy level required by the air vehicle system.

For turboprop and turboshaft engines: a maximum value of 2.5 percent or the accuracy level required by the air vehicle system.

Depending on the complexity of the monitoring system, the following is recommended to tailor the specification paragraph:

(a): Provide flight line indication of onboard diagnostic system failures (self-test).

(b): Provide oil quantity indications or readings at a readily accessible location and at the servicing location for the engine oil. A reliable oil quantity indication or reading should be retained after engine shutdown and electrical power is off. The reading should be retained for status display until the engine is restarted, the engine is replaced, or maintenance is performed on the oil system.

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(c): Provide failure identification to the individual sensor or associated circuit.

(d): Provide propulsion data and advisory status to the aircraft per the ICD.

(e): Acquire vibration data to provide indication of engine mechanical health.

(f): Oil and fuel filter impending and actual bypass. Acquire quantitative oil debris parameters including small, large and total debris. Oil debris analysis should include hit trending (debris accumulation rate) along with total and individual particle counts.

(g): Transmit engine data to aircraft flight data recorder for use in characterizing mission usage.

(h): Hydraulic fluid working pressure and temperature indications.

Background:

The following further defines the information that the contractor should provide the Using Service:

a. The logic, algorithms, and parameters used to measure and calculate engine performance trending should be reviewed with the contractor at Design Reviews or other technical interchange meetings.

b. The structures engineers should review the proposed list of parts life tracking parameters and the algorithms used to calculate cycles or life.

c. The EHMS should record all parameters that exceed their limits, and any mission or safety critical parameter whose limit has been exceeded should also be displayed in the cockpit. Limit exceedance may also include engine stalls or blowouts. For stall events, the FADEC may provide a signal to the EHMS that instructs the EHMS to record the appropriate parameters. Combustor blowout events may be recorded in a similar manner.

d. The pilot should have the capability to initiate a data record any time during the mission. The pilot initiated data should include all parameters normally recorded by the EHMS unless the Using Service has a unique set of parameters that it wants recorded.

e. The pilot must be able to check takeoff thrust or power level and to be able to assess whether the engine is mission capable, especially for carrier operations. Adequate thrust or power is essential for successful launch from a carrier, and notification of a component or system failure which would jeopardize completion of a mission should be provided

f. The EHMS needs to diagnose and report any malfunctions of itself.

g. Oil quantity is a safety critical parameter and should be recorded. It is desired to have the oil quantity available with the engine not running without having to open any access panels to reduce aircraft turnaround times.

h. The onboard diagnostic system may not be sophisticated enough to pinpoint a failure to a specific part or component. However, the onboard system can, as a minimum, find a failure in an electrical circuit (i.e.; wiring, connector, or sensor) if it cannot fault isolate to a specific sensor or component.

i. The EHMS may be required to provide data to the aircraft for either the aircraft monitoring system or for cockpit display of engine parameters. The ICD or other interface document should list which parameters need to be provided to the aircraft.

j. The EHMS may need to provide data to the aircraft's flight data recorder in lieu of a monitoring system. The EHMS design should be flexible enough to accommodate data transfer to several aircraft systems.

k. Vibration data is needed to detect early bearing failures and imbalance conditions. The number of sensors proposed for the engine should be sufficient to detect compressor and turbine imbalance, impending bearing failures, and gearbox problems. Depending on the complexity of the onboard system,

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the vibration frequency analysis and Fast Fourier Transform analysis may or may not be part of the onboard system. Analysis of the data should be part of the EHMS, but whether it's part of the onboard or ground support system should be made on a system level cost effectiveness basis. All turboshaft and turboprop engines should have gearbox-mounted sensors.

l. The EHMS should also maintain the engine serial number, FADEC serial number, FADEC software version number, EMS serial number, and EHMS software version number. Crash survivability may or may not be an EHMS requirement. New air systems may be required to store data in a crash survivable component and this may require a similar component on the engine. Conversely, the aircraft may only require that certain parameters be provided to the aircraft for storage in the air vehicle's system.

REQUIREMENT LESSONS LEARNED (A.3.7.6.2)

Go or No-Go indications may not be acceptable for Power Assurance Check (PAC). The Army has indicated that temperature margin measurement is more desirable than a PAC Failed indication for Engine E's FADEC. The temperature margin measurement allows for mission decisions to be made versus the Go or No-Go indicator.

A.4.7.6.2 Onboard engine diagnostic functions.

The requirement of 3.7.6.2 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.7.6.2)

The EHMS data must be reviewed and analyzed during all engine testing to ensure the EHMS is recording the proper data accurately and that the data can be retrieved easily.

VERIFICATION GUIDANCE (A.4.7.6.2)

Past verification methods have included inspection, analysis, demonstration, and test.

Background:

It is important that the EHMS be tested on the qualification engines to ensure the system will work properly. An important function of the tests will be to establish thresholds for determining out of range faults, verifying when false faults occur, determining if signal noise is present, and revealing other problems that may occur. Flight-testing is needed since that is when the true EHMS environment and usage will occur.

VERIFICATION LESSONS LEARNED (A.4.7.6.2)

EHMS testing and operation has revealed that the expected inputs used in design are actually not received during actual operation due to signal noise, unstable readings due to increased sensitivity, unexpected pilot and maintainer inputs which has affected the operation of the EHMS.

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A.3.7.6.2.1 Built-in-test and inspectability (BITI).

Built-in-Test (BIT) shall be implemented to accomplish failure detection and failure isolation to the WRA and LRU without the assistance of support equipment at the organizational level. The testability of the engine system electronics shall be per _____. If required by the air vehicle, failure status to determine mission capability shall be provided to the air vehicle. A nonvolatile record of BIT failure data shall be retained in memory. The BIT information shall include provisions to allow both Read Only Memory (ROM) and Random Access Memory (RAM) contents of the subject computer to be inspected (monitored).

REQUIREMENT RATIONALE (A.3.7.6.2.1)

BIT reduces maintenance downtime, cost, complexity, and personnel requirements and provides the ground crew and pilots the capability to assess the operational usability of the system.

Nozzle rate and position must be measured to determine nozzle vectoring repeatability and nozzle stability. These parameters must be recorded to detect and quantify degradation in the thrust vectoring nozzle system. The simulation of a thrust vectoring nozzle failure event provides a quick and easy means for the ground crew to assess the fail-safe nozzle position due to hydraulic or electrical failure, or battle damage to the vectoring system and its accompanying components.

REQUIREMENT GUIDANCE (A.3.7.6.2.1)

The following should be used to tailor the specification paragraph: Tasks 103, 104, and 105 of MIL-HDBK-2084 should be used for guidance. MIL-HDBK-2084 should be tailored by the contractor to the requirements of the engine control system (e.g.; false alarm rate, isolation levels, and test time).

Background:

BIT may operate in a periodic mode, an initiated mode, or in a combination of both modes. The periodic mode is the primary mode of BIT and is intended to provide maximum operational capability and to support rapid aircraft turnaround. Periodic BIT should automatically operate when energized with electrical power. The initial periodic BIT at power-up should be a one time comprehensive system performance check intended to verify the system's capability to perform its operational requirements. The initiated mode should accomplish that portion of the failure detection and failure isolation capability that the periodic mode is unable to achieve. Initiated BIT should begin operation upon receipt of an external command, may interrupt normal operation and require operator participation.

REQUIREMENT LESSONS LEARNED (A.3.7.6.2.1)

None.

A.4.7.6.2.1 Built-in-test and inspectability (BITI).

The requirements of 3.7.6.2.1 shall be verified by: (a) . The equipment fault detection and isolation capabilities shall be verified per: (b) .

VERIFICATION RATIONALE (A.4.7.6.2.1)

There is a need to verify BIT and inspectability requirements.

VERIFICATION GUIDANCE (A.4.7.6.2.1)

(a) Past verification methods have included analysis, demonstration, and test.

(b) MIL-HDBK-2165 should be used for guidance.

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The following should be used to tailor the specification paragraph: Appendix A, Task 203 of MIL-HDBK-2165 should be used for guidance.

VERIFICATION LESSONS LEARNED (A.4.7.6.2.1)

None.

A.3.7.7 Optical systems.

Switching from the engine to the air vehicle or external (ground) electrical power system shall not affect the operation of the optical system. Optical sensors (e.g.; augmentor flame detectors, optical pyrometers) shall maintain accuracy and calibration, within established limits, during the life of the engine module in which it is installed.

REQUIREMENT RATIONALE (A.3.7.7)

The Using Service needs to know if a fiber optic system is being used on the engine and for which systems. Optical systems (pyrometers and flame sensors) are used by engine controls to schedule engine functions and must be capable of working properly when using any electrical power. If the engine is required to operate on air vehicle or external electrical power, then the optical systems must function during switching to this power. An optical sensor must not be changed during the life of the module it is installed on, so maintenance is kept to a minimum.

REQUIREMENT GUIDANCE (A.3.7.7)

The following military specifications have been cancelled and can be used for reference only:

MIL-A-24726	Attenuators
MIL-M-24731	Multiplexers, Demultiplexers, Freq. Division
MIL-T-24735	Transmitters, Analog
MIL-R-24737	Receivers, Analog
MIL-C-24621	Couplers, Passive
MIL-T-24721	Transmitter, Digital
MIL-R-24720	Receiver, Digital
MIL-M-24736	Multiplexers, Demultiplexers, Time Division

Background:

In a fiber optic engine control system optical components perform functions that are normally performed by electrical or electronic components. For example, fibers replace wires as the signal transmission medium of a data link. The overall functionality of the control system is left unchanged. The performance requirements for the overall engine control system should therefore be the same whether the system contains fiber optic components or electrical components, or both.

In the simplest configuration a fiber optic data link consists of a single transmitter, a single fiber and a single receiver. The transmitter utilizes a light emitting diode (LED) or laser diode (LD) to convert an electrical signal to an optical signal. The optical signal is transmitted through the fiber to a receiver, which

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utilizes a detector to convert the optical signal back into an electrical signal. System modularity requirements require the addition of fiber optic connectors, which facilitate repair and maintenance.

Fiber networks of increasing complexity contain additional fiber optic components. Passive couplers allow optical signals to be divided (or combined) among two or more pathways without the need to convert them to electrical signals. Multiplexers allow several distinct signals to simultaneously occupy a single fiber element. These include time-division and frequency-division multiplexers, which are analogous to the electrical devices of the same name. Additionally, wavelength-division multiplexers and demultiplexers combine and divide signal channels that are distinguishable by their wavelengths. To control signal levels fiber optic attenuators are used. When damage to optical fibers occurs, repair rather than replacement may be elected, in which case fiber optic splices are implemented to join two fiber ends together.

The use of fiber optics offers significant advantages over electrical wires in several areas. Examples are: (1) weight reduction, (2) signal bandwidth, and (3) electromagnetic environment performance, among others. On the other hand, fiber optics has some unique disadvantages as well. For example, light sources are sensitive to temperature with respect to output power and bandwidth drift. Light emitting diodes typically do not produce high output power, while the higher-powered laser diodes generally require temperature stabilization, have a shorter life, and may raise eye safety concerns. Optical fibers are subject to micro cracking due to mechanical stresses and bending losses. Connectors are not always repeatable in mating and unmating with respect to insertion loss, are easily contaminated, and susceptible to rough handling by untrained personnel. These issues and others are addressed by the requirements and verification methods set forth in this specification.

In order to realize certain advantages that fiber optic systems have over electronic systems (for example, high bandwidth, EMI immunity, low volume and weight) control system designs that include fiber optic components need to be developed. The performance and reliability of the engine control system should be paramount in this development; hence, the engine control system requirements must be met at all stages of this development. Indeed, the control system performance requirements should guide this development. Along with advantages, fiber optic components may bring with them inherent weaknesses that could compromise the performance or reliability of the engine control system. A comprehensive set of requirements and verification procedures for fiber optic systems that take into account these weaknesses will ensure that the integrity of the engine control system is maintained.

Transmitters:

Optical transmitters interface with the electrical system and supply the optical signal energy for all fiber optic engine control and data communication system functions. Therefore, they may be considered as both electrical and optical components. Maintaining certain characteristics of the optical transmitters within strict design tolerances is essential for proper operation of the engine control system.

The referenced fiber optic transmitter general specifications (MIL-T-24735 and MIL-T-24721) have been cancelled and may be used for reference only.

The key differences between shipboard and aircraft applications pertain to the section on environmental requirements, specifically to the temperature extremes and fluid exposure. Laser diodes are particularly sensitive to high temperatures, such that they typically require thermoelectric cooling to sustain reliable operation.

The performance requirements specified for the transmitter will depend to a large extent on its role in the system. The peak emission wavelength, spectral width, and power levels will depend on the type of fiber used (e.g., single mode or multi-mode), which, in turn, will depend on bandwidth and power requirements of the data link. These parameters should be considered carefully in the early stages of system design. The electro-optical properties of the transmitter, on the other hand, are dependent on its role as an electronics component, affecting the overall system transfer function. In this regard the signal-to-noise

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ratio (SNR), frequency range, distortion, and group delay of the optical transmitter should be included as parameters in the electronics system analysis.

Receivers:

Like Optical Transmitters, Optical Receivers can be considered as both optical and electrical components. As with optical transmitters, the optical, electro-optical, and electrical performance parameters, along with their respective tolerances are defined during the fiber optic system design process.

The fiber optic engine control system manufacturer should specify these performance requirements, to include spectral response, dynamic range, sensitivity, signal-to-noise ratio, frequency ranges, signal distortion, group delay, and output signal interface requirements.

MUL/DEM:

The engine control system manufacturer should specify as part of the control system design the physical and performance requirements of all multiplexers and demultiplexers used. These are to be identified as frequency-division or time-division multiplexers (FDM or TDM), demultiplexers (FDDM or TDDM), or MUL/DEMS. The referenced specifications (MIL-M-24731 and MIL-M-24736) have been cancelled and may be used for reference only. They were identified as specific to shipboard applications, but were sufficiently generic as to be readily adopted for aircraft applications as well.

Frequency-division multiplexers and demultiplexers:

The engine manufacturer should specify as part of the control system design the classification of the frequency-division multiplexers and demultiplexers. The designations used to classify these units are Function, Multiplexed Signal Type, Channel Configuration, Analog Subchannel Class, Digital Subchannel Class, and Style. The Function designation defines the unit as a FDM, FDDM, or a MULDEM. The Multiplexed Signal Type designation defines the signal outputs and/or inputs as electrical or optical. The Channel Configuration designation defines the number of input or output channels. The Analog Subchannel designation defines the highest required analog signal subchannel bandwidth. The Digital Subchannel Class designation defines the highest equivalent digital data rate. The Style designation defines the package configuration. See cancelled MIL-M-24731 for additional reference material.

The engine manufacturer should specify quantitative reliability requirements expressed in mean-time-between-failures (MTBF), mean-cycle-between failures, or other quantitative measure of reliability. The engine manufacturer should specify performance requirements on transmitters, receivers, and connectors built into the multiplexers as part of the multiplexer design. Also specified should be dimensions, weight, solder ability, source of prime power, and interface parameters.

Time-division MUL/DEMs:

The engine manufacturer should specify as part of the system design the classification of the time-division multiplexers and demultiplexers. The designations used to classify these units are Function, Multiplexed Signal Type, Channel Configuration, Analog Subchannel Class, Digital Subchannel Class, and Style. The Function designation defines the unit as a TDM, TDDM, or a MULDEM. The Multiplexed Signal Type designation defines the signal outputs and inputs as electrical or optical. The Channel Configuration designation defines the number of input or output channels. The Analog Subchannel designation defines the highest required analog signal subchannel bandwidth. The Digital Subchannel Class designation defines the highest equivalent digital data rate. The Style designation defines the package configuration. See cancelled MIL-M24731 for additional reference material.

The engine manufacturer should specify quantitative reliability requirements expressed in mean-time-between-failures (MTBF), mean-cycle-between failures, or other quantitative measure of

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reliability. The engine manufacturer should specify performance requirements on transmitters, receivers, and connectors built into the multiplexers as part of the multiplexer design. Also specified should be dimensions, weight, solder ability, source of prime power, and interface parameters.

Attenuators:

The engine manufacturer should specify the design and performance requirements of all fiber optic attenuators used in the fiber optic system. Fiber optic attenuators are classified according to Type, Port Construction, and Package. The Type designation defines whether the level of optical power reduction is fixed or variable. The Port Construction designation defines the method of interfacing attenuators to the fiber optic cables. The Package designation defines the type of packaging as either stand-alone or printed circuit board mounted. See cancelled MIL-A-24726 for additional reference material.

The specification of attenuator requirements should include design and performance requirements on the fiber optic pigtails and connectors integral to the attenuator package.

REQUIREMENT LESSONS LEARNED (A.3.7.7)

USAF fighter engine optical pyrometers, used to sense augmentor light off, have experienced false indications from dirty lenses. Their operating environment requires cleansing air to be recirculated over the viewing lenses. Inadequate cleansing airflow resulted in dirty lenses, augmentor shutdowns and increased maintenance. Exercise care in sizing the cleansing airflow.

A.4.7.7 Optical systems.

The requirements of 3.7.7 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.7.7)

Component requirements unique to fiber optic engine control systems must be verified to ensure proper operation of the engine control system under extreme operating conditions.

VERIFICATION GUIDANCE (A.4.7.7)

Past verification methods have included analysis, demonstration, and test.

Background:

The specification documents referenced in 3.7.7 and in following paragraphs are sufficiently comprehensive and generic to be applicable in their totality to the various components of the fiber optic control system. Some were developed with shipboard applications in mind. They are augmented here by the appropriate paragraphs to include the severe operating conditions unique to aircraft engine applications. These conditions include, most notably, temperature extremes, vibration, and chemical exposure conditions. Tailored shipboard documents may be used until aircraft fiber optic specifications are written because no set of documents exists for airborne applications.

A fiber optic engine control system utilizes components that operate on different physical principles than electronic components and thus may behave very differently under extreme operating conditions.

Transmitters:

The most critical area in the verification of optical transmitters is that of reliability, especially for laser diodes in high temperature environments or high power applications. Lifetimes of laser diodes are in the range of 40,000 to 50,000 hours, at room temperature, while those of LEDs may exceed 200,000 hours. The advantage of laser diodes over LEDs is the higher power and efficiency that they afford.

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For LEDs and LDs, spectral peak output power shifts approximately 0.25 NM per °C. Laser diodes generally require thermoelectric cooling, which itself is subject to reliability problems at elevated temperatures.

Receivers:

The engine manufacturer should specify the ranges of temperature (both operating and nonoperating) and vibration, and the nature of the chemical exposure that the optical receiver is expected to undergo while in service. This information will be used to determine test conditions during the verification of performance under the effects of temperature extremes, thermal shock, and nuclear radiation resistance.

Several noise components of the optical receiver are functions of temperature. Thermal noise is the dominant noise source at 25°C, and increases slowly with temperature. Shot noise (the statistical fluctuation of photon arrival rate) and diffusion current noise are usually small at room temperature, but increase rapidly with temperature. Temperature thus affects the SNR, and hence the dynamic range and sensitivity. Nuclear radiation may lead to ionization-induced photocurrents and likewise reduce the SNR. In addition, nuclear radiation may induce transient absorption and luminescence in the fiber pigtails attached to the receiver. These are temporary effects that usually recover soon after the radiation field is removed; however, the recovery rate is a function of temperature (faster at higher temperatures) and the luminescence intensity may, in some cases, be high enough to permanently damage a detector element.

Optical MUL/DEMs:

The performance requirements for optical multiplexers and demultiplexers include specifications of the limits of tolerable performance degradation due to manufacturing tolerances and adverse mechanical and environmental conditions. These limits on performance degradation must be adequate to ensure reliable overall performance of the fiber optic system under adverse conditions. The performance requirements should also be specified in the engine specification. Compliance with these requirements should be verified by tests in which the appropriate performance parameters are measured under simulated adverse conditions. The methods for simulating these conditions and for measuring the performance parameters are described in the referenced documents.

VERIFICATION LESSONS LEARNED (A.4.7.7)

None.

A.3.7.7.1 Fiber optic cables and connectors.

Fiber optic cables and connectors shall be specified herein. At the sea level cold temperature of figure 5, it shall be possible to flex fiber optic cables and connect or disconnect optical connectors using normal maintenance procedures, without damage to the fiber cables or connectors.

It shall be possible to remove contaminants from the ends of the transmissive elements without causing damage to them.

REQUIREMENT RATIONALE (A.3.7.7.1)

Fiber optic cables are perhaps the most critical components of a fiber optic system since they convey the signals. Fiber optic connectors allow the fiber optic system to be designed with a degree of modularity, which enhances serviceability. However, connectors potentially represent a weak link in a signal channel because of contamination, damage, or lack of repeatability.

REQUIREMENT GUIDANCE (A.3.7.7.1)

The following should be used to tailor the specification paragraph:

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See MIL-PRF-85045 or an equivalent commercial standard for guidance.

The contractor should provide the maximum allowable change in optical transmittance induced by installation. Generally, accepted values should be 0.5 dB (multi-mode fibers) or 0.2 dB (single mode fibers).

Cabling of optical fibers allows several to be combined into a single modular unit, simplifying installation, maintenance, and repair. The fibers within a given cable need not be related to the same functional unit, nor even be of the same class. The fiber optic engine control system manufacturer should specify the class (single mode or multi-mode) of each fiber and the total number of fibers used in each cable of the fiber optic engine control system. It should also specify the design, construction, and physical dimensions of the complete cable. The engine manufacturer should specify the operating and storage temperature ranges to which the fiber optic cable will be exposed during its service life, as well as methods employed for routing of cables within the air vehicle.

Specific optical performance requirements will depend on the system design. The engine control system manufacturer should specify the maximum allowable attenuation rate, the maximum change in optical transmittance of each fiber, and far end crosstalk between any two fibers. The change in optical transmittance of a cable due to mechanical stresses in installation may lead to unsatisfactory performance of the optical system. The referenced document (MIL-C-85045E, requirements 3.5.1 and 3.5.2) contains default values for the attenuation rate and change in optical transmittance requirements, if these are not specified.

The contractor should provide the connector type and applicable military or commercial specification. MIL-STD-1353B (Electrical, Connectors, Plug-in Sockets and Associated Hardware, Selection and Use of) has been cancelled but was previously used for electrical connector design and verification guidance. MIL-STD-1353B should only be used for reference material.

REQUIREMENT LESSONS LEARNED (A.3.7.7.1)

None.

A.4.7.7.1 Fiber optic cables and connectors.

The requirements of 3.7.7.1 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.7.7.1)

Compliance of fiber optic cables to the specified performance requirements must be verified. Contamination, damage, and lack of repeatability can have severe deleterious effects on the function and reliability of a fiber optic connector. These effects can only be characterized through verification procedures that accurately simulate the operational conditions within which the connector is designed to operate.

VERIFICATION GUIDANCE (A.4.7.7.1)

Past verification methods have included analysis, demonstration, and test.

All cables should be tested for change in optical transmittance per EIA-455-20 following installation on the engine. In addition, the mating durability test of EIA-455-21A should be performed on connectors that have been contaminated with the applicable contaminants specified in section 3.1.1.7.1 and any expected engine bay contaminants and subsequently cleaned.

Background:

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MIL-C-85045 (Specification for Fiber Optics Cables) has been cancelled but was the approved document covering all types and classes of fiber optic cable for military applications and contained verification methodologies for fiber optic connectors. It did not include tests of the combined effects of contamination and repeated connector mating and unmating. MIL-C-85045 should only be used for reference material. Cables intended for use in aircraft are designated as application A (Airborne). The equipment manufacturer should specify the appropriate specification sheet or particular design aspects for each cable selected for the environment in which it is used.

For this test, a representative sample of connectors should be exposed to the specified contaminants in a manner that necessitates subsequent removal of the contaminants from the ends of the transmissive elements. This procedure is followed by the mating durability test as described in EIA 455-21A.

VERIFICATION LESSONS LEARNED (A.4.7.7.1)

None.

A.3.7.8 Lubrication system.

A.4.7.8 Lubrication system.

A.3.7.8.1 Lubrication oil.

The engine shall meet the requirements of the specification when using the following lubricating oil(s):

REQUIREMENT RATIONALE (A.3.7.8.1)

Lubricating oil selection is critical to the design and important for compatibility with logistical support of field operations. The oils must be specified.

REQUIREMENT GUIDANCE (A.3.7.8.1)

The following should be used to tailor the specification paragraph:

Lubricating oils conforming to MIL-PRF-7808 and MIL-PRF-23699 should be used as standard lubricants. Special lubricants may be required for unusual or nonstandard situations. If special or non-standard lubricants are selected, the engine manufacturer should provide compelling data describing why the non-standard product is superior to the standard lubricating oils.

Background:

The oil selected must be able to withstand the operational environment for the full oil change period without loss of properties and capability to lubricate and dissipate the engine heat.

MIL-PRF-23699 is limited to 13,000 centistoke (139,931 ft^2/s) at -40°C (-40°F) for new oil and therefore it is not recommended for use below -40°C (-40°F). MIL-PRF-7808 is currently recommended as the standard oil for use down to -51°C (-60°F). The limitation with MIL-PRF-23699 oil is based on the fact that MIL-PRF-7808 (lighter oil) originally had a new oil viscosity limit of 17,000 centistokes (182,987 ft^2/s) at -54°C (-65°F). In contrast, MIL-PRF-23699 was limited to 13,000 centistoke (139,931 ft^2/s) at -40°C (-40°F) for new oil.

MIL-PRF-7808 and MIL-PRF-23699 oils have very similar performance characteristics. MIL-PRF-7808 oil has superior cold start capability, less load carrying capability, lower viscosity, and is more volatile (higher vapor pressure). MIL-PRF-23699 has higher load carrying capability, lower consumption, higher ambient temperature capability, and better oxidation and thermal stability.

Selection of oil type should be based on factors such as bearing design, mission cycle mix, and operating environment and should be per MIL-HDBK-838.

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Use of specially-developed lubricants may be needed in order to achieve design objectives but must be incorporated into logistic planning early in the design phase.

Any coke and sludge buildup can not be used as a pretext for the engine not meeting specification requirements. The engine should minimize coking and sludge formation because of potential seal and bearing damage and blocked breather tubes and oil filters. As systems mature with increased performance requirements, lubrication systems may become marginal providing increased coke and sludge buildup.

REQUIREMENT LESSONS LEARNED (A.3.7.8.1)

Turbojet Engine W (Aircraft O) requires the use of specially developed lubricant MIL-PRF-87100 lubricating oil.

Wherever possible, propulsion systems should be designed to use standard lubricants as noted in MIL-HDBK-275 and related NATO STANAG's 1135 and 1414. Use of standard lubricants provides streamlined logistics planning and minimizes the number of products needed to support the aircraft. This is particularly true in the tight confines of our ships. In the case of NATO operations, the Standardization Agreements are items of our national defense policy. These agreements provide for interoperability of NATO aircraft during joint operations and transient aircraft movement by limiting aircraft propulsion system lubricants to only mutually agreed upon product specifications. Deviations from the use of such standard products should require a comprehensive cost-benefit analysis from the propulsion system component supplier to justify its need for a special lubricant.

A.4.7.8.1 Lubrication oil.

The requirement of 3.7.8.1 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.7.8.1)

Lubricating oil must be verified to ensure the engine is able to operate at the conditions defined in the specification.

VERIFICATION GUIDANCE (A.4.7.8.1)

Past verification methods have included analysis, tests including endurance, altitude, and starting with post test dirty and clean inspections of the lubrication system, bearings, seals, etc.

Background:

The contractor may procure high quality oil for use during development testing. The government agency responsible for procuring oil stocks will probably buy for the field based on the lowest priced oil. While both products may be on the Qualified Products List (QPL), the oil used by the contractor may be superior to government supply.

VERIFICATION LESSONS LEARNED (A.4.7.8.1)

In the late 1980's engine development testing used a superior (low coking) MIL-PRF-23699 engine oil to provide additional margin for higher lubrication system temperatures. This provided inaccurate information on oil coking in several critical locations. The field will have available the lowest cost QPL that could be the minimum property oil. The engine should operate on the minimum performance product.

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A.3.7.8.1.1 Oil pressure and temperature limits.

The maximum and minimum operating oil pressure limits and maximum transient and maximum allowable steady-state oil temperature limits shall be listed in table IV. The maximum and minimum oil pressures during starting and initial operation at the sea level cold atmospheric temperature of figure 5 or (a) centistoke oil viscosity, whichever is worse, shall be listed in table IV.

Oil system temperatures and pressures following in-flight shutdown and subsequent in-flight air re-starts shall not be detrimental to the lubrication system components nor the system components requiring oil.

REQUIREMENT RATIONALE (A.3.7.8.1.1)

The engine oil system must effectively lubricate the engine within the specified limits.

REQUIREMENT GUIDANCE (A.3.7.8.1.1)

The following should be used to tailor the specification paragraph:

(a): The centistoke oil viscosity specified should depend on the sea level cold day temperature of figure 5.

Background:

The minimum oil temperature should be consistent with figure 5. The maximum transient and steady-state oil temperature should be determined by a combination of environmental requirements and the oil system capability. Minimum and maximum oil pressures during starting and initial operation should not persist for more than 2.5 minutes.

The oil system pressure in conventional designs is a function of oil temperature and may vary significantly, particularly during cold start and warm up. The allowable operating pressures should be related to oil temperature. The acceptable range of pressures with the allowable range of temperatures should be specified. If the minimum specified oil pressure corresponding to the maximum allowable oil temperature is unacceptable at the minimum operating oil temperature, such should be identified in the specification and installation requirements.

The contractor should note the temperature as ‘bulk’ or ‘local’ temperature for clarity.

REQUIREMENT LESSONS LEARNED (A.3.7.8.1.1)

Lubrication system designs that function well under static conditions have failed to function satisfactorily under dynamic conditions of flight.

Designs that specify a low viscosity (MIL-PRF-7808) oil, primarily for low temperature starting considerations, have encountered low oil pressures and high temperature problems after the startup when reaching the normal operating conditions. Conversely the same design could not start properly using the higher viscosity (MIL-PRF-23699) oil due to low oil pressures persisting longer than allowed. Contractors should design their systems to prevent these occurrences.

USAF fighter engines have experienced excessively high lubrication system pressures resulting from aircraft maneuvers such as knife edges (90 degrees of bank). These high operating pressures resulted in pump failure, overboard oil loss and subsequent non-recoverable in-flight shutdowns. The problem was solved by redesigning the pump and incorporating a high pressure relief (burp) valve.

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A.4.7.8.1.1 Oil pressure and temperature limits.

The requirement of 3.7.8.1.1 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.7.8.1.1)

The adequacy of the oil system must be verified.

VERIFICATION GUIDANCE (A.4.7.8.1.1)

Past verification methods have included analysis, tests including endurance, altitude and starting with post test dirty and clean inspections of the gears, bearings, seals, etc.

Background:

Engine testing and oil wetted component inspections should be performed to verify effective oil system operation. At high operating turbine temperatures or high Mach number flight, oil can coke in internal engine passages, which is detrimental to the lubrication system.

VERIFICATION LESSONS LEARNED (A.4.7.8.1.1)

During cold temperature starting tests, large oil pressure spikes have occurred. In some cases the cold temperature starting pressure was great enough to crack a filter housing on one application and cause internal supply pump failures on another.

A.3.7.8.1.2 Oil consumption limits.

The maximum oil depletion rate shall be specified in table IV.

REQUIREMENT RATIONALE (A.3.7.8.1.2)

The oil consumption limit must be established to meet mission requirements within system weight restraints.

REQUIREMENT GUIDANCE (A.3.7.8.1.2)

The following should be used to tailor the specification paragraph:

The contractor should provide the value. For turbofan and turbojet applications, consideration should be given to specifying a mean consumption rate, based on mission profiles. The mean consumption rate may be useful for sizing the oil reservoir. For turboshaft applications, consider using the maximum consumption rate.

Background:

Oil consumption rate should include all forms of oil loss from the engine. Oil seal carbonization, vent loss and leakage are means by which oil is lost from the lubricating system. Lubrication system operating temperature may affect the rate at which oil is lost and should be considered.

REQUIREMENT LESSONS LEARNED (A.3.7.8.1.2)

Oil consumption can impact oil reservoir, seal, air-oil separator, venting, and other components with a compounding effect on engine weight and subsequently air vehicle structure and weight.

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A.4.7.8.1.2 Oil consumption limits.

The requirement of 3.7.8.1.2 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.7.8.1.2)

Oil consumption rates must be verified.

VERIFICATION GUIDANCE (A.4.7.8.1.2)

Past verification methods have included analysis, and engine test using the 4.12.3.4.1 test condition

Background:

Consideration should be given to the margin between average and maximum rates to address problems such as wear and seal manufacture variations.

VERIFICATION LESSONS LEARNED (A.4.7.8.1.2)

Oil consumption can vary based on the oil used.

A.3.7.8.1.3 Oil flow interruption or depletion.

The engine shall meet the requirements of the specification during and after operation at (a) thrust or power for a period of at least (b) seconds with no oil (only air) supplied to the oil pump inlet.

Engines in fixed wing air vehicle applications shall be capable of operating without oil for (c) minutes at (d) thrust or power without rotor seizure.

For rotary wing air vehicle applications, the engine shall be capable of operation at (e) power for a period of at least (f) minutes with loss of all oil supplied to the engine bearing sumps followed by (g) minutes of operation at (h) power with lubrication restored.

REQUIREMENT RATIONALE (A.3.7.8.1.3)

Weapon systems operating over hostile environments or water must be given every opportunity to maximize pilot survivability. Getting the pilot away from any immediate threat and into a more survivable environment is of primary concern. Oil flow interruption can occur due to failure, maneuvers, combat damage, or improper servicing, so the interruption duration capability must be specified. The engine must operate satisfactorily without oil for the period of time specified in order to egress the hostile area, initiate engine shutdown, or any other required action. Most fighter and attack engines need the capability to withstand multiple brief oil interruptions without damage.

REQUIREMENT GUIDANCE (A.3.7.8.1.3)

The contractor should consider the following to tailor the specification paragraph:

(a): Intermediate thrust for turbofans and turbojet engines and intermediate power for turboprops and turboshafts.

(b): 30 seconds

(c): 5 - 30 minutes

(d): 30% Intermediate SLS power

(e): 75% maximum continuous power

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(f): 6 minutes

(g): 30 minutes

(h): 75% maximum continuous power

Prior to defining the duration of engine operation without oil (c), consideration should be given to the vulnerability requirements for the aircraft for various threats and kill levels. For single engine aircraft the engine oil system is a primary contributor to the aircraft vulnerability. The longer an engine can operate (even at partial power as indicated above) significantly reduces the aircraft's vulnerability to the defined threats.

Background:

Three minutes at intermediate is sufficient time to regain speed and altitude following an engagement. Thirty minutes is usually sufficient time to permit egress from heavily defended areas where combat damage is likely to be rendered (e.g., 100 nm at a "limp home speed" of 200 kts). Typical fighter or attack aircraft can maintain flight at moderate altitudes with thrust levels generally well under 30% of intermediate SLS thrust. Thirty-percent intermediate SLS thrust is typically a comfortable operating condition for the engine, with low temperatures and rotor thrust loads. Since the main function of oil lubrication in rolling element bearings is heat removal, reduction of bearing loads and surrounding metal temperatures affected by throttle reduction should enable the bearings to survive for extended periods. A number of relatively low penalty design possibilities exist such as auxiliary lubrication from a shielded or protected sump.

An engine intended for use in a single engine aircraft should also be capable of operating for longer periods without oil until rotor seizure occurs, and a requirement describing this capability should be stated in this paragraph.

High-agility fighter or attack aircraft may experience repeated transient negative "g" conditions that can interrupt oil supply.

This requirement can add complexity to the bearing cavity design by requiring reservoirs for oil supply out operation. This can cause a need to fill these reservoirs during engine starting when there may be a need for oil-dampened bearings to have oil supply priority.

The requirement to run an additional 30 minutes after the loss of lube for 6 minutes is an acknowledgement that helicopter gearboxes and engine requirements should be capable of some additional time of operation while transmitting power to permit the safe landing and/or recovery of the aircraft..

REQUIREMENT LESSONS LEARNED (A.3.7.8.1.3)

Tests demonstrated Engine A's capability to operate for up to 8 minutes, without oil being supplied to the main bearings. These tests were accomplished at various simulated flight conditions, to assess Aircraft A's mishap risk, in the event of an oil system failure. An Aircraft A pilot will attempt to gain maximum altitude when serious engine trouble is suspected. He then sets up for the optimum glide slope at a constant power setting to yield maximum ground distance in case of actual engine failure. The greater the ability of an engine to operate without oil, the greater survivability for the pilot and air system.

In some current and past Army helicopter programs, engines had to operate satisfactorily for a minimum of six minutes at 75% maximum continuous power after permanent loss of all oil supply to the engine oil pump, followed by 30 seconds of successful operation at intermediate power before engine shutdown with no engine failure. When oil flow was restored, the engines had to operate satisfactorily for 30 minutes at 75 percent maximum continuous power.

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Helicopter steady cruise power testing for loss-of-lube was deemed inadequate in the mid-80's. Helicopters don't have as much latitude as fixed wing applications to fly constant power. Based on the current lubrication system reliability numbers (for component failures and maintenance induced system leaks), the USN's current Loss-of-Lube test spectrum proposes the loss of oil pressure in the main/primary system will be due to combat (projectile) damage during a hover condition.

The requirement for endurance of the engine without lubricating oil has permitted recovery of aircraft that would otherwise have been destroyed. Fleet experience has shown some systems cause problems during high angle climbs, negative g and minimum oil level conditions. Operation down to the oil level defined as “unusable” is intended to prevent oil system pressure fluctuations that might occur in a marginal design when the oil level drops significantly. The capability to operate without oil for a significant period, even at reduced thrust, would prevent most oil depletion aircraft losses.

Engine and VTOL drive systems considered a 30-minute requirement for both engine and drive system components but the concept was not pursued long for engines because of complexity and weight. The 30 minute engine concept included a separate tank, etc., [a backup system], and perhaps armor. The next thought was 10 minutes and the concept that was thought feasible was an internal reservoir, etc., as in Engine C. However, Engine C did not demonstrate the 10 minutes loss of lubrication requirement successfully.

Experience has shown that sudden rotor seizure due to lack of lubrication can be violent enough to break the engine out of its mounts and cause significant damage to air vehicle and test cells.

The USN’s Engine P2 oil interruption requirement was that the engine shall operate at maximum continuous power for a period of 30 seconds during which no oil is supplied to the engine oil pump inlet. As a result of this operation, there shall be no detrimental effects to the engine during the oil flow interruption period or engine operation thereafter. Engine P3 has a specification requirement to demonstrate 30 seconds of operation without oil flow at combat rating followed by 30 minutes of engine operation with oil flow restoration and no damage to the engine. Engine P3 completed an oil interruption test on engine 964/2 reported in E/VDCB/135483 dated 20/8/90 (Part of compliance). The engine continued running satisfactorily after 30 seconds with the oil turned off at combat rating.

A.4.7.8.1.3 Oil flow interruption or depletion.

The requirement of 3.7.8.1.3 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.7.8.1.3)

The tolerance of the engine to oil flow interruption must be verified.

VERIFICATION GUIDANCE (A.4.7.8.1.3)

Past verification methods have included analysis, engine test and posttest dirty and clean inspection of oil wetted components.

After the turboshaft oil depletion test, the analysis and inspection should not indicate pending failure.

Since thrust bearings tend to be most sensitive to oil-out, rig testing on one or both thrust bearings to demonstrate multiple interruption capability should be conducted.

If a full up engine test is required, careful consideration should be given to the condition of the hardware (run time, cycles, etc) and when the test is scheduled. Testing a new engine may give different results than one with accumulated time and cycles. Test scheduling has associated risk due to the potential for damage to the oil wetted components.

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VERIFICATION LESSONS LEARNED (A.4.7.8.1.3)

The USN's Engine H ran two oil flow interruption tests. The first test was a "temporary" thirty second loss of oil test conducted at intermediate power. This test was conducted in the vertical attitude or helicopter mode. The second test was a "permanent" 4.5-minute loss of oil test conducted at 75% MC power. This test was conducted in the horizontal attitude or airplane mode. The 4.5-minute test was reduced from the original specification requirement of 6 minutes due to a failure of a bearing. The specification was subsequently changed.

A.3.7.8.2 Lubrication system components and features.

A.4.7.8.2 Lubrication system components and features.

A.3.7.8.2.1 Oil reservoir.

Unvented oil reservoirs shall meet the factor of safety pressure requirements of 3.4.1.6.1. If the lubrication and hydraulic systems share a common reservoir, the lubrication system shall be protected from total depletion in the event of a hydraulic failure.

The capacity of the oil reservoir shall be as shown in table XXVII.

REQUIREMENT RATIONALE (A.3.7.8.2.1)

Structural integrity of the oil reservoir is mandatory to avoid the fire hazards, which can accrue from rupture or leakage thereof, and to avoid oil depletion. Deaeration and oil baffling is required to minimize foaming and provide proper inlet flow.

The usable oil supply must be compatible with the oil consumption rate and the aircraft mission to ensure an adequate supply of oil.

REQUIREMENT GUIDANCE (A.3.7.8.2.1)

The following should be used to tailor the specification paragraph:

The contractor should provide all the required oil reservoir capacities. Usable oil quantity should, as a minimum, allow the engine to operate for the longest aircraft missions at the maximum hourly oil consumption specified in 3.7.8.1.2. For turbofan and turbojet applications, if maximum hourly consumption cannot be supported, consideration should be given to using the mean consumption rate.

If the engine does not have a separate hydraulic system, delete the second sentence in the main paragraph.

Background:

Engine mounted oil reservoir(s) and mountings should be capable of withstanding, without permanent deformation or fatigue damage, the stresses imposed by reservoir pressurization, engine vibration, landing loads, and cyclic stresses imposed by variations in ambient pressure. The oil reservoir should withstand a differential proof pressure of 100 kPa (15 psi)(pressurized systems) positive and negative or twice the maximum reservoir operating differential pressure within the operating envelope, whichever is greater, without visible leakage or deformation.

The reservoir should satisfactorily withstand 10,000 cyclic pressure reversals at the maximum reservoir operating pressure without visible leakage or permanent deformation. The reservoir internal design should be suitably baffled or use other means to prevent oil sloshing, should contain provisions to facilitate deaeration of return oil, and should include an inverted breathing system to ensure satisfactory oil system operation for applications where inverted flight will be encountered.

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High reservoir capacity should be considered for aircraft with long range, high sortie mission requirements or sealed installations.

The usable and unusable quantities determine the true operating volume of oil and the minimum quantity required to sustain oil pressure. The gulping volume shows how much of the initial capacity will be pumped into the system on startup. Expansion space above the maximum oil level is required to provide for thermal expansion, deaeration, foam buildup in the tank, and a free air space at the tank vent outlet to prevent oil spewing and tank over pressurization. Total enclosed reservoir volume relates to the physical size of the reservoir, and the quantity of oil available for removing heat. A small capacity reservoir causes the oil to be worked harder since for a given flow rate, the bulk quantity of oil sees more passes per minute through the lubrication system circuit.

REQUIREMENT LESSONS LEARNED (A.3.7.8.2.1)

Several engines have had oil tank cracking problems. Inverted flight has caused oil tank over pressurization. Low altitude, high Mach, and inverted (negative g) operation has been found to be the most critical condition for over pressurizing oil tanks in certain engine designs. Redesign has been required even when high altitude inverted flight had been demonstrated. Internal oil tank tubing and baffling have failed due to resonant vibrations in the engine vibration environment.

Inadequate oil tank capacity has resulted in mission aborts and in-flight shutdowns. Proper design and test of the deaeration system in the reservoir can enhance the usable oil quantity.

Usable oil quantity equal to a minimum of 12 times the maximum hourly oil consumption was provided so that the aircraft could fly two 6-hour missions without replenishing the oil supply. Turboshaft engines have used 12 times the maximum hourly oil consumption rate.

Secondary contamination of engine lubrication systems has occurred on turboshaft engines due to failure of accessories on common reservoir systems. Engine AM has shown a history of reduction gearbox contamination due to failure of the air vehicle installed Western gearbox that shared common lubrication.

On Aircraft AF, contamination of the engine oil system with transmission oil will result from a failure of the engine to transmission interseal. Helicopter transmission oil (DOD-L-85734) is authorized for use in the transmission but not the engine

A.4.7.8.2.1 Oil reservoir.

The requirements of 3.7.8.2.1 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.7.8.2.1)

The structural integrity and performance of the oil reservoir must be demonstrated.

VERIFICATION GUIDANCE (A.4.7.8.2.1)

Past verification methods have included analyses and tests which have included cyclic fatigue, proof pressure, valve, and vibration with posttest inspection

The following procedure may be used as a verification method:

“The filler cap and other fittings shall be installed, the tank mounted in a manner similar to that as found on the engine, and the entire test assembly shall be subjected to the following tests:

a. Cyclic Fatigue Test. The oil tank shall be cycled between the minimum and maximum differential pressure limits up to the design service life of 3.4. The differential pressure to be used in the cyclic fatigue test shall be the absolute value of the difference between external and internal pressure of the oil

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tank. The reservoir temperature shall be kept at the maximum oil operating temperature for a length of time per the missions and environments of the specification.

Throughout this cycling, no leakage or permanent deformation of the oil reservoir, filler cap, or fittings shall occur.

a. Proof Pressure Test. Upon successful completion of the cyclic fatigue test, the same oil reservoir assembly shall be subjected to a proof pressure test. The proof pressure shall be held for a minimum of six minutes with the oil reservoir at the maximum oil operating temperature.

No leakage or permanent deformation of the reservoir, filler cap, or fittings shall occur.

a. Valve Tests. If the oil reservoir assembly incorporates a pressurizing valve or pressure relief valve, the assembly shall be tested to demonstrate proper functioning."

The oil tank capacity requirements should be verified through bench testing prior to engine test. Demonstration should be accomplished by determining the quantity of oil that is required to fill the oil reservoir when it contains only the minimum amount of oil necessary to provide oil to the pump at the maximum operating speed of the engine (i.e., unusable oil plus the gulping volume specified.) This quantity of oil must be at least equal to the amount specified as usable oil.

VERIFICATION LESSONS LEARNED (A.4.7.8.2.1)

Cyclic fatigue tests have been conducted with half of the cycles at nominal oil operating temperature and half at maximum oil operating temperature.

A.3.7.8.2.1.1 Oil reservoir external features.

The oil reservoir shall contain features to determine the oil level in the reservoir for the attitudes of 3.2.2.2. An oil quantity signal __(a)__ be provided to the air vehicle. The reservoir shall have a __(b)__ system(s) and shall have features for overfill protection. Oil servicing shall be possible with the engine in a range of positions from 15 degrees nose-up to 20 degrees nose-down.

REQUIREMENT RATIONALE (A.3.7.8.2.1.1)

Oil level indicators are required to facilitate proper oil servicing. Oil quantity indicators may be required to warn the pilot of impending oil depletion.

REQUIREMENT GUIDANCE (A.3.7.8.2.1.1)

The following should be used to tailor the specification paragraph:

(a): The contractor should put "shall" or "shall not" in the blank.

(b): The contractor should put "pressure", "gravity fill", or "both pressure and gravity fill" in the blank.

The quantity indicator can be either a visual indicator or a sensed oil level that is sent to the monitoring system. The requirement for servicing at 15 degrees nose up to 20 degrees nose down ensures the reservoir supply can be replenished in any military environment including on an aircraft carrier.

Background:

An indicating device should be provided on or in the oil reservoir to notify the maintenance personnel when the tank needs servicing. Graduated markings should be clearly legible and should be spaced such that overfill conditions can be easily recognized. Air vehicle and engine manufacturer coordination should be conducted to ensure oil checking and servicing can be performed easily. Sight gage glasses should

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be removable for cleaning of the glass and background. Consideration should be given to chemical effects that may cloud the glass or darken the background. A high-contrast background is desirable.

A continuous oil quantity gaging system should be provided to indicate oil loss rate; however, reliability and maintenance requirements should be given special attention if a gage is required. Signal accuracy should be specified such as ±-4 percent of the usable oil volume.

A low-level warning signal should be provided to the aircraft when there are two hours of usable oil left.

Oil port caps should be of the quick-disconnect type with positive closure feature and should be oversized to permit operation with gloved hands. Caps should be attached to the fill port by a plastic covered steel cable. The caps and adapter should be such that water can not collect and drain into the reservoir.

The reservoir should prevent filling the expansion space when servicing. Servicing capability should be designed to prevent oil overflow of the reservoir from spilling on engine or aircraft components. The pressure fill connector and the overflow return connection should be per MIL-PRF-25676. A convenient method should be provided for servicing the oil reservoir for those aircraft installations where the reservoir is not readily accessible.

See NATO STANAG 3595 for guidance on standard pressure fitting connections for adding oil to the engine.

REQUIREMENT LESSONS LEARNED (A.3.7.8.2.1.1)

Considerable difficulty has been encountered in attempts to devise a remote reading oil quantity system for gas turbine engines. This was largely due to the relatively small quantities of oil used and the short dwell times. Nucleonic continuous reading systems have been incorporated but difficulties have been encountered. Overfill scenarios can present potentially serious hazards. On one engine, before the oil level was checked, the engine was supposed to be motored to scavenge the oil from the engine gearbox to the oil tank. The motoring was not done and the oil tank was inadvertently overfilled. This happened twice to the engine. On a subsequent flight the scavenge pump in the engine gearbox could not pump the excess oil into the tank and the pump failed. One such occurrence resulted in an in-flight lubrication pump failure that caused the engine rotor to seize forcing the pilot to make an unpowered emergency landing.

A.4.7.8.2.1.1 Oil reservoir external features.

The requirements of 3.7.8.2.1.1 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.7.8.2.1.1)

The oil reservoir external features must be verified to insure compliance with the lubrication system requirements.

VERIFICATION GUIDANCE (A.4.7.8.2.1.1)

Past verification methods have included analysis, inspection, and endurance testing.

Background:

The oil reservoir external features should be examined and operated to verify their capability to display the quantity of oil in the tank, to service, and to drain the reservoir.

VERIFICATION LESSONS LEARNED (A.4.7.8.2.1.1)

None.

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A.3.7.8.2.2 Oil drains.

Drain ports shall be provided at appropriate low points in the oil system for draining the oil and taking oil samples with the engine in a range of positions from (a) degrees nose-up to (b) degrees nose-down.

REQUIREMENT RATIONALE (A.3.7.8.2.2)

Drains which completely drain the lubrication system must be provided so the engine can be effectively and easily maintained.

REQUIREMENT GUIDANCE (A.3.7.8.2.2)

(a): Past programs have used 15

(b): Past programs have used 20

Background:

If spectrometric oil analysis is to be part of the engine system maintenance plan, then a readily accessible, self closing drain valve should be provided at an optimum location in the oil system for obtaining representative oil samples for analysis. The oil samples used for the oil spectrometric analysis must be free of contamination from other fluids or dirt. Since maintenance technicians must perform oil sample tests on a frequent basis, the design of the oil drain should be such that operator safety is ensured while also allowing ease of operation of the oil drain valve. The oil drain valve should be constructed to prevent accidental opening during routine maintenance operations or during severe engine vibration cycles. One oil drain should be placed in the lowest practical point in the oil system to ensure all oil is removed from the engine and sump during periodic maintenance oil changes. Complete drainage must be possible during varying climatic conditions as specified on figure 5.

The engine should be designed so that repair and maintenance can be done in the vertical position without oil spillage or leakage.

REQUIREMENT LESSONS LEARNED (A.3.7.8.2.2)

Experience has shown that the oil spectrometric analysis is one of the best indicators of turbine engine condition. Some lubricating systems have incorporated oil sample drains that were inconvenient to use or had limited accessibility, preventing operating personnel from obtaining oil samples on a scheduled basis. Other oil sample drain valves have been located in such a manner that the samples contained an unrealistic quantity of sludge and contamination, thereby misrepresenting the actual condition of the engine. Self-locking drain valves reduced the probability of maintenance errors.

Spectrometric analysis has been ineffectual with fine filtration and has not detected imminent failures. Spectrometric analysis has been useful in development testing and as a post qualification activity to determine oil-sampling intervals in service.

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A.4.7.8.2.2 Oil drains.

The requirements of 3.7.8.2.2 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.7.8.2.2)

The adequacy of the oil system drains and the ease of servicing must be demonstrated.

VERIFICATION GUIDANCE (A.4.7.8.2.2)

Past verification methods have included test and inspection.

Verification should be accomplished on endurance test and attitude test engines and the maintainability demonstration of 4.5.2.

VERIFICATION LESSONS LEARNED (A.4.7.8.2.2)

None.

A.3.7.8.2.3 Oil filters.

Oil filter type(s), micron size, capacity, and filtration ratio rating shall be specified herein. Primary filter assemblies shall incorporate a pressure relief bypass and (a) provide direct and/or remote indication of filter capacity used and/or remaining. The bypass indicators shall incorporate provisions to preclude activation during low temperature starting. Bypass shall remain indicated until reset. The oil filter element shall have a life of at least (b) hours prior to impending bypass indication and at least (c) hours between impending bypass and bypass.

REQUIREMENT RATIONALE (A.3.7.8.2.3)

Adequate oil filtration must be provided to reduce the wear of oil wetted parts of the engine. Filter operation must continue under conditions of severe wear, partial failure, and progressive failure.

REQUIREMENT GUIDANCE (A.3.7.8.2.3)

The following should be used to tailor the specification paragraph:

(a) Visual (pop-up button) and electronic (connected to the engine monitoring system) are two methods currently used. The contractor should provide the method(s) to be used.

(b) The contractor should provide the value and is dependent upon maintenance approach.

The time interval used to oil filter impending bypass has been 500 hours with a 3-micron filtration rating.

(c) The time from impending bypass to bypass should be sufficient to allow completion of two combat missions.

The oil filter impending bypass indicator should be manually reset internally, requiring both disassembly and inspection of the filter.

A method of defining filter performance can be found in SAE AIR1666 Jul 94, “Performance Testing of Gas Turbine Lube Oil Filter Elements”.

Background:

Some prognostics health monitoring systems may provide direct and/or remote indication of filter capacity used and/or remaining. The filter elements should be accessible and easily removed and the filter

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housing should have provisions for draining the element prior to its removal and incorporate a self-closing shutoff valve to prevent system drainage after its removal.

The filter assembly should incorporate a pressure relief bypass and be of a design that will prevent the discharge of filter contaminant through the bypass. At a minimum, the filter assembly should provide a visual indication of both impending and actual bypass. Engines with monitoring systems may provide a means to indicate impending and actual bypass electronically. Whenever activation of the indicator is a result of the exceedance of the differential pressure limit, the indicator should remain in the fault mode until removal, cleaning, or replacement of the filter element is accomplished.

The manual reset requirement is necessary to prevent maintenance personnel from resetting the indicator without removing, cleaning, or replacing the filter element, which would defeat the purpose of the warning indicator. The impending bypass differential pressure level should be such that an impending bypass is detected through pre and postflight inspection thereby minimizing the likelihood of operating in bypass. Pressure surges or cold weather starts and operation should not cause false activation of the indicators. The filter assembly should also provide a signal for remote indication of both impending and actual bypass.

Scavenge side filtration may be accomplished by strainers or screens, preferably located at the outlet from each bearing compartment. Each strainer or screen should provide sufficient capacity to collect anticipated debris and to prevent clogging due to buildup of debris.

Experience has shown that sequence filtration systems with a high efficiency three micron scavenge filter provides excellent protection and virtually eliminates bearing wear during normal operation.

REQUIREMENT LESSONS LEARN (A.3.7.8.2.3)

Engines have used a wide variation of filtration systems, which was a cause for many maintenance and logistic problems. Some engines did not have a filter or a screen at the scavenge portion of the oil system, which allowed broken bearing tabs, seals, and washers to damage the oil scavenge pump.

The Army has experienced a large number of nuisance faults during cold weather operations due to false high filter pressure drop (filter bypass) indications.

Numerous false “impending bypass” indications have been experienced in naval aircraft due to normal carrier landing shock. Designs should withstand false activation.

A.4.7.8.2.3 Oil filters.

The requirements of 3.7.8.2.3 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.7.8.2.3)

The suitability of the oil filters to meet the engine lubrication system requirements must be verified.

VERIFICATION GUIDANCE (A.4.7.8.2.3)

Past verification methods have included analysis, component test to verify characteristics, and repeatability and inspection

Background:

Oil filter performance should be verified by bench test prior to engine test. Engine filter elements should be inspected during engine tests to ensure that the oil filter configuration, filtration capacity, and servicing features are per the specification.

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VERIFICATION LESSONS LEARNED (A.4.7.8.2.3)

None.

A.3.7.8.2.4 Oil debris monitor(s).

Oil debris monitor(s) shall be installed in locations where wear particles in the oil would most likely be detected and shall detect oil debris as specified herein.

REQUIREMENT RATIONALE (A.3.7.8.2.4)

An oil debris monitor or monitor system may consist of one or more parts and indicators. The oil debris monitor(s) or systems are placed within the oil wetted portions of the engine to facilitate detection of impending failures of bearings, gears, pumps, and associated parts.

REQUIREMENT GUIDANCE (A.3.7.8.2.4)

Some oil debris monitors can detect wear particles and isolate their origin to a specific area of the engine. The contractor should provide the capabilities of the oil debris monitor.

If an oil debris monitor is not required the text should be replaced with “Not Applicable”.

Background:

Oil debris monitors strategically located throughout the lubrication system serve as diagnostic maintenance aids to permit early detection of impending failures of oil wetted parts.

The master chip detector should be exposed to the entire scavenge oil flow, along with magnetic plugs in individual scavenge lines for fault isolation.

Oil debris monitors should be designed for easy access by maintenance personnel and incorporate an electrical continuity checkout feature. The design should be such as to prevent the loss of oil and introduction of contamination into the oil system when the detector is removed for inspection. These features should be accomplished by using self closing port designs or incorporating other design considerations that will perform these functions.

Allowance should be made for either visual (i.e., remove and check) or indicating (via immediate access or remote indicator) monitors. These monitors should be designed to minimize indications caused by normal wear particles (meaningless fuzz) and should be designed to provide a high probability of indicating impending engine failures.

REQUIREMENT LESSONS LEARNED (A.3.7.8.2.4)

Oil debris monitors have been clearly responsible for detecting engine failures that were not detectable by the spectrometric analysis. Monitors are frequently the sole means of identifying particles too large to be held in suspension in the oil.

The design features should permit the removal of the detector without the loss of oil or introduction of contamination into the oil system. There has also been good experience with properly designed chip detectors that are not self-closing.

The absence of a cockpit warning light in some aircraft is the result of many “false alarm” warnings caused by metal “fuzz” on the detector, resulting in mission aborts. For this reason development of improved monitors, such as the “burn off” type, is continuing.

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A.4.7.8.2.4 Oil debris monitor.

The requirements of 3.7.8.2.4 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.7.8.2.4)

The capabilities of the oil debris monitor must be demonstrated to show that early detection of impending failures of oil wetted components is possible.

VERIFICATION GUIDANCE (A.4.7.8.2.4)

Past verification methods have included analyses, component and engine tests, and inspections.

Oil debris monitor performance should be verified by bench test prior to engine test. Demonstration of the oil debris monitor capability should be accomplished by accumulating oil samples and periodically inspecting the oil debris monitor throughout the qualification testing. Details on the nature of the deposits and the amount and rate of accumulation of deposits on the oil debris monitor should be recorded and included in the test reports for use in demonstrating the adequacy of the monitor configuration and location. Monitor data should be correlated with spectrometric oil analysis, oil filter inspection, and engine teardown results.

VERIFICATION LESSONS LEARNED (A.4.7.8.2.4)

There are many different types of oil monitoring systems available for use in gas turbine engines. Bench testing of specific types may be indicative of the effectiveness and potential of a given design for use but cannot verify acceptability in the actual engine. Location of the monitoring system(s) is critical for effective operation. In a recent engine development program, using a specific debris monitoring system, the effectiveness of the sensors was not satisfactory and the initial concept replaced with more conventional “chip detectors”. Along with the bench test strong emphasis should be placed on actual on-engine demonstration to verify design effectiveness.

A.3.7.8.3 Breather mist.

Lubrication system breather exhaust shall not inhibit ground handling or maintenance personnel from performing tasks around and underneath the installed engine during ground operations at approved ground maintenance power settings. If oil mist is emitted, it shall not exceed the limits specified in the ICD and the quantity shall not exceed the current OSHA health and safety Threshold Limit Values (TLV).

The oil mist exhaust ports shall preclude foreign material (rain, particulates, debris matter, etc) from intruding back into the system during both operation and non-operation.

REQUIREMENT RATIONALE (A.3.7.8.3)

Engine lubrication systems typically vent breather air from the bearing compartments and gearbox to prevent these systems from over pressurization. When such systems vent breather air overboard, limits should be placed on the amount of oil particles that are carried with the air such that maintenance personnel can perform routine ground maintenance tasks with the engine operating and installed.

REQUIREMENT GUIDANCE (A.3.7.8.3)

If applicable the ICD should define the effects of misting (for example wetting of aircraft surfaces). The Threshold Limit Value (TLV) was set by the American Conference of Governmental Industrial Hygienists (ACGIH). These values are considered safe levels for continuous exposure of up to eight hours. Current recommended TLV limits are 5 mg/cubic meter (per the current ACGIH requirement).

Although some synthetic oils do not represent a health hazard to maintenance personnel, excessive amounts of unburned oil particles carried overboard with breather air can inhibit routine tasks from being

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performed. Every effort must be made to design a system that does not emit oil mist. If allowed, this would result in the inability to perform routine maintenance tasks, such as leak checks, safety hazards due to the oil particles settling on the ground creating a slippery surface, and permanent soiling of personnel clothing. Breather exhaust levels should be limited to that which allows routine maintenance tasks to take place. If the engine lubrication system does not vent breather air external to the engine, this requirement should be deleted from the specification.

Additional guidance on oil drains and breather ports may be found in section 3.1.1.8.

Background:

The previous OSHA limit for oil mist was 3-gm/m^3 and the latest values being considered are of the order of 0.03-gm/m^3.

REQUIREMENT LESSONS LEARNED (A.3.7.8.3)

Counter-rotating engines present difficult challenges in terms of seals. Intershaft seal locations operate at relative speeds much greater than those in co-rotating engines do. Engine O on Aircraft H used labyrinth seals in a counter-rotating location which resulted in greater air leakages into bearing compartments than had been experienced on previous co-rotating engines. This led to breather flows several times higher than those in previous applications did. These higher breather flows carried microscopic oil particles overboard with the breather air resulting in oil streaking and staining along the bottom of the aircraft, and a cloud of oil mist under and around the aircraft when the engine was operated which inhibited maintenance personnel from performing routine tasks. Although oil was exiting the engine, the size of the particles was small enough that the quantity was not enough to prevent the engine from meeting its oil consumption requirement. With no specification requirement present for breather mist levels and no other specification violations, a redesign of the bearing compartment was undertaken at an increased cost to the government.

A.4.7.8.3 Breather mist.

The requirements of 3.7.8.3 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.7.8.3)

The requirements for breather mist must be verified to ensure compliance with current OSHA requirements.

VERIFICATION GUIDANCE (A.4.7.8.3)

The breather mist requirements should be verified during engine testing by determining if the amount of oil particles in the breather air being dumped overboard allows uninhibited activity around the point of discharge. Methods of verification include analysis, inspection and test. Analysis and inspection may include review of drawings and oil system test data. Testing may include measuring the actual concentration of oil mist exhaust.

VERIFICATION LESSONS LEARNED (A.4.7.8.3)

None.

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A.3.7.9 Hydraulic system.
The hydraulic system fluid(s) shall be (a) . The hydraulic system shall withstand a proof pressure of (b) times the maximum working pressure. Hydraulic systems that are integral to the fuel system shall meet the requirements of 3.7.3.

REQUIREMENT RATIONALE (A.3.7.9)

The hydraulic fluid must be identified so that the appropriate safety and supportability requirements can be established. The proof pressure must be stipulated so that inadvertent pressure spikes can be tolerated by the system.

REQUIREMENT GUIDANCE (A.3.7.9)

The following should be used for tailoring the specification paragraph:

(a): The contractor should provide the fluid type.

(b): The value should be at least 1.5.

If the engine does not have a separate hydraulic system, replace the paragraph wording with “Not Applicable.”

Background:

The hydraulic system should be able to withstand operational stresses and vibration without failure and there should be no permanent deformation, collapse, or seam cracking when exposed to these stresses and vibrations. The system must be designed to prevent leakage from a removable joint since this could result in a fire.

System pumps should utilize a pressure regulating device and an independent means of limiting excessive pressure.

Designing the system with an integral unloading feature prevents heat build up in the hydraulic pump and reduces the power required by the system from the engine.

The hydraulic system should utilize hydraulic fluid (e.g., MIL-PRF-83282), engine fuel, or oil.

REQUIREMENT LESSONS LEARNED (A.3.7.9)

Aircraft fires have resulted from leakage at hydraulic fluid system joints and faulty maintenance.

A.4.7.9 Hydraulic system.

The requirements of 3.7.9 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.7.9)

The physical and functional characteristics of the hydraulic system must be verified to assure compatibility with the engine and its application. Hydraulic system performance and limits must be verified.

VERIFICATION GUIDANCE

Past verification methods have included analysis and tests.

The following procedure may be used as a verification method:

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The hydraulic system testing may be conducted either in a complete system rig test or installed on an engine substantially identical to the endurance test engine. The hydraulic system high pressure side shall be subjected to proof pressure for one or two successive times and held two minutes for each pressure application. Hydraulic test fluid shall be maintained at maximum system working temperature. The equipment shall be operated in its normal function between applications of the test pressure. At test completion, there shall be no leakage, excessive distortion or permanent set. Components that require varying test pressures in different elements may have these pressures applied either separately or simultaneously.

All physical features of the hydraulic system shall be examined and compared to the specification and installation drawing.

Functional characteristics shall be verified by demonstrations and tests. Test cycles shall be consistent with the environment and design usage specified in section 3.4.1.2. Tests shall be accomplished at both sea level and simulated altitude test conditions. Maximum loads on the system shall be based on an analysis of flight conditions. The hydraulic fluid reservoir shall be subjected to a cyclic fatigue test.

VERIFICATION LESSONS LEARNED (A.4.7.9)

Engine testing at high Mach number, low altitude conditions frequently gave maximum loads on hydraulically-actuated exhaust nozzles.

A.3.7.9.1 Hydraulic ground test provisions.

Ground test provisions shall be provided if required by the Using Service. Fluid leakage shall not occur during ground test.

REQUIREMENT RATIONALE (A.3.7.9.1)

Ground test provisions are needed for easier troubleshooting capabilities.

REQUIREMENT GUIDANCE (A.3.7.9.1)

This requirement may be deleted here and included in the Reliability and Maintainability section as part of a list that describes maintenance and troubleshooting capabilities.

Background:

Ground test points should be easily accessible and compatible with ground support test equipment. Ground test provisions should be shown on the installation drawing.

Self-sealing ground test couplings should be provided with the hydraulic system. These couplings should be secured with sealing caps when not used

REQUIREMENT LESSONS LEARNED (A.3.7.9.1)

Sealing caps have been required to prevent spillage of flammable fluid when self-sealing couplings were not used.

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A.4.7.9.1 Hydraulic ground test provisions.

The requirements of 3.7.9.1 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.7.9.1)

Hydraulic test provisions must be verified for accessibility and sealing capabilities.

VERIFICATION GUIDANCE (A.4.7.9.1)

Past verification methods have included analysis and demonstrations.

Ground test equipment compatibility should be tested concurrent with the maintainability demonstration.

VERIFICATION LESSONS LEARNED (A.4.7.9.1)

None.

A.3.7.9.2 Hydraulic system air removal.

Entrapped air shall not cause sustained loss of system pressure or degradation of system operational performance. There shall be provisions for bleeding air from the system at critical points for maintenance purpose.

REQUIREMENT RATIONALE (A.3.7.9.2)

The operating characteristics of the hydraulic system must not be affected by entrapped air.

REQUIREMENT GUIDANCE (A.3.7.9.2)

This requirement is not necessary if oil or fuel is used as the hydraulic fluid.

Background:

Disconnection of lines or loosening of fittings is not an acceptable method for bleeding air because it is impossible to control or direct the bleed fluid. Reconnection of lines often causes entrapment of air into the system. The system must be designed to automatically scavenge free air from the system to minimize the need for manual bleeding.

Critical point for bleeding air from the system includes fluid filter locations because these filters are replaced during regularly scheduled maintenance.

REQUIREMENT LESSONS LEARNED (A.3.7.9.2)

In the past, engines that tried to save weight by eliminating air removal devices usually required the devices once the engine was put into service.

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A.4.7.9.2 Hydraulic system air removal.

The requirements of 3.7.9.2 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.7.9.2)

The ability of the hydraulic system to remove entrapped air must be verified.

VERIFICATION GUIDANCE (A.4.7.9.2)

Past verification methods have included analysis and demonstrations.

Background:

Entrapped air removal should be verified in a complete rig test. Engine testing should also be used to verify hydraulic system functional characteristics both at sea level and simulated altitude test conditions.

VERIFICATION LESSONS LEARNED (A.4.7.9.2)

None.

A.3.7.9.3 Hydraulic fluid filters.

Filter(s), type(s), location(s), and filtration rating(s) in microns (nominal and absolute particle sizes) shall be specified herein. Each filter housing shall incorporate an automatic shutoff device to prevent fluid drainage during filter element removal. Each filter element shall be accessible, easily removed and provide (visual/engine monitoring system) indication of both impending and actual bypass. The bypass indicators shall incorporate provisions to preclude activation during low temperature operation. Whenever activation of the indicator is a result of the exceedance of the differential pressure limit, the indicator shall remain in the fault mode until removal, cleaning, or replacement of the filter element is accomplished.

The filter shall have a life of at least (a) hours to impending bypass indication and at least (b) hours between impending bypass and bypass.

REQUIREMENT RATIONALE (A.3.7.9.3)

Filters are necessary to protect critical components such as hydraulic pumps, seal mechanisms, and valves from particulate contamination.

REQUIREMENT GUIDANCE (A.3.7.9.3)

The following should be used to tailor the specification paragraph:

(a): The contractor should provide the value and is dependent upon the maintenance approach.

(b): The time from impending bypass to bypass should be sufficient to allow completion of two combat missions.

If the fluid being used for the hydraulic system is fuel or oil, then the requirement paragraphs for these filters can be referenced instead of using the requirements shown above.

Background:

The filter housings should have an automatic shutoff device because spilled fluid can be a fire hazard and maintenance nuisance.

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Filters for the hydraulic system should provide the following protection to the system:

Supply line:

a. Filters should be located such that all fluid from the engine pumps and ground test equipment pressure connections will be filtered prior to entering components of the system.

Return line:

a. Filters should be located such that all fluid will be circulated through the filter prior to entering the return line relief valve, pumps, or reservoir.

Other:

a. Additional filters at critical component locations should be provided where filter protection is necessary.

Filters are normally located in the supply and return lines because in these locations the flow is reduced to a single line and the necessary filtering can be easily accomplished before the fluid reaches critical components in the system. Filters should be made of a material resistant to media migration.

Most filter assemblies incorporate an integral differential pressure activated pop-out device that will give visual warning by raising a red indicator when the differential pressure across the element exceeds a specified value. Once activated, the red indicator should remain extended until manually reset internally after filter removal.

REQUIREMENT LESSONS LEARNED (A.3.7.9.3)

Pressure ripples occurring during field operation have caused false activation of indicators.

A.4.7.9.3 Hydraulic fluid filters.

The requirements of 3.7.9.3 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.7.9.3)

The physical and functional characteristics of the fluid filters must be verified to assure compatibility with the hydraulic system and its application.

VERIFICATION GUIDANCE (A.4.7.9.3)

Past verification methods have included analysis, demonstrations and tests.

Background:

The capabilities of the hydraulic fluid filters should be verified during all engine tests at sea level and altitude conditions. Maintainability features of the filters should be verified during the maintainability demonstration of 4.5.2.

VERIFICATION LESSONS LEARNED (A.4.7.9.3)

None.

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A.3.7.10 Exhaust nozzle system.

Any exhaust nozzle system failure shall not prevent safe return.

Failure of any single part of the actuating mechanism on a controllable nozzle shall not cause asymmetric thrust relative to the nozzle. The operational mode, and limits resulting from any single part failure in the actuating mechanism of a variable nozzle, shall be specified herein. Provisions shall be made to actuate the nozzle for ground maintenance checkouts, rigging, and adjustment without the engine running. Life of the engine exhaust system, special components, features, materials, and applied coatings shall be specified.

REQUIREMENT RATIONALE (A.3.7.10)

Exhaust nozzles have an appreciable effect on engine capability and aircraft performance. This is based on factors such as augmentor characteristics, discharge coefficients, thrust coefficients, cooling requirements, and internal drag. Life of the exhaust system, components, and features must equal or exceed engine hot parts life to reduce the number of maintenance actions, cost, and aircraft down time.

REQUIREMENT GUIDANCE (A.3.7.10)

Background:

If the air vehicle contractor supplies the exhaust nozzle, the performance, physical characteristics, loads, and method of attachment, installation, and removal should be defined.

The method of nozzle actuation, type and number of actuators, control sensing parameters, provisions for nozzle position feedback, cooling provisions and method of cooling employed, fail-safe characteristics, degree of vectoring (if directionally controlled), nozzle throat and exit area shape deformations, external geometric contour and surface anomalies which may affect aerodynamic drag or gross thrust of the nozzle, and any other features unique to the particular design, including protective and low observable coatings and features.

Variable nozzle fail-safe and manual or emergency features including actuator redundancy should be compatible with the intended application.

Mission requirements should dictate the nozzle design criteria. Data to be used in the design process should include the tradeoffs between thrust, weight, size, drag, stability, external flow field, low observables, fuel puff, ultraviolet, infrared radiation or other visual signature, and cooling requirements. Nozzle design and operation should include consideration of airflow match with upstream components to ensure sufficient engine stall margin. Engine and nozzle flow matching should consider the effects of internal cooling flows, flow area effectiveness, and nozzle throat and exit area deformation.

Exhausts with special technology features and coatings may need additional, even frequent, maintenance actions. The exhaust system should be repairable at “I” or “O” level. Shelf life of stored or spare exhaust systems (metallic or composite structure or parts) should exceed ten years (problem, moisture pickup). Nozzle structure, ducts, and components should be repairable (versus “throwaway”). Exhaust system life should be achievable in all engine-operating environments (e.g., acoustics) with a specified duty cycle.

REQUIREMENT LESSONS LEARNED (A.3.7.10)

None.

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A.4.7.10 Exhaust nozzle system.

The requirements of 3.7.10 shall be verified by:

Actuator scraper and seal design shall be verified for effectiveness and wear from engine generated debris in a heated environment.

VERIFICATION RATIONALE (A.4.7.10)

The exhaust nozzle system must be verified.

VERIFICATION GUIDANCE (A.4.7.10)

Past verification methods have included analysis, inspections of parts and tests. If testing is used for verification, then nozzle structural integrity should be tested during operation at maximum ram pressure conditions.

Background:

Engine tests, to verify exhaust nozzle matching and functional characteristics, should be accomplished at sea level and altitude flight conditions. Engine tests at envelope extremes should be used to ensure nozzle stability and freedom from vibration. Engine testing at maximum inlet pressure is desirable to demonstrate structural capability. Component tests may be used to demonstrate maximum structural loads, mounting and air vehicle forces and external fields. External flow field effects on the nozzle should be considered in the nozzle test program. External exhaust nozzle wear debris has caused actuator leakage due to ineffective rod scraper design. Exhaust systems with high aspect ratio or non-round shapes have nozzle area deformations that affect the engine flow match, performance, and stall margin. Multifunction exhaust systems have added weight, complexity, and wear. Exhaust systems with unconventional construction or materials need backup approaches (some nozzle designs have been abandoned in major programs). Life and repair of special coatings and features should be demonstrated. The effectiveness and integrity of the exhaust cooling system should be demonstrated.

VERIFICATION LESSONS LEARNED (A.4.7.10)

Accelerated Mission Tests (AMTs) typically do not reveal all nozzle wear modes due to the interaction between the air vehicle, engine and the external flow fields. Flight testing of variable geometry nozzles has uncovered unexpected nozzle vibrations and wear problems.

A.3.7.10.1 Exhaust nozzle external asymmetrical air pressure loads.

The engine shall meet the requirements of the specification when subject to the asymmetrical and all other air pressure loads resulting from the external flow field pressure distribution on the nozzle and exhaust systems. The limits shall be specified: _____.

REQUIREMENT RATIONALE (A.3.7.10.1)

The air vehicle manufacturer must provide the external flow field pressure distribution to assure the engine is compatible with the air vehicle. The engine must be compatible with the air pressure loads external to the air vehicle.

REQUIREMENT GUIDANCE (A.3.7.10.1)

The following should be used to tailor the specification paragraph:

The limits can be shown in the Interface Control Document (ICD), or for programs that do not have an ICD, the limits will be put into the specification.

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Background:

The requirement is intended primarily for engines with variable or vectoring exhaust nozzles. The fact that an exhaust nozzle is variable lends itself to deformation by asymmetrical air pressure loads. Limit information is needed to prevent any permanent deformation. Nozzles with high aspect ratio, nonround shapes can have structural area deformations with changing pressure loads. These area and shape deformations can affect the engine flow match, performance, and stall margin.

REQUIREMENT LESSONS LEARNED (A.3.7.10.1)

None.

A.4.7.10.1 Exhaust nozzle external asymmetrical air pressure loads.

The requirements of 3.7.10.1 shall be verified for each engine and air vehicle combination by: _____.

VERIFICATION RATIONALE (A.4.7.10.1)

The exhaust nozzle external asymmetrical air pressure loads must be verified to assure compatibility with the air vehicle and its mission.

VERIFICATION GUIDANCE (A.4.7.10.1)

Past verification methods have included analysis and tests.

Background:

The initial verification must be conducted by analysis. Later the engine contractor should demonstrate the load limits by simulated loading.

VERIFICATION LESSONS LEARNED (A.4.7.10.1)

On one vehicle program, the nozzle area deformation caused an SFC degradation of about 5% at low altitude due to engine down matching. On another vehicle program, the nozzle exit had to be reconfigured to reduce an aft surface-heating problem. The selection of the engine fan operating line with deforming nozzles should consider stall margin at the smallest area and performance degradations at the largest areas. The effective flow area of the nozzle throat and exit with large or elaborate cooling schemes can be hard to determine.

A.3.7.10.2 Vectoring nozzle.

The engine thrust vectoring system shall revert to the fail-safe mode after receiving a fail-safe or standby signal. Nozzle feedback signals, if required, shall be provided to the air vehicle and shall include: _____.

REQUIREMENT RATIONALE (A.3.7.10.2)

Thrust vectoring is considered an aircraft performance enhancement feature, and it can improve aircraft handling qualities. The fail-safe signal requirement provides a degree of pilot control and safety for the mission, aircraft, and equipment.

REQUIREMENT GUIDANCE (A.3.7.10.2)

The following should be used to tailor the specification paragraph:

The contractor should identify the feedback signals and they should be coordinated in the ICD.

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Background:

The thrust vectoring nozzle should be designed for the mission of the aircraft and its on board weapon systems, not designed for distinct engine types.

At design reviews the Using Service should analyze the following features of the design: the actuation system, including type and number of actuators, performance, loads, interface attachments, sensing parameters, weight, nozzle position feedback, cooling provisions, geometric contours, fail-safe mode, manual operation, emergency features, materials, and coatings that are used. The contractor who is responsible for the nozzle should be specified.

Nozzle failure modes and their effects should also be addressed. Mission requirements should dictate the vectoring nozzle criteria. Data used in the design process should include the tradeoffs between thrust, weight, size, drag, stability, external flow field, exhaust cooling flows, and infrared radiation or other visual signatures. Vectoring nozzle design and operation should include consideration of airflow match with upstream components to ensure sufficient engine stall margin with deformed throat or exit areas.

The exhaust nozzle should not be considered an independent aircraft subsystem, but should be synthesized into the aircraft flight control laws. This will permit development of a truly integrated and optimized aircraft, engine and nozzle control system. Failure of one segment of a multi-segmented nozzle should not prevent operation of the nozzle. Nozzle structural integrity must be ensured during operation at maximum ram pressure conditions.

A thrust vector nozzle operating in a “transparent to the pilot” design configuration has been determined to be a more safe, reliable, stable method of controlling the vectoring nozzle operation. Similar to the other primary flight control surfaces, the vectoring nozzle does not require, but should allow, the engine nozzle control system to be commanded by the aircraft flight control computer. This allows a central location for all flight control commands and permits the vectoring nozzle to be an integral part of the flight control laws and control surfaces.

To allow successful integration to occur, several items need to be specified by the integrator:

a. Data bus transmission rate
b. Total time delay through the nozzle control unit
c. Nozzle actuator spectral characteristics
d. Expected failure coverage levels of nozzle failures
e. A definition of the failure hierarchy and failure annunciation
f. A definition of critical parameters used for inner-loop control

The thrust-vectoring nozzle may or may not operate in the critical mode depending on the design. Critical mode of operation is the maneuvering of the aircraft using the thrust from the vectoring nozzle(s) only.

If nozzle signal feedback is required, the engine contractor should provide the following information based on the air vehicle contractor’s requirement: feedback may include current pitch and yaw position, current pitch and yaw maximum limit, calculated gross thrust, engine and nozzle status words, and Vertical Parity Check (VPC) word to monitor health of the system. A vectoring nozzle system should be prime reliable if vectoring operation is necessary for vehicle flight safety and control. The nozzle should be returnable to the neutral position for horizontal flight in event of control failure.

REQUIREMENT LESSONS LEARNED (A.3.7.10.2)

During development of Aircraft H, while flight testing the thrust vectoring nozzle performance at Edwards Air Force Base, the aircraft was approaching the runway to perform a go-around for photo session

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purposes. While in this attitude the test pilot had difficulty controlling the aircraft and a mishap followed. It was later determined the mode switch, which was located in the cockpit and regulated the position of the vectoring nozzle, was placed in the wrong position. The mode switch was only for the purpose of performing flight test maneuvers and was supposed to be used only in distinct flight regimes. The flight control laws and piloted simulators had been developed using generic engine and nozzle representations instead of specific representations.

A.4.7.10.2 Vectoring nozzle.

The requirements of 3.7.10.2 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.7.10.2)

The requirements of the thrust-vectoring nozzle must be verified during qualification.

VERIFICATION GUIDANCE (A.4.7.10.2)

Past verification methods have included analysis and tests.

Background:

Engine tests, to verify nozzle matching and vectoring functional characteristics, should be accomplished at sea level and at altitude flight conditions. Engine tests at envelope extremes should be used to ensure nozzle stability and freedom from vibration. Engine testing at maximum inlet pressure is desirable to demonstrate structural capability. Component test may be used to demonstrate maximum structural loads, mounting and air vehicle forces, and external fields.

Flight-testing should incorporate a double check system to ensure proper execution of each element of testing and each flight test program should be flown on a piloted simulator prior to actual flight. Additionally, each flight test element affecting stability should be initiated at altitude, leaving altitude margin for recovery.

The thrust-vectoring nozzle is unique to the aircraft for it is an item that is shared by both the air vehicle and the engine, and should be verified in the “installed” configuration.

Testing of engine and aircraft integrated control system rigs and a hot bench should be conducted to verify the integrated system control laws, transfers, and failures.

The flight control computer should control the vectoring nozzle like an integral segment of the entire primary flight control surface thus making it a transparent command function to the pilot.

Nozzles which vector, transition, block, or reverse flow for Vertical and Short Take-off and Landing (V/STOL) operation should be thoroughly analyzed and demonstrated on both a steady-state and transient basis for effects on performance, stall margin, and operability.

VERIFICATION LESSONS LEARNED (A.4.7.10.2)

None.

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A.3.7.10.2.1 Vectoring nozzle angle and rate.

The vectoring nozzle shall have a maximum slew rate of at least (a) degree/sec at (b) flight condition. The nozzle shall provide a maximum vectored deflection angle of (c) degrees from the symmetric axial thrust position with an accuracy of (d) degrees over the entire deflection range. The nozzle shall vector without interruption, in a constant, steady movement.

REQUIREMENT RATIONALE (A.3.7.10.2.1)

The slew rate and maximum deflection angle must be specified to assure proper vectoring nozzle operability and for aircraft design information. The air vehicle contractor needs to be informed of the nozzle positioning accuracy.

REQUIREMENT GUIDANCE (A.3.7.10.2.1)

The following should be used to tailor the specification paragraph:

(a): The maximum slew or thrust-vectoring rate should be negotiated by both the engine and air vehicle contractors to match vehicle mission and stability requirements. The maximum slew rate capability should approach that of the vehicle control surfaces while assuring vehicle stability. The specified value for the “maximum slew rate of at least” requirement should be the fastest rate the nozzle can move under any condition. The nozzle must be able to move faster than this rate under all conditions, but the specification value is the fastest rate that can always be provided by the engine.

(b): The contractor should provide the flight altitude and Mach number or flight dynamic pressure.

(c): The contractor should provide the angle.

(d): The contractor should provide the accuracy.

Background:

Some uses and benefits of vectoring nozzles are: improved nose pointing at low speed, reduced size of aircraft control surfaces, improved combat maneuvering, improved control and trim with inoperative engine(s), and steering potential.

One of the vectoring nozzle goals is to improve upon the slew rate and develop a rate that will offer good stability and vectoring performance. This rate is dependent on the desired achievable envelope, the aircraft static instability, and aircraft performance considerations. Statically stable aircraft or aircraft with sizable pitching moments of inertia may require a higher maximum nozzle slew rate. Based on experience, the nozzle is seldom driven at its maximum slew due to fan or compressor instability (i.e., stall margin).

The flight condition value determines actuator loads and should correspond to a value of constant maximum “q” for vectoring operation such as low altitude, high airspeed and extending to higher Mach numbers at higher altitudes.

Nozzles being developed at this date have a specification deflection of +20 degree pitch, and ±20 degree yaw maximum for the axisymmetric nozzle but this is simply a mechanical range. The requirement driving this parameter is total side force generated, or the maximum pitch and yaw acceleration desired. These parameters will determine aircraft fuselage requirements and engine case loads. The lower bound on side force is determined by controllability and the upper bound is determined by weight and CG limitations. The vertical force generated is a function of Gross Thrust x Sine (vector angle) plus induced effects. Engine size, aircraft and nozzle configuration, and desired performance will drive this requirement.

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For the weapon system to function in a complete arena of battle scenarios, the vectoring deflection angle requirement should permit the aircraft to maneuver, without danger of a stall, spin, or an out-of-control situation, to meet the demands of the mission. The normal aircraft systems check should be built into the control system Built-In-Test (BIT) and on ground panel-up checks. This will allow the pilot to verify that the vectoring capability is delivering the proper amount of vector angle on pilot command. This is a safety requirement as well as a functional test of the system. When the vectoring nozzle system begins to lose its accuracy, the components that makeup the vectoring system will begin to indicate a possible malfunction, or begin to show a future component problem. This requirement has been put in place to offer both an indication of a decline in component quality and to establish a safety margin for component wear. As of 1993, manufacturers were capable of producing nozzles with ±1.0 degree position accuracy. Maximum internal nozzle performance is of secondary importance during vectoring operation; because the deflection and deflection rate capability have priority over the nozzle performance.

The Using Service should ensure that the ICD addresses the maximum and minimum dynamic response characteristics of the vectoring nozzle. The frequency response should be consistent at all power levels and vector positions. If the nozzle is designed to be an integral part of the flight control system, the response characteristics must not show a salient amount of lead or lag, or out of phase relationship between the vectoring of the nozzle to the aircraft control commands.

The flight control computer has a design characteristic that causes the nozzle vectoring command to either lead, or lag, the aircraft primary flight control surface command input. The computer processing time (signal response) is known as the “throughput” of the computer control signal. The lead and lag condition occurs at specific flight regimes. Until such time that the aircraft flight computer throughput time can be improved to cause the response of the vectoring nozzle to match the response characteristics of the primary flight control surfaces, an out-of-phase relationship between these two features will undoubtedly occur in some flight regimes.

Slew rates have been conducted up to 60 degrees/sec, but even higher rates have been tested without any aircraft instability or performance difficulty.

The nozzle slew rate required should be specified at a given flight condition or flight dynamic pressure to avoid over or under designing nozzle actuators. The nozzle actuation system should have sufficient control power to achieve the desired vectoring at all flight conditions.

Vectoring requirements for nozzle design should be stated as slew rate and delta force (in pounds rather than percent) at the most important flight conditions and including those at maximum flight dynamic pressure; otherwise, the result is over design with added weight and cost.

REQUIREMENT LESSONS LEARNED (A.3.7.10.2.1)

During the early development of the Thrust Vectoring Nozzle, it was learned that the ±20 degrees pitch and yaw full deflection angle was only occasionally needed. To achieve rapid high AOA (Alpha), or yaw (Beta), it was only necessary to introduce 5-10 degrees of nozzle vectoring for short periods of time to assist the aircraft primary flight control surfaces.

The vectoring nozzle slew rate has been a constantly changing parameter with each new vectoring nozzle development. This rate continues to be increased. As of 1993 the rate is up to 60 degrees/sec. The engine manufacturers seem to be satisfied with this figure.

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A.4.7.10.2.1 Vectoring nozzle angle and rate.

The requirements of 3.7.10.2.1 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.7.10.2.1)

The maximum thrust-vectoring rate and vectoring angle are considered high-priority characteristics for thrust vectoring and must be verified. The nozzle position accuracy must be verified to detect the level of component wear.

VERIFICATION GUIDANCE (A.4.7.10.2.1)

Past verification methods have included analysis and tests.

Background:

The testing of the slew rate should address the issue of nozzle and system deterioration. The systems check should verify whether degradation is present.

Since the thrust vectoring nozzle angle is essential in the makeup of the aircraft maneuverability and aircraft trim characteristics, this parameter must be tested to ensure that continuous proper angle positioning is achievable.

Similar to the primary flight control surfaces, the pilot should have the capability to test the vectoring nozzle angle prior to aircraft launch. This is a safety requirement as well as a functional test of the system.

VERIFICATION LESSONS LEARNED (A.4.7.10.2.1)

Engine off testing of the thrust vectoring nozzle system has required a remote hydraulic power source.

A.3.7.10.2.2 Vectoring nozzle failure accommodation.

Failure accommodation for the nozzle system shall be described. Any system failure shall not preclude the nozzle from reverting to a fail-safe position.

REQUIREMENT RATIONALE (A.3.7.10.2.2)

For both safety of crew and equipment, the thrust-vectoring nozzle must fail to a fail-safe position. Without this ability, the aircraft could possibly assume a vulnerable flight attitude and result in aircraft loss.

REQUIREMENT GUIDANCE (A.3.7.10.2.2)

The contractor should describe the failure accommodation for each type and number of failures. That is, one electrical failure might not affect nozzle operation, but two electrical failures may result in degraded operation (i.e., no augmentor operation or reduced thrust) or one hydraulic failure may result in fail-safe accommodation.

Background:

For naval aircraft carrier operation, it is essential that the thrust-vectoring nozzle deploy to a fail-safe nozzle position. This is because of the short and narrow runways and other space constraints including elevators on board the carrier, the aircraft is left with little maneuvering room to counter any abnormal thrust vectoring angle attitude. The landing approach and arresting of the aircraft would be unsafe if the nozzle was permitted to retain a vectoring angle other than the fail-safe or neutral position.

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The weapon delivery capability would be in jeopardy without the ability to deploy, automatically or manually, to the fail-safe configuration.

REQUIREMENT LESSONS LEARNED (A.3.7.10.2.2)

The Multi-Axis Thrust Vectoring (MATV) program aircraft (VISTA), has the ability to revert to a fail-safe condition using the fail-safe or standby switch in any power setting, which avoids nozzle instability.

A.4.7.10.2.2 Vectoring nozzle failure accommodation.

The requirement of 3.7.10.2.2 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.7.10.2.2)

The nozzle failure accommodations must be verified to ensure the nozzle will operate safely in the event of any failures. Verification of the thrust vectoring nozzle failure mode(s) should provide the Using Service with a valuable tool to analyze worst-case scenarios in vectored flight.

VERIFICATION GUIDANCE (A.4.7.10.2.2)

Past verification methods have included analysis and tests.

The contractor should provide a means to test the thrust vectoring nozzle failure mode accommodation. This may be incorporated into the BIT system, if required by the Using Service.

VERIFICATION LESSONS LEARNED (A.4.7.10.2.2)

None.

A.3.7.11 Augmentation system.

The engine shall provide thrust or power augmentor operation, as required, including initiation and termination, throughout the augmentor operating envelope. The amount of augmentation shall be controllable, and transition from nonaugmented to augmented operation shall be free of large abrupt thrust or power changes. The augmentor shall not induce an unacceptable loss of engine operability, either in operation or during initiation or cancellation over the required envelope.

REQUIREMENT RATIONALE (A.3.7.11)

Augmentation, when necessary to fulfill mission requirements, offers a high thrust-to-weight ratio method for obtaining limited time high thrust or power ratings. Use of the system must not result in premature degradation of engine life.

REQUIREMENT GUIDANCE (A.3.7.11)

At design reviews the Using Service should analyze the following features of the design: For augmentation systems using fuel, the system may include components such as an augmentor case, liner, flameholder or spraybar system, mixer or diffuser and centerbody.

For augmentation systems using water, the Using Service should analyze the system flow limits, accuracies and system limitations.

The augmentation transition from on to off should prevent a large step change in thrust. The allowable step change from augmentation off to on and from on to off should be defined in the specification section 3.7.2.1.

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Background:

Augmented engines typically increase thrust 40-80% and weight 20-25% relative to a conventional nonaugmented engine. The added weight, cost, and complexity of the augmented engine are often preferable to oversizing a nonaugmented engine for the same application. Developing an augmented derivative for increased thrust needs may also be less costly than developing a growth version of the original engine.

For engines that must operate at extreme altitudes, the use of an automatic relight system may be essential to prevent flameouts. Ignition sources should not induce fan or compressor stall. Pilot burner stabilized flame, catalytic, or pyrotechnic igniters may be used to provide a continuous reignition source to reduce the possibility of augmentor flameout.

The Using Service should ensure the aircraft system specification has a requirement that the engine should not cause degradation to the air vehicle due to augmentation destructive operating modes (rumble, screech, etc.).

Development elapsed time for augmented exhaust systems with unconventional features can equal or exceed the elapsed time for turbomachinery. Test hours required may only be 1/4 to 1/2 of the hours for turbomachinery. Some future engine programs may be paced by the augmentor exhaust system.

For augmentors using fuel:

Nozzle liners are used to distribute internal nozzle cooling flows and control combustion instabilities. The cooling flows are needed for material survivability. The forward portion of the augmentor liner should serve as a “screech liner” to control destructive combustion instabilities.

Flameholders should have sufficient blockage to provide stable burning throughout their operating envelope. In turbofan engines, mixing of the hot and cold streams prior to burning has improved high altitude combustion stability.

The augmentor should not exhibit either low (rumble) or high (screech) frequency instabilities at a level which induces unacceptable impacts on augmentor and engine performance, operability, and life. Rumble is the instability phenomena most frequently encountered. This axial mode discrete frequency instability is sufficiently low in frequency to cause fan stalls (usually 40-100 Hz, which is less than one-per-rev frequency of the fan) and burner blowouts.

Augmentor flameholder blockage can be increased to improve combustion stability at a cost of increased pressure loss during dry operation of the engine.

For augmentors using water:

Thrust augmentation using a water injection system is normally used for takeoff. A water injection system is used to add mass and cool the airflow after the compression process. This in turn produces an increase in thrust or a decrease in specific fuel consumption. Demineralized water is required to minimize formation of internal engine deposits.

There are inherent disadvantages to the use of a water injection system, including increased weight, susceptibility to freezing conditions, and the nonavailability of usable water, especially for Naval operations.

Water injection into the combustor should be provided for lift power settings of V/STOL engines, if appropriate.

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REQUIREMENT LESSONS LEARNED (A.3.7.11)

Successful augmentor designs have incorporated light off detectors in the control system to prevent hard lights and flameout and relight stalls from occurring. Augmentor design and tuning has been highly iterative and early development activity is recommended. Water injection has required specialized logistics servicing and intensive field maintenance. It is not practical at low temperatures. Water injection systems have been employed on V/STOL aircraft, such as Aircraft Z, and many older engines.

A.4.7.11 Augmentation system.

The requirement of 3.7.11 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.7.11)

The operational characteristics of thrust augmentation devices must be fully verified.

VERIFICATION GUIDANCE (A.4.7.11)

Past verification methods have included analysis and tests.

Background:

Augmentors should be tested as part of the total engine in an altitude facility to demonstrate satisfactory augmentor initiation, operation, and termination throughout the augmentor flight envelope.

Testing should cover the augmentor operating envelope and include takeoff, key operating points, and low Mach number at high altitude flight conditions. Attention should be on the high altitude, low Mach number portion of the flight envelope for operability, and on the low altitude high Mach number portion of the flight envelope for highest loads.

Water injection systems are usually tested at sea level takeoff conditions.

VERIFICATION LESSONS LEARNED (A.4.7.11)

Testing has shown that augmentor operational characteristics can be adversely affected by relatively small changes to the engine configuration and control system. Every effort should be made to ensure the test engine is representative of the intended production configuration.

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A.3.7.11.1 Water injection system fluid.

For engines that use a water injection system, the engine shall meet all requirements specified herein with the water solution consisting of distilled or demineralized water and contaminated up the following levels:

Impurities	Quality
Total solids (max) PPM	35
Dissolved solids (max) PPM	25
pH Factor	6.5 - 7.5

REQUIREMENT RATIONALE (A.3.7.11.1)

The engine must be capable of operating with slightly-contaminated water. Deposits, coatings, and corrosion may form on the engine and affect its performance.

REQUIREMENT GUIDANCE (A.3.7.11.1)

Background:

The purity table is related to the water quality of military demineralization and distillation equipment. The Using Service may tailor these requirements when the state-of-the-art of the equipment changes. Engine performance may also deteriorate due to deposits or coatings on the engine blades and stators due to the water.

REQUIREMENT LESSONS LEARNED (A.3.7.11.1)

Engine K's model specification lists these same limits for the wash water to be used with the engine internal water wash system.

A.4.7.11.1 Water injection system fluid.

The requirements of 3.7.11.1 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.7.11.1)

The performance and operability of the injection system must be verified. Engine tests may be needed to determine if deposits and coatings deteriorate engine performance, operability, and life.

VERIFICATION GUIDANCE (A.4.7.11.1)

Past verification methods have included tests and inspection of the parts following the tests. The purity of the water should be checked if it is not provided by the source. The required engine tests and teardown inspections should determine if deposits form and affect engine performance.

VERIFICATION LESSONS LEARNED (A.4.7.11.1)

None.

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A.3.7.12 Wash system.

An integral wash system shall be provided with the engine if required by the Using Service. The wash system shall not allow standing water to accumulate in the engine. Wash system interface connections shall be compatible with existing Using Service support equipment and be described in the _____.

REQUIREMENT RATIONALE (A.3.7.12)

Integral wash systems at the engine inlet are needed for cleaning of internal engine parts to restore performance. Washing an engine by flushing cleaning solution into the air inlet may not be adequate; there may be a need to introduce the solution (directly) further down stream. An integral wash system improves maintenance and reduces the inventory of maintenance items; this is especially important for shipboard aircraft where storage is at a premium. Standing water may freeze and damage the engine upon ingestion.

REQUIREMENT GUIDANCE (A.3.7.12)

The interfaces can be shown in the ICD, or for programs that do not have an ICD, the interfaces can be put into the specification.

Turbine engine gas path cleaner (MIL-C-85704) has been used as a washing medium on previous programs.

Background:

During design reviews, the contractor should provide a description of the wash system that includes information such as required pressure, temperature, flow, water quality and procedure to be used for field service.

The manufacturer is responsible for developing the wash procedure for field service maintenance. In certain theaters of operation, the availability of water may be limited. The effectiveness of the wash system should also take into account the economical use of water.

All military aircraft during normal service operation encounter unusual environmental contaminants such as salt water, sand and dust particles, armament gases, and engine exhaust gas plumes that are ingested by the engine. These contaminants often accumulate as deposits on various engine internal parts that can result in performance deterioration, corrosion, erosion, etc. Aircraft engines operating in a salt laden atmosphere often develop heavy internal deposits of sea salts. Installation of an integral wash system on the engine inlet permits spraying of a washing medium of a prescribed composition into the engine for cleaning of internal parts. This cleaning action will remove all or most of the contaminants thus restoring the engine to acceptable performance levels and reducing corrosion of internal engine parts.

Standing water in the plumbing of the wash system may cause damage from freezing. The system should be designed to avoid standing water, and a method of purging the water from the plumbing should be provided.

The wash system should be able to remove foreign matter accumulations (such as, residue from sand, dust, and salt laden air) from the inside of the engine in a timely and effective manner. The wash system can be used to restore performance during the calibration and recalibration of the endurance test.

REQUIREMENT LESSONS LEARNED (A.3.7.12)

During Desert Shield and Desert Storm, the deposit due to the ingestion of very fine sand and soil had varying effects on the performance, time between maintenance intervals, parts life, and overall life of many turbine engines. Depending on the temperature of the combustion, some of the sand and soil either glazed over or encrusted the combustion chamber and turbine section, resulting in loss of

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performance, corrosion, and erosion. Loose sand clogged nozzles (resulting in heat stresses) and threatened to compromise seals and bearings. In at least one aircraft type a special cleaning procedure had to be implemented, a cleaning wand had to be fabricated and was inserted via the engine igniter plug opening into the combustion chamber where it introduced multiple streams of washing medium to remove loose sand from the combustor and stage one nozzle components. This procedure was used as often as every 20 hours when operating in heavy sand, and brownout conditions, and was effective in minimizing the damage caused by fine sand accumulation, and extended the cleaning interval.

A.4.7.12 Wash system.

The requirement of 3.7.12 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.7.12)

When engine performance deteriorates due to contamination or after operation in an adverse environment, the wash system must be verified to see if it removes a suitable amount of foreign matter to restore performance and preserve engine life.

VERIFICATION GUIDANCE (A.4.7.12)

Past verification methods have included a test conducted with the official AMT for turbofan and turbojet engines or with the official missionized durability testing for turboprop and turboshaft engines.

VERIFICATION LESSONS LEARNED (A.4.7.12)

None.

A.3.7.13 Brake system.

For turboprop engines, if required by the air vehicle, when the air vehicle is parked and the engine is shut off, the brake shall prevent (a) when exposed to winds up to (b) from any direction with the lower half of the propeller area blanked off. If required for air vehicle mission performance during flight, the brake shall be capable of being shut off during flight and shall prevent rotation at (c). For free turbine engines, the brake shall be capable of preventing rotation of the power turbine at the highest ground idle engine torque, if required by the Using Service.

REQUIREMENT RATIONALE (A.3.7.13)

A brake system may be needed to prevent autorotation of parts, which may induce excessive wear, as a result of improper lubrication. A rotating propeller on a parked turboprop aircraft may present a safety hazard to ground personnel.

REQUIREMENT GUIDANCE (A.3.7.13)

The following should be used to tailor the specification paragraph:

(a): For engines (or for designs for which the engine contractor is responsible for the engine and power absorber gearbox combination) which are directly coupled to the propeller, then the blank should be completed with “rotation of the propeller”. If the engine can’t control propeller rotation, then the first sentence should be removed from the requirement paragraph.

(b): The velocity should be the same as required in the air vehicle specification.

(c): For engines (or for designs for which the engine contractor is responsible for the engine and power absorber gearbox combination) which are directly coupled to the propeller such that it controls propeller rotation when the engine is shut off, then the airspeed required in the air vehicle specification (requirement 3.12.2) should be put in the blank.

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If the turboprop engine and propeller can be decoupled so that the propeller can rotate without turning the engine, and for turbofan, turbojet and turboshaft engines, the requirement should be replaced with “Not Applicable”.

Background:

When the engine is shutdown there is not sufficient lubrication supplied to the bearings, gears and seals. If the rotating parts are allowed to move freely with inadequate oil pressure, the bearing surfaces may experience excessive wear, the oil seals may eventually fail, and leakages may result.

REQUIREMENT LESSONS LEARNED (A.3.7.13)

None.

A.4.7.13 Brake system.

The requirement of 3.7.13 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.7.13)

Verification is needed to show the propeller will not rotate under the required conditions.

VERIFICATION GUIDANCE (A.4.7.13)

Past verification methods have used tests.

Background:

The ideal test for this requirement is to use an aircraft for both flight test and ground test. For the ground test the wind speed specified in 3.7.13 from any direction should be simulated by a wind tunnel.

VERIFICATION LESSONS LEARNED (A.4.7.13)

None.

A.3.7.14 Negative torque limiter.

For turboshaft or turboprop engines, following an engine failure or malfunction, a means to automatically prevent excessive negative torque shall be provided. The propulsion system shall return to normal operation when the malfunction is corrected and positive torque is requested.

REQUIREMENT RATIONALE (A.3.7.14)

A limiter is needed to prevent excessive negative torque that may cause engine damage.

REQUIREMENT GUIDANCE (A.3.7.14)

This requirement is not applicable for systems that incorporate an overrunning clutch type device between the engine and drive systems or for turboshaft or turboprop engines systems not capable of generating negative torque.

Background:

The description of the negative torque limiter should include the method used, the amount and duration of negative torque, and the shaft power absorber signal (torquemeter, strain gauge, etc.).

Negative torque reduces the life of the gearbox since it loads the gears, gear teeth, bearings, and shafts in the opposite direction. With the Using Service approval, the engine contractor should set the negative

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torque for the minimum damage to the gearbox. Typical helicopters do not generate negative torque as rotor blades cannot go into negative pitch and blades do not reverse direction, even in autorotation.

REQUIREMENTS LESSONS LEARNED (A.3.7.14)

None.

A.4.7.14 Negative torque limiter.

The requirement of 3.7.14 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.7.14)

The negative torque limiter must be verified that it will prevent excessive negative torque and return the engine to normal operation under the required conditions.

VERIFICATION GUIDANCE (A.4.7.14)

Past verification methods have included analysis and test.

Background:

The test should include negative torque values of magnitude greater than the specified value (example: if the specified negative torque value is -50 ft-lbs, then the test should include negative values such as -60, -70 ft-lbs, etc.)

The negative torque tested should be increased to the specified value. The duration of the negative torque should be varied from below the specified time to include negative torque duration after limiter activation. This test procedure is to determine the response of the system.

VERIFICATION LESSONS LEARNED (A.4.7.14)

None.

A.3.7.14.1 Multiple power section engines.

For multiple power section turboshaft engines, a means shall be provided to automatically prevent an inoperative power section from absorbing power from the remaining operating power section(s) and the shaft power absorber at any flight condition or power setting. Inoperative power sections shall not result in the operating power section exceeding design or operating limits. An override mechanism shall be provided. A signal shall be provided to identify the section that has been separated. Separation and reconnection of a power section shall not cause any adverse affects on the performance and durability of the engine(s) and power train, and shall not affect safety of flight.

Multiple power section engines shall provide for load sharing and torque matching required by the engine control system in 3.7.2.

When re-starting an inoperative power section with the other power section(s) running, the subsequent reconnection shall be automatic.

REQUIREMENT RATIONALE (A.3.7.14.1)

In a multiple power engine application it is necessary to separate an inoperative power section from the operating section(s) to maintain safety of flight. An easily distinguishable signal or indicator of which engine is separated may help prevent the aircrew from shutting down the wrong engine.

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REQUIREMENT GUIDANCE (A.3.7.14.1)

Background:

If possible, the means to de-couple/ disengage the inoperative power section should be a line replaceable unit. This is necessary to keep the level of maintenance at organizational level.

The requirement applies to “twin-pack” engines. Automatic activation is necessary because manual activation would impose an unwanted burden on the flight crew.

The inoperative power section should be able to be restarted and reconnected with the engine-combining gearbox at any flight condition and power setting, without affecting safety of flight or durability of the operating engine(s). If power from the operating power section is used for the restart, there should be enough power remaining to maintain safety of flight.

It should not be necessary to reconnect or decouple multiple power section engines at extremely low power condition that may affect safety of flight due to insufficient power necessary to sustain flight.

Helicopter engines are typically qualified as single engine installations whether used in single or multiengine aircraft. The engine control system has logic for torque matching and operation with inoperative engine(s).

REQUIREMENT LESSONS LEARNED (A.3.7.14.1)

A twin-engine helicopter has had numerous high speed shaft failures. The high-speed shaft (between the engine and the gearbox) vibrated at a frequency and amplitude that made it difficult for the aircrew to read the existing instrumentation and determine which engine was disconnected. The crew’s vision was blurred from the vibration and the good, connected engine was shut down by mistake.

A.4.7.14.1 Multiple power section engines.

The requirement of 3.7.14.1 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.7.14.1)

Verification of the means to assure proper operation of remaining operating power section(s) when individual power section(s) become inoperative is required.

VERIFICATION GUIDANCE (A.4.7.14.1)

Past verification methods have included analysis and test.

A component test will demonstrate the feasibility of the method. Actual engine tests will give the assurance that the method is acceptable for the aircraft.

VERIFICATION LESSONS LEARNED (A.4.7.14.1)

A past aircraft with interconnected engines and propellers, crashed when a propeller gearbox failed and the engines rapidly overtorqued the remaining gearboxes, popping off each propeller in turn.

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A.3.7.15 Power absorber (engine supplied).

The engine supplied power absorber, which is used to propel the air vehicle as installed in the aircraft, shall meet the requirements of this specification. The power absorber shall include any remote engine supplied mechanism that absorbs power from the engine cycle.

REQUIREMENT RATIONALE (A.3.7.15)

The engine must be capable of operating with the power absorber so that engine operation can be verified.

REQUIREMENT GUIDANCE (A.3.7.15)

The type of power absorber (i.e.; propeller, remote fan, gas ducting, gearbox, transmission, etc.) should be specified by the contractor.

REQUIREMENT LESSONS LEARNED (A.3.7.15)

Dissimilar power absorber characteristics for engines with multiple vehicle application may result in development of separate or modified engine control systems.

A.4.7.15 Power absorber (engine supplied).

The requirement of 3.7.15 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.7.15)

Operation of the power absorber and engine must be verified.

VERIFICATION GUIDANCE (A.4.7.15)

Past verification methods have included analysis and tests.

The intended power absorber should be used for the test if possible. When a dynamometer is used to simulate the power absorber, not all of the loads such as thrust and vibration are reproduced.

VERIFICATION LESSONS LEARNED (A.4.7.15)

Simulation of Aircraft C and Aircraft D vibration loads using Engine C was expensive and unproductive since primarily the test stand and aircraft parts failed. The simulation was performed without engine supplied power absorbers.

A.3.7.15.1 Propeller feathering and reverse pitch.

Engines using propellers shall not exceed any operating limit when the propeller is operated in the feather or reverse pitch mode. If fuel shutoff is required, it shall be accomplished automatically by the control system. If engine and gearbox lubrication oil shutoff is required, it shall not occur until rotation has ceased.

REQUIREMENT RATIONALE (A.3.7.15.1)

The engine must be capable of withstanding the loads and temperatures and must remain within all operating limits when the propeller is in the feathered or reverse pitch position. Engine durability and operability must not be adversely affected.

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REQUIREMENT GUIDANCE (A.3.7.15.1)

Background:

The direction of the loads (thrust and torsion) on the drive shaft of a forward pitched and a feathered propeller are in opposite directions. All drive train components forward of the brake(s) will experience loading on the backside during feathering. The reversed pitch propeller will experience vibration that may be completely different than normal operation, and the characteristics of the vibration may vary with the velocity vector. There is also the possibility of ingesting hot exhaust gases.

When a propeller feathering command has been issued, the engine should be shutdown simultaneously either manually or via an air vehicle signal. If the engine is not stopped when the propeller is feathered, the engine may exceed its temperature limit or other damage may occur.

The Using Service should ensure the reverse pitch operating limits and requirements are in agreement with the limits and requirements of the propeller and aircraft system specification.

REQUIREMENT LESSONS LEARNED (A.3.7.15.1)

None.

A.4.7.15.1 Propeller feathering and reverse pitch.

The requirement of 3.7.15.1 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.7.15.1)

Engine compatibility with manual feathering operation must be verified to ensure the engine won't exceed any limits. Operation in the reverse pitch mode must be verified to ensure there is no adverse impact on the engine during reverse pitch operation.

VERIFICATION GUIDANCE (A.4.7.15.1)

Past verification methods have included analysis and tests.

An analysis of the loads on the engine by a feathered and a reversed pitch propeller should be made. This should be verified by flight or ground tests. Flight tests are preferred since the airflow will be affected by the air vehicle and the engine nacelle.

VERIFICATION LESSONS LEARNED (A.4.7.15.1)

None.

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A.3.7.16 Gearbox.

The gearbox shall provide mechanical power to components required for engine and aircraft operation and, if applicable, as a means to transmit torque to the engine for starting. The mechanical power characteristics at each pad shall be specified in table I.

The gearbox shall be capable of simultaneous operation of all the drives when each drive is subjected to (a) times the maximum permissible torque or power rating specified for the individual drive.

All gears shall be free from damaging resonance at all speeds up to (b) percent of maximum allowable steady-state speed.

REQUIREMENT RATIONALE (A.3.7.16)

Physical and performance characteristics of the gearbox must be established to ensure satisfactory operation of critical engine and aircraft accessories. Peak transient loads are needed to design engine power take-off structure and components.

REQUIREMENT GUIDANCE (A.3.7.16)

The following should be used to tailor the specification paragraph:

(a): A value of 1.25 to 1.50.

(b): The contractor should specify a value such that the resonance will not adversely effect the overspeed requirements of 3.4.1.6.5c and the design service life requirements of 3.4.1.1. A value of 105% has been used in the past.

The gearbox should supply mechanical power to the following engine and aircraft components: generators, fuel, lubrication, and hydraulic pumps; and a means to transmit torque to the engine for starting. The configuration, ratings, and alignment requirements should be included in table I.

The 1.25 value is specified to provide a margin of safety.

The Using Service should ensure that the following is specified in the air vehicle system specification or ICD: “Power Take-Off (PTO) failure and accessory drive failure shall not cause air vehicle or engine damage.” If the engine manufacturer supplies the (PTO) shaft, then this requirement should be placed in the engine specification.

The Using Service should check that the aircraft systems specification defines all the steady-state and transient requirements for PTO shaft power extraction from the engine and the proper limits are in the engine specification.

The gearbox design or the mounted accessories should incorporate shear sections that would allow disengagement of seized components, promoting safe engine operation, or safe engine shut down to prevent further secondary damage to the gearbox and components. Replacement of shear sections after failure should not require removal of the gearbox from the engine nor the engine from the air vehicle.

The gearbox housing should be designed to minimize external surface areas that trap water, which can lead to the formation of salt deposits, which causes or accelerates corrosion.

Table I pad ratings must be based on actual duty cycle, including PTO duty cycle from the air vehicle contractor. All steady-state and transient torque conditions, including one engine only requirements (if applicable), should be reflected in the table I data.

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Background:

PTOs are being used more often on modern engine designs in lieu of a single gearbox that mounts both aircraft accessories and engine components. This permits faster engine removal and installation, as the aircraft systems are not disturbed. Also, to simplify engine removal, aircraft mounted accessories with an independent lubricating system are preferable to engine mounted aircraft accessories. The PTO drive should incorporate a shear section to control the point of failure in the event of an aircraft accessory gearbox lockup.

REQUIREMENT LESSONS LEARNED (A.3.7.16)

Failure of gearbox drives has resulted in aircraft losses on single-engine aircraft.

Provisions must be made to prevent a post-failure PTO shaft from damaging surrounding hardware as happened during one event when a flailing PTO shaft severed an actuator fuel line and flooded the engine.

Engines with long PTO shafts, which have deflected excessively during hard carrier landings, have caused operational problems with the aircraft. Transient peak loads were a problem for Engine J's tower shaft on one aircraft application. Some aircraft power demands are really work or power times time. For example, raising a landing gear may be 500 hp-sec. The delta power demand is thus 500 hp for one second or 50 hp for ten seconds. Aircraft with short moment arms for control surfaces have high actuator loads and a high transient peak duty cycle.

Accessory drive shaft spline bearing air leakage should be quantified and accommodated in the design of the attached housing so that venting requirements can be determined and accommodated in the design when there is negative pressure during shutdown.

A significant life cycle cost driver on many military aircraft has been spline wear. In the past to preclude excessive fretting drive splines were positively lubricated by engine oil or designed for use with nonmetallic spline adapters per MS14169 or MS14184. Use of nonmetallic spline adapters in lubricated accessory pads is permitted. Nonmetallic spline inserts conforming to MS14169 or MS14184 were considered in applications where excessive fretting occurred.

A.4.7.16 Gearbox.

The requirement of 3.7.16 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.7.16)

Examination of the gearbox must be conducted to ensure that the gearbox characteristics conform to the installation drawing. The operational and load carrying properties of the gearbox must be verified.

VERIFICATION GUIDANCE (A.4.7.16)

Past verification methods have included analysis and tests.

The following procedure has been used as a verification method:

A gearbox endurance test using design duty cycles developed from the engine missions specified in 3.4.1.2 and table XI should be accomplished. During the endurance test, the dynamic interface characteristics of the actual aircraft installation should be simulated, including support stiffness and PTO alignment. The endurance test should be run on the gearbox with all accessories and components installed. During the test, all drives should be loaded to at least the maximum overhung moment rating of the drive and subjected to at least the maximum spline misalignment allowed. Elements or components of the engine lubrication system such as filters and oil reservoirs may be incorporated in the test rig to simulate system characteristics. Where applicable, pad seal drain leakage should be monitored

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throughout the test. The power extraction values for the various endurance test segments should be selected to ensure representation of time at maximum steady-state and transient torque values expected in the duty cycle over the life of the gearbox. Some of these transients may be 2X steady torque values in magnitude, but are only a few seconds in duration.

Prior to endurance running, the gearbox should undergo a static torque test wherein the input shaft is held stationary and all drive pads except the starter pad are simultaneously loaded to a static torque value equivalent to the maximum operating torque, at any drive speed, to which the drive pad is subjected during engine operation for a period of five seconds. With this loading maintained, each drive pad, singularly, should be subjected to 150 percent of its maximum shear torque rating for a period of five seconds. In an additional test, with all drive pads loaded to the maximum torque value the pad would be subjected to during starting, the starter drive pad should be loaded to 250 percent of the maximum starter torque or to 120 percent of the maximum starter shear torque rating (the maximum torque that can be transmitted prior to shear section failure), which ever is greater, in a direction which will not unload any other component of the accessory drive train for a period of 5 seconds.

Following the static torque test, the gearbox with all accessories and components installed, should be subjected to a vibratory scan and resonant search, while operating throughout the speed range from idle to the maximum speed specified in 3.7.16 under varying loads including maximum rated pad loads.

After the last maximum speed run, at least ten minutes of operation should be at maximum speed, maximum oil inlet temperatures, and all pads loaded to maximum torque values. This is to check the gears for scoring which will occur under these conditions. Scoring is load and temperature driven. Ten minutes of testing will allow load and temperature stabilization of the system.

During and following the test, scavenge oil filter and magnetic plug residue should be collected and separated into magnetic and nonmagnetic material. This material should then be weighed, analyzed, and the results reported. At the completion of the test, the gearbox should be completely disassembled, inspected dirty then cleaned, and reinspected. There should be no evidence of material defects, undue wear, or impending failure. Conformance with the fluid leakage requirements should be demonstrated during this test.

VERIFICATION LESSONS LEARNED (A.4.7.16)

In the past, gearbox testing on the engine did not expose all the problems. Also, gearbox qualification by similarity did not expose the component to all the loads.

Past gear resonance testing was performed at 130% of normal operating speed that equates to 121% of maximum allowable steady-state speed or 116% maximum transient speed.

Field experience has shown that gear resonance is a major problem on engines due to the high speed and lightweight of the gears. Normal engine testing does not show that gears are free of damaging resonances.

Field experience has also shown that PTO systems should be exposed to maneuver and landing loads during qualification testing to expose any potential problem. Using in-house analytical tools that accurately predict the strength of shafts transmitting torque, coupled with component level testing, NAVAIR engineers saved the Aircraft F program $1.5 million, without impacting the program's return to flight efforts by proving that Aircraft F's engine torquemeter meets all operational requirements.

During flight testing at Naval Air Station Patuxent River, Md., the cause of unmatched torque output between the two Aircraft F engines could not be readily identified. A comprehensive root cause analysis proved that a costly redesign effort was not needed and that the current torquemeter shaft is strong enough to handle aircraft loads in all operating conditions.

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A.3.8 Software resources.

A.4.8 Software resources.

A.3.8.1 Software performance and design.

Software shall perform all computations, data processing, and interfacing functions necessary for the engine to meet the performance and operability requirements of the specification. All programming language shall be per _____.

REQUIREMENT RATIONALE (A.3.8.1)

Following proper software engineering techniques will insure a sound software product. This includes a complete detection of faults, timing analysis, and reduced supportability costs. The final product should result in effective and efficient software operation.

REQUIREMENT GUIDANCE (A.3.8.1)

The contractor should specify the programming language.

Further guidance on developing digital computer software can be found in EIA/IEEE 12207.0, "Software Life Cycle Processes", EIA/IEEE 12207.1, "Software Life Cycle Processes – Life Cycle Data", and EIA/IEEE 12207.2, Software Life Cycle Processes - Implementation Considerations" and RTCA DO-178 “Software Considerations in Airborne Systems and Equipment Certification”. FAA Advisory Circular No: 20-115 advises that RTCA DO-178 can be used as a means to secure FAA approval of digital computer software.

Software design requirements are included as part of the DOD regulations and includes digital engine controls, engine monitoring systems, and support equipment containing software. The use of assembly language must be approved by the Using Service.

REQUIREMENT LESSONS LEARNED (A.3.8.1)

None.

A.4.8.1 Software performance and design.

The requirements of 3.8.1 shall be verified by: _____. The functional performance of the complete computer system shall be demonstrated after integration into the computer system hardware during all engine testing.

VERIFICATION RATIONALE (A.4.8.1)

The design of software programs for the engine computer resource must be verified to ensure that all requirements have been met.

VERIFICATION GUIDANCE (A.4.8.1)

Past verification methods have included inspection, analysis, demonstration, and test.

The design verification must be accomplished in the early stages of development and must be completed prior to the beginning of overall computer program testing.

Verification of software design should be accomplished through the contractor’s quality assurance. Plans, procedures, and test results should be reviewed by the Using Service. The Higher Order Language (HOL) and structure may be verified by inspection. Real-time processing should be verified by demonstration and the Built-in-Test (BIT) by test.

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Any software that affects safety of flight should be thoroughly verified through both component and systems rig testing. The FADEC software is an example of software that should be fully verified prior to first flight.

DO-254 includes procedures for programming chips and is a requirement for control hardware that carries embedded software. DO-254 may be used for FAA-approved airworthiness certification for logic devices.

VERIFICATION LESSONS LEARNED (A.4.8.1)

None.

A.3.8.1.1 Computer reprogramming.

The engine system shall have the computer capability for making software changes without requiring removal of memory hardware. Safety design provisions shall be provided to prevent unauthorized reprogramming. Each software version shall be identified by a unique software version number, which shall be verified during the loading process.

REQUIREMENT RATIONALE (A.3.8.1.1)

The ability to reprogram computer equipment via the WRA connector is needed to reduce maintenance workload and component removal rates.

REQUIREMENT GUIDANCE (A.3.8.1.1)

Background:

Reprogramming capability may be through the air vehicle or engine WRA/LRU. A description of the reprogramming procedure should be presented at design reviews and analyzed by the Using Service.

REQUIREMENT LESSONS LEARNED (A.3.8.1.1)

Original Aircraft F reprogramming design required removal and disassembly of the FADEC which required excessive labor. Aircraft I and Aircraft Z upgrades to EMS software required block upgrades to the mission computer software. This complication has prevented EMS software upgrades that incorporate lessons learned from being implemented.

A.4.8.1.1 Computer reprogramming.

The requirements of 3.8.1.1 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.8.1.1)

The computer equipment reprogramming via the WRA connector, and safety design provisions preventing unauthorized reprogramming, must be verified.

VERIFICATION GUIDANCE (A.4.8.1.1)

Past verification methods have included demonstration and test.

VERIFICATION LESSONS LEARNED (A.4.8.1.1)

None.

A.3.8.2 Spare resources.
Prior to the completion of the (a) engine milestone, the electronic fuel control and engine monitoring system shall utilize no greater than (b) and (c) percent, respectively, of the available throughput during

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the worst case (most heavily loaded) execution of the operational software for each minor frame. The memory reserve shall be at least (d) percent for control systems, and at least (e) percent for engine monitoring systems.

REQUIREMENT RATIONALE (A.3.8.2)

In order to be the most cost effective, computer reserve capability must be provided early in the development stages of engine software.

REQUIREMENT GUIDANCE (A.3.8.2)

The following should be used to tailor the specification paragraph:

(a): The term IFR, FFR, ISR or OCR (depending on milestone selected).

(b): A value from 33 to 50 percent (depending on milestone selected).

(c): A value from 33 to 50 percent (depending on milestone selected).

(d): A value of at least 50 percent (depending on the milestone selected).

(e): A value of at least 50 percent (depending on the milestone selected).

Background:

The required Milestone for memory and throughput reserve capacity should be late in the development program, to allow for as many software changes as possible. Memory and throughput reserve capacity should be incorporated in the design to avoid costly changes or additions to the control microprocessor and memory chips. It may be advisable to specify the reserve capacity for individual memory devices (i.e.; Erasable Electronically Programmable Read Only Memory (EEPROM), RAM, etc.).

The 33 to 50 percent reserve capacity for both memory and throughput should assure reprogramming without disassembly of the control units. Deviation from this requirement requires justification.

Experience indicates that as a system reaches approximately 92% throughput utilization, the software architecture is often compromised, sometimes with negative results, in order to account for a margin of uncertainty in the worst case throughput estimates. A process which evaluates all safety risks should be definitized for safety critical systems approaching maximum throughput utilization. Recent Aircraft I Mission Computer (MC) experience confirms the need for this. When the MC approached throughput saturation, it started unpredictably dropping messages to the FADEC. This caused the FADEC to declare a false 1553 bus failure.

REQUIREMENT LESSONS LEARNED (A.3.8.2)

In a late-1980's USAF program, the engine control system had a requirement for 33 1/3% growth capability after ISR. The need for electronic fuel control computational capability has increased considerably over the years. The control has been used as an engine condition monitor to reduce engine power during exhaust gas ingestion, monitor stall margin, control the aircraft inlet ramps during supersonic flight, and control vector nozzle thrust.

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A.4.8.2 Spare resources.

The requirements of 3.8.2 shall be verified by: _____.

VERIFICATION RATIONALE (A.4.8.2)

There is a need to confirm the software reserve capacity to ensure expandability for the resolution of field problems.

VERIFICATION GUIDANCE (A.4.8.2)

Past verification methods have included demonstration and test.

VERIFICATION LESSONS LEARNED (A.4.8.2)

None.

A.3.9 Production facilities, capabilities, and processes.

The engine design shall be producible using the contractor's manufacturing system. The manufacturing system shall have the facilities, capabilities, processes, and process controls to provide products of consistent quality that meet performance requirements. Production processes shall have the stability, capability, and controls to maintain product characteristics within design tolerances and to reduce the variability of key product characteristics.

REQUIREMENT RATIONALE (A.3.9)

To maintain design tolerances during manufacturing as well as control product cost in production, every production process must be identified, developed, and controlled.

REQUIREMENT GUIDANCE (A.3.9)

Product and process performance will benefit from reduction of variability in key characteristics, so special attention should be paid to key processes and opportunities for reducing variability of those processes. This requirement is intended to ensure the contractor applies the same discipline and effort to the qualifications of the production processes as previously done for performance of primary mission equipment. By identifying and qualifying key production processes up front, production will be smoother and subsequent process improvement efforts can be directed to control cost and quality.

Supplier should have a recognized concurrent engineering approach to match design requirements to processes and a recognized Quality System (such as ISO 9001 to ensure parts and processes conform to requirements.

REQUIREMENT LESSONS LEARNED (A.3.9)

In the past, programs have experienced severe problems in production. Under past practices, development was primarily oriented to the demonstration of product performance with little attention to the ability to consistently and predictably produce the required product characteristics in a cost effective manner. In many cases, the product designs were completed and then turned over to manufacturing who attempted to optimize the production implementation within existing plant capabilities. Little or no effort had been made during development to address producibility as part of the design process. In addition, process control is not a norm within the current aerospace industry, therefore process capability is not known, and isn't matched to product requirements. Mismatches between design limits and process capabilities are discovered too late. Resulting design or process changes are generally sub-optimal.

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A.4.9 Production facilities, capabilities, and processes.

The requirements of 3.9 shall be verified by:

a. The following Incremental Verifications shall be accomplished prior to the Preliminary Design Review (PDR):

b. The following Incremental Verifications shall be accomplished prior to Critical Design Review (CDR):

c. The following Incremental Verifications shall be accomplished prior to Initial Service Release (ISR):

d. The following Incremental Verifications shall be accomplished prior to Operational Capability Release:

VERIFICATION RATIONALE (A.4.9)

Verification is employed to mitigate the risk associated with production, to ensure the ability to maintain design tolerances during the manufacturing process, and to confirm that the contractor has a process to control production cost and quality in production.

VERIFICATION GUIDANCE (A.4.9)

Past verification methods have included analyses of the manufacturing process, demonstrations of the manufacturing method, inspections of the manufactured parts and tests of the components and engine.

Verification at the engine level verifies that the contractor has established a disciplined approach with a process development strategy that (1) includes pre-planned process improvement and evolutionary strategies, (2) provides for the identification and risk abatement of high risk production processes within the program's overall risk management process, (3) includes the identification and control of key processes, (4) ensures consistency between process performance, product performance and cost, (5) defines quality assurance requirements consistent with product performance and cost requirements, (6) flows these requirements to appropriate subtier contractors, and (7) is consistent with the approach at the weapon system level.

Key product characteristics are those measurable design details whose variations have the greatest influence on the product meeting its requirements (form, fit, function, cost, service life, etc.) and are documented in a manner determined by the contractor on the drawings and supporting Technical Data Packages in the "Build-To" and "Support" Packages. Key production processes follow logically from the identification of key product characteristics and the selection of production processes. Key production processes are those processes that produce and control those key product characteristics.

The identification of the key product characteristics and key processes, and the establishment of process capability and process control requirements occur at the engine and subtier levels through the system engineering process. In general, production cost risk can be controlled by demonstrating the key process requirements which include the establishment of design limits and process capabilities. Process capability is typically defined in terms of the statistical probability of non-conformance, such as defects per million or Cp, which is the ratio of design limits to the process variation. Once process capability requirements are established and the capability of the key processes verified, the process controls are established for use during production.

Tasks essential to accomplishing this are (1) identification of high risk production processes with appropriate risk abatement activities, (2) identification and documentation of key product characteristics, (3) identification of key production processes and their key process parameters, (4) establishing the process requirements, which include both the design limits and the process capability, (5) determination of the actual process capability, (6) establishing the process control requirements and (7) flow-down of these requirements to the suppliers whose products will have an effect on the system's attainment of

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performance requirements. Therefore, verification at the engine level confirms compliance with requirements at the engine level, that appropriate requirements are flowed down to the subtier level and that essential tasks have been accomplished at the appropriate engine and subtier level

VERIFICATION GUIDANCE (A.4.9.a)

The following procedure may be used as the verification method:

The Using Service should examine and analyze documentation and design trade study reports to confirm the following have been accomplished at the appropriate engine and subtier levels:

(1) Identification of key product characteristics and the documentation of those characteristics on drawings including appropriate geometric tolerancing and datum control.

(2) Identification of key processes derived from the key product characteristics.

(3) Establishment of process capability requirements, which include both the design limits and process capabilities (C_{pk}, defects per million, etc.)

(4) Evaluation of key process capabilities.

(5) Flow-down of key process requirements.

(6) Assessment of risk abatement status on high risk production processes and appropriate action taken if needed

As a minimum the items listed above should be placed in the verification paragraph. The Using Service may also want to ensure the contractor has completed a manufacturing feasibility assessment. Also, the contractors drawing system, drawing standards and drawing release criteria should be defined prior to the start of detailed design

Usually, the fidelity of the design at PDR is such that all key product characteristics and key production processes are not yet identified. However, based on historical data and the existing level of design, an initial identification and assessment of key production processes can be accomplished and initial capability requirements should be established. In addition, sufficient information exists to assess the progress of risk abatement activities for high risk production processes.

VERIFICATION GUIDANCE (A.4.9.b)

The following procedure may be used as the verification method:

The Using Service should examine and analyze documentation and design trade study reports to confirm the following have been accomplished at the appropriate engine and subtier levels:

(1) Completion of preliminary specifications for key processes.

(2) Completion of preliminary process control plans.

(3) Documentation of rationale to support the detailed design (product, special tooling, special test equipment and support equipment) including key product characteristic's design limit sensitivity to off nominal production. Details should include the results of key suppliers' efforts.

(4) Documentation of rationale to support selection of production processes, including comparison of required process capabilities to documented capabilities and selection of process control criteria with the associated process control plan for achieving required product quality.

(5) Definition of verification requirements for key processes including facility capabilities.

As a minimum the items listed above should be placed in the verification paragraph. The Using Service may also want to ensure the contractor has refined the effort of the verification done at PDR to reflect expected design maturity at CDR. The Using Service should also ensure the contractor has completed the manufacturing feasibility assessment including key process capabilities.

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Usually the fidelity of the design at CDR is such that all key product characteristics and key production processes are identified, capability requirements established, process capabilities verified and preliminary process control plans completed. In addition, sufficient information exists to assess the progress of risk abatement activities including the demonstration of process capability for high risk production processes.

VERIFICATION GUIDANCE (A.4.9.c)

The following procedure may be used as the verification method:

The Using Service should examine and analyze documentation and design trade study reports to confirm the following have been accomplished at the appropriate engine and subtier levels:

(1) Identification of all key product characteristics and the documentation of those characteristics on drawings including appropriate geometric tolerancing and datum control.

(2) Establishment of process capability requirements, which include both the design limits and process capabilities.

(3) Verification of key process capabilities complete including validated process control plans.

(4) Completion of final process control plans.

(5) Proof of final manufacturing feasibility including facility capability.

(6) Completion of final specifications for all key production processes.

(7) Completion of contractor build-to documentation.

As a minimum the items listed above should be placed in the verification paragraph.

The fidelity and stability of the design at ISR is such that all key product characteristics and key production processes are identified, capability requirements established, process capabilities verified, process control plans completed and contractor build-to package completed. Risk abatement activities should have lowered key production process risk to an acceptable level for start of production.

VERIFICATION GUIDANCE (A.4.9.d)

The following should be used to tailor the specification paragraph.

The Using Service should examine documentation to confirm that the adequacy and completeness of the build-to documentation was verified at the engine and appropriate subtier level.

VERIFICATION LESSONS LEARNED (A.4.9.a-d)

None.

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A.3.10 Production cost requirement (PCR).

The production cost shall not exceed that specified in the table:

Subsystem or Equipment	Cost to be Controlled	Dollar Amount	Type of Dollars	Volume	Rate	Elements

REQUIREMENT RATIONALE (A3.10)

The need for a production cost requirement during SDD is driven by the increasing importance of affordability and the need to improve the DOD and industry performance in predicting and meeting cost, schedule and performance requirements. Balancing cost, performance and schedule is an integral part of the Integrated Product Development (IPD) concept. To balance cost, it must be defined and must play an equal role in the systems engineering process. Total Ownership Cost (TOC) is the ultimate cost, which defines true affordability, but TOC is difficult to verify with any confidence during SDD. Therefore, a Production Cost Requirement was selected as a more verifiable cost element. When combined with development cost, the PCR can provide the baseline cost against which design trades can be accomplished. Operational support cost is equally important, but there are other product performance requirements (e.g.; reliability, maintainability, availability) which can be used to assess progress.

REQUIREMENT GUIDANCE (A3.10)

The government must define specific parameters to be used. These include constant versus then year dollars, production volume and rates, and any known fiscal year budget constraints. The production volume and rates are important in defining the return on investment for capital equipment costs and other potential manufacturing investments which have a strong influence on product design. Additionally these factors impact efficiencies such as “the learning curve” and manufacturing manpower utilization. To avoid a "point" design solution, the production rates may be defined as ranges with the target rate identified, and overall production volume may be similarly defined. This would avoid large production cost changes for small changes in quantities. However, it must be clearly understood that with few exceptions, the rate and volume can have a significant impact on the final design and production cost. Therefore, the assumptions used in defining the PCR should be as realistic as possible and the rate and volume ranges should be as small as realistically possible.

Subsystem and Equipment. Specify the subsystem and equipment to which the production cost requirement applies

Cost to be Controlled. Specify the cost to be controlled for each of the subsystems or equipment identified. The cost to be controlled should be selected to satisfy specific program requirements, and may be, for example, unit flyaway, procurement, or average unit production cost. In most cases, it will be important to account for Special Tooling and Special Test Equipment (ST and STE) and Support Equipment (SE). Warranty costs should also be considered. Identify specific warranty cost included in the requirement. The actual selection of the cost definition must be made to control those costs considered most important. For example controlling specific cost drivers versus total production cost.

Dollar Amount. Specify the threshold dollar amount for each subsystem or equipment identified.

Type of Dollars. Specify whether constant or then year dollars are used for each subsystem or equipment identified.

Volume. Specify the overall production volume or a volume for specific lot buys during early production for each of the subsystems or equipment identified. It should also be recognized that costs during rate build up will be higher than for steady-state production. Therefore, it may be more advantageous to

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define the production cost requirement in terms of specific lot buys expected during early production rather than as a single term applicable to the average cost over the entire production volume. PCR may be specified as a range with a target volume. In some instances, it may be necessary to establish individual cost requirements for equipment that have different production quantities and rates, such as equipment that may be common to more than one system or subsystem or have multiple applications within one subsystem.

Rate. Specify the production rate amount for each of the subsystems or equipment identified. The production rate may be specified as a range with a target rate. Specify in terms of number per calendar time, for example, 2 per month or 50 per year. In some instances, it may be necessary to establish individual cost requirements for equipment that have different production quantities and rates. Examples of this are for equipment that may be common to more than one system or subsystem or have multiple applications within one subsystem.

Elements. Identify and define cost elements included, or the explicitly excluded amount for each system, subsystem or equipment identified.

Estimates are usually broken down by functional cost elements such as labor, material, overhead, subcontracted effort, etc. Cost requirements should include all deliverable hardware and software. It is often best to include both Government Furnished Equipment (GFE) and Contractor Furnished Equipment (CFE) in the requirement to provide proper visibility of system options. It is also appropriate to include "overhead" and "investment" costs such as sustaining engineering and rate tooling in the requirement if these are likely to be cost-drivers. In addressing production cost, it is essential that the cost of Special Test (ST), Special Test Equipment (STE) and Support Equipment (SE) also be addressed, as these elements contribute to the overall system cost.

The intent of the production cost requirement is to provide a tool for predicting and controlling design-related cost. This would include the facilities and equipment required to implement the selected production processes. It is not intended that this requirement include indirect costs not controlled by the design, such as impacts of the overall business base. The specific cost components of the requirement which form the basis of the cost requirement must be sufficiently documented to provide an audit trail for any subsequent adjustments and for verification.

Allocated cost requirements must be established in each lower level specification regardless of whether these specifications are developed by the government (for application internally or to the prime contractor) and the prime contractor (for application internally or to subcontractors).

Cost elements included in the cost requirement number must be clearly identified to preclude any misunderstanding. An absolute requirement should be identified which represents the program affordability threshold. If an incentive structure is used, the government must construct the incentive to prioritize the areas where the contractor should expend the greatest effort; i.e., cost versus schedule or specific product performance elements. Any incentive approach must clearly state that the integrated definition and design approach is critical to achievement of the optimal design, cost, schedule, and performance—which is part of the Integrated Plan. This could imply a major funding milestone decision at CDR for the continuation of the development program.

REQUIREMENT LESSONS LEARNED (A3.10)

None.

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A.4.10 Production cost requirement (PCR).

This requirement shall be verified by analysis using a Government agreed to production cost estimating method. The production cost estimating method consists of the procedures, assumptions, tools, database(s), and model(s) agreed to by the government and contractor and is hereafter referred to as the “Production Cost Model” (PCM).

a. The following Incremental Verifications shall be accomplished prior to the Preliminary Design Review:

b. The following Incremental Verifications shall be accomplished prior to Critical Design Review:

c. The following Incremental Verifications shall be accomplished prior to Initial Service Release (ISR):

VERIFICATION RATIONALE (A.4.10)

The requirement must be verified using a PCM that is as accurate as possible.

VERIFICATION GUIDANCE (A.4.10)

The contractor is responsible for obtaining or developing a PCM and maintaining the PCM. All of the assumptions and basic definitions should be contained in a PCM. The agreement on these procedures and assumptions represents the "validation" of the cost model. This model must be agreed to (determined to be a reasonably accurate predictor of production cost and relative cost impacts due to design changes) by the government and used for the final verification of the requirement. The contractor's PCM should be evaluated during source selection and agreed to by the government prior to PDR.

Any appropriate analysis procedure may be used (e.g.; parametric, historical, analogy, or detailed engineering estimates) depending on data availability and the maturity of candidate designs. However, a complete cost model, which contains all the appropriate data and relationships should be developed and maintained by the contractor. This will include factors which account for inspection, test, scrap, and rework. Cost analysis is based on the most current hardware and software configuration. When government data is needed for this analysis, the contractor shall acquire it through the government contracting activity.

This model should form the basis for the verification of the production cost requirement. The primary use of the PCM in the SDD phase is to verify that the most probable cost of the applicable cost elements meet or exceed (are less than) the production cost requirement. The PCM would also form the baseline for follow-on production pricing with the addition of the in-direct costs not contained in the cost model and profit along with a recognition of the current status of the cost risk of the estimate produced by the cost model. Recognizing that it is the intent to define most probable cost, there will always be some uncertainty interval associated with the resultant estimate. This uncertainty interval will be relatively large early in the SDD phase, and should continuously shrink as the design and process capabilities solidify. Properly utilized, the cost model should play a significant role in the overall risk management effort.

In defining this requirement, it must be recognized that the ability to model production cost with the fidelity required does not currently exist. Especially with respect to predicting costs accurately at the start of SDD. Until this capability is developed, it may be necessary to refine the cost requirement at the completion of the Preliminary Design Review and at the completion of detailed design (CDR). Such refinements should always be considered at those times, and should include adjustments among the costs of those system elements assigned separate costs by the specifications that would avoid costly pursuits of lower costs for some subsystems when they are or can be offset by lower costs of other subsystems to achieve overall system cost objectives. Such adjustments are part of the design process as contracted, and should not require adjustments of cost targets of the cost-reimbursement type contracts normally used for SDD.

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VERIFICATION GUIDANCE (A.4.10.a)

The following procedures may be used as the verification method:

Inspection and analysis of documentation should be used to confirm the accomplishment at all levels of the following:

(1) Contractor has established the relationship between design, key production processes, material and production cost.

(2) Contractor correlated the PCM against their historical data and deficiencies identified and corrected.

(3) Contractor defined a process for updating and changing the model and data base (configuration control).

(4) Cost allocation complete and rationale documented.

(5) Preliminary cost estimate complete and consistent for the level of design detail on the preliminary design using the PCM.

(6) Individual subsystem design in agreement with the model assumptions.

As a minimum the items listed above should be placed in the verification paragraph.

The contractor should also ensure the bottoms up assessment of the preliminary design supports the allocated PCR and their roll-up to the system or subsystem PCR, and that the cost estimates are correlated with cost risk abatement and integrated into the production cost estimate.

VERIFICATION GUIDANCE (A.4.10.b)

The following procedures may be used as the verification method:

Inspection and analysis of documentation should be used to confirm the accomplishment at all levels of the following:

(1) Cost estimate completed and consistent for the level of design using the PCM.

(2) Individual subsystem or equipment design in agreement with the model assumptions.

As a minimum the items listed above should be placed in the verification paragraph.

The contractor should also ensure the bottoms up assessment of the detailed design supports the allocated PCR and their roll-up to the system or subsystem PCR, and that the cost estimates are correlated with cost risk abatement and integrated into the production cost estimate.

VERIFICATION GUIDANCE (A.4.10.c)

The following procedures may be used as the verification method:

Inspection and analysis of documentation should be used to confirm the accomplishment at all levels of the following:

(1) Final cost estimate correlated with development test results, test article [and Low Rate Initial Production] build experience, and any remaining risk abatement effort to be completed prior to production and integrated into the production cost estimate.

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(2) Verification of model consistency and completeness.

(3) Analysis using verified data and the agreed to model confirm achievement of the requirement.

As a minimum the items listed above should be placed in the verification paragraph.

The contractor should also ensure the bottoms up assessment of the design supports the allocated PCR and their roll-up to the system or subsystem PCR.

VERIFICATION LESSONS LEARNED (A.4.10.a-c)

None.

A.3.11 Engine system and controls and externals verification.

For the purposes of qualifying the engine system and Controls and Externals (C&E) during System Design and Development (Engineering and Manufacturing Development), the engine system and controls and externals shall be verified per 4.11.

If testing is used as the verification method, then the verification methods described in 4.11 shall be used.

REQUIREMENT GUIDANCE (A.3.11)

Note: 3.11 should not contain requirements. 4.11 should contain verification requirements for engine systems and controls and externals that are not specified in other paragraphs of this specification.

Under some conditions, tests could be fully or partially waived, due to the similarity of previously qualified parts with similar uses and designs. Some environmental tests like explosion proof, humidity and sand and dust are also waived if parts can be shown to have hermetic seals, but a waiver for a hermetic seal should not be based on similarity. All candidates for these types of qualification relief are treated on a case by case basis.

A.4.11 Engine system and controls and externals verification.

The engine system and controls and externals (C&E) that require a test for verification shall be verified by the test procedures per the following paragraphs.

VERIFICATION RATIONALE (A.4.11)

Full up engine tests must be conducted to assess engine performance, operability, and durability capabilities. These tests are conducted to verify particular requirements in paragraph 3 and can also be used to help validate or gain confidence in simulation tools and analytical models. C&E tests must be specified to verify C&E performance and integrity.

VERIFICATION GUIDANCE (A.4.11)

Generally, individual engine component tests are conducted to help ensure full up engine testing can be conducted safely and with a high degree of confidence. C&E component verification testing is usually conducted off engine via bench or system test rigs. At the appropriate point in the program, engine and C&E components are tested together to verify total system integrity.

Additional guidance on engine testing and program risk may be found in a paper presented at the October 13, 2000 Conference on Evolutionary Acquisition. A copy of this paper may be found on the Internet at URL http://www.dau.mil/pubs/pm/pmpdf00/coyn-d.pdf.

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VERIFICATION LESSONS LEARNED (A.4.11)

Thorough test planning and early preparation assures timely execution of tests. It has been a good practice to get the test and evaluation team in place early on to ensure test and verification goals are met.

A.4.11.1 Engine systems tests.

A.4.11.1.1 Endurance tests.

The endurance tests in the subparagraphs below shall be required for the class of engines specified herein.

VERIFICATION RATIONALE (A.4.11.1.1)

Endurance tests include the Accelerated Mission Test (AMT) and durability testing to verify flight test safety, engine life, and effects due to thermal cycling and to simulate accelerated deterioration.

VERIFICATION GUIDANCE (A.4.11.1.1)

The following should be used to tailor 4.11.1.1:

Requirement 4.11.1.1 requires an AMT for turbofan or turbojet engines and durability tests for all classes of engines. Turbofan or turbojet engines should be tested to verify full life requirements of 3.4.1. Additionally, those engines should be required to be tested per the procedures in the missionized Durability Test (4.11.1.1.2.a), to verify the effects of accelerated deterioration at higher than average turbine temperatures.

Turboprop and turboshaft engines should be tested per the missionized Durability Test, and the Cyclic Durability Test (4.11.1.1.2.b). Turboprop and turboshaft engines should be tested per the procedures of 4.11.1.1.2.b, in addition to 4.11.1.1.2.a, for “time-at-temperature” testing of flight regimes and power settings not included in the missionized Durability Test. 4.11.1.1.2b includes the incremental torque runs of table XXIX to insure vibration-free operation.

Background:

The intent of durability tests at IFR and FFR is to ensure flight safety in limited and unlimited flight envelopes and developmental tests for production configurations. The durability tests for ISR and OCR should be sufficient to prove durability and life throughout the flight envelope and worthiness for low and high rate production.

VERIFICATION LESSONS LEARNED (A.4.11.1.1)

None.

A.4.11.1.1.1 Accelerated mission test (AMT).

For turbofan and turbojet engines the following tests shall apply: _____.

For IFR:

An engine conforming to the IFR configuration shall be subjected to an AMT of at least _(a)_ hours or equivalent cycles. The test shall consist of the engine accelerated mission duty cycles defined on figure 21 and in table XL and table XXVIII.

A minimum of _(b)_ starts shall be made on the engine. There shall be _(c)_ false starts and _(d)_ restarts. All test starts shall be performed with a starter as specified in 4.12.3.3.4. At least _(e)_ starts shall be regulated to provide shutdown periods between starts of _(f)_ minutes increasing at _(g)_ minute intervals. Each of the regulated shutdown periods shall be preceded by immediate engine shutdown, without being

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held at idle, after operation for a duration of not less than 30 seconds at intermediate power at the discretion of the Using Service. The remaining starts shall be conducted per the temperature stabilization procedure specified below, but otherwise the shutdown period need not be controlled. The test shall be preceded and followed by the stair-step (HCF) test schedule as specified in 4.4.1.5.1.

For FFR:

An engine conforming to the FFR configuration shall be subjected to an AMT of at least (a) hours or equivalent cycles. The test shall consist of the accelerated mission duty cycles defined on figure 21 and in table XL and table XXVIII. A minimum of (b) starts shall be made on the engine. There shall be (c) false starts and (d) restarts. All test starts shall be performed with a starter as specified in 4.12.3.3.4. At least (e) starts shall be regulated to provide shutdown periods between starts of (f) minutes increasing at (g) minute intervals. Each of the regulated shutdown periods shall be preceded by immediate engine shutdown, without being held at idle, after engine operation for a duration of not less than 30 seconds at intermediate power at the discretion of the Using Service. The remaining starts shall be conducted per the temperature stabilization procedure specified below, but otherwise the shutdown period need not be controlled.

The test shall be preceded and followed by the stair-step (HCF) test schedule as specified in 4.4.1.5.1.

For ISR:

An engine conforming to the ISR configuration shall be subjected to an AMT of at least (a) hours or equivalent cycles. The test shall consist of the accelerated mission duty cycles defined on figure 21 and in table XL and table XXVIII. A minimum of (b) starts shall be made on the engine. There shall be (c) false starts and (d) restarts. All test starts shall be performed with a starter as specified in 4.12.3.3.4. At least (e) starts shall be regulated to provide shutdown periods between starts of (f) minutes increasing at (g) minute intervals. Each of the regulated shutdown periods shall be preceded by immediate engine shutdown, without being held at idle, after engine operation for a duration of not less than 30 seconds at intermediate power at the discretion of the Using Service. The remaining starts shall be conducted per the temperature stabilization procedure specified below, but otherwise the shutdown period need not be controlled.

The test shall be preceded and followed by the stair-step (HCF) test schedule as specified in 4.4.1.5.1.

For OCR:

An engine conforming to the OCR configuration, but containing the same cold parts used in the ISR test, shall be subjected to an AMT of at least (a) hours or equivalent cycles. The test shall consist of the accelerated mission duty cycles defined on figure 21 and in table XL and table XXVIII. A minimum of (b) starts shall be made on the engine. There shall be (c) false starts and (d) restarts. All test starts shall be performed with a starter as specified in 4.12.3.3.4. At least (e) starts shall be regulated to provide shutdown periods between starts of (f) minutes increasing at (g) minute intervals. Each of the regulated shutdown periods shall be preceded by immediate engine shutdown, without being held at idle, after engine operation for a duration of not less than 30 seconds at intermediate power at the discretion of the Using Service. The remaining starts shall be conducted per the temperature stabilization procedure specified below, but otherwise the shutdown period need not be controlled.

The test shall be preceded and followed by the stair-step (HCF) test schedule as specified in 4.4.1.5.1.

The test operation for all engine milestones shall be conducted as follows:

A fixed level of engine inlet air temperature, fuel inlet temperature, and pressure shall be set for each mission cycle as defined by the Strength and Life Analysis Report and defined on figure 21 and in table XL and table XXVIII.

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Air passing over the engine shall correspond to the engine inlet air temperature for that mission condition, within facility constraints.

Upon completion of each mission cycle, the engine shall be shutdown prior to the start of a subsequent cycle. During the shutdown period the engine shall be motored or wind-milled until the (h) measured temperature stabilizes below (i) °C ((i) °F) when the engine is not rotating.

Condition monitoring and diagnostic system installed engine hardware and software shall be in the production configuration. The condition monitoring and diagnostic system shall be operated in a manner analogous to its proposed service operation throughout the test and the output data shall be correlated to observations made during the test.

The customer power takeoff drives and customer bleed air extraction level shall be representative of the maximum required at that condition.

The engine anti-icing or de-icing system shall be actuated at periods during the test to simulate mission operation.

The exhaust duct, inlet duct, and bleed air connections shall be loaded to the maximum allowable loads specified in 3.1.1.3 and in the pretest data of 4.12.4.1.1 respectively, at the discretion of the Using Service.

Fuel inlet temperature shall be maintained at the maximum fuel inlet temperature specified in 3.7.3.2.2 and 3.7.3.2.3 for at least (j) percent of the duration of the mission oriented schedule.

The thrust or power demand shall be advanced or retarded in 0.50 seconds or less unless otherwise specified. The time for changing thrust shall be charged to the duration of the lower thrust setting.

The oil pressure, if adjustable, shall be adjusted at the beginning of the test to the minimum steady-state value specified in 3.7.8.1.1. No further adjustments shall be permitted during the test except when authorized by the Using Service.

The total quantity of fluid leakage from the engine drains shall be recorded and shall not exceed values specified in 3.1.1.8.

Special engine control system features (e.g.; overspeed, overtemperature) shall be cycled through their functions periodically as specified in the pre-test data of 4.12.4.1.1.

At the completion of the test, the engine oil drain provisions specified in 3.7.8.2.2 shall be demonstrated.

If the engine provides special features (i.e.; IR suppressors, fuel heaters, etc.), these items shall be activated during the test to simulate mission operation.

This test is to be run using the distortion pattern(s) of 3.2.2.11.

The test exhaust system (if different from vehicle flight exhaust) shall produce the same aerodynamic and thermodynamic cycle match as the flight exhaust of the primary using application.

VERIFICATION RATIONALE (A.4.11.1.1.1)

Accelerated mission testing is needed to verify life, performance retention, and operability requirements.

VERIFICATION GUIDANCE (A.4.11.1.1.1)

The following should be used to tailor the specification paragraph:

IFR

(a): Two times the number of hours or equivalent cycles of the high time flight test engine in the IFR Milestone.

(b): The number of starts should be based on the flight test schedule.

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(c): A value between or including 5% - 10% of the total number of starts.

(d): A value between or including 5% - 10% of the total number of starts.

(e): A value between or including 5% - 10% of the total number of starts.

(f): A value of at least 5 minutes.

(g): A value between or including 5 minutes and 15 minutes.

(h): T5 or T4.5 (measured or harness temperature).

(i): A temperature of 100°C (212°F).

(j): A value of at least 10%.

FFR

(a): The number of hours equivalent to one hot section life.

(b) through (j): see above in IFR.

ISR & OCR

(a) The number of hours equivalent to one hot section life.

(b): The number of starts should be based on the engine design life.

(c) through (j): see above in IFR.

In table XL the column labeled “Max. Turbine Temp.” may either be deleted or made a reference value as directed by the Using Service.

Background:

A mission oriented cyclic endurance test, also known as an Accelerated Mission Test (AMT) by the Air Force and Accelerated Simulated Mission Endurance Test (ASMET) by the Navy or flight verification test by the Army, should be conducted on flight test configuration as well as production configuration engines. The purpose of the AMT is to demonstrate that the engine meets the durability and life requirements for the milestone. That is, the flight test configured engine should be tested to the life and durability required for flight test, and the production configured engine should be tested to the full life requirements of the specification. Normally the life required for flight test is much less than full engine life.

The AMT exposes durability and other structural problems so they can be addressed and solved before a large number of engines enter fleet service. The flight test AMT ensures that the engine is structurally safe to fly.

Each proposed aircraft mission should be analyzed to define the corresponding engine duty cycles for each mission. Fighter aircraft traditionally perform 8-11 missions. The engine cycles for each mission are then analyzed and the non-damaging cycles are discarded as are the dwell times at part power and idle. Because the lengthy non-damaging dwell times are eliminated, the number of engine cycles accumulated per AMT test hour is higher than the number of cycles accumulated per flight hour. This is the origin of the term “accelerated”. For example, with a “2:1 acceleration factor” the test time would be one-half of hot section life. Test time may also be expressed in equivalent TAC cycles. Engine durability and life can be verified in a shorter time using the AMT than if the specification mission cycles were tested exactly as

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they are in the specification. Dwell times at intermediate power or other power settings that cause damage are retained. Augmentor ignition and augmentor dwell times should be included to obtain the proper number of augmentor ignition cycles and amount of time at temperature on the augmentor hardware. To ensure the engine is fully verified during the AMT, aircraft bleed and horsepower should be extracted at levels appropriate for the test condition.

Engines with short period ratings (time limit <30 minutes) should include the appropriate time for short period power settings if they are above intermediate power. The engine missions and engine life should be analyzed to ensure the AMT includes the appropriate number of starts, hot starts, and false starts. After the test engine has been stopped it should be motored or otherwise cooled to simulate the thermal cycle a field engine would experience after it has been stopped, allowed to cool, and restarted.

VERIFICATION LESSONS LEARNED (A.4.11.1.1.1)

None.

A.4.11.1.1.2a Missionized durability test.

For turbofan, turbojet, turboprop and turboshaft engines, the following tests apply:

For IFR:

An engine conforming to the IFR configuration shall be tested to the missionized cycles of figure 21 and table XL, for _(a)_ hours or cycles. The test shall be conducted so that the _(b)_ turbine temperature shall be equal to _(c)_ °C (_(c)_ °F) at thrust or power settings of intermediate power and above.

For turboprop and turboshaft engines: not applicable

For FFR:

An engine conforming to the FFR configuration shall be tested to the missionized cycles of figure 21 and table XL, for _(d)_ hours or cycles. The test shall be conducted so that the _(e)_ turbine temperature shall be equal to _(f)_ °C (_(f)_ °F) at thrust or power settings of intermediate power and above.

For turboprop and turboshaft engines: not applicable

For ISR:

An engine conforming to the ISR configuration shall be tested to the missionized cycles of figure 21 and table XL, for _(g)_ hours or cycles. The test shall be conducted so that the _(h)_ turbine temperature shall be equal to _(i)_ °C (_(i)_ °F) at thrust or power settings of intermediate power and above.

For turboprop and turboshaft engines: not applicable

For OCR:

An engine conforming to the OCR configuration shall be tested to the missionized cycles of figure 21, table XL, and table XXIX for _(j)_ hours or cycles. The test shall be conducted so that the _(k)_ turbine temperature shall be equal to _(l)_ °C (_(l)_ °F) at thrust or power settings of intermediate power and above.

For turboprop and turboshaft engine Life Verification Test. Two engines each conforming to the OCR configuration shall be tested to the missionized cycles of figure 21 and table XL for _(m)_ hours. The test shall be conducted so that the _(k)_ turbine temperature shall be equal to _(l)_ °C (_(l)_ °F) at maximum continuous and above.

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VERIFICATION RATIONALE (A.4.11.1.1.2a)

All engine classes require durability testing at operating temperatures that are higher than average to accelerate the effects of deterioration.

VERIFICATION GUIDANCE (A.4.11.1.1.2a)

The following should be used to tailor the specification paragraph:

(a): The Using Service should determine the number of hours.

(b): T4.1 (TRIT), T4.5 (LTIT or FTIT), T5 (TET).

(c): At least the maximum allowable temperature of 3.2.2.12.

(d): The Using Service should determine the number of hours.

(e): T4.1 (TRIT), T4.5 (LTIT or FTIT), T5 (TET).

(f): The maximum allowable temperature of 3.2.2.12.

(g): The Using Service should determine the number of hours.

(h): T4.1 (TRIT), T4.5 (LTIT or FTIT), T5 (TET).

(i): The maximum allowable temperature of 3.2.2.12.

(j): The Using Service should determine the number of hours.

(k): T4.1 (TRIT), T4.5 (LTIT or FTIT), T5 (TET).

(l): At least the maximum allowable temperature of 3.2.2.12.

(m): 1200 hours minimum for each engine.

For table XL the values in the column labeled “Max. Turbine Temp.” should be the same values as specified in blanks “c”, “f”, “i”, and “l”, above.

Once confidence is obtained in the performance data sea level correction factors or the contractor verifies the sea level performance correction program, sea level unconditioned air may be used to run the durability test as opposed to more costly conditioned air to obtain development program cost savings.

VERIFICATION LESSONS LEARNED (A.4.11.1.1.2a)

The Navy calls this test a Durability Proof Test (DPT), and is typically a 300-hour test. A 330-hour DPT was done during Limited Production Qualification (LPQ) for the Engine J1. The engine and components were operating satisfactorily at the end of the test. Steady-state performance, transient performance (except for decel from IRP - FI) and the performance deterioration levels were within the allowable limits of the model specification.

For the Aircraft F (Engine H) program, the durability test turbine temperature was achieved by using bleed air to maintain the turbine temperature during durability testing.

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A.4.11.1.1.2b Cyclic durability test.

For turboshaft and turboprop engines, the following tests shall apply:

For IFR:

An engine conforming to the IFR configuration shall be tested for (a) hours of the durability cycles of table XXIX. The engine shall be tested such that the (b) turbine temperature shall be equal to the maximum operating or control limiting temperature for each power setting.

The durability test shall be performed on one engine using the fuels and oils of 4.12.3.4.2.1 and 4.12.3.4.2.2 respectively, and the shaft torque absorber characteristics of 3.7.16.

For FFR:

An engine conforming to the FFR configuration shall be tested for (c) hours of the durability cycles of table XXIX. The engine shall be tested such that the (d) turbine temperature shall be equal to the maximum operating or control limiting temperature for each power setting.

The durability test shall be performed on one engine using the fuels and oils of 4.12.3.4.2.1 and 4.12.3.4.2.2, respectively, and the shaft torque absorber characteristics of 3.7.16.

For ISR:

An engine conforming to the ISR configuration shall be tested for (e) hours of the durability cycles of table XXIX. The engine shall be tested such that the (f) turbine temperature shall be equal to the maximum operating or control limiting temperature for each power setting.

The durability test shall be performed on two engines, using the fuels and oils of 4.12.3.4.2.1 and 4.12.3.4.2.2, respectively, and the shaft torque absorber characteristics representative of different vehicle applications, as applicable.

For OCR:

An engine conforming to the OCR configuration shall be tested for (g) hours of the durability cycles of table XXIX. The engine shall be tested such that the (h) turbine temperature shall be equal to the maximum operating or control limiting temperature for each power setting.

The durability test shall be performed on one engine using the fuels and oils of 4.12.3.4.2.1 and 4.12.3.4.2.2 respectively, and the shaft torque absorber characteristics representative of 3.7.16.

VERIFICATION RATIONALE (A.4.11.1.1.2b)

In addition to durability missionized testing required in 4.11.1.1.2a, all turboshaft and turboprop engines must also be tested for “time-at-temperature” durability and vibration within specification limits.

VERIFICATION GUIDANCE (A.4.11.1.1.2b)

The following should be used to tailor the specification paragraph:

(a): 30 hours minimum.

(b): T4.1 (TRIT), T4.5 (LTIT or FTIT), or T5 (TET).

(c): 60 hours minimum.

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(d): T4.1 (TRIT), T4.5 (LTIT or FTIT), or T5 (TET).

(e): 300 hours minimum.

(f): T4.1 (TRIT), T4.5 (LTIT or FTIT), or T5 (TET).

(g): 600 hours minimum.

(h): T4.1 (TRIT), T4.5 (LTIT or FTIT), or T5 (TET).

Background:

The minimum durability test time to obtain flight clearance in a limited flight test envelope for IFR is 30 hours. The minimum durability test time for flight clearance in the full flight envelope is 60 hours. The minimum durability test time to qualify a production configuration for ISR is 300 hours. The minimum durability test time to qualify for unlimited production at OCR is 600 hours. These times correspond to 60 hours and 300 hours used on previous (US Army) Preliminary Flight Rating Test (PFRT) and Qualification Test (QT) milestones. Note: Current DODI 5000.2 milestones should be used.

Two engines should be tested for ISR. These engines should have different fuels (e.g., JP-8 or JP-5) and oils and different shaft torque absorber characteristics if intended for multiple vehicles. The torque absorber should represent the upper and lower range of rotor inertia, torque limit, and other characteristics for vehicles of primary interest. Multiple absorber characteristics are primarily for verifying transient load conditions and developing a common control system.

The durability test in table XXIX essentially tests time-at-temperature for all power settings at maximum continuous and above, incremental torque runs for critical vibrations from maximum to idle, and transients from no-load to maximum and maximum to no-load. Approximately 60% of the test time is at maximum continuous and above. Contingency and other short period emergency power settings are tested at the end of the durability test. Starts and restarts are included in the test.

LCF testing should be accomplished for all engines at each milestone, either by the AMT test of 4.11.1.1.1, the missionized durability test of 4.11.1.1.2a, or the LCF engine test of 4.4.1.5.2 in addition to the component testing of critical parts/critical safety items done from PDR through OCR. See page 4-2 Naval Aviation Systems Team, Fundamentals of Propulsion System Life Management.

VERIFICATION LESSONS LEARNED (A.4.11.1.1.2b)

Turboprop and turboshaft engines have been tested for many engine programs using the durability test cycle requirements of 4.11.1.1.2b. This method of testing has been considered appropriate and time proven for helicopters, small engines, and multiple vehicle applications (commercial and military). The durability test of table XXIX also has some commonality with commercial FAA tests, an important consideration for this engine class.

A.4.11.1.2 Altitude test.

The performance requirements of 3.2 shall be verified by test. An engine conforming to the same parts list and configuration as the endurance test engine of 4.11.1.1, per 4.12.3.2, shall be subjected to altitude tests at several selected conditions within the operating limits envelope specified in figure 6 (P_{T1} and T_{T1}). Pre-test data shall be per 4.12.4.1.1. The test points shall include the effects of power extraction, bleed air extraction, inlet recovery, inlet distortion, anti-icing or de-icing, and windmilling on engine performance and operability. No adjustments shall be made to the engine without approval of the Using Service. The altitude tests shall be accomplished using the fuels and oils of 3.7.3.1.1 and 3.7.8.1, respectively. Fuel temperature shall be varied over a range sufficient to encompass all anticipated engines operating environments. If the engine has a continuous ignition system, it shall be in operation at all times after the

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start sequence has been completed. Data to be taken and recorded during the test shall be as specified in 4.12.4. Overall true RMS velocity measurements and acceleration spectrograms shall be obtained for each transducer mounted on the engine case and engine gearbox case at the engine speeds and thrusts or power settings selected for the test. The points selected shall include at least the altitude representative points and the point in the complete environmental conditions and operating envelope where the highest engine vibration levels are generated. Critical components of the engine shall be identified on each spectrogram.

The following altitude test procedures apply to each of the following engine qualification milestones.

For IFR:

a. Altitude Performance Points. The requirements of 3.2.1.1 and 3.2.1.2 shall be verified at no fewer than _(a)_ flight altitude condition performance points selected by the Using Service. An adequate number of engine thrust or power settings as determined by the Using Service shall be selected for each specified altitude test condition to establish operating and performance characteristics at each flight altitude condition tested. Operation at each test point shall be of sufficient duration to stabilize the engine.

b. Transient Operation and Functional Test. The requirements of 3.2.2.6 and 3.2.2.7 shall be verified at no fewer than _(b)_ flight altitude conditions selected by the Using Service. Effects of bursts, chops, and bodies throughout the thrust or power range shall be demonstrated. The time elapsed versus engine speeds, measured pressures and temperatures, fuel flows, thrust or power and facility operating conditions shall be obtained for stability verification over the entire thrust or power range. Transients shall be repeated at selected flight conditions to explore control system lags, repeatability, and performance droop. Augmentor ignition limits of 3.2.2.1 shall be tested.

c. Inlet distortion. The requirements of 3.2.2.11 shall be verified by engine operation with simulated inlet airflow distortion at selected operating conditions chosen by the Using Service. Engine stability shall be verified and stall margin remaining defined by engine test at a minimum of _(c)_ operating conditions and _(d)_ distortion patterns selected by the Using Service that combine worst case engine destabilizing influences. Effects of bursts, chops, and bodies throughout the thrust or power range shall be demonstrated.

d. Starts and Restarts. Sufficient engine start and restart testing shall be accomplished to verify the starting requirements of 3.2.2.3 and 3.2.2.3.2. If the automatic relight feature of 3.2.2.3.5 is provided, it shall be verified. The automatic relight feature may be disabled to allow verification of restart requirements with the approval of the Using Service. Each spooldown airstart test condition shall be accomplished at no fewer than _(e)_ core turnaround speeds as defined by the Using Service. A success rate of _(f)_ shall be used to establish successful air starting and restarting at any given condition.

e. Windmilling Test. The requirements of 3.2.2.8 shall be verified at _(g)_ altitude and Mach number condition(s) as selected by the Using Service.

f. Stopping. The requirements of 3.2.2.4 shall be verified each time the engine is stopped and at _(h)_ flight test condition(s) selected by the Using Service.

For FFR (If there is no configuration change from IFR, this test may be waived by the Using Service.)

a. Altitude Performance Points. The requirements of 3.2.1.1 and 3.2.1.2 shall be verified at no less than _(aa)_ flight altitude condition performance points selected by the Using Service. An adequate number of engine thrust or power settings as determined by the Using Service shall be selected for each specified altitude test condition to establish operating and performance characteristics at each flight altitude condition tested. Operation at each test point shall be of sufficient duration to stabilize the engine.

b. Transient Operation and Functional Test. The requirements of 3.2.2.6 and 3.2.2.7 shall be verified at no fewer than _(bb)_ flight altitude conditions selected by the Using Service. Effects of bursts, chops, and bodies throughout the thrust or power range shall be demonstrated. The time elapsed versus engine speeds, measured pressures and temperatures, fuel flows, thrust or power, and facility operating

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conditions shall be obtained for stability verification over the entire thrust or power range. Transients shall be repeated at selected flight conditions to explore control system lags, repeatability, and performance droop. Augmentor ignition limits of 3.2.2.1 shall be tested.

c. Inlet distortion. The requirements of 3.2.2.11 shall be verified by engine operation with simulated inlet airflow distortion at selected operating conditions chosen by the Using Service. Engine stability shall be verified and stall margin remaining defined by engine test at a minimum of (cc) operating conditions and (dd) distortion patterns selected by the Using Service that combine worst case engine destabilizing influences. Effects of bursts, chops, and bodies throughout the thrust or power range shall be demonstrated.

d. Starts and Restarts. Sufficient engine start and restart testing shall be accomplished to verify the starting requirements of 3.2.2.3 and 3.2.2.3.2. If the automatic relight feature of 3.2.2.3.5 is provided, it shall be verified. The automatic relight feature may be disabled to allow verification of restart requirements with the approval of the Using Service. Each spooldown airstart test condition shall be accomplished at no fewer than (ee) core turnaround speeds as defined by the Using Service. A success rate of (ff) shall be used to establish successful air starting and restarting at any given condition.

e. Windmilling Test. The requirements of 3.2.2.8 shall be verified at (gg) altitude and Mach condition(s) as selected by the Using Service.

f. Stopping. The requirements of 3.2.2.4 shall be verified each time the engine is stopped and at (hh) flight test condition(s) selected by the Using Service.

For ISR (If there is no configuration change from FFR, this test may be waived by the Using Service)

a. Altitude Performance Points. The requirements of 3.2.1.1 and 3.2.1.2 shall be verified at no less than (aaa) flight altitude condition performance points selected by the Using Service. An adequate number of engine thrust or power settings as determined by the Using Service shall be selected for each specified altitude test condition to establish operating and performance characteristics at each flight altitude condition tested. Operation at each test point shall be of sufficient duration to stabilize the engine.

b. Transient Operation and Functional Test. The requirements of 3.2.2.6 and 3.2.2.7 shall be verified at no fewer than (bbb) flight altitude conditions selected by the Using Service. Effects of bursts, chops, and bodies throughout the thrust or power range shall be demonstrated. The time elapsed versus engine speeds, measured pressures and temperatures, fuel flows, thrust or power, and facility operating conditions shall be obtained for stability verification over the entire thrust or power range. Transients shall be repeated at selected flight conditions to explore control system lags, repeatability, and performance droop. Augmentor ignition limits of 3.2.2.1 shall be tested.

c. Inlet distortion. The requirements of 3.2.2.11 shall be verified by engine operation with simulated inlet airflow distortion at selected operating conditions chosen by the Using Service. Engine stability shall be verified and stall margin remaining defined by engine test at a minimum of (ccc) operating conditions and (ddd) distortion patterns selected by the Using Service that combine worst case engine destabilizing influences. Effects of bursts, chops, and bodies throughout the thrust or power range shall be demonstrated.

d. Starts and Restarts. Sufficient engine start and restart testing shall be accomplished to verify the starting requirements of 3.2.2.3 and 3.2.2.3.2. If the automatic relight feature of 3.2.2.3.5 is provided, it shall be verified. The automatic relight feature may be disabled to allow verification of restart requirements with the approval of the Using Service. Each spooldown airstart test condition shall be accomplished at no fewer than (eee) core turnaround speeds as defined by the Using Service. A success rate of (fff) shall be used to establish successful air starting and restarting at any given condition.

e. Windmilling Test. The requirements of 3.2.2.8 shall be verified at (ggg) altitude and Mach condition(s) as selected by the Using Service.

f. Stopping. The requirements of 3.2.2.4 shall be verified each time the engine is stopped and at (hhh) flight test condition(s) selected by the Using Service.

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For OCR (If there is no configuration change from ISR, this test may be waived by the Using Service)

a. Altitude Performance Points. The requirements of 3.2.1.1 and 3.2.1.2 shall be verified at no fewer than (aaaa) flight altitude condition performance points selected by the Using Service. An adequate number of engine thrust or power settings as determined by the Using Service shall be selected for each specified altitude test condition to establish operating and performance characteristics at each flight altitude condition tested. Operation at each test point shall be of sufficient duration to stabilize the engine.

b. Transient Operation and Functional Test. The requirements of 3.2.2.6 and 3.2.2.7 shall be verified at no fewer than (bbbb) flight altitude conditions selected by the Using Service. Effects of bursts, chops, and bodies throughout the thrust or power range shall be demonstrated. The time elapsed versus engine speeds, measured pressures and temperatures, fuel flows, thrust or power, and facility operating conditions shall be obtained for stability verification over the entire thrust or power range. Transients shall be repeated at selected flight conditions to explore control system lags, repeatability, and performance droop. Augmentor ignition limits of 3.2.2.1 shall be tested.

c. Inlet distortion. The requirements of 3.2.2.11 shall be verified by engine operation with simulated inlet airflow distortion at selected operating conditions chosen by the Using Service. Engine stability shall be verified and stall margin remaining defined by engine test at a minimum of (cccc) operating conditions and (dddd) distortion patterns selected by the Using Service that combine worst case engine destabilizing influences. Effects of bursts, chops, and bodies throughout the thrust or power range shall be demonstrated.

d. Starts and Restarts. Sufficient engine start and restart testing shall be accomplished to verify the starting requirements of 3.2.2.3 and 3.2.2.3.2. If the automatic relight feature of 3.2.2.3.5 is provided, it shall be verified. The automatic relight feature may be disabled to allow verification of restart requirements with the approval of the Using Service. Each spooldown airstart test condition shall be accomplished at no fewer than (eeee) core turnaround speeds as defined by the Using Service. A success rate of (ffff) shall be used to establish successful air starting and restarting at any given condition.

e. Windmilling Test. The requirements of 3.2.2.8 shall be verified at (gggg) altitude and Mach condition(s) as selected by the Using Service.

f. Stopping. The requirements of 3.2.2.4 shall be verified each time the engine is stopped and at (hhhh) flight test condition(s) selected by the Using Service.

VERIFICATION RATIONALE (A.4.11.1.2)

Altitude testing is required to verify engine performance and operating capability within the specified engine operating envelope for each milestone.

VERIFICATION GUIDANCE (A.4.11.1.2)

The following should be used to tailor the specification paragraph:

The contractor should complete the blanks. The large range of values for the blanks takes into account the wide variety of engine types and vehicle applications for which the specification is written (i.e.; trainer, transport, turboshaft or turboprop, and augmented turbofans). A fighter engine, for example, will have more flight conditions and distortion patterns than would a trainer engine.

The following are expected input for tailoring of the specification paragraph by the contractor:

IFR: (Determine performance, operability and safety in selected portion of flight envelope.)

(a): 10 - 20 flight altitude, Mach number and inlet temperature combinations should be tested for performance. Turboshaft applications should use a minimum of 3 - 5 conditions.

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(b): 10 - 20 altitude and Mach number conditions should be tested for transients and operability.

(c): 12-21 flight altitude, Mach number, distortion and power demand.

(d): 4 - 7 distortion patterns should be tested at the pattern's design flight condition and other minimum surge margin conditions. Aircraft F (engine H) used five distortion patterns, as an example, and this was negotiated based on flight regime, installation effects, and mission requirements.

(e): 2 - 4 core turnaround speeds should be identified.

(f): A success rate of 2 out of 3 or 3 out of 4 should be required.

(g): A minimum of one (1) altitude and Mach number condition should be tested for windmilling.

(h): 1 - 2 test condition(s) including a high flight Mach number, high and low inlet ram temperature, and high and low inlet ram pressure, as appropriate, should be tested.

FFR: (Determine performance, operability and safety throughout full flight envelope.)

(aa): 15 - 30 flight altitude, Mach number and inlet temperature combinations shall be tested for performance. Vary number with engine type and application. Turboshaft applications should use a minimum of 3 - 5 conditions.

(bb): 10 - 20 altitude and Mach number condition points should be tested for transients and operability.

(cc): 3 operating conditions per distortion pattern.

(dd): All distortion patterns specified by 3.2.2.11 that have not been tested should be tested at the pattern's design flight condition and other minimum surge margin conditions.

(ee): 2 - 4 core turnaround speeds should be identified.

(ff): A success rate of 2 out of 3 or 3 out of 4 should be required.

(gg): A minimum of one (1) altitude and Mach number for windmilling. (A flight condition different from the IFR condition should be used.)

(hh): 1-3 test condition(s) should be tested at high flight Mach numbers, high and low inlet ram temperatures, and high and low inlet ram pressures.

ISR: (Repeat altitude performance and operability testing for engines having changes in configuration, materials, exhaust system, etc. Omit altitude performance and operability testing, if the ISR configuration or other changes do not affect performance or operability.)

(aaa): 15-30 flight altitude, Mach number and inlet temperature combinations should be tested for performance. Turboshaft applications should use a minimum of 5 - 10 conditions.

(bbb): 15-30 altitude and Mach number condition performance points should be tested for transients and operability.

(ccc): 3 operating conditions per distortion pattern.

(ddd): All distortion patterns specified by 3.2.2.11 that have not been tested should be tested at the pattern's design flight condition and other minimum surge margin conditions.

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(eee): 2 - 4 core turnaround speeds should be identified.

(fff): A success rate of 2 out of 3 or 3 out of 4 should be required.

(ggg): One (1) altitude and Mach number condition should be tested for windmilling (only if redesign of oil sump and bearing cavity areas has occurred). A flight condition different from the other than IFR and FFR conditions should be used).

(hhh): 1-2 conditions at high flight Mach number, high and low inlet ram temperature and high and low inlet ram pressure condition(s) should be tested.

OCR: Altitude testing should not be needed in this milestone as performance and operability should be completely defined in prior milestones. However, if numerous design changes are made during the ISR Milestone, another altitude test may be required.

(aaaa) - (hhhh): See and use values from ISR above.

Background:

The intent of the qualification milestones is to have performance and operability of the engine defined throughout the flight envelope by FFR. Subsequent milestones, ISR and OCR, are to qualify the engine configuration for production. Engines that change configuration, materials, parts, devices or exhaust systems between milestones may need to repeat altitude testing on later milestones to reconfirm or determine performance. Flight envelope conditions should consider alternate flight vehicles or applications at the discretion of the Using Service.

The flight conditions selected should serve to define engine operating line performance and operability over a range of altitude, Mach number, and inlet ram temperature conditions. The number of conditions can vary with engine type or vehicle application. The flight conditions should include, but are not limited to table III performance points.

The number of flight conditions shown for subparagraphs a. and b. (above) at each milestone are not cumulative, but are the recommended number for that particular milestone, taking into consideration the number of conditions tested for a previous milestone. For example, at IFR, 15 flight conditions may be tested. At FFR a trainer or turboprop engine may only need an additional 10 conditions, whereas a fighter engine may need an additional 15 or 20. The number of flight conditions at ISR may also be related to the number of configuration changes made between FFR and ISR. If the ISR engine is essentially the same as the FFR engine, little or no additional testing may be needed, whereas an engine with several major configuration changes may need testing throughout the entire flight envelope.

A similar reasoning was used for the distortion patterns. For IFR the patterns that most affect safety or patterns that may occur during flight test should be tested. During FFR, the number of additional patterns that may need testing is related to the engine type, size of the flight envelope and the extent of flight testing that is proposed (i.e., severe maneuvers that generate high distortion). For ISR, most, if not all, of the patterns described and shown in the specification should be tested. Exceptions may be patterns that are less stringent or patterns that have been tested for IFR and FFR and no major engine configuration changes have occurred between FFR and ISR.

Transient performance tests should include the following:

(a) Accels from various low power settings to high power at low and high altitude.

(b) Decels from high power setting(s) to idle at low and high altitude.

(c) Perturbations at various low power settings at low altitude.

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(d) Other transients as needed for in-flight maneuvering or vehicle control.

(e) Transients as needed for augmentor ignition.

(f) Transients as needed for loaded and unloaded turboshaft rotors. Transient performance should be repeated at selected flight conditions to investigate control system lags, repeatability, and performance droop (e.g.; helicopter rotor, cold engine launch, hot day takeoff, etc.). Altitude test cells cannot replicate helicopter inertias using rotor heads in the altitude test cell, but the rotor loads are simulated using power absorbers during testing. In the event that the altitude test cell installation cannot simulate helicopter inertias with the power absorption system, measured transient performance may not be representative of actual vehicle performance. Accuracy of the simulation of external load input to the propulsion system under test may be limited by the characteristics or capabilities of the facility load absorption system, however, gas generator (or core) acceleration and decelerations schedules and power or free turbine control and limit functions should be fully evaluated and verified under load conditions throughout the envelope.

The number of start and restart flight conditions (including spooldown airstarts) should continually increase through ISR. IFR tests may not include much airstart testing, and FFR tests may only contain a few points at low risk flight conditions. ISR tests should map out the full airstart region to generate a database that will be used to develop the aircraft flight manual. For airstart tests not only will the flight condition change, but so will the turnaround speeds (i.e.; 25%, 40%, and 50% core rotor speed).

The windmilling test may take a long time to accomplish depending on the specification requirement. Windmilling may not be required for IFR depending on the program and safety requirements at the test site. Some commands require that initial flights occur within gliding distance of the airfield in case the engine fails. Depending on the number of windmilling flight conditions required in the specification, each milestone test should occur at different flight conditions, as long as major configuration changes have not occurred.

The test to verify stopping will occur during each test session, but may only be critical at a high altitude high Mach number condition where rotor seizure may occur. Depending upon the thoroughness of the flight test program, an in-flight stopping test may not be accomplished until FFR or even ISR. Flight test facility safety guidelines may require an in-flight stopping test at an early milestone.

Demonstration of engine operation with the control system in control failure modes should be accomplished.

The stabilization period may be anywhere from 5 to 10 minutes depending on the size of the engine. The small turboshaft engine class may need 5 minutes whereas the large turbofan engine class may need the full 10 minutes to stabilize.

VERIFICATION LESSONS LEARNED (A.4.11.1.2)

Extensive lessons learned databases have been developed and may be referenced directly through systems program offices at NAVAIR, USAF, and AMCOM. Current best test practices may also be available through AEDC.

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A.4.11.1.3 Engine systems tests success criteria.

The test shall be satisfactorily completed when the following conditions are met:

a. The engine shall meet the pass or fail criteria of the test plan.

b. The engine shall meet the inspection criteria of 4.12.3.5.4.

c. The engine and engine components meet the service limits of 4.12.3.5.3.

d. There are no part failures or any parts failing integrity requirements of 3.4.

e. All failures and problems encountered during the test and any additional testing that may be required has been analyzed and resolved to the satisfaction of the Using Service (i.e., penalty runs).

f. The test engine and all initial (start of test) components are operated satisfactorily to the end of the test and an analysis utilizing engine calibration results establishes that the performance characteristics of 3.2 and the starting requirements of 3.2.2.3 have been met.

g. Inspection of engine parts and engine components throughout the test, including the final inspection, shows that the parts life requirements of the milestone have been met.

h. Test results indicate compliance with the reliability and maintainability requirements specified in 3.5.

i. The engine weight meets the requirements of 3.1.2.1.

j. The engine has not exceeded any limits of table IV.

VERIFICATION RATIONALE (A.4.11.1.3)

Success criteria are needed to establish a condition by which to approve test results.

VERIFICATION GUIDANCE (A.4.11.1.3)

Where the altitude test has been successful, the engine has not been disassembled for inspection.

VERIFICATION LESSONS LEARNED (A.4.11.1.3)

Extensive lessons learned databases have been developed and may be referenced directly through systems program offices at NAVAIR, USAF, and AMCOM. Current best test practices may also be available through AEDC.

A.4.11.2 Controls and externals component tests.

A.4.11.2.1 Controls and externals component operational tests.

Controls and external component specification compliance shall be verified by analysis, test, and inspection. Table XXX specifies required component tests.

VERIFICATION RATIONALE (A.4.11.2.1)

Verification of component subassemblies and components capability and reliability must be accomplished to satisfy weapon system mission requirements.

VERIFICATION GUIDANCE (A.4.11.2.1)

Past verification methods have included analysis, tests, and inspections. The contractor should complete table XXX.

Under some conditions, tests could be fully or partially waived, based on analysis of similarity of previously qualified components with similar uses and designs. Some environmental tests like explosion

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proof, humidity and sand and dust are also waived if parts can be shown to have hermetic seals, but a waiver for a hermetic seal should not be based on similarity. All candidates for these types of qualification relief are treated on a case by case basis.

Simulated mission operational tests to verify the life specified in 3.4.1.1 should be conducted on all C&E systems and components. The ignition system should be tested per 4.7.5. All C&E tests should be conducted on test assemblies that have the same part number, consisting of groups of related components arranged and interconnected to simulate their normal relationship and function on the engine. However, subassemblies or components of a system may be tested separately if such separation does not prevent simulation of the complete function of the components or subassemblies. If practical, components should be mounted in their normal position as mounted on the engine. No adjustments should be made subsequent to the component calibration.

VERIFICATION LESSONS LEARNED (A.4.11.2.1)

Past programs (Engine D) have used the following:

"COMBINED US ARMY/FAA QUALIFICATION TEST REQUIREMENTS FOR DIGITAL ENGINE CONTROLS," AMSAT-R-EPT - 14 NOV 94, UPDATED 9/96

TEST	ECU	HMU	SENSORS	WIRE HARN	PMA	FUEL SYS
DO 160C/ADS37A EMI	x	x	x	x	x	
DO 160C LIGHTNING	x	x	x	x	x	
ENG ENDUR TEST W/CALS	x	x	x	x	x	x
8593 SOT (160C SECT. 4/5)	x	x	x	x	x	x
8593 FUEL PUMP CAV		x				x
8593 LOW LUBR FUEL		x				x
DO160C 10/11 FLUIDS/WATER	x	x	x	x	x	
8593 HUMIDITY	x	x	x	x	x	
8593 ENG ALT W. HOT/COLD STARTING	x	x	x			x
DO160C FUNGUS	x	x	x	x	x	x
8593 EXPLOSION. PROOF	x	x	x	x	x	x
DO160C SAND & DUST	x	x	x	x	x	
8593 SUSTAINED ACCEL.	x	x	x		x	
8593 IMPACT	x	x	x		x	
8593 VIBRATION	x	x	x	x	x	x
8593 PMA CONTAINMENT					x	
FAA FIRE (FAA AC No: 20-135)		x		x		x
DO 178B SOFTWARE VER./ VAL.	x					
DO 160 SALT SPRAY	x	x	x	x	x	x

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Notes:

1. 8593 tests are based on qualification tests previously performed for Engine D AND E FADECs using AV-E-8593D. The DO160, DO178 and FAA AC 20-135 tests are based on FAR 33 required tests for the FADEC system of the Commercial Engines D, N, and other engines. See ADS-37A-PRF for EMC testing.

2. Sensors include P_3, Torque, N_G, N_P, Compressor Inlet Temperature (CIT), $T_{4.5}$ sensors or any other air vehicle or engine mounted sensors that have an electrical interface with the ECU or HMU.

3. Fuel system. Includes all fuel-wetted components (lines, filters, pumps, coolers, and nozzles). Fuel pump cavitation test applies to pumps only. Explosion proof test for electrical components only. DO160 and DO 178 are RTCA documents. AV-E-8593 is a USA document."

(ECU is Engine Control Unit; HMU is Hydro-Mechanical Unit; PMA is Permanent Magnetic Alternator.)

A.4.11.2.1.1 Controls and externals component calibration.

If C&E testing is performed, these components shall be calibrated and the calibration shall be recorded.

VERIFICATION RATIONALE (A.4.11.2.1.1)

Components and their subassemblies must be calibrated to ascertain accuracies and establish proper input and output relationships prior to using these systems in a test.

VERIFICATION GUIDANCE (A.4.11.2.1.1)

Prior to the initiation of the C&E tests, each component that has input-output relationships should be subjected to a calibration. The calibration should be extensive enough to cover the engine steady-state and dynamic ranges of operation of the component and should indicate conformance with the design tolerance range of the component. The engine control components should be shown to conform to accuracy, stability, and response requirement(s) stated in the component test procedure. Prior to initiation of the calibration, the contractor should develop a list of components that do not require calibration. This list should be subject to approval by the Using Service. Components not requiring calibration should be operated under normal operating conditions to demonstrate satisfactory functioning and compatibility with other system components.

VERIFICATION LESSONS LEARNED (A.4.11.2.1.1)

Calibration has been performed prior to testing to ensure proper component function.

A.4.11.2.1.2 Controls and externals component test procedures.

If component testing is required, then prior to the start of any testing, all components shall be cleaned of any corrosion preventive compounds used for storage protection to simulate the condition of the part as installed on the engine. Test assemblies or components shall be subjected to operating loads simulating those encountered on the engine. Instrumentation shall be provided to indicate the performance of each component and to indicate that the functional relationships of components are maintained as required by the applicable test schedule. Functional checks shall be performed at the end of each test or group of tests and at other times, at the option of the Contractor, to indicate that no calibrated component has changed its calibration beyond allowable service limits and that the function of uncalibrated components is unimpaired. All components shall be tested with the primary operating fluids of 3.7.3.1.1, 3.7.8.1, and 3.7.9. All items driven by the engine gearbox shall be operated at maximum allowable axial and angular misalignment conditions at the drive pad. Component subassemblies or components shall be the correct part number intended for use during the appropriate milestone.

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VERIFICATION RATIONALE (A.4.11.2.1.2)

The component should be tested in the as-installed condition and exposed to engine operating loads and temperatures. This is to ensure the component test simulates the engine operating environment.

VERIFICATION GUIDANCE (A.4.11.2.1.2)

The test plan should contain the test procedures and describe the instrumentation necessary for the tests. Table XXXII should be used for guidance for preparing a test plan.

VERIFICATION LESSONS LEARNED (A.4.11.2.1.2)

None.

A.4.11.2.1.3 Controls and externals component missionized tests.

If C&E component missionized tests are required, then all engine components, shall be subjected to a simulated mission operational test (or cycles) intended to produce the same equivalent component damage as would be achieved by real time tests based on the air system mission and mission mix of table XI. The simulated mission operational test cycle shall be defined by the contractor. The test shall include maximum and minimum fluid temperatures expected during engine operation and shall include maximum hot and cold soak temperatures of figure 5.

Components utilizing bleed air or requiring pneumatic input signals shall be subjected to air at pressure and temperature values corresponding to those occurring throughout the range of engine operation. _____ test cycles, interspersed throughout the test, shall include air contaminated as follows:

a. The amount of engine lubricating oil shall be consistent with what has leaked into the flow path in previous engine testing, but not less than 3 parts per million engine lubricating oil by mass.

b. A salt concentration of 200-ppb (0.2-ppm) salt in air injected into the engine using a 4-percent salt-water solution per table VI.

c. Humidity levels consistent with 3.3.1.1

d. Sand and dust composed of quartz and other material per table XXXV with a particle size range of 0-80 micron (0.00315-inches).

VERIFICATION RATIONALE (A.4.11.2.1.3)

Verification that the components can operate reliably throughout all the missions is needed to preclude failures that may occur in service. Missionized verification methods more accurately simulate the expected operating environment than non-missionized verification methods. Pneumatic air C&E components must not be affected by contaminants in the gas path flow.

VERIFICATION GUIDANCE (A.4.11.2.1.3)

The contractor should provide the number of cycles that contain contaminated air based upon the anticipated usage. If the usage is not defined, then ten percent of the cycles should include contaminated air. The following procedure may be used as a verification method:

“Test cycles shall be consistent with the following requirements:

a. Components in test assemblies shall function in their normal sequence of operation on the engine.

b. Cycling shall be controlled by varying simulated inputs to the test assembly or component. Pilot controlled inputs, such as power settings, shall be varied continuously as required by the test cycle and without regards to variables having reached the demanded values. Engine supplied inputs shall be varied in their usual relations to component outputs. For Army turboshaft and turboprop engines: the

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components should pass through their maximum range of functions at least once during each cycle and pilot controlled inputs such as power settings, should be varied in single step changes over their entire range and should not be changed again until all variables have reached the demanded values.

c. Input variables substantially independent of other control inputs, such as altitude pressure, shall be cycled at a rate faster or slower than the basic functional cycle so every component shall eventually have accomplished each part of its function at each value of the independent variables.

d. When a component provides for a manual or automatic transfer from one mode of operation to another, the manual or automatic means shall be used to transfer an appropriate number of times during the test.

e. Components designed to prevent the engine from exceeding its operating limits, but are not actuated in normal operation, shall be actuated an appropriate number of times with their input variables set to reach the necessary range of values.

Salt should be introduced using four-percent salt water solution."

The following sand and dust requirement has been used in the past:

"Crushed quartz, 146 mg of quartz per kg of air (0.0146 weight percent quartz), of the following distribution:

Microns (inches)	Percent
0-5 (0-0.000197)	37-41
5-10 (1.97×10^{-4}-3.94×10^{-4})	15-21
10-20 (3.94×10^{-4}-7.87×10^{-4})	13-19
20-40 (7.87×10^{-4}-1.57×10^{-3})	15-21
40-80 (1.57×10^{-3}-3.15×10^{-3})	6-12 "

Background:

The test should include high temperature, room temperature, low temperature, fuel pump cavitation, and low lubricity that must be conducted on the same serial number component(s).

VERIFICATION LESSONS LEARNED (A.4.11.2.1.3)

Compressor discharge and combustor pressures used in hydromechanical fuel control schedules have experienced contamination problems that caused incorrect scheduling of fuel to the engine. Contaminated bleed air has caused compressor bleed actuator failures.

Past Army programs have used the following test procedure updated with the following superseding specifications and standards:

"High Temperature. The necessary test assemblies shall be assembled and each test assembly or component shall be operated as specified in the following subparagraphs. Components normally in contact with fuel shall be supplied with the fuel specified in the engine specification but with an aromatic content of at least 25 percent. Toluene may be added to the fuel to attain the required aromatic content. Each test assembly or component shall be functionally cycled per the appropriate component test cycle

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for 100 hours or 600 cycles, whichever represents the longer period. During this cycling, ambient and fluid temperatures shall be maintained as follows:

a. The ambient temperature shall be maintained at 71°C (159.8°F) for 60 minutes. The ambient temperature shall then be increased, within five minutes to the maximum temperature for the component specified in 3.2.2.13.1 and maintained at this temperature for 120 minutes. The ambient temperature shall then be returned to 71°C (159.8°F) within five minutes. This procedure shall be repeated until completion of the test.

b. Components normally in contact with fuel shall be supplied with the fuel controlled to the maximum temperature specified in 3.7.3.2.2.

c. Other fluids used for cooling or control purposes shall be maintained at their maximum allowable temperatures.

Room Temperature. Each test assembly or component shall undergo functional cycling per the following subparagraphs. During the above testing, the fuel filters, if furnished with the engine fuel system, shall be serviced as recommended by the engine manufacturer, but at intervals representing a cumulative fuel flow equivalent to not less than that obtained in 12 hours continuous operation at intermediate rated power. Components utilizing bleed air or requiring pneumatic input signals shall be subjected to air at pressure and temperature values corresponding to those occurring throughout the range of engine operation. During the first hour and each succeeding tenth hour of testing, this air shall be contaminated as follows:

a. Three parts per million engine lubricating oil by mass.

b. A salt concentration of 0.2 parts salt (NaCl) per million parts of air by mass (salt shall be introduced using 4.0 percent water solution).

c. Distilled water to saturate the air at 52°C (125.6°F) at an ambient absolute pressure of 101.3 kPa (14.69 psi).

d. Crushed quartz, 146 mg of quartz per kg of air (0.0146 percent by weight), of the following distribution:

Microns (inches)	Percent
0-5 (0-0.000197)	37-41
5-10 (1.97×10^{-4}-3.94×10^{-4})	15-21
10-20 (3.94×10^{-4}-7.87×10^{-4})	13-19
20-40 (7.87×10^{-4}-1.57×10^{-3})	15-21
40-80 (1.57×10^{-3}-3.15×10^{-3})	6-12

Each test assembly or component shall undergo functional cycling per the appropriate component test cycle for at least 300 hours or 1800 cycles, whichever represents the longer period. Control of ambient or fluid temperatures shall not be required during this test. Test assemblies normally in contact with fuel shall be supplied with fluid conforming to ASTM D471. The functional cyclic test shall be run in six 50-hour segments. During the last five hours of each segment, the fluid shall be contaminated downstream from the fluid tank with at least the amount of contaminant specified in table XXVc part a and with at least the amount of contaminant specified in table XXVc part b for the remainder (first 45 hours) of each segment. The solid contaminant shall not be recirculated. The test assembly shall remain idle for at least 18 hours following the second and fifth segments and for at least 6 hours following the first, third, and

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fourth segments. During these idle times, the assembly shall contain the contaminated residual test fluid from the previous test segment. The idle time shall not count as part of the cyclic test time.

Low Temperature. Upon completion of the room temperature test, each test assembly or component shall be soaked in an ambient temperature of lower than -54°C (-65.2°F) for a minimum period of ten hours. Upon completion of soaking, the soaking temperature of -54°C shall be maintained while each test assembly or component is operated as detailed below. During the entire low temperature test, fluid conforming to ASTM D471, Type I shall be present in each test assembly or component part normally coming in contact with fuel. Prior to each cycling period the test fluid temperature shall be reduced to below -54°C (-65.2°F). Other fluid temperatures may rise as anticipated in service operation under similar ambient conditions. If -34°C (-29.2°F) is reached before completion of a cycling period, the cycling shall be stopped and restarted when the fluid temperature has been reduced to below -54°C (-65.2°F).

Each assembly or component shall undergo a test of functional cycling for at least a total of 20 hours or 120 cycles, whichever represents the longer period. The test shall consist of at least ten separate runs with the cycle sequence of operation in each run as defined in the component test cycle.

Fuel Pump Cavitation. See 4.7.3.2.3.

Low Lubricity Fuel Test. See 4.7.3.2.6."

U.S. Navy letter AE 419: DRB: cts S/N 2257 Nov. 1967 contains the research data for the salt concentration in the air contamination requirement.

A.4.11.2.1.4 Controls and externals component recalibration, disassembly, and inspection.

Upon completion of the preceding tests, component calibrations of 4.11.2.1.1 shall be repeated. Components not subjected to calibration shall be operated under normal operating conditions to demonstrate satisfactory functioning and compatibility with other system components. During recalibration, the same fluids, inputs, and operating cycles shall be used as in the calibration. Each recalibration shall be recorded. All components shall then be completely disassembled and inspected for indications of failure or excessive wear.

VERIFICATION RATIONALE (A.4.11.2.1.4)

Component recalibrations are necessary to ensure no component has changed its calibration beyond allowable service limits.

VERIFICATION GUIDANCE (A.4.11.2.1.4)

When reviewing the recalibration test results, all out of limit conditions should be addressed. Compare the recalibration test results to those of the initial calibration and identify any instances where large shifts have occurred. These shifts may still be within acceptable limits at post test recalibration but may also indicate design or verification issues requiring action.

VERIFICATION LESSONS LEARNED (A.4.11.2.1.4)

Extensive lessons learned databases have been developed and may be referenced directly through systems program offices at NAVAIR, USAF, and AMCOM.

A.4.11.2.1.5 Controls and externals component environmental test.

All engine components or sub-components shall be subjected to environmental component testing per table XXX and as noted herein. These tests may be conducted on test assemblies or individual components, either of which may be new or not previously subjected to any testing. When a single hermetically sealed component is used in a series of the tests below, the component need not be disassembled for inspection until the last test of such series has been completed. In addition, hermetically sealed components need not be subjected to the explosion-proof, sand and dust, and fungus tests. Prior to disassembly, a test to determine hermetic seal integrity shall be performed. Failure of the

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hermetic seals during any test shall disqualify that component. The components shall be inspected for defects or damage that may have been incurred during any of the tests performed.

VERIFICATION RATIONALE (A.4.11.2.1.5)

Environmental testing must be conducted on components to ensure they will survive real world environments.

VERIFICATION GUIDANCE (A.4.11.2.1.5)

The same serial number component should be subjected to all the environmental component tests namely: humidity, fungus, explosion proof, sand and dust, sustained acceleration, shock, and vibration.

For component humidity, fungus, explosion proof, sustained acceleration, sand & dust, shock, and vibration test guidance, see 4.3.1.1, 4.3.1.2, 4.1.8.3, 4.4.1.2.1, 4.11.2.1.5.1, 4.4.1.2.1, and 4.11.2.1.5.2, respectively.

VERIFICATION LESSONS LEARNED (A.4.11.2.1.5)

Past programs have used the following test procedures updated with the current standards and procedures listed herein:

> "Component Test Procedures. Prior to the start of testing, all components shall be cleaned of any corrosion-preventive compounds used for storage protection. Components shall be subject to operational loads simulating the maximum operating loads encountered on the engine. Power supply transients, when required herein, shall consist of the application of overvoltages and the application of spike voltages. The application of spike voltages shall consist of five spikes of +600 volts for 10 microseconds each, five spikes of -600 volts for 10 microseconds each, five spikes of +200 volts for 50 microseconds each, and five spikes of -170 volts for 50 microseconds each. The application of overvoltages shall consist of voltages of 180 volts RMS for at least 0.10 seconds, of 160 volts RMS for at least 0.50 seconds, of 140 volts RMS for at least 2.0 seconds, and 125 volts RMS for at least 6.0 seconds for alternating current systems or of 80 volts for at least 0.050 seconds, of 60 volts for at least 0.50 seconds, of 40 volts for at least 2.0 seconds, and of 35 volts for a least 3.0 seconds for direct current systems. These transients will be only applied to all components that are in direct connection with the 28 VDC power lines. Submission of an electrical analysis, which proves that components can in no way be affected by MIL-STD-704 power sources and transients, is required for consideration of waiver of these requirements by the Using Service. The hermetically sealed component is used in a series of the tests below; the component need not be disassembled for inspection until the last test of such series has been completed. At this time, the components will be inspected for defects or damage that may have been incurred during any of the tests performed. In addition, hermetically sealed components need not be subjected to the explosion-proof, sand and dust, and fungus tests. Prior to disassembly, a test to determine hermetic seal integrity shall be performed. Failure of the hermetic seals during any test shall disqualify that component. Sustained acceleration, impact, and vibration shall not be performed on the wiring harness, thermocouple harness, and ignition lead.
>
> Humidity. Components shall be subjected to an atmosphere of clear water vapor under the conditions and cycle as specified herein for a period of 15 days. Immediately after 12 hours of the first test cycle, all components shall be checked for proper operation with the maximum specified voltage. During the last cycle of the 15-day test each component shall be operated throughout its entire range of operation under each of the ambient conditions specified. At the end of the 15 days of test, the components shall be dried without forced convection for 12 hours in air at less than 21°C (69.8°F) with not less than 50 percent relative humidity. At the conclusion of this period, power supply transients shall be applied to the components. After recalibration,

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components shall be disassembled and inspected. There shall be no corrosion or other defects present that affect the function or structural strength of the component.

Humidity Conditions Test Cycle

Period	Time Period(hours)	Relative Humidity (Percent)	Ambient Temperature degrees C (deg F)
1	8	95 ± 5	54 (129°F) ± 3 (5°F)
2	4	Holding chamber moisture content constant	54 (129°F) ± 3 (5°F) gradually down to 21 ± 3
3	8	Holding chamber moisture content constant	21 (69.8°F) ± 3 (5°F)
4	4	Up to 95 ± 5, holding chamber moisture content constant or adding vapor to reach required value	21 (69.8°F) ± 3 (5°F) uniformly up to 54 (129°F) ± 3 (5°F)

Fungus. Components shall be subjected to a fungus test per MIL-STD-810, Method 508. Evidence must be provided in a fungus analysis report which indicates all material used are fungus inert per MIL-HDBK-454A, Guideline 4 or have been previously tested per MIL-STD-810. Components shall be subject to pre- and posttest calibrations or functional checks as applicable but need not be operated during the test. Criteria for passing the test shall be as defined in MIL-STD-810, Method 508. Test reports documenting MIL-STD-810F notice 3 and report approval letter must be included in the Fungus Analysis Report.

Explosion-Proof. All electrical components not hermetically sealed shall be subjected to explosion-proof testing per MIL-STD-810F notice 2, Method 511, Procedure I. The test shall be conducted as specified at the following test altitudes of sea level to 5000 feet (1,524-m), 10,000 feet (3,048-m), 15,000 feet (4,572-m), 20,000 feet (6,096-m), 25,000 feet 7,620-m), and 30,000 feet (9,144-m). During the test, components shall have maximum input voltage applied to them and shall be operated continuously at their maximum loads. During each altitude condition, all make and break contacts shall be operated at least ten times. A power supply transient shall be applied to the components during each altitude condition. At least four of these power supply transients shall be applied during operation of make and break contacts. Power supply transients shall be as stated in component test procedures above. Ignition components or systems shall be operated continuously. Electrodes of spark ignitors shall be mounted in such a manner that the explosive vapor in the test chamber shall not be contacted. Electrically self-sufficient ignition systems shall be exempted from the application of power supply transients. Failure criteria shall be as defined in MIL-STD-810F, Notice 2, Method 511 (i.e., ignition of test fuel vapor constitutes test item failure for procedure I).

Sand and Dust. Components shall be subjected to a sand and dust test per MIL-STD-810F Notice 3, Method 510. Components need not be operated during the test; however, at the conclusion of the test, and prior to any cleaning of the components, four power supply transients, covering the transient range, shall be applied to the component. Criteria for passing the test shall be as defined in MIL-STD-810F, Method 510.

Sustained Acceleration. Components shall be subjected to an acceleration test per MIL-STD-810F change notice 2 dated 1 Jan 2000, Method 513. The test item shall be subjected only to the operational test per Procedure II and table II of MIL-STD-810F for the aircraft or helicopter vehicle category as applicable. The value of "A" (for use with Method 513) shall be 4g (see table II note 3) consistent with figure 8a. Test time may be increased beyond the minimum specified if necessary to determine proper operation.

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Shock. Components shall be subjected to a functional test per MIL-STD-810F, Method 516, Procedure I. The shock pulse shape shall be per figure 1 of MIL-STD-810F of amplitude "A_p" and time duration "T_e". Tests shall be conducted under room ambient conditions. Components shall be subjected to pre- and post-test calibrations or functional checks as applicable but need not be operated during the test. Post-test inspection procedures shall be per MIL-STD-810F, Method 516.

Vibration. Components shall be subjected to a vibration test per MIL-STD-810F change 3, Method 514, Procedure I. Tests shall be conducted on engine mounted equipment per tables III and IV, and curve "L" of figure 16 of MIL-STD-810F. Components may be tested in test assemblies or as individual units. During the test the component shall be subjected to its maximum limiting component temperature specified in the engine specification."

A.4.11.2.1.5.1 Controls and externals component sand and dust.

Components shall be subjected to a sand and dust test per MIL-STD-810, Method 510. Components need not be operated during the test; however, at the conclusion of the test, and prior to any cleaning of the components, four power supply transients, covering the transient range, shall be applied to the component. Criteria for passing the test shall be as defined in MIL-STD-810, Method 510.

VERIFICATION RATIONALE (A.4.11.2.1.5.1)

Components must be exposed to a sand and dust test to ascertain the ability of the component to resist particles which may penetrate into cracks, crevices, and joints and to assess the degradation of the component performance, reliability, and maintainability.

VERIFICATION GUIDANCE (A.4.11.2.1.5.1)

Past verification tests have been per MIL-STD-810.

Past power supply transients have been per MIL-STD-704

Past criteria for passing the test was defined in MIL-STD-810.

Components may be operated during the test based on the use requirements of the specific test item.

SAE AIR 4246 may be used as guidance for fluid contamination descriptions.

VERIFICATION LESSONS LEARNED (A.4.11.2.1.5.1)

Extensive lessons learned databases have been developed and may be referenced directly through systems program offices at NAVAIR, USAF, and AMCOM.

A.4.11.2.1.5.2 Controls and externals component vibration.

All components shall be verified to be operational when exposed to vibrations as when the components are installed and in an operational environment. During the verification the component shall be subjected to its maximum limiting component temperature specified in table V.

VERIFICATION RATIONALE (A.4.11.2.1.5.2)

Verification of C&E components operation with vibrations must be made to determine the resistance of components to vibration stresses expected in its shipping and operating environments.

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VERIFICATION GUIDANCE (A.4.11.2.1.5.2)

Past verification methods have included vibration tests that were per MIL-STD-810.

The engine D program used the following procedure:

> "MIL-STD-810C, Method 514, Procedure I. Tests shall be conducted on engine mounted equipment per table II, and curve "L" of figure 2 of MIL-STD-810C. Components may be tested in test assemblies or as individual units. During the test the component shall be subjected to its maximum limiting component temperature specified in the engine specification."

Components may be tested in test assemblies or as individual units.

VERIFICATION LESSONS LEARNED (A.4.11.2.1.5.2)

Extensive lessons learned databases have been developed and may be referenced directly through systems program offices at NAVAIR, USAF, and AMCOM.

A.4.11.2.1.6 Controls and externals component test success criteria.

The component tests shall be considered to be satisfactorily completed when, in the judgment of the Using Service:

a. During the tests, component performance and function were within established limits.

b. During the tests, there was no fluid leakage from any component other than that of a nature and rate specified in the specification.

c. During the tests, there was no binding or hesitation of any component.

d. Recalibrations indicate that no component has changed its calibration beyond allowable service limits.

e. The component teardown inspection shows no indication of failed, excessively worn, and distorted parts. Measurements shall be taken and compared with the contractor's drawing dimensions and tolerances or with similar measurements made prior to the test.

VERIFICATION RATIONALE (A.4.11.2.1.6)

The success or failure of the test needs to be assessed per pass or fail criteria.

VERIFICATION GUIDANCE (A.4.11.2.1.6)

None.

VERIFICATION LESSONS LEARNED (A.4.11.2.1.6)

Extensive lessons learned databases have been developed and may be referenced directly through systems program offices at NAVAIR, USAF, and AMCOM.

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A.3.12 Engine qualification.

For the purposes of qualifying the engine during System Design and Development (formerly Engineering and Manufacturing Development), Production and Deployment and Operations and Support (formerly Production, Fielding/Deployment and Operational Support), the engine shall be verified per 4.12.

Note: 3.12 shall not contain requirements. 4.12 shall contain verification requirements for engine tests that have not been specified in other sections of this guide.

REQUIREMENT RATIONALE (A.3.12)

To ensure that the engine design satisfies minimum requirements in various performance, life and operability characteristics so that confidence is gained to progress into the next phase of the acquisition process, engine qualification is for the following purposes: 1) to define the qualification milestones in engine development, 2) to define what the contractor must accomplish to progress through each milestone, 3) to provide a table that clearly outlines types of verifications required in each milestone, and 4) to provide generic verification test methodology to be used, where applicable, for tests throughout section 4.

REQUIREMENT GUIDANCE (A.3.12)

Engine development test hours has ranged on various past programs anywhere from 8,677 hours for the Engine O program to 15,000 hours for the Engine J program. Note, out-year advanced integrated propulsion technology are looking at reducing development costs by 50% using Numerical Propulsion System Simulator (NPSS) in design and simulation. This software is being developed with the intent to transition it to Industry for use on future SDD programs. Until that time, engine testing to the levels identified above is considered reasonable.

REQUIREMENT LESSONS LEARNED (A.3.12)

Engine development is expensive but there can be significant cost avoidance through development program identification and correction of failure modes. Otherwise, there will be early failures in the fielded systems that are expensive to correct and impact system readiness.

A.4.12 Engine qualification.

The engine shall be verified during each engine milestone per this section, and table XXXI. Each verification in each milestone must pass its respective criteria and gain approval of the Using Service prior to proceeding to the next milestone. All verifications shall have quantified success criteria and these criteria shall be approved by the Using Service prior to initiation of each milestone.

VERIFICATION RATIONALE (A.4.12)

This section and tables XXXI a and b establish the test methods, procedures, and criteria that must be satisfactorily achieved for each engine milestone for the contractor to qualify the engine and to allow the program to proceed through the acquisition phases.

VERIFICATION GUIDANCE (A.4.12)

Selection of table XXXI a or b depends on the type of acquisition program approach being used. Table XXXIa, Spiral Development Matrix, reflects an example of a generic Spiral Development acquisition approach. Spiral Development is intended to allow earlier delivery of increased capability to the warfighter than traditional acquisition approaches. Each “spiral” includes increased system capabilities along with their associated increased verification requirements. Programs with small production lots or utilizing Commercial Off The Shelf systems as a baseline (i.e., unmanned air vehicles), may want to consider using a Spiral Development approach to help minimize development cost and schedule. The

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tailoring of table XXXIa allows for flexibility in the concept refinement and technology development phases for evolving system capabilities without overstating development requirements, recognizing the system requirement has not been fully established. Table XXXIb, Incremental Verification Matrix for System Development and Demonstration (SDD), reflects a more traditional acquisition approach where the system requirement has been defined through Milestone B (authorization for SDD), but tailored to accommodate the 5000 Series Acquisition requirements. Table XXXIb can be applied to each acquisition phase and each spiral within the phase depending on program requirements. Typically, in Concept Refinement only PDR is used, in Technology Development PDR, CDR, IFR are used, and in the last three acquisition phases all six milestones are used.

Regardless of which table is selected, it should be completed such that the verification methods shown match those specified in each applicable section 4 paragraph. Each square of each milestone in the table identifies the verification method to be used. The verification methods presented in these tables do not account for any configuration changes that may occur throughout the development cycle. During the course of the program the table should be updated to reflect any changes in requirements or system capabilities.

The following are examples of tailoring table XXXI for Fighter, UAV, and Cargo/Tanker applications:

Fighter Aircraft:

The complex mission of a fighter aircraft imposes a more stringent requirement verification approach than most other systems (i.e.; transports, cargo).

In the Technology Demonstration phase, a conscientious decision should be made to limit the verification of the design requirements to those based on safety of flight, structural integrity, and performance. Separate tables can be used to define design requirements verification versus design requirements that are directly related to airworthiness and flight clearance and those that are goal oriented. These three tables comprise the total set of design information for the program. Examples of these tables are provided in tables XLVIII a, b, and c. Additional guidance for these three tables are below.

Table XLVIIIa (Design Requirements Verification Matrix): Engine development and verification activities should be accomplished by analysis, inspection, demonstration, and/or testing as specified in this table. This table identifies the methodology by which the requirement should be verified. It is the responsibility of the program to define how the methodology will be applied.

Table XLVIIIb (Flight Clearance Configuration Matrix): This table contains preliminary identification of the requirements and their planned verification approach (analysis, inspection, demonstration, and/or testing) that will be part of a Flight Clearance Plan. These preliminary requirements are needed to support the air vehicle design activity and for the engine program to assess progress towards satisfying airworthiness criteria of MIL-HDBK-516.

Table XLVIIIc (Design to Goal Characteristics Matrix): In a Technology Demonstration phase, using spiral development, this table contains goals that help establish the capability of the maturing engine technology. These goals are assessed using the verification approach shown in the table and may transition to specification requirements in SDD and later phases or spirals.

Unmanned Air Vehicles (UAV):

Since UAV's are unmanned and their system capabilities are continually evolving through the Air Force CONOPS and the Navy's Sea Power 21 vision processes, they may not require the same level of verification requirements in the initial spiral phases that manned systems do.

Using the lessons learned from the fighter engine example, table XXXI a and b may be used as a template and non-applicable requirements deleted (i.e.; 4.1.1.7.1, 4.1.8.1). As the weapon system

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evolves and capability requirements are added, previously non-applicable or new requirements may have to be included in the verification matrix.

An example of a verification table that could be used to tailor the column titled Concept Refinement and Technology Development may be found in table XLIX.

Commercial Off The Shelf (COTS) Cargo/Tanker:

The mission of a cargo/tanker aircraft is generally less complex than other weapon systems. This may permit a COTS approach that considers civilian airworthiness authority approval and fleet experience in the development of the requirement verification matrix. The use of COTS systems may rely heavily on alternate verification and certification approaches such as those used by the FAA or Joint Aviation Authority (JAA). Every effort should be made to ensure all qualified missions of COTS applications meet military applications. Differences may exist between commercial and military missions and usage. These differences may force re-verifications of some JSSG-2007A requirements reflected in table XXXI. Differences may force the user to use information available other that what was used by the civilian airworthiness authority to meet military requirements. Fleet usage information in addition to the qualification information should be utilized as an aide to evaluate the differences for applicable requirements in table XXXI with moderate/high risk.

An example of a verification table that could be used to tailor the column titled Technology Development may be found in table L.

VERIFICATION LESSONS LEARNED (A.4.12)

Fighter: Concept Development Aircraft (CDA) for Aircraft AH (Engine AN) successfully used the Spiral Development approach shown above.

UAV: None

Cargo/Transport (COTS): Aircraft L (Engine S) and Aircraft AE (Engine AP) have both successfully used a COTS approach.

Spiral Development in the 5000 Series Acquisition requirements and UAV development are relatively new concepts with minimal practical experience.

A.4.12.1 Qualification milestones.

The acceptability of the engine design for each milestone shall be predicated on satisfactory completion of the verification requirements of section 4 per table XXXI a or b and the approval of the Using Service.

VERIFICATION RATIONALE (A.4.12.1)

Each milestone must be defined and the verification method for each milestone should be identified to ensure the contractor understands the requirements and to ensure the proper method is being used for the milestone.

VERIFICATION GUIDANCE (A.4.12.1)

The definitions for the milestones and verification methods are contained in 6.3 Definitions.

VERIFICATION LESSONS LEARNED (A.4.12.1)

Extensive lessons learned databases have been developed and may be referenced directly through systems program offices at NAVAIR, USAF, and AMCOM.

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A.4.12.2 Verification responsibility.

The contractor shall be responsible for the performance of all verification methods specified herein.

VERIFICATION RATIONALE (A.4.12.2)

The Department of Defense policy is to ensure that responsibility rests upon the contractors for controlling product quality.

VERIFICATION GUIDANCE (A.4.12.2)

The contractor may use any facility suitable for the performance of the verification specified, unless disapproved by the Using Service. The Using Service reserves the right to perform any of the verifications set forth in this specification when necessary to ensure supplies and services conform to prescribed provisions. Specific verifications required may be waived by the Using Service provided the contractor submits documentation with appropriate justification for approval.

VERIFICATION LESSONS LEARNED (A.4.12.2)

Parts have been accepted by similarity and Government test results have sometimes differed from those of the contractor.

Extensive lessons learned databases have been developed and may be referenced directly through systems program offices at NAVAIR, USAF, and AMCOM.

A.4.12.3 Manner of testing.

Unless otherwise noted, the following testing methods, conditions, and procedures shall be required for full engine system verification and qualification per table XXXI a or b.

The test procedures shall be in compliance with table XXXII.

VERIFICATION RATIONALE (A.4.12.3)

Test activities and test designers performing under the JSSG must be held to the requirements contained herein. Manner of testing provides the baseline expectation for the team that performs the qualification testing to ensure testing is uniform and consistent throughout the entire SDD and Production programs.

VERIFICATION GUIDANCE (A.4.12.3)

Prior to the initiation of each separate engine or component test, demonstration or consecutive group of tests or demonstrations conducted on any single assembly or component, a test procedure should be submitted. Using Service approval of this procedure will authorize test initiation. Each test procedure should contain but is not limited to the items identified in table XXXII.

VERIFICATION LESSONS LEARNED (A.4.12.3)

In the past, disagreements occurred between the Using Service and Contractor on test success or failure. These requirements have helped to bring successful resolution.

Extensive lessons learned databases have been developed and may be referenced directly through systems program offices at NAVAIR, USAF, and AMCOM.

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A.4.12.3.1 Test surveillance.

Each test and demonstration herein shall be subject to witnessing by authorized Using Service representatives. At convenient times prior to the tests and during teardown inspections, the engine and components shall be examined to determine if they conform to all requirements of the contract and specifications under which they were built. At no time shall any part of the engine or component be disassembled, adjusted, cleaned, replaced, or removed without prior approval of the Using Service representative.

VERIFICATION RATIONALE (A.4.12.3.1)

Using Service representatives insure that all the requirements for testing are carried out per the model specification. The Using Service representatives are to perform an independent examination of engine parts at the designated inspection periods. This also prevents any unauthorized actions that could invalidate test results.

VERIFICATION GUIDANCE (A.4.12.3.1)

The Using Service should provide surveillance of the contractor for proper conduct of a test or demonstration. There should be a Using Service representative witnessing all tests and engine inspections. The Using Service can perform its own independent inspections, if appropriate.

VERIFICATION LESSONS LEARNED (A.4.12.3.1)

On some occasions, engine parts have been replaced during a test that resulted in parts that were not fully tested as required; however, this was in concurrence with the Using Service.

Extensive lessons learned databases have been developed and may be referenced directly through systems program offices at NAVAIR, USAF, and AMCOM.

A.4.12.3.2 Test article configuration.

Test article configuration and the differences from the milestone parts list of 3.1.9.1 shall be identified and justified in the pretest data of 4.12.4.1.1 and shall be included in the detailed test procedure. The mixing of parts of segmented stator assembly or different designed blades in a disc is not allowed without Using Service approval. All parts shall be considered as having zero time at the start of a test, as appropriate, unless otherwise specified by the Using Service. While all parts do not have to be new, any part that fails the success criteria during the test shall be cause for rejection of that part or test article as determined by the Using Service.

VERIFICATION RATIONALE (A.4.12.3.2)

A test article configuration and parts list is needed for each milestone. This gives the Using Service a record of test article changes made during the development phase. The contractor is prohibited from mixing parts for comparison purposes because the tests and demonstrations are not meant for the contractor to perform parts development. Deviation parts list is required to tell how configurations differ at all milestones. This includes any software version changes.

VERIFICATION GUIDANCE (A.4.12.3.2)

The test article configuration and parts list should always include all the latest proposed parts for the particular milestone. There should not be any interim parts from previous models or milestones just to maintain the development schedule, without Using Service approval. For some tests, software may need to be modified to permit operation at or greater than limit conditions, for example endurance testing at limit temperatures.

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VERIFICATION LESSONS LEARNED (A.4.12.3.2)

Extensive lessons learned databases have been developed and may be referenced directly through systems program offices at NAVAIR, USAF, and AMCOM.

A.4.12.3.2.1 Test article.

The particular article intended for a specific milestone test or demonstration shall be officially designated by the contractor prior to the start of the test.

VERIFICATION RATIONALE (A.4.12.3.2.1)

There is a need to know a particular article configuration for a specific test because it establishes the baseline configuration for the milestone. This allows the Using Service to review the test article and determine its suitability for the test.

VERIFICATION GUIDANCE (A.4.12.3.2.1)

Engine parts can be selected within their manufacturing tolerance range to increase the engine capability to pass a given test. For example, proper selection of first stage turbine area can optimize the compressor operating line to give additional stall margin by using excess compressor efficiency.

VERIFICATION LESSONS LEARNED (A.4.12.3.2.1)

Extensive lessons learned databases have been developed and may be referenced directly through systems program offices at NAVAIR, USAF, and AMCOM.

A.4.12.3.2.2 Installation interfaces.

Inlet, exhaust, and air bleed connections shall be as defined in 3.1.1.1. The engine gearbox shall duplicate the overhung moments of table I.

VERIFICATION RATIONALE (A.4.12.3.2.2)

The test article interfaces and environment must be representative of the installed environment.

VERIFICATION GUIDANCE (A.4.12.3.2.2)

Interfaces should conform to the installation drawing and when required, accessory pads should be subjected to rated loads and overhung moments as specified in table I. The actual torque loading and overhung moments imposed during the endurance test should be stated in the test report. During the test, the exhaust duct, intake duct, and bleed air duct connections should be loaded as specified in 3.1.1.3. Also see JSSG-2001 Engine compatibility and installation verification.

VERIFICATION LESSONS LEARNED (A.4.12.3.2.2)

Extensive lessons learned databases have been developed and may be referenced directly through systems program offices at NAVAIR, USAF, and AMCOM.

A.4.12.3.3 Test equipment.

A.4.12.3.3.1 Automatic recording equipment.

Automatic steady-state and transient recording equipment shall be used to record data during the execution of those parts of the tests required to meet the objectives of each verification. Automatic continuous recording equipment shall be used for tests requiring the verification of test article variables versus time.

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VERIFICATION RATIONALE (A.4.12.3.3.1)

It is necessary to use automatic continuous recording equipment when a measured parameter changes with time. This is to verify transient operability requirements. Steady-state data is needed for calculations to verify test article performance requirements.

VERIFICATION GUIDANCE (A.4.12.3.3.1)

Background:

The word continuous should not be taken literally. Only analog equipment monitors continually. Digital equipment is considered continuous when it uses a high sampling rate. The sampling rate is determined by the type of test. A very high sampling rate may pick up extraneous measurements while another test may require a very high rate or electrical scanning circuitry.

VERIFICATION LESSONS LEARNED (A.4.12.3.3.1)

Most automatic recording equipment used presently is digital equipment. This equipment works with a scanning valve or electrical, scanning circuitry. More parameters result in a smaller sampling rate or longer time between consecutive samples. This method is fine during an engine acceleration that may take four or more seconds. A test that may measure engine parameters during an engine surge will require a very high sampling rate. This reduces the number of parameters that can be recorded.

A.4.12.3.3.2 Vibration measuring equipment and response characteristics.

The vibration measurement and analysis equipment shall operate over a frequency band of at least _(a)_ to _(a)_ and produce acceleration spectrograms with a confidence level of _(b)_.

VERIFICATION RATIONALE (A.4.12.3.3.2)

Engine vibration measurement is one of the necessary parameters used to monitor the health of the engine. Response characteristics are needed to control the type of measurement equipment used.

VERIFICATION GUIDANCE (A.4.12.3.3.2)

The following should be used to tailor the specification paragraph:

(a): 3 Hz – 10 kHz

(b): 95%

For turbojet and turbofan engines the maximum allowable effective filter bandwidth of the spectrum analysis equipment should be 30 Hz. For turboshaft and turboprop engines, the maximum allowable effective filter bandwidth of the spectrum analysis equipment should be 3 Hz up to 1,000 Hz and 6 Hz above 1,000 Hz. The equipment should be calibrated as a complete system. The system frequency response, when calibrated by applying a known sinusoidal motion to the pickup, should not deviate by more than ±3 dB from the known sinusoidal input at frequencies from 3 Hz to 10 kHz. If high bypass filters are required when measuring overall velocity levels, the filters should not exceed 3 dB down at frequencies of 30, 70, or 110 Hz, as appropriate, with a roll-off of at least 18 dB per octave. High pass filters should not be used to produce velocity and acceleration spectrograms.

Background:

A sudden increase in the level of vibration can signal an existing or an impending engine failure. A gradual increase in vibration level may mean parts wearing.

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The different filter characteristics for each engine class assumes turbofan and turbojet engines have larger physical core sizes and lower rotational speeds than turboshaft or turboprop engines.

VERIFICATION LESSONS LEARNED (A.4.12.3.3.2)

Monitoring the vibration level of engines during a test has saved many engines from catastrophic failure.

A.4.12.3.3.3 Tests stand dynamic characteristics.

The test stand shall not induce damaging or detrimental resonances into the engine at any test or operating condition.

VERIFICATION RATIONALE (A.4.12.3.3.3)

Test stand natural frequencies must not damage the engine or adversely affect the test instrumentation accuracy.

VERIFICATION GUIDANCE (A.4.12.3.3.3)

The test stand should have enough mass to ensure that the test stand natural frequency is not within the starting to maximum rotor speed range. If the engine has multiple rotors, all rotors should be considered when analyzing the test stand characteristics.

VERIFICATION LESSONS LEARNED (A.4.12.3.3.3)

Rotor unbalance can be excited by a test stand where the natural frequency is in the rotor speed range. This has caused problems during the test.

A.4.12.3.3.4 Starter torque characteristic.

Starting shall be performed with a starter that has torque characteristics within ____ percent of the minimum required torque shown on figure 2.

VERIFICATION RATIONALE (A.4.12.3.3.4)

This requirement determines the maximum starter size used in testing by specifying its torque characteristics. A starter with “below specification” performance can bias test results.

VERIFICATION GUIDANCE (A.4.12.3.3.4)

The following should be used to the tailor the specification paragraph:

The contractor should provide a value between 0-5 percent. A low value is preferred.

A starter with more than 5% above minimum torque characteristics also biases the test results.

VERIFICATION LESSONS LEARNED (A.4.12.3.3.4)

To prevent field starting problems it is necessary to ensure that the engine starts with the specified minimum starting torque characteristics.

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A.4.12.3.4 Test conditions.

A.4.12.3.4.1 Oil servicing and consumption.

The oil system shall be serviced at the start of each test or demonstration. All additions of oil added after initial servicing shall be recorded. Oil shall be drained only when authorized by the Using Service. The use of external oil filters shall not be permitted. Oil consumption shall be measured and the average consumption rate determined.

VERIFICATION RATIONALE (A.4.12.3.4.1)

This requirement is needed to control the test conditions when servicing the engine oil system. Addition of oil is recorded for oil consumption determination. It is important to control the conditions for servicing the oil in the engine and the fluid that is used. Each test should start with new oil since oil is monitored during the engine test to determine test article deterioration.

VERIFICATION GUIDANCE (A.4.12.3.4.1)

The following should be used to tailor the specification paragraph:

Oil consumption should be determined and reported after a specific number of cycles or hours. Values that have been used are 60 to 75 hours. Oil should be added on an as required basis as indicated by the oil tank sight glass or low oil sensor. Oil drained for analysis should not be charged to engine oil consumption and should be replaced by an equivalent amount of new oil. JOAP samples should be taken at intervals consistent with the contractors recommended maintenance interval.

VERIFICATION LESSONS LEARNED (A.4.12.3.4.1)

Oil samples for an oil spectrum analysis are used to determine condition of engine parts that are exposed to the oil. Using clean new oil at the start of the test provides a base line for the analysis. Starting a test with used oil can bias results because used oil may not meet the specification set forth for oil characteristics. Past experience has shown that engine oil can be a source of concern with test article deterioration.

A.4.12.3.4.2 Fluid properties for test.

A.4.12.3.4.2.1 Fuel properties.

Each test and demonstration shall be conducted using a primary or alternate fuels specified in 3.7.3.1.1 and 3.7.3.1.2, unless otherwise specified. If more than one primary fuel is specified, the fuel used shall be the fuel, which makes the test the most difficult as determined by the Using Service. As directed by the Using Service and consistent with the model specification, testing will be repeated with other fuels. Fuel samples shall be taken at the start and completion of Verification and Qualification tests, and other tests as applicable. The fuel samples shall be analyzed for physical and chemical properties to determine conformance with applicable specifications. The results of these analyses shall be included as part of the applicable reports.

VERIFICATION RATIONALE (A.4.12.3.4.2.1)

Fuel properties must be determined because significant batch to batch property variations can exist between fuels procured to the same specification. Their properties directly affect the results of the test. Fuel properties are required because certain fuel properties are used during testing (e.g.; heating value, density, viscosity) to calculate certain real time engine performance parameters (e.g.; fuel flows, specific fuel consumption, etc.) and because fuel properties can directly affect engine durability. Testing with a fuel having properties which exceeds the specification requirements can lead to erroneous results since these test results may not accurately depict real field performance when operating with fuel that

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marginally meets the specification. Knowing the fuel's properties will allow a better assessment of test results and their correlation with expected worst-case field behavior.

VERIFICATION GUIDANCE (A.4.12.3.4.2.1)

Fuel properties may require adjustment to achieve test or demonstration goals.

VERIFICATION LESSONS LEARNED (A.4.12.3.4.2.1)

JP-4 has a lower flashpoint than JP-5 and JP-8. JP-5 and JP-8 are more viscous than JP-4 at the same temperature. For starting and windmill relights, JP-4 has given better results than JP-5 and JP-8. These results are affected by fuel system design such as pressure, nozzle configuration, and fuel flow.

Low lubricity fuel has caused problems with fuel controls and pumps, yet the fuel was within the specification limits. Fuels high in sulfur content have caused problems with certain alloys and salt air. For emergency fuel test using NATO Code No. F-54 DF-2, sulfur content has been adjusted on prior tests to reflect high field levels experienced.

A.4.12.3.4.2.2 Oil properties.

Each test and demonstration shall be conducted using the oils specified in 3.7.8.1. If more than one oil is specified, the oil used will be the oil that makes the test most difficult as determined by the Using Service. Oil samples shall be taken at the start and completion of Verification and Qualification tests and other tests as applicable. The oil samples shall be analyzed for physical and chemical properties to determine conformance with applicable specifications. The results of these analyses shall be included as part of the applicable reports.

VERIFICATION RATIONALE (A.4.12.3.4.2.2)

Oil properties must be determined since they can directly affect the test results. The oil viscosity must be measured since it will directly effect the starting characteristics of the engine.

VERIFICATION GUIDANCE (A.4.12.3.4.2.2)

The type of oil will determine the extreme temperature at which the engine can start and operate.

MIL-PRF-23699 is a 5 cSt (54 ft^2/s) oil (at 100°C) and approximately 13,000 cSt (139,931 ft^2/s) at -40°C. Past requirements have been that the engine be able to start when the oil is 13,000 cSt (139,931 ft^2/s). MIL-PRF-7808, which is classified as a 3 cSt (32 ft^2/s) oil, reaches 13,000 cSt (139,931 ft^2/s) at approximately - 54°C (-65°F). So the starting temperature of an engine is dependent on the viscosity of the oil being used.

VERIFICATION LESSONS LEARNED (A.4.12.3.4.2.2)

MIL-PRF-23699 oil has been tested with JP-5 fuel and MIL-PRF-7808 has been used with JP-4 fuel; this is because of the oil viscosity properties at various operating temperatures. JP-4 and MIL-PRF-7808 are matched to give better starting capabilities at lower temperatures. Some oils that meet QPL have much higher temperature capabilities and test results may be misleading. The USAF has switched from the use of JP-4 to JP-8 along with MIL-PRF-7808 and no problems have been reported.

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A.4.12.3.4.2.3 Hydraulic fluid properties.

Each test and demonstration shall be conducted using the hydraulic fluids specified in 3.7.9. Hydraulic fluid samples shall be taken at the start and completion of Verification and Qualification tests and other tests as applicable. The fluid samples shall be analyzed for physical and chemical properties to determine conformance with applicable specifications. The results of these analyses shall be included as part of applicable [test] reports.

VERIFICATION RATIONALE (A.4.12.3.4.2.3)

Physical and chemical properties of hydraulic fluid must be determined since their properties directly affect the results of the test.

VERIFICATION GUIDANCE (A.4.12.3.4.2.3)

None.

VERIFICATION LESSONS LEARNED (A.4.12.3.4.2.3)

Aircraft A (Engine A) utilizes a hydraulic actuation system for its nozzle. Lessons learned on this system can be obtained from the USAF propulsion program office.

A.4.12.3.5 Test calibrations and procedures.

A.4.12.3.5.1 Calibration.

A.4.12.3.5.1.1 Control and engine monitoring system calibration.

Prior to the initiation of the engine calibration all components necessary for proper control and engine monitoring systems operation and scheduling shall be checked on the bench and on the engine to establish proper functioning and calibration over the applicable ranges of engine operating conditions.

VERIFICATION RATIONALE (A.4.12.3.5.1.1)

Sensing system(s) require a calibration to ascertain accuracies before attempting to use these systems prior to test.

Fuel system calibration should be conducted to insure the engine fuel system is supplying the required amount of fuel at the required pressures for operation throughout the environmental conditions and operating envelope of the engine. This includes both starting and augmentation operation.

VERIFICATION GUIDANCE (A.4.12.3.5.1.1)

The following should be used to tailor the specification paragraph:

If thermocouples are used, the relationship between temperatures and output signal should be per a DoD approved standard. Previous programs used NBS Monograph 125. The accuracy of the signals in relation to the actual measured temperature and transient time response characteristics should be specified. All fuel nozzles and fuel carrying components of the engine control system should undergo bench calibrations.

Background:

Thermocouples are presently the most common temperature sensing devices. They can be calibrated by either using the ice bath (thermistor (RTD) integrated circuit sensor) or the cold junction compensation (using thermistors) method of calibration.

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The thermocouple designations E, J, T, and K all refer to specific calibration curves with established limits of error but do not refer to the specific wire alloy used. Any metal or alloy that fits the curve within the specified limits of error is classified according to that curve. The types E and T thermocouples are useful for low temperature measurement, while type K is suited for high temperature measurement.

The following fuel flow calibration limits may be used:

(1) Acceptance Limit - Criterion of acceptance at bench test after calibration of any new or overhauled component. This criterion should apply before shipment from the component manufacturer or overhaul facility.

(2) Audit Limit - Criterion of acceptance at any bench test run after the vendor or overhaul facility acceptance and before any significant engine test hours have been accumulated. It can also be the criterion of acceptance at bench test following any engine milestone tests.

(3) Ultimate Service Limit - The extreme limit of component performance beyond which satisfactory engine operation cannot be assured. It is the criterion of acceptance at bench test following Qualification Test (QT) simulated operational component tests or equivalent component tests.

The acceptance criteria must be determined by test on new or overhauled components.

VERIFICATION LESSONS LEARNED (A.4.12.3.5.1.1)

None.

A.4.12.3.5.1.2 Engine calibration.

The engine calibration shall be accomplished to establish the performance characteristics of the complete engine. Prior to the beginning of the calibration the engine may be cleaned using the wash procedure specified in 3.7.12 and engine controls may be adjusted but shall not be readjusted throughout the calibration. Calibrations shall be made initially with no customer power extraction and no bleed air extraction other than that required for continuous engine operation. Data required in 4.12.4.1.3 and 4.12.4.1.4 shall be recorded. During calibration, conformance with the leakage and shutdown drainage requirement of 3.1.1.8 shall be demonstrated. The engine shall be calibrated using fuel per 3.7.3.1.1 and oil per 3.7.8.1. The following shall be obtained:

a. Data to establish compliance with 3.2.1 and sea level performance of (a). The use of outside ambient temperature (OAT), to calibrate the engine in lieu of (a) conditions is acceptable with Using Service approval. (b) power settings shall be included in the performance calibration. For turboprop and turboshaft engines, sufficient data to establish the accuracy of the torque sensor signals shall be recorded.

b. Data to establish compliance with 3.2.2.6, 3.2.2.7 and 3.2.2.3 requirements at sea level static conditions. All transient performance data shall be used to update the transient status performance computer program of 3.2.1.2 and 3.2.1.3.

c. Repeat items “a” and “b” conditions with maximum permissible bleed air extraction for that condition. The maximum permissible bleed air extraction includes customer and anti-icing bleed airflow, if applicable.

d. Repeat item “c” with customer or accessory power extraction as specified by (a), except for starting.

e. Data required characterizing engine performance as a function of time as defined by the Using Service.

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f. Data showing adequate distorted stall or surge margin remaining with (c) distortion pattern (screen) selected by the Using Service. In addition, data establishing the distorted stall or surge margin shall be determined at (d) engine thrust setting selected by the Using Service.

Additionally, for turboprop engines, the following shall be obtained with the engine fitted with the power-absorbing device specified in the specification:

g. Data to establish compliance with 3.7.15.1 and load and no load operation.

VERIFICATION RATIONALE (A.4.12.3.5.1.2)

The engine must be calibrated to establish a baseline of performance characteristics. The engine must be fitted with the intended power absorber or equivalent and operated throughout its operating modes to establish the operability of the propulsion system.

VERIFICATION GUIDANCE (A.4.12.3.5.1.2)

The following should be used to tailor the specification paragraph:

(a): The blank should be completed with “tables IIa, b, and c”, or “the computer performance program”. Whichever is specified in this paragraph should be consistent with what is specified in 3.2.1.1.

(b): The contractor should provide the number of power settings. The power settings should include maximum and minimum augmentation, intermediate power and a sufficient number of part power settings to verify any shifts in component efficiencies during development, including idle, to accurately characterize the performance for turbojet and turbofans.

For engines having short period ratings above intermediate power (example turboprop and turboshaft, or V/STOL applications) those power settings should also be tested. Examples are contingency, maximum or combat. Thrust augmentation by other means such as water injection should also be tested as applicable.

(c): The contractor should provide the number of screens. One to two screens are recommended.

(d): The contractor should provide the number of thrust or power settings. One to three thrust settings are recommended, especially at intermediate power or maximum continuous, and a few part power settings where the surge margin “pinch point” may occur.

For sub-paragraph “e”, the Using Service should provide guidance on whether the data should be collected as a function of hours, TAC’s, or some other parameter or format. This data is useful when establishing maintenance inspections or inspection intervals. Tables and curves derived from this data have been put into maintenance manuals. The data is also useful in case the engine fails or repairs are made which negate performance deterioration up to that point. The last good performance calibration can be used as a basis to establish the performance at the time of the failure.

Background:

The calibration will probably take place at sea level using non-conditioned air (ambient pressure and temperature) at the contractor’s facility. If an altitude facility is used, then several flight conditions could be tested instead of just one sea level static point. Having conditioned air available at altitude test cell facility offers more test flexibility and better baseline data can be obtained.

For turboprop engines, the manual feather and reverse pitch calibration should be done with the specification power absorber (propeller). If that propeller has not been specified or is not available, a different power absorption mechanism, such as a club prop should be used. The full range of propeller travel from feathering to reverse pitch should be tested.

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Stall and Surge margin check may be accomplished with the use of Bodie Runs. This check should be performed on tests deemed appropriate by the Using Service.

Bodie Run. This run should consist of 180 minutes of thrust transients as follows:

Twelve periods of four minutes at idle thrust position and twelve periods of 30 +3 seconds at the maximum thrust position for a total of 54 minutes.

Ten periods, each consisting of three minutes of operation, making a total of 30 minutes. Each period should be run as follows: with the thrust or power demand in the idle thrust position, advance toward intermediate thrust, and as soon as the engine reaches the condition of minimum surge margin (i.e., rpm or gas temperature), decrease the thrust or power demand to the idle thrust position.

Ten periods, each consisting of three minutes of operation with the final period followed by an additional two minutes at idle making a total of 32 minutes. Each period should be run as follows: with the thrust or power demand in idle thrust position, advance toward maximum thrust, and as soon as the engine reaches the condition of minimum surge margin (i.e., rpm or gas temperature), decrease the thrust or power demand to the idle thrust position. (For augmenting engines, as soon as the engine reaches the condition of minimum surge margin after lighting the augmentor, decrease the thrust or power demand to the idle thrust position.) (If the engine does not have a maximum rating, intermediate thrust should be substituted for maximum thrust.)

Ten periods, each consisting of three minutes of operation with the final period followed by an additional two minutes at maximum thrust making a total of 32 minutes. Each period should be run as follows: with the thrust or power demand in the maximum thrust position, retard toward idle, as soon as the engine reaches the condition of minimum surge margin (i.e., rpm or gas temperature), increase the thrust or power demand to the maximum thrust position. (For augmenting engines, as soon as the engine reaches the condition of minimum surge margin after lighting the augmentor, increase the thrust or power demand to the maximum thrust position.) (If the engine does not have a maximum thrust rating, intermediate thrust should be substituted for maximum thrust.)

Ten periods, each consisting of three minutes of operation with the final period followed by an additional two minutes at intermediate thrust making a total of 32 minutes. Each period should be run as follows: with the thrust or power demand in the intermediate thrust position, retard toward the idle thrust position, and as soon as the engine reaches the condition of minimum surge margin (i.e., rpm or gas temperature), increase the thrust or power demand to the intermediate thrust position.

VERIFICATION LESSONS LEARNED (A.4.12.3.5.1.2)

These Bodie tests in the calibration phase are used primarily to see if the engine is a “staller” before other tests, i.e. durability and altitude tests are run. Bodie runs were used during surge and stall margin testing in performance verification testing of the Engine J and during exploratory altitude development and qualification testing for the Engine S.

A.4.12.3.5.1.2.1 Customer bleed air analysis calibration.

Prior to and during the _____ test, a customer bleed air analysis shall be performed. The customer bleed air shall be sampled from each bleed air outlet during an intermediate or maximum continuous power run. A sample of air entering the engine inlet shall be taken at the same time the bleed air samples are obtained. The samples shall be properly identified and processed through laboratory analysis to determine whether the contaminant levels are within the limits specified in 3.1.1.7.1.

VERIFICATION RATIONALE (A.4.12.3.5.1.2.1)

See 4.1.1.7.1, Customer bleed air contamination.

VERIFICATION GUIDANCE (A.4.12.3.5.1.2.1)

The following should be used to the tailor the specification paragraph:

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The contractor should specify which test will be used; i.e., altitude or endurance.

Background:

The Using Service should be aware that engine cleaning compounds could be a substance that could cause bleed air contamination. Analysis should be completed to detect its presence.

The test should be conducted at power settings that will generate the highest levels of contaminants. Results, methods, and apparatus should be detailed in the test report.

Acrolein is difficult to detect during the customer bleed air analysis. The Using Service should request that the engine contractor provide the method(s) used to detect this substance.

VERIFICATION LESSONS LEARNED (A.4.12.3.5.1.2.1)

None.

A.4.12.3.5.2 Inspection, maintenance and calibration.

The following inspections, maintenance, and calibrations shall be performed:

a. Using established procedures, inspections shall be performed at intervals equivalent to the inspection periods specified in 3.4.1.7.3. Inspections and partial disassembles at other times during the test are permitted when warranted by performance deterioration and after approval from the Using Service is obtained.

b. Prior to and following each inspection interval, an abbreviated engine calibration may be performed. If performance deterioration exceeds 3.2.1.4, the engine may be cleaned. If cleaning is necessary, the washing system and procedure specified in 3.7.12 shall be used prior to initiation of the next test segment.

c. Maintenance shall be performed on a basis equivalent to the maintenance procedures specified in 3.5.2.3. Parts replaced during the test, except for consumables defined by maintenance instructions, shall not be permitted without approval of the Using Service. A failure analysis shall be performed for any failures resulting in the replacement or repair of non-consumable parts. Non-consumable parts replaced during the test shall require retest equivalent to the severity and test hours of the full engine test.

VERIFICATION RATIONALE (A.4.12.3.5.2)

Inspections are necessary to ensure no imminent failures are present. Maintenance should be done per the established procedures to ensure the proposed reliability and maintainability projections are met.

VERIFICATION GUIDANCE (A.4.12.3.5.2)

A record of all consumable and non-consumable parts replaced during the test should be maintained by the contractor.

VERIFICATION LESSONS LEARNED (A.4.12.3.5.2)

None.

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A.4.12.3.5.3 Recalibration.

After completion of the tests specified in 4.11.1, a recalibration per the requirements of 4.12.3.5.1.2 shall be made on the engine. The recalibration may be preceded by a run during which the cleaning procedure of 3.7.12 may be used. The fuel and oil used shall be the same as those used during the initial calibration.

After completion of the engine recalibration, all components necessary for proper control and engine monitoring systems operation and scheduling shall be rechecked to establish proper functioning and calibration per 4.12.3.5.1.1.

The performance of these components shall meet their design service limits.

VERIFICATION RATIONALE (A.4.12.3.5.3)

See Rationale for 4.12.3.5.1.1 through 4.12.3.5.1.2.

VERIFICATION GUIDANCE (A.4.12.3.5.3)

Background:

See Guidance for 4.12.3.5.1.1 and 4.12.3.5.1.2.

VERIFICATION LESSONS LEARNED (A.4.12.3.5.3)

None.

A.4.12.3.5.4 Engine disassembly and inspection.

For tests requiring engine disassembly the Using Service shall be notified of the inspection commencement date prior to each inspection. Prior to cleaning, the engine parts shall be given a “dirty” inspection for abnormal conditions. The dirty inspection shall be completed before any parts are cleaned. Parts shall then be cleaned and a “clean” inspection shall be performed. Part measurements shall be taken, as necessary, to determine excessive wear and distortion. Those measurements shall be compared with the manufacturer’s drawing dimensions and tolerances and with similar measurements made prior to the test. Non-Destructive Inspection (NDI) techniques shall be per 3.4.1.7.2. During the clean inspection a visual examination and condition assessment shall be conducted. The contractor shall maintain all data and results of non-destructive inspection tests. All NDI inspection data and findings shall be made available for Using Service review during the inspections. All data of 4.12.4 shall be made available for Using Service review upon request.

As a result of the inspection, parts shall be categorized as follow:

Cat. A. No repair required.

Cat. B. Repair required before further use.

Cat. C. Outside repair limits.

VERIFICATION RATIONALE (A.4.12.3.5.4)

All engine parts must be inspected to assure durability and safety requirements of the specification are met. Engine test data must be available for review during inspection to aid in diagnosing any inspection findings.

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VERIFICATION GUIDANCE (A.4.12.3.5.4)

Background:

The Using Service should ensure that the contract requires at least 21 days notification of a scheduled inspection.

An example of some of the data the Using Service may want to review during the inspection is listed below.

a. Inspection forms filled out by the contractor listing all deficiencies and NDI data and findings.
b. Tabulation of all parts not meeting success criteria and reasons for failure.
c. The detailed configuration list of 4.12.3.2.
d. Test summary, test logs, and list of test events.
e. Spectrometric oil analysis results.
f. Chip detector or oil debris analysis and results.
g. Tabulation of all parts which required replacement prior to the test completion.

VERIFICATION LESSONS LEARNED (A.4.12.3.5.4)

Advanced NDI techniques have revealed fatigue and stress cracks that normally may have gone undetected. It is common for NDI inspections to detect flaws consisting of microcracks that could propagate.

A.4.12.4 Data and reporting.

Data and reporting for tests conducted in this specification shall be per table XXXIII and as required below.

VERIFICATION RATIONALE (A.4.12.4)

Proper data and information must be recorded for the Using Service to verify requirements. There must be common understanding between the contractor and the government on data requirements that will result in productive testing, improved first time data submission acceptability, and reduced data expenditures.

VERIFICATION GUIDANCE (A.4.12.4)

Table XXXIII contains guidance on test reporting. See American National Standard developed by the National Information Standards Organization and approved by the American National Standards Institute, ANSI Z39.18-1995 for alternate guidance. Confer with the Using Service to determine which report format is acceptable.

VERIFICATION LESSONS LEARNED (A.4.12.4)

None.

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A.4.12.4.1 Types of data.

A.4.12.4.1.1 Pretest data.

Pretest data shall be made available to the Using Service for approval prior to the initiation of each test or demonstration. The data shall include:

a. Detailed test and maintenance plans. The detailed test and maintenance plans to be used by test and maintenance personnel during the conduct of the test. Test procedures shall be per the test plan.

b. Service and wear limits. Service limits to be used by maintenance personnel in the field and repair facilities.

c. The parts list and configuration identification of the engine, component or assembly and justification for any differences per 4.12.3.2.

VERIFICATION RATIONALE (A.4.12.4.1.1)

Before the test, there is a need to require a test procedure and test article configuration. Standardized test plans and article configurations are needed to ensure adequate testing on appropriate test articles.

VERIFICATION GUIDANCE (A.4.12.4.1.1)

The test and maintenance plans should be reviewed to determine if the test will meet the specification requirements. The parts list and configuration must be reviewed to determine if the test will be conducted on the test article described in the model specification. The Using Service should not accept contractor justification of parts substitution without a thorough analysis.

VERIFICATION LESSONS LEARNED (A.4.12.4.1.1)

Contractors have substituted parts during qualification or verification tests. Tests have been performed prior to test plan approval causing test repetition to address all requirements. Turbine blade rainbow type tests should not be attempted during qualification.

A.4.12.4.1.2 Preliminary data.

Photographs and other pertinent data shall be obtained and recorded at the time the test article is prepared for the test. For engines, dry mass (weight), as specified in 3.1.2.1 and center of gravity as specified in 3.1.2.2 shall be provided by the contractor.

VERIFICATION RATIONALE (A.4.12.4.1.2)

A record of the data is needed to track the progress of the test article development though each milestone.

VERIFICATION GUIDANCE (A.4.12.4.1.2)

Preliminary data should be submitted as part of the test reports.

VERIFICATION LESSONS LEARNED (A.4.12.4.1.2)

None.

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A.4.12.4.1.3 Data - steady-state.

During operation at each specified steady-state condition and after performance stabilization, as a minimum the data specified in table XXXIV shall be recorded.

VERIFICATION RATIONALE (A.4.12.4.1.3)

The engine must reach a fully stabilized condition before steady-state data is recorded.

VERIFICATION GUIDANCE (A.4.12.4.1.3)

The following should be used to tailor the specification paragraph:

The contractor should fill in the table.

Background:

Steady-state data can only be taken after performance stabilization, which can take as long as ten minutes of engine operation.

VERIFICATION LESSONS LEARNED (A.4.12.4.1.3)

The stabilization time periods have been dependent on the size of the engine and the magnitude of rotor speed change. In the past, data was recorded before stabilization that resulted in faulty performance determination.

A.4.12.4.1.4 Data - transient.

As a minimum, the data specified in table XXXIV shall be recorded.

VERIFICATION RATIONALE (A.4.12.4.1.4)

Transient data is needed to verify transient operability and performance of the engine. Transient performance response also affects engine life.

VERIFICATION GUIDANCE (A.4.12.4.1.4)

The following should be used to tailor the specification paragraph:

The contractor should fill in the table.

Background:

It is better to obtain more than one sample of transient data since the accuracy is usually less than steady-state data. Transient runs should be repeated at selected conditions to investigate control system lags, repeatability, and performance droop.

VERIFICATION LESSONS LEARNED (A.4.12.4.1.4)

As the name implies, transient data is a record of the dynamic response of a given change in an engine performance parameter. Since it is a dynamic response, it does not always repeat exactly each time. Therefore, several samples may be needed to determine the average response.

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A.4.12.4.1.5 Starting data.

During each start, to the extent defined in the individual test verification requirement derived from table XXXIV, data shall be recorded to verify the requirements of 3.1.1.11.1, 3.2.2.3.3, and 3.2.2.3, as a minimum.

VERIFICATION RATIONALE (A.4.12.4.1.5)

Starting data is recorded to keep a history of the engines health. This requirement specifies when and what starting data should be recorded.

VERIFICATION GUIDANCE (A.4.12.4.1.5)

The maximum turbine temperature and time to reach idle speed should be recorded during the start. Include time, temperature and rotor speeds vs. starting event (i.e., initiation of start, starter cutout, stabilized engine speed). Note flat spots on acceleration curves.

VERIFICATION LESSONS LEARNED (A.4.12.4.1.5)

Any increase in turbine temperature or time to reach idle speed indicates a change in engine characteristics with consistent starter and drivetrain performance. Comparisons should be done at the same inlet conditions. Colder inlet temperature will require higher turbine temperature and longer time to reach idle speed and could cause a hot start.

A.4.12.4.1.6 Creditable test time.

Total test time for a period less than fifteen minutes shall not be credited to the official run time, except when shorter periods are a test requirement.

VERIFICATION RATIONALE (A.4.12.4.1.6)

This requirement prevents the engine contractor from counting minimal test periods as creditable test time that can result in reduced test fidelity to the requirements.

VERIFICATION GUIDANCE (A.4.12.4.1.6)

Background:

Most required test runs last longer than 15 minutes. Tests should be credited for the exact time of run. For tests started and stopped in less than 15 minutes for reasons other than required by the test plan, no time credit should be given.

VERIFICATION LESSONS LEARNED (A.4.12.4.1.6)

In past engine tests, contractors have added many periods of less than 15 minutes run time to accumulate significant number of total test hours. These test hours do not represent a significant accomplishment in the test plan.

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A.4.12.4.1.7 Miscellaneous data and notes.

The date, test title, test article designation, and serial number shall be recorded on each page of the log sheet. Notes shall be added to the log sheets to record all incidents and irregular functioning of the test article and equipment and the corrective measures taken. All test article software changes shall be recorded on the log sheets.

VERIFICATION RATIONALE (A.4.12.4.1.7)

There is a need to identify each page of the log with all the pertinent data that will identify a particular test and test article. Test and test cell anomalies need to be recorded and reviewed to assure test conduct complies with test intent.

VERIFICATION GUIDANCE (A.4.12.4.1.7)

Background:

Most log sheets are preprinted. They do not allow for any unusual incidence. The test notes are the place to record any irregular engine problems (such as leaks, vibrations, and “flat spots” in acceleration) or facility problems.

VERIFICATION LESSONS LEARNED (A.4.12.4.1.7)

Past engine test records have had missing pages, because there was no identification to show where to reinsert any unattached pages. Test notes record anomalies and servicing, maintenance, adjustments, etc. which all must be reviewed to verify test compliance with the test plan.

A.4.12.4.2 Accuracy of data.

For all test article calibrations and tests or demonstrations, the reported data shall have steady-state accuracy within the tolerances specified herein. The accuracy of transient data and the corresponding instrument calibration methods shall be described in the test plans. No credit or debit shall be permitted for measurement uncertainty to establish compliance with the requirements of this specification. The instrument calibration methods shall be subject to Using Service approval prior to the initiation of any testing. The “as tested” accuracies as defined by the test facility shall be included in the test report.

VERIFICATION RATIONALE (A.4.12.4.2)

Accuracy of data is a factor in the interpretation of test data, and determination of requirement compliance. Obtaining accurately measured parameters is an important step toward quantifying derived results such as specific fuel consumption, gross thrust, and fan efficiency.

VERIFICATION GUIDANCE (A.4.12.4.2)

The contractor should insert the accuracies in the specification, test plans and reports.

The following values should be considered when tailoring the requirement:

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DATA ITEM	TOLERANCE
Rotor speed(s)	±0.2 percent of the value obtained at maximum
Metered thrust or power	±0.5 percent of the value measured - for minimum augmented and above ±0.5 percent of the value measured at intermediate - for intermediate and below
Fuel Flow	±0.5 percent of the value measured - for intermediate and above ±0.5 percent of the value measured at intermediate - for all values below intermediate
Torque	±0.5 percent of the value measured - for intermediate and above ±0.5 percent of the value measured at intermediate - for all values below intermediate
Airflow	±1.0 percent of the value measured for intermediate and above ±1.0 percent of the value measured at intermediate - for all values below intermediate
Temperatures	±1.0° up to 200°C ±3.0° between 200°C and 800°C ±4.0°C above 800°C
Engine Weight	±1.0 lb. or ±0.1 percent of the weight being determined, whichever is greater
Pressure	±0.5 to ±1.0 percent of the value measured at intermediate thrust or power
Vibration Velocity	±5.0 percent of specified engine limit during 4.4.1.8 vibration survey, vibration scan, resonant search, and vibration & stress test. ±10.0 percent of specified engine limit for all other tests
All other data	±2.0 percent of the value obtained at maximum thrust or power

Background:

The Using Service should use the accuracy characteristics of those instrumentation and data systems used in performing the qualification tests. Steady-state accuracy is easier to control than transient accuracy.

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VERIFICATION LESSONS LEARNED (A.4.12.4.2)

The following accuracies have been used on commercial engine development testing:

Item of Data	Uncertainty (U95)
Rotor Speeds (rpm)	± 0.1% of reading
Metered Thrust (lbs)	± 0.2% of reading
Fuel Flow (Volumetric - gph)	± 0.25% of reading down to 20% of engine maximum flow rate
Fuel Density	± 0.1% of reading
Fuel Lower Heating Value	± 0.2% of reading
Engine Air Inlet Temperature	± 0.5C
Fuel Temperature at Flowmeters	± 0.5C
All Other Temperatures	± 1.0C for readings up to 200°C ± 3.0C for readings between 200°C and 800°C ± 4.0C for readings above 800°C
Air Pressures (psia)	± 0.015 psi for readings 5 to 20 psia ± 0.25% of reading for readings >20 psia
Wet Pressures (psig)	± 0.5% of reading
Barometric Pressure (psia)	± 0.02% of reading
Engine Vibration (ips rms)	± 10% of engine limit

A.4.12.4.2.1 Measurement uncertainty analysis.

A detailed measurement uncertainty analysis addressing all elemental random and systematic errors shall be performed per: _____.

As a minimum, the analysis shall cover the three categories of errors: calibration errors, data acquisition errors, and data reduction errors. Analysis will be performed for the following measured parameters: airflow, fuel flow, thrust, torque, temperature, pressure, rotor speeds (i.e., Ng & Np), SFC, output power, engine weight, and vibration.

VERIFICATION RATIONALE (A.4.12.4.2.1)

The estimation of uncertainty is an essential requirement for all measurements. Without such an estimate, interpretation of the measurement may be difficult or impossible. Estimates of uncertainty can strongly influence the methods used to test engines, components, and the interpretation of results.

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VERIFICATION GUIDANCE (A.4.12.4.2.1)

The following should be used to tailor the specification paragraph:

AEDC-TR-73-5 or ISO 10012-1 should be used for guidance.

Background:

Methods, procedures, and documentation that substantiates the uncertainties in measuring and calculating airflow, net thrust, and thrust specific fuel consumption during steady-state testing conditions should be provided. Analysis information should include, but not limited to, the following:

a. All basic measurements and calculated parameters based on these measurements should have the uncertainty reported as combination of random index and system as specified in the handbook. All three values, uncertainty, random index, and system should accompany each measurement and calculated parameter in the uncertainty documentation.

b. A hierarchy of uncertainty should be provided for all the pressure transducers, thermocouples, flowmeters, load cells, and areas, or whatever equipment is used to calibrate or measure the performance parameters used in the calculation of airflow, net thrust, and thrust specific fuel consumption. The hierarchy should trace the uncertainty from the in-place-measuring device back to the primary measurement standards traceable to the National Institute of Standards and Technology.

c. The set of performance calculation equations used to calculate airflow, net thrust, and thrust specific fuel consumption should be provided. The partial derivative functions determined from the performance equations should also be given.

VERIFICATION LESSONS LEARNED (A.4.12.4.2.1)

None.

A.4.12.4.3 Atmospheric data.

A.4.12.4.3.1 Barometer reading and relative humidity.

The barometric reading shall be corrected for temperature and shall be recorded at intervals of _(a)_ hours. Relative humidity data readings shall be recorded at intervals of _(b)_ if required for performance calculations. The relative humidity shall be measured in the vicinity of the test article.

VERIFICATION RATIONALE (A.4.12.4.3.1)

All the engines gas pressures are dependent on the barometric pressure. Relative humidity affects quality of the air that is used by the engine and can affect engine performance.

VERIFICATION GUIDANCE (A.4.12.4.3.1)

The following should be used to tailor the specification paragraph:

(a): The contractor should provide the time interval, but it should not exceed three hours.

(b): A wet and dry thermometer or a sling psychrometer should be used for these humidity measurements.

Background:

The barometric pressure should be measured in the vicinity of the test and its accuracy should be comparable to that used for measuring gas pressure in the engine. The standard should be a mercury barometer with all its corrections.

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Relative humidity data should be recorded more frequently on days when weather conditions are rapidly changing or the temperature is hot (>80°F). Relative humidity should be measured where the engine is operating, not in the control room where the humidity may be different.

VERIFICATION LESSONS LEARNED (A.4.12.4.3.1)

Some test facilities have been known to use the local airport or weather station for the information, which has led to inaccurate data. Weather stations do not update these readings as frequently as needed. Others have used an aneroid barometer of questionable accuracy. Aneroid barometers require frequent recalibration.

A.5 PACKAGING.

A.5.1 Packaging.

For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

A.6 NOTES.

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

A.6.1 Intended use.

Engines covered by this specification are intended for military-unique air vehicle propulsion. Fighter and attack missions, combat environment, carrier suitability, rotary wing, transports, UAVs, etc., are a few examples of the different design requirements for military engines.

A.6.2 Acquisition requirements.

Acquisition documents should specify the following:

a. Title, number, and date of the specification.

b. Preparation of the engine for storage and shipment: the level of preservation, the type of shipping container, and delivery of engine historical records and a packing list for each engine which will include all components and tools which are not installed but are shipped with the engine will be provided per contract requirements. Packaging guidance may be found in MIL-STD-2073.

Acquisition documents will be tailored based on the information in the JSSG-2007A tailoring handbook (Appendix A) and peculiar program requirements, which may include spiral development.

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A.6.3 Definitions.

Definitions for terms used in this appendix follow.

a. General definitions:

Audit Limit. Criterion of acceptance at any bench test run after the vendor or overhaul facility acceptance and before any significant engine test hours have been accumulated. It can also be the criterion of acceptance at bench test following any engine milestone tests.

B1.0 life. This is the age at which 1.0% of the population will fail.

Bodie Transient. A rapid transient in which the engine power is commanded from intermediate for TJ/TF and maximum for TP/TS to idle, then as the engine is decelerating, the power is commanded to intermediate for TJ/TF and maximum for TP/TS.

Capture Area. Capture area is the projected physical area of the aircraft inlet.

Class A Incident. An accident that involves $1 million in damages, a fatality, or a destroyed aircraft.

Closed Loop Control Logic. A control system that uses a feedback control system to exert continuous control and ensures that a specific desired status is maintained. Uses continuous feedback as opposed to discrete sequential control logic.

Cold Engine (for starting only). A ‘‘cold” engine is defined as one which has been allowed to windmill or “motor” (rollover on the starter) until the low pressure turbine exit gas temperature is within 55°C (100°F) of the engine compressor inlet temperature before a start is attempted.

Contractor Furnished Equipment. Equipment provided by the contractor.

Depot, Intermediate, or Base Level Inspectable Structure. Structure is depot, intermediate, or base level inspectable if the nature and extent of damage will be detected utilizing one or more selected NDI procedures. See MIL-HDBK-1783.

Deteriorated Minimum Engine. Minimum engine after operation for the time period and environment specified in 3.2.1.4.

Dissimilar Control Technology. Various technologies, such as hydromechanical, fluidic, analog, used to control an engine.

Domestic Object Damage. Domestic object damage (DOD) is caused by internal or external engine parts coming loose and striking the engine. See foreign object damage (FOD).

Fail Operational. The ability to detect any single or multiple failure(s) and to continue to operate the engine after detection and still be capable of meeting all defined mission and performance requirements.

Fail Degraded. The ability to continue to operate the engine after any single or multiple failure(s) and retain a defined minimum operational capability as defined in the ICD and the definition of levels of Flying Qualities.

Fail Safe. The ability to continue to operate the engine safely or safely shutdown following the failure or partial failure of one or more elements.

Frequency Response. The steady-state output of the system to input sinusoids of varying frequency. The output for a linear system can be completely described in terms of the amplitude ratio of the output sinusoid to the input sinusoid. The amplitude ratio, gain, and phase are functions of the frequency of the input sinusoid.

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Government Furnished Equipment. Equipment provided by the government.

Hot Engine (for starting only). A "hot" engine is defined as one where a start is attempted within 10 seconds after a flameout or shutdown.

Hot Parts. Parts exposed to the hot gas stream.

Initial Flight Release (IFR). This milestone is conducted to provide the Using Service and the contractor limited performance and durability verification of the engine for initial flight testing. The engine must meet limited performance, operability or durability requirements prior to the completion of this milestone. The initial flight release engine should complete a minimum durability demonstration as defined in 4.11.1.1.1. (See DODI 5000.2).

Installed Configuration. When the engine is tested in the air vehicle of the air system, this is known as the installed configuration.

Latent Failure. Any undetected condition which prevents the part or component from performing its intended function. This includes undetected conditions between missions and during missions.

Maximum Operating Temperature. The highest temperature the engine can obtain in normal operation at a commanded power setting considering all effects due to engine-to-engine variation, deterioration, installation factors, control limits and tolerances, environment, and operating point.

New Minimum Engine. The minimum engine is the engine at the time of government acceptance.

Non-Operating Environment. The environment to which the engine is exposed during logistics, storage, maintenance and transportation activities.

Operational Life. The life of the engine when exposed to operational usage.

Operational Usage. The usage that the engine is exposed to during actual service operation.

Primary Flight Control Surfaces. Movable aircraft surfaces which control the direction of flight. In most cases, they are airfoils that control the pitch, yaw, and lateral stability of the aircraft in flight. Rudders, ailerons, elevators, stabilators, rotors, etc. are considered primary flight control surfaces.

Process Quality Control. Established manufacturing procedures that ensure repeatability within specific process limits.

Reliability Critical Items. Items whose failure would critically affect the weapon system safety, availability, ability to meet mission objectives, or cause extensive or expensive repair.

Safety Limit. The predicted typical propagation life required to grow an assumed initial flaw to its critical size.

Similarity Analysis. A verification approach that studies or reviews an existing or modified component or part to determine if a new application or environment for the component or part is within the design limits.

Small Engines. A small engine is an engine with fan inlet corrected flow of 100 pps or less or compressor inlet corrected flow of 20 pps or less.

Total Ownership Cost. Total Ownership Cost is the Program Office cost plus Life Cycle Cost.

Transparent to the Pilot. This is any event the pilot cannot detect or can be informed about via the EHMS.

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Ultimate Load. The maximum load that a material, part, component, or engine must withstand without catastrophic failure. (See MIL-HDBK-1783.)

Ultimate Service Limit. The Ultimate Service Limit is the extreme limit of component performance beyond which satisfactory engine operation cannot be assured. It is the criterion of acceptance at bench test following Qualification Test (QT) simulated operational component tests or equivalent component tests.

Unweathered fuel. Unweathered fuel is fuel with chemical and physical properties that have not been degraded by environmental effects such as temperature and pressure cycles.

Uprating. Uprating is using a part beyond its design limits.

Usable Life. Usable Life is life without repair or replacement.

Usage Parameters. (See MIL-HDBK-1783.)

Weapons Replaceable Assembly. A generic term which includes all the replaceable packages of an avionic equipment, pod, or engine system as installed in an air system with the exception of cables, mounts, and fuse boxes or circuit breakers.

b. The following definitions apply for turboshaft engines:

Contingency Test Temperature Setting. This is an operating condition at which the engine is operated during the durability test to assure an operational capability for at least an incremental time duration specified in this specification for contingency power.

Emergency Test Temperature Setting. This is an operating condition at which the engine is operated during the durability test to assure an operational capability for at least an incremental time duration specified in this specification for emergency power.

Rated Gas Temperature. Rated Gas Temperature is the maximum turbine rotor inlet temperature (TRIT) permitted to attain a given table IIA, IIB, IIC, or III rating, at or below the maximum rated gas generator speed.

Rated Torque. Rated torque is the delivered output shaft torque at rated delivered shaft speed when the engine is operating at a table IIA, IIB, IIC, or III power rating.

A.6.4 Symbols, subscripts, abbreviations, and acronyms.

The symbols, subscripts, abbreviations and acronyms used in this appendix are listed below and should be per SAE AS 681 and SAE AS 755.

ACGIH	American Conference of Government Industrial Hygienists
ADS	Automated Data System
AEDC	Arnold Engineering and Development Center
AEMS	Aircraft Engine Management System
AIA	Aerospace Industries Association
AMS	Aerospace Material Specification
AOA	Angle of Attack
APC	Aircraft Power Control
API	Armor Piecing Incendiary

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APU	Auxiliary Power Unit
ARD	Aerospace Resource Document
ARP	Aerospace Recommended Practices
ASCII	American Standard Code for Information Interchange
ASIP	Aircraft Structural Integrity Program
ASME	American Society of Mechanical Engineers
ASMET	Accelerated Simulated Mission Engine Test
ASW	Anti Submarine Warfare
ATF	Advanced Tactical Fighter
Aviation 3M	Aviation Maintenance-Material-Management System
AVIP	Avionics Integrity Program
BITI	Built In Test and Inspectability
BPS	Bytes Per Second
CAMS	Core Automated Maintenance System
CDA	Concept Development Aircraft
CDRL	Contractor Data Requirements List
CEMS	Comprehensive Engine Management System
CERT	Combined Environmental Reliability Test
CFE	Contractor Furnished Equipment
CIT	Compressor Inlet Temperature
CL	Confidence Level
CLS	Contractor Logistics Support
CND	Can Not Duplicate
COMNAVAIRLANT	Commander Naval Air Atlantic Fleet
COMNAVAIRPAC	Commander Naval Air Pacific Fleet
CONUS	Continental United States
CR	Component Removal
CSCIs	Computer Software Configuration Items
CSI’s	Critical Safety Items
cSt	Centistoke
dBsm	Decibels per sq. meter
DDR&E	Deputy Director of Research and Engineering
DECU	Digital Electronic Control Unit
DEM/EVAL	Demonstration/Evaluation
DEW	Directed Energy Weapons
DID	Data Item Description
DOD	Domestic Object Damage

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DPT	Durability Proof Test
DT&E	Development Tests and Evaluation
DT&V	Development Tests and Verification
DTD	Damage Tolerant Design
E&MD	Engineering and Manufacturing Development
ECC	Engine Cycle Counter
ECOMTRAK	Engine Component Tracking System
ECP	Engineering Change Proposal
ECU	Engine Control Unit
EDSA	European Distribution System Aircraft
EEPROM	Erasable Electronically Programmable Read Only Memory
EGT	Exhaust Gas Temperature
EIA	Electronic Industries Association
EMD	Engineering and Manufacturing Development
EME	Electromagnetic Environment
EMP	Electromagnetic Pulse
EMV	Electromagnetic Vulnerability
EOH	Engine Operating Hours
EPR	Engine Pressure Ratio
ER	Engine Removal
ESS	Environmental Stress Screening
FDDM	Frequency Division Demultiplexer
FDM	Frequency Division Multiplexer
FER	Failure Engine Removals
FETT	First Engine To Test
FITS	Fuel Ingestion Tolerance Specification
FMECA	Failure Modes, Effects, and Criticality Analysis
FRACAS	Failure Reporting and Corrective Action System
FSD	Full Scale Development
FTIT	Fan Turbine Inlet Temperature
GdB	Giga decibel
GFE	Government Furnished Equipment
GGT	Gas Generator Temperature
HC	Hydrocarbon
HEMP	High Energy Electromagnetic Pulse
HMU	Hydro-Mechanical Unit
HOL	Higher Order Language

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HPT	High Pressure Turbine
IAS	Indicated Airspeed
IBIT	Integrated Built In Test
IEC	International Electrotechnical Commission
IECMS	In-Flight Engine Condition Monitoring System
IFSD	In-Flight Shut Down
IGV	Inlet Guide Vanes
ILS	Integrated Logistics Support
IRTP	Integrated Reliability Test Program
ISA	International Standard Atmosphere
JBD	Jet Blast Deflector
JLFT	Joint Live Fire Test
JOAP	Joint Oil Analysis Program
JTCG/AS	Joint Tactical Coordination Group on Aircraft Survivability
LCC	Life Cycle Costs
LD	Laser Diode
LE	Leading Edge
LED	Light Emitting Diode
LFT&E	Live Fire Test and Evaluation
LHV	Lower Heating Value
LI	Lubricity Improver
LO	Low Observable
LORA	Level of Repair Analysis
LPQ	Limited Production Qualification
LPT	Low Pressure Turbine
LTIT	Low Pressure Turbine Inlet Temperature
LWC	Liquid Water Content
MA	Maintenance Action
MATV	Multi Axis Thrust Vectoring
MECSIP	Mechanical Equipment Structural Integrity Program
MGT	Measured Gas Temperature
MMH/EFH	Maintenance Man Hours/Engine Flight Hours
MNS	Mission Need Statement
MTTR	Mean Time to Repair
MULDEM	Multiplexer/Demultiplexer
NACA	Nation Advisory Committee for Aeronautics
NADEP	Naval Air Depot

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NALDA	Naval Engine Logistics Data Analysis
NAPC	Naval Air Propulsion Center
NAPTC	Naval Air Propulsion Test Center
NASA	National Aeronautics and Space Administration
NAVAIR	Naval Air Systems Command
NAVSEA	Naval Sea Systems Command
NAWC	Naval Air Warfare Center
NAWCWD	Naval Air Warfare Center Weapons Division
NBCCS	NBC Contamination Survivability
N_g	Gas Generator Speed
NO_X	Oxides of Nitrogen including Nitric Oxide plus Nitrogen dioxide
N_p	Power Turbine Speed
OASPL	OverAll Sound Pressure Level
OMC	Organic Matrix Composites
OPNAVINST	Naval Operations Instruction
P_3	Compressor exit pressure
PAC	Power Assurance Check
PFR	Preliminary Flight Release
PFRT	Preliminary Flight Rating Test
PMA	Permanent Magnetic Alternator
PNdB	Perceived Noise Level
POD	Probability of Detection
PPS	Pounds Per Second
QAP	Quality Assurance Provision
QEC	Quick Engine Change
QPL	Qualified Products List
QT	Qualification Test
RCI	Reliability Critical Items
RFP	Request for Proposal
SAM’s	Surface to Air Missiles
SCN	Specification Change Notice
SDIP	Software Development Integrity Program
SE	Support Equipment
SNR	Signal to Noise Ratio
SOO	Statement of Objective
SOW	Statement of Work
SRA	Service Replaceable Assembly

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SRU	Service Replaceable Unit
ST	Special Tooling
STE	Special Test Equipment
STR	System Threat Report
SURVIAC	Survivability and Vulnerability Information Analysis Center
$T_{4.5}$	Power Turbine Inlet Temperature
TBO	Time Between Overhauls
TDDM	Time Division Demultiplexer
TDM	Time Division Multiplexer
TEMS	Turbine Engine Monitoring System
TET	Turbine Entry Temperature
TOC	Total Ownership Cost
TSFC	Thrust Specific Fuel Consumption
USAF	United States Air Force
USC	United States Code
UV	Ultraviolet
VEN	Variable Exhaust Nozzle
VISTA	Variable-Stability In-Flight-Simulator Test Aircraft
V/L	Vapor-Liquid Ratio
VLF	Very Low Frequency
VPC	Vertical Parity Check
WOW	Weight of Wheels

A.6.5 Material Safety Data Sheet.

Contracting officers should identify those activities requiring copies of completed Material Safety Data Sheets.

A.6.6 Subject term (key word) listing.

Bearings
Bird Ingestion
Bleed Air
Commercial Off The Shelf (COTS)
Domestic Object Damage (DOD)
Foreign Object Damage (FOD)
Fuel
Gas
Ice
Idle
Ignition System
Mount

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Oil
Propulsion
Sand
Smoke
Spiral Development
Stall
Thrust
Torque
Turbofan
Turbojet
Turboprop
Turboshaft
Windmilling

A.6.7 Metrication.

The engine will be of metric design per Public Laws 94-168 and 100-418 unless stated otherwise in the contract. The engine will conform to the SI standard described in IEEE/ASTM SI 10, American National Standard for Use of the International System of Units (SI): The Modern Metric System, as listed in the DOD Index of Specifications and Standards. All new metric items (requiring a new part identification) or any new design practice created or designed to replace any customary system of measurement will be integrated into the SI Metric System of Measurement per Public Laws 94-168 and 100-418, and Executive Order 12770. The specification will contain both metric and English units. The metric units will be primary. English units will be provided within parenthesis.

A.6.8 International standardization agreement implementation.

This specification implements NATO STANAG 1135: Interchangeability Chart of Fuels, Lubricants and Associated Products used by the Armed Forces of the North Atlantic Treaty Nations; NATO STANAG 1414: Guidelines to Ensure that Contractors Design and Supply New Equipment Capable of Using Standardized Lubricants; and NATO STANAG 3595: Aircraft Fitting for Pressure Replenishment of Gas Turbine Engines with Oil. When amendment, revision, or cancellation of this specification is proposed, the preparing activity must coordinate the action with the U.S. National Point of Contact for the international standardization agreement, as identified in the ASSIST database at www.dodssp.daps.mil.

A.6.9 Responsible Engineering Office.

The Air Force office responsible for the development and technical maintenance of this specification is ASC/ENF, 2530 LOOP ROAD WEST, WRIGHT-PATTERSON AFB OH 45433-7101. Requests for additional information or assistance can be obtained from this office by writing to the address provided or by calling Commercial (937) 255-8604 or DSN 785-8604.

Any information relating to Government contracts must be obtained through the contracting officers. The responsible Navy engineering office is the Naval Air Systems Command, AIR-4.4.7, 22195 Elmer Road, Patuxent River MD 20670-1534; DSN 757-0509; Commercial (301) 757-0509. The responsible Army engineering office is the USA AMCOM, Engineering Directorate, AMSAM-AR-E, Redstone Arsenal AL 35898-5000.

A.6.10 Changes from previous issue.

Marginal notations ar not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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APPENDIX B

PRODUCTION ACCEPTANCE TEST HANDBOOK

B.1 SCOPE.

Appendix B of JSSG-2007A provides guidance to the Using Service for establishing an engine model specification for the production phase of the engine program. The Tri-Service working group along with engineers in the propulsion communities of the Army, Navy, and Air Force developed, coordinated, and approved this appendix. Government engineers should use the appendix as a guide for adapting and modifying JSSG-2007A based SDD model specifications for use as a production specification. This Appendix is not a mandatory part of the specification. The information contained herein is intended for guidance only.

RATIONALE (B.1)

The scope of JSSG-2007A main body, and Appendix B and all specifications based on their requirements, and guidance must state the intent of that document.

GUIDANCE (B.1)

Use the following text to prepare paragraph 1 of the production specification:

This model specification for a production engine contains military-unique, performance-based requirements and parameters for the (put engine model number here) engine model. This specification contains only requirements and corresponding verification criteria that are essential for defining the acceptability of production engines for delivery to the Using Service. Section 3 of this specification contains the minimum performance-based requirements of the model engine for use by the Using Service. This specification also states the approved verification procedures and criteria the model engine passed during the SDD phase in Section 4. In addition, section 4.13 contains minimum acceptance test requirements that production engines must meet before delivery to the Using Service.

LESSONS LEARNED (B.1)

None.

B.1.1 Structure.

Appendix B maintains consistency with the structure of the SDD specifications based on JSSG-2007A. Section 1 provides information regarding Appendix B, and Sections 2 through 6 provide applicable direct tailoring guidance for corresponding paragraphs in the SDD document.

RATIONALE (B.1.1)

The structure for the main document comes from government standards. The structure of Appendix B provides easiest referencing for tailoring the main document.

GUIDANCE (B.1.1)

Use the following text to prepare paragraph 1.1 of the production specification:

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Structure the engine model specification with the same paragraph numbering and titles as contained in this document. Number and title paragraphs that are not applicable to the application as contained in this document with the words "Not Applicable" in place of the requirement.

Background:

The Using Service, when developing a production specification based on an existing JSSG-2007A based SDD specification, should fill in blanks in the main paragraphs. Another option is to leave the blanks open for contractor input when they respond to an RFP. Make every effort to leave blanks open, when appropriate, to prevent over-specifying. Make clear to the contractor when negotiating a production specification that they are to fill in all blanks contained therein. The paragraph numbers and titles should always be consistent with JSSG-2007A to preserve commonality among Joint Service engine specifications. In cases where a requirement from JSSG-2007A does not apply for the specification application, the Using Service or the contractor should retain the paragraph number and title followed by the words "Not Applicable."

LESSONS LEARNED (B.1.1)

None.

B.1.2 Handbook.

This document (Appendix B) does not have a handbook.

RATIONALE (B.1.2)

The main document, JSSG-2007A, has a requirement to explain the purpose of the handbook; see A.1.2. This paragraph provides a summary of the handbook.

GUIDANCE (B.1.2)

Use the following text to prepare paragraph 1.2 of the production specification:

"Not Applicable."

Background:

The handbook (appendices A and B) provides assistance to tailor the Engine Specification. The Rationale paragraph for each requirement provides justification for that requirement. Guidance gives recommendations on how to tailor the specification paragraph. In cases where blanks appear in the specification paragraph, the Handbook guidance paragraph provides recommended values or text that the Using Service may use to insert in the blank. The Guidance paragraph states when contractors must fill in the blanks and the Using Service makes the final decision. Background, in some Guidance paragraphs, includes additional information on the specification paragraph or on the recommendations in guidance. Finally, Lessons Learned provides valuable insight to events in the past that may impact the tailoring of the Specification.

LESSONS LEARNED (B.1.2)

None.

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B.1.3 Deviations.

There will be no deviations from the structure or content of Appendix B unless approved by the Using Service.

RATIONALE (B.1.3)

This appendix provides the structure, format, and content for the production model specification. To maintain consistency among Tri-Service specifications make no deviation from this format.

GUIDANCE (B.1.3)

Use the following text to prepare paragraph 1.3 of the production specification:

Use text from the SDD model specification.

LESSONS LEARNED (B.1.3)

None.

B.1.4 Interoperability and Joint Technology Architecture.

See A.1.5.

RATIONALE (B.1.4)

Engine operability with international allies, for example, establishes this requirement.

GUIDANCE (B.1.4)

Possible engine compatibility areas to review are engine controls, maintenance (tooling), fuels, and lubricants.

LESSONS LEARNED (B.1.4)

None.

B.2 APPLICABLE DOCUMENTS.

B.2.1 General.

Use the documents listed in this section to meet the requirements specified in sections 3, 4, and 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, and 5 of this specification, whether or not they are listed.

B.2.2 Government documents.

B.2.2.1 Specifications, standards, and handbooks.

The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

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DEPARTMENT OF DEFENSE SPECIFICATIONS

DOD-L-85734	Lubricating Oil, Helicopter, Transmission System, Synthetic Base

(Copies of this document are available online at http://assist.daps.mil/quicksearch/ or www.dodssp.daps.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia PA 19111-5094.)

B.2.2.2 Other Government documents, drawings, and publications.

Not applicable.

B.2.3 Non-Government publications.

The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract..

SAE INTERNATIONAL

SAE ARP 741	Turbofan and Turbojet Gas Turbine Engine Test Cell Correlation
SAE ARP 4755	Turboshaft/Turboprop Gas Turbine Engine Test Cell Correlation

(Application for copies should be addressed to SAE International, 400 Commonwealth Drive, Warrendale PA 15096-0001; www.sae.org.)

B.2.4 Order of precedence.

In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

B.3.13 Acceptance Test.

Test the engine for the purposes of accepting engines for production, using Section 4.13.

Note: Section 3.13 shall not contain requirements other than requiring an acceptance test. Section 4.13 shall contain test requirements for production engines unique from those specified in other sections of the SDD specification.

VERIFICATION RATIONALE (B.3.13)

Delivered engine performance and operability characteristics must be verified before Using Service acceptance of each engine.

VERIFICATION GUIDANCE (B.3.13)

None.

VERIFICATION LESSONS LEARNED (B.3.13)

None.

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B.4.13 Acceptance Test.

Each production engine submitted for delivery shall have an acceptance test conducted on it.

VERIFICATION RATIONALE (B.4.13)

These tests are necessary to ensure all production engines meet the sea level static and altitude performance and operability requirements of the model specification.

VERIFICATION GUIDANCE (B.4.13)

Background:

The purpose of Section 4.13 is to define a standard set of production tests and procedures that all Services can use to measure the production engine performance and operability characteristics relative to the engine model specification. The purpose also includes providing generic test methodologies for engine servicing, data gathering and rejection and retesting criteria. The final purpose is to establish a minimum acceptable level of recording equipment and accuracies. Tailoring these procedures for a specific application, assures each Using Service adequate testing of the production engines. To maintain continuity between the SDD specification and the production specification, some of the paragraphs from Section 4.12 of the acquisition specification are the same. Also, accuracies of the measuring equipment and data will be consistent between the development and production specification requirements.

VERIFICATION LESSONS LEARNED (B.4.13)

The Engine V Production Specification was the first Production Specification for a major program that AIR-4.4 has developed and incorporated into a production contract since standing up the Competency Aligned Organization. As a program moves from development into production, it faces differing requirements. In development, the requirements center around qualification of the system for production through the verification of salient engine characteristics. In production, the requirements center around acceptance testing to check performance characteristics and fleet readiness requirements that can be substantiated with easily obtainable field metrics.

It might be beneficial to run some number of Accelerated Mission Test (AMT) cycles on a limited number of engines early in the production phase as a risk reduction activity. Some programs have done this (Engine O on Aircraft H) and others have not. Discovering infant mortality failures is a potential benefit of this type of testing. AMT testing is very expensive and consumes engine life at a substantial rate. In view of this, it would be best to run these tests early in a program on a limited number of engines.

B.4.13.1 Test equipment.

B.4.13.1.1 Automatic recording equipment.

See A.4.12.3.3.1.

B.4.13.1.2 Vibration measuring equipment and response characteristics.

The vibration measurement and analysis equipment shall operate over a frequency band of: _____. Calibrate the equipment as a complete system.

VERIFICATION RATIONALE (B.4.13.1.2)

Engine vibration measurement is one of the necessary parameters used to monitor the health of the engine.

VERIFICATION GUIDANCE (B.4.13.1.2)

Use the following to tailor the specification paragraph:

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The contractor should provide the frequency band.

Background:

A sudden increase in the level of vibration can signal an existing or an impending engine failure. A gradual increase in vibration level may mean parts are wearing.

Engine acceptance testing may not have the same vibration measurement and analysis equipment requirements as development testing has (A.4.12.3.3.2).

Past turboshaft engine programs have used the following:

> "The engine vibration shall be measured with sensors. The vibration measurement and analysis equipment shall operate over a frequency band of at least 15 Hz to 5000 Hz and produce acceleration spectrograms with a demonstrated accuracy confidence level of 95 percent. The maximum allowable effective filter bandwidth of the spectrum analysis equipment shall be 3.0 Hz up to 1000 Hz and 6 Hz above 1000 Hz. The vibration measuring equipment shall be calibrated as a complete system. The frequency response of the system, when calibrated by applying a known sinusoidal motion to the sensor shall not deviate by more than 3.0 dB from the known sinusoidal input at frequencies from 15 Hz to 5000 Hz. If filters are required when measuring overall velocity levels, they shall be not more than three dB down at the cutoff frequencies, which shall be 100 Hz high pass and they shall have a roll-off rate of at least 18 dB per octave."

VERIFICATION LESSONS LEARNED (B.4.13.1.2)

None.

B.4.13.1.3 Test cell and test equipment.

Acceptance testing of engines shall be conducted in a test cell and with test equipment that is acceptable to the Using Service. Any natural frequencies of the test stand shall not induce detrimental nor damaging resonances to the engine when installed on the test stand at any test or operating condition. Derive the test cell "cell factor" from engine test data by a method satisfactory to the Using Service. Use the power absorber characteristics of 3.1.1.9, for turboprop and turboshaft engines. The referee nozzle for turboshaft engines shall be a straight circular duct for consistency of uninstalled performance comparisons.

VERIFICATION RATIONALE (B.4.13.1.3)

The test cell must be adequate for the engine being tested and must be free of damaging natural frequencies. Establish "cell factors" for new engines when first testing in production cells.

The test cell power absorber characteristics need to meet the requirements of 3.1.1.9 so that engine and control response characteristics and control limits can be adequately tested.

VERIFICATION GUIDANCE (B.4.13.1.3)

Use the following documents to establish "cell factors" for gas turbine engines:

a. SAE ARP 4755, Turboshaft/Turboprop Gas Turbine Engine Test Cell Correlation

b. SAE ARP 741, Turbofan and Turbojet Gas Turbine Engine Test Cell Correlation.

The test cell "cell factor" should be derived from engine test data acquired from both outdoor and indoor testing.

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VERIFICATION LESSONS LEARNED (B.4.13.1.3)

The Army has encountered situations where low dynamometer polar moments of inertia had prevented the safe testing of the lockout mode on Engine C.

B.4.13.1.4 Starter.

To perform starts, use a starter that has torque characteristics within ___ percent of the minimum required torque shown on figure 2.

VERIFICATION RATIONALE (B.4.13.1.4)

This requirement limits the size of the starter used in testing by specifying its torque characteristics. A starter with performance characteristics that vary significantly from specification requirements can bias test results.

VERIFICATION GUIDANCE (B.4.13.1.4)

Use the following to the tailor the specification paragraph:

The contractor should provide a value between 0-5 percent, preferably, a low value.

VERIFICATION LESSONS LEARNED (B.4.13.1.4)

None.

B.4.13.2 Servicing.

The fuel used for acceptance testing shall be a primary fuel as specified in the engine model specification. Use the oil as specified in 3.7.8.1 for acceptance testing. Record all oil filter inspection results in the engine records. At the start of each test, service the oil system. Record all additions of oil added after initial servicing. Drain oil only when authorized by the Using Service. Do not use external oil filters.

VERIFICATION RATIONALE (B.4.13.2)

The fuel and oil used for acceptance testing should not be special blends. The fuels and oils must be one of the fuels and oils required by the specification. Using the oil servicing requirement controls the test conditions when servicing the engine oil system. Record the addition of oil to determine oil consumption.

VERIFICATION GUIDANCE (B.4.13.2)

The engine model specification prescribes using JP-5 fuel with its specified values for minimum and maximum physical and chemical fuel properties.

VERIFICATION LESSONS LEARNED (B.4.13.2)

The Navy has used the lower heating value of 18,400 BTUs per lb. for JP-5 fuel in the Engine J and V programs. The JP-5 fuel that is available in the field is usually above the minimum value required in the fuel specification of 18,300 BTUs per lb. The Navy and Company A, after analysis of the fuel available in the field, agreed upon the value of 18,400 BTUs per lb. The Engine J data reduction program uses the value of lower heating value in the calculation of specific fuel consumption. The actual test value of fuel lower heating value is ratioed to the agreed upon value of 18,400 BTUs per lb. in the data reduction program calculation to correct the engine specific fuel consumption back to minimum engine performance. In the Engine V engine program, the cycle deck is run with the as tested lower heating value and thrust, fuel flow and high pressure turbine rotor inlet temperature are compared to the value obtained running the specification deck with as tested conditions.

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B.4.13.3 Test records.

B.4.13.3.1 Acceptance test log sheet.

Record the following information on the test log sheets for each engine run: _(a)_.

The contractor shall retain copies of acceptance test log sheets for each engine for _(b)_.

Record all test article software changes on the log sheets. Copies of test sheets shall be provided to the Using Service upon request.

VERIFICATION RATIONALE (B.4.13.3.1)

Test log sheets are used for all pertinent information not recorded by the automatic recording equipment.

VERIFICATION GUIDANCE (B.4.13.3.1)

Use the following to tailor the specification paragraph:

(a) The contractor and Using Service should negotiate recorded parameters. Previous programs have recorded the following parameters:

(i) date,

(ii) engine type and model,

(iii) engine serial number,

(iv) cell number,

(v) bellmouth serial number and area, sq. m. (sq. in.),

(vi) exhaust nozzle serial number and area, sq. m. (sq. in.),

(vii) type of fuel used,

(viii) type of oil used,

(ix) fuel lower heating value,

(x) total running time and total number of starts,

(xi) vibration (maximum recorded in requirement 4.13.4.1.2).

Use table XXXIV as further guidance to determine additional log sheet parameters.

(b) The Using Service should provide the time interval. Previous programs have required that the records be retained: (1) two years, (2) until final test report publication, or (3) until engine overhaul at the depot.

VERIFICATION LESSONS LEARNED (B.4.13.3.1)

None.

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B.4.13.4 Data.

B.4.13.4.1 Test data.

Take the data in the following subparagraphs during the acceptance test and record it in the acceptance test log sheets.

VERIFICATION RATIONALE (B.4.13.4.1)

Measure and record parameters necessary to ensure that the engine meets the minimum level of performance.

VERIFICATION GUIDANCE (B.4.13.4.1)

None.

VERIFICATION LESSONS LEARNED (B.4.13.4.1)

None.

B.4.13.4.1.1 Preliminary data.

Measure and record the engine weight. If the engine weight is measured after the engine has been serviced with fuel and oil, and after subsequent draining, calculate the dry weight by subtracting the weight of residual fluids specified in section 4.1.2.1 from the measured engine weight.

VERIFICATION RATIONALE (4.13.4.1.1)

The engine weight must be measured and recorded to ensure that the engine meets the specification requirement. This will also ensure that minor changes made to the production configuration do not allow the weight to exceed the limit.

VERIFICATION GUIDANCE (B.4.13.4.1.1)

None.

VERIFICATION LESSONS LEARNED (B.4.13.4.1.1)

None.

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B.4.13.4.1.2 Steady-state data.

During operation at each specified steady-state condition and after performance stabilization, record the following data:

VERIFICATION RATIONALE (B.4.13.4.1.2)

A specific set of parameters must be measured and recorded to confirm the engine meets the minimum level of specification performance. The list required here will ensure that the proper parameters are being recorded.

VERIFICATION GUIDANCE (B.4.13.4.1.2)

Use the following to tailor the specification paragraph:

Use table XXXIV as a guide to determine which parameters to measure and record.

VERIFICATION LESSONS LEARNED (B.4.13.4.1.2)

Past programs have recorded these types of parameters:

a. time of day,
b. total running time,
c. power lever position,
d. engine rotor(s) speed(s), rpm,
e. fuel consumption, kg/hr (lbm./hr),
f. data for determining airflow,
g. inlet total pressure, mm (in.) Hg abs. kPa (psi),
h. engine inlet total temperature, °C (°F),
i. turbine discharge total pressure, mm (in.) Hg abs. kPa (psi),
j. exhaust static pressure, mm (in.) Hg abs. kPa (psi),
k. breather pressure, mm (in.) Hg abs. kPa (psi),
l. oil temperature at point shown on installation drawing °C (°F),
m. oil pressure at point shown on installation drawing, mm (in.) Hg kPa (psig),
n. fuel pressure at fuel system inlet, mm (in.) Hg, kPa (psig),
o. fuel pressure at point shown on installation drawing, kPa (psig),
p. measured steady-state gas temperature, °C (°F),
q. augmentor fuel flow, kg/hr (lbm./hr),
r. liquid injection flow, kg/hr (lbm./hr),
s. engine case vibration at points shown on installation drawing, cm (inches) per second, true root mean square velocity,
t. thrust, N (lbf),
u. customer bleed air extraction airflow, kg (lbm) per sec, m
v. first stage turbine rotor inlet gas temperature, °C (°F),
w. oil leakage at accessory pads and drains, cc/hr (oz/hr).,

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x. torque, N-m (ft-lbf)

y. shaft horsepower, kw (ship)

z. power turbine inlet temperature, T4.5 °C (°F),

aa. additional data as required by the Using Service.

B.4.13.4.1.3 Transient data.

For each power or thrust transient performed and if applicable, reverse thrust operation, continuously record the following parameters versus time: _____.

VERIFICATION RATIONALE (B.4.13.4.1.3)

A specific set of parameters must be measured and recorded to confirm the engine meets the minimum level of specification performance. This list will ensure that the proper parameters are being recorded.

VERIFICATION GUIDANCE (B.4.13.4.1.3)

Use the following to tailor the specification paragraph:

Use table XXXIV as a guide to determine which parameters to measure and record.

VERIFICATION LESSONS LEARNED (B.4.13.4.1.3)

Acceleration and deceleration times should be verified during the acceptance tests on all engines delivered to the Using Service.

B.4.13.4.1.4 Starting data.

For each start performed, the time required from initiation of the start to ignition, to starter cutout, and to stabilized engine idle speed, record the maximum measured gas temperature and any noticeable flat spots in the acceleration curve.

VERIFICATION RATIONALE (B.4.13.4.1.4)

Engine start times must be recorded to ensure the engine meets specification requirements and to ensure the control system is functioning properly.

VERIFICATION GUIDANCE (B.4.13.4.1.4)

None.

VERIFICATION LESSONS LEARNED (B.4.13.4.1.4)

None.

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B.4.13.4.1.5 Miscellaneous data.

All stops and spool down times shall be measured and recorded. At least once during each test period, readings shall be taken of barometric pressure, ambient air temperature, and fuel specific gravity.

Oil consumption for the entire test run shall be measured and recorded. Notes shall be placed on the log sheets of all incidents of the run, such as leaks, unusual vibrations, and other irregular functioning of the engine together with corrective measures taken.

VERIFICATION RATIONALE (B.4.13.4.1.5)

Miscellaneous test cell data must be recorded to ensure the proper corrections are made to the performance data. Miscellaneous engine data must be recorded to ensure that oil consumption and vibration levels are within limits.

VERIFICATION GUIDANCE (B.4.13.4.1.5)

Oil analysis samples should be required for acceptance testing. The sampling program should be coordinated with the Joint Oil Analysis Program (JOAP) Manager of the Using Service to determine the sampling plan and applicable oil analysis tools for the engine and filtration system under test.

JOAP sampling encompasses several tools including spectroscopy, ferrography, filter debris analysis, Fourier transforms infrared spectrogram analysis, etc. Several factors must be considered when designing a sampling program for acceptance testing. These include the engine system and engine filtration system, the oil used and, the oil supply system. A set of at least test two samples should be obtained for an acceptance test, one at the beginning of the test and one prior to completion.

Trending is a much more valuable tool for determining suspect engines than absolute wear limits. The difference for wear metal readings should be less than 3 PPM; the field wear metal concentration limits (service limits) should not be used.

VERIFICATION LESSONS LEARNED (B.4.13.4.1.5)

The Army has had experience with fine filtration systems depleting oil additives.

New oil particulate levels can vary. For engines with fine filtration (< 3 micron (1.18×10^{-4} in.)) spectroscopy may not be applicable. Spectroscopy measures particulates in the 0 - μm to 10 - μm range; however, particulates in the 0 - μm to 2 - μm range have not been correlated with impending failures. The Army has had success in correlating ferrographs and filter debris analysis results with failures on fine filtration engines.

The Army has had experience with DOD-L-85734 oils skewing oil analysis results due to its detergent action cleaning out built up deposits.

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B.4.13.4.1.6 Accuracy of data.

For engine acceptance testing, all steady-state and transient data accuracy shall be within the following tolerances:

All instruments and equipment shall be calibrated as necessary to insure that the required degree of accuracy is maintained.

Parameter	Steady-state Accuracy (%)	Transient Accuracy (%)

VERIFICATION RATIONALE (B.4.13.4.1.6)

The accuracy of the measurands must be established to ensure the resulting test data and correction factors are reliable and precise.

VERIFICATION GUIDANCE (B.4.13.4.1.6)

The following should be used to tailor the specification paragraph:

The following steady-state accuracies have been used on previous programs:

Rotor speed(s)	±0.2 percent of the value obtained at maximum thrust or power
Metered Thrust or power	±0.5 to ±1.0 percent of the value measured
Fuel Flow	±0.5 to ±1.0 percent of the value measured – for minimum augmented thrust and above. ±0.5 to ±1.0 percent of the value measured at intermediate – for all values at or below intermediate thrust or power
Torque	±0.5 percent of the value measured
Airflow	±1.0 to ±1.5 percent of the value measured
Temperature	±1°C up to 200°C ±3°C between 200°C and 800°C ±4°C above 800°C
Engine weight	±1.0 to ±4.0 lbs. or ±0.1 percent of the weight being determined, whichever is greater
Vibration velocity	±10.0 percent of specified engine limit or ±7.5 percent of the full scale reading
Pressure	±0.5 to ±1.0 percent of the value measured at intermediate thrust or power
All other data	±2.0% of the value obtained at maximum thrust or power

VERIFICATION LESSONS LEARNED (B.4.13.4.1.6)

None.

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B.4.13.5 Test procedure.

The acceptance test shall consist of the initial and final runs specified below.

VERIFICATION RATIONALE (B.4.13.5)

A baseline standard set of test procedures must be established to provide consistency throughout the Military Services.

VERIFICATION GUIDANCE (B.4.13.5)

When using special engine features, which would not function under the following test schedules, the following should be accomplished in a manner approved by the Using Service:

a. Record time at each thrust or power condition beginning at the completion of the power demand signal change necessary to obtain that thrust or power condition.

b. If applicable, prior to initiation of the final run, the engine controls should be adjusted while installed on the engine. Use only routine field adjustments to produce the standard sea level static condition thrust or power or higher and specific fuel consumption or lower. Ensure the adjustments are within the limits of the measured gas temperatures and rotor speeds. If any further adjustment to the engine or its components becomes necessary, rerun those portions of the test already completed.

c. When directed by the Using Service, test a production engine at altitude conditions.

VERIFICATION LESSONS LEARNED (B.4.13.5)

None.

B.4.13.5.1 Initial run.

The engine shall be subjected to an initial run using __(a)__.

All power demand signal changes shall be accomplished in __(b)__ seconds.

The sequence of testing procedures shall be __(c)__.

VERIFICATION RATIONALE (B.4.13.5.1)

Detailed test procedures must be established to ensure each production engine will be tested thoroughly.

VERIFICATION GUIDANCE (B.4.13.5.1)

The following should be used to tailor the specification paragraph:

(a) The contractor shall specify the nature and extent of checks, adjustments, and running prior to the initial run.

Typical initial runs consist of the following:

Intermediate and maximum thrust or power transient runs.

Reverse thrust and pitch-maximum runs.

Performance runs.

Slow transient runs.

Other transient runs as required by the Using Service.

(b) Typical times are one-half second or less.

(c) The sequence for each procedure should be put in the specification paragraph and the time for each dwell specified. Use the following as guidance in determining the sequence of each procedure:

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The contractor should propose the test sequence. This sequence should consist of a series of power demand signal changes and power settings that ensure the engine performance meets the requirements of requirement 3.2.2.7 Transients.

This subparagraph should be deleted if the engine does not have a reverser or propeller. The contractor should propose the test sequence. This sequence should consist of a series of power demand signal changes into and out of maximum reverse thrust or power. Use thrust or power settings that duplicate the anticipated in-service usage.

The contractor should propose the test sequence. This sequence should consist of dwells at the various thrust or power settings from idle to intermediate/maximum continuous and maximum augmentation from table II. The dwell times should be of sufficient duration such that the engine is fully stabilized before any data is recorded.

The contractor should provide the time. This sequence should consist of a dwell at idle followed by a slow transient (i.e. 80 -100 seconds) at a uniform rate of power demand signal change from idle to maximum thrust or power. After a short dwell at maximum thrust or power, the engine should be decelerated to idle at the same power lever rate that was used for the acceleration, followed by a dwell at idle.

Turboshaft/turboprop engines may require special transients, or the specific application may require a special transient.

Background:

It is desirable to perform the transient runs before the steady-state performance runs to ensure that the engine is more fully broken in. An engine that has tight clearances when the performance test is run may demonstrate high positive SFC or temperature margin, but this performance will be lost after only a few flights. Tight engines that barely meet performance requirements and have low margins may become unacceptable after a short operating time in the field.

The time for the dwell is different in each of the above paragraphs; e.g., some dwell time is needed to achieve temperature stabilization at the beginning and the end of a transient series. Sometimes the dwell will not be long enough to reach thermal stabilization, but just long enough to reach the maximum transient speed or temperature to check for surge or stall during Bodies. Another example is to dwell long enough to verify that the engine will stabilize at a control point with the performance parameters not oscillating outside of the specification limits. The longest dwell is needed to attain the steady-state performance levels in the performance run. The times the contractor specifies for the dwell are approximate and are not the specification minimum times.

VERIFICATION LESSONS LEARNED (B.4.13.5.1)

During Engine F1 production, component efficiencies shifted which resulted in stalls during accelerations during acceptance testing.

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B.4.13.5.1.1 Inspection after initial run.

When invoking a disassembly-sampling plan, after completing the initial run, disassemble the engines chosen for the plan. The extent of disassembly shall be ___.

If any part is found to be defective, an approved part shall be supplied to replace it and at the discretion of the Using Service, a penalty run of suitable duration shall be made.

After successful completion of the inspection, the engine shall be reassembled and prepared for the final acceptance test of 4.13.5.2.

VERIFICATION RATIONALE (B.4.13.5.1.1)

The level of disassembly inspection for new engines must be established by the Using Service to ensure proper assembly procedures were used when the engine was built.

VERIFICATION GUIDANCE (B.4.13.5.1.1)

Use the following to tailor the specification paragraph:

Establish a sampling plan to determine the number of engines in the production lot that will be disassembled and to what extent the disassembly should be. If a sampling plan is used it should be referenced in this paragraph.

The contractor should provide the details on the extent of disassembly, but must be sufficient to allow a detailed inspection of all vital working parts.

A study should be accomplished to determine the effectiveness of a disassembly-sampling program. If a disassembly-sampling plan is implemented, the number of engines to be included and the level of disassembly should be established. The initial phase of the plan may require a complete disassembly of a specific percentage of the first few production engines. As fewer defects are discovered, the level of disassembly and number of engines disassembled should be reduced. The manufacturing group in the program office should have information from past programs on the effectiveness of disassembly inspections.

Other considerations are whether the engine model is new or if it is a derivative, or if new production techniques are being used. The detrimental effect of a complete disassembly is that more parts are worked on and different faults than those noted during the disassembly may be introduced during reassembly.

VERIFICATION LESSONS LEARNED (B.4.13.5.1.1)

During an inspection after the initial run the first engine off the production line was found to have large dents in the ball elements of the number 1 thrust bearing. The problem was determined to be the bearing assembly tool. The tool design did not retain the ball element in the race during assembly and as a consequence ball elements were dented when forced against the bearing during assembly.

It took several iterations of the tool design before the engines were assembled without damage to the bearings. The requirement to conduct post engine "green-run" teardowns to ensure the manufacturing process was proper was critical to this program. Several faulty engines would have been fielded without this process in place.

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B.4.13.5.1.2 Penalty run.

The penalty run shall consist of the following: _____.

VERIFICATION RATIONALE (B.4.13.5.1.2)

The duration and severity of a penalty run must be established. The Using Service and contractor must know to what extent and under what conditions the penalty run will be accomplished.

VERIFICATION GUIDANCE (B.4.13.5.1.2)

Use the following to tailor the specification paragraph:

The maximum penalty run should be a complete repetition of the initial run. The minimum penalty run should repeat the test sequence that was accomplished immediately prior to stopping the test.

Additional running prior to the penalty run may be required to test replaced parts.

The length of the penalty run should be proposed by the contractor and included in the specification.

VERIFICATION LESSONS LEARNED (B.4.13.5.1.2)

None.

B.4.13.5.2 Final run.

Engines that have been disassembled and inspected per 4.13.5.1.1 shall be final tested using _____.

A final run shall be accomplished after any penalty run or as defined in the Acceptance Test Procedure.

VERIFICATION RATIONALE (B.4.13.5.2)

The procedures for the final test run must be established so the Using Service and contractor understand what is required after a penalty run or after the disassembly and inspection.

VERIFICATION GUIDANCE (B.4.13.5.2)

Use the following to tailor the specification paragraph:

The final run can be a complete repetition of the initial run, or an abbreviated initial run to check specific steady-state or transient performance parameters.

The extent of the final engine run should be based on the level of disassembly and inspection. An extensive level of disassembly should require rigorous testing since more parts have been disturbed.

A final run may also be required after a penalty run depending on the extent of the penalty run tests. The final run after the penalty run should include the performance run and a spot check of some of the other runs. This is to ensure the engine is not affected by the changes made to the engine prior to the penalty run.

VERIFICATION LESSONS LEARNED (B.4.13.5.2)

None.

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B.4.13.5.3 Rejection and retest.

Whenever there is evidence that the engine is malfunctioning or is not meeting engine specification requirements, the difficulty shall be investigated and its cause corrected to the satisfaction of the Government representative before the test is continued.

If such investigation requires disassembly of the engine or any of its components, this shall be considered a rejection of the engine and a penalty run shall be made using 4.13.5.1.2.

VERIFICATION RATIONALE (B.4.13.5.3)

If an engine malfunction occurs during the acceptance test, the criteria for rejection and retest and the resultant actions must be established to avoid disagreements and other problems.

VERIFICATION GUIDANCE (B.4.13.5.3)

All foreseen rejection criteria as well as penalty run success criteria should be documented in the Acceptance Test Procedure, prior to the beginning of any testing.

VERIFICATION LESSONS LEARNED (B.4.13.5.3)

None.

B.4.13.5.3.1 Engine vibration.

The engine shall meet the vibration requirements of 3.4.1.8.1 .

VERIFICATION RATIONALE (B.4.13.5.3.1)

Criteria for rejection of the engine must be established to avoid disagreements over whether the engine should be accepted by the Using Service.

VERIFICATION GUIDANCE (B.4.13.5.3.1)

The vibration equipment of 4.13.1.2 should be used to continuously record the engine vibration throughout this run. The Acceptance Test Procedure should clearly document engine vibration success criteria prior to the beginning of testing.

VERIFICATION LESSONS LEARNED (B.4.13.5.3.1)

None.

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B.4.13.5.3.2 Over-temperature.

The engine _____ temperature shall not exceed the maximum allowable measured temperatures of 3.4.1.6.5.

VERIFICATION RATIONALE (B.4.13.5.3.2)

Criteria for rejection of the engine must be established to avoid disagreements over whether the engine should be accepted by the Using Service.

VERIFICATION GUIDANCE (B.4.13.5.3.2)

Use the following to tailor the specification paragraph:

The contractor should specify the parameter that is being monitored for exceedance, i.e., hydraulics, oil, gas path, etc.

Temperatures that have been used are:

a. high-pressure turbine first stage rotor inlet ($T_{4.1}$)
b. low pressure/power turbine rotor inlet temperature ($T_{4.5}$)
c. turbine blade temperature (T_{4B}) or exhaust gas temperature (T_5).

VERIFICATION LESSONS LEARNED (B.4.13.5.3.2)

None.

B.4.13.5.3.3 Stoppage.

Interruptions or stoppage from any cause other than an engine malfunction shall require a repetition of the particular period during which the interruption or stoppage occurred.

VERIFICATION RATIONALE (B.4.13.5.3.3)

The consequence of stopping the acceptance test must be established and the necessary remedial actions must be identified.

VERIFICATION GUIDANCE (B.4.13.5.3.3)

None.

VERIFICATION LESSONS LEARNED (B.4.13.5.3.3)

None.

B.4.13.5.3.4 Fluid leakage.

The engine shall not exceed the fluid leakage requirements of 3.1.1.8.

VERIFICATION RATIONALE (B.4.13.5.3.4)

The consequence of leaking fluids must be established and the necessary remedial actions must be identified.

VERIFICATION GUIDANCE (B.4.13.5.3.4)

The cause of the leak should be documented. Once the leak is corrected, a penalty run may be required.

VERIFICATION LESSONS LEARNED (B.4.13.5.3.4)

None.

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B.4.13.5.3.5 Maximum hours of running.

Any engine exceeding a total of (a) hours or more of operation, or a total of (b) or more attempted starts, shall have Using Service approval prior to the initial run.

VERIFICATION RATIONALE (B.4.13.5.3.5)

The Using Service must be informed when the total test time or number of starts exceeds the specified values.

VERIFICATION GUIDANCE (B.4.13.5.3.5)

Use the following to tailor the specification paragraph:

The contractor should propose values for (a) and (b) and document them in the Acceptance Test Procedure.

The total amount of running time and attempted starts should consider all previous runs, checks, and adjustments made prior to the initial run.

Background:

Some augmented turbofan engines have used values of 25/25, 30/25, 30/30, and 20/15 for hours/starts, depending upon the length of the acceptance tests. Some engines were subjected to a sampling plan where the engine went through the acceptance test, was disassembled then retested, thus the operating hours and number of starts were higher. Some sampling plans may require a portion of the engines to receive more extensive operability and surge margin testing that will increase the number of test hours.

The Army and the Navy have both used 16 hours and 25 starts for the maximum number of test hours and starts during the entire acceptance test. The maximum number of "attempted" starts is limited, since if the engine is having trouble starting this may indicate compressor or starter system problems.

VERIFICATION LESSONS LEARNED (B.4.13.5.3.5)

None.

B.4.13.6 Test completion.

The acceptance test shall be completed when the conditions of 4.13.5 have been met satisfactorily and the acceptance test data demonstrates compliance with the engine specification requirements.

VERIFICATION RATIONALE (B.4.13.6)

Criteria for successful completion of the acceptance test must be established so the Using Service and contractor understand when the engine is acceptable for delivery.

VERIFICATION GUIDANCE (B.4.13.6)

The following areas should be considered as minimum success criteria to be evaluated:

a. Performance at standard day Sea Level, Static (SLS) Conditions (table II).
b. Oil Pressure and Temperature Limits
c. Oil Consumption
d. Vibration Limits
e. Stopping
f. Stability
g. Thrust or power Transients
h. Dry Weight of Complete Engine
i. Fluid Leakage

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j. Starting

VERIFICATION LESSONS LEARNED (B.4.13.6)

None.

B.4.13.7 Sampling tests.

When sampling tests are required, a sampling plan with _____ number of engines and detailed acceptance test procedures shall be provided.

VERIFICATION RATIONALE (B.4.13.7)

The sampling plan must be designed to provide reasonable confidence that the sampled engines are representative of the population. The acceptance tests that are to be used in conjunction with the sampling must be established.

VERIFICATION GUIDANCE (B.4.13.7)

Use the following to tailor the specification paragraph:

The contractor should provide the number of engines and the extent of acceptance testing required.

If there is large variation in a measured characteristic of the engine between the sampled engines then a larger sample will be required of the population. For example, if the engine is required to have a certain amount of stall margin, sample a significant number of engines to determine that this margin exists.

The sampling procedures for an augmented turbofan have included the following operability tests:

a. Decel Flameout Mapping Test : 1 of every 60 engines (1 of 40 originally proposed)
b. Fan Stall Margin Check: 1 of every 120 engines
c. Idle Bodie Stall Mapping Test: 1 of every 120 engines
d. Start Stall Mapping Test: 1 of every 120 engines (1 of 80 originally proposed)

These tests may require extra time, slave hardware, and cell reconfiguration. The slave hardware may include:

a. extra air bleed
b. fuel bleed
c. remote control false signals (pressure and temperature)
d. and even a special control

In view of this, it may be best to run these tests prior to high rate production, or when it is believed low margins exist (i.e.; SFC, temperature, compressor surge).

Use test-sampling procedures if problems are anticipated or if they are unlikely to be found during development (i.e., by testing a single engine). In addition, sampling procedures should be used throughout production. For example, a problem could be predicted when it is known that an operability characteristic is not tolerant to manufacturing variability. Flameout margin tests may be useful for diffuser, combustor, fuel nozzle variability, and bodie stall margin tests may be useful for compressor clearance variability.

In addition to Sea Level Static (SLS) testing, a Performance Verification Test (PVT) should be conducted on a production engine under altitude conditions once every two years minimum. The test plan for the PVT should be a scaled-down version of the altitude qualification test plan. The PVT includes steady-state performance, functional points, transients, augmentor lights, and auto relights. Trending steady-state performance versus time is the primary reason for running PVTs. The engine should meet specification requirements and any adverse trends in margins versus previous PVTs should be noted, especially adverse trends in SFC. Adverse trends can indicate the build quality is slipping or can be the cumulative effect of CIP design changes.

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VERIFICATION LESSONS LEARNED (B.4.13.7)

The operability tests that are run as part of the sampling tests should be "test run" during EMD. The reasons for this are so that no "new" requirements are imposed during production that were never checked during development, and to work out any problems in the test procedures. On one augmented turbofan several procedural changes were made until the tests were acceptable. Many engines were mis-run first.

B.5 PACKAGING.

(See A.5.)

B.6 NOTES.

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

B.6.1 Intended use.

(See A.6.1.)

B.6.2 Acquisition requirements.

(See A.6.2.)

B.6.3 Definitions.

Acceptance Test. Ground tests and inspections conducted on engines submitted for production acceptance to demonstrate proper fabrication, adjustment, assembly techniques, and performance.

Dwell. Amount of time the engine spends at a constant power setting.

B.6.4 Material Safety Data Sheet.

(See A.6.5.)

B.6.5 Subject term (key word) listing.

(See A.6.6.)

B.6.6 Metrication.

(See A.6.7.)

B.6.7 International standardization agreement implementation.

(See A.6.8.)

B.6.8 Responsible Engineering Office.

(See A.6.9.)

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TABLE I. Gearbox pads and drives.

Accessory or Component	Type of Drive 1/	Ratio Pad to Rotor rpm 2/	Rotation Direction Facing Pad	Torque N-mm (lb_f-in)			Overhung Moment N-mm (lb_f-in) 4/	Maximum Spline Misalignment mm (in)
				Steady State	Transient			
				Maximum Continuous	Over-load 3/	Static		

NOTES:
1/ Give the type of drive including AN or MS number and type or other commercial equivalent.
2/ Ratio of speeds based on 100 percent (of the driving gas generator) rotor speed of _____ RPM.
3/ Specify duration and frequency of overload.
4/ Overhung moment of accessory or component must include max allowable load permissible from attachments; for example, fuel pump must include aircraft fuel connection load.

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TABLE IIa. Turbofan/turbojet performance at standard day, 15°C (59°F), sea level, static conditions[1/] with _____ nozzle.

Power Setting[2/]	Minimum Thrust N (lb_f)		Maximum SFC kg/(N-hr) ($lb_m/(lb_f$-hr))		Max Engine Rotor Speed (rpm)		Maximum Gas Temp[3/] °C(°F)		Maximum Measured Temp[4/] °C(°F)		Total Airflow[5/] kg/sec (lb_m/sec)	
	New	Det.[7/]	New	Det.[7/]	New	Det.[7/]	New	Det.[7/]	New	Det.[7/]	New	Det.[7/]
Max. Aug. (if applicable)												
Min. Aug. (if applicable)												
Int.												
90% Int.												
75% Int.												
40% Int.												
Idle	(Max)	(Max)	[6/]	[6/]	(Min)	(Min)						

NOTES:

[1/] The engine performance values shown shall be from the engine performance computer program of 3.2.1.3.

[2/] Power settings and columns shall be added or deleted as required by the Using Service (e.g., customer bleed air/power extraction).

[3/] Defined at the first stage turbine rotor inlet location (Contractor shall parenthetically insert, in column heading, the station designation).

[4/] Contractor shall parenthetically insert, in column heading, the measurement plane station designation.

[5/] Total engine airflow.

[6/] Maximum fuel consumption - kg/hr (lb_m/hr).

[7/] Deteriorated engine performance is specified for ____ TACs (or hours) of use.

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TABLE IIb. Turbofan/turbojet performance at non-standard hot ambient, sea level, static conditions[1/] with _____ nozzle.

Power Setting[2/]	Minimum Thrust N (lb_f)		Maximum SFC kg/(N-hr) ($lb_m/(lb_f$-hr))		Max Engine Rotor Speed (rpm)		Maximum Gas Temp[3/] °C(°F)		Maximum Measured Temp[4/] °C(°F)		Total Airflow[5/] kg/sec (lb_m/sec)	
	New	Det.[7/]	New	Det.[7/]	New	Det.[7/]	New	Det.[7/]	New	Det.[7/]	New	Det.[7/]
Max. Aug. (if applicable)												
Min. Aug. (if applicable)												
Int.												
90% Int.												
75% Int.												
40% Int.												
Idle	(Max)	(Max)	[6/]	[6/]	(Min)	(Min)						

NOTES:

[1/] The engine performance values shown shall be from the engine performance computer program of 3.2.1.3.

[2/] Power settings and columns shall be added or deleted as required by the Using Service (e.g., customer bleed air/power extraction).

[3/] Defined at the first stage turbine rotor inlet location (Contractor shall parenthetically insert, in column heading, the station designation).

[4/] Contractor shall parenthetically insert, in column heading, the measurement plane station designation.

[5/] Total engine airflow.

[6/] Maximum fuel consumption - kg/hr (lb_m/hr).

[7/] Deteriorated engine performance is specified for ____ TACs (or hours) of use.

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TABLE IIc. Turbofan/turbojet performance at non-standard cold ambient, sea level, static conditions 1/ with _____ nozzle.

Power Setting 2/	Minimum Thrust N (lb_f)		Maximum SFC kg (N-hr) (lb_m/(lb_f -hr))		Max Engine Rotor Speed (rpm)		Maximum Gas Temp 3/ °C(°F)		Maximum Measured Temp 4/ °C(°F)		Total Airflow 5/ kg/sec (lb_m/sec)	
	New	Det. 7/	New	Det. 7/	New	Det. 7/	New	Det. 7/	New	Det. 7/	New	Det. 7/
Max. Aug. (if applicable)												
Min. Aug. (if applicable)												
Int.												
90% Int.												
75% Int.												
40% Int.												
Idle	(Max)	(Max)	6/	6/	(Min)	(Min)						

NOTES:

1/ The engine performance values shown shall be from the engine performance computer program of 3.2.1.3.

2/ Power settings and columns shall be added or deleted as required by the Using Service (e.g., customer bleed air/power extraction).

3/ Defined at the first stage turbine rotor inlet location (Contractor shall parenthetically insert, in column heading, the station designation).

4/ Contractor shall parenthetically insert, in column heading, the measurement plane station designation.

5/ Total engine airflow.

6/ Maximum fuel consumption - kg/hr (lb_m/hr).

7/ Deteriorated engine performance is specified for ____ TACs (or hours) of use.

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TABLE IIa. Turboprop/turboshaft engine performance at standard day, 15°C (59°F), sea level static conditions[6/] with _____ nozzle.

Power setting[1/]	Max. gas generator speed(s) (rpm)		Max. gas temp.[2/] (°C(°F))		Max. measured temp.[3/] (°C(°F))		Total airflow[4/] kg/sec (lb_m/sec)		Max. SFC kg/kw-hr (lb_m/(shp-hr))		Min. shaft power kw (shp)		Rated Output torque N-m (lb_f-ft)		Rated Output shaft speed (rpm)		Resid. jet thrust N (lb_f)	
	New	Det[6/]	New	Det[6/]	New	Det[6/]	New	Det[6/]	New	Det[6/]	New	Det[6/]	New	Det[6/]	New	Det[6/]	New	Det[6/]
Contingency (if applicable)																		
Maximum (if applicable)																		
Intermediate																		
Maximum Continuous																		
75% Maximum Continuous																		
50% Maximum Continuous																		
25% Maximum Continuous																		
No load (if applicable)																		
Idle	Min.	Min.							5/	5/	Max	Max						
Reverse 7/																		

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TABLE IIa: Turboprop/turboshaft engine performance at standard day, 15°C (59°F), sea level static conditions 6/ with _____ nozzle - continued.

NOTES:

1/ Parameters and columns shall be added or deleted as required by the Using Service (e.g., customer bleed air).

2/ Defined at the first stage high pressure turbine rotor inlet location (contractor shall parenthetically insert, in column heading, the station designation).

3/ Contractor shall parenthetically insert, in column heading, the measurement plane station designation.

4/ Total engine airflow.

5/ Maximum fuel consumption - kg/hr (lb_m/hr).

6/ Deteriorated engine performance is specified for ____ TACs (or hours) use.

7/ Turboprop only.

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TABLE IIb. Turboprop/turboshaft engine performance at non-standard hot ambient sea level static conditions with _____ nozzle.

Power setting 1/	Max. gas generator speed(s) (rpm)		Max. gas temp. 2/ (°C(°F))		Max. measured temp. 3/ (°C(°F))		Total airflow 4/ kg/sec (lb_m/sec)		Max. SFC kg/kw-hr (lb_m/(shp-hr))		Min. shaft power kw (shp)		Rated Output torque N-m (lb_f-ft)		Rated Output shaft speed (rpm)		Resid. jet thrust N (lb_f)	
	New	Det 6/	New	Det 6/	New	Det 6/	New	Det 6/	New	Det 6/	New	Det 6/	New	Det 6/	New	Det 6/	New	Det 6/
Contingency (if applicable)																		
Maximum (if applicable)																		
Intermediate																		
Maximum Continuous																		
75% Maximum Continuous																		
50% Maximum Continuous																		
25% Maximum Continuous																		
No load (if applicable)																		
Idle	Min.	Min.							5/	5/	Max	Max						
Reverse 7/																		

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TABLE IIb. Turboprop/turboshaft engine performance at non-standard hot ambient sea level static conditions with _____ nozzle - continued.

NOTES:

1/ Parameters and columns shall be added or deleted as required by the Using Service (e.g., customer bleed air).

2/ Defined at the first stage high pressure turbine rotor inlet location (contractor shall parenthetically insert, in column heading, the station designation).

3/ Contractor shall parenthetically insert, in column heading, the measurement plane station designation.

4/ Total engine airflow.

5/ Maximum fuel consumption - kg/hr (lb_m/hr).

6/ Deteriorated engine performance is specified for ____ TACs (or hours) use.

7/ Turboprop only.

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TABLE IIc. Turboprop/turboshaft engine performance at non-standard cold ambient sea level static conditions with _____ nozzle.

Power setting 1/	Max. gas generator speed(s) (rpm)		Max. gas temp. 2/ (°C(°F))		Max. measured temp. 3/ (°C(°F))		Total airflow 4/ kg/sec (lb_m/sec)		Max. SFC kg/kw-hr (lb_m/(shp-hr))		Min. shaft power kw (shp)		Rated Output torque N-m (lb_f-ft)		Rated Output shaft speed (rpm)		Resid. jet thrust N (lb_f)	
	New	Det 6/	New	Det 6/	New	Det 6/	New	Det 6/	New	Det 6/	New	Det 6/	New	Det 6/	New	Det 6/	New	Det 6/
Contingency (if applicable)																		
Maximum (if applicable)																		
Intermediate																		
Maximum Continuous																		
75% Maximum Continuous																		
50% Maximum Continuous																		
25% Maximum Continuous																		
No load (if applicable)																		
Idle	Min.	Min.							5/	5/	Max	Max						
Reverse 7/																		

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TABLE IIc. Turboprop/turboshaft engine performance at non-standard cold ambient sea level static conditions with _____ nozzle – continued.

NOTES:

1/ Parameters and columns shall be added or deleted as required by the Using Service (e.g., customer bleed air).

2/ Defined at the first stage high pressure turbine rotor inlet location (contractor shall parenthetically insert, in column heading, the station designation).

3/ Contractor shall parenthetically insert, in column heading, the measurement plane station designation.

4/ Total engine airflow.

5/ Maximum fuel consumption - kg/hr (lb_m/hr).

6/ Deteriorated engine performance is specified for ____ TACs (or hours) use.

7/ Turboprop only.

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TABLE III. Turbofan/turbojet performance at altitude conditions[4/] with _____ nozzle.

Power setting[1/]	Alt. m (ft)	Amb. temp.	Mach No.	Minimum net thrust N (lb_f)		Maximum SFC $\frac{kg}{hr\text{-}N}$ ($lb_m/(hr\text{-}lb_f)$)		Maximum engine rotor speed (rpm)		Maximum gas temp.[2/] (°C(°F))		Maximum measured temp.[3/] (°C(°F))		Total airflow kg/sec (lb_m/sec)		Cust. bld. Air kg/sec (pps)	Cust. pwr. ext. kw (SHP)
				New	Det[5/]	New	Det[5/]	New	Det[5/]	New	Det[5/]	New	Det[5/]	New	Det[5/]		

NOTES:

1/ Power settings shall be as required by the Using Service to cover the operating envelope and to be compatible with mission requirements.

2/ Defined at the first stage turbine rotor inlet location (contractor shall parenthetically insert, in column heading, the station designation).

3/ Contractor shall parenthetically insert, in column heading, the measurement plane station designation.

4/ The engine performance values shown shall be from the engine computer program of 3.2.1.3.

5/ Deteriorated engine performance is specified for _____ TACs (or hours) of use.

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TABLE III. Turboprop/turboshaft performance at altitude conditions with _____ nozzle.

Power Setting 1/	Alt. m (ft)	Amb. Temp.	Mach No.	Max. Gas Gen. Speed (rpm)		Max. Gas Temp. 2/ (°C(°F))		Max. Meas. Temp. 3/ (°C(°F))		Total Airflow kg/sec (lb_m/sec)		Max. SFC kg kw-hr (lb_m/(shp-hr))		Min. shaft power kw (shp)		Output torque N-m (ft-lb_f)		Output shaft speed (rpm)		Resid. jet thrust N (lb_f)		Cust. bld. Air kg/sec (pps)		Cust. pwr. ext. kw (SHP)	
				New	Det 4/	New	Det 4/	New	Det 4/	New	Det 4/	New	Det 4/	New	Det 4/	New	Det 4/	New	Det 4/	New	Det 4/	New	Det 4/	New	Det 4/

NOTES:

1/ Power settings shall be as required by the Using Service to cover the operating envelope and to be compatible with mission requirements.

2/ Defined at the first stage high pressure turbine rotor inlet location (contractor shall parenthetically insert, in column heading, the station designation).

3/ Contractor shall parenthetically insert, in column heading, the measurement plane station designation.

4/ Deteriorated engine performance is specified for _____ TACs (or hours) of use.

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TABLE IV. Limits.

Parameter	Steady State		Transient	
	Minimum	Maximum	Minimum	Maximum
Rotor Speed				
Turbine Temperature				
Burner Pressure				
Airflow (see 3.2.2.10)				
Fuel Flow				
Vibration				
Etc.				

TABLE V. Component limiting temperature.

Component	Maximum Ambient Air Temperature	Limiting Surface Temperature	Time Limit for Temperature	Limiting Cooling Fluid Inlet Temperature
Fuel Pump				
Oil Pump				
FADEC				
Actuators				
Etc.				

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TABLE VI. Schedule of salt spray injection endurance cycles.

Phase No. 1/	Duration of Phase (hours)	Test Engine Operation	Salt Solution Parts per Billion (PPB)	Engine Ambient Air	
				Temperature	Relative Humidity
1	3 2/, 3/	Operating	200±40 4/	10°C min. 50°F min.	73% min.
2	2 5/	Not operating	0	Atmospheric	Atmospheric
3	7 5/	Not operating	200±40 4/, 6/	10°C min. 50°F min.	73% min.
4	12 5/	Not operating	0	43±5°C 109±9°F	90% min.

NOTES:

1/ Engine inlet and exhaust openings shall remain open for all phases of the test cycle.

2/ During shutdown, while the engine is decelerating from idle, the salt solution shall continue to be sprayed into the engine until the rotor has come to rest.

3/ An engine operating cycle shall be selected from the mission mix which has at least six power levels with dwell times of at least 5 minutes per level. The power levels shall roughly be equally spaced throughout the cycle. If a suitable cycle cannot be identified, then the cycle of figure 23 shall be used. The cycle shall be constructed so that the metal temperature in the hot gas flow path is between 760°C and 980°C (1400°F and 1800°F). This shall be accomplished by varying the turbine inlet temperature from 760°C (1400°F) to the maximum temperature.

4/ Salt solution ingested by the engine shall conform to that specified in Note 7/ below, and shall be regulated to provide a concentration of 200±40 PPB by weight of salt in air during those phases of each cycle which requires shalt ingestion. A salt sampling system shall be employed to determine the concentration level during each cycle of operation.

5/ The duration of phases 2, 3, and 4 can vary up to 10 percent of the time for the respective phase for any particular cycle. However, the total time for phases 1 - 4 (1 cycle) shall be 24 hours. Down time shall be considered as phase 2. Down time shall include time for engine facility maintenance, weekends, holidays, etc. The total down time shall not be less than 50 hours for 25 cycles. The maximum down time is negotiable. The minimum duration for phases 3 and 4 is 175 and 300 hours, respectively.

6/ The test facility shall provide the flow of salt-laden air through the engine gas flowpath(s) and over the external surfaces of the engine. The external flow velocity shall be between 4.6-6.1 m/sec (15-20 ft/sec).

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TABLE VI. Schedule of salt spray injection endurance cycles - continued.

7/ The basic salt formulation shall be composed of the following materials dissolved with sufficient distilled water to make one liter of salt solution. Additional distilled water may be added to the salt solution, as required, to provide a uniform salt aerosol profile across the face of the engine. The maximum dilution shall not exceed 40:1. The aerosol droplet size shall not exceed 25 microns.

Chemical Designation	Quantity per liter of stock solution
NaCl (c.p.)	23 grams
$Na_2SO4\ 10H_2O$	8 grams
Stock solution (see note 8/)	20 milliliters

8/ The stock solution shall be composed of the following materials dissolved with sufficient distilled water to make one liter of stock solution:

Chemical Designation	Quantity per liter of stock solution
KCl (c.p.)	10 grams
KBr	45 grams
$MgCl_2\ 6H_2O$ (c.p.)	550 grams
$CaCl_2\ 6H_2O$ (c.p.)	110 grams

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TABLE VII. Sea level anti-icing conditions.

	Part 1			Part 2
Ambient Temperature	-21 to -19°C (-5 to 3°F)	-10 to -9°C 1/ (14 to 16°F)	-6 to -4°C (22 to 24°F)	-6 to -4°C (22 to 24°F)
Velocity	0 to 31 m/sec (0 to 60 knots)	0 to 31 m/sec (0 to 60 knots)	0 to 31 m/sec (0 to 60 knots)	0 to 31 m/sec (0 to 60 knots)
Altitude	0 to 150 m (0 to 500 ft)	0 to 150 m (0 to 500 ft)	0 to 150 m (0 to 500 ft)	0 to 150 m (0 to 500 ft)
Mean Effective Drop Diameter	20 ±5 microns	20 ±5 microns	20 ±5 microns	30 ±5 microns
Liquid Water Content (Continuous)	1.25 to 1.75 gm/m^3	2.85 to 3.35 gm/m^3	2.85 to 3.35 gm/m^3	0.6 to 0.8 gm/m^3

NOTE:
1/ This condition applies only to turbofan engines.

TABLE VIII. Bird ingestion.

Bird Size	Number of Birds	Bird Velocity	Thrust/ Power Setting	Percent Thrust/ Power Retention	Thrust/Power Recovery Time	Damage
100 gm (3.5 oz)		Takeoff =				Blendable
100 gm (3.5 oz)		Cruise =				Blendable
100 gm (3.5 oz)		Low Level Hi-Speed =				Blendable
100 gm (3.5 oz)		Descent =				Blendable
1 kg (2.2 lbs)		Takeoff =				Minor
1 kg (2.2 lbs)		Cruise =				Minor
1 kg (2.2 lbs)		Low Level Hi-Speed =				Minor
1 kg (2.2 lbs)		Descent =				Minor
2 kg (4.4 lbs)		Takeoff or Low Level Hi-Speed =				Contain Failure

Each line of the table must be satisfied to comply with the requirements of 3.3.2.1.

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TABLE IX. Ice ingestion.

Dimensions	Number	Velocity km/hr (mph)	Thrust/ Power Setting	Percent Thrust/ Power Retention	Thrust/ Power Recovery Time	Specific Gravity g/cm^3 (lbm/ft^3)
Hailstone Diameter mm(in)						
Sheet Ice mm x mm x mm (in x in x in)						

Inlet area: _____ m^2 (_____ in^2)

TABLE X. Hot parts.

HOT PARTS	
Component	**Life**
(This list is an example and is not all inclusive. The contractor should list all applicable parts.)	
Combustor	
Turbine blades	
Turbine vanes	
Turbine disks	
Augmentor	

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TABLE XI. Design duty cycle summary.

(See MIL-HDBK-1783 for guidance and example table.)

System Category	Parts Classification	Total flight hours	Eng optg time (hrs)	Gnd run time (hrs)	No. of flts and gnd runs	Type I Cycs	Type III Cyc	Type IV Cycles	Minor cycles	Augmentor Lights or Thrust Reversing (cycles/hrs)	Vector Cycles (hrs)	Time at IRP and Above (hrs)
						0ff-Int-0ff	Idle-IRP-Idle	Cruise-IRP-Cruise				
Fighter	Cold Parts	4,000	5,400	1,400	2,973	2,973	503	22,074		20,329/186	108,938	684
	Hot Parts	2,000	2,700	700	1,487	1,487	10,252	11,037		10,165/93	54,469	342
Bomber	Cold Parts	10,000	11,000	1,000	2,700	2,700	30,000	30,000		16,000	N/A	180
	Hot Parts	4,000	4,400	400	1,350	1,350	15,000	15,000		8,000	N/A	90
Cargo	Cold Parts	20,000	22,000	2,000	10,000	10,000	14,000	TBD		N/A	N/A	6,300
	Hot Parts	10,000	11,000	1,000	5,000	5,000	7,000	TBD		N/A	N/A	3,000
Trainer	Cold Parts					15,000	150K	150K		TBD	TBD	3,600
	Hot Parts					7,500	75,000	75,000		TBD	TBD	1,800
Helicopter	Cold Parts					15,000	N/A	N/A		N/A	N/A	
	Hot Parts					15,000	N/A	N/A		N/A	N/A	

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TABLE XII. Parts classification.

(See MIL-HDBK-1783 for guidance and example table.)

Engine Part or Component	Safety Critical	Mission Critical	Durability Critical	Durability Non-critical

TABLE XIII. Reliability quantitative requirements.

Reliability parameters	Checkpoint values			Requirement	Operational Goals	
	IFR	FFR	OCR	At ISR	100K EFH	500K EFH
1. Mean Time Between Removals (MTBR) (in Engine Flight Hours EFH)						
2. Mean Time Between Failure (MTBF)						
3. In flight shutdown rate (events/1000 EFH)						
4. Mean Time Between Maintenance, Inherent (MTBMI)[1]						
5. Mean Engine Flight Hours between Maintenance Actions Unscheduled (MEFH/MAU) (USN)						
6. Mean Engine Flight Hours between Maintenance Actions Scheduled (MEFH/MAS) (USN)						
7. Line replaceable unit removal rate (events/10^3 EFH)						
8. Total engine removal rate[2] (events/10^3 EFH)						
9. Engine removal rate for exceedance of performance deterioration (event/10^3 EFH) (if applicable)						

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TABLE XIII. Reliability quantitative requirements - continued.

Reliability parameters	Checkpoint values			Requirement	Operational Goals	
	IFR	FFR	OCR	At ISR	100K EFH	500K EFH
9a. Mean Time Between Essential Maintenance Action (MTBEMA) (USA only)						
10. Mission Affecting Failure (MAF) (USA only)						
11. Mean Time Between Mission Abort (MTBMA) (USA and USN only)						

NOTES:
[1] The value specified here, in EFH, should be for inherent maintenance, which is maintenance caused by defects in design or manufacturing. The parameter includes both scheduled and unscheduled maintenance.
[2] The value should include both scheduled and unscheduled removals.

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TABLE XIV. Maintainability quantitative milestone requirements.

	Milestones				Operational Goals	
	IFR	FFR	OCR	Requirement at ISR	100K EFH	500K EFH
1. Total Maintenance ManHour per Engine Flight Hour scheduled (MMH/EFH)						
Organizational (O-level)						
Intermediate (I-level)						
Depot (D-level)	X	X				
2. Total Maintenance ManHours per Engine Flight Hour unscheduled (MMH/EFH unscheduled)						
Organizational (O-level)						
Intermediate (I-level)						
Depot (D-level)	X	X				
3. Total Maintenance ManHour per Maintenance Action (MMH/MA)						
Organizational (O-level)						
Intermediate (I-level)						
Depot (D-level)	X	X				
4. Total Engine Flight Hours per Maintenance Action (MTBMA) (EFH/MA)						
Organizational (O-level)						
Intermediate (I-level)						
Depot (D-level)	X	X				
5. Mean Time To Repair (MTTR)						
Organizational (O-level)						
Intermediate (I-level)						
Depot (D-level)	X	X				
6. Engine replacement time						
(1) Total man hours (min)						
(2) Elapsed Maintenance Time (EMT) (CLOCK-min)						
7. Weapon Replacement Assembly (WRA) replacement time-engine installed (minutes)						
(1) Weighted average						
(2) Not to exceed						
Diagnostics						
8. Percent of success that diagnostic testing can isolate faulty WRA components within X minutes.						
In-flight systems						
Ground-based systems						

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TABLE XIV. Maintainability quantitative milestone requirements - continued.

	Milestones				Operational Goals	
	IFR	FFR	OCR	Requirement at ISR	100K EFH	500K EFH
9. Percent of success that diagnostics testing can isolate faulty replaceable components within WRA assemblies.						
In-flight systems						
Ground-based systems						
10. Percent of success that diagnostic testing correctly diagnosed fault by retest.						
In-flight systems						
ground based systems						
11. Percent false alarms from diagnostic system						
In-flight systems						
Ground-based systems						
12. % WRA fault detection						
In-flight systems						
Ground-based systems						
13. % WRA component fault detection						
In-flight systems						
Ground-based systems						
14. Mean Flight hour between false alarm rate						
In-flight systems						
Ground-based systems						

TABLE XV. Maintainability quantitative requirement.

Systems	Requirements	
	Percent ATE fault isolation to faulty replaceable subassemblies within WRA/LRU assemblies	Percent retest OK to specified subassemblies

TABLE XVI. Excluded maintenance functions.

Function	Reason for Exclusion

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TABLE XVII. Maintainability qualitative checklist.

Qualitative	Requirement

TABLE XVIII. Modules.

Module Removal	
Module Name	Removal Time (hrs)

TABLE XIXa. Time to remove and replace modules.

Module or Item	Total Time	Elapsed time

TABLE XIXb. Removal and replacement times of externally-mounted components.

Item	Total Time	Elapsed time

TABLE XIXc. Module inspection period.

Module	Module Inspection Interval	Mean Time Between Repairs	
		ISR	500K EFH

TABLE XX. Tools.

Tool Set	Maintenance Level

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TABLE XXI. IR signature specification matrix.

	AZIMUTH (degrees)																		
ELEVATION	0	10	20	30	40	50	60	70	80	90	100	110	120	130	140	150	160	170	180
90																			
45																			
30																			
20																			
10																			
5																			
0																			
-5																			
-10																			
-20																			
-30																			
-45																			
-90																			

NOTES:
Signature: (Source, Apparent or contrast) Intensity (W/SR) in the ___ um to ___ um band
Operating conditions: (e.g.; Mach No, Alt, PLA, vehicle configuration, etc.)
Environmental Conditions: (e.g.; Sun AZ/El, US STD ATMOS 1976, Vis, T_{Earth}=290 K, Earth emissivity=0.7, etc.)

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TABLE XXII. RCS specification matrix (example of an estimate of the first moment at a given frequency and polarization).

	Azimuth Sector			
Elevation	**-45° to 45°**	**45° to 135°,**	**-45° to -135°**	**-135° to 135°**
-20°				
-10°				
-5°				
0°				
5°				
10°				
20°				

NOTE:
Values in table are Pcum 0 in dBsm for Given Azimuth Sector at the Given Elevation for a Frequency of _____ and _____ Polarization.

TABLE XXIII. RCS specification matrix (example of an estimate of the second moment at a given frequency and polarization).

	Azimuth Sector			
Elevation	**-45° to 45°**	**45° to 135°**	**-45° to -135**	**-135° to 135°**
-20°				
-10°				
-5°				
0°				
5°				
10°				
20°				

NOTE:
Values in table are Pcum 90 in dBsm for Given Azimuth Sector at the Given Elevation for a Frequency of _____ and _____ Polarization.

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TABLE XXIVa. Option 1 (see 3.7.3.1).

The altitude relight envelope for each major fuel-type shall be identified.

Primary fuels

Military			Commercial[1]		Fuel Type
US	UK	NATO Code	US	UK	
MIL-DTL-5624 Grade JP-5	DEF STAN 91-86 AVCAT/FSII	F-44	None	None	High Flash Kerosene
MIL-DTL-83133 Grade JP-8	DEF STAN 91-87 AVTUR/FSII	F-34	ASTM D1655 Jet A-1 with Additives[2]	None	Kerosene
Grade JP-8 +100	DEF STAN 91-87 AVTUR/FSII	F-37	ASTM D1655 Jet A-1 with Additives[2, 4]	None	Kerosene
None	None	F-35	ASTM D1655 Jet A-1 w/o Additives[3]	DEF STAN 91-91	Kerosene
None	None	None	ASTM D1655 Jet A With Additives[2]	None	Kerosene
None	None	None	ASTM D1655 Jet A w/o Additives[3]	None	Kerosene
None	None	None	GOST 10227 TS-1 w/o Additives[3]	None	Kerosene
MIL-DTL-5624 Grade JP-4	DEF STAN 91-88 AVTAG/FSII	F-40	ASTM D6615 Jet B with Additives[2]	None	Wide-Cut
None	None	None	ASTM D6615 Jet B w/o Additives[3]	None	Wide-Cut

NOTES:
1. Once treated with the additives listed in note 2 below, the commercial fuel listed in this column is virtually identical to the corresponding Military fuel.
2. The following additives must be injected into the fuel at the concentrations specified in MIL-DTL-5624 or MIL-DTL-83133:
   a. corrosion inhibitor/lubricity improver (CI/LI) (MIL-PRF-25017)
   b. fuel system icing inhibitor (FSII) (MIL-DTL-85470)
   c. an approved antioxidant (AO) material listed in section 3 of MIL-DTL-5624 or MIL-DTL-83133
   d. an approved static dissipator additive (SDA) listed in section 3 of MIL-DTL-5624 or MIL-DTL-83133
3. Commercial fuels do not normally contain any additives.
4. Use of engine company A's +100 thermal stability additive is prohibited on most Navy aircraft since shipboard fuel handling systems are not designed to handle fuel containing the +100 additive.

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TABLE XXIVb. Option 2 (see 3.7.3.1).

Primary fuels

Military			Commercial[1]		Fuel Type
US	UK	NATO Code	US	UK	
MIL-DTL-5624 Grade JP-5	DEF STAN 91-86 AVCAT/FSII	F-44	None	None	High Flash Kerosene
MIL-DTL-83133 Grade JP-8	DEF STAN 91-87 AVTUR/FSII	F-34	ASTM D1655 Jet A-1 with Additives[2]	None	Kerosene
Grade JP-8 +100	DEF STAN 91-87 AVTUR/FSII	F-37	ASTM D1655 Jet A-1 with Additives[2, 4]	None	Kerosene
None	None	F-35	ASTM D1655 Jet A-1 w/o Additives[3]	DEF-STAN 91-91	Kerosene
None	None	None	ASTM D1655 Jet A with Additives[2]	None	Kerosene
None	None	None	ASTM D1655 Jet A w/o Additives[3]	None	Kerosene
None	None	None	GOST 10227 TS-1 w/o Additives[3]	None	Kerosene

NOTES:
1. Once treated with the additives listed in note 2 below, the commercial fuel listed in this column is virtually identical to the corresponding Military fuel.
2. The following additives must be injected into the fuel at the concentrations specified in MIL-DTL-5624 or MIL-DTL-83133:
   a. corrosion inhibitor/lubricity improver (CI/LI) (MIL-PRF-25017)
   b. fuel system icing inhibitor (FSII) (MIL-DTL-85470)
   c. an approved antioxidant (AO) material listed in section 3 of MIL-DTL-5624 or MIL-DTL-83133
   d. an approved static dissipator additive (SDA) listed in section 3 of MIL-DTL-5624 or MIL-DTL-83133
3. Commercial fuels do not normally contain the mandatory Military additives.
4. Use of engine company A's +100 thermal stability additive is prohibited on most Navy aircraft since shipboard fuel handling systems are not designed to handle fuel containing the +100 additive.

Alternate fuel(s): None required.

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TABLE XXIVb. Option 2 (see 3.7.3.1) - continued.

Restricted fuels

Military			Commercial[1]		Fuel Type
US	UK	NATO Code	US	UK	
MIL-DTL-5624 Grade JP-4	DEF STAN 91-88 AVTAG/FSII	F-40	ASTM D6615 Jet B with Additives[2]	None	Wide-Cut
None	None	None	ASTM D6615 Jet B w/o Additives[3]	None	Wide-Cut

NOTES:
1. Once treated with the additives listed in note 2 below, the commercial fuel listed in this column is virtually identical to the corresponding Military fuel.
2. The following additives must be injected into the fuel at the concentrations specified in MIL-DTL-5624 or MIL-DTL-83133:
   a. corrosion inhibitor/lubricity improver (CI/LI) (MIL-PRF-25017)
   b. fuel system icing inhibitor (FSII) (MIL-DTL-85470)
   c. an approved antioxidant (AO) material listed in requirement 3.3.1.1 of MIL-DTL-5624 or MIL-DTL-83133
   d. an approved static dissipator additive (SDA) listed in requirement 3.3.5 of MIL-DTL-5624 or requirement 3.3.3 of MIL-DTL-83133
3. Commercial fuels do not normally contain any additives.

Emergency fuel(s): see requirement in 3.7.3.1.4.

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TABLE XXIVc. Option 3 (see 3.7.3.1).

Primary fuels

Military			Commercial[1]		Fuel Type
US	UK	NATO Code	US	UK	
MIL-DTL-5624 Grade JP-5	DEF STAN 91-86 AVCAT/FSII	F-44	None	None	High Flash Kerosene
MIL-DTL-83133 Grade JP-8	DEF STAN 91-87 AVTUR/FSII	F-34	ASTM D1655 Jet A-1 with Additives[2]	None	Kerosene
Grade JP-8 +100	DEF STAN 91-87 AVTUR/FSII	F-37	ASTM D1655 Jet A-1 with Additives[2, 4]	None	Kerosene
None	None	None	ASTM D1655 Jet A with Additives[2]	None	Kerosene

NOTES:
1. Once treated with the additives listed in note 2 below, the commercial fuel listed in this column is virtually identical to the corresponding Military fuel.
2. The following additives must be injected into the fuel at the concentrations specified in MIL-DTL-5624 or MIL-DTL-83133:
   a. corrosion inhibitor/lubricity improver (CI/LI) (MIL-PRF-25017)
   b. fuel system icing inhibitor (FSII) (MIL-DTL-85470)
   c. an approved antioxidant (AO) material listed in requirement 3.3.1.1 of MIL-DTL-5624 or MIL-DTL-83133
   d. an approved static dissipator additive (SDA) listed in requirement 3.3.5 of MIL-DTL-5624 or requirement 3.3.3 of MIL-DTL-83133
3. Commercial fuels do not normally contain any additives.
4. Use of engine company A's +100 thermal stability additive is prohibited on most Navy aircraft since shipboard fuel handling systems are not designed to handle fuel containing the +100 additive.

Alternate fuels

Military			Commercial[1]		Fuel Type
US	UK	NATO Code	US	UK	
None	None	F-35	ASTM D1655 Jet A-1 w/o Additives[1]	DEF-STAN 91-91	Kerosene
None	None	None	ASTM D1655 Jet A w/o Additives[1]	None	Kerosene
None	None	None	GOST 10227 TS-1 w/o Additives[1]	None	Kerosene

NOTES:
1. Commercial fuels do not normally contain any additives.

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TABLE XXIVc. Option 3 (see 3.7.3.1) - continued.

Restricted fuels

Military			Commercial[1]		Fuel Type
US	UK	NATO Code	US	UK	
MIL-DTL-5624 Grade JP-4	DEF STAN 91-88 AVTAG/FSII	F-40	ASTM D6615 Jet B with Additives[2]	None	Wide-Cut
None	None	None	ASTM D6615 Jet B w/o Additives[3]	None	Wide-Cut

NOTES:
1. Once treated with the additives listed in note 2 below, the commercial fuel listed in this column is virtually identical to the corresponding Military fuel.
2. The following additives must be injected into the fuel at the concentrations specified in MIL-DTL-5624 or MIL-DTL-83133:
   a. corrosion inhibitor/lubricity improver (CI/LI) (MIL-PRF-25017)
   b. fuel system icing inhibitor (FSII) (MIL-DTL-85470)
   c. an approved antioxidant (AO) material listed in requirement 3.3.1.1 of MIL-DTL-5624 or MIL-DTL-83133
   d. an approved static dissipator additive (SDA) listed in requirement 3.3.5 of MIL-DTL-5624 or requirement 3.3.3 of MIL-DTL-83133
Commercial fuels do not normally contain any additives.

Emergency fuel(s): See requirement in 3.7.3.1.4.

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TABLE XXVa. Fuel contaminants for continuous operation.

Contaminant	Particle Size Microns (inches)	Quantity mg/liter (g/1,000 gal)
Ferroso-Ferric Iron oxide (Fe_3O_4, Black color)	0-5 (0-1.97 x 10^{-4})	0.264 (1.0)
Magnetite	0-5 (0-1.97 x 10^{-4})	1.321 (5.0)
Ferric Iron Oxide (Fe_2O_3, Hematite)		
Prepared dirt conforming to (coarse Arizona road dust)	Mixture as follows: 0 - 5 (0-1.97 x 10^{-4}) (12%) 5 - 10 (1.97 x 10^{-4}-3.94x10^{-4}) (12%) 10 - 20 (3.94x10^{-4}-7.87 x10^{-4}) (14%) 20 - 40 (7.87 x10^{-4} – 1.57 x 10^{-3}) (23%) 40 - 80 (1.57 x 10^{-3} – 3.15 x 10^{-3}) (30%) 80 - 200 (3.15 x 10^{-3} x 7.87 x10^{-3}) (9%)	0.528 (2.0)
Cotton Linters	Staple below 7 (USDA) as ground in a Wiley mill and screened through a 4 mm (0.157 in) screen	0.0264 (0.1)
Salt water prepared by dissolving salt in distilled water or other water containing not more than 200 parts per million of total solids	4 parts by weight of NaCl 96 parts by weight of H_2O	0.01 percent by volume entrained

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TABLE XXVb. Fuel contaminants for equivalent mission time.

Contaminant	Particle Size Microns (inches)	Quantity mg/liter (g/1,000 gal)
Ferroso-Ferric Iron oxide (Fe_3O_4, Black color)	0-5 (0-1.97 x 10^{-4})	0.396 (1.5)
Magnetite	0-5 (0-1.97 x 10^{-4})	7.66 (29.0)
Ferric Iron Oxide (Fe_2O_3, Hematite)		
Crushed Quartz (turbojet, turbofan & turboprop engines)	300 - 420 (1.18 x 10^{-2}–1.65 x 10^{-2}) 150 - 300 (5.91 x 10^{-3}-1.18 x 10^{-2})	0.528 (2.00) 0.528 (2.00)
Prepared dirt conforming to (coarse Arizona road dust)	Mixture as follows: 0 - 5 (0-1.97 x 10^{-4}) (12%) 5 - 10 (1.97 x 10^{-4}-3.94x10^{-4}) (12%) 10 - 20 (3.94x10^{-4}-7.87 x10^{-4}) (14%) 20 - 40 (7.87 x10^{-4} – 1.57 x 10^{-3}) (23%) 40 - 80 (1.57 x 10^{-3} – 3.15 x 10^{-3}) (30%) 80 - 200 (3.15 x 10^{-3} x 7.87 x10^{-3}) (9%)	2.11 (8.00)
Cotton Linters	Staple below 7 (USDA) as ground in a Wiley mill and screened through a 4 mm (0.157 in) screen	0.0264 (0.1)
Salt water prepared by dissolving salt in distilled water or other water containing not more than 200 parts per million of total solids	4 parts by weight of NaCl 96 parts by weight of H_2O	0.01 percent Volume entrained

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TABLE XXVc. Fuel contamination.

Contaminant	Particle Size Microns (inches)	Quantity (gm/1,000 gal)
(a)		
Iron Fe_3O_4	0-5 (0-1.97 x 10^{-4})	14.0
Iron Fe_2O_3	0-5 (0-1.97 x 10^{-4})	14.5
Iron Fe_2O_3	5-10 (1.97 x 10^{-4}-3.94x10^{-4})	1.5
Crushed Quartz	1000-1500 (3.94x10^{-2} – 5.91 x 10^{-2})	0.25
Crushed Quartz	420-1000(1.65 x 10^{-2} - 3.94x10^{-2})	1.75
Crushed Quartz	300-420 (1.18x10^{-2}–1.65x10^{-2})	1.00
Crushed Quartz	150-300 (5.91x10^{-3}-1.18x10^{-2})	1.00
Prepared Dirt (Arizona road)	0- 5 (0-1.97 x 10^{-4}) ((12%) 5-10 (1.97 x 10^{-4}-3.94x10^{-4}) (12%) 10-20 (3.94x10^{-4}-7.87 x10^{-4}) (14%) 20-40 (7.87x0^{-4} – 1.57x10^{-3}) (23%) 40-80 (1.57x10^{-3} – 3.15x10^{-3}) (30%) 80-200 (3.15x10^{-3} x 7.87x10^{-3}) (9%)	8.00
Cotton Linters	Staple below 7	0.10
Napthenic Acid	-	0.03% by vol.
Salt Water	-	0.01% by vol.
(b)		
Iron Fe_3O_4	0-5 (0-1.97 x 10^{-4})	0.90
Iron Fe_2O_3	5-10 (1.97 x 10^{-4}-3.94x10^{-4})	4.73
Prepared Dirt (Arizona road)	Same as above	1.90
Cotton Linters	Staple below 7	0.02

NOTES:
(a) Helicopter engines shall operate satisfactorily with these amounts with filter not in bypass, use for first 5 hours of the 50-hour fuel system test.
(b) Use these amounts for last 45 hours of the 50-hour fuel system test.

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TABLE XXVI. Fuel filter filtration rating.

Filter		Filtration Microns or Mesh	Liter (gal) to Impending Bypass	Liter (gal) Impending Bypass to Bypass	Capacity
Type	Location				

TABLE XXVII. Oil reservoir capacities.

Usable Oil	liters	(_____quarts)
Gulping Volume Oil	liters	(_____quarts)
Unusable Oil	liters	(_____quarts)
Expansion Space (equivalent oil)	liters	(_____quarts)
Total Enclosed Reservoir Volume (equivalent oil)	liters	(_____quarts)

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TABLE XXVIIIa. Accelerated mission duty cycles.

Endurance test cycle profiles	TOTALS PER MISSION								
	Type I cycle	Type III cycle	Type IV cycle	A/B lights	A/B time (min.)	Int time (min.)	Hot time (min.)	Total time (min.)	TAC per cycle
Mission #1									
Mission #2									
Mission #3									
Mission (etc.)									
Composite									
Ground Test									

NOTE:
For turboprop and turboshaft engines, the ‘‘A/B lights” and ‘‘A/B time” columns should be deleted. The ‘‘Hot time” column includes time at intermediate and above augmented for turbofan and turbojet engines, and includes the contingency ratings for turboshaft and turboprop engines.

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TABLE XXVIIIb. Accelerated mission duty cycle.

Endurance test mission profiles	TOTAL AMTs FOR _____ HOURS								
	# AMT missions per ___ hrs	Total TAC	Type I cycle	Type III cycle	Type IV cycle	Augmentor lights	Augmentor time (hrs)	TAIAA (hrs)	Total run time (hrs)
Mission #1									
Mission #2									
Mission #3									
Mission (etc.)									
Composite									
Ground test									
Totals									

NOTE:
For turboprop and turboshaft engines, the ‘‘Augmentor lights” and ‘‘Augmentor time” columns should be deleted. The TAIAA (Time at Intermediate and Above) column for turbofan and turbojet engines includes time at intermediate and above (augmented), for turboshaft and turboprop engines it includes contingency ratings.

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TABLE XXIX. Durability test runs for turboprop or turboshaft engines.

The engine shall be tested with turbine temperature equal to the maximum operating or control limiting value at each power setting. The engine control shall be adjusted to permit operation at maximum allowable steady-state gas temperature at the first stage turbine rotor inlet. The relationship between measured gas temperature and T4.1 shall be determined prior to test and rechecked every tenth cycle. The control shall be operated in failure mode during specified runs and cycles. Each cycle shall consist of the following runs a -k:

a. Maximum-idle run (60-min). -Six periods of ten minutes each, (five minutes at maximum power test temperature and delivered shaft speed for maximum power in table IIa, followed by five minutes at idle). If engine has anti-icing, it is operated at maximum anti-icing bleed for one minute at the end of each period with maximum power test temperature. On each fifth cycle of this test run, the control is operated in the failure mode for the last three minutes of each five minute period at maximum power test temperature.

b. Incremental Torque Run. This run shall consist of 96 minutes including:

(1) Eight minutes at the maximum delivered shaft speed and the delivered shaft torque available at maximum test temperature setting or at the maximum continuous torque limit, whichever is less.

(2) Eight minutes at the minimum delivered shaft speed and the delivered shaft torque available at maximum test temperature setting or at the maximum continuous delivered shaft torque limit, whichever is less.

(3) Eight minutes at the minimum delivered shaft speed and 90 percent of the maximum continuous delivered shaft torque limit.

(4) Eight minute at the maximum delivered shaft speed and 90 percent of the maximum continuous delivered shaft torque limit.

(5) Eight minutes at the maximum delivered shaft speed and 80 percent of the maximum continuous delivered shaft torque limit.

(6) Eight minutes at the minimum delivered shaft speed and 80 percent of the maximum continuous delivered shaft torque limit.

(7) Four minutes at the minimum delivered shaft speed and 60 percent of the maximum continuous delivered shaft torque limit.

(8) Four minutes at the maximum delivered shaft speed and 60 percent of the maximum continuous delivered shaft torque limit.

(9) Four minutes at the maximum delivered shaft speed and 40 percent of the maximum continuous delivered shaft torque limit.

(10) Four minutes at the minimum delivered shaft speed and 40 percent of the maximum continuous delivered shaft torque limit.

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TABLE XXIX. Durability test runs for turboprop or turboshaft engines - continued.

(11) Four minutes at the minimum delivered shaft speed and 20 percent of the maximum continuous delivered shaft torque limit.

(12) Four minutes at the maximum delivered shaft speed and 20 percent of the maximum continuous delivered shaft torque limit.

(13) Four minutes at the maximum delivered shaft speed and 10 percent of the maximum continuous delivered shaft torque limit.

(14) Four minutes at the minimum delivered shaft speed and 10 percent of the maximum continuous delivered shaft torque limit.

(15) Four minutes at the minimum delivered shaft speed and at zero delivered shaft torque.

(16) Four minutes at the maximum delivered shaft speed and at zero delivered shaft torque.

(17) Four minutes at idle.

(18) Four minutes at idle and at zero delivered shaft speed.

If the test data indicates the existence of critical compressor or turbine vibration conditions within the operating speed range of the engine between idle and maximum test temperature setting, at the option of the Government, the following shall be substituted for 48 minutes of the incremental torque run of each cycle to be chosen by the Government.

(1) Sixteen minutes at gas generator rotor and delivered shaft speeds which produce the critical vibration conditions.

(2) Eight minutes at gas generator rotor and delivered shaft speeds 2.0 percent more than those which produce the critical vibration conditions.

(3) Eight minutes at a gas generator rotor speed 2.0 percent more and a delivered shaft speed 2.0 percent less than those which produce the critical vibration conditions.

(4) Eight minutes at gas generator rotor and delivered shaft speeds 2.0 percent less than those which produce the critical vibration conditions.

(5) Eight minutes at gas generator rotor speed 2.0 percent less and delivered shaft speed 2.0 percent more than those which produce the critical vibration conditions.

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TABLE XXIX. Durability test runs for turboprop or turboshaft engines - continued.

c. Power transient run (39-min) -
(1) Four minutes at no-load followed by 1 minute at maximum power test temperature and delivered shaft speed for maximum power in table IIa, repeat for 20-minutes.
(2) One minute at maximum test temperature and delivered shaft speed for maximum power in table IIa, followed by immediate decrease to no-load, followed by an increase to speed and temperature of previous condition, hold for one minute, repeat sequence for 19-minutes.

d. Reverse pitch operation (9-min) -
One minute at idle, three minutes at maximum reverse, one minute at maximum power test temperature and delivered shaft speed for maximum power in table IIa, three minutes at maximum reverse, and 1 minute at idle. If engine does not have reverse, substitute intermediate power test temperature for maximum reverse.

e. Intermediate run (30-min) -
(1) Fifteen minutes at intermediate test temperature at maximum allowable output shaft speed.
(2) Fifteen minutes at intermediate test temperature at minimum delivered shaft speed and delivered torque as high as intermediate power in table IIa.

On every fifth cycle of this run, the first 26-minutes shall be with the control in the failure mode.

f. Maximum continuous run (20-min) -
(1) Ten minutes at maximum continuous test temperature at maximum allowable output shaft speed.
(2) Ten minutes at maximum continuous test temperature at minimum delivered shaft speed and delivered torque as high as maximum continuous power in table IIa.

g. Intermediate run (15-min) -
Fifteen minutes at intermediate test temperature and intermediate power delivered shaft speed.

h. Maximum continuous run (15-min) -
Fifteen minutes at maximum continuous test temperature and maximum continuous power delivered shaft speed in table IIa.

i. Intermediate-maximum run (30-min) -
Five minutes at intermediate test temperature and intermediate power delivered shaft speed in table IIa followed by five minutes at maximum test temperature and maximum power delivered shaft speed in table IIa, repeat sequence for 30-minutes.

j. Maximum continuous run (16-min) -
Sixteen minutes at maximum continuous test temperature and maximum continuous delivered shaft speed in table IIa.

k. Intermediate-maximum run (30-min) -
(1) Fifteen minutes at intermediate test temperature and intermediate power delivered shaft speed in table IIa.

(2) Ten minutes at maximum power test temperature and maximum power delivered shaft speed in table IIa, followed by five minutes at intermediate test temperature and Intermediate power delivered shaft speed in table IIa. At five minute intervals, anti-icing bleed shall be operated at maximum rate for one minute.

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TABLE XXIX. Durability test runs for turboprop or turboshaft engines - continued.

l. Contingency power qualification (67-min) (if applicable) -
5.6-minutes at maximum continuous followed by 5.6-minutes at contingency followed by 5.6- minutes at maximum continuous, repeat sequence four times.

Notes:

(1) Test turbine temperatures used in the Durability Test are based on the maximum operating or control limiting temperature for each power setting. For Army helicopter engines, this is equivalent to the rating temperature values plus a gas temperature margin for control tolerances and deterioration. The test temperature at the highest steady-state power setting should be equal to maximum allowable temperature. Following the calibration run, the engine shall be adjusted to permit operation at the maximum allowable temperature at the first stage turbine rotor inlet. This value shall be reestablished at the beginning of each cycle. The number of adjustments required shall be recorded.

(2) The Contingency Qualification test is performed after completion of the Durability test.

(3) Air passing over the engine shall correspond to the engine inlet air temperature for that mission condition, within facility constraints.

(4) Idle power, unless otherwise specified, is with minimum self-susaining rotor speeds.

(5) Power setting load demand changes shall be made within 0.5 seconds, with engine control and fuel flow schedules set for maximum acceleration. The time for changing power shall be charged to the duration of the lower setting.

(6) Each cycle shall be preceded by a 2-hour shutdown period. The test cycle shall be run in the order above for ‘‘a-k”.

(7) If the engine does not have maximum power, then intermediate shall be substituted for maximum power on those runs.

(8) Unless otherwise specified, the engine control shall be operated in the failure mode on every fifth repetition of the test sequence, each channel in a multi-channel or multi-unit electronic control shall be tested.

(9) Internal water wash provisions shall be demonstrated on every tenth repetition of the test sequence.

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TABLE XXIX. Durability test runs for turboprop or turboshaft engines - continued.

(10) For the purpose of the endurance test, minimum delivered shaft speed shall be interpreted as that delivered shaft speed at which the delivered shaft torque is not less than the torque limit for the applicable power condition or the minimum delivered shaft governing speed, whichever occurs first. Maximum delivered shaft speed shall be interpreted as the delivered shaft maximum speed limit or the delivered shaft maximum governing speed, whichever occurs first. Maximum allowable speed is defined in 3.2.1.4.6. Maximum continuous torque limit and maximum continuous delivered shaft torque limit shall be interpreted as the maximum allowable torque, defined in 3.2.1.4.11.

(11) The Durability Test shall be run with the inlet air temperature producing the highest operating temperature for each power setting. If this does not occur with table IIa conditions, the air temperature shall be the lowest value producing the highest operating temperatures for the test sequence. A single air temperature shall be used for test runs ‘‘a-k” with another value for ‘‘1” if needed. Alternately, the test sequence may use a table II condition specified by the Using Service.

(12) Power settings not corresponding to those above may be qualified separately like contingency, or substituted for other power settings as appropriate. Some examples are various power settings for OEI (One Engine Inoperative). The amount of test time and the test plan shall be approved by the Using Service.

(13) The engine shall operate at the measured gas temperature that corresponds to the T4.1. The relationship between measured gas temperature and T4.1 shall be established via a 4-point performance calibration at the contingency, maximum, intermediate and maximum continuous ratings, prior to the test and checked at the conclusion of every 10th cycle (+/-2 cycles). A base-level installed performance calibration shall be conducted at 15°C (59°F) if either the following conditions exist when run a., in the procedure above maximum power data points are compared with run i., above in this table maximum power data points.

- SHP at measured gas temperature decreases greater than or equal to 1.0 percent
- gas generator speed at measured gas temperature changes greater than or equal to 1.0 percent

(14) The fuel manifold and combustible fluid drainage provisions shall be demonstrated after a false start and also after a shutdown from normal operation. Inlet fuel shall be maintained at minimum specified fuel pressure throughout the test. During five successive cycles prior to the tenth cycle and five successive cycles after the 40th cycle of the endurance test, the temperature of the fuel shall be maintained at the maximum temperature specified in table IV. The inlet fuel temperature and pressure conditions shall be within the operating limits of table IV.

(15) The amount of bleed air required for continuous engine operation and the maximum permissible customer and anti-icing bleed air flow during the cycles, including bleed air temperatures and pressures, shall be stated in the test report.

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TABLE XXIX. Durability test runs for turboprop or turboshaft engines - continued.

(16) For all operations during cycles 1-5, 11-15, 21-30, 41-50, 51-60, 71-80, 91-100, the oil temperature shall be maintained at no less than the maximum oil temperature specified in table IV. The oil pressure, if adjustable, shall be adjusted at the beginning of the test to the minimum steady-state value specified in table IV for the test in which MIL-PRF-23699 oil will be used and the maximum steady-state value specified in table IV for the test in which MIL-PRF-7808 oil will be used. No further adjustments shall be permitted during the test except when authorized by the Using Service. Oil consumption shall be determined and reported after each block of 10 cycles (or at the end of the test if the test is less then 10 cycles). Oil shall be added on an as needed basis as indicated by the oil tank sight glass or the low oil sensor. Oil drained for analysis shall not be charged to engine oil consumption and shall be replaced by an equivalent amount of new oil.

(17) The exhaust duct, inlet duct, and bleed air connections shall be loaded to the maximum allowable loads specified in 3.1.1.9 and in the pretest data of 4.12.4.1.1; respectively; at the discretion of the Using Service.

(18) Accessory pads shall be subjected to rated loads and overhung moments as specified in table I. The actual torque loading and overhung moments imposed during the endurance test shall be stated in the test report.

(20) If the engine is supplied with an infrared suppression system, it shall operate continuously during runs "e" and "f".

(21) If the engine provides special features such as fuel heaters, indicator lights, and switching functions, these items shall be actuated during selected test runs.

(22) The angular misalignment of the power absorber drive shaft to the engine output shaft shall not be less than the maximum allowable angular misalignment specified in 3.1.1.9.

(23) The engine internal washing provisions shall be demonstrated once every 10 cycles. The procedure for demonstration of the internal washing provisions shall be as specified in the pretest data of 4.12.4.1.1.

(24) At the completion of the endurance runs, the engine oil drain provisions specified in 3.7.8.2.2 shall be demonstrated. This demonstration shall also verify the adequacy and the locations for the magnetic chip detectors.

(25) At the end of each cycle, special engine control system features (e.g.; overspeed, overtemperature, torque matching) shall be cycled throughout their functions (full range to the extent feasible).

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TABLE XXIX. Durability test runs for turboprop or turboshaft engines - continued.

(26) Starts:

For IFR:

A minimum of 50 starts, commencing with the first test cycle shall be made on each endurance test engine. In addition to the 50 endurance test starts, there shall be five false starts (a starting sequence without benefit of light-off followed immediately after the permissible engine draining time by a successful start), and ten restarts (a start within a maximum of 14 minutes time from shutdown). Starts shall be performed with a starter that performs within 5% of minimum starter torque curve on figure 2. The engine shall be started and shutdown not less than six times each test cycle. Of the at least 50 endurance starts, 25 shall be accomplished following varied regulated shutdown periods. Those starts at the beginning of each endurance cycle shall follow a shutdown period of at least two hours. The shutdown period for nine starts shall be regulated to provide intervals between starts of 5.0 minutes, 10 minutes, and increasing, thereafter, by 5.0 minute increments up to and including 45 minutes for the 9th start. Each of the 9 regulated shutdown periods shall be preceded by immediate engine shutdown without being held at idle, after engine operation for a duration of not less than two minutes at intermediate test temperature setting. For eleven starts, the shutdown period shall be regulated to provide an interval between starts of no less than 45 minutes. The shutdown period for the remaining at least 25 starts need not be controlled. During all endurance starts, immediately after the engine has reached stabilized idle speed, an acceleration to the next scheduled endurance test condition shall be accomplished by a control power signal change from idle to the appropriate test temperature setting.

For FFR:

A minimum of 100 starts, commencing with the first test cycle, shall be made on each endurance test engine. In addition to the 100 endurance test starts, there shall be ten false starts (a starting sequence without benefit of light-off followed immediately after the permissible engine draining time by a successful start), and ten restarts (a start within a maximum of 14 minutes time from shutdown). Starts shall be performed with a starter that performs within 5% of minimum starter torque curve. The engine shall be started and shutdown not less than six times each test cycle. Of the at least 100 endurance starts, 50 shall be accomplished following varied regulated shutdown periods. Those starts at the beginning of each endurance cycle shall follow a shutdown period of at least two hours. The shutdown period for 18 starts shall be regulated to provide intervals between starts of 5.0 minutes, 10 minutes, and increasing, thereafter, by 5.0 minute increments up to and including 90 minutes for the 18th start. Each of the 18 regulated shutdown periods shall be preceded by immediate engine shutdown without being held at idle, after engine operation for a duration of not less than two minutes at intermediate test temperature setting. For 22 starts, the shutdown period shall be regulated to provide an interval between starts of no less than 45 minutes. The shutdown period for the remaining at least 50 starts need not be controlled. During all endurance starts, immediately after the engine has reached stabilized idle speed, an acceleration to the next scheduled endurance test condition shall be accomplished by a control power signal change from idle to the appropriate test temperature setting.

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TABLE XXIX. Durability test runs for turboprop or turboshaft engines - continued.

For ISR:

A minimum of 600 starts, commencing with the first test cycle shall be made on each endurance test engine. In addition to the 600 endurance test starts, there shall be 20 false starts (a starting sequence without benefit of light-off followed immediately after the permissible engine draining time by a successful start) and 20 restarts (a start within a maximum of 14 minutes time from shutdown). Starts shall be performed with a starter that is acceptable to the Using Service. The engine shall be started and shutdown not less than six times each endurance test cycle. Of the at least 600 endurance starts, 220 shall be accomplished following varied regulated shutdown periods. Those starts at the beginning of each endurance cycle shall follow a shutdown period of at least two hours. The shutdown period for 36 starts shall be regulated to provide intervals between starts of 2.5 minutes, 5.0 minutes, 7.5 minutes, and increasing, thereafter, by 2.5-minute increments up to and including 90 minutes for the 36th start. Each of the 36 regulated shutdown periods shall be preceded by immediate engine shutdown, without being held at idle, after engine operation for a duration of not less than two minutes at intermediate test temperature setting. For 134 starts, the shutdown period shall be regulated to provide an interval between starts of no less than 45 minutes. The shutdown period for the remaining at least 380 starts need not be controlled. During all endurance starts, immediately after the engine has reached stabilized idle speed, an acceleration to the next scheduled endurance test condition shall be accomplished by a control power signal change from idle to the appropriate test temperature setting.

For OCR:

A minimum of 1200 starts, commencing with the first test cycle shall be made on each endurance test engine. In addition to the 1200 endurance test starts, there shall be 40 false starts (a starting sequence without benefit of light-off followed immediately after the permissible engine draining time by a successful start) and 40 restarts (a start within a maximum of 14 minutes time from shutdown). Starts shall be performed with a starter that is acceptable to the Using Service. The engine shall be started and shutdown not less than six times each endurance test cycle. Of the at least 1200 endurance starts, 440 shall be accomplished following varied regulated shutdown periods. Those starts at the beginning of each endurance cycle shall follow a shutdown period of at least two hours. The shutdown period for 72 starts shall be regulated to provide intervals between starts of 2.5 minutes, 5.0 minutes, 7.5 minutes, and increasing, thereafter, by 2.5 minute increments up to and including 90 minutes for the 72nd start. Each of the 72 regulated shutdown periods shall be preceded by immediate engine shutdown, without being held at idle, after engine operation for a duration of not less than two minutes at intermediate test temperature setting. For 268 starts, the shutdown period shall be regulated to provide an interval between starts of no less than 45 minutes. The shutdown period for the remaining at least 760 starts need not be controlled. During all endurance starts, immediately after the engine has reached idle speed, an acceleration to the next scheduled endurance test condition shall be accomplished by selecting the control power signal for the appropriate test temperature setting.

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TABLE XXIX. Durability test runs for turboprop or turboshaft engines - continued.

If the engine has a contingency power rating, this rating will be verified by four periods of operation at contingency power test temperature. Each operation at contingency power will be preceded by and followed by a period of operation at maximum continuous power. The duration of each period of operation will be $3(t^{0.677})$ where "t" is the rating time for contingency power. All times are in minutes. (Therefore, a 5.6 minute period for a 2.5 minute rating). This will be demonstrated on the turboprop and turboshaft durability test engine after completion of the turboprop and turboshaft engine durability test runs. (The period of operation defined by $3(t^{0.677})$ is based on engine testing.)

If the engine has an emergency power capability, this capability will be verified by two periods of operation at emergency power test temperature. Each operation at emergency power will be preceded by and followed by a period of operation at maximum continuous power. The duration of each period of operation will be $3(t^{0.677})$ where "t" is the capability time for emergency power. All times are in minutes. This will be demonstrated on the turboprop and turboshaft durability test engine after completion of the turboprop and turboshaft engine durability test runs.

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TABLE XXX. Controls and Externals (C&E) component test.

Assembly	Missionized/ operational test	Environmental tests						
		shock	vibration	humidity	fungus	sand and dust	Sustained acceleration	explosion proof
variable vanes assembly								
load control valve								
surge control valve								
bleed flow sensor assembly								
starter/-generator								
alternator								
Ignition system								
fuel control assembly								

(T) = test, (TA) = test conducted as part of an assembly, (S) = similarity,
(H) = hermetically sealed, (N/A) = not applicable

NOTE: Environmental tests must be conducted in sequence from left to right as presented on the table. The same physical component must be used for the entire sequence of environmental tests.

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TABLE XXX. Controls and Externals (C&E) component test - continued.

Assembly	Individual Tests			Reliability and Maintenance Test			
	fire	cavitation	low lubricity	sneak circuit analysis	environmental stress screening	reliability growth test	built in test
variable vanes assembly							
load control valve							
surge control valve							
bleed flow sensor assembly							
starter/ generator							
alternator							
Ignition system							
fuel control assembly							

(T) = test, (TA) = test conducted as part of an assembly, (S) = similarity,
(H) = hermetically sealed, (N/A) = not applicable

NOTE: Environmental tests must be conducted in sequence from left to right as presented on the table. The same physical component must be used for the entire sequence of environmental tests.

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TABLE XXX. Controls and Externals (C&E) component test - continued.

Assembly	Missionized /operational test	Environmental tests						
		shock	vibration	humidity	fungus	sand and dust	sustained acceleration	explosion proof
fuel distribution system								
oil system								
hydraulic system								
Engine history recorder or counter								
miscellaneous controls and accessories								
Accessory drive/PTO								
Anti-icing system								
Inlet Particle Separator								
Plumbing								

(T) = test, (TA) = test conducted as part of an assembly, (S) = similarity,
(H) = hermetically sealed, (N/A) = not applicable

NOTE: Environmental tests must be conducted in sequence from left to right as presented on the table. The same physical component must be used for the entire sequence of environmental tests.

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TABLE XXX. Controls and Externals (C&E) component test - continued.

Assembly	Individual Tests			Reliability and Maintenance Test			
	fire	cavitation	low lubricity	sneak circuit analysis	environmental stress screening	reliability growth test	built in test
fuel distribution system							
oil system							
hydraulic system							
Engine history recorder or counter							
miscellaneous controls and accessories							
Accessory drive/PTO							
Plumbing							

(T) = test, (TA) = test conducted as part of an assembly, (S) = similarity, (H) = hermetically sealed, (N/A) = not applicable

NOTE:
Environmental tests must be conducted in sequence from left to right as presented on the table. The same physical component must be used for the entire sequence of environmental tests.

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TABLE XXXIa. Spiral development matrix.

(To be completed by the specification developer; entries represent a sample guidance.)

Reqt. No.	Reqt. Title	Concept Refinement	Technology Development	System Development & Demonstration (SDD)	Production & Deployment	Operations and Support
¾	VERIFICATION AND QUALIFICATION (TITLE)	Spiral 1, 2, … (See tables XLVIII and XLIX)	Spiral 1, 2, … (See tables XLIX and L)	Spiral 2, 3, … (See table XXXIb)	Spiral 3, 4, … (See table XXXIb)	Spiral 4, 5, …
3/4.1	Design and construction (title)					
3/4.1.1	Item and interface definition (title)					
3/4.1.1.1	Interface and installation					
3/4.1.1.2	Installation hardware					
3/4.1.1.3	Interface loads					
3/4.1.1.4	Mounts (title)					
3/4.1.1.4.1	Main mounts					
3/4.1.1.4.2	Ground handling mounts					
3/4.1.1.4.3	Engine stiffness					
3/4.1.1.5	Control lever and signals (title)					
3/4.1.1.5.1	Power lever and signal					
3/4.1.1.5.2	Load demand command and signal					
3/4.1.1.5.3	Output speed lever and signal					
3/4.1.1.5.4	Fuel shutoff lever and signal					
3/4.1.1.5.5	Lever torque					
3/4.1.1.6	Sub- system interfaces					
3/4.1.1.7	Bleed air interface					
3/4.1.1.7.1	Customer bleed air contamination					

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¾	VERIFICATION AND QUALIFICATION (TITLE)	Spiral 1, 2, … (See tables XLVIII and XLIX)	Spiral 1, 2, … (See tables XLIX and L)	Spiral 2, 3, … (See table XXXIb)	Spiral 3, 4, … (See table XXXIb)	Spiral 4, 5, …
3/4.1.1.8	Drains and fluid collection devices					
3/4.1.1.9	Power absorber interface					
3/4.1.1.10	Power take-off (PTO)					
3/4.1.1.11	Starting drivetrain					
3/4.1.1.11.1	Starting torque and speed					
3/4.1.1.12	Thrust reverser interface					
3/4.1.1.13	Exhaust system interface					
3/4.1.1.14	Compatibility - Engine Airframe					
3/4.1.1.15	Control and External (C&E) component list					
3/4.1.2	Physical characteristics (title)					
3/4.1.2.1	Dry mass (weight) of complete engine					
3/4.1.2.2	Mass moment of inertia of complete engine					
3/4.1.3	Materials, processes and parts					
3/4.1.3.1	Hazardous Materials					
3/4.1.4	Fasteners					
3/4.1.5	Nameplate and product marking					
3/4.1.5.1	Engine data plate marking					

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¾	VERIFICATION AND QUALIFICATION (TITLE)	Spiral 1, 2, … (See tables XLVIII and XLIX)	Spiral 1, 2, … (See tables XLIX and L)	Spiral 2, 3, … (See table XXXIb)	Spiral 3, 4, … (See table XXXIb)	Spiral 4, 5, …
3/4.1.5.2	Critical parts identification and tracking					
3/4.1.6	Transportability					
3/4.1.7	Interchangeability					
3/4.1.8	Safety (title)					
3/4.1.8.1	Flammable fluid systems					
3/4.1.8.2	Fire prevention					
3/4.1.8.3	Explosion-proof					
3/4.1.8.4	Combustible fluid drains					
3/4.1.8.5	Air and gas leakage					
3/4.1.8.6	Ground Safety					
3/4.1.8.7	Jet Wake					
3/4.1.9	Design control (title)					
3/4.1.9.1	Parts list					
3/4.1.9.2	Assembly of components and parts					
3/4.1.9.3	Changes in vendors or fabrication process					
3/4.1.9.4	Standardization					
3/4.2	Performance and operability (title)					
3/4.2.1	Performance characteristics (title)					
3/4.2.1.1	Steady-state performance					
3/4.2.1.2	Transient performance					

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¾	VERIFICATION AND QUALIFICATION (TITLE)	Spiral 1, 2, … (See tables XLVIII and XLIX)	Spiral 1, 2, … (See tables XLIX and L)	Spiral 2, 3, … (See table XXXIb)	Spiral 3, 4, … (See table XXXIb)	Spiral 4, 5, …
3/4.2.1.3	Performance computer program					
3/4.2.1.4	Performance retention					
3/4.2.2	Operating characteristics and limits					
3/4.2.2.1	Operating envelope					
3/4.2.2.2	Operating attitude and conditions					
3/4.2.2.3	Starting					
3/4.2.2.3.1	Ground starts					
3/4.2.2.3.2	Air starts					
3/4.2.2.3.3	Starting limits					
3/4.2.2.3.4	Starting procedure					
3/4.2.2.3.5	Automatic relight					
3/4.2.2.4	Stopping					
3/4.2.2.5	Low power conditions (title)					
3/4.2.2.5.1	Idle thrust/power - (ground/flight idle)					
3/4.2.2.5.2	No load condition					
3/4.2.2.6	Stability					
3/4.2.2.7	Transients					
3/4.2.2.7.1	Overshoot and undershoot					
3/4.2.2.7.2	Droop					
3/4.2.2.8	Windmilling					
3/4.2.2.9	Reverse thrust					
3/4.2.2.10	Steady-state and transient airflow					

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¾	VERIFICATION AND QUALIFICATION (TITLE)	Spiral 1, 2, … (See tables XLVIII and XLIX)	Spiral 1, 2, … (See tables XLIX and L)	Spiral 2, 3, … (See table XXXIb)	Spiral 3, 4, … (See table XXXIb)	Spiral 4, 5, …
3/4.2.2.11	Inlet airflow distortion					
3/4.2.2.11.1	Pressure and temperature rate of change					
3/4.2.2.11.2	Engine Recoverability					
3/4.2.2.12	Gas path and measurement plane temperature limits					
3/4.2.2.13	Surface temperature and heat rejection					
3/4.2.2.13.1	Controls and external component limiting temperature					
3/4.3	Environmental conditions [title only]					
3/4.3.1	Atmospheric conditions [title only]					
3/4.3.1.1	Humidity					
3/4.3.1.2	Fungus					
3/4.3.1.3	Corrosive atmosphere					
3/4.3.1.4	Icing conditions					
3/4.3.2	Ingestion capability (hazard resistance) [title only]					
3/4.3.2.1	Bird ingestion					
3/4.3.2.2	Foreign object damage (FOD)					
3/4.3.2.3	Ice ingestion					

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¾	VERIFICATION AND QUALIFICATION (TITLE)	Spiral 1, 2, … (See tables XLVIII and XLIX)	Spiral 1, 2, … (See tables XLIX and L)	Spiral 2, 3, … (See table XXXIb)	Spiral 3, 4, … (See table XXXIb)	Spiral 4, 5, …
3/4.3.2.4	Sand and dust ingestion					
3/4.3.2.5	Atmospheric liquid water ingestion					
3/4.3.2.6	Armament gas ingestion					
3/4.3.2.7	Steam ingestion					
3/4.3.3	Electromagnetic environmental effects (E^3)					
3/4.3.3.1	Electromagnetic interference (EMI)					
3/4.3.3.2	Intrasystem electromagnetic compatibility (EMC)					
3/4.3.3.3	Intersystem electromagnetic compatibility (EMC)					
3/4.3.3.4	Lightning					
3/4.4	Integrity (title only)					
4.3/4.4.4.1	Structural integrity (title only)					
3/4.4.1.1	Design service life					
3/4.4.1.2	Design Usage					
3/4.4.1.2.1	Externally applied forces					
3/4.4.1.3	Material characterization					
3/4.4.1.4	Parts classification					
3/4.4.1.5	Durability					

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¾	VERIFICATION AND QUALIFICATION (TITLE)	Spiral 1, 2, … (See tables XLVIII and XLIX)	Spiral 1, 2, … (See tables XLIX and L)	Spiral 2, 3, … (See table XXXIb)	Spiral 3, 4, … (See table XXXIb)	Spiral 4, 5, …
3/4.4.1.5.1	High cycle fatigue (HCF) life					
3/4.4.1.5.2	Low cycle fatigue (LCF) life					
3/4.4.1.5.3	Creep					
3/4.4.1.6	Strength					
3/4.4.1.6.1	Factors of safety					
3/4.4.1.6.2	Blade and disk deflection					
3/4.4.1.6.3	Containment					
3/4.4.1.6.4	Blade out					
3/4.4.1.6.5	Overspeed and overtemperature					
3/4.4.1.6.6	Disk burst speed					
3/4.4.1.6.7	Output shaft torque limits					
3/4.4.1.6.8	Output shaft speed limits					
3/4.4.1.6.9	Pressure balance					
3/4.4.1.6.10	Gyroscopic moments					
3/4.4.1.7	Damage tolerance					
3/4.4.1.7.1	Residual Strength					
3/4.4.1.7.2	Initial production and in-service flaw size					
3/4.4.1.7.3	Flaw growth and inspection intervals					
3/4.4.1.8	Vibration and dynamic response					
3/4.4.1.8.1	Vibration limits					
3/4.4.1.8.2	Critical speeds					

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Reqt. No.	Reqt. Title	Concept Refinement	Technology Development	System Development & Demonstration (SDD)	Production & Deployment	Operations and Support
¾	VERIFICATION AND QUALIFICATION (TITLE)	Spiral 1, 2, … (See tables XLVIII and XLIX)	Spiral 1, 2, … (See tables XLIX and L)	Spiral 2, 3, … (See table XXXIb)	Spiral 3, 4, … (See table XXXIb)	Spiral 4, 5, …
3/4.4.1.9	External surface foreign object damage					
3/4.4.2	Mechanical equipment and subsystem integrity					
3/4.4.3	Avionic and electronic integrity					
3/4.5	R&M [title only]					
3/4.5.1	Reliability [title only]					
3/4.5.1.1	Reliability quantitative requirements					
3/4.5.2	Maintainability [title only]					
3/4.5.2.1	Maintainability quantitative requirements					
3/4.5.2.1.1	Excluded maintenance functions					
3/4.5.2.2	Maintainability qualitative requirements					
3/4.5.2.3	Maintenance [title only]					
3/4.5.2.3.1	Modules					
3/4.5.2.3.2	Maintenance, inspection and repair cycle					
3/4.5.2.3.2.1	Maintenance inspection techniques					
3/4.5.2.3.3	Tools					
3/4.5.2.4	Battle damage repair					

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Reqt. No.	Reqt. Title	Concept Refinement	Technology Development	System Development & Demonstration (SDD)	Production & Deployment	Operations and Support
¾	VERIFICATION AND QUALIFICATION (TITLE)	Spiral 1, 2, … (See tables XLVIII and XLIX)	Spiral 1, 2, … (See tables XLIX and L)	Spiral 2, 3, … (See table XXXIb)	Spiral 3, 4, … (See table XXXIb)	Spiral 4, 5, …
3/4.5.3	Human performance and human engineering					
3/4.6	Combat survivability (title only)					
3/4.6.1	Susceptibility (title only)					
3/4.6.1.1	Noise					
3/4.6.1.2	IR radiation (Note: F to obtain platform signature, A to backout engine contribution)					
3/4.6.1.3	Smoke					
3/4.6.1.4	Gaseous emissions					
3/4.6.1.5	Fuel streaming and vapor puffing					
3/4.6.1.6	Water vapor contrails					
3/4.6.1.7	Radar cross section (RCS)					
3/4.6.1.8	Radar absorbent materials (RAMs) and coatings					
3/4.6.2	Vulnerability (title only)					
3/4.6.2.1	Ballistic weapons					
3/4.6.2.1.1	Static structure					
3/4.6.2.2	Nuclear weapons effects					
3/4.6.2.3	Chemical and biological agent effects					

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Reqt. No.	Reqt. Title	Concept Refinement	Technology Development	System Development & Demonstration (SDD)	Production & Deployment	Operations and Support
¾	VERIFICATION AND QUALIFICATION (TITLE)	Spiral 1, 2, … (See tables XLVIII and XLIX)	Spiral 1, 2, … (See tables XLIX and L)	Spiral 2, 3, … (See table XXXIb)	Spiral 3, 4, … (See table XXXIb)	Spiral 4, 5, …
3/4.6.2.4	Fuel ingestion effects					
3/4.6.2.4.1	Fuel ingestion - steady-flow					
3/4.6.2.4.1.1	Fuel ingestion - cooling air contamination					
3/4.6.2.4.2	Fuel ingestion - transient (“quick dump”)					
3/4.7	Subsystems (title)					
3/4.7.1	Anti-icing and de-icing system					
3/4.7.2	Control system					
3/4.7.2.1	Control system performance					
3/4.7.2.1.1	Backup control					
3/4.7.2.2	Control system adjustments					
3/4.7.2.3	Overspeed protection system					
3/4.7.3	Fuel System (title only)					
3/4.7.3.1	Fuels					
3/4.7.3.1.1	Primary fuel (repeat test if engine configuration changes which impact fuel performance)					
3/4.7.3.1.2	Alternate fuel (repeat test if engine configuration changes which impact fuel performance)					

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Reqt. No.	Reqt. Title	Concept Refinement	Technology Development	System Development & Demonstration (SDD)	Production & Deployment	Operations and Support
¾	VERIFICATION AND QUALIFICATION (TITLE)	Spiral 1, 2, … (See tables XLVIII and XLIX)	Spiral 1, 2, … (See tables XLIX and L)	Spiral 2, 3, … (See table XXXIb)	Spiral 3, 4, … (See table XXXIb)	Spiral 4, 5, …
3/4.7.3.1.3	Restricted fuel (repeat test if engine configuration changes which impact fuel performance)					
3/4.7.3.1.4	Emergency fuel					
3/4.7.3.2	Fuel system performance (title only)					
3/4.7.3.2.1	Fuel contamination					
3/4.7.3.2.2	Fuel system performance with external assistance					
3/4.7.3.2.3	Fuel system performance with no external assistance					
3/4.7.3.2.4	Fuel system performance under conditions of excessive fuel vapor					
3/4.7.3.2.5	Fuel pump priming					
3/4.7.3.2.6	Fuel lubricity					
3/4.7.3.2.7	Fuel system performance with water saturated fuel					
3/4.7.3.2.8	Fuel filter					
3/4.7.3.2.9	Fuel flow limit					
3/4.7.4	Electrical system [title only]					
3/4.7.4.1	Electrical power					
3/4.7.4.1.1	Generator					

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Reqt. No.	Reqt. Title	Concept Refinement	Technology Development	System Development & Demonstration (SDD)	Production & Deployment	Operations and Support
¾	VERIFICATION AND QUALIFICATION (TITLE)	Spiral 1, 2, … (See tables XLVIII and XLIX)	Spiral 1, 2, … (See tables XLIX and L)	Spiral 2, 3, … (See table XXXIb)	Spiral 3, 4, … (See table XXXIb)	Spiral 4, 5, …
3/4.7.4.2	Alternate electrical power					
3/4.7.4.3	Electrical connectors and cables					
3/4.7.4.4	Electrical bonding					
3/4.7.4.5	Ground isolation					
3/4.7.4.6	Potting compounds					
3/4.7.5	Ignition system					
3/4.7.5.1	Ignition system fouling [title only]					
3/4.7.5.1.1	Carbon fouling					
3/4.7.5.1.2	Water fouling					
3/4.7.6	Engine monitoring system (EMS)					
3/4.7.6.1	EMS fault detection and isolation					
3/4.7.6.2	On-board engine diagnostic function					
3/4.7.7	Optical systems					
3/4.7.7.1	Fiber optic cables and connectors					
3/4.7.8	Lubrication system [title only]`					
3/4.7.8.1	Lubrication oil					
3/4.7.8.1.1	Oil pressure and temperature limits					
3/4.7.8.1.2	Oil consumption limits					
3/4.7.8.1.3	Oil flow interruption or depletion					

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Reqt. No.	Reqt. Title	Concept Refinement	Technology Development	System Development & Demonstration (SDD)	Production & Deployment	Operations and Support
¾	VERIFICATION AND QUALIFICATION (TITLE)	Spiral 1, 2, … (See tables XLVIII and XLIX)	Spiral 1, 2, … (See tables XLIX and L)	Spiral 2, 3, … (See table XXXIb)	Spiral 3, 4, … (See table XXXIb)	Spiral 4, 5, …
3/4.7.8.2	Lubrication system components and features [title only]					
3/4.7.8.2.1	Oil reservoir					
3/4.7.8.2.1.1	Oil reservoir external features					
3/4.7.8.2.2	Oil drains					
3/4.7.8.2.3	Oil filters					
3/4.7.8.2.4	Oil debris monitor(s)					
3/4.7.8.3	Breather mist					
3/4.7.9	Hydraulic system					
3/4.7.9.1	Hydraulic ground test provisions					
3/4.7.9.2	Hydraulic system air removal					
3/4.7.9.3	Hydraulic fluid filters					
3/4.7.10	Exhaust nozzle system					
3/4.7.10.1	Exhaust nozzle external asymmetrical air pressure loads					
3/4.7.10.2	Vectoring nozzle					
3/4.7.10.2.1	Vectoring nozzle angle and rate					
3/4.7.10.2.2	Vectoring nozzle failure accommodation					
3/4.7.11	Augmentation system					
3/4.7.11.2.1	Water injection system fluid					
3/4.7.12	Wash system					

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Reqt. No.	Reqt. Title	Concept Refinement	Technology Development	System Development & Demonstration (SDD)	Production & Deployment	Operations and Support
¾	VERIFICATION AND QUALIFICATION (TITLE)	Spiral 1, 2, … (See tables XLVIII and XLIX)	Spiral 1, 2, … (See tables XLIX and L)	Spiral 2, 3, … (See table XXXIb)	Spiral 3, 4, … (See table XXXIb)	Spiral 4, 5, …
3/4.7.13	Brake system					
3/4.7.14	Negative torque limiter					
3/4.7.14.1	Multiple power section engines					
3/4.7.15	Power absorber (engine supplied)					
3/4.7.15.1	Propeller feathering and reverse pitch					
3/4.7.16	Gearbox					
3/4.8	Software resources (title)					
3/4.8.1	Software performance and design					
3/4.8.1.1	Built-in-test and inspectability					
3/4.8.1.2	Computer reprogramming					
3/4.8.2	Spare resources					
3/4.9	Production facilities, capabilities, and processes					
3/4.10	Production cost requirement (PCR)					

NOTE:

Reverification of the engine is required when the configuration changes impact the engine requirement.

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TABLE XXXIa. Incremental verification matrix continuation.

Verification Methods	
Code	**Description**
I	Inspection
X	Contractor/Agency Agreement
A	Analysis
S	Simulation/Modeling
D	Demonstration
De	Development Test
C	Component Test
E	Engine System Level Ground Test
F	Flight Test
Q	Process/Quality Control
P	Documentation
AMT/2	1/2 of the AMT (one hot section life)

Milestones	
Code	**Description**
PDR	Preliminary Design Review
CDR	Critical Design Review
DRR	Design Readiness Review
IFR	Initial Flight Release
FFR	Full Flight Release
ISR	Initial Service Release
OCR	Operational Capability Release

UAV example legend.

Verification Methods	
Code	**Description**
A	Analysis
C	Component Test
D	Demonstration
N/A	Not Applicable
E	Engine Test
I	Inspect Job Folder
G	Goal
ICD	Include req't in Interface Control Document
P	Propulsion System Test
S	Similarity
W	Weapon System Test

Milestones	
Code	**Description**

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TABLE XXXIb. Incremental verification matrix for System Development & Demonstration (SDD).

(To be completed by the specification developer; entries represent a sample guidance.)

Reqt. No.	Reqt. Title	System Development & Demonstration						Production and Deployment
¾	VERIFICATION AND QUALIFICATION (TITLE)	PDR	CDR	DRR	IFR	FFR	ISR	OCR
3/4.1	Design and construction (title)	-	-		-	-	-	-
3/4.1.1	Item and interface definition	I	I	I				
3/4.1.1.1	Interface and installation	I	I	I	D			
3/4.1.1.2	Installation hardware	I	I	I	D			
3/4.1.1.3	Interface loads	A	A	A	E	F		
3/4.1.1.4	Mounts (title)	-	-		-	-	-	-
3/4.1.1.4.1	Main mounts	A	A	A	De, E	F		
3/4.1.1.4.2	Ground handling mounts	A	A	A	D, E			
3/4.1.1.4.3	Engine stiffness	A	A	A	De, E	F		
3/4.1.1.5	Control lever and signals (title)							
3/4.1.1.5.1	Power lever and signal				D, E			
3/4.1.1.5.2	Load demand command and signal				D, E			
3/4.1.1.5.3	Output speed lever and signal				D, E			
3/4.1.1.5.4	Fuel shutoff lever and signal				D, E			
3/4.1.1.5.5	Lever torque				D, E			
3/4.1.1.6	Sub- system interfaces				D, E	F		
3/4.1.1.7	Bleed air interface				D, E			
3/4.1.1.7.1	Customer bleed air contamination				A, E			
3/4.1.1.8	Drains and fluid collection devices		I	I	E			
3/4.1.1.9	Power absorber interface		I	I	De, E			
3/4.1.1.10	Power take-off (PTO)				D, E			
3/4.1.1.11	Starting drivetrain	A	De, A	A, De	E			
3/4.1.1.11.1	Starting torque and speed	A	De, A		E			
3/4.1.1.12	Thrust reverser interface		I	I	D			
3/4.1.1.13	Exhaust system interface	A	A	A	E	F		
3/4.1.1.14	Compatibility - Engine Airframe	A	A	A	D, De, E			
3/4.1.1.15	Control and External (C&E) component list	A	A	A	I			

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TABLE XXXIb. Incremental verification matrix for System Development & Demonstration (SDD) - continued.

Reqt. No.	Reqt. Title	System Development & Demonstration						Production and Deployment
	VERIFICATION AND QUALIFICATION (TITLE)	PDR	CDR	DRR	IFR	FFR	ISR	OCR
3/4.1.2	Physical characteristics (title)	-	-		-	-	-	-
3/4.1.2.1	Dry mass (weight) of complete engine	A	A	A	E			
3/4.1.2.2	Mass moment of inertia of complete engine	A	A	A	E			
3/4.1.3	Materials, processes and parts	A	A	A	D, C, E			
3/4.1.3.1	Hazardous Materials	A	A	A	D, C, E			
3/4.1.4	Fasteners	A	A	A	E			
3/4.1.5	Nameplate and product marking				I			
3/4.1.5.1	Engine data plate marking				I			
3/4.1.5.2	Critical parts identification and tracking				I			
3/4.1.6	Transportability	A	A	A	D	D		
3/4.1.7	Interchangeability	A	A	A	I	D	E	
3/4.1.8	Safety (title)	-	-	A	-	-	-	-
3/4.1.8.1	Flammable fluid systems	A	A	A	C, I			
3/4.1.8.2	Fire prevention	A	A	A				
3/4.1.8.3	Explosion-proof		A	A	C			
3/4.1.8.4	Combustible fluid drains	A	A	A	A, I, E	A, E, F		
3/4.1.8.5	Air and gas leakage	A	A	A	E			
3/4.1.8.6	Ground Safety	A	I	A, I				
3/4.1.8.7	Jet Wake	A	A	A			E	
3/4.1.9	Design control (title)	-	-		-	-	-	-
3/4.1.9.1	Parts list	I	I	I	I, P	I, P	I, P	I, P
3/4.1.9.2	Assembly of components and parts		D	D	D			
3/4.1.9.3	Changes in vendors or fabrication process	I	A, C	I, A, C	A, C	A, C	P, A, C	I, C
3/4.1.9.4	Standardization	I	I	I	D	D	D	D

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TABLE XXXIb. Incremental verification matrix for System Development & Demonstration (SDD) - continued.

Reqt. No.	Reqt. Title VERIFICATION AND QUALIFICATION (TITLE)	System Development & Demonstration PDR	CDR	DRR	IFR	FFR	ISR	Production and Deployment OCR
3/4.2	Performance and operability (title)							
3/4.2.1	Performance characteristics (title)							
3/4.2.1.1	Steady-state performance	A	S	A, S	S, E	E		E
3/4.2.1.2	Transient performance	A	S	A, S	S, E	E		
3/4.2.1.3	Performance computer program	S, I	S, I	S, I	S, E	E		
3/4.2.1.4	Performance retention	A	S	A, S	E	E		
3/4.2.2	Operating characteristics and limits	A	A	A	A, E	A, E		
3/4.2.2.1	Operating envelope	A	A	A	A, E	A, E, F		
3/4.2.2.2	Operating attitude and conditions	A	A	A	A, E	A, E, F		
3/4.2.2.3	Starting	A	De	A, De	D, E			E
3/4.2.2.3.1	Ground starts	A	De	A, De	E		E	
3/4.2.2.3.2	Air starts	A	De	A, De	E			
3/4.2.2.3.3	Starting limits	A	A	A	E			
3/4.2.2.3.4	Starting procedure	A	De	A, De	D, E			
3/4.2.2.3.5	Automatic relight		De	De	D, E			
3/4.2.2.4	Stopping	A	A	A	E	E		
3/4.2.2.5	Low power conditions (title)	-	-		-	-	-	-
3/4.2.2.5.1	Idle thrust/power - (ground/flight idle)	A	A	A	E	E		
3/4.2.2.5.2	No load condition	A	A	A	E	E, F		
3/4.2.2.6	Stability	A	A	A	A, E	A, E		
3/4.2.2.7	Transients	A	A	A	E	E		E
3/4.2.2.7.1	Overshoot and undershoot	A	A	A	E	E		
3/4.2.2.7.2	Droop	A	A	A	D, E	E, F		
3/4.2.2.8	Windmilling	A	A	A	E	E		
3/4.2.2.9	Reverse thrust	A	A	A	E	E		
3/4.2.2.10	Steady-state and transient airflow	A	A	A	E	E		
3/4.2.2.11	Inlet airflow distortion	A	A	A	A, E	A, E		
3/4.2.2.11.1	Pressure and temperature rate of change	A	A	A	A	A, E		
3/4.2.2.11.2	Engine Recoverability	A	A	A	A	A, E		

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TABLE XXXIb. Incremental verification matrix for System Development & Demonstration (SDD) - continued.

Reqt. No.	Reqt. Title	System Development & Demonstration						Production and Deployment
	VERIFICATION AND QUALIFICATION (TITLE)	PDR	CDR	DRR	IFR	FFR	ISR	OCR
3/4.2.2.12	Gas path and measurement plane temperature limits	A	A	A	A, E	A, E		
3/4.2.2.13.1	Surface temperature and heat rejection	A	A	A	A, E	A, E		
3/4.3	Controls and external component limiting temperature	A	A	A	A, E	A, E		
3/4.3.1	Environmental conditions [title only]	-	-		-	-	-	-
3/4.3.1.1	Atmospheric conditions [title only]	-	-		-	-	-	-
3/4.3.1.2	Humidity	A	A	A		A, C, D		
3/4.3.1.3	Fungus	A	A	A		A, C, D		
3/4.3.1.4	Corrosive atmosphere	A	A	A		A, C, D, E, I		
3/4.3.2	Icing conditions	A	A	A		E, I		
3/4.3.2.1	Ingestion capability (hazard resistance) [title only]	-	-		-	-	-	-
3/4.3.2.2	Bird ingestion	A	A	A		D, E		
3/4.3.2.3	Foreign object damage (FOD)	A	A	A		A, D, E		
3/4.3.2.4	Ice ingestion	A	A	A		D, E		
3/4.3.2.5	Sand and dust ingestion	A	A	A		E, I		
3/4.3.2.6	Atmospheric liquid water ingestion	A	A	A		A, E, I		
3/4.3.2.7	Armament gas ingestion	A	A	A		E, F		
3/4.3.3	Steam ingestion	A	A	A		E		
3/4.3.3.1	Electromagnetic environmental effects (E^3)	-	-		-	-	-	-
3/4.3.3.2	Electromagnetic interference (EMI)	A	A	A	C			
3/4.3.3.3	Intrasystem electromagnetic compatibility (EMC)	A	A	A	E	F		
3/4.3.3.4	Intersystem electromagnetic compatibility (EMC)	A	A	A	E	F		
3/4.4	Lightning	A	A	A	C			
4.3/4.4.4.1	Integrity (title only)							
3/4.4.1	Structural integrity (title only)							
3/4.4.1.1	Design service life	A	A	A	E	E	E	

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TABLE XXXIb. Incremental verification matrix for System Development & Demonstration (SDD) - continued.

Reqt. No.	Reqt. Title	System Development & Demonstration						Production and Deployment
	VERIFICATION AND QUALIFICATION (TITLE)	PDR	CDR	DRR	IFR	FFR	ISR	OCR
3/4.4.1.2	Design Usage		A	A	A	E	E	
3/4.4.1.2.1	Externally applied forces		A, C	A, C	C, E			
3/4.4.1.3	Material characterization		P, A	P, A	C			
3/4.4.1.4	Parts classification		A	A				
3/4.4.1.5	Durability		A	A	C, E	A AMT/2	A	A, AMT
3/4.4.1.5.1	High cycle fatigue (HCF) life		A	A	C, E	A, AMT/2	A	A, AMT
3/4.4.1.5.2	Low cycle fatigue (LCF) life		A	A	C, E	A, AMT/2	A	A, AMT
3/4.4.1.5.3	Creep		A	A	C, E	A, AMT/2	A	A, AMT
3/4.4.1.6	Strength		A	A	C, E	A, AMT/2	A	A, AMT
3/4.4.1.6.1	Factors of safety		A	A	C			
3/4.4.1.6.2	Blade and disk deflection		A	A	C, De	A, AMT/2		AMT
3/4.4.1.6.3	Containment		A	A	C		E	
3/4.4.1.6.4	Blade out		A	A	C		E	
3/4.4.1.6.5	Overspeed and overtemperature		A	A			E	
3/4.4.1.6.6	Disk burst speed		A	A	C			
3/4.4.1.6.7	Output shaft torque limits		A	A	C		E	
3/4.4.1.6.8	Output shaft speed limits		A	A	E			
3/4.4.1.6.9	Pressure balance	A	A	A	De	A, AMT/2		A, AMT
3/4.4.1.6.10	Gyroscopic moments		A	A			E	
3/4.4.1.7	Damage tolerance		A, P, Q	A, P, Q	C	A	A	A
3/4.4.1.7.1	Residual Strength		A	A			E	
3/4.4.1.7.2	Initial production and in-service flaw size		A	A	I, C, D	A	A, I	A
3/4.4.1.7.3	Flaw growth and inspection intervals		A		C, I	A, I, AMT/2	A	A, I, AMT

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TABLE XXXIb. Incremental verification matrix for System Development & Demonstration (SDD) - continued.

Reqt. No.	Reqt. Title	System Development & Demonstration						Production and Deployment
	VERIFICATION AND QUALIFICATION (TITLE)	PDR	CDR	DRR	IFR	FFR	ISR	OCR
3/4.4.1.8	Vibration and dynamic response		A	A	C, De			
3/4.4.1.8.1	Vibration limits		A	A	C, De			
3/4.4.1.8.2	Critical speeds		A	A	C, De			
3/4.4.1.9	External surface foreign object damage		A	A	C			
3/4.4.2	Mechanical equipment and subsystem integrity	A	A, I	A, I	D, E	E		
3/4.4.3	Avionic and electronic integrity	A	A, I	A, I	D, E	E		
3/4.5	R&M [title only]	-	-		-	-	-	-
3/4.5.1	Reliability [title only]	-	-		-	-	-	-
3/4.5.1.1	Reliability quantitative requirements	A	A	A		A, C, D,E		
3/4.5.2	Maintainability [title only]	-	-		-	-	-	-
3/4.5.2.1	Maintainability quantitative requirements	A, D	A, D	A, D		A, C, D,E		
3/4.5.2.1.1	Excluded maintenance functions	A, D	A, D	A, D		A, D		
3/4.5.2.2	Maintainability qualitative requirements	A, D, I	A, D, I	A, D, I		A, D, I		
3/4.5.2.3	Maintenance [title only]	-	-		-	-	-	-
3/4.5.2.3.1	Modules	A	D	A, D		D		
3/4.5.2.3.2	Maintenance, inspection and repair cycle	A, D	A, D	A, D		A, D		
3/4.5.2.3.2.1	Maintenance inspection techniques		D, I	D, I		D, I		
3/4.5.2.3.3	Tools	D	D, I	D, I		D, I		
3/4.5.2.4	Battle damage repair	A	A	A		A, D		
3/4.5.3	Human performance and human engineering	A	A	A		A, D		
3/4.6	Combat survivability (title only)	-	-		-	-	-	-
3/4.6.1	Susceptibility (title only)	-	-		-	-	-	-
3/4.6.1.1	Noise	A	A	A			E	
3/4.6.1.2	IR radiation (Note: F to obtain platform signature, A to backout engine contribution)	A	A	A		I, A, F, S	A, E	

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TABLE XXXIb. Incremental verification matrix for System Development & Demonstration (SDD) - continued.

Reqt. No.	Reqt. Title	System Development & Demonstration						Production and Deployment
	VERIFICATION AND QUALIFICATION (TITLE)	PDR	CDR	DRR	IFR	FFR	ISR	OCR
3/4.6.1.3	Smoke	A	A	A		D, F	A, E	
3/4.6.1.4	Gaseous emissions	A	A	A			A, E	
3/4.6.1.5	Fuel streaming and vapor puffing	A	A	A		F	A, E	
3/4.6.1.6	Water vapor contrails	A	A	A		F	A	
3/4.6.1.7	Radar cross section (RCS)	A	A	A		D, E	A	
3/4.6.1.8	Radar absorbent materials (RAMs) and coatings	A	A	A			A, E	
3/4.6.2	Vulnerability (title only)	-	-		-	-	-	-
3/4.6.2.1	Ballistic weapons	A	A	A			A, E	
3/4.6.2.1.1	Static structure	A	A	A			D, E	
3/4.6.2.2	Nuclear weapons effects	A	A	A			D, C	
3/4.6.2.3	Chemical and biological agent effects	A	A	A			D, A	
3/4.6.2.4	Fuel ingestion effects	-	-		-	-	-	-
3/4.6.2.4.1	Fuel ingestion - steady-flow	A	A	A			A, E	
3/4.6.2.4.1.1	Fuel ingestion - cooling air contamination	I, A	I, A	I, A			I, A	
3/4.6.2.4.2	Fuel ingestion - transient (“quick dump”)	A	A	A			A, E	
3/4.7	Subsystems (title)	-	-		-	-	-	-
3/4.7.1	Anti-icing and de-icing system	A	C, D	A, C, D	E			
3/4.7.2	Control system	A	C, De	A, C, De	D, E,	E		
3/4.7.2.1	Control system performance	A	D, C, De	A, D, C, De	E	E		
3/4.7.2.1.1	Backup control	A	C	A, C	E	F	D	
3/4.7.2.2	Control system adjustments		C, D	C, D	D, E			
3/4.7.2.3	Overspeed protection system	A	D	A, D	E			
3/4.7.3	Fuel System (title only)	-	-		-	-	-	-
3/4.7.3.1	Fuels	-	-		- -	-	-	-
3/4.7.3.1.1	Primary fuel (repeat test if engine configuration changes which impact fuel performance)				E	E	E	
3/4.7.3.1.2	Alternate fuel (repeat test if engine configuration changes which impact fuel performance)				E	E	E	

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TABLE XXXIb. Incremental verification matrix for System Development & Demonstration (SDD) – continued.

Reqt. No.	Reqt. Title	System Development & Demonstration						Production and Deployment
	VERIFICATION AND QUALIFICATION (TITLE)	PDR	CDR	DRR	IFR	FFR	ISR	OCR
3/4.7.3.1.3	Restricted fuel (repeat test if engine configuration changes which impact fuel performance)				E	E	E	
3/4.7.3.1.4	Emergency fuel				E	E		
3/4.7.3.2	Fuel system performance (title only)	-	-		-	-	-	-
3/4.7.3.2.1	Fuel contamination	A	C	A, C	E	E	E	
3/4.7.3.2.2	Fuel system performance with external assistance		C, D	C, D	E	E	E	
3/4.7.3.2.3	Fuel system performance with no external assistance		A	A	C	A, C	A, C	
3/4.7.3.2.4	Fuel system performance under conditions of excessive fuel vapor		C	C	C	C	C	
3/4.7.3.2.5	Fuel pump priming		C	C	C, E	C, E		
3/4.7.3.2.6	Fuel lubricity		A	A	A, D	A, C	C	
3/4.7.3.2.7	Fuel system performance with water saturated fuel		A	A	E			
3/4.7.3.2.8	Fuel filter		A	A	D	A, D	A, E	
3/4.7.3.2.9	Fuel flow limit		A	A	C, E	E	E	
3/4.7.4	Electrical system [title only]	-	-		-	-	-	-
3/4.7.4.1	Electrical power	A	A	A	D, E	C, E		
3/4.7.4.1.1	Generator	A	A	A	C, D, E	C		
3/4.7.4.2	Alternate electrical power	A	C	A, C	D, E	E		
3/4.7.4.3	Electrical connectors and cables	A	A, D, I	A, D, I	C, I	C, I		
3/4.7.4.4	Electrical bonding	A	A	A	C, E, I	C, E, I		
3/4.7.4.5	Ground isolation				C, E, I	C, E, I		
3/4.7.4.6	Potting compounds				D, I	C, E		
3/4.7.5	Ignition system				C, E	C, E		
3/4.7.5.1	Ignition system fouling [title only]	-	-		-	-	-	-
3/4.7.5.1.1	Carbon fouling					C		
3/4.7.5.1.2	Water fouling					C		

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TABLE XXXIb. Incremental verification matrix for System Development & Demonstration (SDD) - continued.

Reqt. No.	Reqt. Title	System Development & Demonstration						Production and Deployment
	VERIFICATION AND QUALIFICATION (TITLE)	PDR	CDR	DRR	IFR	FFR	ISR	OCR
3/4.7.6	Engine monitoring system (EMS)	A	A	A	A, D, I, E	A, E		
3/4.7.6.1	EMS fault detection and isolation	A	A	A	A, D, I, E	A, E		
3/4.7.6.2	On-board engine diagnostic function	A	A	A	A, D, I, E	A, E		
3/4.7.7	Optical systems		A, De	A, De	C, D, E	F		
3/4.7.7.1	Fiber optic cables and connectors		A, De	A, De	C, D, E	F		
3/4.7.8	Lubrication system [title only]`	-	-		-	-	-	-
3/4.7.8.1	Lubrication oil	A	A	A	E, I			
3/4.7.8.1.1	Oil pressure and temperature limits		A	A	E, I			
3/4.7.8.1.2	Oil consumption limits		A	A	E			E
3/4.7.8.1.3	Oil flow interruption or depletion		A	A		E, I		
3/4.7.8.2	Lubrication system components and features [title only]	-	-		-	-	-	-
3/4.7.8.2.1	Oil reservoir	A	A	A	C, I			
3/4.7.8.2.1.1	Oil reservoir external features	A	A	A	E, I			
3/4.7.8.2.2	Oil drains				E, I			
3/4.7.8.2.3	Oil filters	A	A	A	C, I			
3/4.7.8.2.4	Oil debris monitor(s)	A	A, C	A, C	E, I			
3/4.7.8.3	Breather mist	A	A	A	I			
3/4.7.9	Hydraulic system	A	C	A, C	E			
3/4.7.9.1	Hydraulic ground test provisions	A	A	A	D			
3/4.7.9.2	Hydraulic system air removal	A	A	A	D			
3/4.7.9.3	Hydraulic fluid filters	A	D	A, D	C, E			
3/4.7.10	Exhaust nozzle system	A	De	A, De	I, E	F		
3/4.7.10.1	Exhaust nozzle external asymmetrical air pressure loads	A	C, De	A, C, De	C, De	F		

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TABLE XXXIb. Incremental verification matrix for System Development & Demonstration (SDD) - continued.

Reqt. No.	Reqt. Title	System Development & Demonstration						Production and Deployment
	VERIFICATION AND QUALIFICATION (TITLE)	PDR	CDR	DRR	IFR	FFR	ISR	OCR
3/4.7.10.2	Vectoring nozzle	A	C, De	A, C, De	E	F		
3/4.7.10.2.1	Vectoring nozzle angle and rate	A	C, De	A, C, De	E	F		
3/4.7.10.2.2	Vectoring nozzle failure accommodation	A	C, De	A, C, De	E	F		
3/4.7.11	Augmentation system	A	C, De	A, C, De	E	F		
3/4.7.11.2.1	Water injection system fluid		C, De	C, De	E, I	F		
3/4.7.12	Wash system						E	
3/4.7.13	Brake system						F	
3/4.7.14	Negative torque limiter				A	E		
3/4.7.14.1	Multiple power section engines		A	A	C, E			
3/4.7.15	Power absorber (engine supplied)				A	E, F		
3/4.7.15.1	Propeller feathering and reverse pitch		A	A	E	F		
3/4.7.16	Gearbox		A	A	C	E		
3/4.8	Software resources (title)	-	-		-	-	-	-
3/4.8.1	Software performance and design		I	I	A, D	E		
3/4.8.1.1	Built-in-test and inspectability		A	A	D	E		
3/4.8.1.2	Computer reprogramming				D		E	
3/4.8.2	Spare resources				D	C		
3/4.9	Production facilities, capabilities, and processes	A	A, De	A, De	C, D, E	F	A	A
3/4.10	Production cost requirement (PCR)	A	A	A	A	A	A	A

NOTE:
Reverification of the engine is required when the configuration changes impact the engine requirement.

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TABLE XXXIb. Incremental verification matrix for System Development & Demonstration (SDD) - continued.

Verification Methods	
Code	**Description**
I	Inspection
X	Contractor/Agency Agreement
A	Analysis
S	Simulation/Modeling
D	Demonstration
De	Development Test
C	Component Test
E	Engine System Level Ground Test
F	Flight Test
Q	Process/Quality Control
P	Documentation
AMT/2	1/2 of the AMT (one hot section life)

Milestones	
Code	**Description**
PDR	Preliminary Design Review
CDR	Critical Design Review
DRR	Design Readiness Review
IFR	Initial Flight Release
FFR	Full Flight Release
ISR	Initial Service Release
OCR	Operational Capability Release

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TABLE XXXII. Test procedure format.

a. Cover (title number and source of test procedure, date, names of the author(s), and contract number).
b. Title page (title number and source of the test procedure, date, names of the author(s), and contract number).
c. Introduction (identifying item under test, test dates, etc. and government witnessing organization)
d. Applicable documents (test procedure related specifications, development plans, parts list documents, inspection specifications, assembly instructions, detailed test directives, acceptance test procedures, etc.; any document without prior government approval must be attached to the procedure for Using Service approval.)
e. Test objective (test purpose, anticipated results, specification requirements to be verified (cite specific paragraph) success criteria, hardware condition, post test performance).
f. Test article description (conformance to official parts list, deviation from parts list with explanation/justification, include schematic, drawings, pictures as required, identify specific build documents by number, revision, and date, state relevant features (e.g.; if engine performance test state build clearances, nozzle areas, etc.) Identify specific test article by part number(s) and serial number(s), procedures for approval of parts replacement).
g. Test requirements (description of test facilities, power absorption, inlet/exhaust, control/inputs, electrical power required, starting system, test conditions/environment, description of special test equipment used, description of instrumentation placement and accuracies, description of data acquisition system, operating parameters (e.g.; fuels, oils, test limits). All analysis or preliminary activity required to establish test conditions (e.g., spin pit speeds or strain gage locations) shall be described or referenced.
h. Test procedures (pretest procedure (initial set-up, calibrations, pretest check-outs)), conduct of test, description of specific actions required by specification (e.g.; cycling, stabilization, starting specifics, bleed air extraction and transient operations), deviations from specification with explanation or justification, any interpretation of specification requirements (e.g.; performance measurement is SHP, in lieu of temperature), post test calibrations, inspections, teardowns).
i. Data (collection and reduction methodology, anticipated data points).

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TABLE XXXIII. Test report format.

Following the completion of each separate engine or component test, demonstration or consecutive group of tests or demonstrations conducted on any single assembly or component, a report should be submitted. These reports, certified by a Government representative, will constitute the basis for approval. Each report should contain, but is not limited to, the following items:

"a. Cover (title number and source of report, date, names of the author(s), and contract number).

b. Title page (title number and source of the report, date, names of the author(s), and contract number).

c. Abstract (a brief statement of the contents of the report, including the objective).

d. Table of contents.

e. List of illustrations (provide figure numbers and captions of all illustrations. Photographs, charts, and graphs should be treated as illustrations and given figure numbers.)

f. Summary (brief resume of the test conducted, including objective, procedure, results, conclusions, and recommendations referencing the applicable paragraph of the engine specification).

g. Body of the report:

(1) Brief general description of the engine or of the component and a detailed description of all features which differ from the previous model, if applicable.

(2) If approval is being requested, without test, based on similarity to a component or assembly for which previous test approval was obtained, any physical or functional dissimilarities or differences in testing requirements with respect to the tested component and reference to the approved component test report shall be included.

(3) Method of test (general description of test facility, equipment and methods used in conducting the test). Include photographs of test article in test facility.

(4) Record of test (chronological history of all events and incidents in connection with all of the testing, including details of all leaks, vibrations and other irregular functioning, and any adjustments, repairs, replacements of parts and the corresponding engine operating time)for each event/incident, cite date and log sheet number.

(5) Analysis of results (a complete discussion of all phases of the test, such as probable reason for failure or unusual wear, comparison in performance with previous models, analysis of general operation, data reduction and analysis, and any items that are significant from an engineering viewpoint). For items not in compliance, cite planned corrective action.

(6) Calibrations and recalibration data including acceptance limits. (Data in uncorrected form and corrected form, if applicable, shall be shown by suitable curves. A plot of vibration characteristics in three planes shall be provided for both calibration and recalibration runs.)

(7) Tabulated data for all pertinent instrument readings and all required instrument readings taken during the test.

(8) Description of the condition of the engine or components at disassembly inspection with photo documentation. A completed disassembly inspection and material discrepancy description shall be provided for each identified discrepancy.

(9) Conclusions and recommendations, with respect to approval of the engine or components tests, supplemented by such discussion as is necessary for their justification.

h. Appendix (final approved test procedure), test log sheets."

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TABLE XXXIV. Data recording requirements.

	Data Recording Frequency		Taken During Tests		
	Start	Once/30 min or once/cycle steady-state	Continuously during transients	Cal/ recal	Thrust
1. Time of day	X	X		X	X
2. Total endurance time					X
3. Power setting	X	X	X	X	X
4. Exhaust nozzle area sq. in.				X	X
5. Variable geometry position		X		X	X
6. Engine rotor(s) speed(s) rpm	X	X	X	X	X
7. Rotor speed(s) at idle rpm	X	X			
8. Rotor speed(s) at ignition rpm	X	X			
9. Rotor speed(s) at starter cutout rpm	X	X			
10. Engine fuel flow kg/hr (lbm/hr)		X	X	X	X
11. Delivered shaft power hp		X	X	X	X
12. Customer power extraction hp		X	X	X	X
13. Data for determining airflow kg/sec (lbm/sec)		X		X	
14. Engine inlet total pressure average kPa (in Hg abs)		X		X	X
15. Engine inlet total temperature average °C (°F)		X		X	X
16. Compressor discharge total pressure kPa (in Hg abs)		X	X	X	X
17. Customer bleed air total pressure kPa (psia)		X	X	X	
18. Compressor bleed air total pressure (acceleration) kPa (psia)			X		
19. Customer bleed air static pressure kPa (psia)		X		X	X
20. Compressor bleed air static pressure (acceleration) kPa (psia)			X		

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TABLE XXXIV. Data recording requirements - continued.

	Data Recording Frequency		Taken During Tests		
	Start	Once/30 min or once/cycle steady-state	Continuously during transients	Cal/ recal	Thrust
21. Compressor bleed air total temperatures (all ports) °C (°F)		X		X	X
22. Compressor bleed airflow rate (all ports) kg/sec (lbm/sec)		X		X	X
23. Turbine discharge total pressure average kPa (in Hg abs)		X		X	X
24. Compressor discharge total temperature °C (°F)		X	X	X	X
25. Oil breather static pressure kPa (psig)		X		X	X
26. Gas generator turbine discharge total temperature °C (°F)	X	X	X	X	X
27. Exhaust nozzle exit total pressure kPa (in Hg abs)		X		X	
28. Exhaust nozzle static pressure kPa (in Hg abs)		X		X	X
29. Exhaust nozzle total temperature °C (°F)		X		X	X
30. Oil flow kg/min (lbm/min)		X		X	X
31. Oil inlet temperature at pressure pump inlet °C (°F)	X	X		X	X
32. Oil pressure at pressure pump inlet kPa (psig)	X	X		X	X
33. Oil pressure at pressure pump outlet kPa (psig)	X	X		X	X
34. Oil pressure at scavenging pump outlet kPa (psig)		X		X	X

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TABLE XXXIV. Data recording requirements - continued.

	Data Recording Frequency		Taken During Tests		
	Start	Once/30 min or once/cycle steady-state	Continuously during transients	Cal/ recal	Thrust
35. Oil temperature at scavenging pump outlet °C (°F)		X		X	X
36. Oil temperature at outlet from fuel/oil cooler °C (°F)		X		X	X
37. Oil consumption for each cycle		X			
38. Fuel pressure at fuel system inlet kPA (psig)	X	X		X	X
39. Fuel pressure at point shown on engine configuration and envelope figure kPa (psig)	X	X		X	X
40. Fuel temperature at fuel system inlet °C (°F)		X		X	X
41. Fuel temperature at outlet from fuel/oil cooler °C (°F)		X		X	X
42. Measured temperature °C (°F) (specify station location)	X	X	X	X	X
43. Maximum measured temperature °C (°F) (specify station location)	X	X			
44. Liquid injection flow kg/hr (lbm/hr (e.g., water)		X		X	X
45. Engine vibration at points shown on engine configuration and envelope figure velocity cm/sec (in/sec)	X	X	X	X	X
46. Accessory compartment temperature(s) where applicable °C (°F)		X		X	X
47. Cooling air inlet temperature°C (°F)		X		X	X
48. Cooling air outlet temperature °C (°F)		X		X	X

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TABLE XXXIV. Data recording requirements - continued.

	Data Recording Frequency		Taken During Tests		
	Start	Once/30 min or once/cycle steady-state	Continuously during transients	Cal/ recal	Thrust
49. Engine condition monitoring system data (list each parameter)		X		X	X
50. Cell static pressure kPa (in Hg abs)	X	X		X	X
51. Ignition source voltage and current (when external power is being used)	X	X			X
52. Oil leakage at accessory pads		X		X	X
53. Start number	X			X	
54. Time to ignition actuation	X				X
55. Time to light-off	X				X
56. Time to starter cutout	X			X	X
57. Time to stabilized idle RPM	X			X	X
58. Time to oil pressure indication (at point shown on engine configuration and envelope figure	X			X	X
59. Time to stabilize to normal oil pressure (at point shown on engine configuration and envelope figure)	X			X	X
60. Engine life counter	X	X		X	X
61. Additional data as required by the Using Service	X	X	X	X	X

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TABLE XXXV. Sand and dust concentration and particle size distribution.

CONCENTRATION GUIDANCE			
	Fix-wing Aircraft	Helicopter	VSTOL
(a): Concentration (mg sand/m^3 air)	53	53	350
(b): Coarse sand Time (hours)	0.5	50	5
(c): Fine sand Time (hours)	1.5	54	5
(d): Thrust or power loss (%)	5	10	5
(e): SFC gain (%)	5	5	5

PARTICLE SIZE DISTRIBUTION GUIDANCE		
	Particle Size Microns (inches)	Cumulative % by Weight
Coarse sand*	1,000 (3.94×10^{-2})	100
	707 (2.78×10^{-2})	95-99
	500 (1.97×10^{-2})	89-93
	354 (1.39×10^{-2})	77-81
	250 (9.84×10^{-3})	60-64
	177 (6.97×10^{-3})	38-42
	125 (4.92×10^{-3})	18-22
	88 (3.46×10^{-3})	6-10
	63 (2.48×10^{-3})	1-5
Fine sand**	1,000 (3.94×10^{-2})	100
	500 (1.97×10^{-2})	85-90
	250 (9.84×10^{-3})	70-75
	125 (4.92×10^{-3})	50-55
	75 (2.95×10^{-3})	25-30
	<75 (2.95×10^{-3})	10-15

* Composition is crushed quartz (SiO_2).
** Composition is 60% quartz (SiO_2), 26% gypsum (hydrated $CaSO_4$), 12% calcite ($CaCO_3$), and 2% soluble salts. Soluble salts shall contain the identical weight fraction of the salts from table VI. Soluble salts shall all be <75 microns in particle size.

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TABLE XXXVIa. Gypsum.

Particle Size - Microns (inches)	% by Weight Finer than Indicated Size
1000 (0.039)	100%
500 (0.020)	15-25%

TABLE XXXVIb. Calcite.

Particle Size - Microns (inches)	% by Weight Finer than Indicated Size
500 (0.020)	100%
250 (0.010)	95-99%
125 (0.005)	80-90%
75 (0.003)	60-75%

TABLE XXXVII. Design service life.

(See MIL-HDBK-1783 for guidance and example table.)

		Service Life			
System Category	Parts	Flight hours	Ground Run hours	Flight missions	Ground Runs missions
Fighter/Attack	Cold Parts	4,000	400	3,000	200
	Hot Parts	2,000	200	1,500	100
Bomber	Cold Parts	10,000	1,000	2,500	200
	Hot Parts	4,000	500	1,250	100
Cargo	Cold Parts	30,000	3,000	9,000	1,000
	Hot Parts	15,000	1,500	4,500	500
Trainer	Cold Parts	18,000	1,800	13,500	1,500
	Hot Parts	9,000	900	6,750	750
Helicopter	Cold Parts	6,000	TBD	3,000	TBD
	Hot Parts	6,000	TBD	3,000	TBD

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TABLE XXXVIII. Current engine service life.

		Service Life			
System Category	Parts	Flight (hours)	Ground Run (hours)	Flight (missions)	Ground Run (missions)
Fighter: (Aircraft H, Engine O)	Cold Parts	4,000	1400 (50 test cell)	2,938	35
	Hot Parts	2,000	700 (25 test cell)	1,469	18
Bomber: (Aircraft K, Engine T)	Cold Parts	10,000	not applicable	not available	not available
	Hot Parts	4,000	TBD	not available	not available
Cargo: (Aircraft L, Engine S)	Cold Parts	30,000	TBD	8,516	TBD
	Hot Parts	15,000	TBD	4,258	TBD
Trainer: (Aircraft M, Engine U)	Cold Parts	9,000	12,600	2,760	TBD
	Hot Parts	4,500	6,300	1,380	TBD
Turboshaft: (Engine D)	Cold Parts	6,000	not applicable	not applicable	not applicable
	Hot Parts	6,000	not applicable	not applicable	not applicable

NOTES:
1. The Ground Run (missions) column is the number of ground runs and should correlate to the test cell hours in the Ground Run (hours) column.
2. The Ground Run (hours) column includes test cell hours, taxi time, pre and post-flight checks and other troubleshooting ground runs.
3. Definition of terminology has led to numerous misunderstandings between design requirements and system capabilities.
4. Before detailed analysis is conducted, define each parameter; confusion has occurred over flight hours, mission hours, engine flight hours, engine operating hours, type cycle categories, etc.

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TABLE XXXIX. Current engine usage parameters.

System Category	Parts Classification	Type I Cycles	Type III Cycles	Type IV Cycles	Augmentor Lights or Thrust Reversing	Augment Time or Vector Cycles (hrs)	Time at IRP and Above (hrs)
		0-max-0	idle-max -idle	cruise-int-cruise			
Fighter (Aircraft H, Engine O)	Cold Parts	2,973	20,503	22,074	20,239	186 (109)	684
	Hot Parts	1,487	10,252	11,037	10,165	93 (54)	342
Bomber (Aircraft K, Engine T)	Cold Parts	2,371	7,113	N/A	N/A	N/A	TBD
	Hot Parts	948	2,844	N/A	N/A	N/A	TBD
Cargo (Aircraft L, Engine S)	Cold Parts	8,516	25,840	17,178	12,700	N/A	TBD
	Hot Parts	4,258	12,920	8,589	6,350	N/A	TBD
Trainer (Aircraft M, Engine U)	Cold Parts	28,000	20,288	3,817	N/A	N/A	885
	Hot Parts	14,000	10,140	1,909	N/A	N/A	443
Helicopter	Cold Parts	15,000	TBD	TBD	0	0	9.1%
	Hot Parts	15,000	TBD	TBD	0	0	9.1%

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TABLE XL. AMT test/cycles.

Endurance Test Mission Profiles	No. of Missions								
	Total Missions	Heated Inlet °C (°F)			Ram Pressure kPa (psi)				Max. Turbine Temp.
		≥ X°	≥ Y°	≥ Z°	101.3 (14.7)	A	B	C	
Mission #1									
Mission #2									
Mission #3									
Mission (etc.)									
Composite									
Ground Test									

NOTE:
The blanks with “X”, “Y”, “Z”, “A”, “B”, and “C” should be completed by the contractor and is based on the temperatures and pressures for the mission. When added together, the number of missions under the Heated Inlet and Ram Pressure columns should equal the value in the Total Missions column.

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TABLE XLI. LCF engine test cycle for turboprop and turboshaft engines.

(See MIL-HDBK-1783 for guidance and example table.)

Time (min)	Event
0.5	Start engine
2.0	Run at idle
0.1	Accel to maximum power
2.5	Run at maximum power
0.1	Decel to idle
3.0	Run to idle
0.1	Accel to max continuous
2.5	Run to max continuous
0.1	Decel to idle
2.0	Run to idle
2.1	Shutdown and cool down
15.0	Total

NOTES:
1. Transient power commands are in 1/2 second or less.
2. Power settings are initial rating values of output thrust or power or specified by Using Service.
3. Engine control and fuel flow schedules are set for maximum acceleration.
4. Output shaft speeds are rated values at high power settings and minimum self sustaining speed at idle.
5. With customer bleed and power extractions.

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TABLE XLII. Reliability quantitative test requirement.

Tests and Analyses	Test Time (hours)
a. Accelerated mission tests	
b. Engine performance (e.g., altitude) tests	
c. Bench qualification tests or subassemblies	
d. FMECA	
e. Etc..	

NOTE:
The results of other engine and component level tests shall be counted providing that the contractor gains agreement with the Government, prior to conducting the tests, that they are pertinent to reliability determination.

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TABLE XLIII. US Navy module experience.

Engine Type	Module	Average FHRS-1st Failure	Hot Section Inspection
Turbofan	Fan	500	None
	Compressor	430	None
	Combustor	525	None
	HPT	720	None
	LPT	490	None
	Augmentor	385	None
Turboshaft	Cold Section	400	None
	Accessory	580	None
	Hot Section (Combustor, G.G.)	875	None
	Power Turbine	710	None
Turboprop	Gearbox	824	Unavailable
	Power Section	1450	Unavailable
	Torque Meter	Unavailable	Unavailable

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TABLE XLIV. Engine B1.0 expected life.

	Design TACs	Life TAC*	Inspection TACs	Interval TAC*
Cold Parts (Durability Critical)	8,000	8,600	4,000	4,300
Cold Parts (Fracture Critical)	TBD	TBD	TBD	TBD
In-service Inspectable	8,000	8,600	4,000	4,300
In-service Non-inspectable	TBD	TBD	TBD	TBD
Deterministic	16,000	17,200	TBD	TBD
Probabilistic	8,000	8,600	TBD	TBD
Bearings (based on average bearing and engine installation tolerances)				
Ball (B1 value based on a material factor = 25)	4,000 EOH		4,000 EOH	4,300 EOH
Roller (B1 value based on a material factor = 6)	4,000 EOH		4,000 EOH	4,300 EOH

(NOTE: B1.0 life is the age at which 1.0% of the population will fail.)

	Design TACs	Life TAC*	Inspection TACs	Interval TAC*
Hot Parts (Durability Critical)	4,000	4,300		
Hot Parts (Fracture Critical)	4,000	4,300	4,000	4,300
Augmentor Module (excl. duct)	1,200	(total operation hr)	1,600	1,720
Ignitor Plugs High Energy	2,000	2,150	2,000	2,150
Low Energy	1,000	1,075	2,000	2,150
Augmentor	2,000	2,150	2,000	2,150
Expendable Items	4,000	4,300	4,000	4,300

* Includes ground operation and test cell and troubleshooting time.

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TABLE XLVa. Start ignition system cycle.

		ON	OFF
(1)	NOM voltage	1 hour	20 minutes
(2)	MAX voltage	40 minutes	20 minutes
(3)	MIN voltage	30 minutes	20 minutes
(4)	NOM voltage	40 minutes	10 minutes

TABLE XLVb. Continuous duty augmentor ignition system cycle.

		ON	OFF
(1)	NOM voltage	5 minutes	20 minutes
(2)	NOM voltage	5 minutes	5 minutes
(3)	NOM voltage	5 minutes	40 minutes
(4)	MAX voltage	5 minutes	15 minutes
(5)	MAX voltage	5 minutes	5 minutes
(6)	MAX voltage	5 minutes	25 minutes
(7)	MIN voltage	5 minutes	10 minutes
(8)	MIN voltage	5 minutes	30 minutes
(9)	MAX voltage	5 minutes	5 minutes
(10)	MAX voltage	25 minutes	15 minutes

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TABLE XLVc. Intermittent ignition system cycle.

CONDITIONS	ON	OFF	REPEAT
(1) MIN voltage	30 seconds	4.5 seconds	4X
(2) MAX voltage	30 seconds	4.5 minutes	4X
(3) MAX voltage	5 minutes	5 minutes	1
(4) NOM voltage	5 minutes	5 minutes	1
(5) MIN voltage	5 minutes	5 minutes	2X
(6) NOM voltage	1 minute	2 minutes	7X
(7) MIN voltage	5 minutes	5 minutes	2X
(8) MIN voltage	5 minutes	4 minutes	1
(9) MAX voltage	30 seconds	4.5 minutes	6X
(10) MIN voltage	5 minutes	5 minutes	1
(11) MAX voltage	5 minutes	5 minutes	1
(12) OFF 10 minutes			
(13) MAX voltage	1 minute	4 minutes	4X
(14) OFF 10 minutes			
(15) NOM voltage	5 minutes	5 minutes	2X

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TABLE XLVIa. Recorded and monitored aircraft parameters.

Monitored Aircraft Parameters	Accuracy
Commands/requests:	
TT2 probe heater request	
Nozzle angle request	
Nozzle exit area request	
Engine IBIT request/in progress	
Pilot initiated data request	
Engine fault reset	
Engine minimum airflow request	
Bleed flow request	
Aircraft sensed data:	
Angle of attack	
Angle of sideslip	
Calibrated airspeed	
Mach number	
Pressure altitude	
Static pressure - ambient press	
Axial acceleration	
Lateral acceleration	
Normal acceleration	
Time (data, hours, minutes, seconds)	
Missile ‘‘armed” index	
Missile ‘‘launched” index	
Weight on gear (nose, left and right)	
Shutoff valve closed - fire (nacelle)	
Fuel supply pressure	
Fuel density	
Fuel temperature at density meter	
Emergency fuel usage	
Cross-engine inlet temperature	
Cross-engine inlet pressure	
A/C boost pumps on-line	
Maneuvering index	

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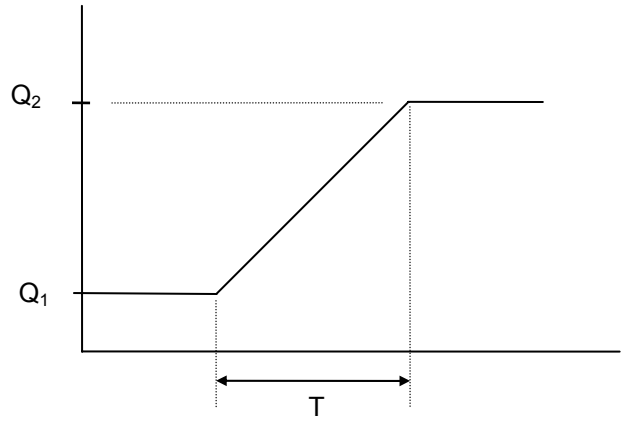
TABLE XLVIb. Recorded and monitored engine parameters.

Monitored Engine Parameters	Accuracy
Rotational speeds:	
Low rotor	
High rotor	
Gas path pressures:	
Turbine exhaust	
Inlet	
Main burner	
Fan duct	
Gas path temperatures:	
Inlet	
Turbine blade	
Exhaust gas	
Oil system:	
Lube debris indication	
Temperature	
Pressure	
Level	
Filter delta press	
Fuel system:	
Fuel filter delta press	
GG fuel temperature	
Actuator pressure	
GG throttle valve delta press	
Gearbox:	
Lube debris indication	
Temperature	
Pressure	
Level	
Filter delta press	
Effector positions:	
Miscellaneous:	

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TABLE XLVII. Turboshaft engine 1.22 km (4000 ft), 35°C (95°F) output shaft speed droop requirements.

Maneuver Endpoints		T - Transient Time (Seconds)						
Q_1	Q_2	1	2	3	4	5	6	8
0	100	(d11)	(d12)	(d13)	(d14)	(d15)	(d16)	(d18)
40	100	(d21)	(d22)	(d23)	(d24)	(d25)	(d26)	(d28)
100	0	(d31)	(d32)	(d33)	(d34)	(d35)	(d36)	(d38)
100	40	(d41)	(d42)	(d43)	(d44)	(d45)	(d46)	(d48)
Percent Aircraft Torque		Percent Maximum Allowable Droop						



NOTES for table XLVII:
(1) Torque shall be applied linearly as in the accompanying figure.
(2) Torque is given as a percentage of AIRCRAFT torque determined using 100% engine output shaft speed and the value of the power required for each engine as specified by the Using Service through coordination with the airframe manufacturer as opposed to available ENGINE torque.
(3) Droop requirement represents a maximum transient speed decrease (excursion from steady state).
(4) A negative droop represents a maximum transient speed increase (excursion from steady state), never to reach the engine's overspeed shutoff threshold.

Zero percent torque condition is with the power turbine clutched to rotor drive system (zero needle split).

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TABLE XLVIIIa. Requirements verification matrix.

Reqt.	*Subject*	*Inspection*	*Analysis*	*Component Test*	*Engine Test*	*Similarity Justification*
3.0	Characteristics (information only)	—	—	—	—	—
3.1	Design Configuration and Construction (title only)	—	—	—	—	—
3.1.1.4.1	Customer bleed air contamination				X	
3.1.3	Materials, processes, and parts	X				X
3.1.4	Fasteners/joint connections	X				X
3.1.6	Safety	X				X
3.1.6.1	Flammable fluid systems		X	X		X
3.1.6.2	Explosion-proof			X		X
3.1.6.3	Air and gas leakage	X			X	
3.1.6.4	Ground safety	X				
3.2	Performance (information only)	—	—	—	—	—
3.2.1	Operating envelope		X		X	
3.2.1.1	Thrust/horsepower extraction		X		X	
3.2.1.2	HPC bleed and PTO power extraction		X		X	
3.2.1.3	Thrust transients		X		X	
3.2.1.4	Stability		X		X	
3.2.1.5	Steady-state and transient airflow		X		X	
3.2.1.6	Inlet airflow distortion		X		X	
3.3.2	Hazard resistance (title only)	—	—	—	—	—
3.3.2.1	Bird ingestion		X			X
3.3.2.2	Foreign object damage		X			X
3.3.2.5	Electromagnetic interference (EMI)		X	X		X
3.4	Integrity (title only)	—	—	—	—	
3.4.1	Structural integrity (information only)	—	—	—	—	—
3.4.1.1	Structural Requirements (full life components designed during CDA)	X	X			
3.4.1.1.1	Durability		X	X	X	X
3.4.1.1.1.1	Low-cycle fatigue (LCF) life		X	X	X	X
3.4.1.1.1.2	Creep		X		X	X
3.4.1.1.2	Damage tolerance (information only)	—	—	—	—	—
3.4.1.1.2.1	Residual strength		X			
3.4.1.1.2.2	Initial flaw size	X				
3.4.1.1.2.3	Inspection Intervals	X				
3.4.1.1.2.4	Flaw growth		X			
3.4.1.1.3	Shock		X			
3.4.1.2	Structural rqmts (CDA components)	X				
3.4.1.2.1	Durability		X		X	X
3.4.1.2.1.1	Low-cycle fatigue (LCF) life		X		X	X
3.4.1.2.1.2	Creep		X		X	X
3.4.1.2.2	Damage tolerance (information only)	—	—	—	—	—
3.4.1.2.2.1	Residual strength (bore only)		X			
3.4.1.2.2.2	Initial flaw size	X				
3.4.1.2.2.3	Flaw growth (bore only)		X			
3.4.1.3	Structural Requirements (Common to All Components) Information only	—	—	—	—	—
3.4.1.3.1	Internal environment		X		X	X
3.4.1.3.2	Material and process characterization			X*		X
3.4.1.4	Parts classification		X			X
3.4.1.5.	High-cycle fatigue life		X	X	X	X

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TABLE XLVIIIa. Requirements verification matrix - continued.

Reqt.	*Subject*	*Inspection*	*Analysis*	*Component Test*	*Engine Test*	*Similarity Justification*
3.4.1.6	Strength		X	X	X	X
3.4.1.6.1	Load factors	X				
3.4.1.6.2	Blade and disk deflection		X		X	X
3.4.1.6.3	Radial containment		X			X
3.4.1.6.4	Bladeout		X			X
3.4.1.6.5	Overspeed/overtemperature		X		X	X
3.4.1.6.6	Disk burst speed		X			X
3.4.1.6.7	Gyroscopic moments		X			X
3.4.1.6.8	Pressure Balance		X		X	
3.4.1.7	Damage tolerance (Ref. Section 3.4.1)	—	—	—	—	—
3.4.1.8	Vibration and dynamic response		X	X	X	X
3.4.1.8.1	Critical speeds		X		X	X
3.4.1.9	Catastrophic failure					X
3.5	Reliability and maintainability (title only)	—	—	—	—	—
3.6	Vulnerability (title only)	—	—	—	—	—
3.7	Subsystems (title only)	—	—	—	—	—
3.7.2	Control system		X	X	X	X
3.7.3.1.1	Primary fuel (JP-8/JP-5)		X		X	
3.7.7.1.2	Oil flow interruption/depletion				X	
3.7.9.2	Air starts				X	
3.7.10.2.1	Vectoring nozzle range and rate		X		X	
3.8	Embedded computer resources (title only)	—	—	—	—	—
3.8.5	Software development and qualification			X	X	

* Coupon testing to compare with existing database on 1st-stage fan disk

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TABLE XLVIIIb. Flight clearance plan matrix.

Reqt.	Subject	Inspection	Analysis	Component Test	Engine Test	Similarity Justification
3.1.1.4.1	Customer bleed air contamination				X	
3.1.3	Materials, processes, and parts	X				X
3.1.4	Fasteners/joint connections	X				X
3.1.6	Safety	X				X
3.1.6.1	Flammable fluid systems		X	X		X
3.1.6.2	Explosion-proof			X		X
3.1.6.3	Air and gas leakage	X			X	
3.1.6.4	Ground safety	X				
3.2	Performance (information only)					
3.2.1	Operating envelope		X		X	
3.2.1.1	Thrust/horsepower extraction		X		X	
3.2.1.2	HPC bleed and PTO power extraction		X		X	
3.2.1.3	Thrust transients		X		X	
3.2.1.4	Stability		X		X	
3.2.1.5	Steady-state and transient airflow		X		X	
3.2.1.6	Inlet airflow distortion		X		X	
3.3.2.1	Bird ingestion		X			X
3.3.2.2	Foreign object damage		X			X
3.3.2.5	Electromagnetic interference (EMI)		X	X		X
3.4.1.1	Structural requirements (full life components designed during CDA)	X	X			
3.4.1.1.1	Durability		X	X	X	X
3.4.1.1.1.1	Low-cycle fatigue (LCF) life		X	X	X	X
3.4.1.1.1.2	Creep		X		X	X
3.4.1.1.2	Damage tolerance (information only)					
3.4.1.1.2.1	Residual strength		X			
3.4.1.1.2.2	Initial flaw size	X				
3.4.1.1.2.3	Inspection Interval	X				
3.4.1.1.2.4	Flaw growth		X			
3.4.1.1.3	Shock		X			
3.4.1.2	Structural rqmts (CDA components)	X				
3.4.1.2.1	Durability		X		X	X
3.4.1.2.1.1	Low-cycle fatigue (LCF) life		X		X	X
3.4.1.2.1.2	Creep		X		X	X
3.4.1.2.2	Damage tolerance (information only)					
3.4.1.2.2.1	Residual strength (bore only)		X			
3.4.1.2.2.2	Initial flaw size	X				
3.4.1.2.2.3	Flaw growth (bore only)		X			
3.4.1.3.1	Internal environment		X		X	X
3.4.1.3.2	Material and process characterization			X*		X
3.4.1.4	Parts classification		X			X
3.4.1.5	High-cycle fatigue life		X	X	X	X
3.4.1.6	Strength		X	X	X	X
3.4.1.6.1	Load factors	X				

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TABLE XLVIIIb. Flight clearance plan matrix - continued.

Reqt.	Subject	Inspection	Analysis	Component Test	Engine Test	Similarity Justification
3.4.1.6.2	Blade and disk deflection		X		X	X
3.4.1.6.3	Radial containment		X			X
3.4.1.6.4	Bladeout		X			X
3.4.1.6.5	Overspeed/overtemperature		X		X	X
3.4.1.6.6	Disk burst speed		X			X
3.4.1.6.7	Gyroscopic moments		X			X
3.4.1.6.8	Pressure Balance		X		X	
3.4.1.8	Vibration and dynamic response		X	X	X	X
3.4.1.8.1	Critical speeds		X		X	X
3.4.1.9	Catastrophic failure					X
3.7.2	Control system		X	X	X	X
3.7.3.1.1	Primary fuel (JP-8/JP-5)		X		X	
3.7.7.1.2	Oil flow interruption/depletion				X	
3.7.9.2	Air starts				X	
3.7.10.2.1	Vectoring nozzle range and rate		X		X	
3.8.5	Software development and qulification			X	X	

* Coupon testing to compare with existing database on 1st-stage fan disk

NOTES:

1\ Analysis. That analytical effort accomplished to show that contractual requirements have been met. This effort may include solution of equations, performance of simulations, evaluation and interpretation of charts, and reduced data, and comparisons of analytical predictions versus test data. The normal reduction of data generated during ground and flight tests is not included.

2\ Component. An integral element. Synonymous with part.

3\ Demonstration. Demonstration is an engineering effort performed to show that the contractual requirements have been met. Compliance or noncompliance is determined by observation only. Fit and function checks may be accomplished as demonstrations.

4\ Engine. A machine that produces thrust for the air vehicle that includes parts from the inlet to the exhaust mounted on the exterior and the interior of the cases. Parts include duct, gears, controls accessories and externals, disks, augmentor, spacers, seals, shrouds, shafts, housings, etc.

5\ Inspection. Inspection is an engineering effort performed to show that the contractual requirements have been met. The visual review and evaluation of the physical items, documentation, drawings, etc., for conformance with the requirements of the procurement and acquisition.

6\ Similarity. The component design and operational environment is similar to that of the referenced engine such that additional verification beyond that being conducted for the referenced engine program is not required for the demonstrator program phase.

7\ Test. A test is an empirical effort performed to show that contractual requirements have been met. Documented procedures, instrumentation, and known environmental conditions are normally applicable or noncompliance is determined by observation, where practical, and evaluation of collected data. Most ground and flight empirical efforts associated with this procurement and acquisition qualify as tests.

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TABLE XLVIIIc. Design-to-goal characteristics verification matrix (table XXXI and 4.12 guidance).

Paragraph	Subject	Verification Process					
		I	*A*	*D*	*CT*	*ET*	*S*
10.3.1	Design (Configuration) and Construction (Title only)	-	-	-	-	-	-
10.3.1.1.	Interface Definition	X					
10.3.1.1.1	Installation hardware	X		X			
10.3.1.1.2	Drains and fluid collection devices		X				X
10.3.1.1.3	Mounts (Title only)	-	-	-	-	-	-
10.3.1.1.3.1	Main mounts		X		X		
10.3.1.1.3.2	Ground handling mounts		X				X
10.3.1.1.4	Bleed air interface		X				X
10.3.1.2	Physical characteristics. (Title only)	-	-	-	-	-	-
10.3.1.2.1	Dry mass (weight) of engine components		X	X			X
10.3.1.2.2	Mass (weight) of residual fluids		X				X
10.3.1.2.3	Mass (weight) of operating fluid		X				X
10.3.1.2.4	Mass moment of inertia of complete engine		X				
10.3.1.3.1	Adhesives and sealants	X					X
10.3.1.3.2	Elastomeric materials, “O” rings, seals, etc.		X				X
10.3.1.5	Transportability		X				X
10.3.1.6.1	Flammable Fluid Systems						X
10.3.1.6.5	Jet wake					X	X
10.3.1.7	Design control. (Title only)	-	-	-	-	-	-
10.3.1.7.1	Parts list	X					X
10.3.1.7.2	Assembly of components and parts	X		X			X
10.3.1.8	Changes in vendors or fab. process	X					
10.3.1.9	Nameplate and product marking	X					
10.3.1.9.1	Critical parts ident. and tracking	X					
10.3.1.10	Interchangeability	X					
10.3.2	Performance and operating characteristics (Title only)	-	-	-	-	-	-
10.3.2.1.1	Thrust/Horsepower Extraction		X			X	
10.3.2.1.1.1	Idle thrust - [ground/flight idle]		X			X	
10.3.2.1.3	Thrust Transients		X			X	
10.3.2.1.7	Operating attitude and conditions		X				X
10.3.2.1.8	Stopping					X	
10.3.2.1.9	Surface temperature and heat rejection		X				
10.3.2.2.1	Performance and presentation - steady-state		X			X	
10.3.2.2.2	Performance and presentation - transient		X			X	
10.3.2.2.3	Performance retention		X			X	
10.3.2.2.4	Performance evaluation		X			X	
10.3.3	Environmental conditions (Title only)	-	-	-	-	-	-
10.3.3.1.1	Humidity						X
10.3.3.1.2	Fungus						X
10.3.3.1.3	Corrosive atmosphere						X
10.3.3.2.3	Sand and dust ingestion						X
10.3.4	Integrity (Title only)	-	-	-	-	-	-
10.3.4.1.7.5	Composites		X				X
10.3.4.1.8.2	Vibration limits		X			X	
10.3.5	Reliability and maintainability (R&M)	X	X				X
10.3.5.1	Reliability	X	X				X
10.3.5.1.1	Reliability quantitative characteristics	X	X				X

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TABLE XLVIIIc. Design-to-goal characteristics verification matrix (table XXXI and 4.12 guidance) - continued.

Paragraph	Subject	Verification Process					
		I	A	D	CT	ET	S
10.3.5.2	Maintainability	X	X				X
10.3.5.2.1	Maintainability quantitative characteristics	X	X				X
10.3.5.2.1.1	Excluded maintenance functions	X	X				X
10.3.5.2.3	Maintenance	X	X				X
10.3.5.3.1	Support concept	X	X	X			X
10.3.5.3.2	Support equipment	X	X	X			X
10.3.5.4	Human perform. and human engrg	X	X				X
10.3.6	Combat survivability (Title only)	-	-	-	-	-	-
10.3.6.1.1	Noise						X
10.3.6.1.2	Vulnerability - components/externals	X	X				X
10.3.7	Subsystems (Title only)	-	-	-	-	-	-
10.3.7.2.1	Control system performance		X		X	X	X
10.3.7.2.2	Control system adjustments	X					X
10.3.7.2.3	Overspeed protection		X			X	X
10.3.7.2.4	Thrust split dynamics (rate and range)		X			X	
10.3.7.2.5	Roll flow dynamics		X			X	
10.3.7.2.6	On-Board diagnostics	X			X		X
10.3.7.2.6.1	Failure detection		X		*	*	X
10.3.7.2.6.2	Diagnostic capabilities	X				*	X
10.3.7.3	Fuel system		X			X	X
10.3.7.3.2.1	Fuel contamination		X				X
10.3.7.3.2.2	Fuel sys perform. with ext. assistance		X				X
10.3.7.3.2.3	Fuel pump priming						X
10.3.7.3.2.4	Fuel lubricity						X
10.3.7.3.2.5	Fuel filters						X
10.3.7.4	Electrical system (Title only)	-	-	-	-	-	-
10.3.7.4.1	Electrical power	X					X
10.3.7.4.1.1	Generator/alternator					X	X
10.3.7.4.2	Electrical connectors and cables		X				X
10.3.7.4.3	Electronic components						X
10.3.7.4.4	Electrical grounding						X
10.3.7.4.5	Electrical bonding				X		X
10.3.7.4.6	Potting compounds						X
10.3.7.5	Ignition system(s)					X	X
10.3.7.6	Engine Diagnostic and Health Management System (DHMS)	X					X
10.3.7.6.1	DHMS Failure Detection/ Fault Isolation				*	*	X
10.3.7.6.3	Test compatibility	X					X
10.3.7.6.4	Testability	X					X
10.3.7.7	Lubrication system		X				X
10.3.7.7.1	Lubrication oil					X	X
10.3.7.7.1.1	Oil consumption limits		X	X			X
10.3.7.7.2.1	Oil reservoir		X				
10.3.7.7.2.1.1	Oil reservoir external features	X					X
10.3.7.7.2.2	Oil drains		X			X	X
10.3.7.7.2.3	Oil filters	X					X
10.3.7.7.2.4	Oil debris monitor	X					X
10.3.7.8	Hydraulic system	X					X
10.3.7.8.1	Hydraulic ground test provisions	X					X
10.3.7.8.2	Hydraulic fluid filters						X

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TABLE XLVIIIc. Design-to-goal characteristics verification matrix (table XXXI and 4.12 guidance) - continued.

Paragraph	Subject	Verification Process					
		I	*A*	*D*	*CT*	*ET*	*S*
10.3.7.9	Starting	X				X	X
10.3.7.9.1	Ground starts					X	
10.3.7.9.3	Starting procedure					X	X
10.3.7.9.4	Automatic relight					X	X
10.3.7.10	Exhaust nozzle system		X				X
10.3.7.10.1	Exhaust nozzle external asymmetrical air pressure loads		X				X
10.3.7.10.2	Vectoring nozzle		X			X	X
10.3.7.11	Augmentation system		X			X	X
10.3.7.12	Gearbox	X				X	X
10.3.7.13	Tubing/plumbing system (Title only)	-	-	-	-	-	-
10.3.7.13.1	Tubing/plumbing identification	X					
10.3.7.13.2	Tubing/plumbing clearances	X	X				X
10.3.8	Embed. Comp. Resources (Title only)	-	-	-	-	-	-
10.3.8.1	System Description	X					
10.3.8.2	Functional Performance		X		X	X	X
10.3.8.3	Computer Software Config. Items	X					
10.3.8.4	High Order Language	X					

Table Key:

I *Inspection*
A *Analysis*
D *Demonstration*
CT *Component Test (* opportunistic)*
ET *Engine Test(* opportunistic)*
S *Similarity (to reference engine)*

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TABLE XLIX. Example of table XXXI tailored for an Unmanned Air Vehicle (UAV) single turbofan engine application.

(See 4.12 guidance.)

(To be completed by the specification developer; entries represent a sample guidance.)

Reqt. No.	Reqt. Title	Concept Refinement	Technology Development		
¾	VERIFICATION AND QUALIFICATION (TITLE)	Spiral 0	Spiral 1	Spiral 2	Spiral 3
3/4.1	Design and construction (title)	G			
3/4.1.1	Item and interface definition (title)	ICD	E	E	
3/4.1.1.1	Interface and installation	ICD	E	E	
3/4.1.1.2	Installation hardware	ICD	E	E	
3/4.1.1.3	Interface loads	ICD	E	E	
3/4.1.1.4	Mounts (title)	ICD	E	E	
3/4.1.1.4.1	Main mounts	ICD	E	E	
3/4.1.1.4.2	Ground handling mounts	ICD	E	E	
3/4.1.1.4.3	Engine stiffness	ICD	E	E	
3/4.1.1.5	Control lever and signals (title)	ICD	E	E	
3/4.1.1.5.1	Power lever and signal	N/A			
3/4.1.1.5.2	Load demand command and signal	N/A			
3/4.1.1.5.3	Output speed lever and signal	N/A			
3/4.1.1.5.4	Fuel shutoff lever and signal	N/A			
3/4.1.1.5.5	Lever torque	N/A			
3/4.1.1.6	Sub- system interfaces	ICD	E	W	W
3/4.1.1.7	Bleed air interface	ICD	E	W	W
3/4.1.1.7.1	Customer bleed air contamination	N/A			
3/4.1.1.8	Drains and fluid collection devices	ICD			
3/4.1.1.9	Power absorber interface	N/A			

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TABLE XLIX. Example of table XXXI tailored for an Unmanned Air Vehicle (UAV) single turbofan engine application - continued.

(See 4.12 guidance.)

(To be completed by the specification developer; entries represent a sample guidance.)

Reqt. No.	Reqt. Title	Concept Refinement	Technology Development		
¾	VERIFICATION AND QUALIFICATION (TITLE)	Spiral 0	Spiral 1	Spiral 2	Spiral 3
3/4.1.1.10	Power take-off (PTO)	N/A			
3/4.1.1.11	Starting drivetrain	E	E	W	W
3/4.1.1.11.1	Starting torque and speed	E	E	W	W
3/4.1.1.12	Thrust reverser interface	N/A			
3/4.1.1.13	Exhaust system interface	ICD	W	W	W
3/4.1.1.14	Compatibility - Engine Airframe	ICD	E	E	W
3/4.1.1.15	Control and External (C&E) component list	ICD	ICD	ICD	ICD
3/4.1.2	Physical characteristics (title)	ICD	ICD	ICD	ICD
3/4.1.2.1	Dry mass (weight) of complete engine	D	D	D	D
3/4.1.2.2	Mass moment of inertia of complete engine	A	A	A	A
3/4.1.3	Materials, processes and parts	G	C	E	E
3/4.1.3.1	Hazardous Materials	G	C	E	E
3/4.1.4	Fasteners	ICD	C	E	E
3/4.1.5	Nameplate and product marking	G	D	D	D
3/4.1.5.1	Engine data plate marking	G	D	D	D
3/4.1.5.2	Critical parts identification and tracking	G	A	A	A
3/4.1.6	Transportability	G	D	W	W

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TABLE XLIX. Example of table XXXI tailored for an Unmanned Air Vehicle (UAV) single turbofan engine application - continued.

(See 4.12 guidance.)
(To be completed by the specification developer; entries represent a sample guidance.)

Reqt. No.	Reqt. Title	Concept Refinement	Technology Development		
¾	VERIFICATION AND QUALIFICATION (TITLE)	Spiral 0	Spiral 1	Spiral 2	Spiral 3
3/4.1.7	Interchangeability	N/A			
3/4.1.8	Safety (title)	G	A	A	A
3/4.1.8.1	Flammable fluid systems	N/A			
3/4.1.8.2	Fire prevention	G	A	W	W
3/4.1.8.3	Explosion-proof	ICD	A	C	C
3/4.1.8.4	Combustible fluid drains	ICD	E	W	W
3/4.1.8.5	Air and gas leakage	ICD	E	E	E
3/4.1.8.6	Ground Safety	G	A	W	W
3/4.1.8.7	Jet Wake	ICD	A	E	W
3/4.1.9	Design control (title)	G	A	A	A
3/4.1.9.1	Parts list	G	A	A	A
3/4.1.9.2	Assembly of components and parts	G	E	E	E
3/4.1.9.3	Changes in vendors or fabrication process	G	C	E	E
3/4.1.9.4	Standardization	G	G	G	G
3/4.2	Performance and operability (title)	G	E	E	E
3/4.2.1	Performance characteristics (title)	G	E	E	E
3/4.2.1.1	Steady-state performance	G	E	E	E
3/4.2.1.2	Transient performance	G	E	E	E
3/4.2.1.3	Performance computer program	A	E	E	E
3/4.2.1.4	Performance retention	G	E	E	E

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TABLE XLIX. Example of table XXXI tailored for an Unmanned Air Vehicle (UAV) single turbofan engine application - continued.

(See 4.12 guidance.)

(To be completed by the specification developer; entries represent a sample guidance.)

Reqt. No.	Reqt. Title	Concept Refinement	Technology Development		
¾	VERIFICATION AND QUALIFICATION (TITLE)	Spiral 0	Spiral 1	Spiral 2	Spiral 3
3/4.2.2	Operating characteristics and limits	G	E	E	E
3/4.2.2.1	Operating envelope	G	E	E	E
3/4.2.2.2	Operating attitude and conditions	G	E	E	E
3/4.2.2.3	Starting	G	E	E	E
3/4.2.2.3.1	Ground starts	G	E	E	E
3/4.2.2.3.2	Air starts	G	E	E	E
3/4.2.2.3.3	Starting limits	G	E	E	E
3/4.2.2.3.4	Starting procedure	G	E	E	E
3/4.2.2.3.5	Automatic relight	G	E	E	E
3/4.2.2.4	Stopping	G	E	E	E
3/4.2.2.5	Low power conditions (title)	G	E	E	E
3/4.2.2.5.1	Idle thrust/power - (ground/flight idle)	G	E	E	E
3/4.2.2.5.2	No load condition	N/A			
3/4.2.2.6	Stability	G	E	E	E
3/4.2.2.7	Transients	G	E	E	E
3/4.2.2.7.1	Overshoot and undershoot	G	E	E	E
3/4.2.2.7.2	Droop	G	E	E	E
3/4.2.2.8	Windmilling	G	E	E	E
3/4.2.2.9	Reverse thrust				
3/4.2.2.10	Steady-state and transient airflow	A	E	E	E
3/4.2.2.11	Inlet airflow distortion	A	E	E	W

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TABLE XLIX. Example of table XXXI tailored for an Unmanned Air Vehicle (UAV) single turbofan engine application - continued.

(See 4.12 guidance.)

(To be completed by the specification developer; entries represent a sample guidance.)

Reqt. No.	Reqt. Title	Concept Refinement	Technology Development		
¾	VERIFICATION AND QUALIFICATION (TITLE)	Spiral 0	Spiral 1	Spiral 2	Spiral 3
3/4.2.2.11.2	Engine Recoverability	A	E	E	E
3/4.2.2.12	Gas path and measurement plane temperature limits	A	E	E	E
3/4.2.2.13	Surface temperature and heat rejection	G	A	E	E
3/4.2.2.13.1	Controls and external component limiting temperature	G	A	E	E
3/4.3	Environmental conditions [title only]	G	A	E	E
3/4.3.1	Atmospheric conditions [title only]	G	A	E	E
3/4.3.1.1	Humidity	G	A	E	E
3/4.3.1.2	Fungus	G	A	E	E
3/4.3.1.3	Corrosive atmosphere	G	A	E	E
3/4.3.1.4	Icing conditions	G	A	E	W
3/4.3.2	Ingestion capability (hazard resistance) [title only]	G	A	E	W
3/4.3.2.1	Bird ingestion	G	A	E	E
3/4.3.2.2	Foreign object damage (FOD)	G	A	E	E
3/4.3.2.3	Ice ingestion	G	A	E	E
3/4.3.2.4	Sand and dust ingestion	G	A	E	E

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TABLE XLIX. Example of table XXXI tailored for an Unmanned Air Vehicle (UAV) single turbofan engine application - continued.

(See 4.12 guidance.)

(To be completed by the specification developer; entries represent a sample guidance.)

Reqt. No.	Reqt. Title	Concept Refinement	Technology Development		
¾	VERIFICATION AND QUALIFICATION (TITLE)	Spiral 0	Spiral 1	Spiral 2	Spiral 3
3/4.3.2.5	Atmospheric liquid water ingestion	G	A	E	E
3/4.3.2.6	Armament gas ingestion	G	A	E	E
3/4.3.2.7	Steam ingestion	G	A	W	W
3/4.3.3	Electromagnetic environmental effects (E^3)	ICD	A	E	E
3/4.3.3.1	Electromagnetic interference (EMI)	ICD	A	E	E
3/4.3.3.2	Intrasystem electromagnetic compatibility (EMC)	ICD	A	E	E
3/4.3.3.3	Intersystem electromagnetic compatibility (EMC)	ICD	A	E	E
3/4.3.3.4	Lightning	ICD	A	E	E
3/4.4	Integrity (title only)	A	A	E	E
4.3/4.4.4.1	Structural integrity (title only)	A	A	E	E
3/4.4.1.1	Design service life	D	D	E	E
3/4.4.1.2	Design Usage	A	A	E	E
3/4.4.1.2.1	Externally applied forces	A	A	E	E
3/4.4.1.3	Material characterization	A	A	E	E
3/4.4.1.4	Parts classification	A	A	E	E
3/4.4.1.5	Durability	A	A	E	E

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TABLE XLIX. Example of table XXXI tailored for an Unmanned Air Vehicle (UAV) single turbofan engine application - continued.

(See 4.12 guidance.)

(To be completed by the specification developer; entries represent a sample guidance.)

Reqt. No.	Reqt. Title	Concept Refinement	Technology Development		
¾	VERIFICATION AND QUALIFICATION (TITLE)	Spiral 0	Spiral 1	Spiral 2	Spiral 3
3/4.4.1.5.1	High cycle fatigue (HCF) life	A	A	E	E
3/4.4.1.5.2	Low cycle fatigue (LCF) life	A	A	E	E
3/4.4.1.5.3	Creep	A	A	E	E
3/4.4.1.6	Strength	A	A	E	E
3/4.4.1.6.1	Factors of safety	A	A	E	E
3/4.4.1.6.2	Blade and disk deflection	A	A	E	E
3/4.4.1.6.3	Containment	N/A			
3/4.4.1.6.4	Blade out	N/A			
3/4.4.1.6.5	Overspeed and overtemperature	N/A			
3/4.4.1.6.6	Disk burst speed	N/A			
3/4.4.1.6.7	Output shaft torque limits	N/A			
3/4.4.1.6.8	Output shaft speed limits	N/A			
3/4.4.1.6.9	Pressure balance	A	A	E	E
3/4.4.1.6.10	Gyroscopic moments	A	A	E	E
3/4.4.1.7	Damage tolerance	A	C	C	E
3/4.4.1.7.1	Residual Strength	A	C	C	E
3/4.4.1.7.2	Initial production and in-service flaw size	A	C	C	E
3/4.4.1.7.3	Flaw growth and inspection intervals	A	C	C	E

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TABLE XLIX. Example of table XXXI tailored for an Unmanned Air Vehicle (UAV) single turbofan engine application - continued.

(See 4.12 guidance.)

(To be completed by the specification developer; entries represent a sample guidance.)

Reqt. No.	Reqt. Title	Concept Refinement	Technology Development		
¾	VERIFICATION AND QUALIFICATION (TITLE)	Spiral 0	Spiral 1	Spiral 2	Spiral 3
3/4.4.1.8	Vibration and dynamic response	E	E	E	E
3/4.4.1.8.1	Vibration limits	E	E	E	E
3/4.4.1.8.2	Critical speeds	E	E	E	E
3/4.5	R&M [title only]	G	A	A	
3/4.5.1	Reliability [title only]	G	A	A	
3/4.5.1.1	Reliability quantitative requirements	G	A	A	
3/4.5.2	Maintainability [title only]	G	D	A	D
3/4.5.2.1	Maintainability quantitative requirements	G	A	A	
3/4.5.2.1.1	Excluded maintenance functions	G	A	A	
3/4.5.2.2	Maintainability qualitative requirements	G	A	A	
3/4.5.2.3	Maintenance [title only]	G	A	D	D
3/4.5.2.3.1	Modules	G	A	D	D
3/4.5.2.3.2	Maintenance, inspection and repair cycle	G	A	D	D
3/4.5.2.3.2.1	Maintenance inspection techniques	G	A	D	D
3/4.5.2.3.3	Tools	G	A	D	D

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TABLE XLIX. Example of table XXXI tailored for an Unmanned Air Vehicle (UAV) single turbofan engine application - continued.

(See 4.12 guidance.)

(To be completed by the specification developer; entries represent a sample guidance.)

Reqt. No.	Reqt. Title	Concept Refinement	Technology Development		
¾	VERIFICATION AND QUALIFICATION (TITLE)	Spiral 0	Spiral 1	Spiral 2	Spiral 3
3/4.5.2.4	Battle damage repair	G	A	D	E
3/4.5.3	Human performance and human engineering	G	A	D	D
3/4.6	Combat survivability (title only)	G	A	A	D
3/4.6.1	Susceptibility (title only)	G	A	A	A
3/4.6.1.1	Noise	G	A	E	W
3/4.6.1.2	IR radiation (Note: F to obtain platform signature, A to backout engine contribution)	G	A	E	W
3/4.6.1.3	Smoke	G	A	E	E
3/4.6.1.4	Gaseous emissions	G	A	E	E
3/4.6.1.5	Fuel streaming and vapor puffing	G	A	E	E
3/4.6.1.6	Water vapor contrails	G	A	E	E
3/4.6.1.7	Radar cross section (RCS)		A	E	W
3/4.6.1.8	Radar absorbent materials (RAMs) and coatings		C	E	W
3/4.6.2	Vulnerability (title only)		A	A	E
3/4.6.2.1	Ballistic weapons		A	A	E
3/4.6.2.1.1	Static structure		A	C	C

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TABLE XLIX. Example of table XXXI tailored for an Unmanned Air Vehicle (UAV) single turbofan engine application - continued.

(See 4.12 guidance.)

(To be completed by the specification developer; entries represent a sample guidance.)

Reqt. No.	Reqt. Title	Concept Refinement	Technology Development		
¾	VERIFICATION AND QUALIFICATION (TITLE)	Spiral 0	Spiral 1	Spiral 2	Spiral 3
3/4.6.2.2	Nuclear weapons effects		A	C	C
3/4.6.2.3	Chemical and biological agent effects		A	A	C
3/4.6.2.4	Fuel ingestion effects		A	E	E
3/4.6.2.4.1	Fuel ingestion - steady-flow	N/A			
3/4.6.2.4.1.1	Fuel ingestion - cooling air contamination	N/A			
3/4.6.2.4.2	Fuel ingestion - transient (“quick dump”)	N/A			
3/4.7	Subsystems (title)	G	E	E	W
3/4.7.1	Anti-icing and de-icing system		A	E	E
3/4.7.2	Control system	ICD	E	E	E
3/4.7.2.1	Control system performance	G	E	E	E
3/4.7.2.1.1	Backup control	N/A			
3/4.7.2.2	Control system adjustments	ICD	E	E	E
3/4.7.2.3	Overspeed protection system	ICD	A	E	E
3/4.7.3	Fuel System (title only)	ICD	E	E	E
3/4.7.3.1	Fuels	G	G	E	E

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TABLE XLIX. Example of table XXXI tailored for an Unmanned Air Vehicle (UAV) single turbofan engine application - continued.

(See 4.12 guidance.)

(To be completed by the specification developer; entries represent a sample guidance.)

Reqt. No.	Reqt. Title	Concept Refinement	Technology Development		
¾	VERIFICATION AND QUALIFICATION (TITLE)	Spiral 0	Spiral 1	Spiral 2	Spiral 3
3/4.7.3.1.1	Primary fuel (repeat test if engine configuration changes which impact fuel performance)	G	G	E	E
3/4.7.3.1.2	Alternate fuel (repeat test if engine configuration changes which impact fuel performance)	G	G	E	E
3/4.7.3.1.3	Restricted fuel (repeat test if engine configuration changes which impact fuel performance)	G	G	E	E
3/4.7.3.1.4	Emergency fuel	G	G	E	E
3/4.7.3.2	Fuel system performance (title only)	ICD	E	E	W
3/4.7.3.2.1	Fuel contamination	ICD	E	E	W
3/4.7.3.2.2	Fuel system performance with external assistance	ICD	E	E	W
3/4.7.3.2.3	Fuel system performance with no external assistance	ICD	E	E	W

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TABLE XLIX. Example of table XXXI tailored for an Unmanned Air Vehicle (UAV) single turbofan engine application - continued.

(See 4.12 guidance.)

(To be completed by the specification developer; entries represent a sample guidance.)

Reqt. No.	Reqt. Title	Concept Refinement	Technology Development		
¾	VERIFICATION AND QUALIFICATION (TITLE)	Spiral 0	Spiral 1	Spiral 2	Spiral 3
3/4.7.3.2.4	Fuel system performance under conditions of excessive fuel vapor	ICD	E	E	W
3/4.7.3.2.5	Fuel pump priming	ICD	E	E	W
3/4.7.3.2.6	Fuel lubricity	ICD	C	E	E
3/4.7.3.2.7	Fuel system performance with water saturated fuel	ICD	E	E	E
3/4.7.3.2.8	Fuel filter	ICD	E	E	E
3/4.7.3.2.9	Fuel flow limit	ICD	E	E	E
3/4.7.4	Electrical system [title only]	ICD	E	E	W
3/4.7.4.1	Electrical power	ICD	E	E	W
3/4.7.4.1.1	Generator	ICD	E	E	W
3/4.7.4.2	Alternate electrical power	ICD	E	E	W
3/4.7.4.3	Electrical connectors and cables	ICD	E	E	E
3/4.7.4.4	Electrical bonding	ICD	C	E	W
3/4.7.4.5	Ground isolation	ICD	E	E	E
3/4.7.4.6	Potting compounds	ICD	C	E	E
3/4.7.5	Ignition system	ICD	E	E	E
3/4.7.5.1	Ignition system fouling [title only]	ICD	E	E	E
3/4.7.5.1.1	Carbon fouling	ICD	E	E	W
3/4.7.5.1.2	Water fouling	ICD	E	E	E
3/4.7.6	Engine monitoring system (EMS)	ICD	E	E	W

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TABLE XLIX. Example of table XXXI tailored for an Unmanned Air Vehicle (UAV) single turbofan engine application - continued.

(See 4.12 guidance.)

(To be completed by the specification developer; entries represent a sample guidance.)

Reqt. No.	Reqt. Title	Concept Refinement	Technology Development		
¾	VERIFICATION AND QUALIFICATION (TITLE)	Spiral 0	Spiral 1	Spiral 2	Spiral 3
3/4.7.6.1	EMS fault detection and isolation	ICD	E	E	W
3/4.7.6.2	On-board engine diagnostic function	ICD	E	E	W
3/4.7.7	Optical systems	N/A			
3/4.7.7.1	Fiber optic cables and connectors	N/A			
3/4.7.8	Lubrication system [title only]`	ICD	E	E	E
3/4.7.8.1	Lubrication oil	ICD	E	E	E
3/4.7.8.1.1	Oil pressure and temperature limits	ICD	E	E	E
3/4.7.8.1.2	Oil consumption limits	ICD	E	E	E
3/4.7.8.1.3	Oil flow interruption or depletion	N/A			
3/4.7.8.2	Lubrication system components and features [title only]	ICD	E	E	E
3/4.7.8.2.1	Oil reservoir	ICD	E	E	E
3/4.7.8.2.1.1	Oil reservoir external features	ICD	E	E	E
3/4.7.8.2.2	Oil drains	ICD	E	E	E
3/4.7.8.2.3	Oil filters	ICD	E	E	E
3/4.7.8.2.4	Oil debris monitor(s)	ICD	E	E	E
3/4.7.8.3	Breather mist	G	G	E	E
3/4.7.9	Hydraulic system	ICD	E	E	E

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TABLE XLIX. Example of table XXXI tailored for an Unmanned Air Vehicle (UAV) single turbofan engine application - continued.

(See 4.12 guidance.)

(To be completed by the specification developer; entries represent a sample guidance.)

Reqt. No.	Reqt. Title	Concept Refinement	Technology Development		
¾	VERIFICATION AND QUALIFICATION (TITLE)	Spiral 0	Spiral 1	Spiral 2	Spiral 3
3/4.7.9.1	Hydraulic ground test provisions	ICD	E	E	E
3/4.7.9.2	Hydraulic system air removal	ICD	E	E	E
3/4.7.9.3	Hydraulic fluid filters	ICD	E	E	E
3/4.7.10	Exhaust nozzle system	N/A			
3/4.7.10.1	Exhaust nozzle external asymmetrical air pressure loads	N/A			
3/4.7.10.2	Vectoring nozzle	N/A			
3/4.7.10.2.1	Vectoring nozzle angle and rate	N/A			
3/4.7.10.2.2	Vectoring nozzle failure accommodation	N/A			
3/4.7.11	Augmentation system	N/A			
3/4.7.11.2.1	Water injection system fluid	N/A			
3/4.7.12	Wash system	ICD	G	E	E
3/4.7.13	Brake system	N/A			
3/4.7.14	Negative torque limiter	N/A			
3/4.7.14.1	Multiple power section engines	N/A			
3/4.7.15	Power absorber (engine supplied)	N/A			
3/4.7.15.1	Propeller feathering and reverse pitch	N/A			
3/4.7.16	Gearbox	ICD	E	E	E

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TABLE XLIX. Example of table XXXI tailored for an Unmanned Air Vehicle (UAV) single turbofan engine application - continued.

(See 4.12 guidance.)

(To be completed by the specification developer; entries represent a sample guidance.)

Reqt. No.	Reqt. Title	Concept Refinement	Technology Development		
¾	VERIFICATION AND QUALIFICATION (TITLE)	Spiral 0	Spiral 1	Spiral 2	Spiral 3
3/4.8	Software resources (title)	G	A	E	E
3/4.8.1	Software performance and design	G	A	E	E
3/4.8.1.1	Built-in-test and inspectability	G	E	E	E
3/4.8.1.2	Computer reprogramming	G	E	E	E
3/4.8.2	Spare resources	G	A	A	A
3/4.9	Production facilities, capabilities, and processes	G	E	E	E
3/4.10	Production cost requirement (PCR)	G	G	G	G

NOTE:

Reverification of the engine is required when the configuration changes impact the engine requirement.

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TABLE L. Example of table XXXI tailored for a Commercial Off The Shelf (COTS) application.

(See 4.12 guidance.)

(To be completed by the specification developer; entries represent a sample guidance.)

Reqt. No.	Reqt. Title	Technology Development				
¾	VERIFICATION AND QUALIFICATION (TITLE)	FAR Paragraph Number	PDR	CDR	DRR	IFR
3/4.1	Design and construction (title)					
3/4.1.1	Item and interface definition (title)					
3/4.1.1.1	Interface and installation	33.5	Similarity		A	
3/4.1.1.2	Installation hardware	33.5	Similarity		A	
3/4.1.1.3	Interface loads	33.5	Similarity		A	
3/4.1.1.4	Mounts (title)					
3/4.1.1.4.1	Main mounts	33.5, 33.23	Similarity		A	
3/4.1.1.4.2	Ground handling mounts	No equivalent		A	A	
3/4.1.1.4.3	Engine stiffness	No equivalent		A	A	
3/4.1.1.5	Control lever and signals (title)					
3/4.1.1.5.1	Power lever and signal	33.5, 33.28	Similarity		A	
3/4.1.1.5.2	Load demand command and signal	33.5, 33.28	Similarity		A	
3/4.1.1.5.3	Output speed lever and signal	33.5, 33.28	Similarity		A	
3/4.1.1.5.4	Fuel shutoff lever and signal	33.5, 33.28	Similarity		A	
3/4.1.1.5.5	Lever torque	No equivalent				C
3/4.1.1.6	Sub- system interfaces	33.5, 33.28	Similarity		A	
3/4.1.1.7	Bleed air interface	33.5, 33.66, 33.89	Similarity		A	
3/4.1.1.7.1	Customer bleed air contamination	33.5	Similarity		A	
3/4.1.1.8	Drains and fluid collection devices	33.5	Similarity		A	

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TABLE L. Example of table XXXI tailored for a Commercial Off The Shelf (COTS) application - continued.

(See 4.12 guidance.)

(To be completed by the specification developer; entries represent a sample guidance.)

Reqt. No.	Reqt. Title	Technology Development				
¾	VERIFICATION AND QUALIFICATION (TITLE)	FAR Paragraph Number	PDR	CDR	DRR	IFR
3/4.1.1.9	Power absorber interface	33.5	Similarity		A	
3/4.1.1.10	Power take-off (PTO)	33.5, 33.25	Similarity		A	
3/4.1.1.11	Starting drivetrain	No equivalent		A	A	
3/4.1.1.11.1	Starting torque and speed	33.5, 33.87, 33.89	Similarity		A	
3/4.1.1.12	Thrust reverser interface	33.97	Similarity		A	
3/4.1.1.13	Exhaust system interface	33.5	Similarity		A	
3/4.1.1.14	Compatibility - Engine Airframe	No equivalent				A, E
3/4.1.1.15	Control and External (C&E) component list	No equivalent	P		P	
3/4.1.2	Physical characteristics (title)					
3/4.1.2.1	Dry mass (weight) of complete engine	33.7	Similarity		A	
3/4.1.2.2	Mass moment of inertia of complete engine	33.7	Similarity		A	
3/4.1.3	Materials, processes and parts	33.15	Similarity		A	
3/4.1.4	Fasteners	33.75	Similarity		A	
3/4.1.5	Nameplate and product marking	No equivalent	X		X	
3/4.1.5.1	Engine data plate marking	45.13	Similarity		A	

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TABLE L. Example of table XXXI tailored for a Commercial Off The Shelf (COTS) application - continued.

(See 4.12 guidance.)
(To be completed by the specification developer; entries represent a sample guidance.)

Reqt. No.	Reqt. Title	Technology Development				
¾	VERIFICATION AND QUALIFICATION (TITLE)	FAR Paragraph Number	PDR	CDR	DRR	IFR
3/4.1.5.2	Critical parts identification and tracking	No equivalent		X	X	
3/4.1.6	Transportability	No equivalent				X
3/4.1.7	Interchangeability	No equivalent				X
3/4.1.8	Safety (title)					
3/4.1.8.1	Flammable fluid systems	33.17	Similarity		A	
3/4.1.8.2	Fire prevention	33.17, 33.21	Similarity		A	
3/4.1.8.3	Explosion-proof	33.91	Similarity		A	
3/4.1.8.4	Combustible fluid drains	33.17	Similarity		A	
3/4.1.8.5	Air and gas leakage	No equivalent		A	A	
3/4.1.8.6	Ground Safety	No equivalent	A	I	A, I	
3/4.1.8.7	Jet Wake	No equivalent	A	A	A	
3/4.1.9	Design control (title)					
3/4.1.9.1	Parts list	21.31, 21.91, 21.93, 21.95, 21.97	Similarity		A	
3/4.1.9.2	Assembly of components and parts	No equivalent		D	D	D
3/4.1.9.3	Changes in vendors or fabrication process	21.139, 21.143	Similarity		A	
3/4.1.9.4	Standardization	No equivalent	I	I	I	D
3/4.2	Performance and operability (title)					
3/4.2.1	Performance characteristics (title)					
3/4.2.1.1	Steady-state performance	33.89	Similarity		A	

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TABLE L. Example of table XXXI tailored for a Commercial Off The Shelf (COTS) application - continued.

(See 4.12 guidance.)

(To be completed by the specification developer; entries represent a sample guidance.)

Reqt. No.	Reqt. Title	Technology Development				
¾	VERIFICATION AND QUALIFICATION (TITLE)	FAR Paragraph Number	PDR	CDR	DRR	IFR
3/4.2.1.2	Transient performance	33.65, 33.73, 33.89	Similarity		A	
3/4.2.1.3	Performance computer program	No equivalent	S, I	S, I	S, I	S, E
3/4.2.1.4	Performance retention	33.87, .90	Similarity		A	
3/4.2.2	Operating characteristics and limits	33.65, 33.71, 33.73, 33.87, 33.89	Similarity		A	
3/4.2.2.1	Operating envelope	33.65, 33.71, 33.73, 33.87, 33.89	Similarity		A	
3/4.2.2.2	Operating attitude and conditions	33.65, 33.71, 33.73, 33.87, 33.89	Similarity		A	
3/4.2.2.3	Starting	33.87, 33.89	Similarity		A	
3/4.2.2.3.1	Ground starts	33.87, 33.89	Similarity		A	
3/4.2.2.3.2	Air starts	33.89	Similarity		A	
3/4.2.2.3.3	Starting limits	33.89	Similarity		A	
3/4.2.2.3.4	Starting procedure	33.89	Similarity		A	
3/4.2.2.3.5	Automatic relight	33.89	Similarity		A	
3/4.2.2.4	Stopping	33.67, 33.87, 33.89	Similarity		A	
3/4.2.2.5	Low power conditions (title)					
3/4.2.2.5.1	Idle thrust/power - (ground/flight idle)	33.89	Similarity		A	
3/4.2.2.5.2	No load condition	33.89	Similarity		A	
3/4.2.2.6	Stability	33.65, 33.73	Similarity		A	
3/4.2.2.7	Transients	33.65, 33.73, 33.89	Similarity		A	
3/4.2.2.7.1	Overshoot and undershoot	33.89	Similarity		A	
3/4.2.2.7.2	Droop	33.89	Similarity		A	

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TABLE L. Example of table XXXI tailored for a Commercial Off The Shelf (COTS) application - continued.

(See 4.12 guidance.)

(To be completed by the specification developer; entries represent a sample guidance.)

Reqt. No.	Reqt. Title	Technology Development				
¾	VERIFICATION AND QUALIFICATION (TITLE)	FAR Paragraph Number	PDR	CDR	DRR	IFR
3/4.2.2.8	Windmilling	33.74	Similarity		A	
3/4.2.2.9	Reverse thrust	33.97	Similarity		A	
3/4.2.2.10	Steady-state and transient airflow	33.89	Similarity		A	
3/4.2.2.11	Inlet airflow distortion	33.5, 33.65	Similarity		A	
3/4.2.2.11.1	Pressure and temperature rate of change	No equivalent		A	A	
3/4.2.2.11.2	Engine Recoverability	No equivalent	A	A	A	A, E
3/4.2.2.12	Gas path and measurement plane temperature limits	33.5, 33.88	Similarity		A	
3/4.2.2.13	Surface temperature and heat rejection	33.5	Similarity		A	
3/4.2.2.13.1	Controls and external component limiting temperature	33.91	Similarity		A	
3/4.3	Environmental conditions [title only]					
3/4.3.1	Atmospheric conditions [title only]					
3/4.3.1.1	Humidity	No equivalent		A	A	
3/4.3.1.2	Fungus	No equivalent		A	A	
3/4.3.1.3	Corrosive atmosphere	No equivalent	A	A	A	
3/4.3.1.4	Icing conditions	33.68, B33.1	Similarity		A	

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TABLE L. Example of table XXXI tailored for a Commercial Off The Shelf (COTS) application - continued.

(See 4.12 guidance.)

(To be completed by the specification developer; entries represent a sample guidance.)

Reqt. No.	Reqt. Title	Technology Development				
¾	VERIFICATION AND QUALIFICATION (TITLE)	FAR Paragraph Number	PDR	CDR	DRR	IFR
3/4.3.2	Ingestion capability (hazard resistance) [title only]					
3/4.3.2.1	Bird ingestion	33.76	Similarity		A	
3/4.3.2.2	Foreign object damage (FOD)	No equivalent	A	A	A	
3/4.3.2.3	Ice ingestion	33.77, 33.88	Similarity		A	
3/4.3.2.4	Sand and dust ingestion	No equivalent	A	A	A	
3/4.3.2.5	Atmospheric liquid water ingestion	33.78, B33.1	Similarity		A	
3/4.3.2.6	Armament gas ingestion	N/A				
3/4.3.2.7	Steam ingestion	N/A				
3/4.3.3	Electromagnetic environmental effects (E^3)	33.28	Similarity		A	
3/4.3.3.3	Intrasystem electromagnetic compatibility (EMC)	33.28	Similarity		A	
3/4.3.3.4	Intersystem electromagnetic compatibility (EMC)	33.28	Similarity		A	
3/4.4	Lightning	33.28	Similarity		A	
4.3/4.4.4.1	Integrity (title only)					
3/4.4.1	Structural integrity (title only)					
3/4.4.1.1	Design service life	33.4, 33.14, 33.19, 33.27	Similarity		A	
3/4.4.1.2	Design Usage			A	A	A
3/4.4.1.2.1	Externally applied forces	33.23		A, C	A, C	C, E

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TABLE L. Example of table XXXI tailored for a Commercial Off The Shelf (COTS) application - continued.

(See 4.12 guidance.)

(To be completed by the specification developer; entries represent a sample guidance.)

Reqt. No.	Reqt. Title	Technology Development				
¾	VERIFICATION AND QUALIFICATION (TITLE)	FAR Paragraph Number	PDR	CDR	DRR	IFR
3/4.4.1.3	Material characterization	33.15	Similarity		A	
3/4.4.1.4	Parts classification	33.75		A	A	
3/4.4.1.5	Durability	33.14, 33.19, 33.90, 33.93	Similarity		A	
3/4.4.1.5.1	High cycle fatigue (HCF) life	33.63	Similarity		A	
3/4.4.1.5.2	Low cycle fatigue (LCF) life	33.14	Similarity		A	
3/4.4.1.5.3	Creep	No equivalent		A	A	C, E
3/4.4.1.6	Strength	33.75, 33.91	Similarity		A	
3/4.4.1.6.1	Factors of safety	33.23, 33.62	Similarity		A	
3/4.4.1.6.2	Blade and disk deflection	No equivalent		A	A	C, De
3/4.4.1.6.3	Containment	33.19, 33.94	Similarity		A	
3/4.4.1.6.4	Blade out	33.94	Similarity		A	
3/4.4.1.6.5	Overspeed and overtemperature	33.27, 33.87(a)(7), 33.88	Similarity		A	
3/4.4.1.6.6	Disk burst speed	33.27	Similarity		A	
3/4.4.1.6.7	Output shaft torque limits	No equivalent		A	A	
3/4.4.1.6.8	Output shaft speed limits	33.27, 33.7	Similarity		A	
3/4.4.1.6.9	Pressure balance	No equivalent	A	A	A	De
3/4.4.1.6.10	Gyroscopic moments	33.5, 33.23	Similarity	A	A	A

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TABLE L. Example of table XXXI tailored for a Commercial Off The Shelf (COTS) application - continued.

(See 4.12 guidance.)

(To be completed by the specification developer; entries represent a sample guidance.)

Reqt. No.	Reqt. Title	Technology Development				
¾	VERIFICATION AND QUALIFICATION (TITLE)	FAR Paragraph Number	PDR	CDR	DRR	IFR
3/4.4.1.7	Damage tolerance	No equivalent		A, P, Q	A, P, Q	C
3/4.4.1.7.1	Residual Strength	No equivalent		A	A	
3/4.4.1.7.2	Initial production and in-service flaw size	No equivalent		A	A	I, C, D
3/4.4.1.7.3	Flaw growth and inspection intervals	No equivalent		A	A	C, I
3/4.4.1.8	Vibration and dynamic response	33.63, 33.83	Similarity		A	
3/4.4.1.8.1	Vibration limits	33.5, 33.83	Similarity		A	
3/4.4.1.8.2	Critical speeds	33.83	Similarity		A	
3/4.4.1.9	External surface foreign object damage	No equivalent		A	A	C
3/4.4.2	Mechanical equipment and subsystem integrity	No equivalent	A	A, I	A, I	D, E
3/4.4.3	Avionic and electronic integrity	No equivalent	A	A, I	A, I	D, E
3/4.5	R&M [title only]					
3/4.5.1	Reliability [title only]					
3/4.5.1.1	Reliability quantitative requirements	No equivalent	A	A	A	
3/4.5.2	Maintainability [title only]					

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TABLE L. Example of table XXXI tailored for a Commercial Off The Shelf (COTS) application - continued.

(See 4.12 guidance.)

(To be completed by the specification developer; entries represent a sample guidance.)

Reqt. No.	Reqt. Title	Technology Development				
¾	VERIFICATION AND QUALIFICATION (TITLE)	FAR Paragraph Number	PDR	CDR	DRR	IFR
3/4.5.2.1	Maintainability quantitative requirements	No equivalent	A, D	A, D	A, D	
3/4.5.2.1.1	Excluded maintenance functions	No equivalent	A, D	A, D	A, D	
3/4.5.2.2	Maintainability qualitative requirements	No equivalent	A, D, I	A, D, I	A, D, I	
3/4.5.2.3	Maintenance [title only]					
3/4.5.2.3.1	Modules	No equivalent	A	D	A, D	
3/4.5.2.3.2	Maintenance, inspection and repair cycle	33.4, 33.90	Similarity	A, D	A, D	
3/4.5.2.3.2.1	Maintenance inspection techniques	33.4, 33.5, 33.90	Similarity		A	
3/4.5.2.3.3	Tools	33.4	D	D, I	D, I	
3/4.5.2.4	Battle damage repair	No equivalent	A	A	A	
3/4.5.3	Human performance and human engineering	No equivalent	A	A	A	
3/4.6	Combat survivability (title only)					
3/4.6.1	Susceptibility (title only)					
3/4.6.1.1	Noise	36.7 No equivalent	A	A	A	E

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TABLE L. Example of table XXXI tailored for a Commercial Off The Shelf (COTS) application - continued.

(See 4.12 guidance.)

(To be completed by the specification developer; entries represent a sample guidance.)

Reqt. No.	Reqt. Title	Technology Development				
¾	VERIFICATION AND QUALIFICATION (TITLE)	FAR Paragraph Number	PDR	CDR	DRR	IFR
3/4.6.1.2	IR radiation (Note: F to obtain platform signature, A to backout engine contribution)	No equivalent	A	A	A	
3/4.6.1.3	Smoke	34.21	Similarity		A	
3/4.6.1.4	Gaseous emissions	34.21	Similarity		A	
3/4.6.1.5	Fuel streaming and vapor puffing	No equivalent	A	A	A	
3/4.6.1.6	Water vapor contrails	No equivalent	A	A	A	
3/4.6.1.7	Radar cross section (RCS)	No equivalent	A	A	A	
3/4.6.1.8	Radar absorbent materials (RAMs) and coatings	No equivalent	A	A	A	
3/4.6.2	Vulnerability (title only)					
3/4.6.2.1	Ballistic weapons	No equivalent	A	A	A	
3/4.6.2.1.1	Static structure	No equivalent	A	A	A	
3/4.6.2.2	Nuclear weapons effects	No equivalent	A	A	A	
3/4.6.2.3	Chemical and biological agent effects	No equivalent	A	A	A	
3/4.6.2.4	Fuel ingestion effects (title only)					
3/4.6.2.4.1	Fuel ingestion - steady-flow	No equivalent	A	A	A	
3/4.6.2.4.1.1	Fuel ingestion - cooling air contamination	No equivalent	I, A	I, A	I, A	

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TABLE L. Example of table XXXI tailored for a Commercial Off The Shelf (COTS) application - continued.

(See 4.12 guidance.)

(To be completed by the specification developer; entries represent a sample guidance.)

Reqt. No.	Reqt. Title	Technology Development				
¾	VERIFICATION AND QUALIFICATION (TITLE)	FAR Paragraph Number	PDR	CDR	DRR	IFR
3/4.7.2.2	Control system adjustments	33.28, 33.91	Similarity		A	
3/4.7.2.3	Overspeed protection system	33.27(b)	Similarity		A	
3/4.7.3	Fuel System (title only)					
3/4.7.3.1	Fuels	33.67	Similarity		A	
3/4.7.3.1.1	Primary fuel (repeat test if engine configuration changes which impact fuel performance)	33.91	Similarity		A	
3/4.7.3.1.2	Alternate fuel (repeat test if engine configuration changes which impact fuel performance)	33.91	Similarity		A	
3/4.7.3.1.3	Restricted fuel (repeat test if engine configuration changes which impact fuel performance)	33.91	Similarity		A	
3/4.7.3.1.4	Emergency fuel	33.91	Similarity		A	

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TABLE L. Example of table XXXI tailored for a Commercial Off The Shelf (COTS) application - continued.

(See 4.12 guidance.)

(To be completed by the specification developer; entries represent a sample guidance.)

Reqt. No.	Reqt. Title	Technology Development				
¾	VERIFICATION AND QUALIFICATION (TITLE)	FAR Paragraph Number	PDR	CDR	DRR	IFR
3/4.7.3.2	Fuel system performance (title only)					
3/4.7.3.2.1	Fuel contamination	33.67	Similarity		A	
3/4.7.3.2.2	Fuel system performance with external assistance	33.67, 33.91, 33.87(a)(7)	Similarity		A	
3/4.7.3.2.3	Fuel system performance with no external assistance	33.67, 33.91, 33.87(a)(7)	Similarity		A	
3/4.7.3.2.4	Fuel system performance under conditions of excessive fuel vapor		A	A	A	C
3/4.7.3.2.5	Fuel pump priming	33.67, 33.91		A	A	C
3/4.7.3.2.6	Fuel lubricity	33.67, 33.91	Similarity		A	
3/4.7.3.2.7	Fuel system performance with water saturated fuel	33.67, 33.91	Similarity		A	
3/4.7.3.2.8	Fuel filter	33.67, 33.91	Similarity		A	
3/4.7.3.2.9	Fuel flow limit	No equivalent		P	P	
3/4.7.4	Electrical system [title only]					
3/4.7.4.1	Electrical power	No equivalent	A	A	A	D, E
3/4.7.4.1.1	Generator	No equivalent	A	A	A	C, D, E
3/4.7.4.2	Alternate electrical power	No equivalent	A	C	A, C	D, E
3/4.7.4.3	Electrical connectors and cables	No equivalent	A	A, D, I	A, D, I	C, I

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TABLE L. Example of table XXXI tailored for a Commercial Off The Shelf (COTS) application - continued.

(See 4.12 guidance.)

(To be completed by the specification developer; entries represent a sample guidance.)

Reqt. No.	Reqt. Title	Technology Development				
¾	VERIFICATION AND QUALIFICATION (TITLE)	FAR Paragraph Number	PDR	CDR	DRR	IFR
3/4.7.4.4	Electrical bonding	33.5		P, A	P, A	
03/4.7.4.5	Ground isolation	No equivalent				C, E, I
3/4.7.4.6	Potting compounds	No equivalent				D, I
3/4.7.5	Ignition system	33.69, 33.89	Similarity		A	
3/4.7.5.1	Ignition system fouling [title only]					
3/4.7.5.1.1	Carbon fouling	No equivalent	Similarity	C	C	
3/4.7.5.1.2	Water fouling	No equivalent	Similarity	C	C	
3/4.7.6	Engine monitoring system (EMS)	33.28		A	A	D, P
3/4.7.6.1	EMS fault detection and isolation	No equivalent		A	A	D, P
3/4.7.6.2	On-board engine diagnostic function	No equivalent				D, P
3/4.7.7	Optical systems	No equivalent		A, De	A, De	C, D, E
3/4.7.7.1	Fiber optic cables and connectors	No equivalent		A, De	A, De	C, D, E
3/4.7.8	Lubrication system [title only]`					
3/4.7.8.1	Lubrication oil	33.71, 33.87, 33.89	Similarity		A	
3/4.7.8.1.1	Oil pressure and temperature limits	33.71, 33.87, 33.89	Similarity		A	
3/4.7.8.1.2	Oil consumption limits	33.5, 33.71, 33.87	Similarity		A	
3/4.7.8.1.3	Oil flow interruption or depletion	No equivalent	A	A	A	D, C
3/4.7.8.2	Lubrication system components and features [title only]					
3/4.7.8.2.1	Oil reservoir	33.71	Similarity		A	

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TABLE L. Example of table XXXI tailored for a Commercial Off The Shelf (COTS) application - continued.

(See 4.12 guidance.)

(To be completed by the specification developer; entries represent a sample guidance.)

Reqt. No.	Reqt. Title	Technology Development				
¾	VERIFICATION AND QUALIFICATION (TITLE)	FAR Paragraph Number	PDR	CDR	DRR	IFR
3/4.7.8.2.1.1	Oil reservoir external features	33.71, 33.87	Similarity		A	
3/4.7.8.2.2	Oil drains	33.71, 33.87	Similarity		A	
3/4.7.8.2.3	Oil filters	33.71, 33.87	Similarity		A	
3/4.7.8.2.4	Oil debris monitor(s)	No equivalent	Similarity A		A	C
3/4.7.8.3	Breather mist	No equivalent	A	A	A	De
3/4.7.9	Hydraulic system	33.72	Similarity		A	
3/4.7.9.1	Hydraulic ground test provisions	No equivalent				C
3/4.7.9.2	Hydraulic system air removal	No equivalent				C
3/4.7.9.3	Hydraulic fluid filters	33.72	Similarity		A	
3/4.7.10	Exhaust nozzle system	No equivalent	A	De	A, De	I, E
3/4.7.10.1	Exhaust nozzle external asymmetrical air pressure loads	No equivalent	A	C, De	A, C, De	C, De
3/4.7.10.2	Vectoring nozzle	No equivalent	A	C, De	A, C, De	E
3/4.7.10.2.1	Vectoring nozzle angle and rate	No equivalent	A	C, De	A, C, De	E
3/4.7.10.2.2	Vectoring nozzle failure accommodation	No equivalent	A	C, De	A, C, De	E
3/4.7.11	Augmentation system	33.79	Similarity		A	
3/4.7.11.2.1	Water injection system fluid	No equivalent		C, De	C, De	E, I
3/4.7.12	Wash system	No equivalent				
3/4.7.13	Brake system	33.92		A	A	De
3/4.7.14	Negative torque limiter	No equivalent				A

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TABLE L. Example of table XXXI tailored for a Commercial Off The Shelf (COTS) application - continued.

(See 4.12 guidance.)

(To be completed by the specification developer; entries represent a sample guidance.)

Reqt. No.	Reqt. Title	Technology Development				
¾	VERIFICATION AND QUALIFICATION (TITLE)	FAR Paragraph Number	PDR	CDR	DRR	IFR
3/4.7.14.1	Multiple power section engines	No equivalent		A	A	C, E
3/4.7.15	Power absorber (engine supplied)	33.95	Similarity		A	
3/4.7.15.1	Propeller feathering and reverse pitch	33.95	Similarity		A	
3/4.7.16	Gearbox	No equivalent		A	A	C
3/4.8	Software resources (title)					
3/4.8.1	Software performance and design	33.28	Similarity		A	
3/4.8.1.1	Computer reprogramming	No equivalent				D
3/4.8.2	Spare resources	No equivalent				D
3/4.9	Production facilities, capabilities, and processes	Part 21, Subpart G	Similarity		A	
3/4.10	Production cost requirement (PCR)	No equivalent	A	A	A	A

Note: Reverification of the engine is required when the configuration changes impact the engine requirement.

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TABLE L. COTS cargo/transport example legend.

Verification Methods	
Code	**Description**
I	Inspection
X	Contractor/Agency Agreement
A	Analysis
S	Simulation/Modeling
D	Demonstration
De	Development Test
C	Component Test
E	Engine System Level Ground Test
F	Flight Test
Q	Process/Quality Control
P	Documentation
AMT/2	1/2 of the AMT (one hot section life)

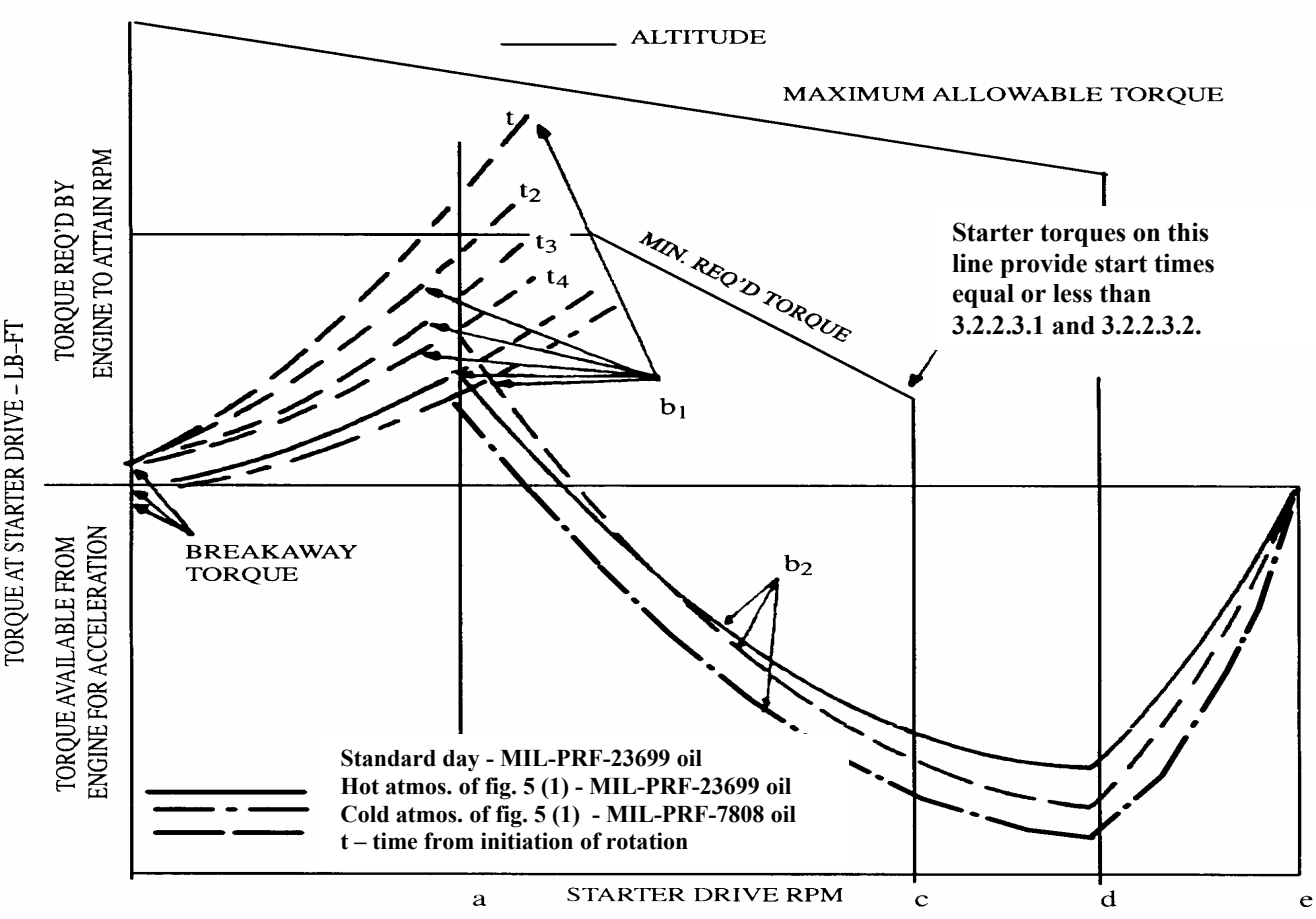
Milestones	
Code	**Description**
PDR	Preliminary Design Review
CDR	Critical Design Review
DRR	Design Readiness Review
IFR	Initial Flight Release
FFR	Full Flight Release
ISR	Initial Service Release
OCR	Operational Capability Release

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TO BE PROVIDED BY THE CONTRACTOR

FIGURE 1. Item and interface definition.

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:

a. Required RPM before firing. (Where applicable the engine manufacturer shall state the minimum time and the number of revolutions or any combination or conditions that must be satisfied before firing.)
   Maximum required cranking time at firing speed _____ sec
b. Steady-state torque at the starter drive.
   1. In an unfired engine. (Curves for cold atmosphere of figure 5 show drag torque after time from initiation of rotation.)
   2. In a fired engine.
c. Minimum starter cutout speed _____ rpm.
d. Maximum starter cutout speed _____ rpm. (Maximum cutout speed should be at least 10 percent above minimum cutout speed.)
e. Engine idle condition.

Notes:
1. Manufacturer shall insert the actual figure 5 temperature here.
2. Torque units are shown as Ft-Lbs; N-m may also be used.

FIGURE 2. Starting torque and speed.

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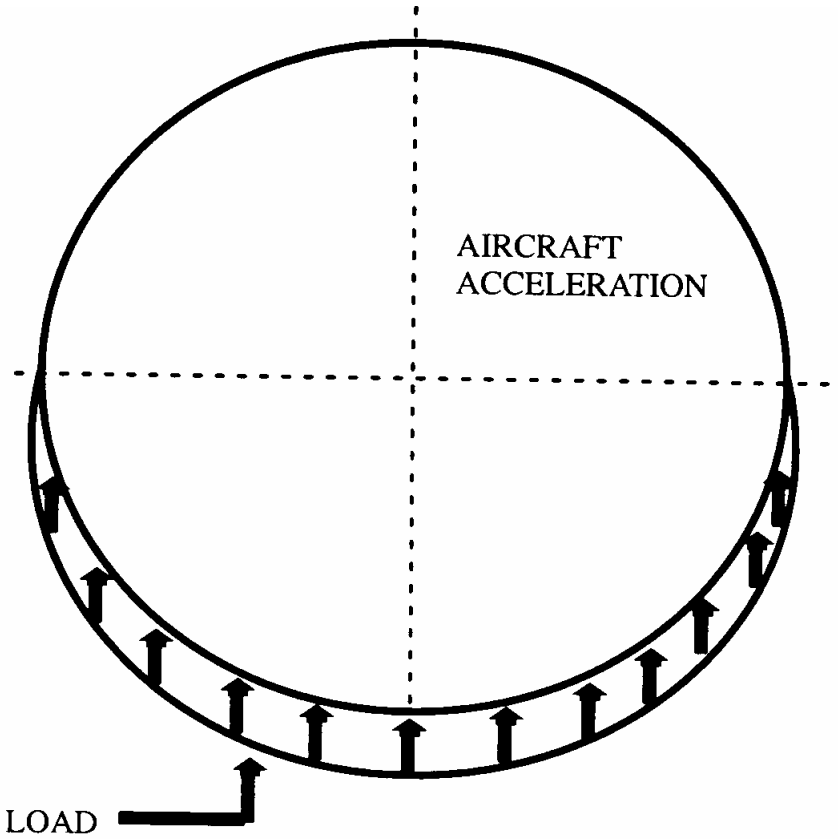


FIGURE 3. Exhaust system interface loads.

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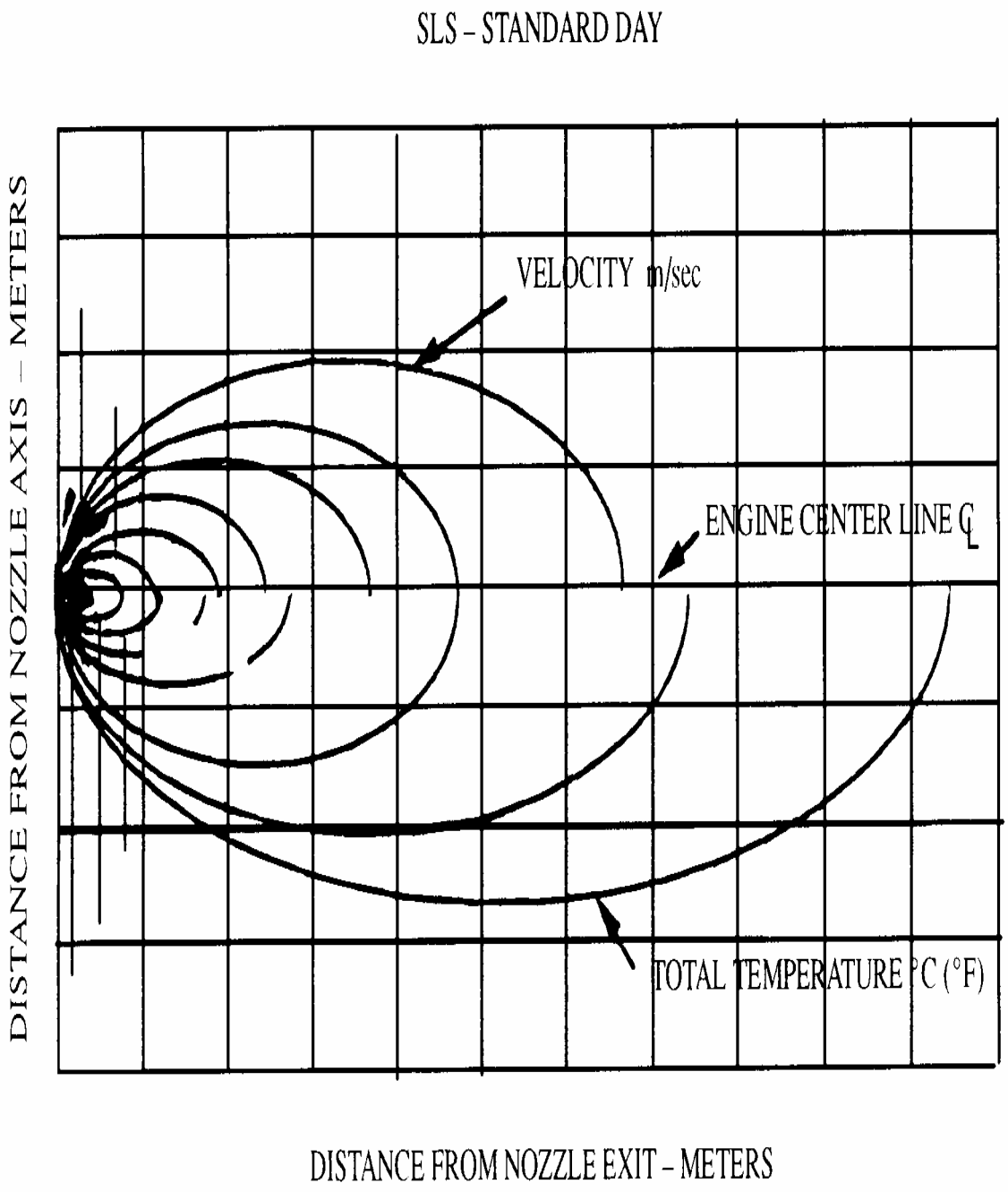
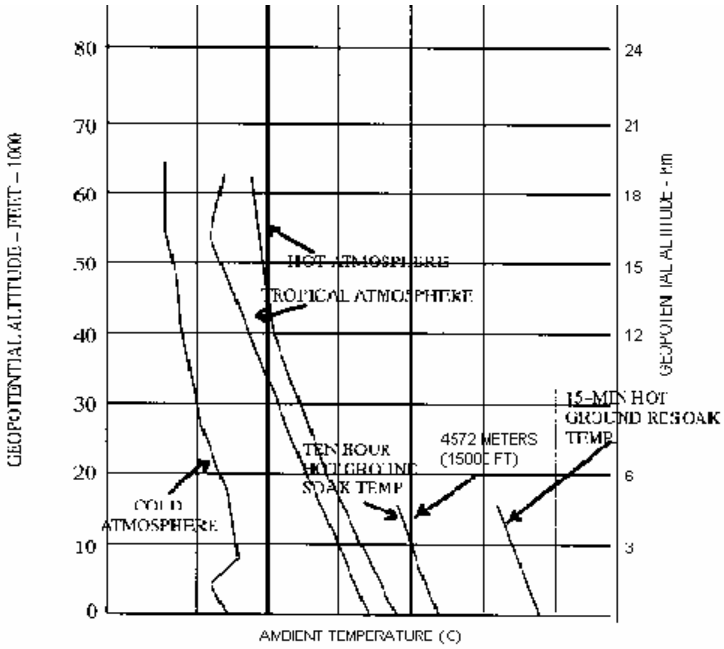


FIGURE 4. Jet wake.

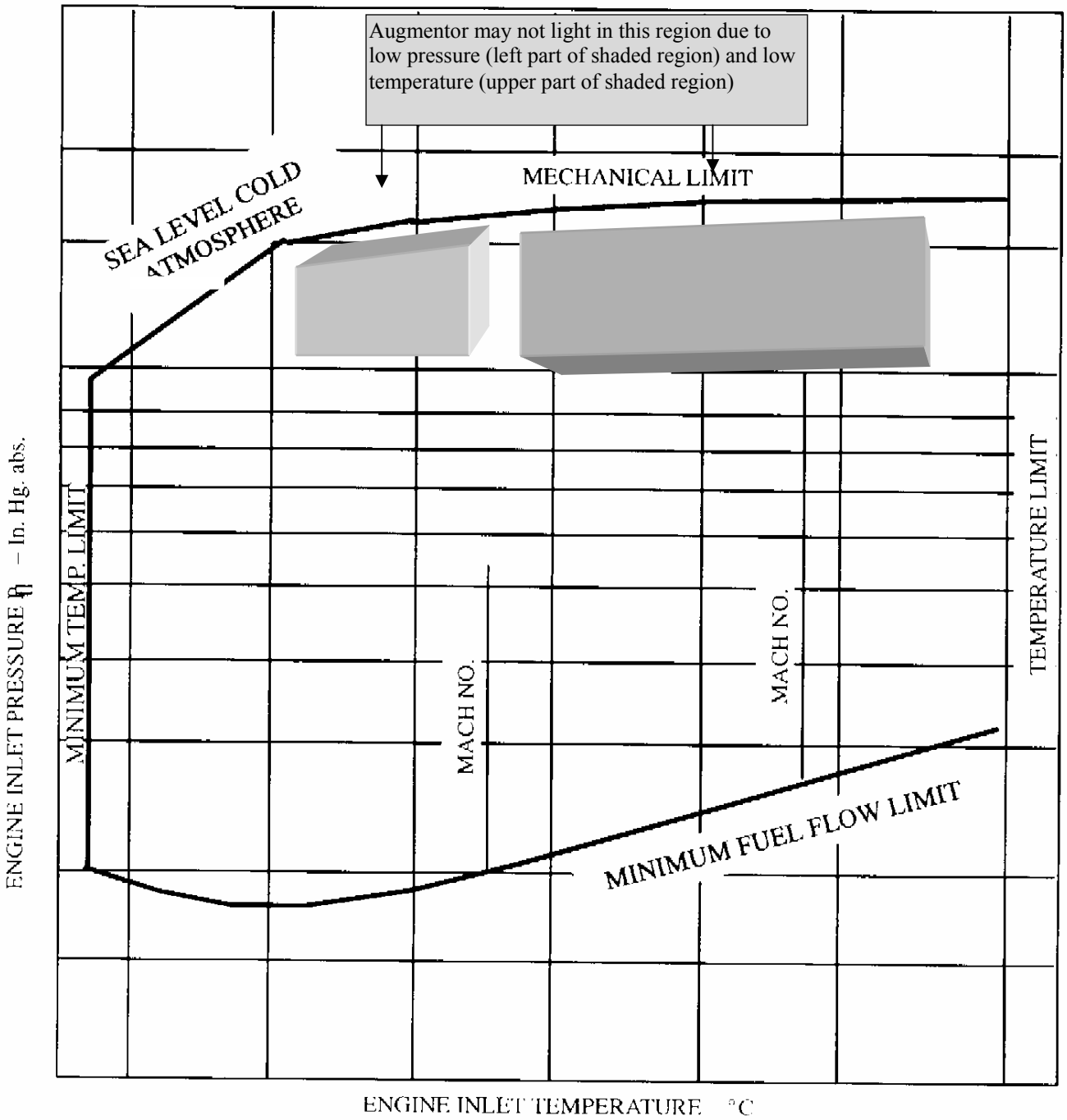
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NOTES:
1. Hot and cold atmosphere, except at sea level, consistent with one percent frequency of occurrence of MIL-HDBK-310, Worldwide Air Environment to 24.4 km (80,000 ft).
2. Sea level hot and cold atmosphere temperatures consistent with one percent and 20 percent frequency of occurrence, respectively consistent with MIL-HDBK-310, Worldwide Surface Environment.
3. Tropical atmosphere consistent with MIL-HDBK-310, table IV for high absolute humidity, at 88°F (surface air temperature).
4. The ambient static pressure variation shall be as defined in U.S. Standard Atmosphere for geopotential altitude.

FIGURE 5. Ambient temperature extremes versus altitude.

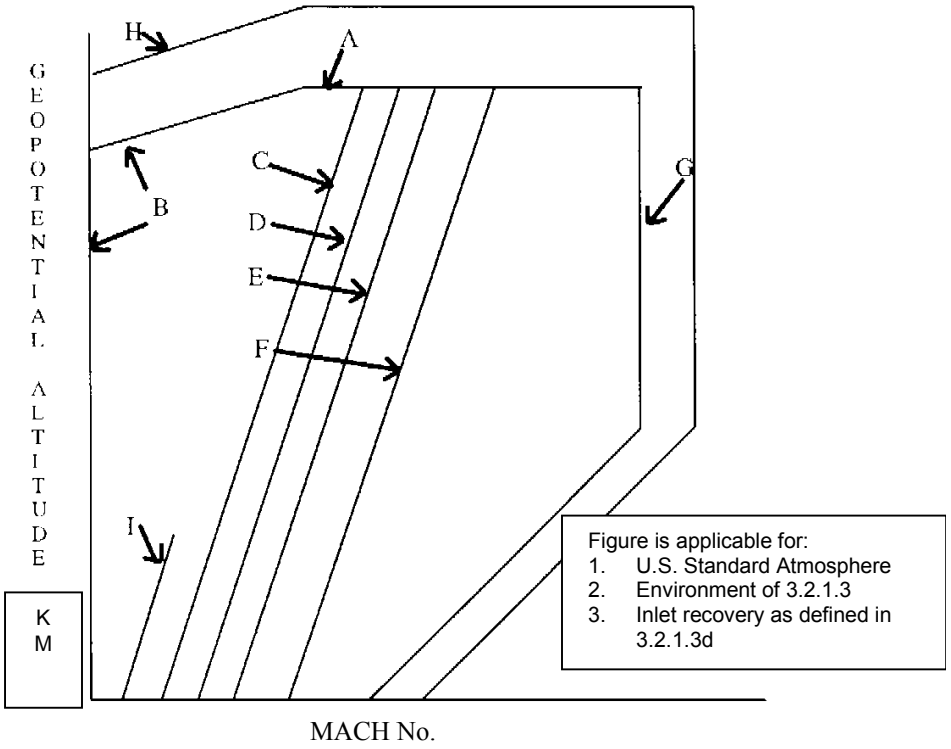
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NOTE:
Inlet pressures are shown in “in Hg abs”; “mm Hg abs” may also be used.

FIGURE 6. Operating limits.

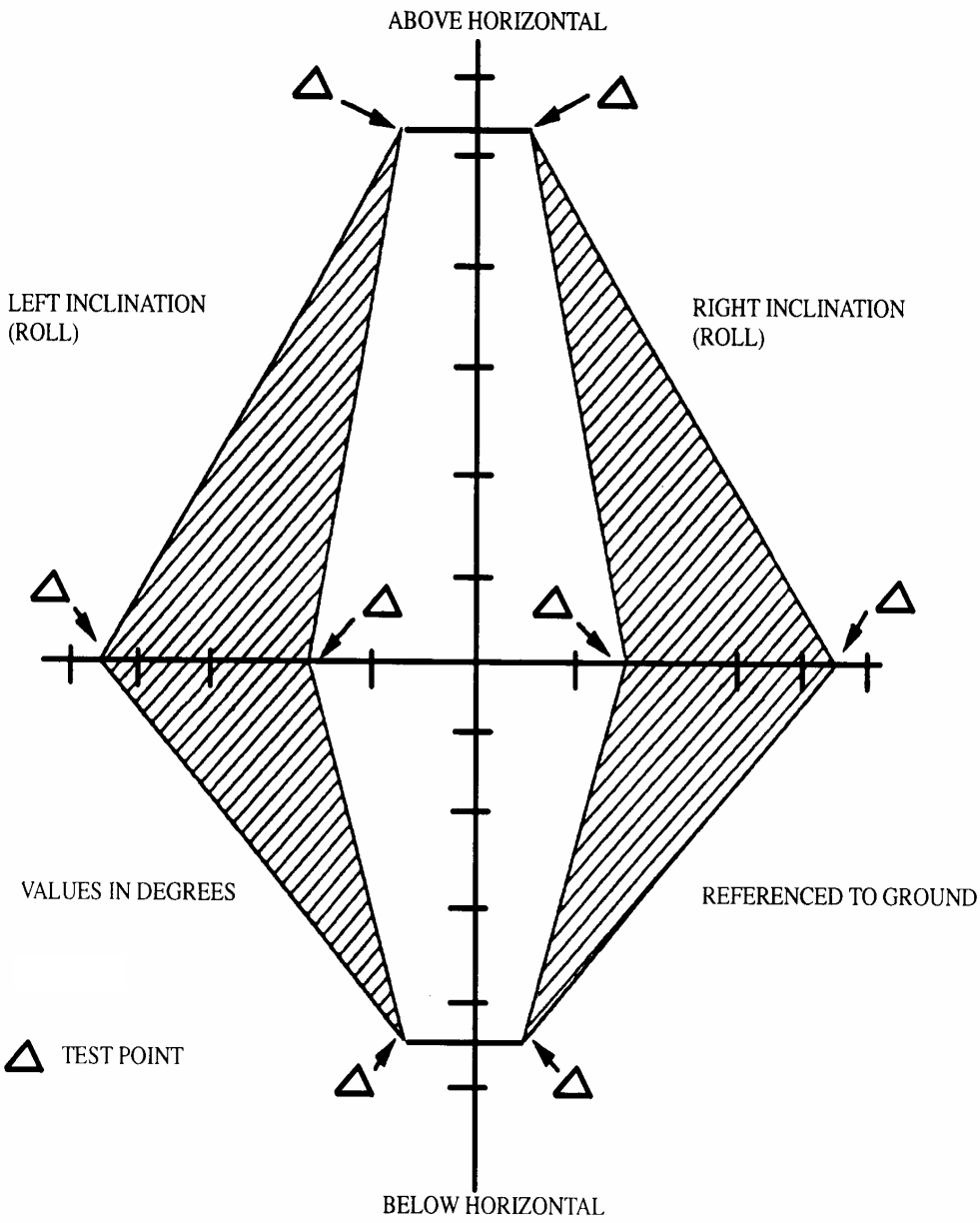
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NOTES:
A. Maximum starting altitude.
B. Minimum Mach No. for starter assisted starting.
C. Minimum Mach No. starting without starter assist, no customer power extraction, no customer bleed air extraction.
D. Minimum Mach No. for starting without starter assist, maximum customer power extraction, no customer bleed air extraction.
E. Minimum Mach No. for starting without starter assist, no customer power extraction, maximum customer bleed air extraction.
F. Minimum Mach No. for starting without starter assist, maximum customer power extraction, maximum customer bleed air extraction.
G. Maximum Mach No. for starting air extraction.
H. Operating envelope.
I. Post loiter starting point.

FIGURE 7. Operating envelope.

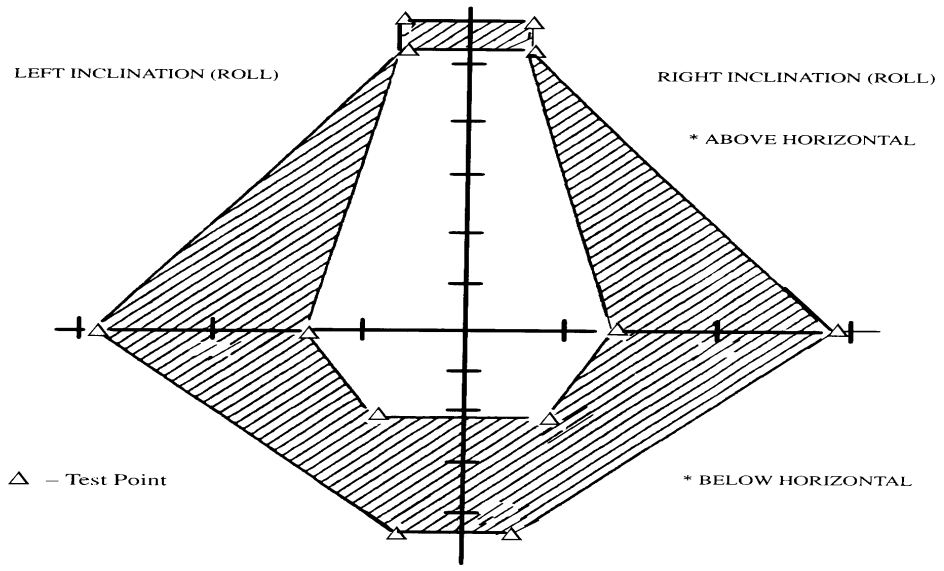
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NOTES:
1. The engine shall be capable of operating at all possible acceleration conditions; however, for the purpose of defining the direction of acceleration vector from the engine CG, the figure assumes no acceleration other than gravity.
2. Engine centerline perpendicular to plane of paper.
3. Continuous operation in clear area
4. _____ second operation in shaded area.

FIGURE 8a. Engine attitude limits (fixed wing aircraft).

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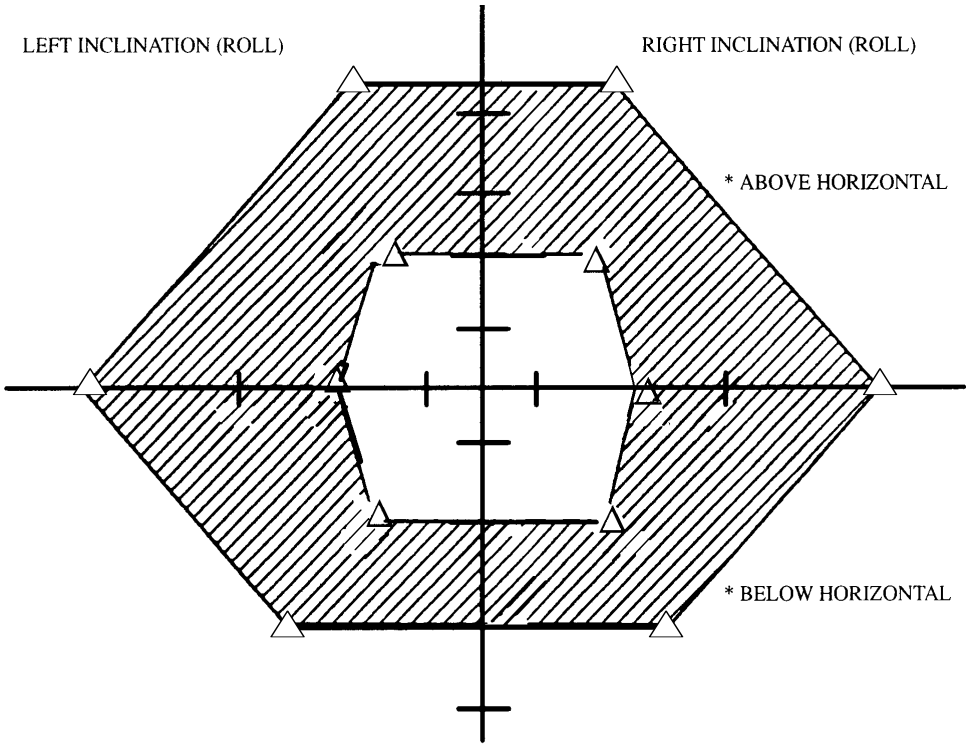


NOTES:

1. The engine shall be capable of operating at all possible acceleration conditions; however, for the purpose of defining the direction of acceleration vector from the engine CG, the figure assumes no acceleration other than gravity.
2. Symbol * indicates referenced to ground.
3. Engine centerline perpendicular to plane of paper.
4. Continuous operation in clear area.
5. _____ second operation in shaded area.

FIGURE 8b. Engine attitude limits (VSTOL aircraft engines).

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NOTES:
1. For the purpose of defining the direction of the acceleration vector from the engine CG, the figure assumes no acceleration other than gravity; however, the engine shall be capable of operating at all possible acceleration conditions.
2. Symbol * indicates referenced to ground.
3. Engine centerline perpendicular to plane of paper.
4. Continuous operation in clear area
5. _____ second operation in shaded area. Symbol Δ indicates points for test.

FIGURE 8c. Engine attitude limits (rotary wing aircraft engines).

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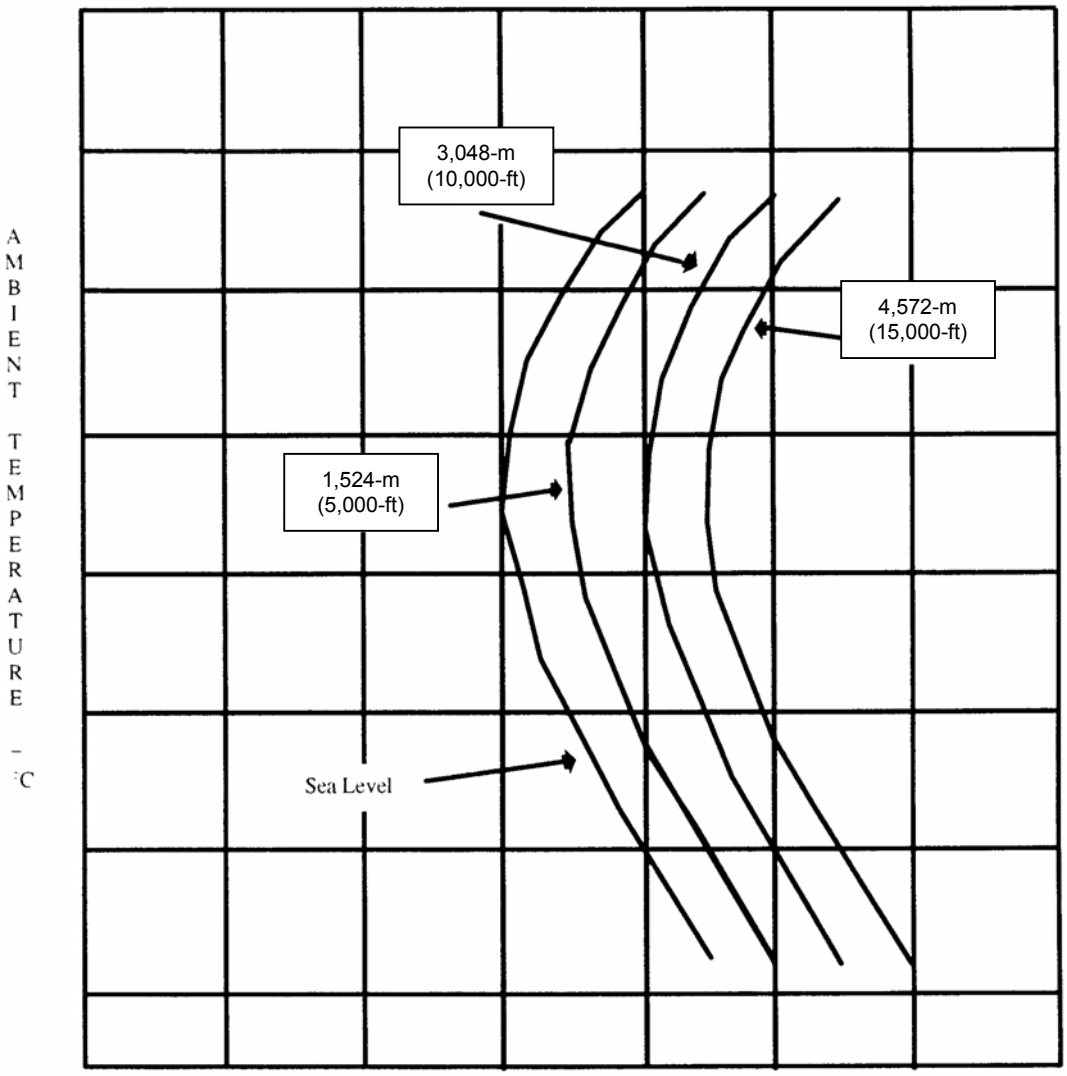
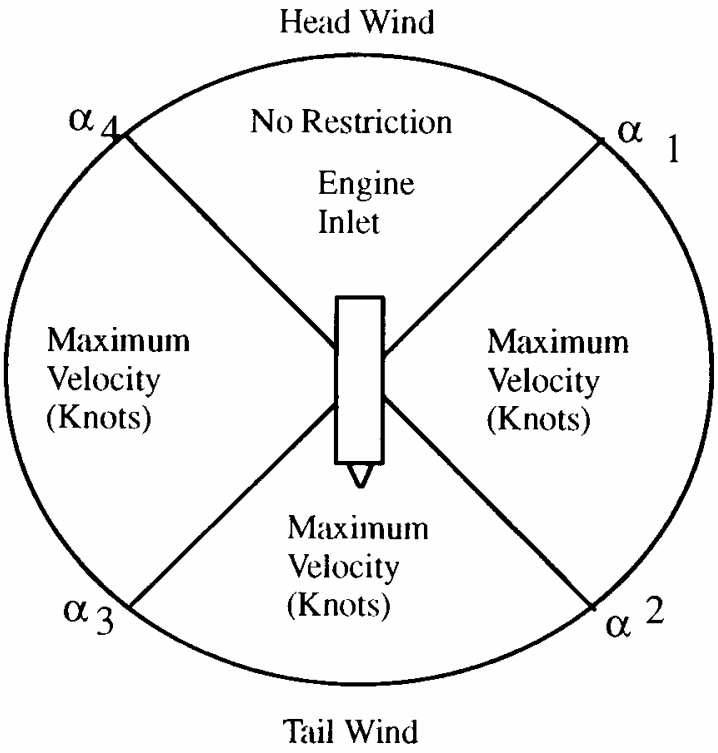


FIGURE 9a. Ground start time.

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RELATIVE WIND CONDITION ENVELOPE STANDARD STARTING PROCEDURE



NOTE:
If wind velocity equals or exceeds the value noted, DO NOT ATTEMPT A START. Refer to Corrective Measures.
α = Angular position from engine inlet centerline in degrees.

FIGURE 9b. Ground start wind envelope.

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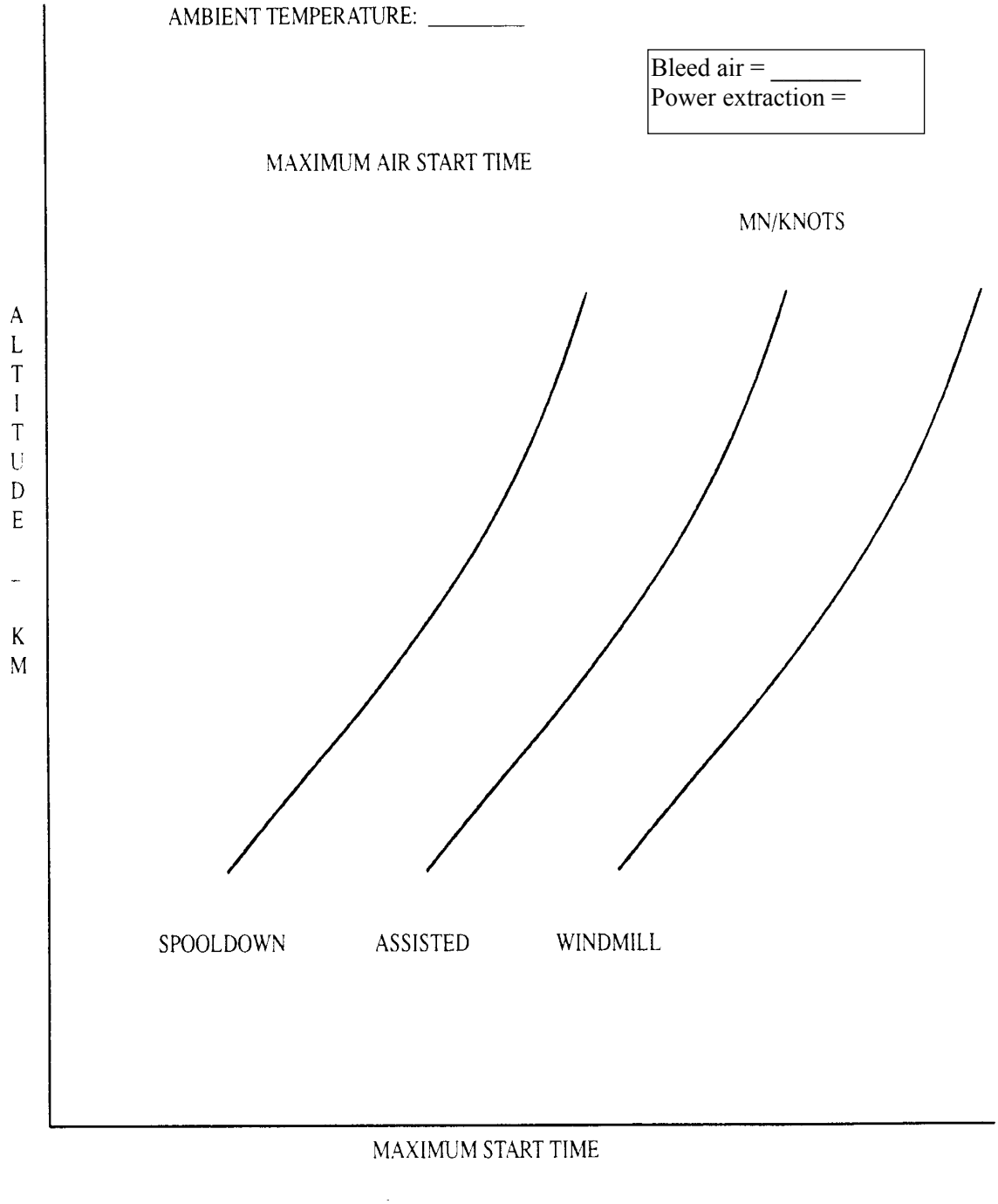


FIGURE 10. Windmill and airstart envelopes.

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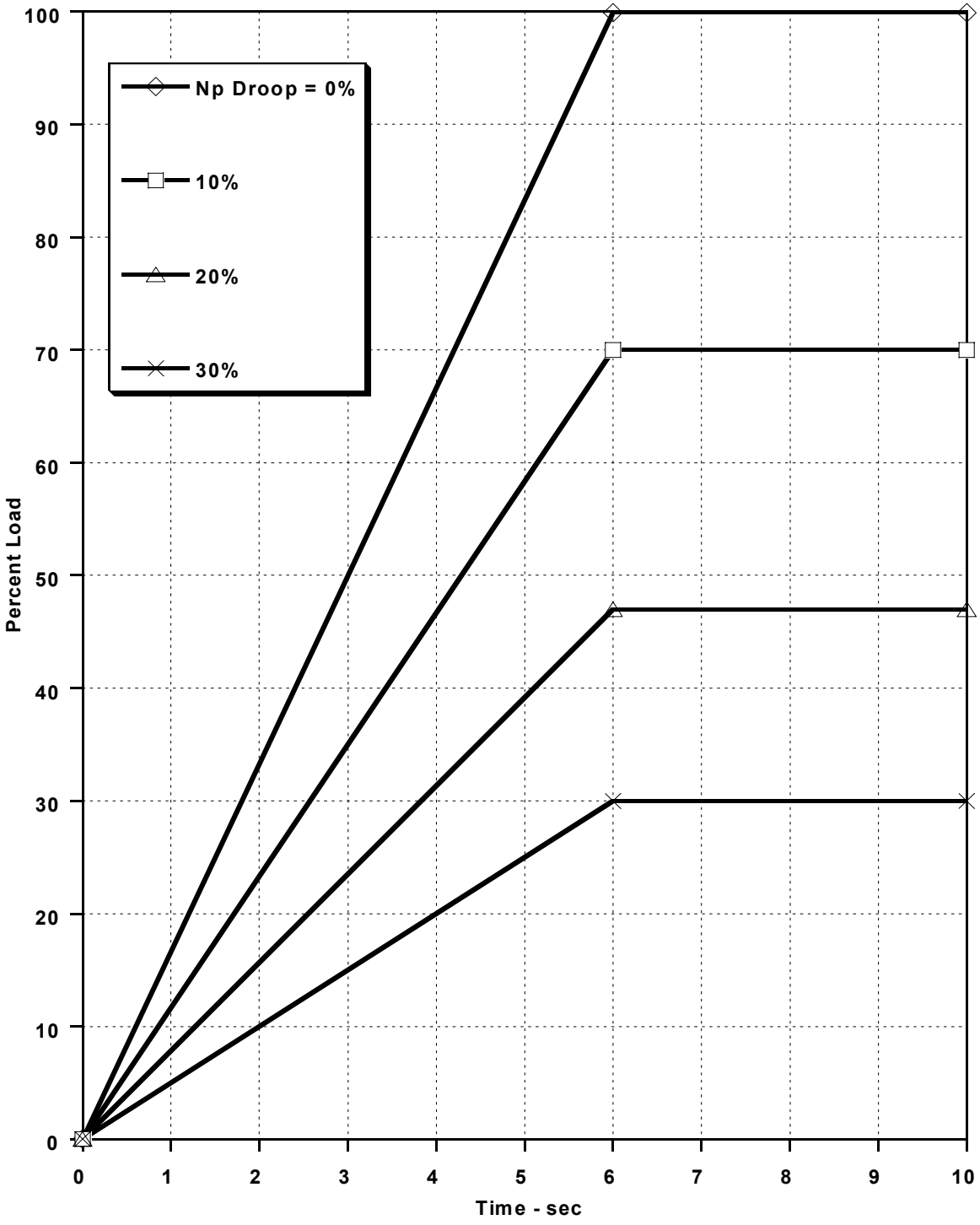


FIGURE 11a. Power absorber loading for acceleration: idle to maximum power, sea level to 3km (example).

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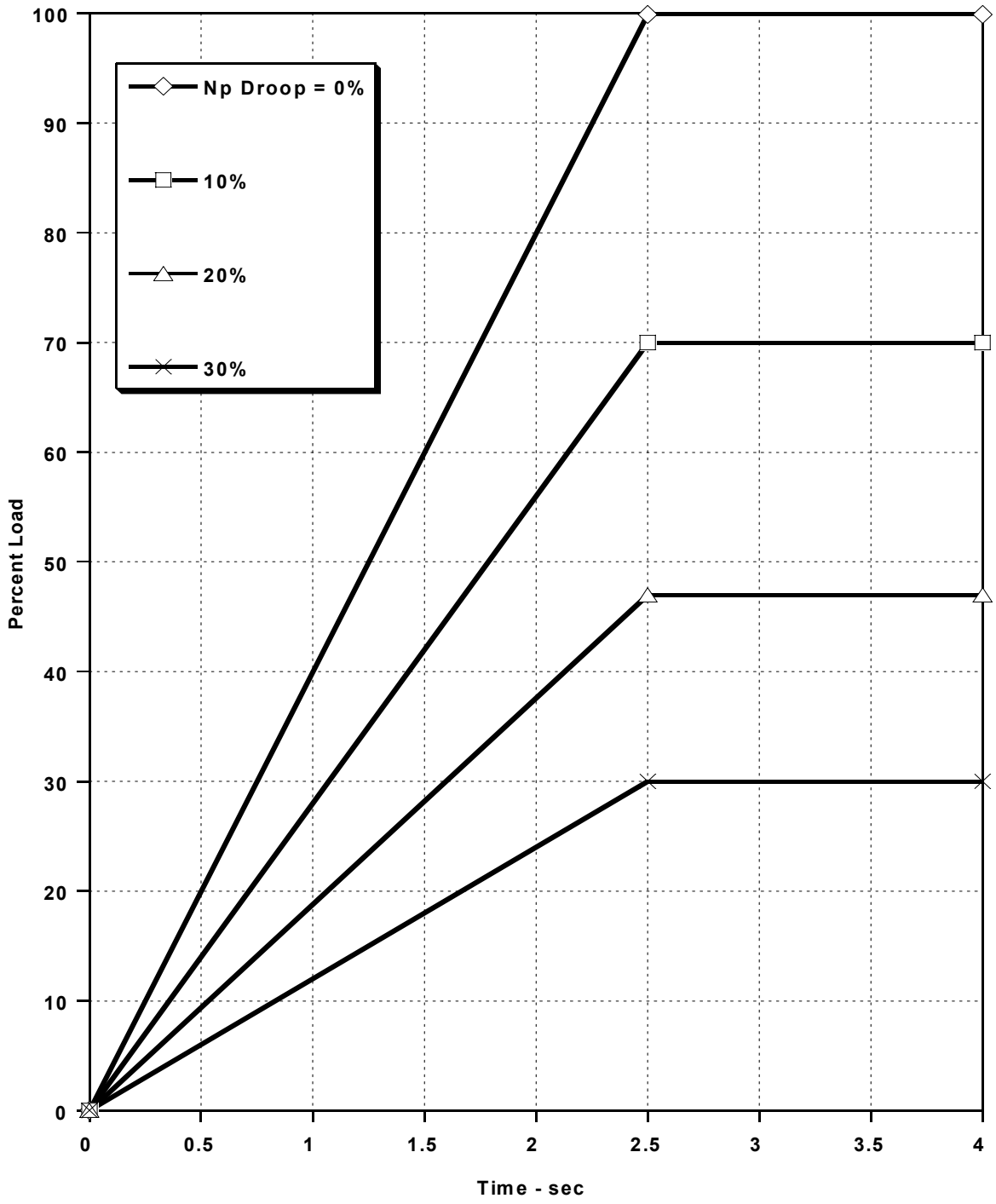


FIGURE 11b. Power absorber loading for acceleration: no load to maximum power, sea level to 3km (example).

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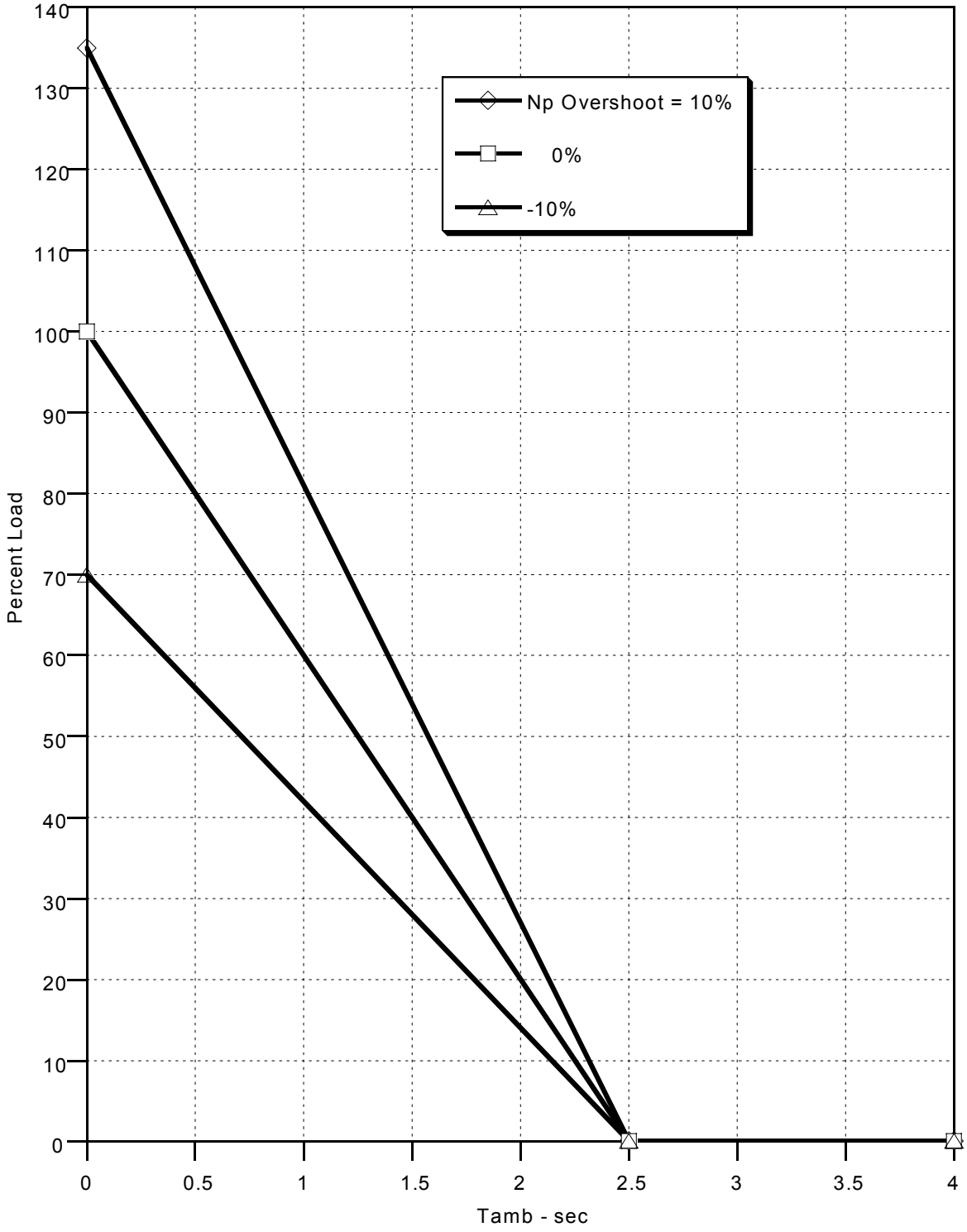


FIGURE 11c. Power absorber loading for deceleration: maximum power to no load, sea level to 3km (example).

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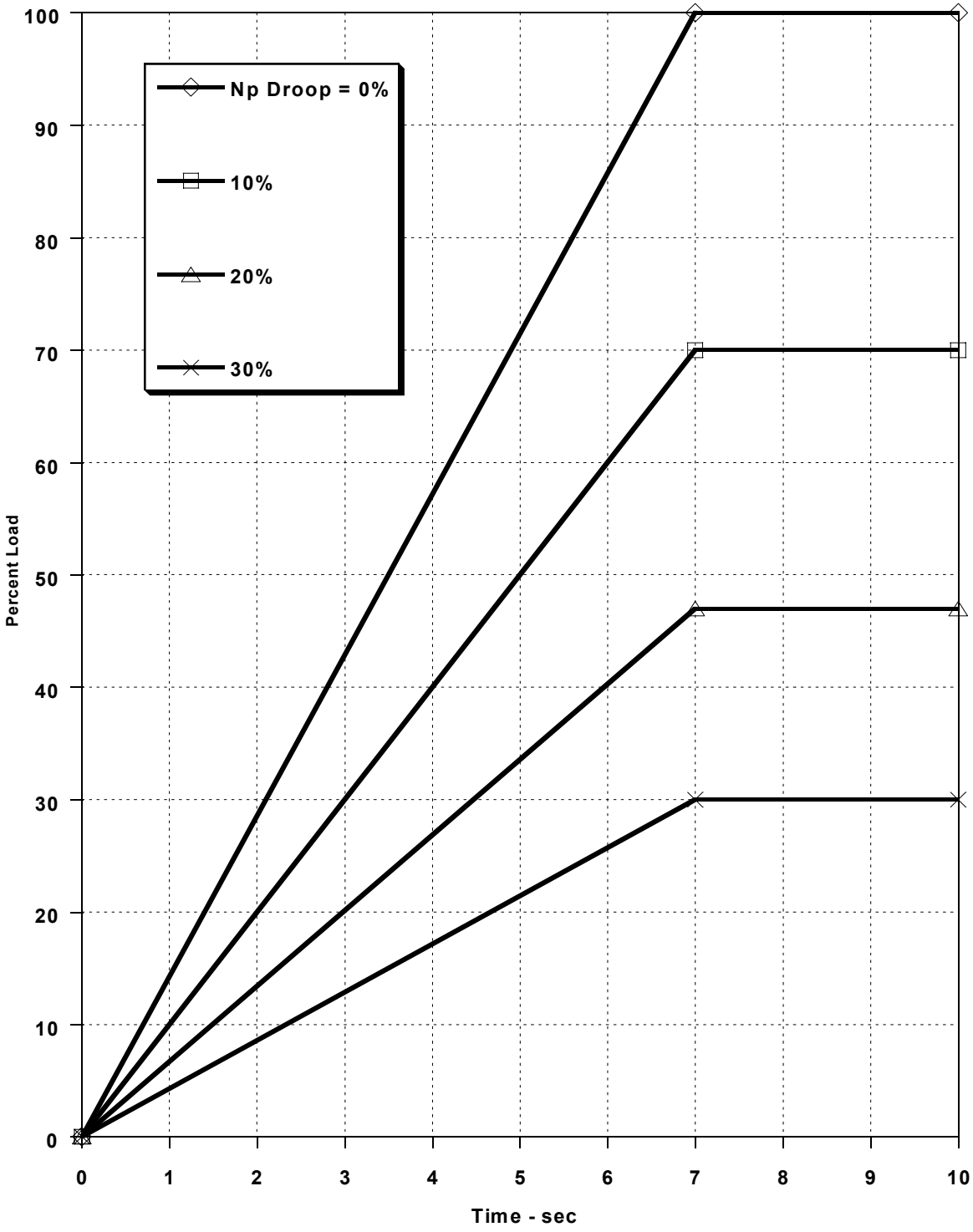


FIGURE 11d. Power absorber loading for acceleration: no load to maximum power, 3km to maximum altitude (example).

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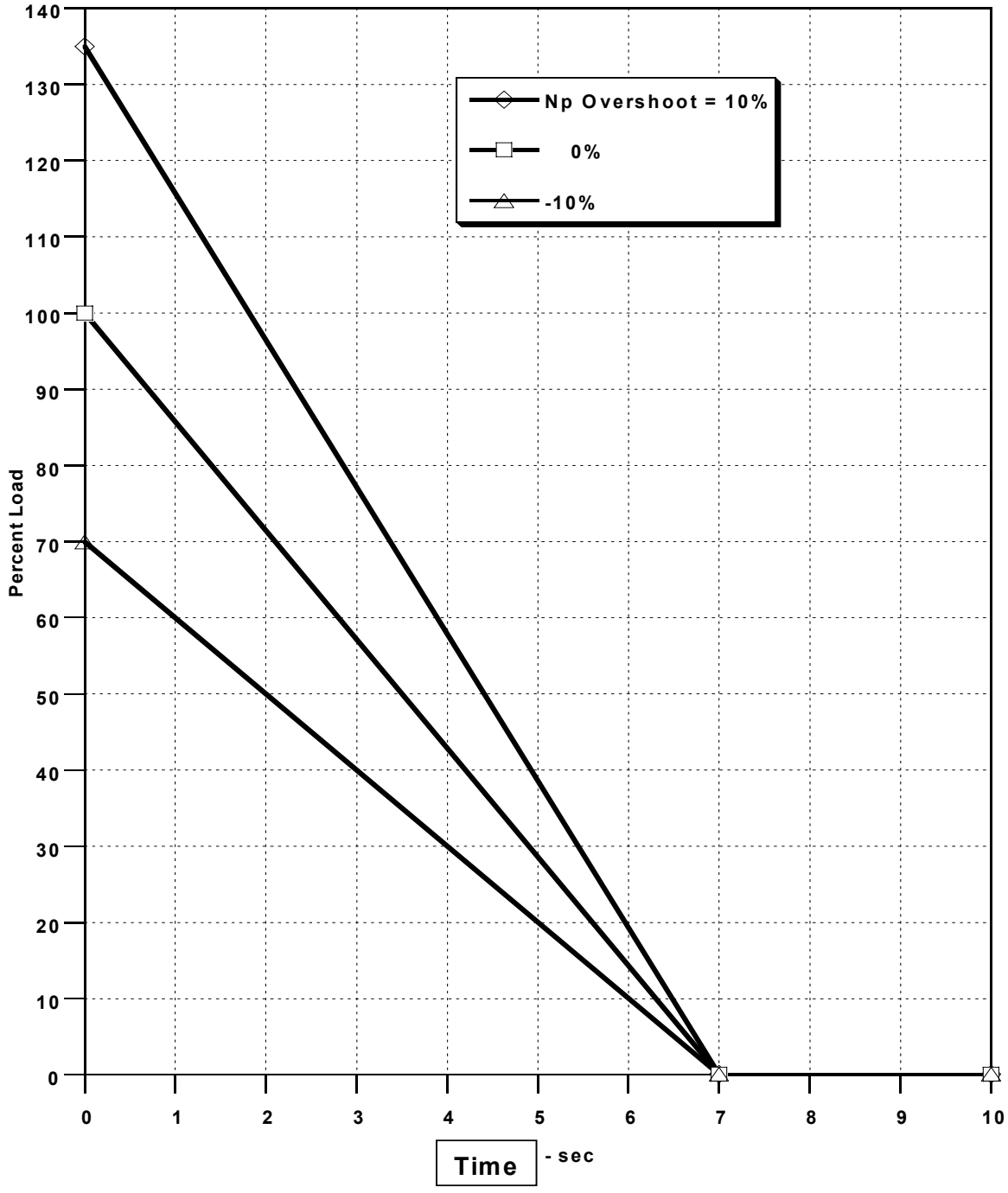


FIGURE 11e. Power absorber loading for deceleration: maximum power to no load, 3km to maximum altitude (example).

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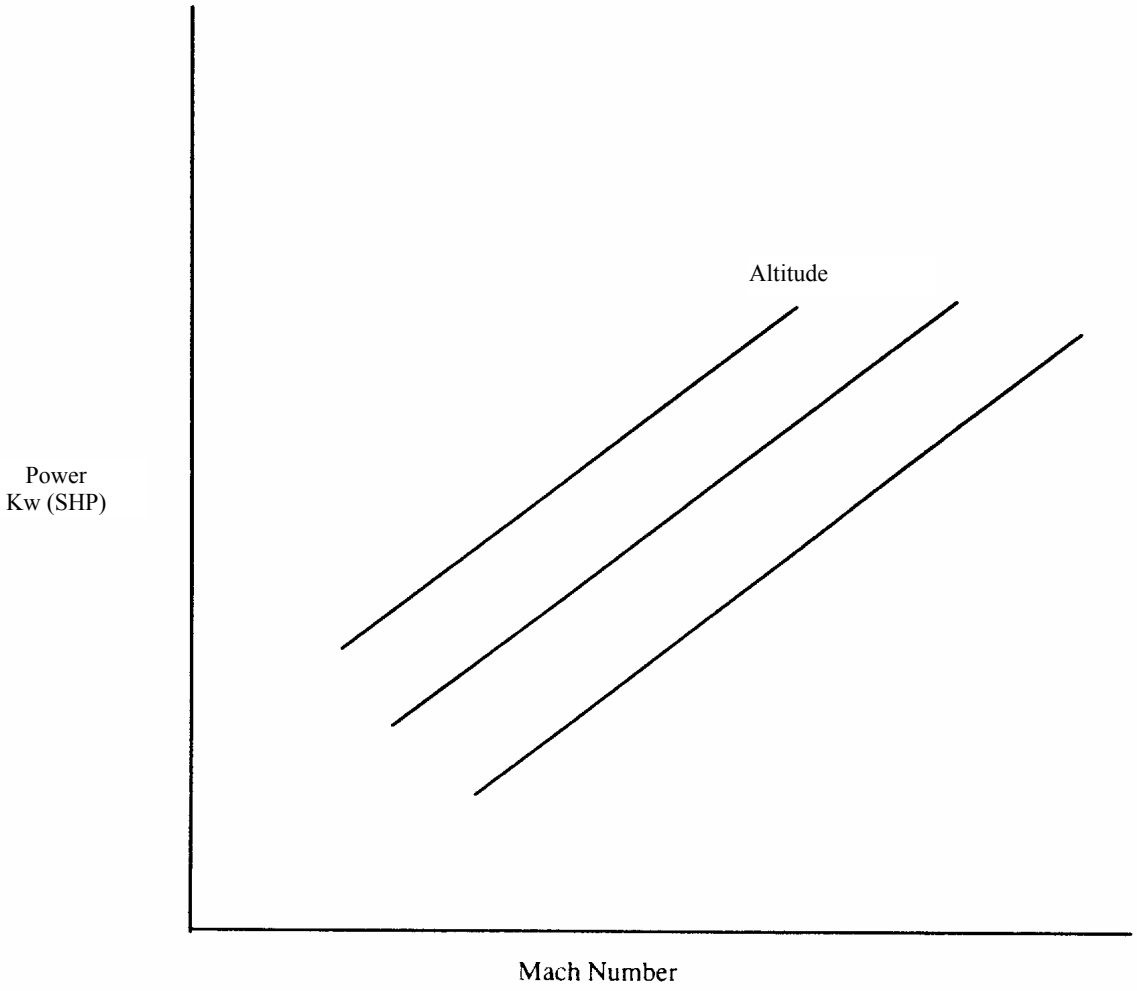


FIGURE 12a. Customer extraction during windmilling (Power).

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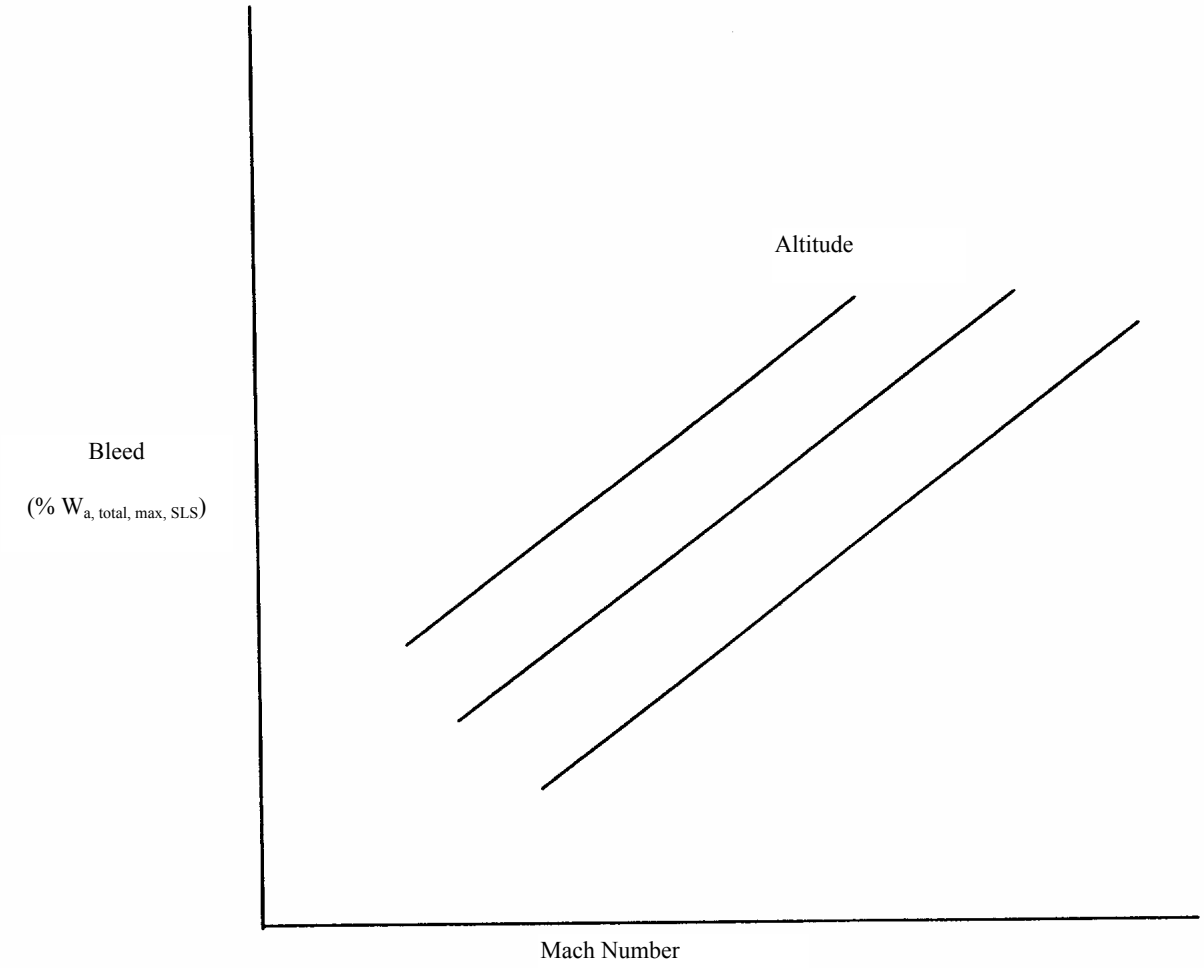


FIGURE 12b. Customer extraction during windmilling (Bleed).

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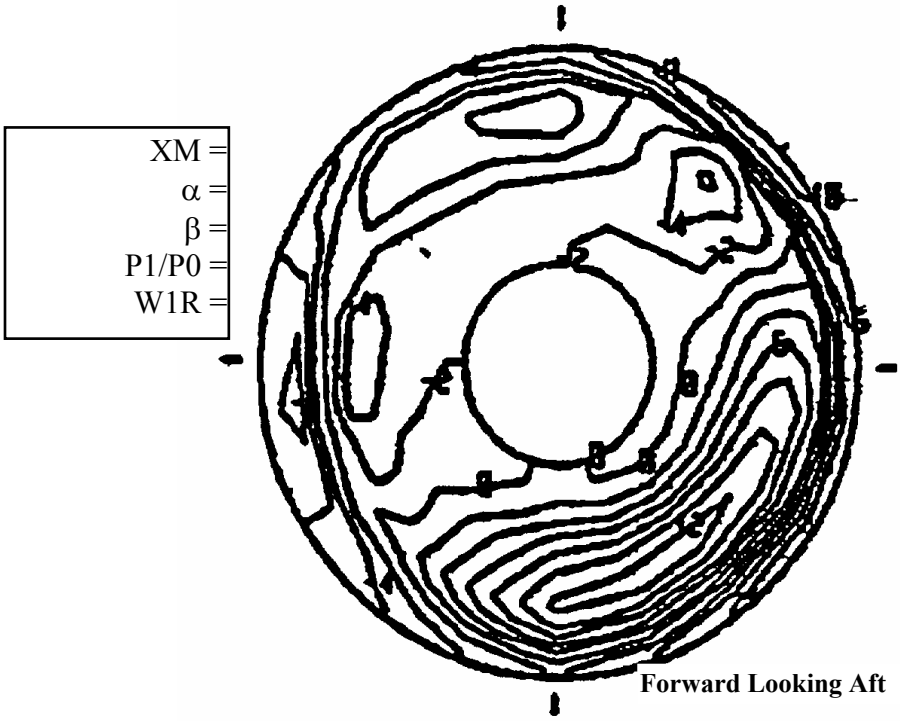


FIGURE 13. Inlet airflow distortion pattern.

Ring Radii	Rake Location (Deg.)	0	45	90	135	180	225	270	315
		Inner Radius =							
		Outer Radius =							

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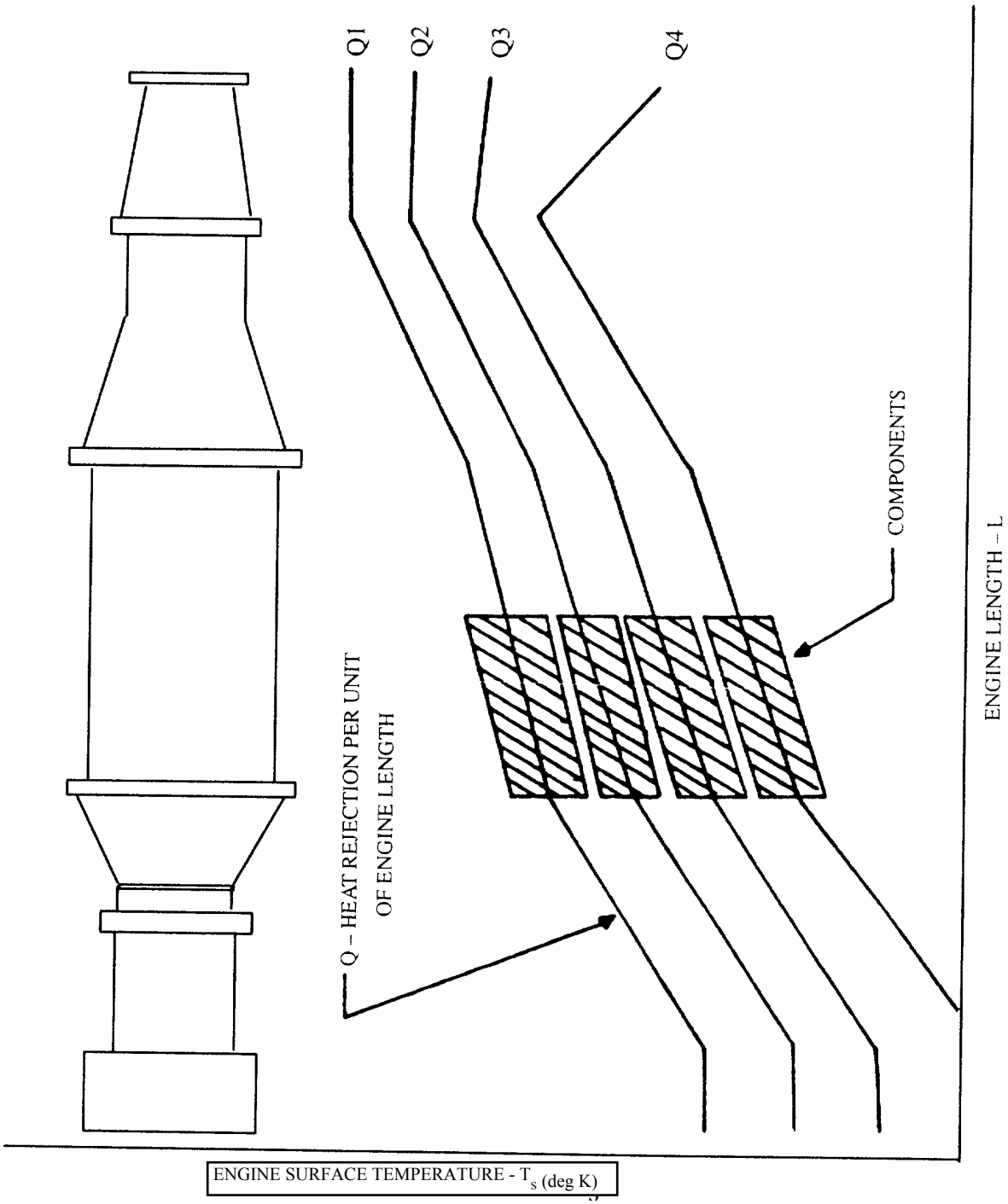


FIGURE 14a. Engine surface temperature.

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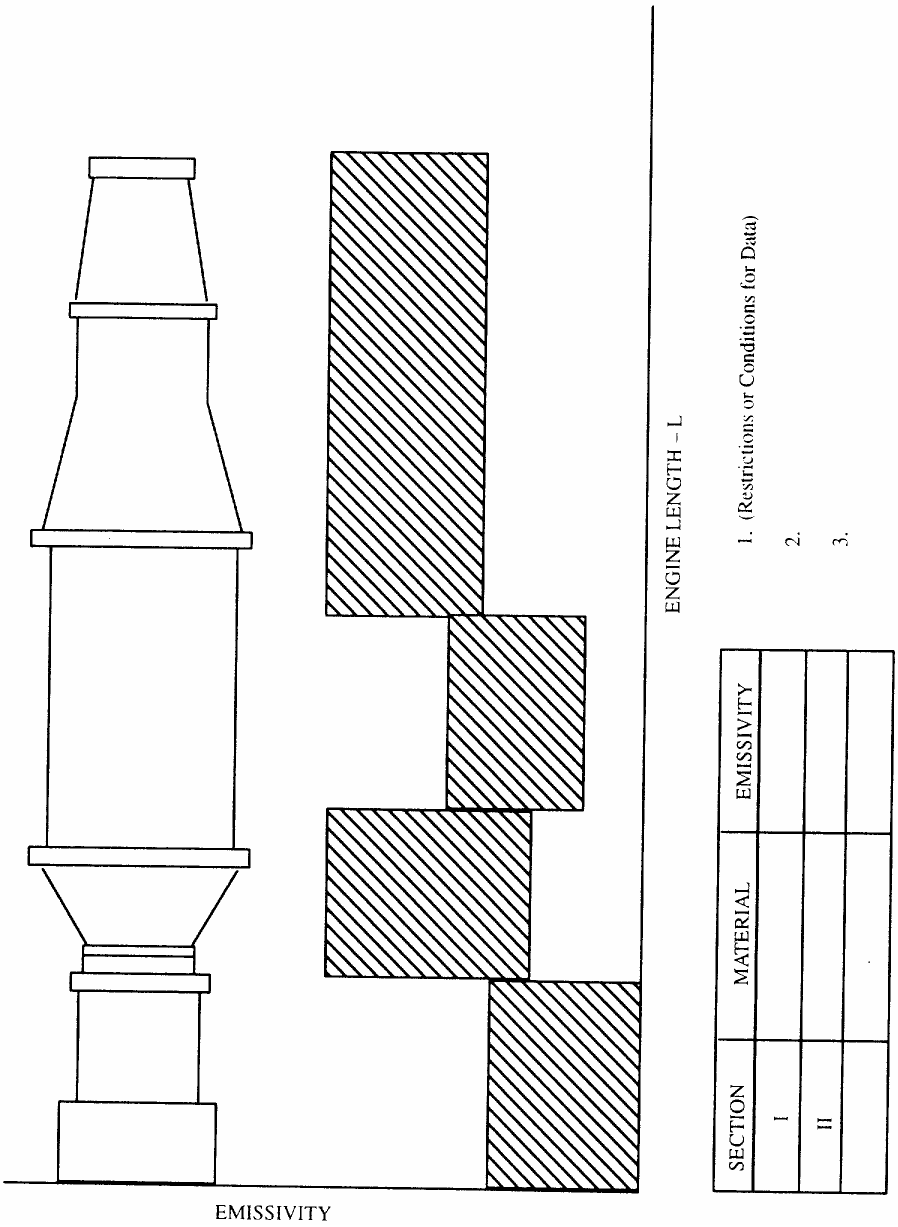
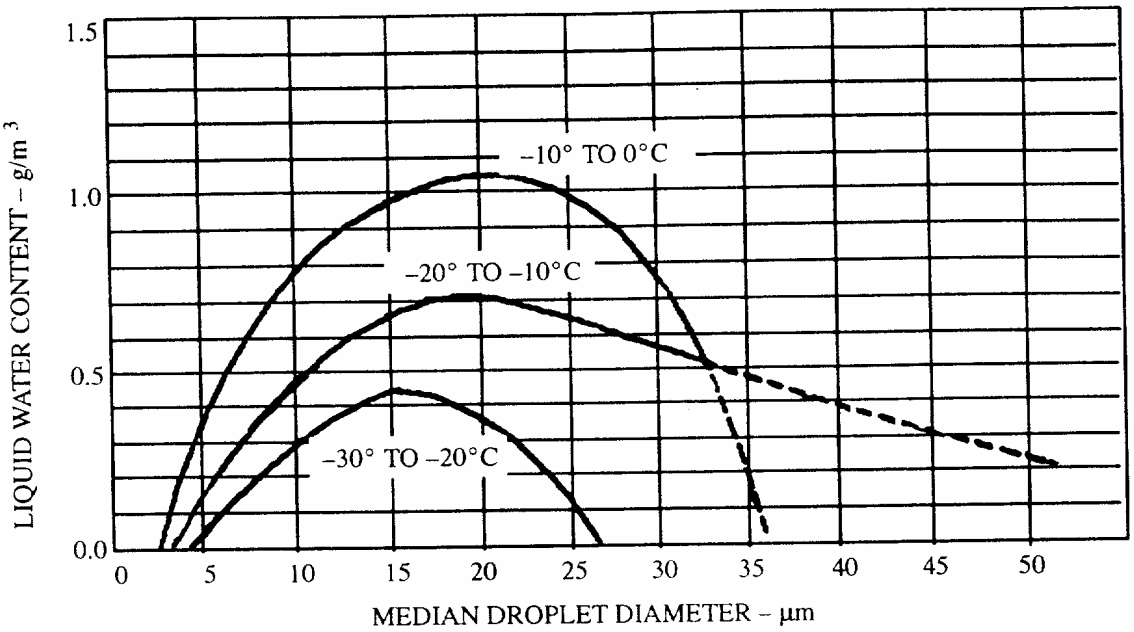


FIGURE 14b. Engine surface emissivity.

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ALTITUDE: Sea Level to 6.7 km (22,000 feet)
MAXIMUM VERTICAL EXTENT: 2.0 km (6,500 feet)



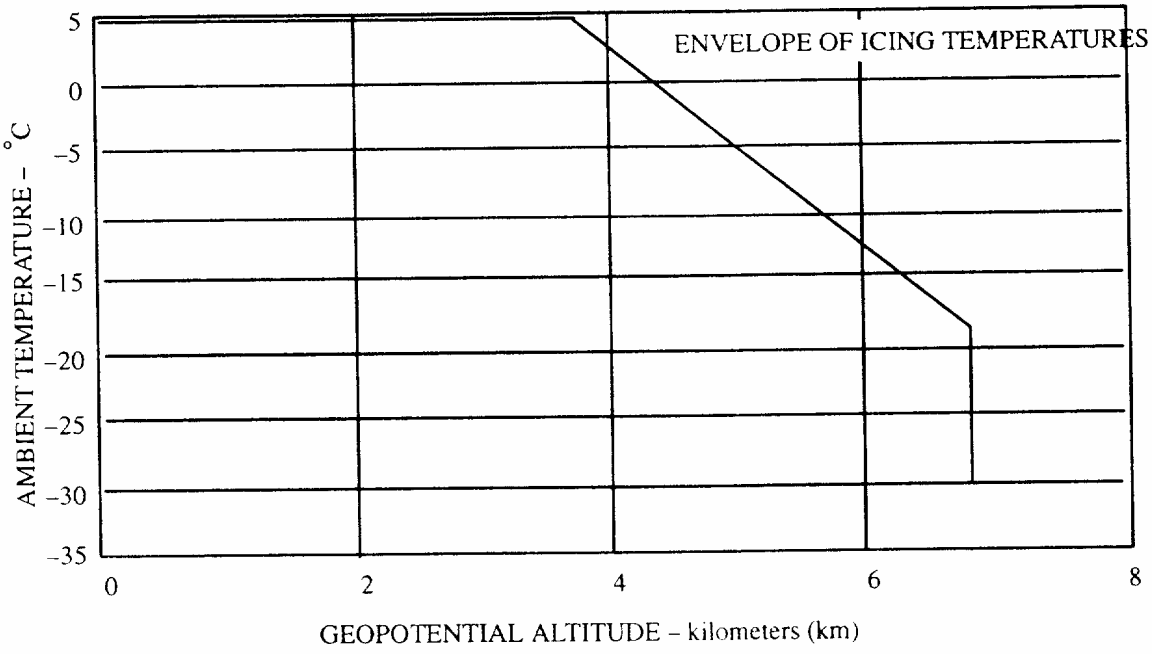


FIGURE 15a. Continuous maximum icing conditions.

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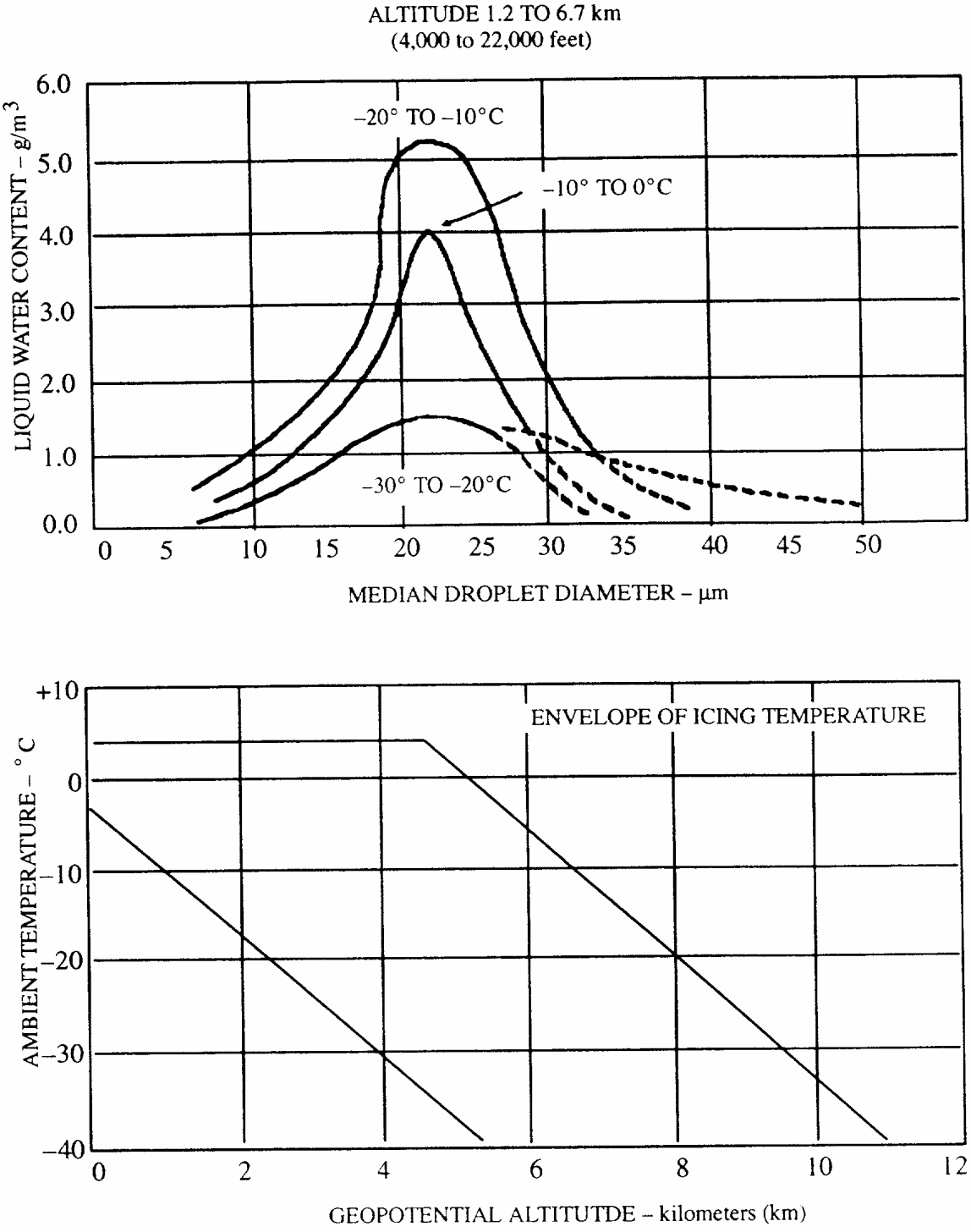


FIGURE 15b. Intermittent maximum icing conditions.

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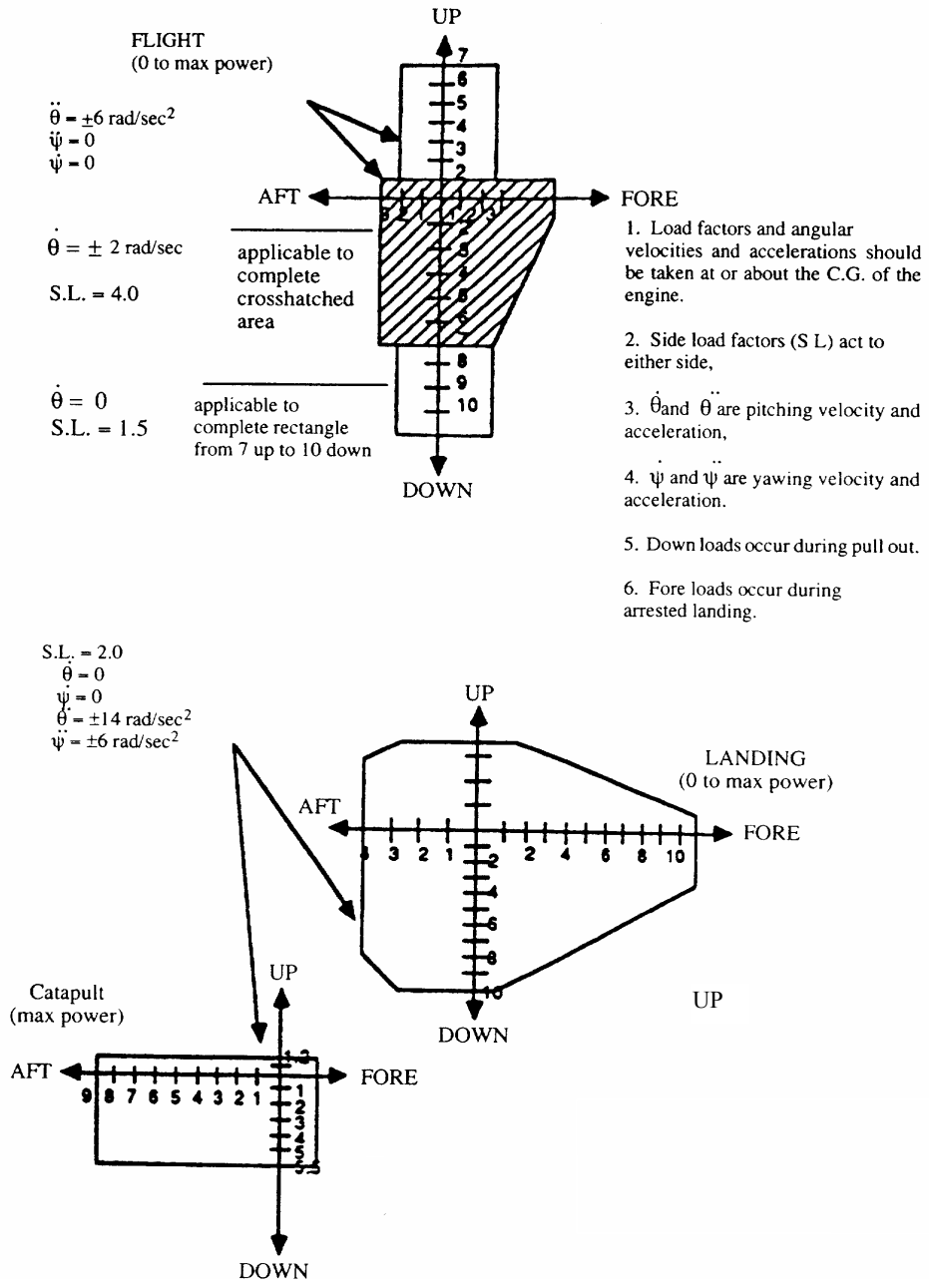
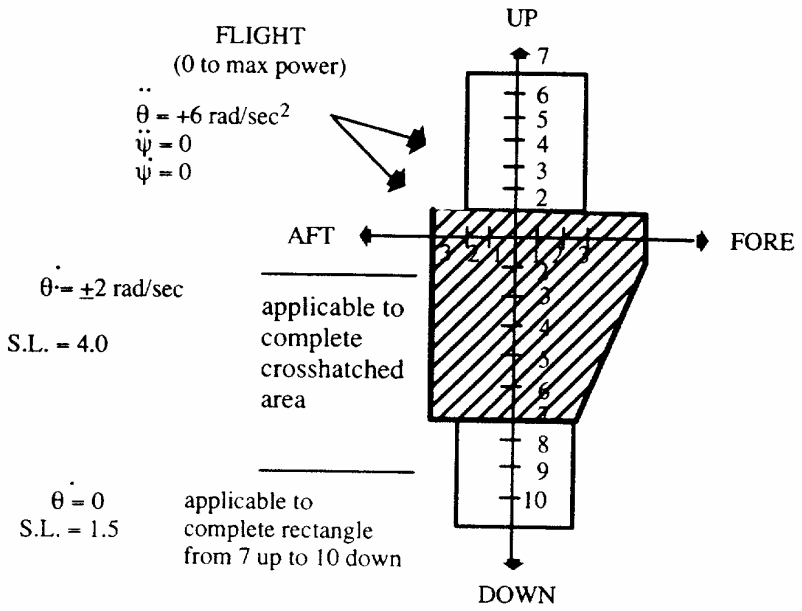


FIGURE 16a. Externally-applied forces (fixed wing).

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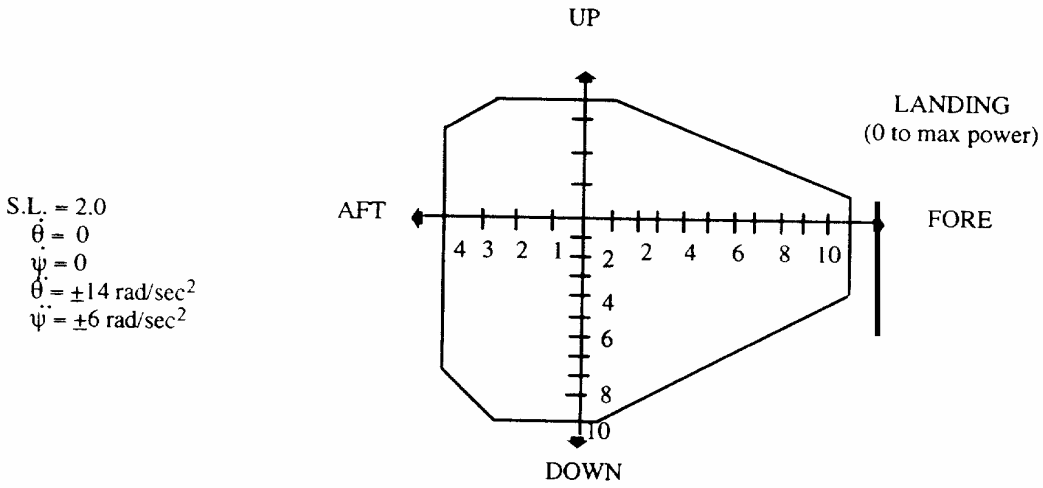
1. Load factors and angular velocities and accelerations should be taken at or about the C.G. of the engine.

2. Side load factors (S.L.) act to either side.

3. $\dot{\theta}$ and $\ddot{\theta}$ are pitching velocity and acceleration.

4. $\dot{\psi}$ and $\ddot{\psi}$ are yawing velocity and acceleration.

5. Down loads occur during pull out.



NOTE: This figure is being revised to reflect current NAVY carrier requirements. Use of this figure should be coordinated with the NAVY.

FIGURE 16b. Externally-applied forces (rotary wing).

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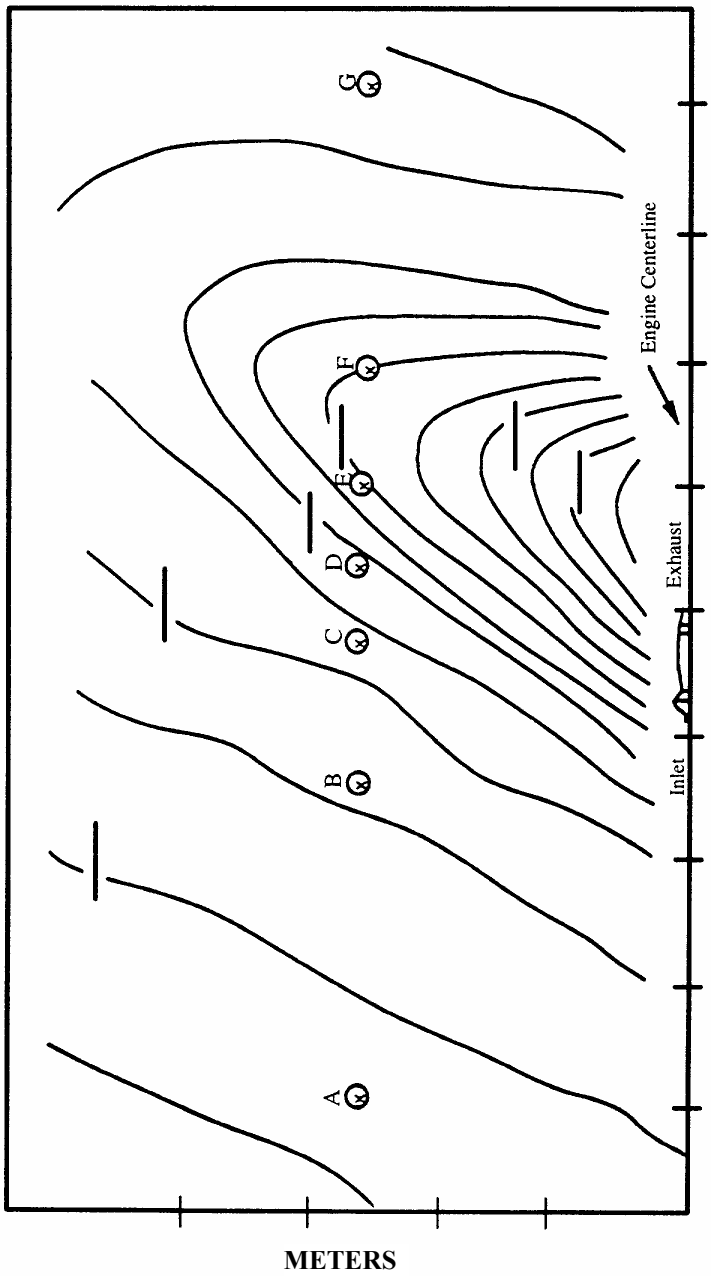


FIGURE 17a. Near field overall sound pressure level contours (dB - ref. 0.0002 μbars).

Near field octave band sound pressure level contours (dB - ref. 0.0002 μbars) center frequency - 250 Hz.

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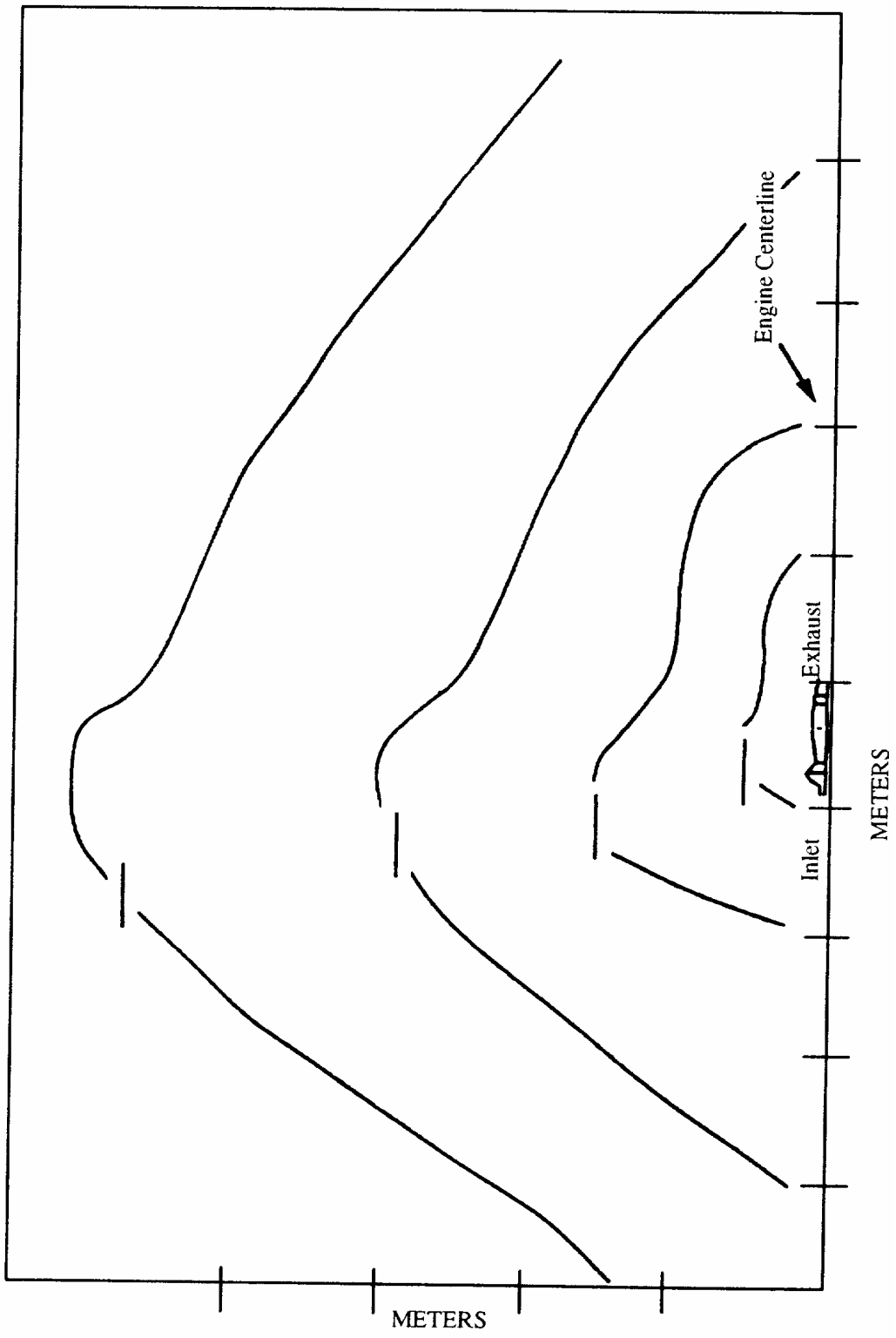


FIGURE 17b. Far field overall sound pressure level contours (dB - ref. 0.0002 μbars).

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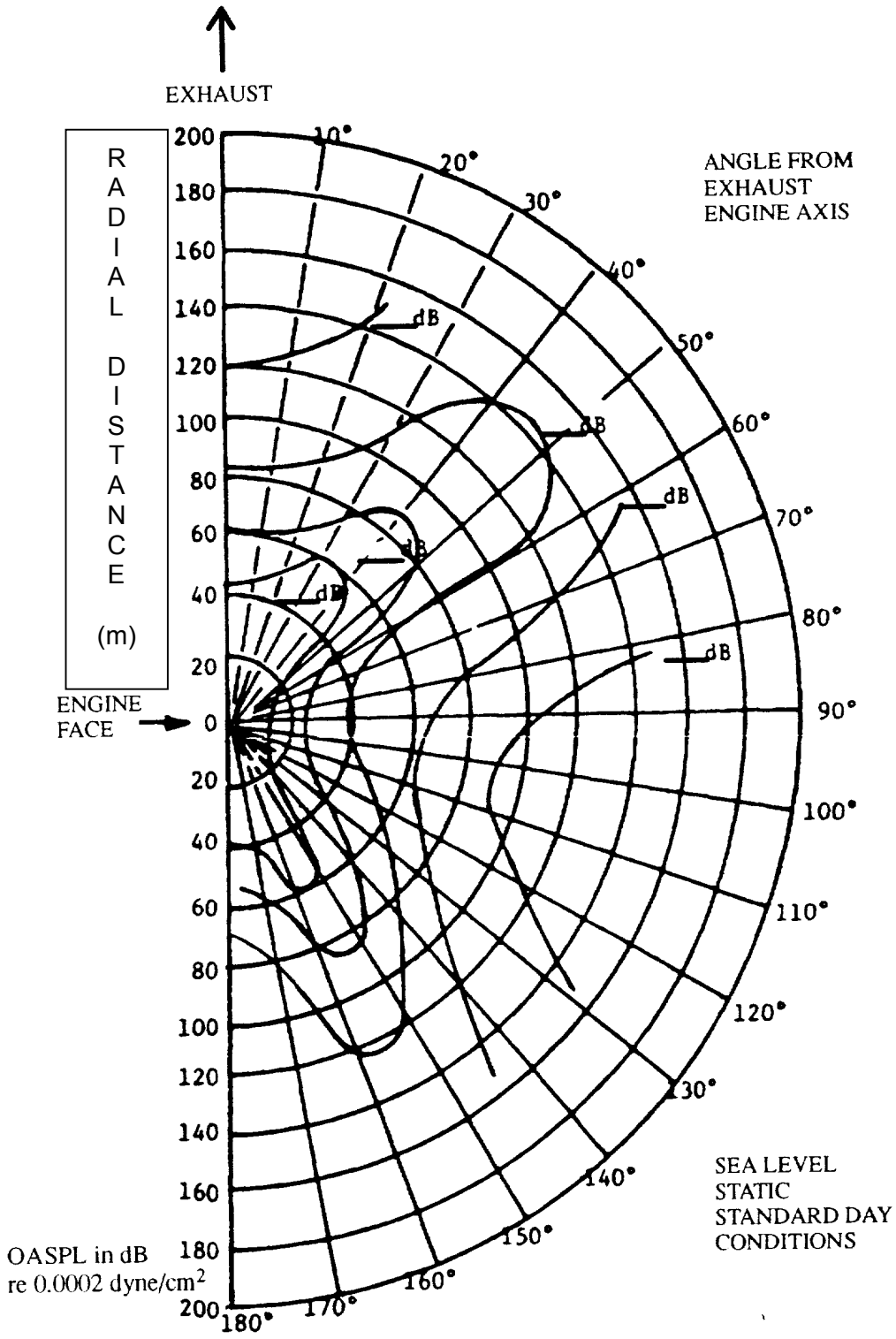


FIGURE 17c. Estimated overall sound pressure level contours at idle.

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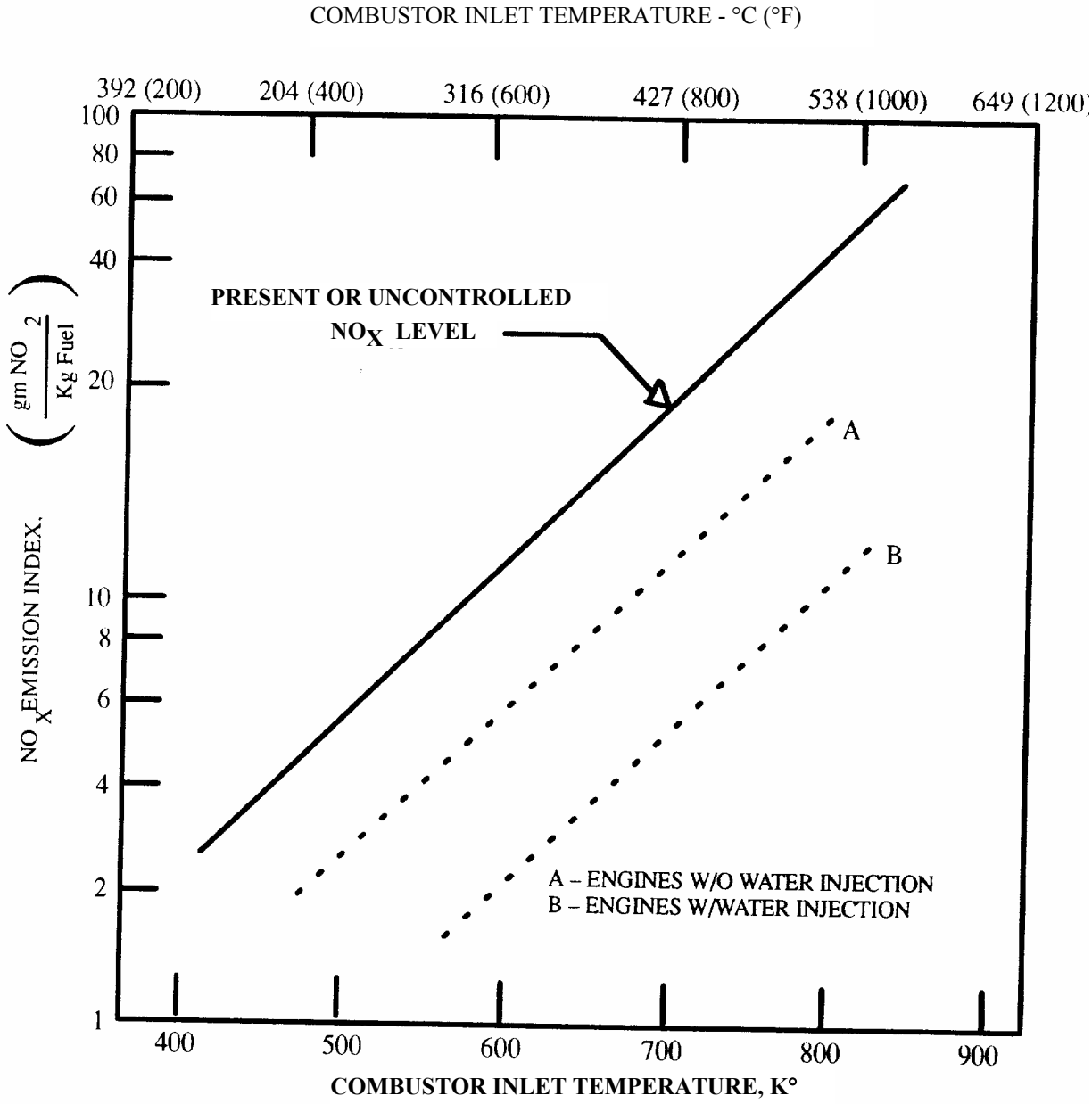


FIGURE 18a. Oxides of Nitrogen.

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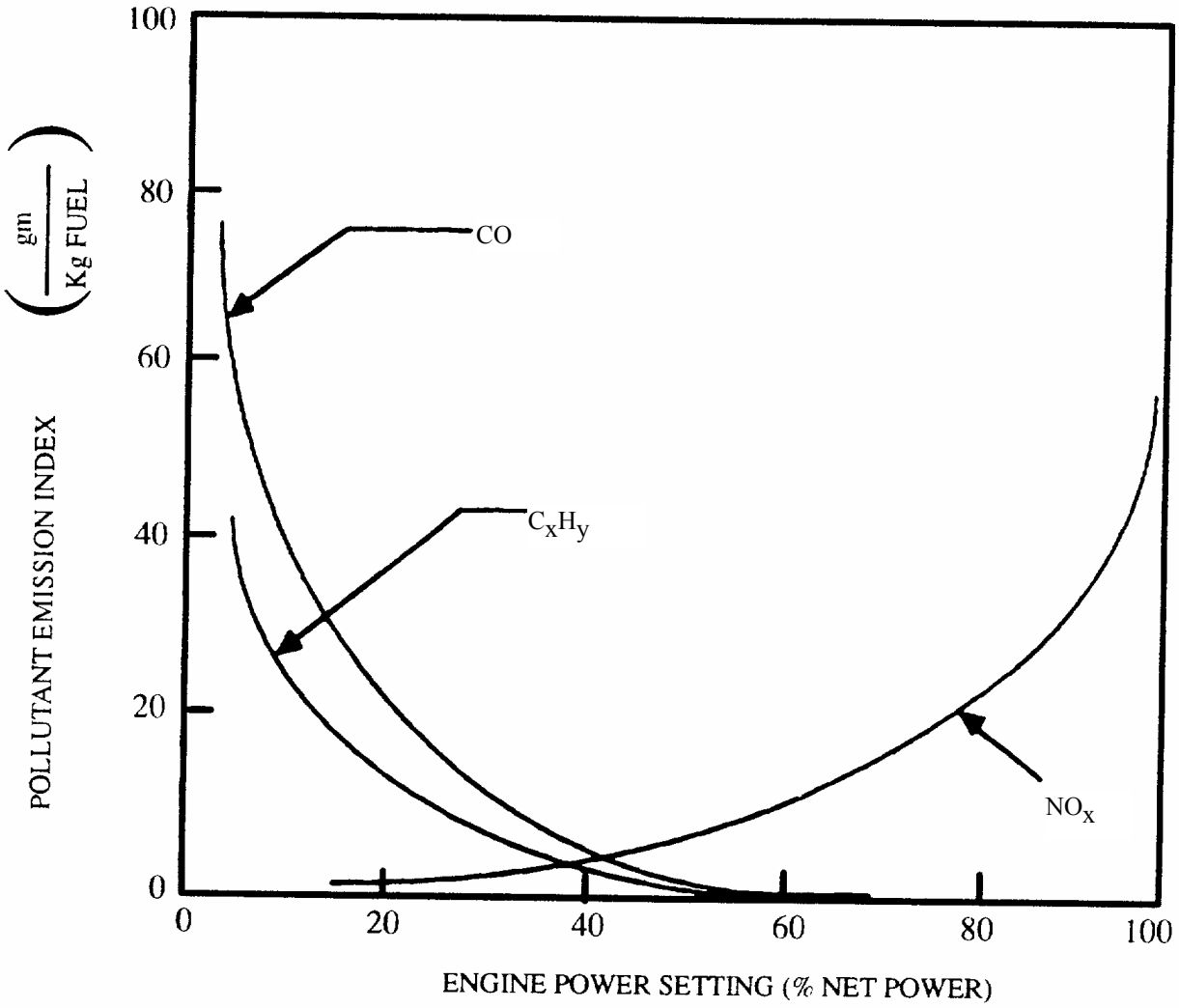


FIGURE 18b. Gaseous emissions.

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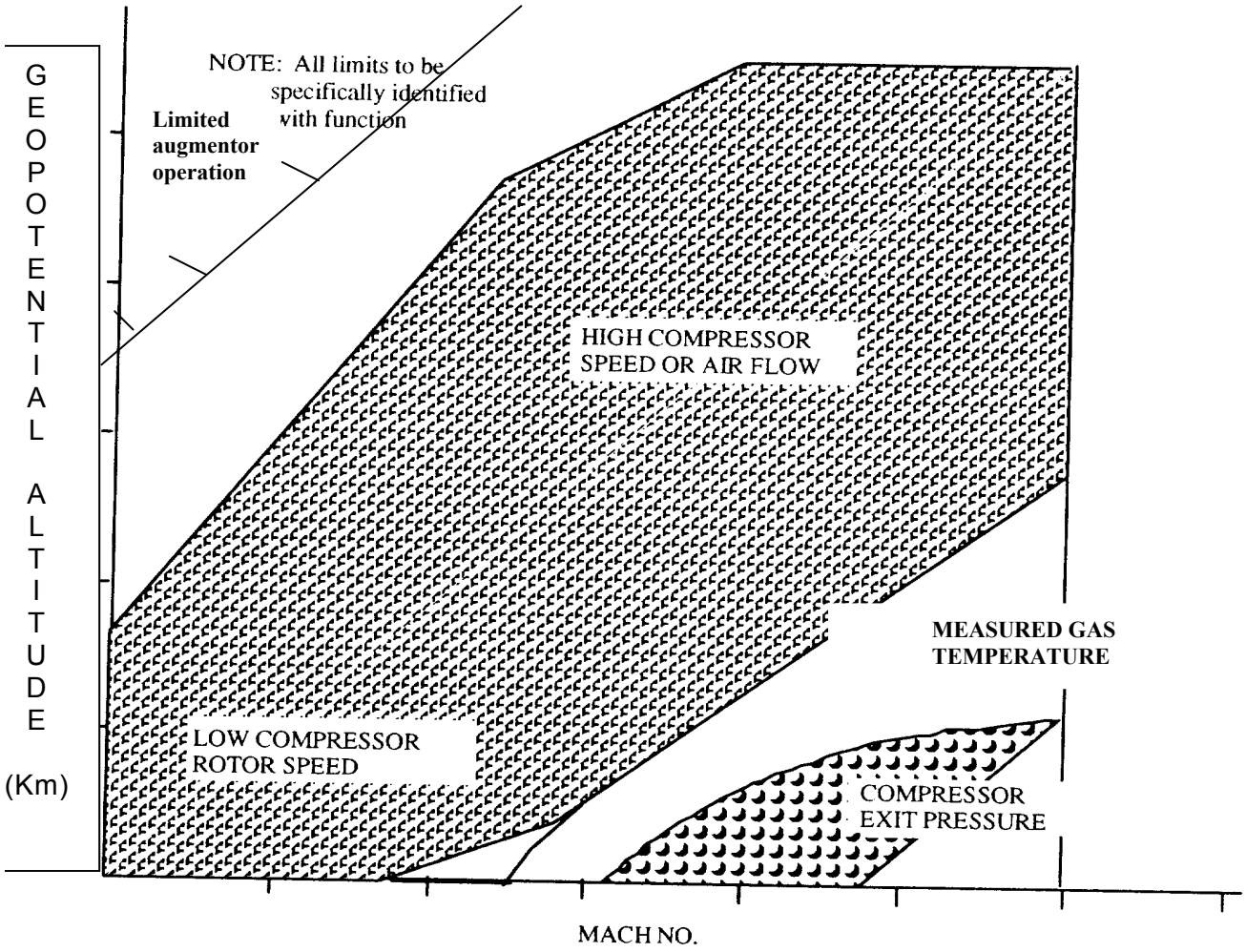


FIGURE 19. Control-limiting regimes.

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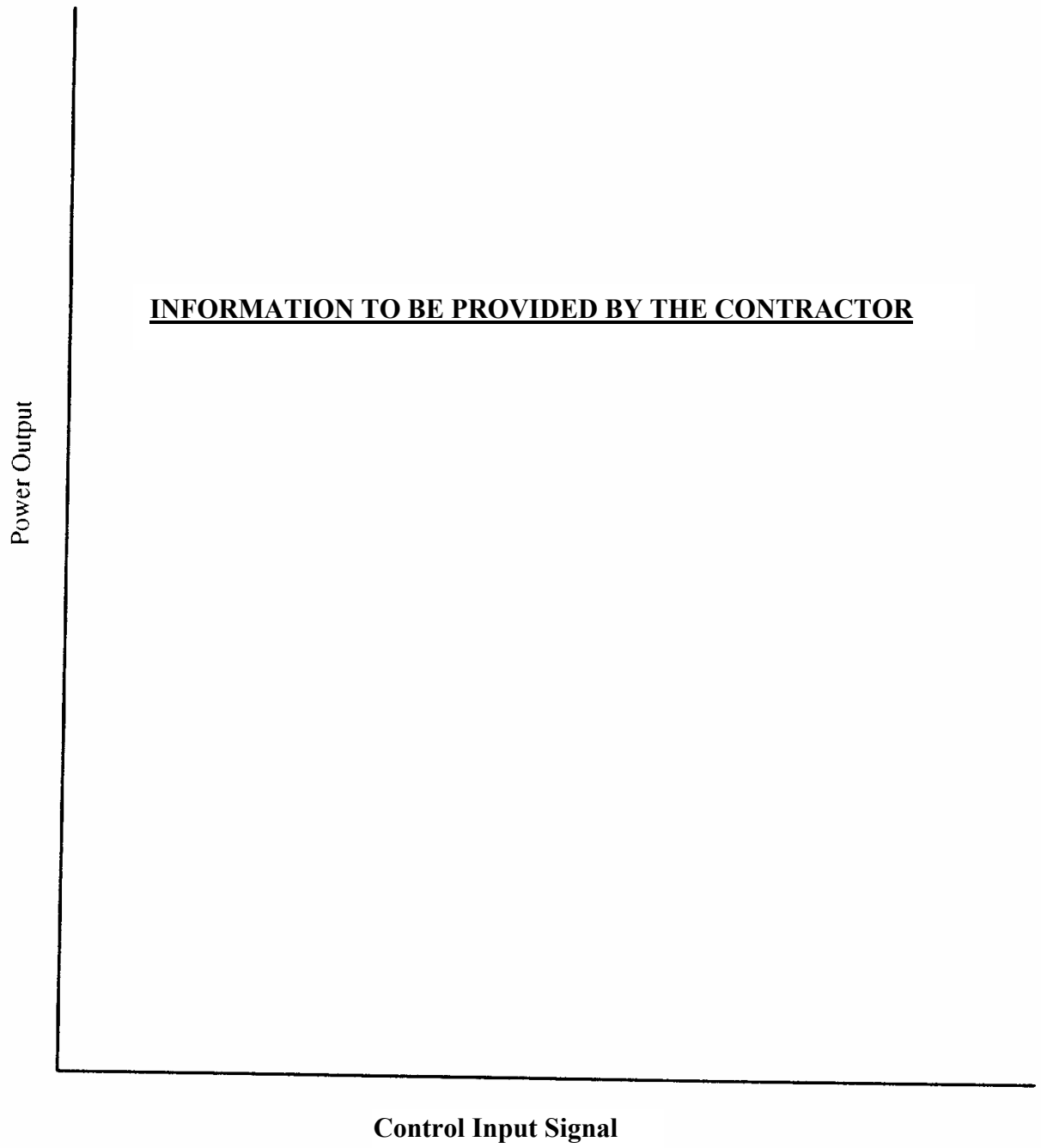


FIGURE 20. Control system performance.

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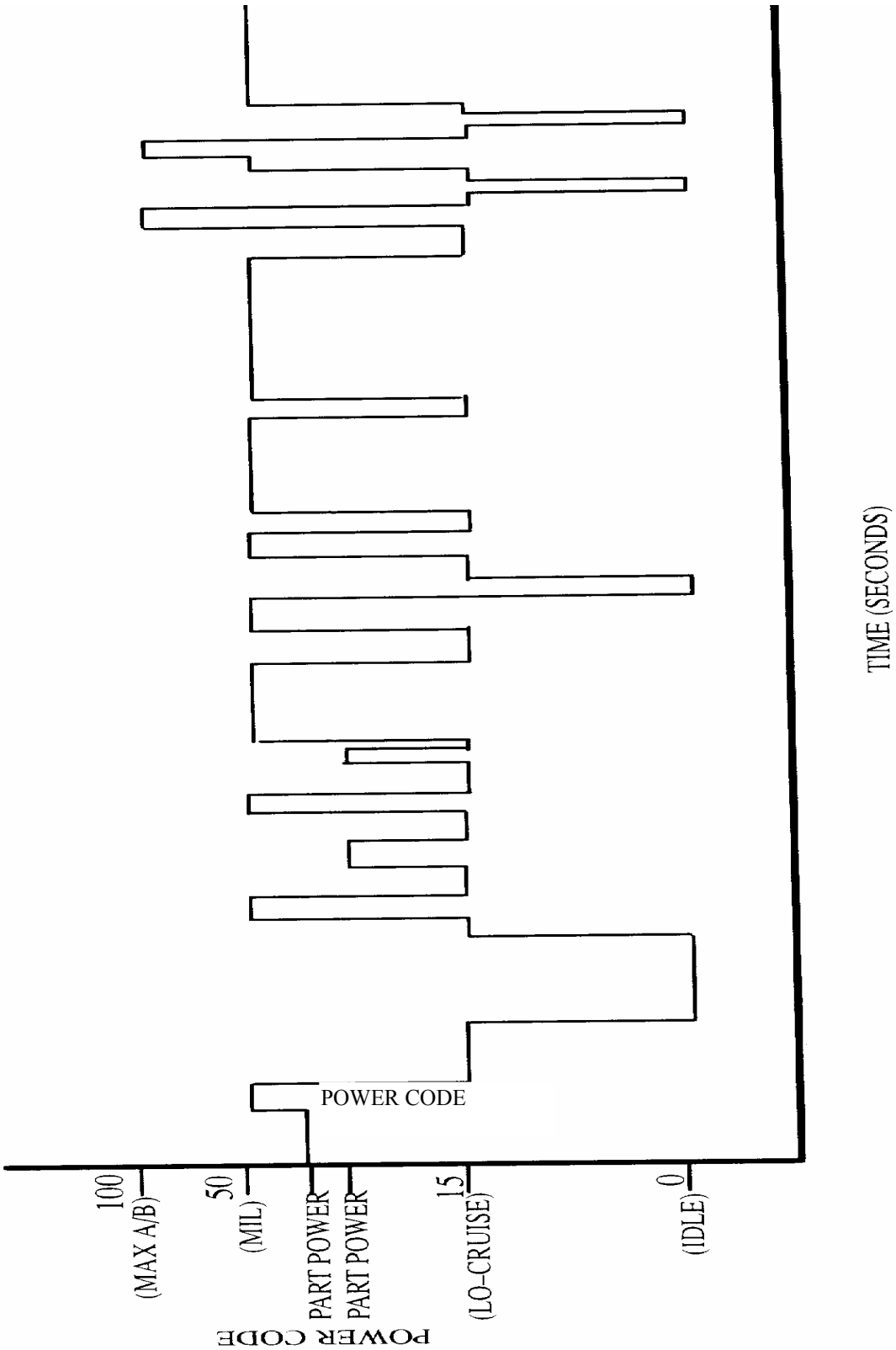


FIGURE 21a. Air to surface mission (example only).

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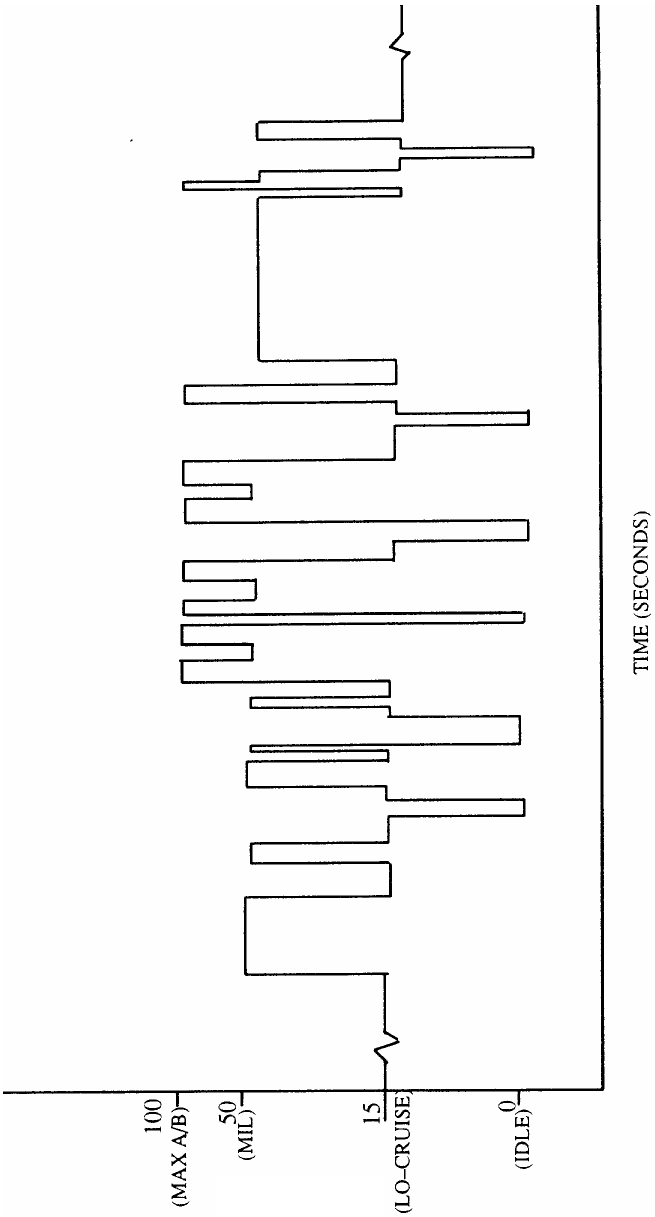


FIGURE 21b. Air combat mission (example only).

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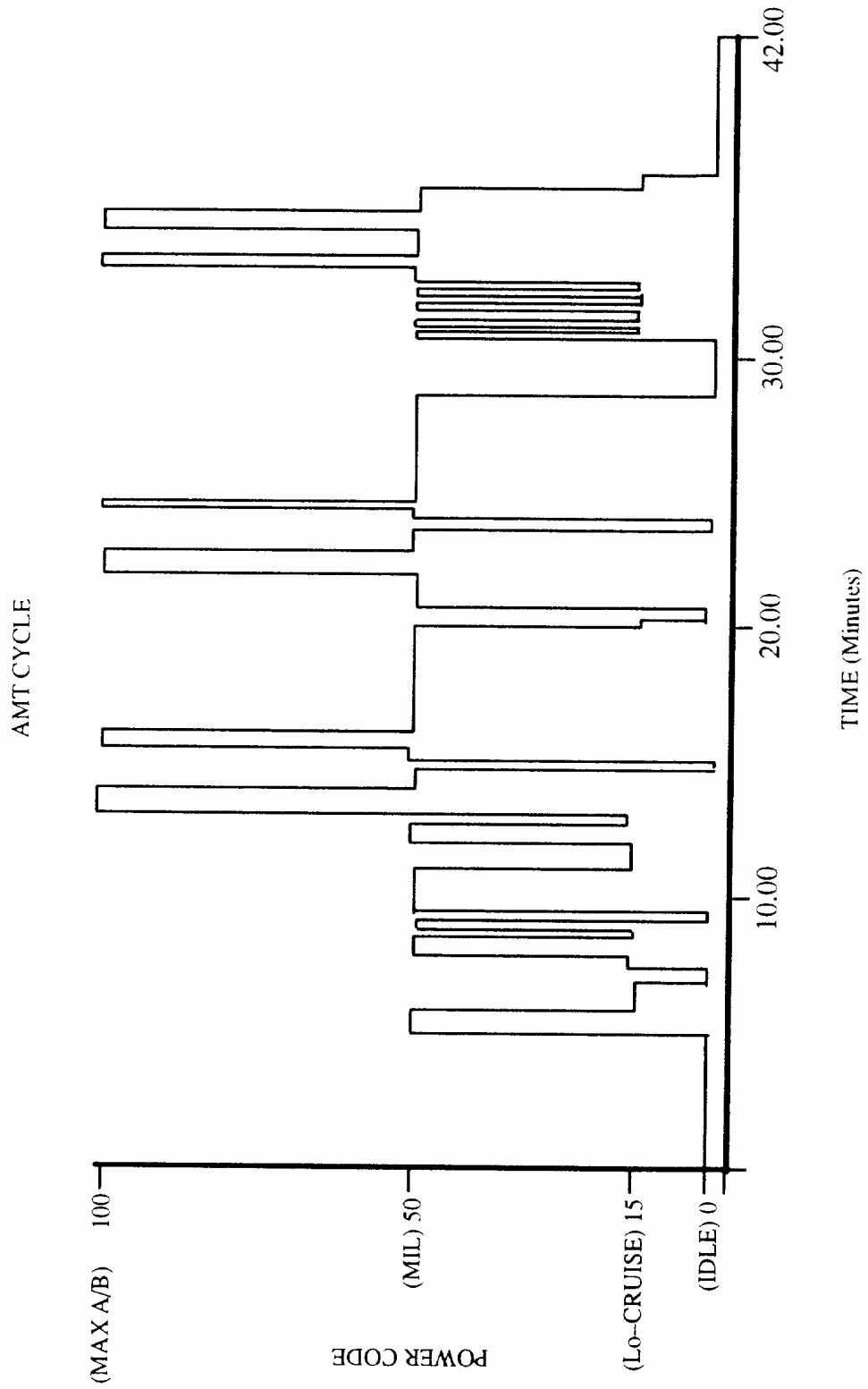


FIGURE 21c. Air combat mission 2 (example only).

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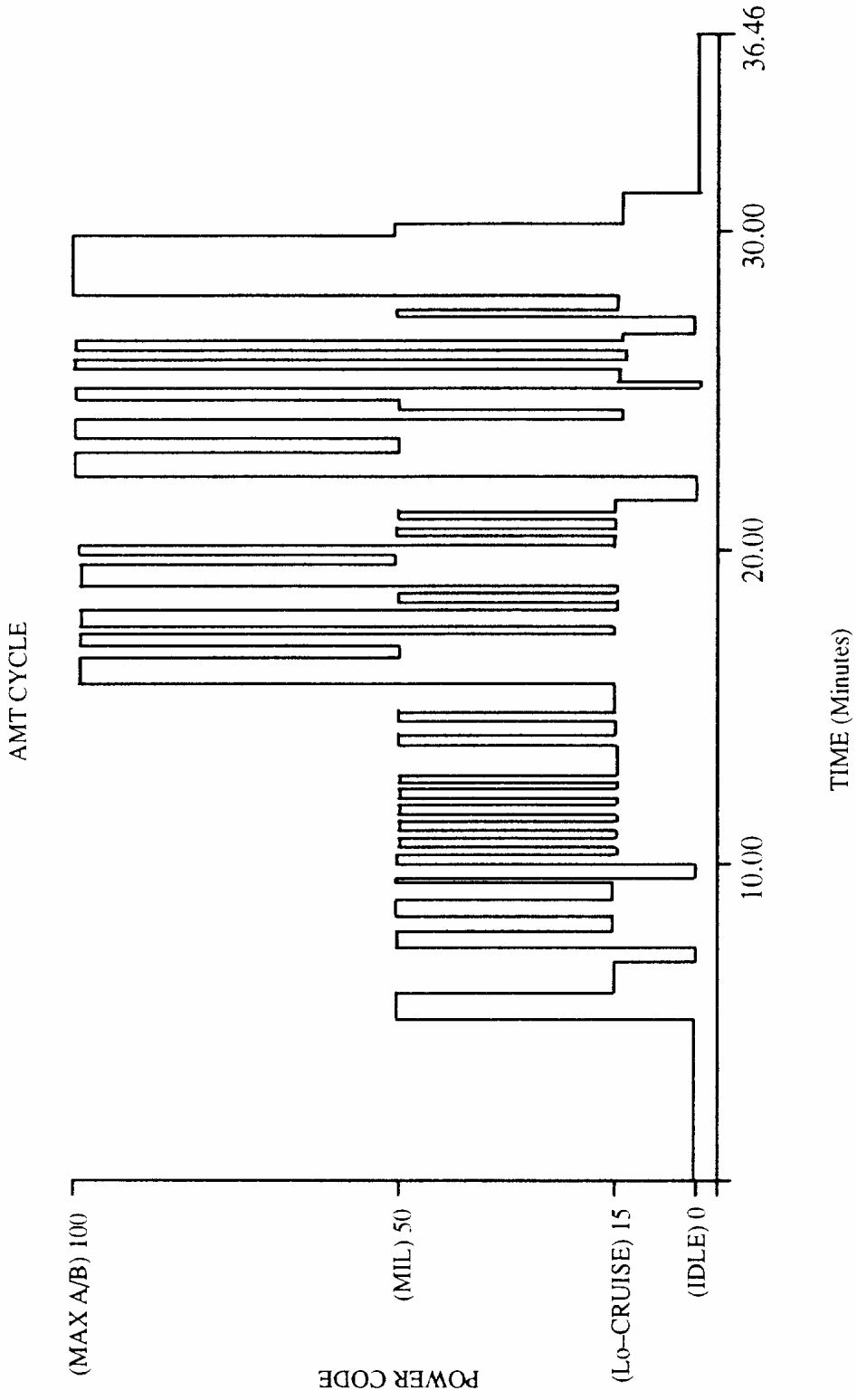


FIGURE 21d. Air combat mission 3 (example only).

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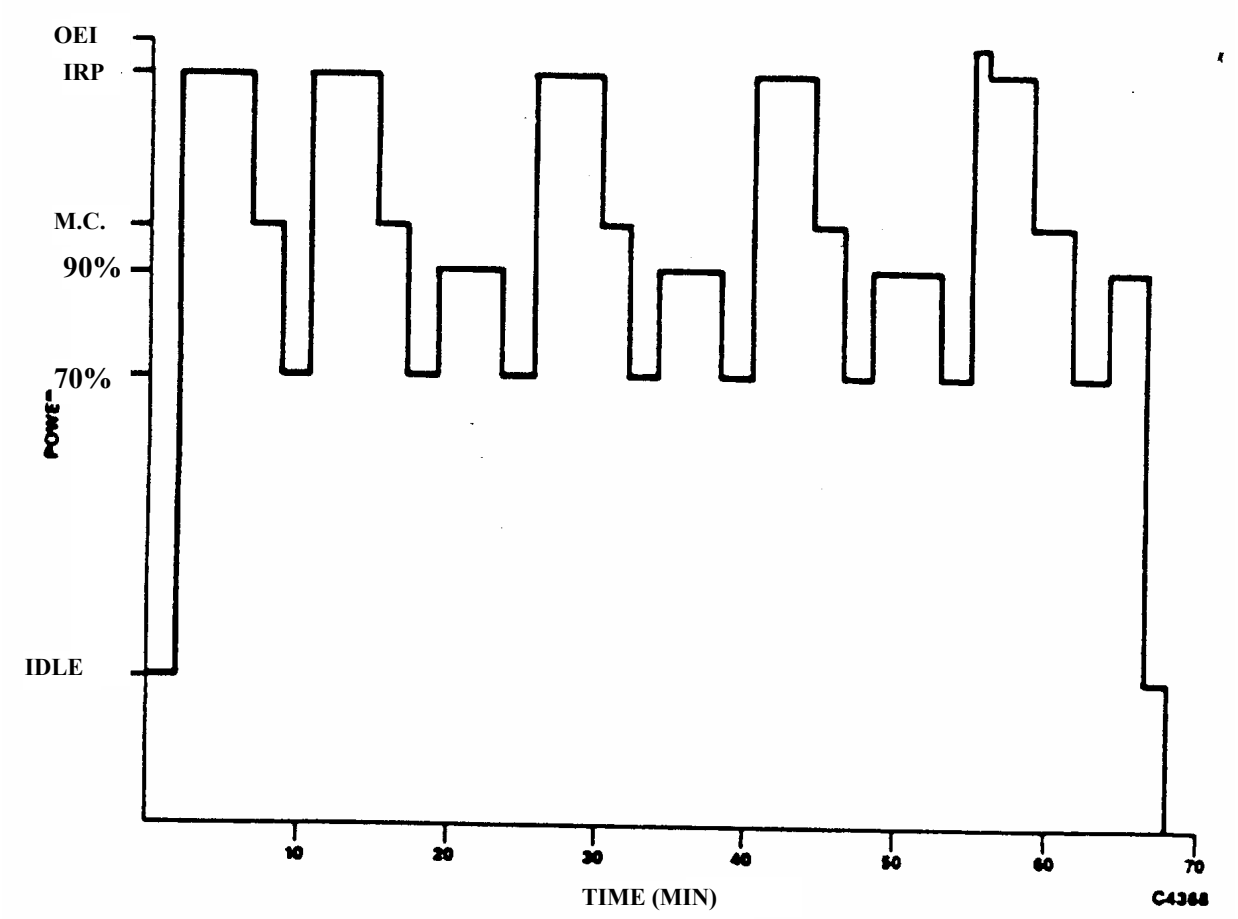


FIGURE 21e. Accelerated composite mission engine duty cycle (example only).

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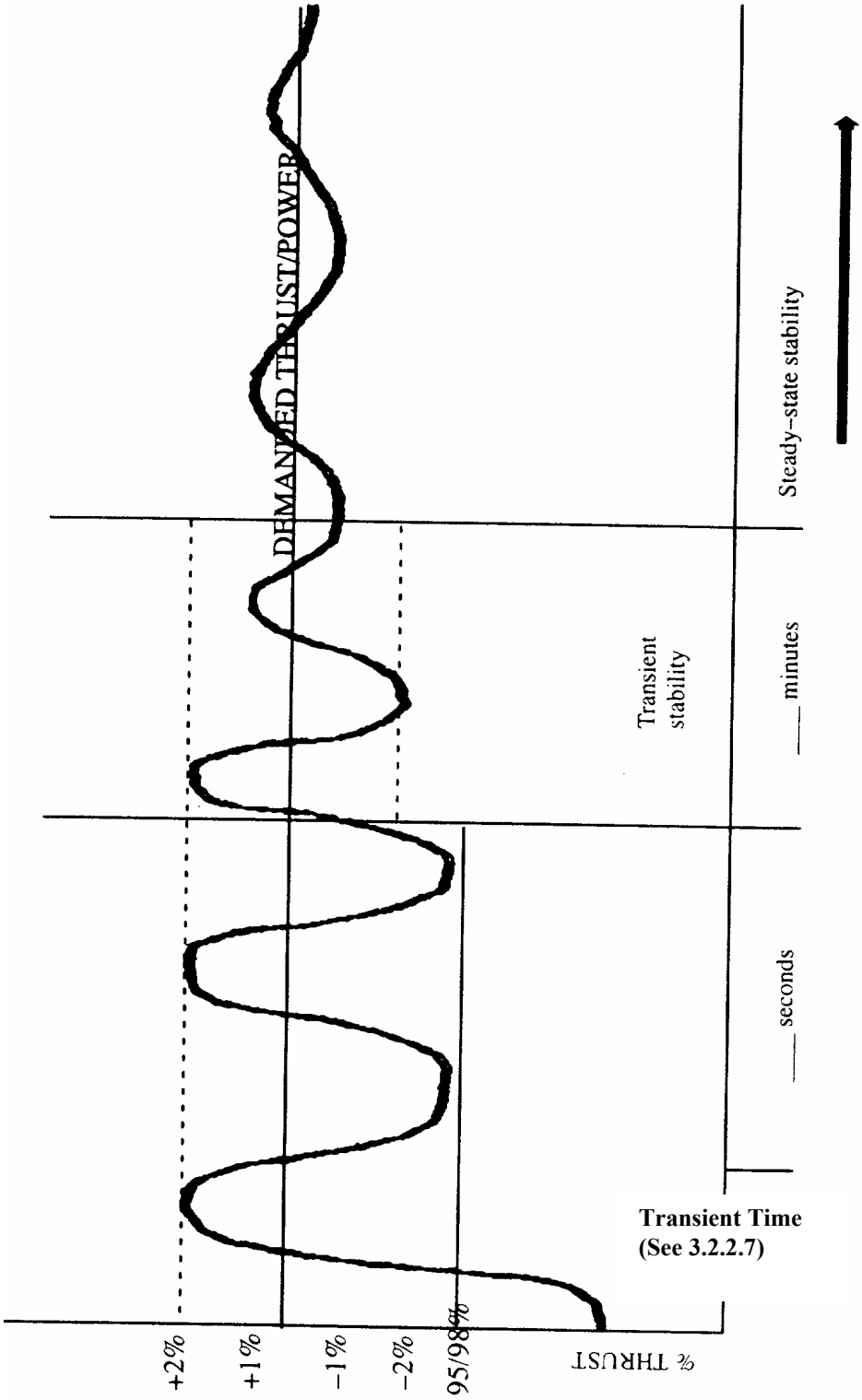
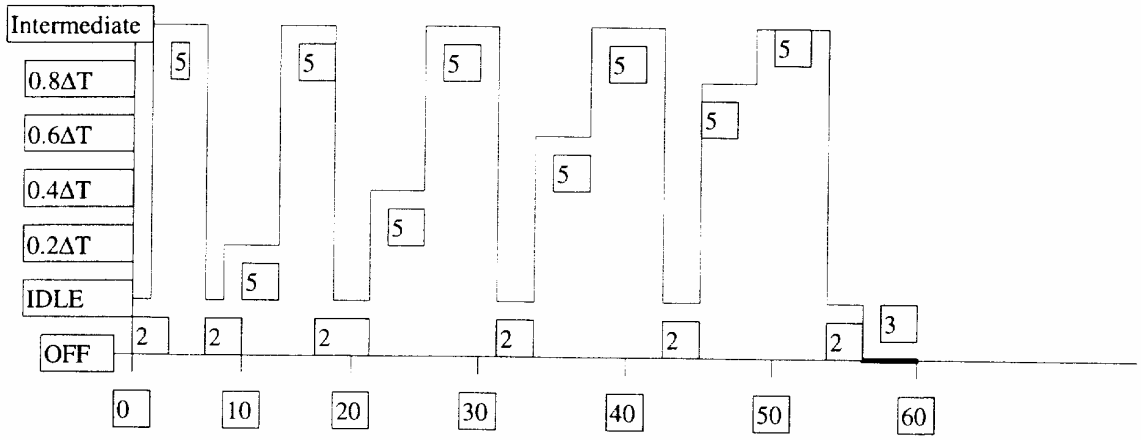


FIGURE 22. Transient.

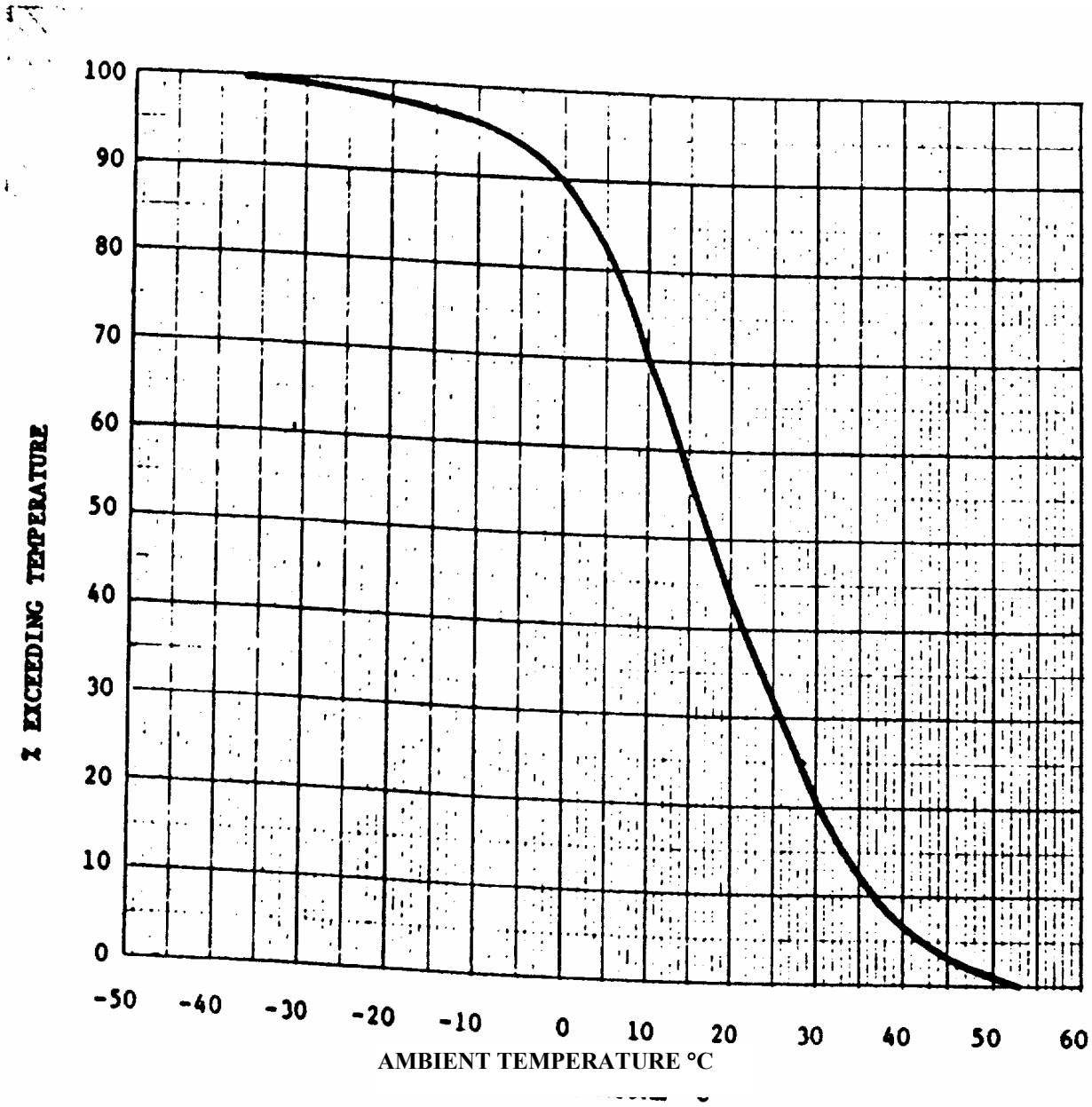
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ΔT = Delta Temperature between Intermediate and Idle

FIGURE 23. Engine corrosion operating cycle.

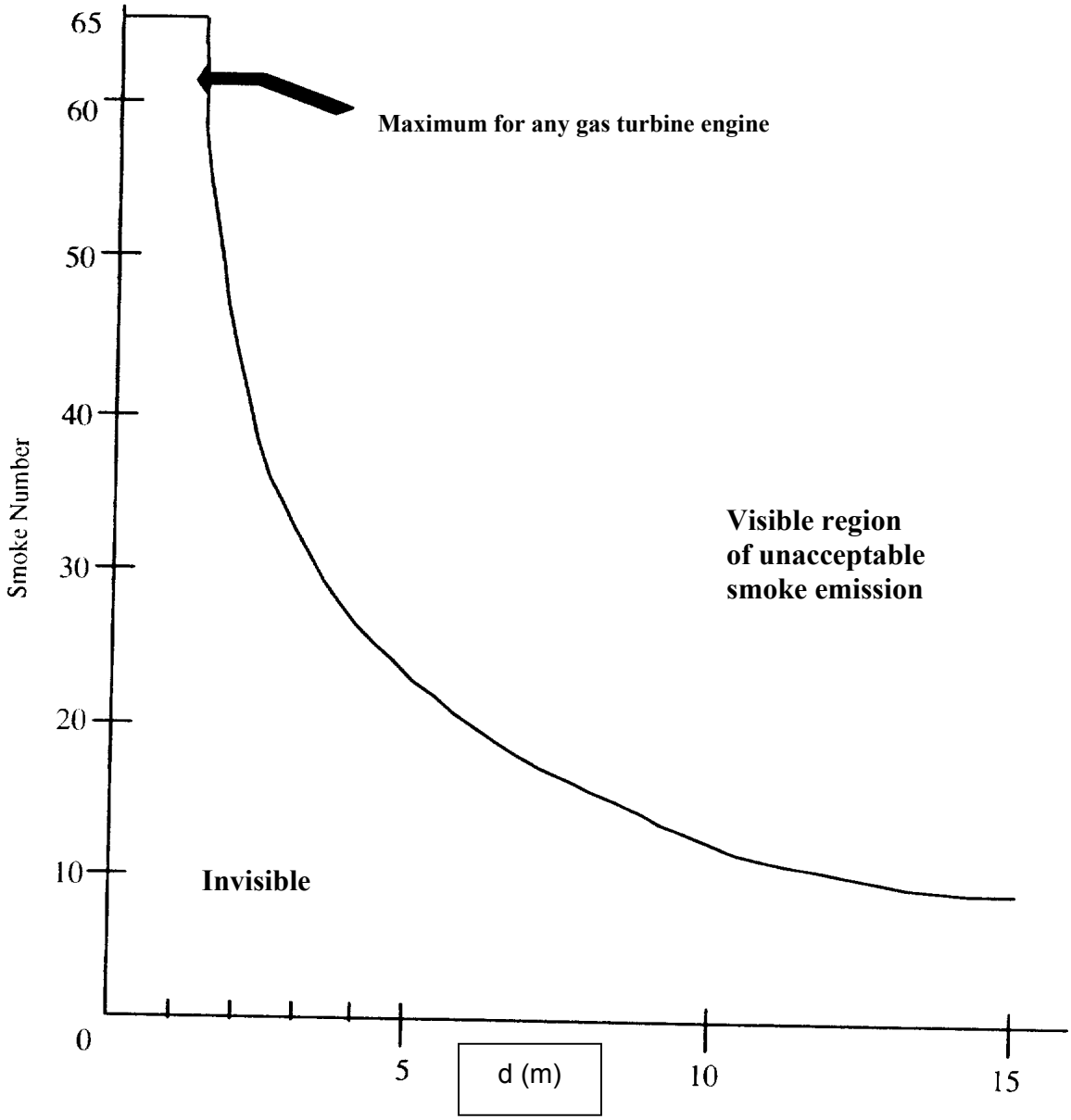
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(See MIL-HDBK-1783 for guidance.)

FIGURE 24. Ambient temperature distribution.

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Diameter, d, is the diameter of the vitiated airflow exhaust nozzle at the engine exhaust exit plane m (ft). The exhaust exit plane is the first downstream plane, normal to the exhaust system that does not contain a solid surface around the stream. For engines with variable area exhaust nozzles, d shall be the mean of values attainable when not using the augmentor. For engines with non-circular exhaust nozzle areas, d shall be the diameter of the smallest circle circumscribing the actual nozzle. For engines with vitiated airflow leaving the engine through an annulus, d shall be the diameter of a circle having the same cross-sectional area as the annular exhaust stream. Smoke may not be visible when viewing the exhaust stream perpendicular to the gas flow.

FIGURE 25. Smoke number.

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Custodians:

Army – AV
Navy – AS
Air Force – 11

Preparing activity:

Air Force – 11
(Project 15GP-0060)

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